GHOST AVIATOR · GROUND SUBJECT

Air Regulations

For the DGCA CPL and ATPL ground examinations. Explained in plain words, with diagrams and official rule references. Questions and recall aids are in the separate companion.

Capt. Pankaj Pahil

Pilot, flight and ground instructor
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Air Regulations

For the DGCA CPL and ATPL ground examinations

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Air Regulations: For the DGCA CPL and ATPL ground examinations
First edition, 2026.

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Important. This book is a study aid. Rules change. Before you fly or sit an examination, check the current text of the Aircraft Rules, the Civil Aviation Requirements (CARs), the AIP India and the ICAO documents. Where this book and an official document differ, the official document is correct.

This edition is still awaiting an independent review of every figure and question. Report any mistake to the author through ghostaviator.com.

Preface

Air Regulations is the subject that many students try to learn by heart and then lose within a week. The reason is simple: the rules are written for lawyers and regulators, not for a student sitting with a book at night. This book tries to do the opposite. Each rule is explained in plain words first, and the exact rule follows, so that you understand why it exists and can then remember it.

The chapters explain Air Regulations and Human Performance topics for student pilots. Where practice questions are provided, use them to test your understanding. Indian rules must be read against the applicable DGCA requirements and AAI AIP India, together with the relevant ICAO provisions. Medical guidance and basic physiology explain how the body works; they are not substitutes for Indian licensing requirements.

When a word sounds difficult, such as rectification, endorsement or cabotage, the book gives its simple meaning beside it, and the glossary at the back collects them all. Read the plain meaning, then the official wording, then try the questions without looking at the answers.

Fly safe, and study with understanding.

Capt. Pankaj Pahil

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How to Use This Book

Each chapter opens with a short box called **This chapter at a glance**. Read it first: it tells you what the chapter is about and what to remember.

Box	What it means
Blue	A definition or explanation.
Green	A procedure or something you do.
Amber	An examination tip or a number to remember.
Red	A limit, a warning, or a common mistake.

Hard words are explained in brackets the first time they appear in a chapter, and all of them are collected in the **Glossary**. The **Index** at the back lists topics with page numbers.

The syllabus for the Air Regulation paper is printed by the DGCA in **CAR Section 7, Series B** (Part IV for the CPL, Part VI for the ATPL). The inside back pages of this book show where each syllabus heading is taught.

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PART I

The Law Behind Flying

How international law, ICAO and the Indian system fit together, and how an aircraft gets its nationality.

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CHAPTER 1

International Organisations and Conventions

01

ARTICLE 29 — ESSENTIAL AIRCRAFT DOCUMENTS (HIGH EXAM PRIORITY)

Every aircraft of a contracting State **engaged in international navigation** shall carry the following **7 documents**:

#	Document	Notes
a)	**Certificate of Registration**	Proof of aircraft nationality
b)	**Certificate of Airworthiness (CofA)**	Aircraft is airworthy
c)	**Appropriate Crew Licenses**	For each member of the crew
d)	**Journey Log Book**	Aircraft journey record
e)	**Aircraft Radio Station License**	Only if equipped with radio apparatus
f)	**Passenger List** (Names, Embarkation & Destination)	Only if it carries passengers
g)	**Cargo Manifest & Declarations**	Only if it carries cargo

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Paris Convention, 1919

DEFINITION / KEY IDENTITY

The Paris Convention was the **first truly international agreement on Civil Aviation**. Its full name is the *Convention Relating to the Regulation of Aerial Navigation*.

DATE **Signed: 13 October 1919**	Signed at the **Peace Conference held in Paris** under the auspices of the **League of Nations**.
BODY **ICAN Established**	Created the **International Commission on Air Navigation (ICAN)** — considered the *forerunner (predecessor) to ICAO*.
LEGACY **What did it do?**	Laid down **preliminary technical standards** for international civil aviation. **Later superseded** by the Convention on International Civil Aviation (Chicago Convention) signed on **7 December 1944**.

EXAM TRAP – ICAN VS ICAO

ICAN (International Commission on Air Navigation) was created by the **Paris Convention 1919**.
ICAO (International Civil Aviation Organization) was created by the **Chicago Convention 1944**.
These are two *different bodies*. ICAN is the predecessor; ICAO replaced it.

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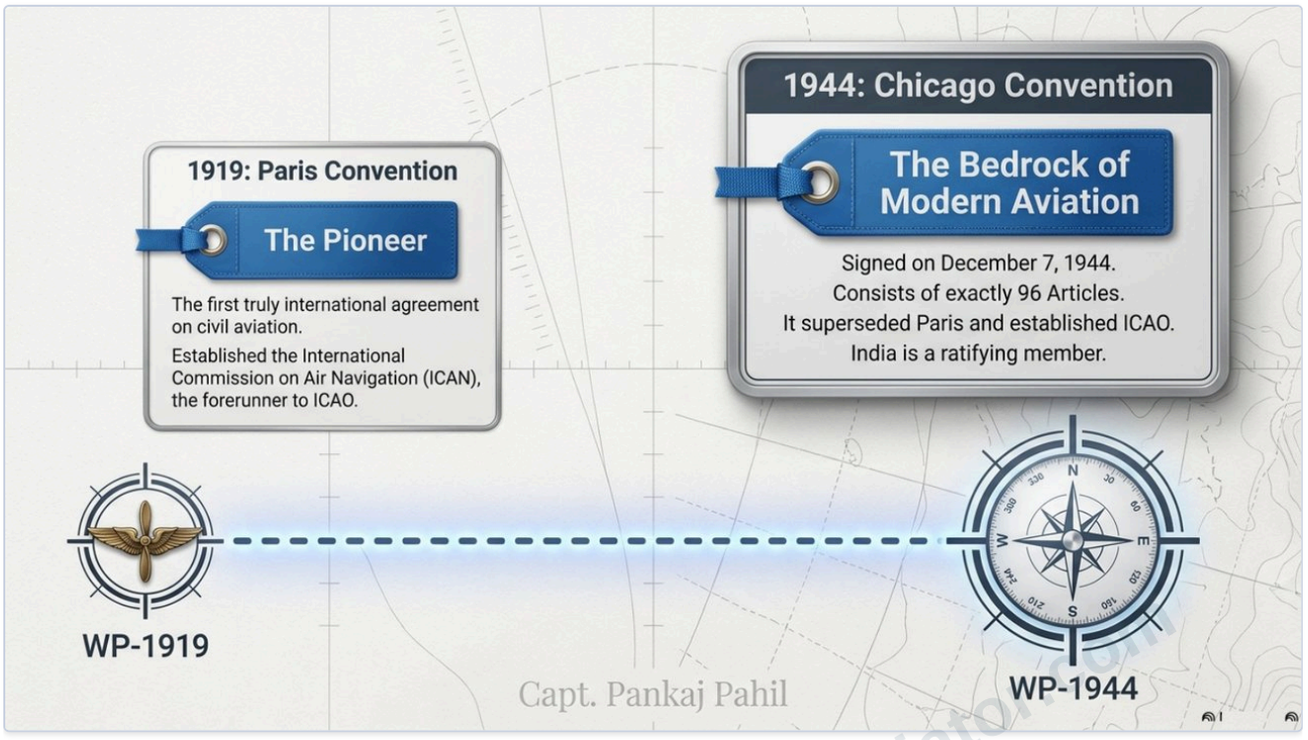

Figure 1.1 *From Paris 1919 to Chicago 1944*

Chicago Convention, 1944 — The Bedrock of Modern Aviation

QUICK IDENTITY FACTS

- **Full Name:** Convention on International Civil Aviation
- **Date Signed: 7 December 1944**
- **Number of Articles: Exactly 96 Articles**
- **India's Status:** India has **RATIFIED** ***(formally approve a treaty so that it binds the State)*** this convention
- **Superseded:** The Paris Convention, 1919
- **Created:** ICAO (International Civil Aviation Organization)

Air Navigation — Articles 1–16

ART. 1 **Sovereignty**

Every State has **complete and exclusive sovereignty** ***(complete power of a State over its own territory)*** over the **airspace above its territory**. This is the foundational principle of international aviation law.

> **Memory:** "Complete AND Exclusive" — both words are examinable. Not "partial", not "shared".

ART. 2 **Territory**	For the purposes of this Convention, the **territory** of a State shall be deemed to be: • The **land areas** • AND the **territorial waters** adjacent thereto *Under the sovereignty, suzerainty, protection or mandate of such State.*

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ART. 3 Civil and State Aircraft

Applies TO (Civil Aircraft):

- All civilian aircraft used for private or commercial purposes.

Does NOT Apply TO (State Aircraft):

- Aircraft used in **military** services
- Aircraft used in **customs** services
- Aircraft used in **police** services

CRITICAL RULE

No state aircraft of a contracting State shall fly over the territory of another State **without authorization by special agreement** or otherwise.

NOTE

When States issue regulations for their state aircraft, they must have due regard for the **safety of navigation** of civil aircraft.

ART. 4 Misuse of Civil Aviation	Each contracting State agrees **not to use civil aviation** for any purpose **inconsistent with the aims** of this Convention.
ART. 5 Right of Non-Scheduled Flight	Aircraft engaged in **non-scheduled flights** enjoy the right to: • Fly into or across the territory of another State • Make stops for **non-traffic purposes** (1st & 2nd Freedom) *However,* the State flown over can require the aircraft to **land** and follow prescribed *(laid down by the authority or the rules)* routes.

ART. 6 Scheduled Air Services

No scheduled international air service may be operated over or into the territory of a contracting State **except with special permission** or other authorization of that State.

Permission = Mandatory for scheduled flights.

ART. 7 Cabotage

Each contracting State has the **right to REFUSE permission** to aircraft of other States to take on passengers, mail, and cargo carried for remuneration **destined for another point within its own territory**.

Cabotage ***(carrying passengers or cargo between two places inside one country)*** **= DOMESTIC air services.**
(Transport within the same State, for hire, by a foreign carrier.)

ART. 8 Pilotless Aircraft (Drones/UAVs)	No aircraft capable of being flown **without a pilot** shall fly over the territory of a contracting State **without special authorization** by that State. Such flights must be **controlled** to avoid danger to civil aircraft.

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ART. 9 **Prohibited Areas**	Each contracting State may **restrict or prohibit** flying over certain areas of its territory for reasons of **military necessity or public safety**. Such areas must be applied **uniformly** to all aircraft.
ART. 10 **Landing at Customs Airport**	The State can require that landing be made at a **designated customs airport**. Similarly, departure from the territory can be required to be from a designated customs airport.
ART. 11 **Applicability of Air Regulations**	All aircraft are to be treated **equally** while operating from or in the territory of a nation, **irrespective of their nationality**.

ART. 12 Rules of the Air

Each contracting State undertakes to adopt measures ensuring:

- Every aircraft flying over or manoeuvring within its territory complies with the **rules and regulations relating to flight and manoeuvre** in force there.
- States keep their own regulations **uniform to the greatest possible extent** with those established under this Convention.

> **HIGH-SEAS RULE (ART. 12)**
>
> **Over the HIGH SEAS, the rules in force shall be those established under THIS Convention.** (ICAO rules govern — not the rules of any individual state.)

ART. 13 **Entry & Clearance Regulations**	Laws relating to **entry, clearance, immigration, passports, customs, and quarantine** shall be complied with by passengers, crew, and cargo upon entrance into or departure from the territory of that State.
ART. 14 **Prevention of Spread of Disease**	Each contracting State agrees to take effective measures to prevent the spread by air navigation of: **cholera, typhus (epidemic), smallpox, yellow fever, plague**, and such other communicable diseases.
ART. 15 **Airport & Similar Charges**	Airport and similar charges will **not be more than** those paid by its national aircraft. Non-discriminatory charges apply.
ART. 16 **Search of Aircraft**	Appropriate authorities have the right, **without unreasonable delay**, to search aircraft of other contracting States on landing or departure, and **inspect certificates and documents** prescribed by this Convention.

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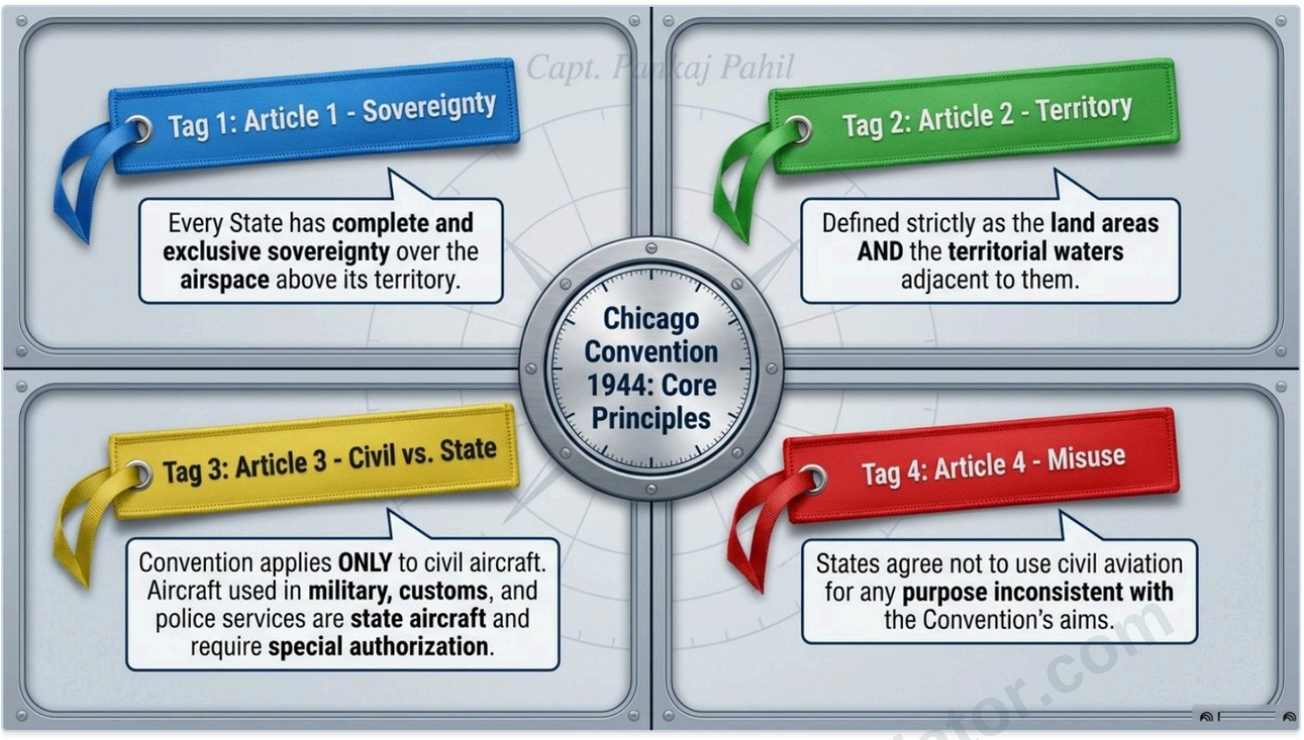

Figure 1.2 *Chicago Convention: Articles 1 to 4*

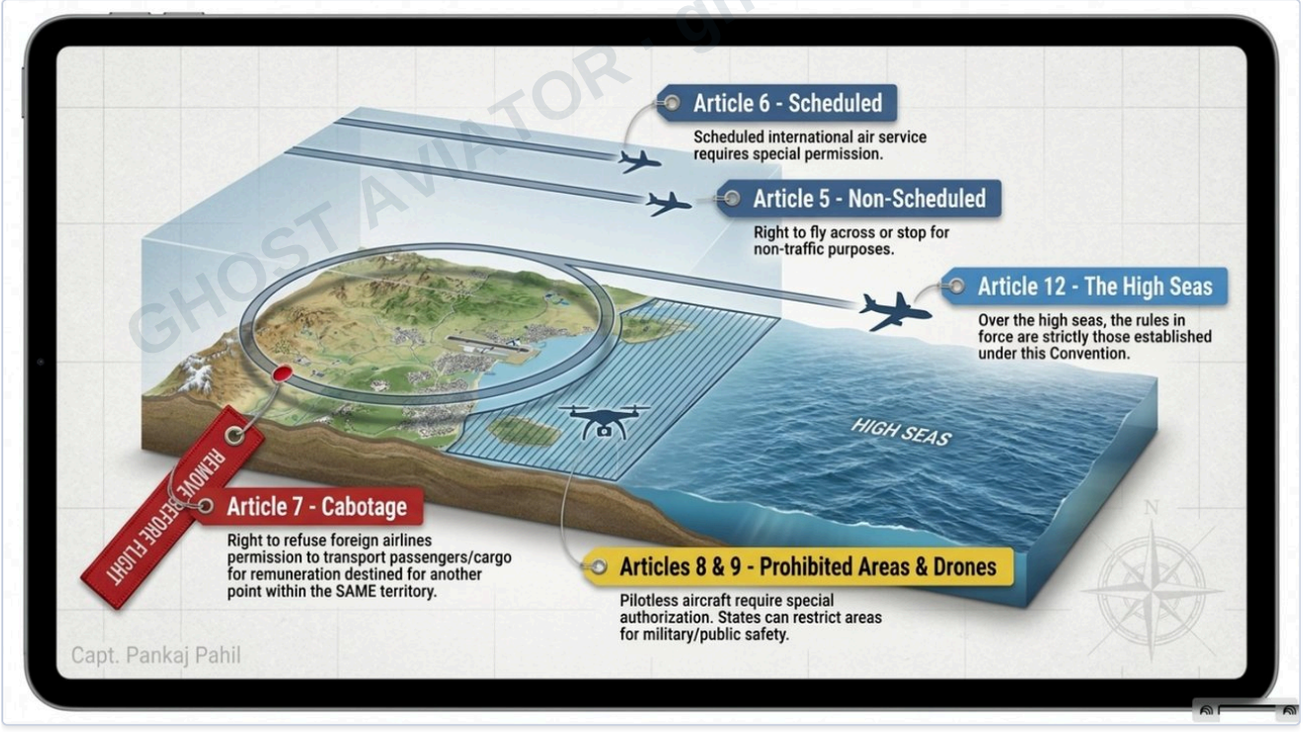

Figure 1.3 *Articles 5, 6, 7, 8, 9 and 12*

Nationality of Aircraft — Articles 17–21

ART. 17 **Nationality of Aircraft**	Aircraft have the **nationality of the State in which they are registered**. (e.g., VT-AXY is Indian because registered in India.)

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ART. 18 **Dual Registration — INVALID**

An aircraft **cannot be validly registered in more than one State at a time**. However, its registration **may be changed** from one State to another.

> DUAL REGISTRATION = INVALID at any given moment.

ART. 19 **National Laws Governing Registration**	Registration or transfer of registration in any contracting State shall be made **in accordance with its law and regulations**.
ART. 20 **Display of Marks**	Every aircraft engaged in **international air navigation** shall bear its appropriate **nationality and registration marks**.
ART. 21 **Report of Registrations**	The registering State must report to ICAO data revealing the **ownership and control** of aircraft it registers. It must also make available to other contracting States, or ICAO, information concerning the registration and ownership of aircraft registered in it, **on demand**.

Measures to Facilitate Air Navigation — Articles 22–28

ART. 22 **Facilitation of Formalities**	Each State agrees to adopt all practicable measures to **facilitate and expedite navigation** between territories — preventing unnecessary delays to aircraft, crews, passengers, and cargo.
ART. 23 **Customs & Immigration Procedures**	Each State undertakes to establish **customs and immigration procedures** affecting international air navigation in accordance with practices established under this Convention.
ART. 24 **Customs Duty**	Aircraft flying to, from, or across the territory shall be admitted **temporarily free of duty**. Also exempt: • Fuel, Oil, Spare parts • Regular equipment & aircraft stores retained on board
ART. 25 **Aircraft in Distress**	Each contracting State undertakes to provide such measures of assistance to **aircraft in distress** in its territory as it may find practicable.
ART. 26 **Investigation of Accidents**	In the event of an accident involving **death, serious injury, or indicating serious technical defect** occurring in the territory of another contracting State: • The **State where the accident occurs** will institute an inquiry. • The **State of registration** shall be given the opportunity to appoint **observers** at the inquiry. • The State holding the inquiry shall communicate the **report and findings** to the State of registration.
ART. 27 **Exemption from Seizure on Patent Claims**	An aircraft **on flight cannot be seized** for violation of patent laws of a State.
ART. 28 **Air Navigation Facilities & Standard Systems**	Each State should provide **airports, navigation facilities, met facilities, radio, charts, maps**, etc., to conform to the convention specifications.

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Conditions to Be Fulfilled With Respect to Aircraft — Articles 29–36

ART. 30 **Aircraft Radio Equipment**	Radios must be **licensed and used in accordance with the regulations of the State of registration**. Radios may only be used by **flight crew members suitably licensed** by the State of registration.
ART. 31 **Certificates of Airworthiness**	Every aircraft engaged in international navigation shall be provided with a CofA **issued or rendered valid by the State in which it is registered**.

ART. 32 Licences of Personnel

(a) The pilot and operating crew of every aircraft engaged in international navigation shall be provided with **certificates of competency and licenses issued or rendered valid by the State of registration**.

> **IMPORTANT CLAUSE (B)**
>
> Each contracting State reserves the right to **refuse to recognize** certificates of competency and licenses granted to **its own nationals** by another contracting State. (State can override for its own pilots flying over its territory.)

ART. 33 **Recognition of Certificates & Licenses**	Certificates and licenses issued by the State of registration shall be **recognized as valid** by other contracting States, **provided** the requirements under which they were issued are **equal to or above the minimum standards** established under the Convention.
ART. 34 **Journey Log Books**	There shall be maintained in respect of every aircraft in international navigation a **journey log book** in which shall be entered particulars of the aircraft, its crew, and of each journey.
ART. 35 **Cargo Restrictions**	No **munitions of war or any item prohibited** to be carried by a State shall be carried.
ART. 36 **Photographic Apparatus**	Each contracting State may **prohibit or regulate the use of photographic apparatus** in aircraft over its territory.

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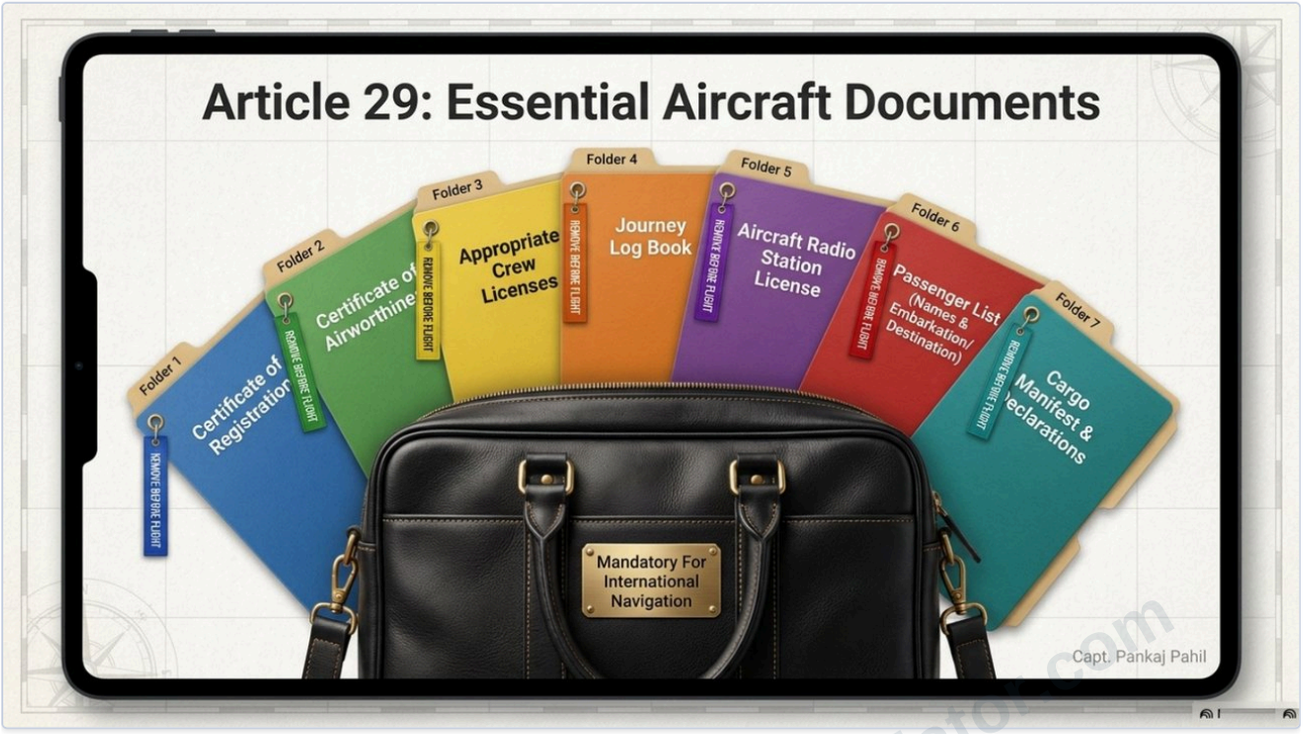

Figure 1.4 *Article 29: the seven documents*

International Standards & Recommended Practices — Articles 37–42

ART. 37 **Adoption of International Standards and Procedures**	Each contracting State undertakes to **collaborate in securing the highest practicable degree of uniformity** in regulations, standards, procedures, and organization in relation to aircraft, personnel, airways, and auxiliary services. To this end, the ICAO shall adopt and amend international standards and recommended practices (SARPs) as necessary.

ART. 38 **Departures from International Standards — THE 60-DAY RULE**

Any State which finds it **impracticable to comply** in all respects with any international standard or procedure, or which **deems it necessary to adopt different regulations or practices**, shall give **immediate notification** to ICAO of differences between its own practice and that established by the international standard.

> **THE CRITICAL NUMBER**
>
> In case of **amendments** to international standards, any State which does not make the appropriate amendments shall give notice to the Council within **SIXTY (60) DAYS** of the adoption of the amendment.

ART. 39 **Endorsement of Certificates & Licenses**	Any aircraft **not satisfying international airworthiness requirements**, or any person not satisfying international licence requirements, shall get the certificate/licence **endorsed** ***(an official entry added to a licence or certificate)*** **giving full details** of the shortfall.
ART. 40 **Validity of Endorsed Certificates**	No aircraft or personnel having **endorsed** certificates or licenses shall participate in international navigation **except with the permission** of the State or States whose territory is entered.

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ART. 41 **Recognition of Existing Standards — Airworthiness**	These Chapter provisions shall not apply to aircraft whose **prototype was submitted for certification prior to 3 years** after the date of adoption of an international standard of airworthiness for such equipment.
ART. 42 **Recognition of Existing Standards — Competency of Personnel**	These Chapter provisions shall not apply to personnel whose licences are originally issued **prior to 1 year** after initial adoption of an international standard of qualification for such personnel. However, they shall apply to all personnel whose licenses remain valid **5 years after the date of adoption** of such standard.

ICAO — International Civil Aviation Organization

KEY IDENTITY FACTS

Full Name	International Civil Aviation Organization
Created By	Chicago Convention, 1944
UN Status	Became a **Specialized Agency of the United Nations** linked to Economic and Social Council in **1947**
Headquarters	Montreal, Canada
Members	**193 members** of the United Nations *(including a non-UN member: Cook Islands)*
Non-members	Liechtenstein, Niue, Tuvalu, Vatican City, and states with limited recognition
Structure	Assembly + Council + such other bodies as may be necessary

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ICAO Objectives (from the Convention)

ICAO's Aims — To develop principles and techniques of international air navigation and foster planning and development of international air transport so as to:

1. Insure the **safe and orderly growth** of international civil aviation throughout the world
2. Encourage the arts of **aircraft design and operation for peaceful purposes**
3. Encourage the development of **airways, airports, and air navigation facilities**
4. Meet the needs of the peoples of the world for **safe, regular, efficient and economical air transport**
5. Prevent **economic waste** caused by unreasonable competition
6. Insure that the **rights of contracting States are fully respected** and that every contracting State has a **fair opportunity to operate international airlines**
7. **Avoid discrimination** between contracting States
8. **Promote safety of flight** in international air navigation
9. Promote generally the development of **all aspects of international civil aeronautics**

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ICAO Organisational Structure

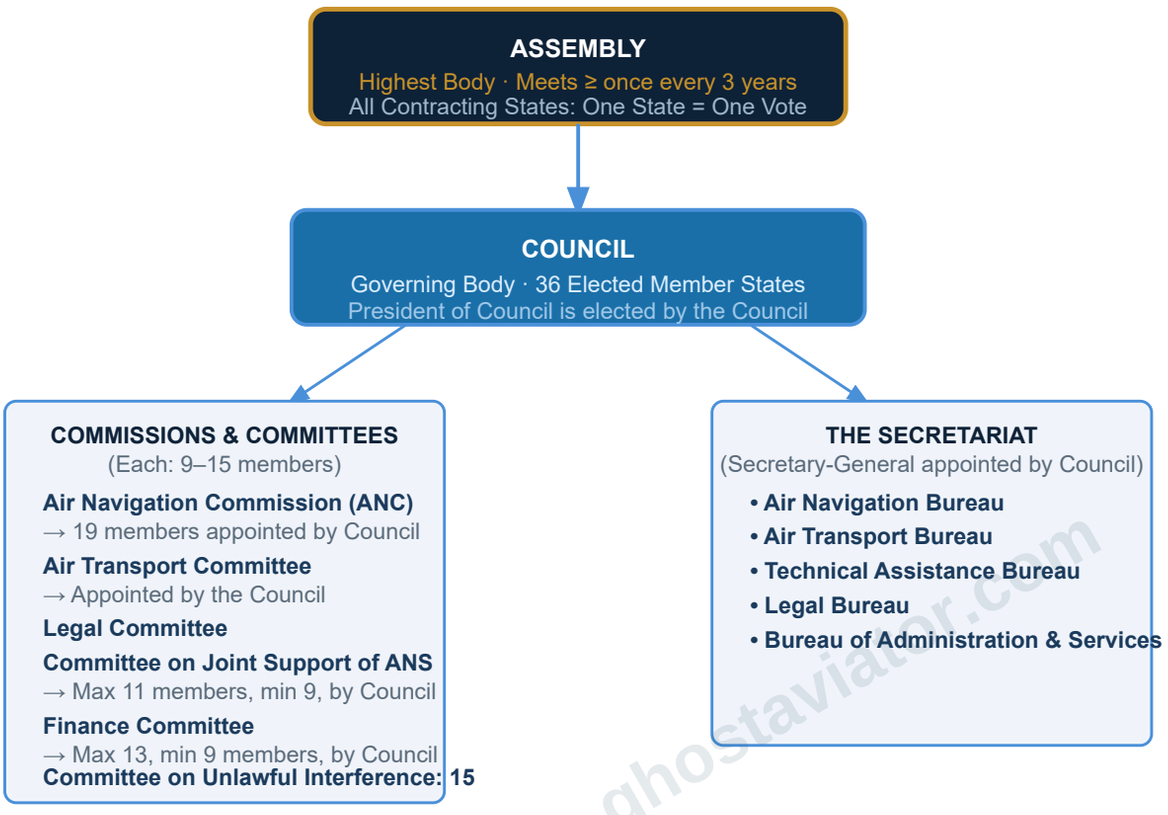

Figure 1.5 *ASSEMBLY*

ICAO Regional Offices (7 offices serving 9 regions)

#	Region	Office Location
1	Asia and Pacific	Bangkok, Thailand
2	Middle East	Cairo, Egypt
3	Western and Central Africa	Dakar, Senegal
4	South America	Lima, Peru
5	North America, Central America and Caribbean	Mexico City, Mexico
6	Eastern and Southern Africa	Nairobi, Kenya
7	Europe and North Atlantic	Paris, France

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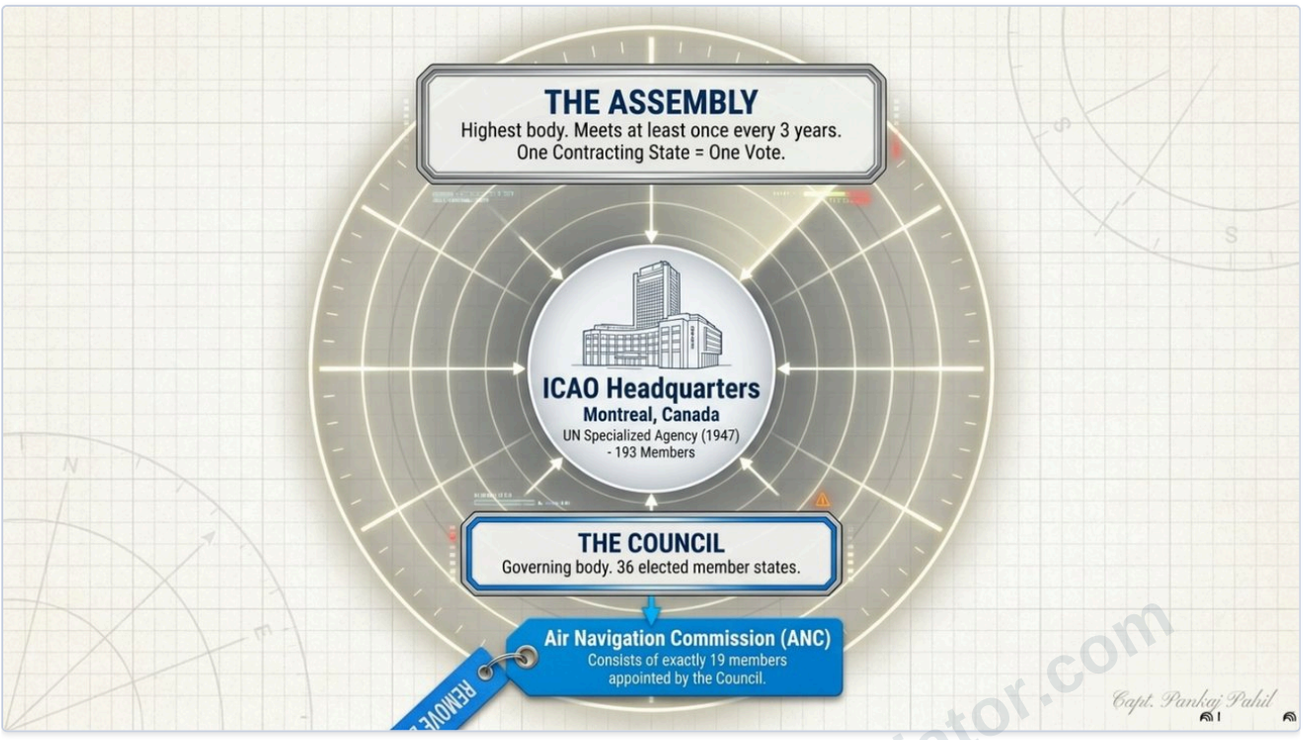

Figure 1.6 *ICAO: Assembly, Council and Air Navigation Commission*

ICAO Technical Publications — SARPs, PANS, SUPPS, Manuals

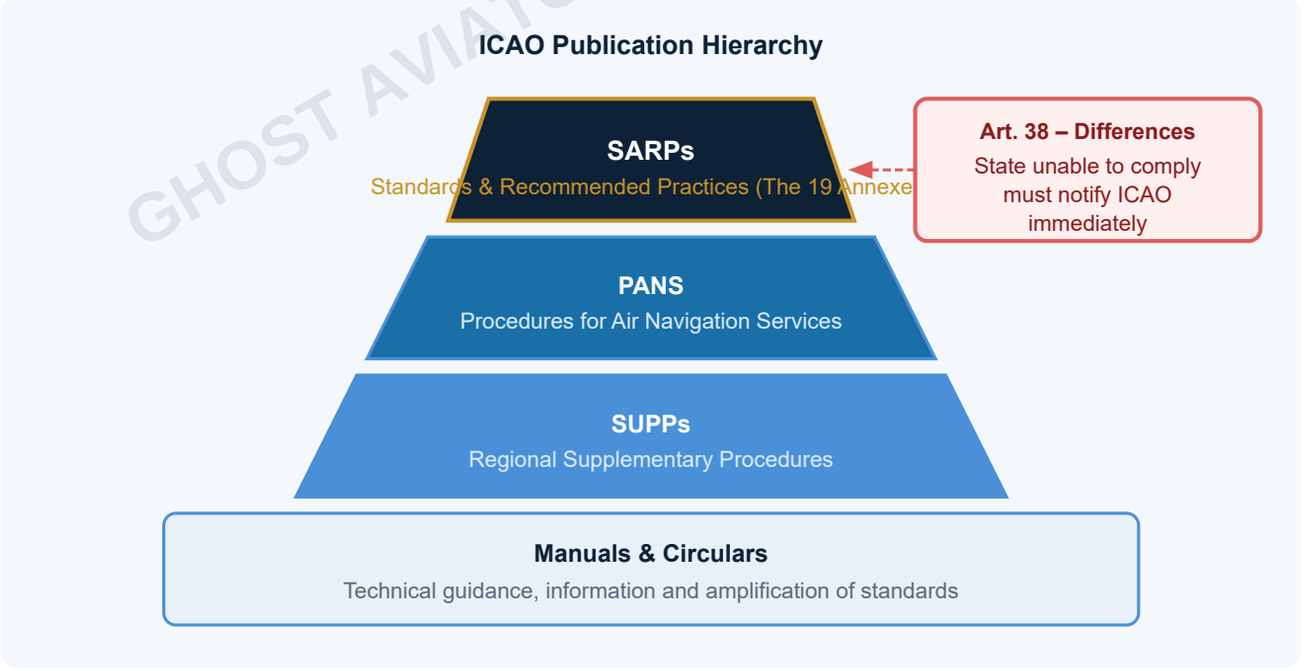

Figure 1.7 *ICAO Publication Hierarchy*

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SARPs — Standards & Recommended Practices	Adopted by the Council under Articles 54, 37, and 90. Designated, for convenience, as **Annexes to the Convention**. • **International Standard:** Uniform application necessary for the *safety or regularity* of international air navigation. **MANDATORY** • **Recommended Practice:** Uniform application desirable in the *interest of safety, regularity or efficiency*. **DESIRABLE**
PANS — Procedures for Air Navigation Services	Approved by the Council for **worldwide application**. Contain operating procedures regarded as *not yet having attained a sufficient degree of maturity* for adoption as SARPs. Also contain material too **detailed for an Annex** or susceptible to frequent amendment.
SUPPs — Regional Supplementary Procedures	Have a status **similar to PANS** in that they are approved by the Council, but only for application in **specific geographic regions**.
Technical Manuals	Provide **guidance and information in amplification** of the International Standards, Recommended Practices, and PANS, the implementation of which they are designed to facilitate.
Air Navigation Plans	Detail requirements for **facilities and services for international air navigation** in the respective ICAO Air Navigation Regions.
ICAO Circulars	Make available **specialized information of interest to Contracting States**. This includes studies on technical subjects.

The 19 Annexes to the Chicago Convention

CRITICAL EXAM FACT — ANNEXES WITH ONLY INTERNATIONAL STANDARDS (NO RECOMMENDED PRACTICES)

Annexes 2, 5, 7 & 8 contain ONLY International Standards — NO Recommended Practices.
The remaining 15 Annexes contain both.

ANNEX 1	Personnel Licensing
ANNEX 2	Rules of the Air *(Standards only)*
ANNEX 3	Meteorological Service for International Air Navigation
ANNEX 4	Aeronautical Charts
ANNEX 5	Units of Measurement to be Used in Air and Ground *(Standards only)*
ANNEX 6	Operation of Aircraft (Parts I, II, III)
ANNEX 7	Aircraft Nationality & Registration Marks *(Standards only)*
ANNEX 8	Airworthiness of Aircraft *(Standards only)*
ANNEX 9	Facilitation
ANNEX 10	Aeronautical Telecommunication (5 Volumes)
ANNEX 11	Air Traffic Services

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ANNEX 12	Search and Rescue
ANNEX 13	Aircraft Accident Investigation
ANNEX 14	Aerodromes (Vol I: Design & Ops; Vol II: Heliports)
ANNEX 15	Aeronautical Information Service
ANNEX 16	Environmental Protection (Vol I: Noise; Vol II: Emissions)
ANNEX 17	Safeguarding International Civil Aviation against Acts of Unlawful Interference
ANNEX 18	Safe Transport of Dangerous Goods by Air
ANNEX 19	Safety Management

= Standards ONLY (no recommended practices). Highlighted = Frequently asked in exams.

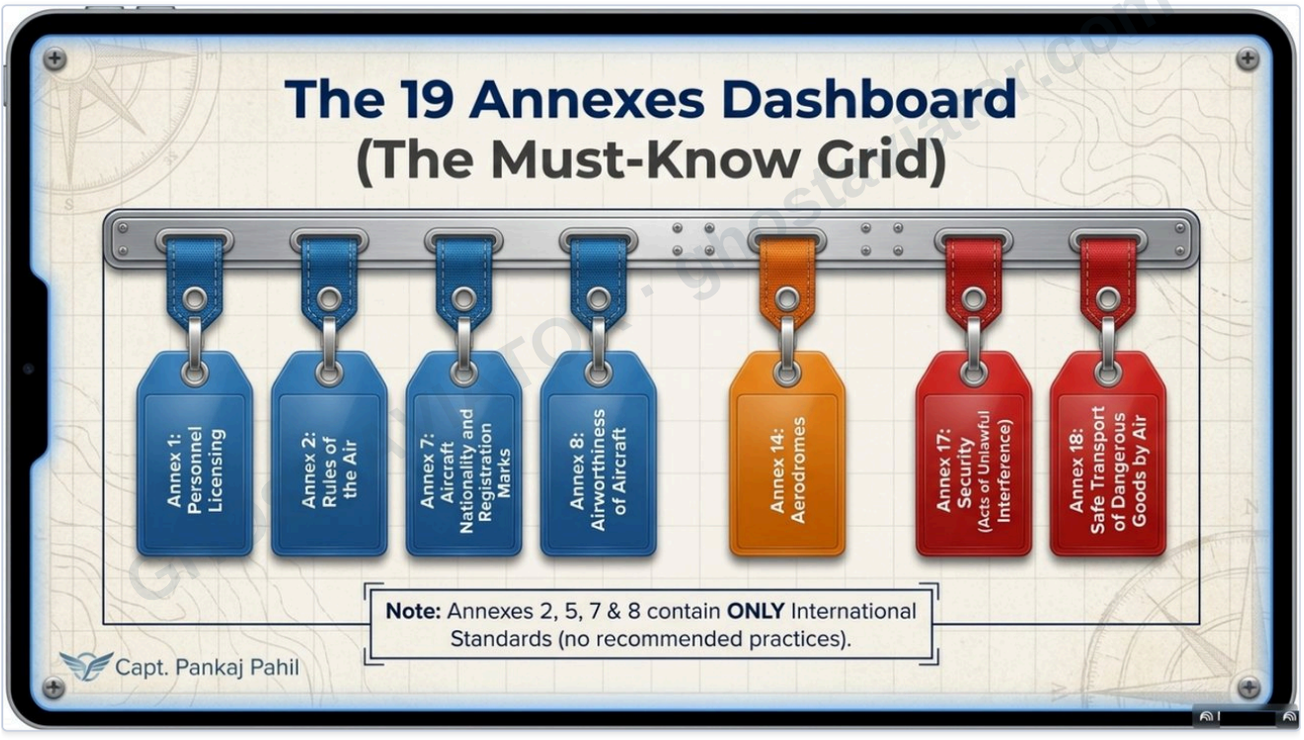

Figure 1.8 *The Annexes to remember*

Freedoms of the Air — Air Services Agreements

INTERNATIONAL AIR SERVICES TRANSIT AGREEMENT (IASTA), 1944

- Signed at Chicago on **7 Dec 1944**
- **India HAS Ratified**
- Grants the **Technical Freedoms** (1st and 2nd)

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INTERNATIONAL AIR TRANSPORT AGREEMENT (IATA), 1944

- Signed at Chicago on **7 Dec 1944**
- **India has NOT Ratified**
- Grants both Technical AND **Commercial Freedoms** (1st–5th)

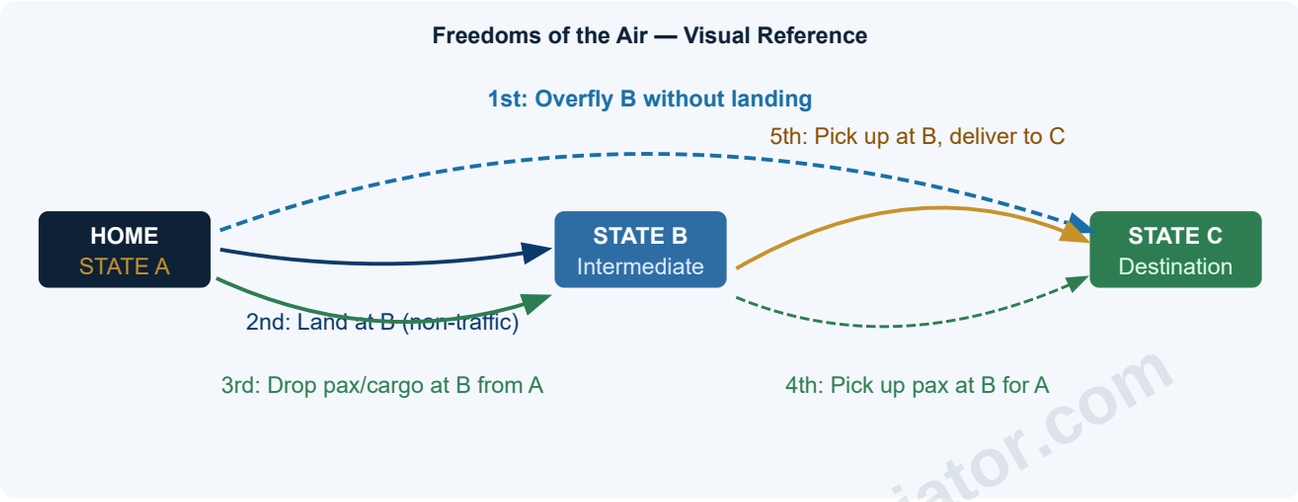

Figure 1.9 *Freedoms of the Air — Visual Reference*

#	Type	Description	Agreement	India Ratified?
1st	TECHNICAL	The privilege *(what a licence allows its holder to do)* to **fly across the territory** of another State **without landing**.	IASTA 1944	**YES**
2nd	TECHNICAL	The privilege to **land for non-traffic purposes** (e.g., fuel stop, technical stop — no pax/cargo loaded or unloaded).	IASTA 1944	**YES**
3rd	COMMERCIAL	The privilege to **put down (drop off)** passengers, mail, and cargo taken on in the territory of the **State whose nationality the aircraft possesses**.	IATA 1944	**NO**
4th	COMMERCIAL	The privilege to **take on (pick up)** passengers, mail, and cargo **destined for the territory of the State** whose nationality the aircraft possesses.	IATA 1944	**NO**
5th	COMMERCIAL	The privilege to take on passengers, mail, and cargo destined for **any other Contracting State**, AND the privilege to put down passengers, mail, and cargo coming from any such territory. *(Between two foreign states.)*	IATA 1944	**NO**

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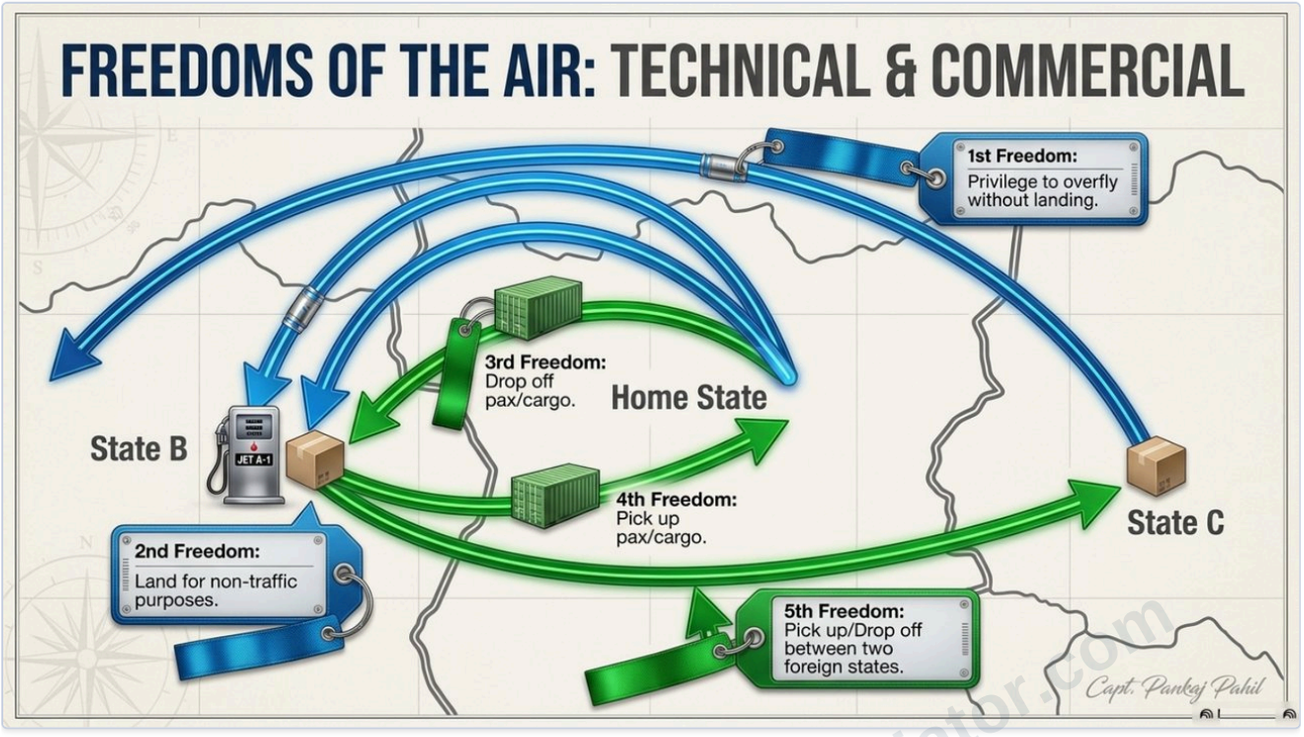

Figure 1.10 *The five freedoms of the air*

Security & Unlawful Acts Conventions

ALL OF THESE CONVENTIONS HAVE BEEN RATIFIED BY INDIA

Tokyo 1963, Hague 1970, Montreal 1971 & 1991, and Montreal Protocol 1988 — all ratified by India.

Convention	Year	Core Focus	Key Provision	India
Tokyo Convention	1963	**Offences on board aircraft**	Grants Aircraft Commander powers to impose **reasonable restraint** on unruly persons	Ratified
Hague Convention	1970	**Hijacking** — Unlawful Seizure of aircraft in flight	Defines act of unlawful seizure; requires **extradition** of offenders; severe punishment	Ratified
Montreal Convention	1971	**Sabotage** — Acts against safety of civil aviation	Acts of violence on board; destroying an aircraft; damaging navigation facilities; transmitting false info	Ratified
Montreal Protocol	1988	**Aerodromes** — Extended Montreal 1971 to airports	Covers violence at aerodromes serving international civil aviation; destruction of airport facilities	Ratified
Montreal Convention	1991	**Plastic Explosives** — Marking for detection	Convention on the *marking of plastic explosives* for the purpose of detection	Ratified

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Tokyo Convention 1963 — Detailed Clauses

FULL NAME & SCOPE

Convention on Offences and Certain Other Acts Committed on Board Aircraft — Signed at Tokyo on 14 September 1963.

- Applies in respect of: (a) offences against penal law; (b) acts which, whether or not they are offences, may do **jeopardize the safety of the aircraft, persons or property therein, or jeopardize good order and discipline on board**.
- Applies to aircraft registered in a contracting State, while **in flight or on the surface of the high seas or any other area outside the territory of any State**.
- Aircraft is considered **"in flight"** from the moment when **power is applied for the purpose of take-off until the moment when the landing run ends**.
- **Does NOT apply** to aircraft used in military, customs, or police services.

Article 2 — No Political Offences	No provision of this Convention shall be interpreted as authorizing or requiring any action in respect of offences against penal laws of a **political nature or based on racial or religious discrimination**.
Article 3 — Jurisdiction	The **State of registration** of the aircraft is competent to exercise jurisdiction *(legal power over a person, place or event)* over offences and acts committed on board. This Convention does not exclude any criminal jurisdiction exercised in accordance with national law.

Article 5 & 6 — Powers of the Aircraft Commander HIGH PRIORITY

The aircraft commander may, when he has **reasonable grounds to believe** that a person has committed, or is about to commit, an offence or act on board, impose upon such person **reasonable measures including restraint** which are necessary:

1. To protect the **safety of the aircraft**, or of persons or property therein
2. To maintain **good order and discipline on board**
3. To enable him to **deliver such person to competent authorities**

ADDITIONAL POWERS

- Commander may **require or authorize** assistance of other crew members.
- Commander may **request but NOT require** assistance of passengers.
- Any crew member or passenger may take **reasonable preventive measures WITHOUT authorization** when immediately necessary.

EXAM GOLDEN RULE

If a question involves the Aircraft Commander's powers of restraint, the answer is ALWAYS the TOKYO CONVENTION.

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Article 7 — Duration of Restraint	Measures of restraint imposed **shall not be continued beyond any point at which the aircraft lands**, UNLESS: • Such point is in a **non-Contracting State** whose authorities refuse to permit disembarkation • The aircraft makes a **forced landing** and the commander is unable to deliver that person to competent authorities • That person **agrees to onward carriage under restraint**

Hague Convention 1970 — Hijacking

FOCUS AREA

Following Tokyo Convention and politically motivated terrorist hijackings, ICAO called a convention hosted by the Dutch government. The *Convention for the Suppression of Unlawful Seizure of Aircraft* defines the act of unlawful seizure and the measures to be taken by contracting states to enforce **severe punishment** upon perpetrators. This agreement specifies **extradition of offenders** and obliges contracting states to extradite offenders.

Montreal Convention 1971 — Sabotage

OFFENCES UNDER MONTREAL 1971 (SUPPRESSION OF UNLAWFUL ACTS AGAINST THE SAFETY OF CIVIL AVIATION)

Makes it an offence to:

- Commit **acts of violence on board aircraft** that endanger people, property, and the safety of the aircraft
- **Destroy an aircraft in service** or cause damage rendering it incapable of flight
- **Place a device on board** an aircraft that is likely to destroy the aircraft or damage it
- **Destroy or damage any navigation facility** or interfere with its correct operation
- **Interfere with aircraft communications** or transmit false information that endangers an aeroplane in flight

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Montreal Protocol 1988 — Extension to Aerodromes

Extended the Montreal Convention 1971 to include offences committed at **aerodromes serving international civil aviation**, including the intentional use of any device, substance, or weapon likely to:

- Cause serious injury or death
- Destroy or seriously damage the **facilities of an airport**
- Destroy or damage **aircraft not in service** at the airport
- Disrupt the **services at an airport**

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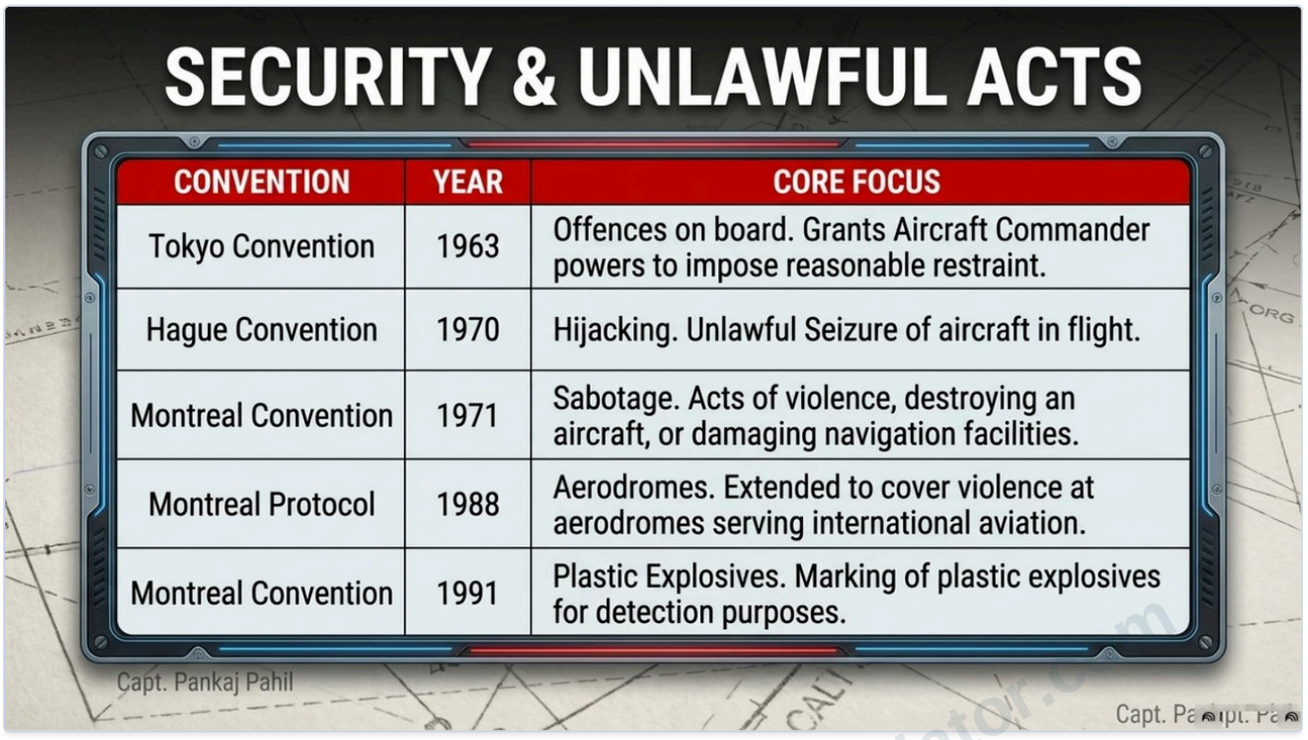

Figure 1.11 *Security conventions*

Liability & Ownership Conventions

Warsaw Convention 1929 — Passenger Liability

FULL NAME & KEY FACTS

Convention for the Unification of Certain Rules Relating to International Carriage by Air — Signed at Warsaw on 12 October 1929

- **India Ratified: Warsaw Convention + Hague Protocol**
- 152 States are party — one of the most widely accepted unifications of private law.
- Applies to all international carriage of **persons, luggage, or goods performed by aircraft for reward**, and to gratuitous carriage by an air transport undertaking.
- Does **NOT apply** to carriage performed under terms of any **international postal Convention**.

Amendments to Warsaw Convention

1929	Warsaw Convention Original treaty signed. India ratified.
1955	Hague Protocol India ratified. Increased liability limits.
1961	Guadalajara Convention India did NOT ratify.

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1971	Guatemala City Protocol India did NOT ratify.
1975	Additional Montreal Protocols Nos. 1–4 India did NOT ratify.
1999	Montreal Convention Aimed at **replacing Warsaw Convention system**. **India is a party** (in force for India from 2009 - the Carriage by Air Act, 1972 was amended in 2009 to give it effect). No financial limits on liability for passenger injury or death. For damages up to the first-tier limit (**113,100 SDRs** when set in 2009; revised by ICAO to 128,821 in 2019 and **151,880 SDRs from 28 Dec 2024**), the air carrier cannot contest claims.

What does Warsaw Unify?

- Documentation on the carriage of **passengers, baggage, and cargo**
- The **financial liability of airlines (operators)**
- The question of **jurisdiction**, by defining the courts before which any action may be brought

Passenger Ticket Requirements (Warsaw)

A PASSENGER TICKET SHALL CONTAIN

- The **place and date of issue**
- An indication of the **place of departure and destination**
- The **agreed stopping places** (carrier may alter, but alteration shall not deprive it of its international character)
- The **name and address of the carrier or carriers**
- A statement that the carriage is subject to the **rules relating to the liability** established by this convention (usually printed on the ticket jacket)

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ABSENCE/IRREGULARITY OF TICKET — CRITICAL RULE

The absence, irregularity, or loss of the passenger ticket **does NOT affect the validity** ***(the period during which a document is accepted)*** **of the contract of carriage**. However, if a carrier accepts a passenger **without a ticket**, the carrier will **NOT be able to fall back on the provisions of the convention that limit liability**.

Baggage Check (Warsaw)

For luggage, other than small personal objects that passengers take themselves, the carrier must issue a **luggage ticket made out in duplicate** — one for the passenger, one for the carrier.

Liability of the Carrier (Warsaw)

LIABILITY LIMITS

- The Treaty imposed **limitations on the liability of the operator**. However, where **gross negligence can be proved, the limit of liability is removed**.
- Currently, the limit of liability for **death of a passenger is SDR 100,000** (under Warsaw system).
- Under **Montreal Convention 1999**: No financial limits to liability for passenger injury or death.

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- For damages up to the first-tier limit — **113,100 SDRs** in the figure most DGCA papers quote (ICAO has since revised it: 128,821 in 2019, **151,880 SDRs from 28 Dec 2024**) — the air carrier **cannot contest** claims for compensation.
- Above that limit — the air carrier **can defend itself** against a claim by proving it was not negligent.

Carriage by Air Act, 1972 (India)

INDIAN LEGISLATION

The rights and liabilities of air carriers are governed by the **Carriage by Air Act, 1972** [as amended in 2009]. **The Act extends to the whole of India.** Consequently, the Act is applicable to Indian citizens involved in domestic carriage by air and in international carriage by air, irrespective of the nationality of the aircraft performing the carriage.

Rome Convention 1952 — Surface Damage

KEY FACTS

- Full Name: *Convention on Damage Caused by Foreign Aircraft to Third Parties on the Surface*
- Replaced Rome Convention 1933 and Brussels Protocol 1938
- The operator is liable for any damage, but liability is **limited to a sum proportionate to the weight of the aircraft**
- Makes it **compulsory to insure** against this liability
- Compulsory recognition and execution of any **foreign judgement on damage to third parties**
- **India has NOT ratified this convention**
- **Montreal Protocol 1978** amended Rome Convention by including operator living in another contracting state.

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Geneva Convention 1948 — Finance Rights

PURPOSE

This convention protects the **rights of seller for aircraft bought on Hire Purchase, Lease, or Mortgage**. Secures rights in aircraft bought on hire purchase, lease, or mortgage.

Cape Town Convention & Protocol, 2001 — Asset Security

KEY FACTS

- Full Name: *Convention on International Interests in Mobile Equipment as applied to Aircraft Objects*
- Signed at **Cape Town on 16 November 2001**
- Aims at introducing a **legally certain, effective, and prompt system of enforcement** that can assure and encourage **investments in Aircraft objects**
- Meets the particular requirements of **aircraft finance** and extends contracts of sale of aircraft equipment
- **India ratified this convention and protocol in 2008**

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Summary Table — All Liability & Ownership Conventions

Convention	Year	Protected Entity	Core Function	India
Warsaw Convention	1929	Passengers, Baggage, Cargo	Defines financial liability for airlines, ticketing, carriage rules	Ratified
Rome Convention	1952	Third parties on the surface	Defines liability for damage caused on ground by foreign aircraft	NOT Ratified
Geneva Convention	1948	Sellers & Financiers	Secures rights in aircraft bought on hire purchase, lease, or mortgage	—
Cape Town Convention	2001	Mobile equipment investments	Secures international interests in aircraft finance and object enforcement	Ratified 2008

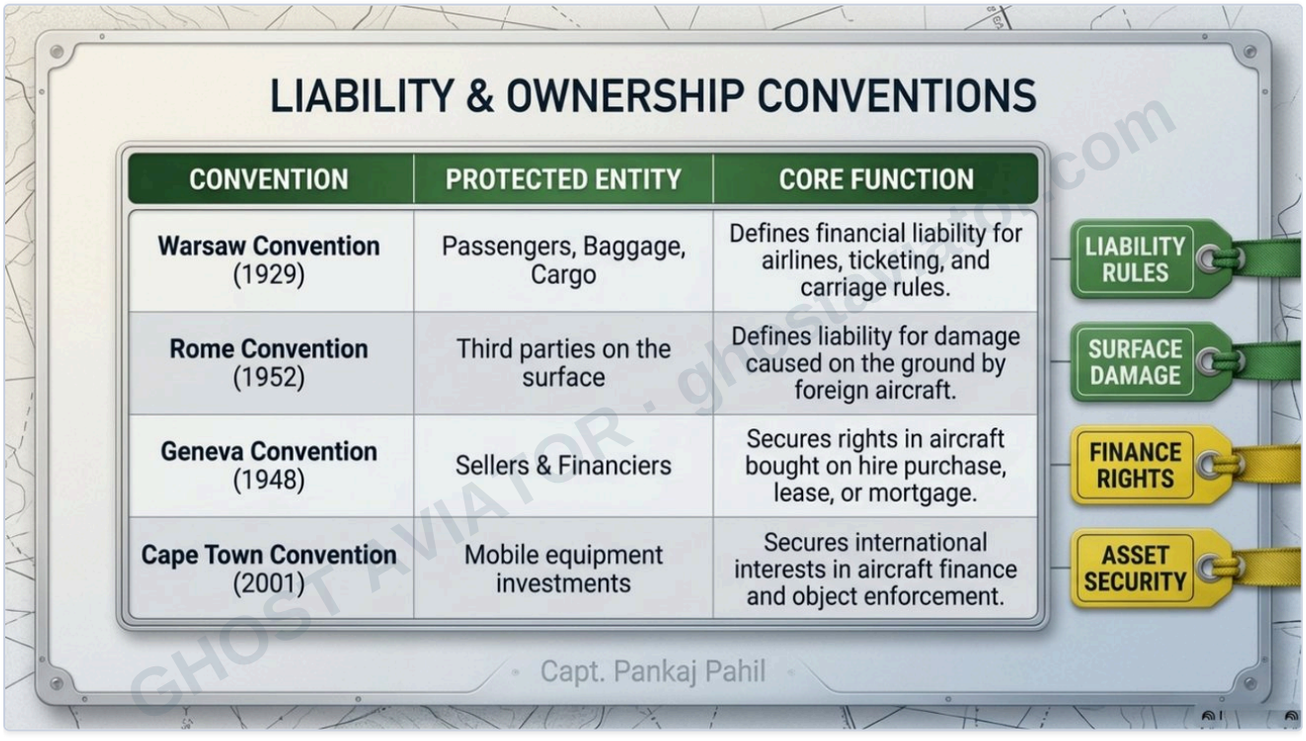

Figure 1.12 *Liability and ownership conventions*

Indian Aviation Organisations — MoCA, DGCA & AAI

Ministry of Civil Aviation (MoCA)

IDENTITY

Located at **Rajiv Gandhi Bhavan, Safdarjung Airport, New Delhi**. Responsible for:

- Formulation of national policies and programmes for the **development and regulation of Civil Aviation**
- Administration of the **Aircraft Act, 1934**, Aircraft Rules, 1937 and various other legislations
- Administrative control over: **DGCA, Bureau of Civil Aviation Security (BCAS), Indira Gandhi Rashtriya Udan Academy (IGRUA), AAI, National Aviation Company of India (Air India), Pawan Hans Helicopters**

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- Commission of Railway Safety (under this Ministry)

Directorate General of Civil Aviation (DGCA)

ROLE

The **primary regulatory body in the field of Civil Aviation, primarily dealing with safety issues**. Responsible for regulation of air transport services to/from/within India, enforcement of civil air regulations, air safety, and airworthiness standards. Coordinates all regulatory functions with ICAO.

Functions of DGCA

DGCA'S FUNCTIONS (12 KEY FUNCTIONS)

1. **Registration of civil aircraft**
2. Formulation of standards of airworthiness for civil aircraft registered in India and grant of **certificates of airworthiness**
3. **Licensing of pilots, aircraft maintenance engineers (AMEs), and flight engineers**; conducting examinations and checks
4. **Licensing of air traffic controllers**
5. Certification of **aerodromes and CNS/ATM facilities**
6. Granting of **Air Operator's Certificates (AOCs)** to Indian carriers; regulation of air transport services operating to/from/within/over India
7. Conducting investigation into **accidents/incidents** and taking accident prevention measures; implementing Safety Aviation Management programmes
8. Carrying out amendments to the **Aircraft Act, the Aircraft Rules** and Civil Aviation Requirements (CARs)
9. Coordination at national level for **flexi-use of air space** by civil and military air traffic agencies
10. Keeping a check on **aircraft noise and engine emissions** in accordance with ICAO Annex 16
11. Promoting **indigenous design and manufacture** of aircraft and aircraft components
12. Approving training programmes for operators for carriage of **dangerous goods**; issuing authorizations for carriage of dangerous goods

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Airports Authority of India (AAI)

FORMATION

Formed on **1st April 1995** by merging the **International Airports Authority of India** and the **National Airports Authority**. Purpose: To accelerate the integrated development, expansion, and modernization of airports in India conforming to international standards.

Functions of AAI (10 Main Functions)

1. Design & Maintenance	Design, Development, Operation, and Maintenance of **international and domestic airports** and civil enclaves.
2. Airspace Management	Control and Management of the **Indian airspace extending beyond the territorial limits** of the country, as accepted by ICAO.

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3. Passenger Terminals	Construction, Modification, and Management of **passenger terminals**.
4. Cargo Terminals	Development and Management of **cargo terminals** at international and domestic airports.
5. Passenger Facilities	Provision of **passenger facilities and information systems** at passenger terminals.
6. Runways, Aprons & Taxiways	Expansion and strengthening of operation areas viz. **Runways, Aprons, Taxiways**, etc.
7. Visual Aids	Provision of **visual aids** (approach lighting, runway lighting, etc.).
8. CNS — Communication & Navigation	Provision of Communication and Navigation aids viz. **ILS, DVOR, DME, Radar**, etc.
9. Air Traffic Services	Provision of **Air Traffic Services** at airports under its jurisdiction.
10. Aeronautical Information	Provision of aeronautical information services: publication of **AIP, NOTAMs, AIRRACs, PIBs**, etc.

Captain's Exam Checklist — Golden Rules to reference

Quick Reference: India's Ratification Status

Convention	Year	Subject	India
Paris Convention	1919	First international aviation agreement	—
Chicago Convention	1944	Core aviation convention; established ICAO	Ratified
IASTA (Technical Freedoms)	1944	1st & 2nd Freedoms (overflight, tech stop)	Ratified
IATA (Commercial Freedoms)	1944	1st–5th Freedoms (commercial traffic)	NOT Ratified
Warsaw Convention	1929	Passenger / cargo liability	Ratified
Hague Protocol	1955	Amendment to Warsaw	Ratified
Montreal Convention 1999	1999	Replace Warsaw system	Party (in force for India 2009)
Rome Convention	1952	Surface damage by foreign aircraft	NOT Ratified
Tokyo Convention	1963	Onboard offences; Commander's powers	Ratified
Hague Convention	1970	Hijacking / unlawful seizure	Ratified
Montreal Convention	1971	Sabotage / acts against civil aviation	Ratified
Montreal Protocol	1988	Offences at aerodromes	Ratified
Montreal Convention	1991	Marking of plastic explosives	Ratified
Cape Town Convention	2001	Aircraft finance / asset security	Ratified 2008

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CHAPTER 2

02

Aircraft Nationality and Registration Marks

HOW TO USE THESE NOTES

The colour code is consistent across the Ghost Aviator library:

- **Blue** — Definitions & core concepts.
- **Green** — Standard procedures & allowances.
- **Red** — Strict limitations, warnings & prohibitions.

Exact regulatory values are pinned like this — **at least 50 cm** — because the examiner tests the number, not the paraphrase.

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Annex 7 — What It Sets Out

Every aircraft has a "name badge" — a nationality mark that identifies the country of registry and a registration mark that identifies the individual airframe. ICAO Annex 7 is the international rulebook that tells States how to pick, size and display those badges.

WHAT ANNEX 7 DOES

Annex 7 sets out procedures for **selection by ICAO Contracting States of nationality marks** from the **nationality symbols included in the radio call signs allocated to the States of Registry by the ITU**.

- It sets **standards for the use of letters, numbers and other graphic symbols** in the nationality and registration marks.
- It spells out **where these characters will be located** on different types of airborne vehicles — lighter-than-air and heavier-than-air aircraft alike.
- It calls for the **registration of the aircraft** and provides a sample **Certificate of Registration** for use by ICAO Contracting States.
- The certificate **must be carried in the aircraft at all times**.

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WHO ALLOTS WHAT — THE THREE BODIES

Three international bodies interact:

- **ITU (International Telecommunication Union)** — allocates the underlying radio call-sign symbols to the States of Registry. It also allots the symbols **4YA to 4YZ** for **common marks** to two or more States carrying out joint operations.
- **ICAO (International Civil Aviation Organisation)** — assigns the common mark to a common-mark registering authority.
- **State of Registry / Common-Mark Registering Authority** — assigns the individual registration mark to each aircraft.

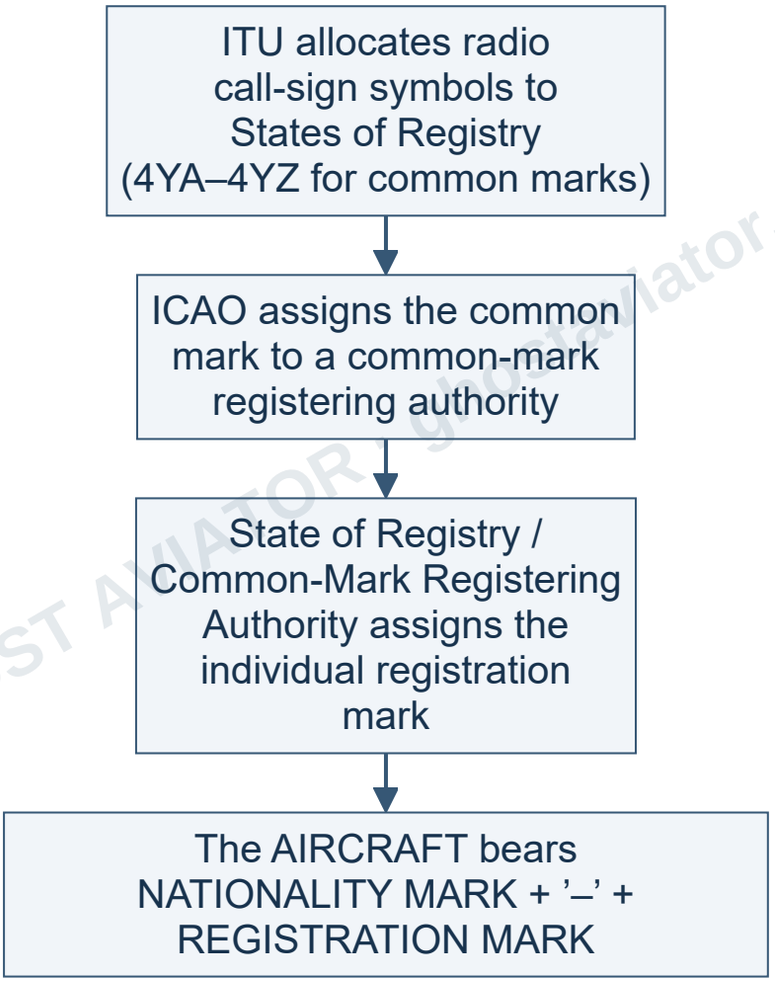

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Figure 2.1 *How a Registration Ends Up on Your Aircraft*

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Identification Plate

THE PLATE — WHAT IT IS

An aircraft **shall carry an identification plate** inscribed with **at least** its:

- Nationality or common mark; and
- Registration mark.

MATERIAL & LOCATION RULE

The plate **shall be made of fireproof metal or other fireproof material of suitable physical properties** and shall be secured to the aircraft in a **prominent position near the main entrance** — or, for special cases:

- **(a) Unmanned free balloon** — **affixed conspicuously to the exterior of the payload**.
- **(b) Remotely piloted aircraft (RPA)** — secured in a prominent position near the main entrance or compartment; or **affixed conspicuously to the exterior of the aircraft** if there is no main entrance or compartment.

THE REGISTRATION MARK ITSELF

The registration mark shall be **letters, numbers, or a combination of letters and numbers** consisting of **1 to 5 digits or characters**, and shall be that **assigned by the State of Registry or common-mark registering authority**.

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The Registration Mark — Assignment & Confusion Rules

A registration mark must never be mistakable for a distress or urgency signal on the radio — otherwise a routine call could be read as a life-and-death broadcast.

COMBINATIONS THAT MUST BE AVOIDED

When letters are used for the registration mark, combinations **shall NOT be used which might be confused with**:

- The **five-letter combinations** used in the **International Code of Signals, Part II**;
- The **three-letter combinations beginning with Q** used in the **Q Code**;
- The **distress signal SOS**, or other similar urgent signals — for example **XXX**, **PAN** and **TTT**.

Table 3.1 — Radio Urgency & Distress Signals To Avoid Confusing With

Signal	Meaning & Where It Comes From
SOS	The classic international **distress signal**.
XXX	An **urgency signal** — the radio-telegraphy equivalent of PAN (very high urgency without being an outright distress).
PAN	The radio-telephony **urgency signal** (spoken "PAN-PAN") — indicates a situation of urgency but not immediate danger.
TTT	The **safety signal** in radio telegraphy — precedes safety messages.

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Signal	Meaning & Where It Comes From
Q-Codes	Three-letter combinations **beginning with Q** (e.g. QNH, QFE, QDM) reserved for aeronautical procedure.

EXAM TRAP

In the exam, the "example" the examiner picks for a forbidden combination will be one of **XXX, PAN, TTT** or **SOS**. Distractors like RCC, LLL, DDD are simply not on the list — but they are only wrong *as examples*, not as legal combinations.

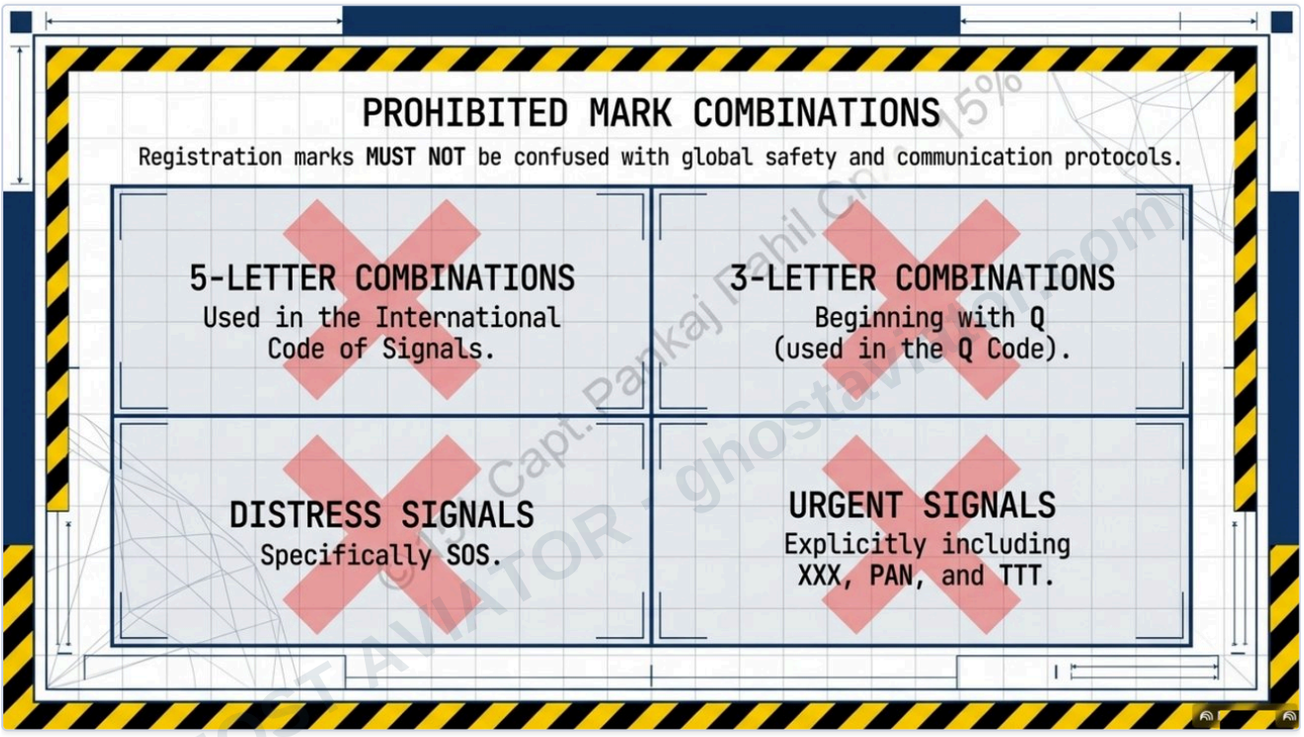

Figure 2.2 *Letter groups that may not be used as registration marks*

Measurements of Nationality, Common & Registration Marks

Annex 7 fixes the minimum character height so that marks stay legible at range. The numbers are simple and heavily tested — learn them cold.

Heavier-Than-Air Aircraft

CHARACTER HEIGHT — HEAVIER-THAN-AIR

- **Wings** — height of marks shall be **at least 50 cm**.
- **Fuselage (or equivalent structure) and vertical tail surfaces** — height of marks shall be **at least 30 cm**.
- **Letters and numbers in each separate group of marks shall be of equal height.**

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Lighter-Than-Air Aircraft

CHARACTER HEIGHT — LIGHTER-THAN-AIR

- Height of marks on lighter-than-air aircraft **other than unmanned free balloons** shall be **at least 50 cm**.
- Measurements for **unmanned free balloons** shall be **determined by the State of Registry**, taking into account the **size of the payload** to which the identification plate is affixed.

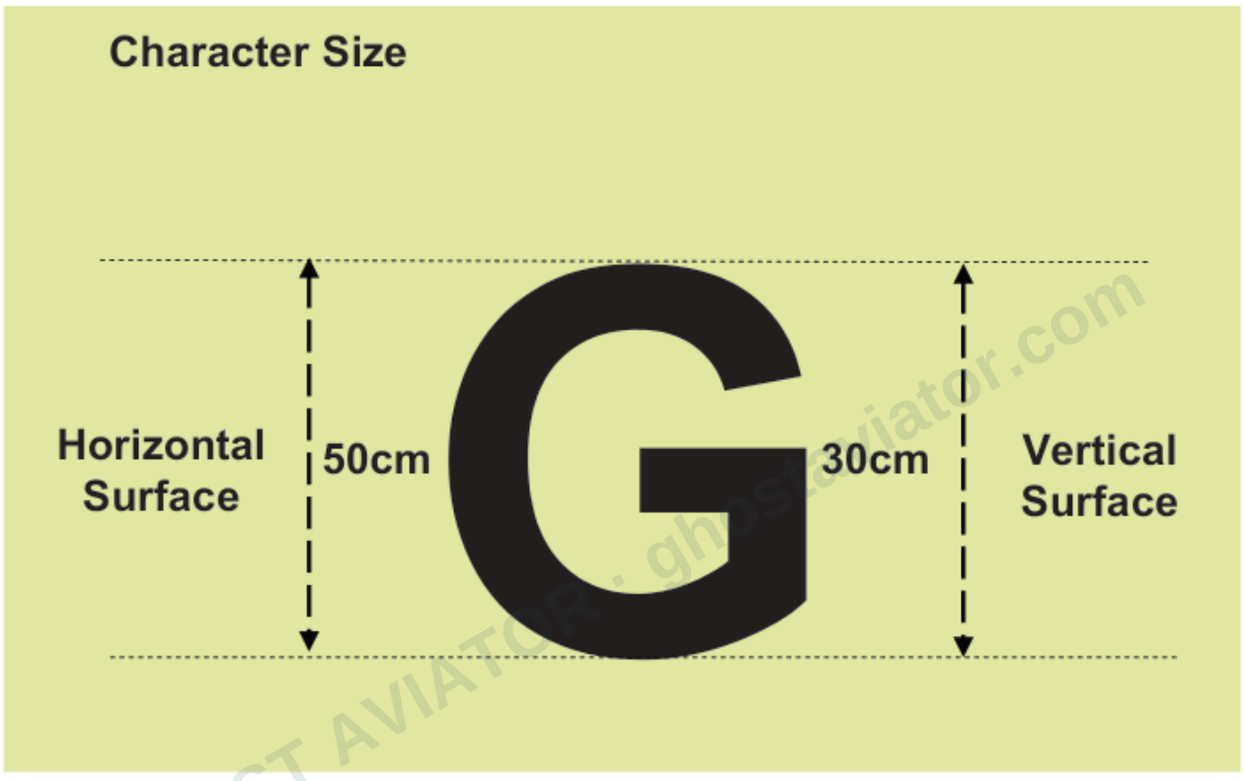

Figure 2.3 *Minimum character height: 50 cm on the wings (lower surface), 30 cm on the fuselage and vertical tail.*

Table 4.1 — Quick-Reference: Minimum Character Heights

Aircraft Type	Location	Minimum Height
Heavier-than-air	Wings	**50 cm**
	Fuselage (or equivalent structure) & vertical tail surfaces	**30 cm**
Lighter-than-air	All types except unmanned free balloons	**50 cm**
	Unmanned free balloons	Determined by the **State of Registry** based on payload size

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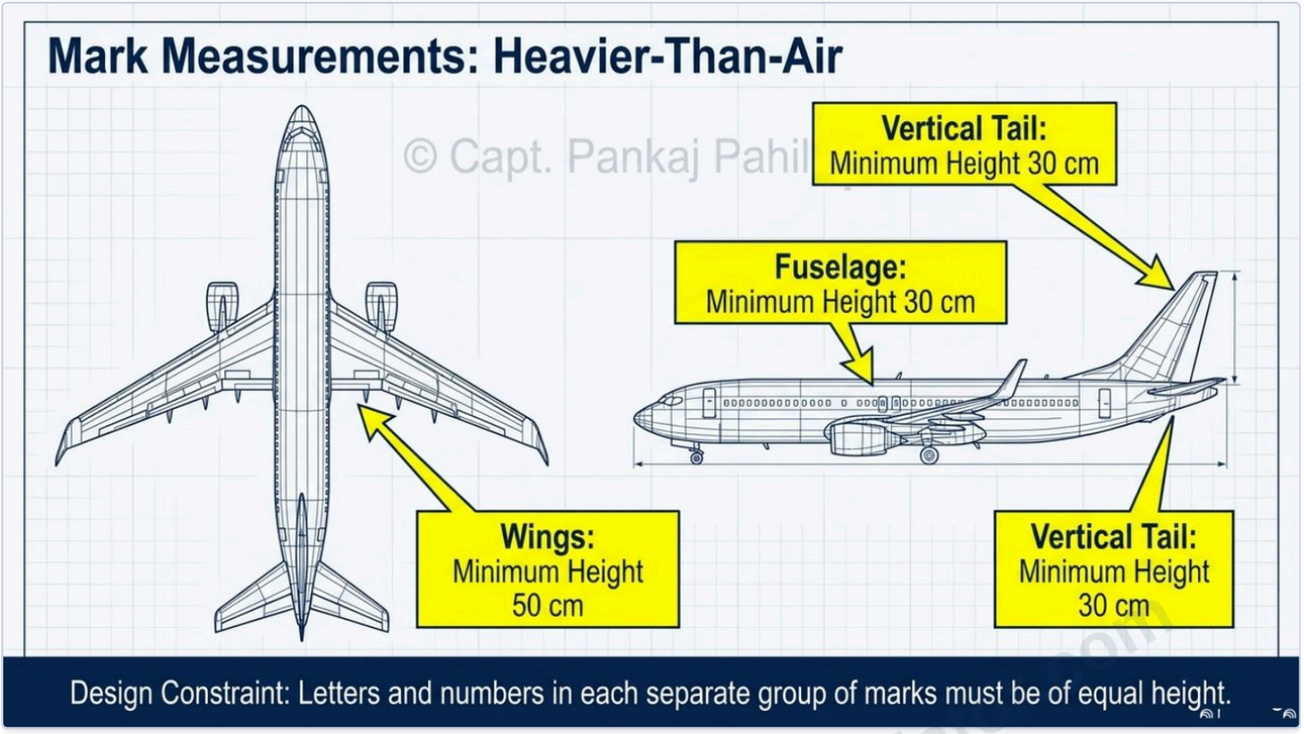

Figure 2.4 *Minimum height of the marks*

National Provisions — Certificate, Cancellation, Register

Annex 7 sets the international rulebook; the CAR then bolts it into Indian law. Three practical items to know: who issues the Certificate of Registration, what it contains, and when the DGCA can pull it.

Certificate of Registration

AUTHORITY TO REGISTER

The authority empowered to **register aircraft and grant the Certificate of Registration in India** shall be the **Central Government**.

PARTICULARS ON THE CERTIFICATE OF REGISTRATION

The certificate shall include the following:

- **Type of aircraft**;
- **Constructor's number**;
- **Year of manufacture**;
- **Nationality and registration marks** (as referred to under these rules);
- **Full name, nationality and address of the owner**;
- **Usual station of aircraft**;
- **Date of registration**;
- **Period of validity** ***(the period during which a document is accepted)***.

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Leased aircraft — additional entries: the certificate shall **also include the validity of the lease** and the **names, nationalities and addresses of the lessor and the lessee**.

Cancellation of Registration

GROUNDS ON WHICH THE DGCA MAY CANCEL

The registration of an aircraft registered in India **may be cancelled at any time by the DGCA**, if satisfied that:

1. Such registration is **not in conformity with the provisions of the rules**; or
2. The registration has been obtained by **furnishing false information**; or
3. The aircraft could **more suitably be registered in some other country**; or
4. The aircraft has been **destroyed or permanently withdrawn from use**; or
5. It is **inexpedient in the public interest** that the aircraft should remain registered in India; or
6. The **lease** in respect of the aircraft: (i) has **expired**, (ii) has been **terminated by mutual agreement** between the lessor and the lessee, or (iii) has been **otherwise terminated** in accordance with the provisions of the Lease Agreement or terms of the lease; or
7. The **Certificate of Airworthiness** in respect of the aircraft has **expired for a period of five years or more**.

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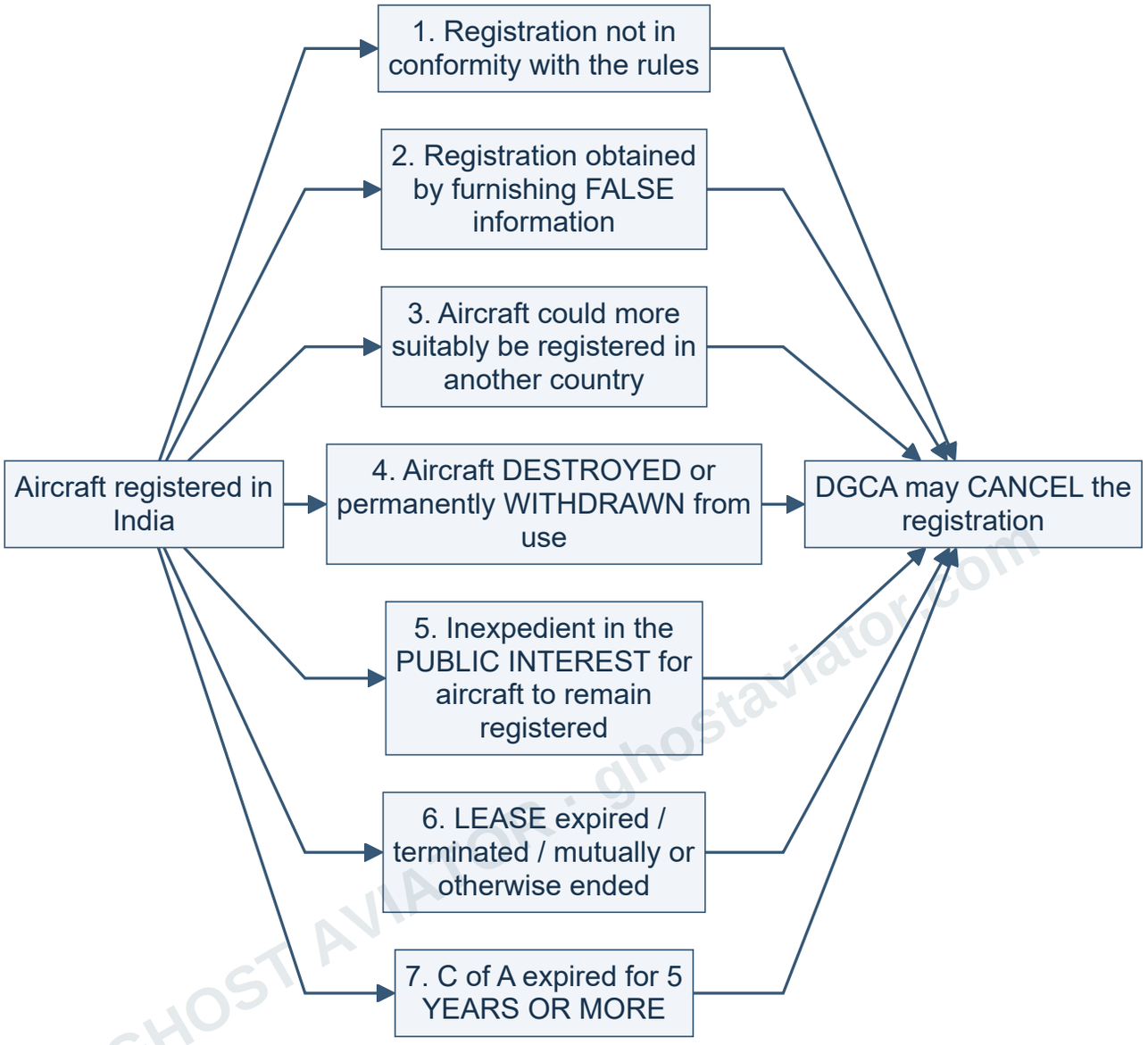

Figure 2.5 *Cancellation of Registration — The Seven DGCA Triggers*

Register of Aircraft

THE REGISTER — PUBLIC RECORD

A **register of aircraft registered in India shall be maintained by the Director-General** and shall include the same particulars as those provided in the Certificate of Registration. The register shall be **open to inspection by members of the public** at such times and subject to such conditions as may be specified by the Director-General. The **Register of Aircraft is also available on the DGCA website**.

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EXAM HOOK — VALIDITY OF REGISTRATION

The Certificate of Registration is valid **from the date of registration until the date shown on the certificate** (CAR Section 2, Series F, Part I, para 7.3). After that date the registration is treated as **expired** and the aircraft may not be flown until the registration is renewed (para 7.5). For an aircraft registered on a lease, the registration lasts **only as long as the lease** (para 7.6). The operator may ask the DGCA to extend the validity (para 7.7). The DGCA may also cancel a registration earlier, on any of the grounds above.

How Indian Marks Are Affixed — the VT Mark

Indian civil aircraft carry the nationality mark **VT**. This part shows how the marks are written and fixed to the aircraft.

ITU Allocation vs. Actual Practice

THE INDIAN NATIONALITY MARK

- The nationality mark on Indian-registered aircraft is the capital letters **VT** in Roman characters (CAR Section 2, Series F, Part I, para 11.1; Aircraft Rules, rule 37).
- The registration mark is a group of **three capital letters** assigned by the DGCA, for example **EPC**.
- A hyphen is placed between the two: **VT-EPC**.
- (The ITU gives each State blocks of call-sign letters. India's blocks include VT to VW and AT to AW, but Indian aircraft use **VT**.)

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Anatomy of an Indian Civil Registration

VT	Nationality Mark (Civil = VT)
–	Hyphen
EPC	Registration (3 Roman letters)

Example: **VT-EPC**

THE RULE — NATIONALITY & REGISTRATION MARKS TOGETHER

The nationality mark is followed by a **hyphen** and a registration mark **consisting of 3 letters** — e.g. **VT-EPC** — in **Roman character without ornamentation**.

Table 6.1 — Indian Nationality Marks At a Glance

Aircraft Type	Nationality Mark	Format	Example
Civil	**VT**	VT – 3 Roman letters, no ornamentation	VT-EPC

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How the Marks Are Affixed

THE MARKS SHALL

a. Be **painted on the aircraft**, or fixed by any other means ensuring a **similar degree of permanency**, in the form and manner specified by the Director-General from time to time;

b. Be **inscribed together with the full name and address of the registered owner** on the owner's name plate, in the form and manner specified by the Director-General from time to time; and

c. **Always be kept clean and visible.**

EXAM TRAP

The nationality mark on Indian-registered aircraft is **VT** (CAR 2-F-I para 11.1). If a question offers more than one letter group as "India's mark", choose **VT**.

Use of State Marks

The exterior of an aircraft is legally regulated real estate. What may — and may not — be displayed on it is spelled out plainly.

THE NO-ADVERTISEMENT RULE

An aircraft **shall not bear on any part of its exterior surface any Advertisement or any sign or lettering except** those under these rules and **as required or permitted by the Director-General**.

WHAT IS PERMITTED

- The **name of the aircraft** and the **name and emblems of the owner** may be displayed **if** the location, size, shape and colour of the lettering and signs:
  - Do **not interfere** with easy recognition of, and
  - Are **not capable of confusion** with, the nationality and registration marks of the aircraft.
- **National flags or colours** may be displayed on the aircraft in such a manner that they are **distinct** and **not likely to create confusion** with the markings used by military aircraft.

STATE AIRCRAFT MARKS ARE OFF-LIMITS

An aircraft **other than a State aircraft shall not bear any mark or sign prescribed** ***(laid down by the authority or the rules)*** **for use by a State aircraft**. Civil aircraft may not "impersonate" military/state markings.

Leasing — Lessor, Lessee, IFSC & GIFT City

Modern airline fleets are largely leased. India's new leasing hub — IFSC/IFSCA at GIFT City — is a favourite recent examiner topic.

Definitions

Table 8.1 — The Four Terms You Must Know

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Term	Definition
Lease	An **agreement by a person (the lessor) to furnish an aircraft to another person (the lessee)** to be used for **compensation or hire purposes**, for a **specified period** or a **defined number of flights**.
Lessor	The party **furnishing** the aircraft under a lease.
Lessee	The party **using** the aircraft under the provisions of a lease.
Leasing Company — GIFT City	Leasing companies/entities **registered with IFSC and approved by IFSCA** for undertaking leasing of aircraft.

Who Operates The Aircraft?

THE OPERATIONAL CONTROL RULE

The **lessee operator** of the aircraft **must hold the necessary economic and operating authority** for the aircraft and **must exercise operational control over the aircraft**. Accordingly, the **lessee must provide**:

- The necessary **flight and cabin crew** members;
- **Ground personnel**;
- **Dispatchers**; and
- **Ground facilities** to operate the aircraft.

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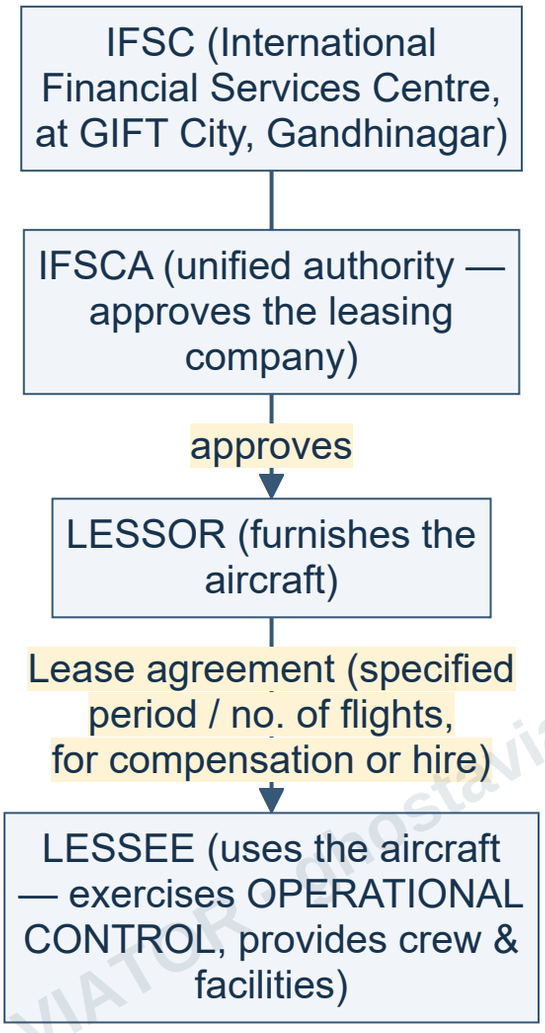

Figure 2.6 *The Lease Relationship — Who Owns, Who Operates, Who Approves*

IFSC, IFSCA & GIFT City

THE THREE INSTITUTIONS

- **IFSC (International Financial Services Centre)** — a financial centre that caters to customers **outside the jurisdiction** ***(legal power over a person, place or event)*** **of the domestic economy**.
- **IFSCA (International Financial Services Centres Authority)** — the **unified authority for the development and regulation of financial products, financial services and financial institutions** in the IFSC in India.
- **GIFT City** — **Gujarat International Finance Tec-city SEZ** is India's **first International Financial Services Centre** under the **Special Economic Zone Act, 2005**. It is headquartered at **GIFT City, Gandhinagar in Gujarat**.

EXAM HOOKS — LEASING

- Operational control over the leased aircraft is exercised by the **lessee**.
- A company undertaking to lease aircraft in India must be approved by the **IFSCA** (not DGCA, not MOCA).
- GIFT City is in **Gandhinagar, Gujarat**, under the **SEZ Act, 2005**.

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PART II

Flying Rules and Air Traffic Services

The rules every pilot follows in the air, and how air traffic control keeps aircraft apart.

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CHAPTER 3

Rules of the Air

03

Essential Aerodrome Definitions

Manoeuvring area — actual regulatory wording

That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

In simple English: The areas used for aircraft take-off, landing and taxiing. The apron is excluded.

Movement area — actual regulatory wording

That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, consisting of the manoeuvring area and the apron(s).

In simple English: Movement area includes the manoeuvring area and the apron or aprons.

Landing area — actual regulatory wording

That part of a movement area intended for the landing or take-off of aircraft.

Source: DGCA CAR, Section 9, Series C, Part I, Chapter 1 — Definitions, printed page 10. These are definitions, not numbered operating paragraphs.

Aerodrome — actual regulatory wording

A defined area on land or water (including any buildings, installations and equipment) intended to be used either wholly or in part for the arrival, departure and surface movement of aircraft.

In simple English: An area used for aircraft arrival, departure and movement on the surface; the definition includes its buildings and equipment.

Source: DGCA CAR, Section 9, Series C, Part I, Chapter 1 — Definitions, printed page 2.

Aerodrome traffic — actual regulatory wording

All traffic on the maneuvering area of an aerodrome and all aircraft flying in the vicinity of an aerodrome.

In simple English: Traffic on the manoeuvring area, together with aircraft flying near the aerodrome.

Source: DGCA CAR, Section 9, Series C, Part I, Chapter 1 — Definitions, printed page 3.

Air traffic — actual regulatory wording

All aircraft in flight or operating on the manoeuvring area of an aerodrome.

In simple English: Aircraft in flight or operating on the manoeuvring area.

Source: DGCA CAR, Section 9, Series C, Part I, Chapter 1 — Definitions, printed page 3.

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Air traffic control clearance — actual regulatory wording

Authorization for an aircraft to proceed under conditions specified by an air traffic control unit.

In simple English: ATC permission to proceed, subject to the stated conditions.

Source: DGCA CAR, Section 9, Series C, Part I, Chapter 1 — Definitions, printed page 4.

Apron — actual regulatory wording

A defined area, on a land aerodrome, intended to accommodate aircraft for purposes of loading or unloading passengers, mail or cargo, fuelling, parking or maintenance.

In simple English: The area used for loading, unloading, fuelling, parking or maintenance.

Source: DGCA CAR, Section 9, Series C, Part I, Chapter 1 — Definitions, printed page 5.

Controlled flight — actual regulatory wording

Any flight which is subject to an air traffic control clearance.

In simple English: A flight that is subject to an ATC clearance.

Source: DGCA CAR, Section 9, Series C, Part I, Chapter 1 — Definitions, printed page 7.

Flight visibility — actual regulatory wording

The visibility forward from the cockpit of an aircraft in flight.

In simple English: What can be seen forward from the cockpit while the aircraft is in flight.

Source: DGCA CAR, Section 9, Series C, Part I, Chapter 1 — Definitions, printed page 8.

Runway — actual regulatory wording

A defined rectangular area on a land aerodrome prepared for the landing and take-off of aircraft.

In simple English: The prepared rectangular area used for landing and take-off.

Source: DGCA CAR, Section 9, Series C, Part I, Chapter 1 — Definitions, printed page 12.

Taxiway and its included types — actual regulatory wording

Taxiway. A defined path on a land aerodrome established for the taxiing of aircraft and intended to provide a link between one part of the aerodrome and another, including:

a) Aircraft stand taxi lane. A portion of an apron designated as a taxiway and intended to provide access to aircraft stands only.

b) Apron taxiway. A portion of a taxiway system located on an apron and intended to provide a through taxi route across the apron.

c) Rapid exit taxiway. A taxiway connected to a runway at an acute angle and designed to allow landing aeroplanes to turn off at higher speeds than are achieved on other exit taxiways thereby minimizing runway occupancy times.

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In simple English: A taxiway links parts of the aerodrome. A stand taxi lane provides access to stands; an apron taxiway gives a route across the apron; a rapid exit taxiway allows aircraft to leave the runway at a higher speed than other exits.

Source: DGCA CAR, Section 9, Series C, Part I, Chapter 1 — Definitions, printed page 12. All three included types are retained above.

Applicability & Territorial Application of Rules of the Air

APPLICABILITY — WHO & WHERE?

These rules apply to:

- All aircraft flying over **Indian Territory**
- Aircraft bearing **Indian nationality and registration marks**, wherever they may be — to the extent that they do not conflict with rules published by the State having jurisdiction *(legal power over a person, place or event)* over the territory overflown.

For flights over the **high seas** where a Contracting State has accepted responsibility for providing ATS (pursuant to a regional air navigation agreement), the "appropriate ATS authority" is the authority designated by that State.

COMPLIANCE WITH RULES OF THE AIR

The operation of an aircraft — either **in flight** or on the **movement area of an aerodrome** — shall be in compliance with:

- The **General Rules**, and in addition when in flight, either:
- a) The **Visual Flight Rules (VFR)**, or
- b) The **Instrument Flight Rules (IFR)**.

> **Note (DGCA):** Local flights as may be exempted by ATC and training flights of Flying Club aircraft cleared by ATC may be operated during night in Visual Meteorological Conditions. A *local flight* is a flight wholly conducted in the immediate vicinity of an aerodrome.

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PIC Authority, Responsibilities & Psychoactive Substances

RESPONSIBILITY FOR COMPLIANCE — PIC

The **Pilot-in-Command (PIC)** of an aircraft shall, whether manipulating the controls or not, be **responsible** for the operation of the aircraft in accordance with the rules of the air.

Exception: The PIC may *depart* from these rules in circumstances that render such departure **absolutely necessary in the interests of safety.**

PRE-FLIGHT ACTION

Before beginning a flight, the PIC shall **become familiar with all available information appropriate to the intended operation.**

For flights **away from the vicinity of an aerodrome**, and for **all IFR flights**, pre-flight action shall include:

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- A careful study of available **current weather reports and forecasts**
- Taking into consideration **fuel requirements**
- An **alternative course of action** if the flight cannot be completed as planned

AUTHORITY OF PIC

The PIC of an aircraft shall have **final authority** as to the disposition of the aircraft **while in command.**

PSYCHOACTIVE SUBSTANCES — STRICT PROHIBITION

No person whose function is **critical to the safety of aviation (safety-sensitive personnel)** shall undertake that function while **under the influence of any psychoactive substance**, by reason of which human performance is impaired.

No such person shall engage in **any kind of problematic use of substances.**

General Rules — Protection of Persons & Property

NEGLIGENT OR RECKLESS OPERATION

An aircraft shall **NOT** be operated in a negligent or reckless manner so as to **endanger life or property of others.**

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Minimum Safe Heights

MINIMUM SAFE HEIGHT — GENERAL RULE

Except when necessary for take-off or landing, or except by permission from the **Director General**, aircraft shall **NOT** be flown over:

- The **congested areas** of cities, towns or settlements
- Over an **open-air assembly** of persons

...unless at such a height as will permit, in the event of an emergency arising, a landing to be made without undue hazard to persons or property on the surface.

Note: The Director-General may, by order in writing, permit — subject to conditions and restrictions — aircraft engaged in **aerial spraying, crop dusting, or mining exploration** to fly at or above such height as specified in the order.

VFR MINIMUM SAFE HEIGHT VALUES (CAR SECTION 9)

Area	Minimum Height AGL
Over congested areas / cities / towns or open-air assembly — within 600m radius of aircraft	300 m (1,000 ft) above the highest obstacle

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Elsewhere (outside congested areas)	150 m (500 ft) above ground or water

> **Note:** These minimums do NOT apply during take-off or landing, or with explicit DG permission.

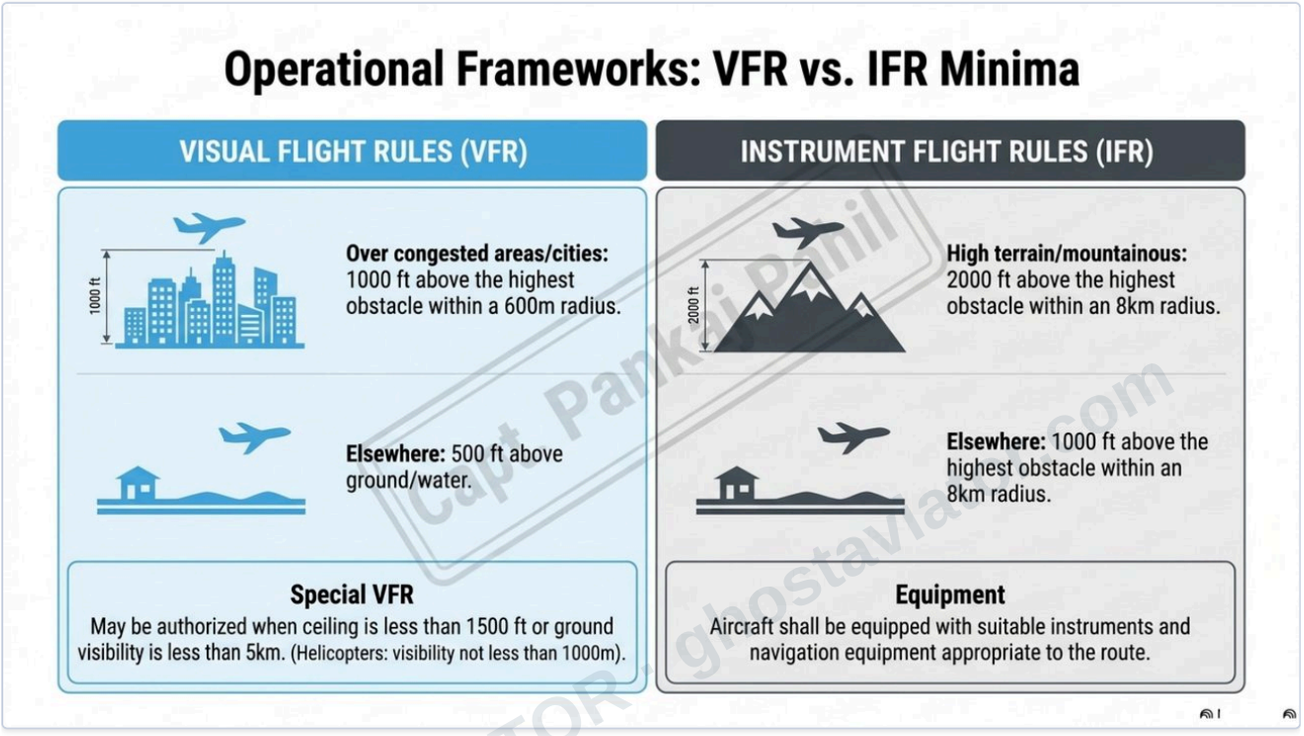

Figure 3.1 *Minimum heights: VFR and IFR*

Cruising Levels

DEFINITION — CRUISING LEVELS

The cruising levels at which a flight or a portion of a flight is to be conducted shall be in terms of:

- a) **Flight levels** — for flights at or above the lowest usable flight level or, where applicable, above the transition altitude
- b) **Altitudes** — for flights below the lowest usable flight level or, where applicable, at or below the transition altitude

STUDENT TIP

Simply put: above transition altitude = use **Flight Levels (QNE)**. At or below transition altitude = use **Altitude (QNH)**. The Semicircular Rule determines which FL/altitude to select based on your magnetic track. See Section 27 for the full Cruising Levels table.

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Dropping / Spraying / Towing / Parachutes / Acrobatic Flight

DROPPING OR SPRAYING

Nothing shall be dropped or sprayed from an aircraft in flight **except** under conditions prescribed *(laid down by the authority or the rules)* in **Rule 26 of the Aircraft Rules, 1937** and as indicated by relevant information, advice and/or clearance from the appropriate ATS unit.

TOWING — INDIA SPECIFIC

No aircraft shall be towed by another aircraft.

Objects other than aircraft shall be towed by an aircraft, in accordance with requirements prescribed by the Director General and as indicated by relevant ATS information/clearance.

TOWING OF AIRCRAFT IS NOT PERMITTED IN INDIA.

Source: DGCA CAR, Section 9, Series C, Part I, §3.1.5, printed page 16.

PARACHUTE DESCENTS

No person shall, **except in an emergency**, descend by means of a parachute from an aircraft, and no person shall drop or cause to be dropped from an aircraft in flight any article, whether attached to a parachute or not, unless:

- The descent/drop is made in accordance with conditions or limitations in general or special order of the **Central Government in writing** (Rule 26 of Aircraft Rules, 1937).

PARACHUTE DESCENT NOT ALLOWED IN NORMAL CONDITIONS IN INDIA.

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Acrobatic Flight — Conditions & Prohibitions

WHEN CAN AN AIRCRAFT BE USED FOR AEROBATICS?

- a) It shall be flown by a **licensed pilot**, or if by a person qualifying for a licence, they must be **accompanied by a licensed pilot instructor**
- b) If passengers are carried, their **previous consent in writing** must be obtained
- c) The pilot must satisfy himself that every person carried is **properly secured by safety belts**
- d) Aerobatics shall be commenced at such a height to permit completion of the manoeuvre at a height of **not less than 600 metres (2,000 ft)** above the ground, or such higher height as specified in the Certificate of Airworthiness for aircraft in the aerobatic category.

NO PERSON SHALL FLY ACROBATICALLY

- So as to **constitute a hazard to air traffic**
- In the vicinity of an aerodrome at a distance of **less than 2 NM** from the nearest point of the perimeter, unless flown at a height greater than **1,800 m (6,000 ft)**
- When flying over any **city, town, village, populous area**, or over any meeting/games/sports/public assembly — *except where a written request has been made to the Director General by the promoters of such meeting.*

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Min. completion height: **600 m / 2,000 ft AGL**
Near aerodrome: **< 2 NM radius prohibited** (unless > 1800m / 6000ft)

Formation Flights

INDIA-SPECIFIC RULE

No civil aircraft shall be flown in formation in India.

ICAO ANNEX 2 — FORMATION FLIGHT PROVISIONS (FOR REFERENCE)

Aircraft shall not be flown in formation except by **pre-arrangement among pilots-in-command**. For formation flight in controlled airspace, it must be in accordance with conditions prescribed by the appropriate ATS authority, which shall include:

- a) The formation operates as a **single aircraft** with regard to navigation and position reporting
- b) Separation between aircraft is the **responsibility of the flight leader** and other PICs, including during transition when aircraft are manoeuvring, join-up, and breakaway
- c) Each aircraft shall maintain a distance **not exceeding 1 km (0.5 NM) laterally and longitudinally** and **30 m (100 ft) vertically** from the flight leader

RPA Formation: A remotely piloted aircraft shall be operated so as to minimize hazards to persons, property or other aircraft and in accordance with conditions specified in Appendix 4 of Annex 2.

Lateral/Longitudinal separation: **≤ 1 km (0.5 NM)**
Vertical separation: **≤ 30 m (100 ft)**
India civil formation: **PROHIBITED**

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Avoidance of Collisions & Right-of-Way Rules

AVOIDANCE OF COLLISIONS — ABSOLUTE RESPONSIBILITY

Nothing in these rules shall relieve the PIC of an aircraft from the responsibility of taking such action, **including collision avoidance manoeuvres based on Resolution Advisories (RA) provided by ACAS equipment**, as well as to avert collision.

PROXIMITY

An aircraft shall **NOT** be operated in such proximity to other aircraft as to **create a collision hazard.**

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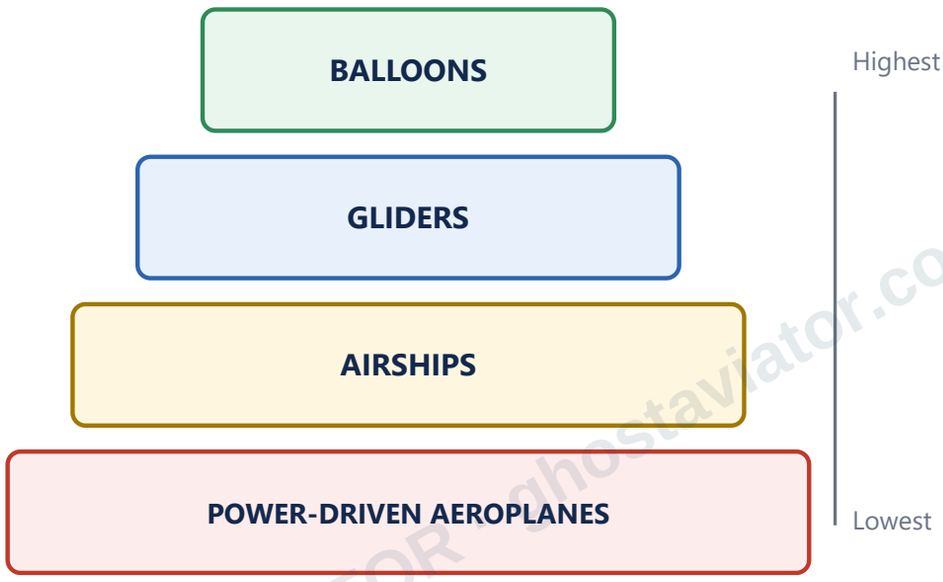
RIGHT-OF-WAY RULES (In Flight)

GENERAL PRINCIPLE

The aircraft that has the **right-of-way shall maintain its heading and speed.**

An aircraft obliged to keep out of the way shall **avoid passing over, under or in front** of the other, unless it passes well clear and takes into account the effect of **aircraft wake turbulence.**

RIGHT-OF-WAY PRIORITY PYRAMID (Give Way ↓ Has Priority ↑)



Priority for right of way, from highest (top) to lowest (bottom)

An aircraft towing another aircraft or an object also has right of way over a power-driven aircraft.

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Figure 3.2 *Order of right of way. A power-driven aircraft gives way to airships, gliders and balloons; an airship gives way to gliders and balloons; a glider gives way to balloons. A power-driven aircraft also gives way to an aircraft towing another aircraft or an object.*

At top = highest priority; power-driven heavier-than-air aircraft gives way to airships, gliders & balloons

Situation	Rule	Key Point
Approaching Head-On	Each shall **alter heading to the RIGHT**	Both aircraft turn right simultaneously
Converging (same level)	Aircraft that has the other on its **RIGHT shall give way**	Except: Power-driven gives way to airships/gliders/balloons; Airships give way to gliders/balloons; Gliders give way to balloons; Power-driven gives way to aircraft towing objects
Overtaking	Overtaking aircraft keeps out of way by **altering heading to the RIGHT**	Overtaking aircraft = approaches from rear at angle <70° to plane of symmetry. Being overtaken = RIGHT OF WAY regardless of climbing, descending or level flight
Landing (general)	Aircraft in flight or on ground/water **gives way to aircraft landing or in final approach**	When two aircraft approaching to land — lower one has priority. But lower one cannot cut in front of another in final approach

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Landing (multiple aircraft)	Aircraft at **higher level gives way** to aircraft at lower level	Power-driven heavier-than-air gives way to gliders; lower aircraft cannot abuse this rule
Emergency Landing	An aircraft aware that another is **compelled to land shall give way** to that aircraft	Absolute priority in emergencies
Taking Off	Aircraft taxiing gives way to aircraft **taking off or about to take off**	Taxiing aircraft yields to aircraft on takeoff roll

STUDENT TIP — OVERTAKING DEFINITION

An **overtaking aircraft** is one approaching another from the **rear on a line forming an angle of less than 70°** with the plane of symmetry of the latter — i.e., positioned such that at night it cannot see either the port (left) OR starboard (right) navigation lights of the other aircraft.

Once overtaking, **no subsequent change in relative positions absolves the overtaking aircraft** from this obligation until it is entirely past and clear.

OFFICIAL STATEMENT

Approaching head-on. When two aircraft are approaching head-on or approximately so and there is danger of collision, each shall alter its heading to the right.

DGCA CAR Section 9, Series C, Part I — Rules of the Air, paragraph 3.2.2.2; PDF page 19. Official document

In plain English: When aircraft approach each other head-on and there is a collision risk, both pilots turn to their own right. This is the aircraft's right, not the reader's right.

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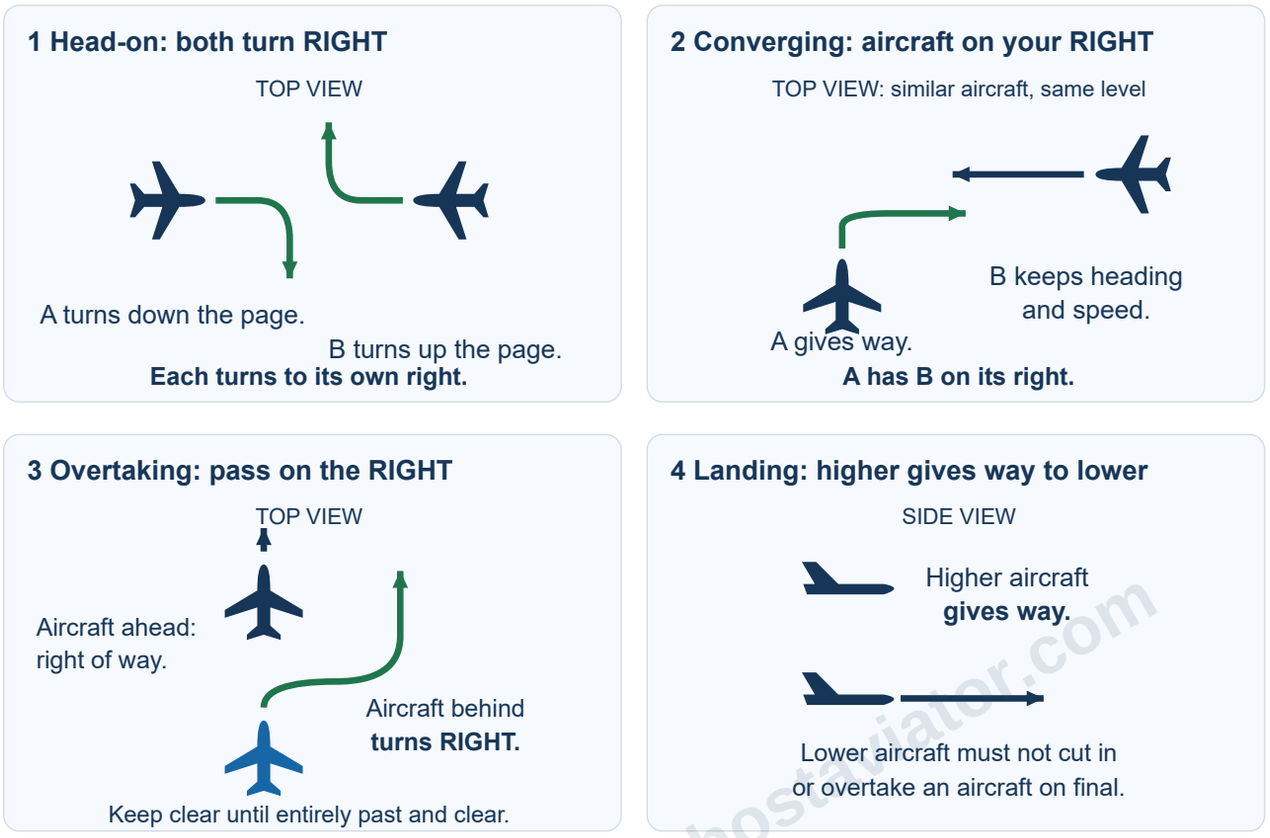

Figure 3.3 *Right of way: use the aircraft's own right, not the reader's right. Converging example assumes similar aircraft at approximately the same level. Aircraft-category exceptions and emergency landing rules are explained in the text. Source: DGCA CAR Section 9, Series C, Part I, §§3.2.2.2–3.2.2.5, printed pages 19–20.*

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Surface Movement of Aircraft

GROUND COLLISION AVOIDANCE — TAXIING RULES

In case of danger of collision between two aircraft taxiing on the movement area of an aerodrome:

- a) **Head-on approach:** Each shall stop or, where practicable, alter course to the **RIGHT** so as to keep well clear
- b) **Converging course:** The one which has the other on its **right shall give way**
- c) **Overtaking:** Aircraft being overtaken has **right-of-way**; overtaking aircraft shall keep well clear

RUNWAY HOLDING POSITION

An aircraft taxiing on the manoeuvring area shall **STOP AND HOLD at all runway holding positions** unless otherwise authorised by the aerodrome control tower.

LIGHTED STOP BARS

An aircraft taxiing on the manoeuvring area shall **STOP AND HOLD at all lighted stop bars** and may proceed further **ONLY when the lights are switched off.**

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Lights to be Displayed by Aircraft

LIGHTS IN FLIGHT — MANDATORY DISPLAY PERIOD

From sunset to sunrise (or during any other period prescribed by the Director General), **all aircraft in flight** shall display:

- a) **Anti-collision lights** — intended to attract attention to the aircraft
- b) **Navigation lights** — intended to indicate the relative path of the aircraft to an observer

Other lights shall NOT be displayed if they are likely to be **mistaken for navigation lights.**

Note: Lights fitted for other purposes (landing lights, airframe floodlights) may be used in addition to anti-collision lights as specified in the Airworthiness Technical Manual (Doc 9051) to enhance aircraft conspicuity.

LIGHTS ON GROUND (MOVEMENT AREA) — FROM SUNSET TO SUNRISE

- a) All aircraft moving on the movement area shall display **navigation lights**
- b) Unless stationary and otherwise adequately illuminated, all aircraft on the movement area shall display lights indicating the **extremities of their structure**
- c) All aircraft operating on the movement area shall display lights to attract **attention**
- d) All aircraft on the movement area whose engines are running shall display lights which **indicate that fact**

ANTI-COLLISION LIGHTS — ALWAYS ON

- All aircraft in flight and fitted with anti-collision lights shall display such lights **also outside the period** specified (sunset to sunrise)
- All aircraft operating on the movement area of an aerodrome and fitted with anti-collision lights shall also display such lights outside the specified period

WHEN MAY FLASHING LIGHTS BE SWITCHED OFF?

A pilot shall be permitted to **switch off or reduce the intensity** of any flashing lights if they:

- a) **Adversely affect the satisfactory performance of duties**, or
- b) **Subject an outside observer to harmful dazzle**

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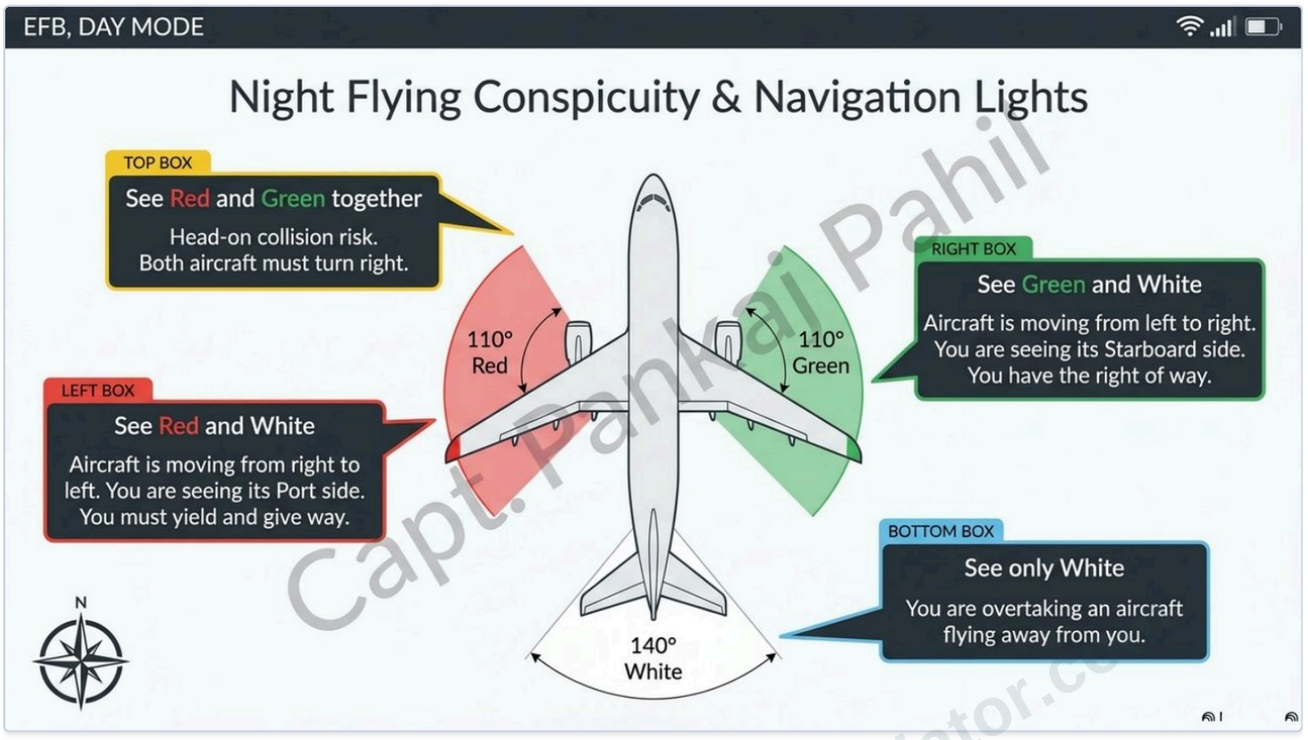

Figure 3.4 *Navigation lights: what you see and what it means*

Simulated Instrument Flights

SIMULATED INSTRUMENT CONDITIONS — CONDITIONS REQUIRED

An aircraft shall **NOT** be flown under simulated instrument flight conditions **UNLESS:**

- a) **Fully functioning dual controls** are installed in the aircraft, AND
- b) A **qualified pilot occupies a control seat** to act as **safety pilot** for the person flying under simulated instrument conditions.

The safety pilot shall have **adequate vision forward and to each side** of the aircraft, OR a **competent observer in communication with the safety pilot** shall occupy a position from which the observer's field of vision adequately supplements that of the safety pilot.

Operation On and In the Vicinity of an Aerodrome

DUTIES OF EVERY AIRCRAFT NEAR AN AERODROME

An aircraft operated on or in the vicinity of an aerodrome shall, whether or not within an aerodrome traffic zone:

- a) **Observe other aerodrome traffic** for the purpose of avoiding collision
- b) **Conform with or avoid the pattern of traffic** formed by other aircraft in operation
- c) Make all turns to the **LEFT** when approaching for a landing and after taking off, **unless otherwise instructed**

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- d) **Land and take off into the wind** unless safety, runway configuration, or air traffic considerations determine a different direction is preferable

WHEN AERODROME CONTROL TOWER IS IN OPERATION — ADDITIONAL DUTIES

The PIC shall also:

- a) Maintain a **continuous listening watch** on the appropriate radio frequency of the aerodrome control tower (unless under approach control service or not possible, in which case keep a watch for **visual signals**)
- b) Obtain either by **radio or visual signals prior authorisation** for any manoeuvre preparatory to or associated with taxiing, landing, or take-off

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The ICAO Flight-Plan Form: Read It Field by Field

A flight plan tells ATS who is flying, the proposed route and level, the timing and the information needed if assistance is required. AAI requires the ICAO model flight plan for the information supplied about an intended flight or part of a flight.

Sources: AAI eAIP ENR 1.10 §1.2; ICAO Doc 4444, Appendix 2 §§1–2.

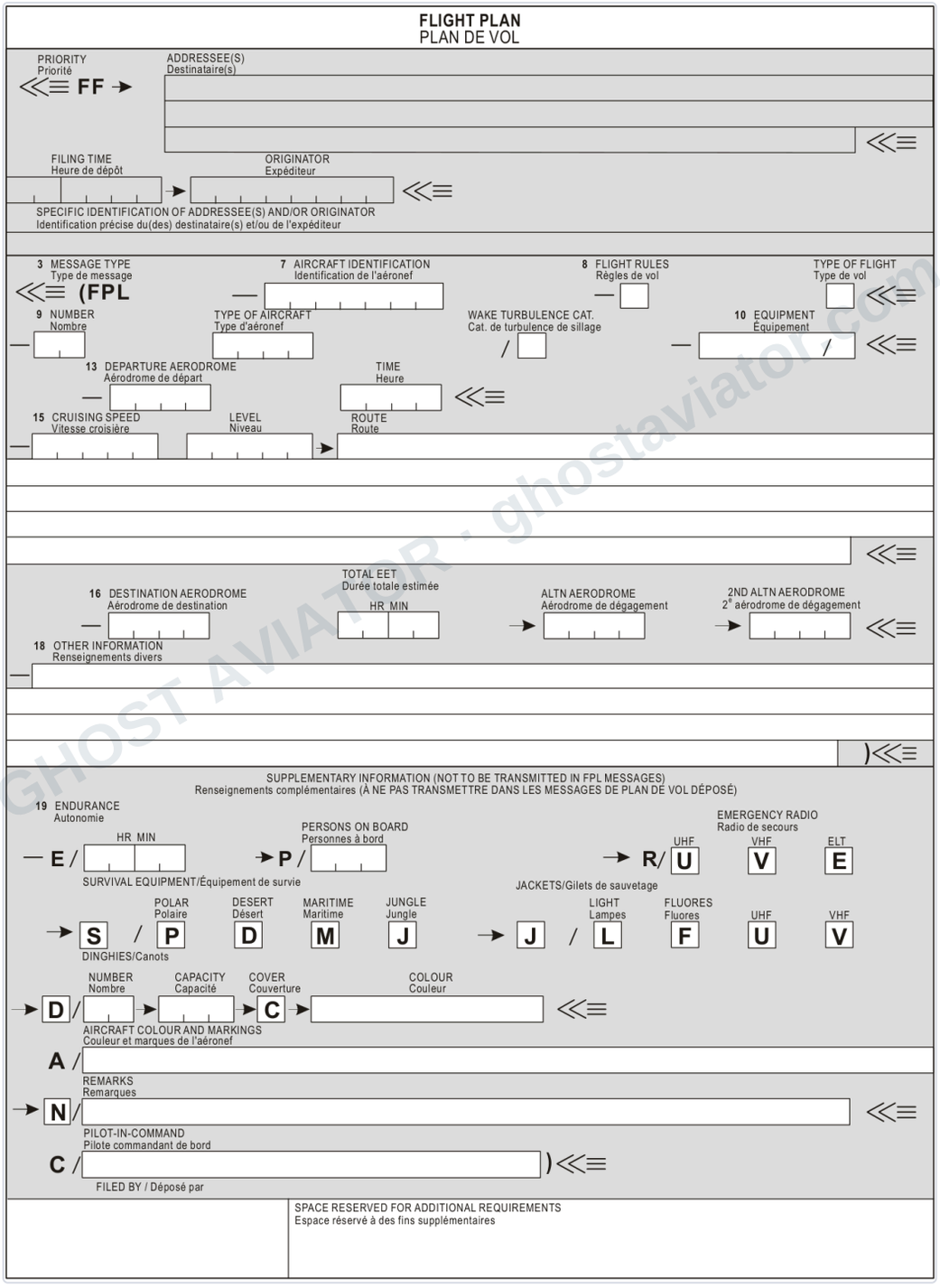

Figure 3.5 *ICAO model flight-plan form — layout reproduced from Doc 4444, Appendix 2 §1, page A2-2. The numbered fields are explained below.*

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Flight-plan terms — regulation and meaning

Flight plan: “Specified information, relative to an intended flight or portion of a flight of an aircraft.” In simple English: the specified information about the flight or the part of the flight concerned.

Filed flight plan (FPL or eFPL): “The latest flight plan as submitted by the pilot, an operator or a designated representative, for use by ATS units.” This is the latest submitted plan.

Current flight plan (CPL): “The flight plan, that reflects changes to the filed flight plan, if any, by subsequent ATC clearances.” This is the submitted plan with changes made through later ATC clearances.

Estimated off-block time: “The estimated time at which the aircraft will commence movement associated with departure.” It concerns departure movement, rather than take-off.

Source: DGCA CAR Section 9, Series C, Part I, Definitions, printed pages 7–8.

Before filling the form

Start at the first available space in each field. Use four figures for clock time in UTC. Use four figures, hours followed by minutes, for elapsed time. The item numbers have gaps because they correspond to ATS message field numbers.

Three different times: `0930` in Item 13 means an off-block clock time of 09:30 UTC for a pre-departure flight plan. `0130` in Item 16 means 1 hour 30 minutes of estimated elapsed time. `E/0430` in Item 19 means fuel endurance of 4 hours 30 minutes. These entries answer different questions.

Source: Doc 4444, Appendix 2 §2.1 and §2.2, Items 13, 16 and 19.

The message header and Item 3

The shaded header contains message priority, addressees, filing time and originator. Addressees are the ATS message addresses that must receive the plan. The originator is the unit or organisation sending the message. Filing time is a message-handling time; it is not the aircraft’s departure time. The header is completed by ATS/COM services unless message origination has been delegated. Item 3 identifies the message as `FPL`, a filed flight plan.

Sources: Doc 4444, Appendix 2 §§1, 2.1 and 2.5; AAI eAIP ENR 1.10 §4 for Indian addressing responsibilities.

Item 7 — Aircraft identification

Enter the aircraft identification, with a maximum of seven characters. This may be the registration, or the ICAO operating-agency designator followed by the flight identification, as applicable to the call sign being used. Keep the flight-plan identification consistent with the identification used by the aircraft’s surveillance equipment.

Sources: Doc 4444, Appendix 2 §2.2, Item 7; AAI eAIP ENR 1.6 §15.8.2.

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Item 8 — Flight rules and type of flight

Flight rules	Meaning	Type of flight	Meaning
I	IFR	S	Scheduled air service
V	VFR	N	Non-scheduled air transport
Y	IFR first; specify the change point in Item 15	G	General aviation
Z	VFR first; specify the change point in Item 15	M / X	Military / other than the listed categories

Enter type of flight when required by the appropriate ATS authority. For example, `IG` identifies IFR general aviation. This describes the proposed flight; it does not itself grant a clearance.

Source: Doc 4444, Appendix 2 §2.2, Item 8. X means other than the listed categories.

Item 9 — Number, aircraft type and wake category

Enter the number of aircraft when there is more than one. Use the ICAO aircraft type designator. If no designator exists, or a formation contains different aircraft types, enter `ZZZZ` and give the required types and numbers after `TYP/` in Item 18. Enter the applicable wake-turbulence category after the slash. Aircraft type and wake category describe different things; use the category assigned under the applicable current ATS provisions.

Source: Doc 4444, Appendix 2 §2.2, Item 9; aircraft type designators are specified in ICAO Doc 8643.

Item 10 — Equipment and capabilities

The part before the slash describes communication, navigation and approach equipment/capabilities. The part after the slash describes surveillance equipment/capabilities. The entry must represent what is serviceable and applicable to this flight. Having a box installed is not the same as holding an operational approval for every use of that box.

For Indian ADS-B flight planning, AAI specifies `B1` for dedicated 1090 MHz ADS-B OUT and `B2` for OUT and IN. AAI also specifies the applicable Mode S designator in the surveillance part. These are examples of surveillance entries, not a complete equipment string to copy into every flight plan.

Sources: Doc 4444, Appendix 2 §2.2, Item 10 for field structure; AAI eAIP ENR 1.6 §§15.2.2, 15.3.1 and 15.8.1 for Indian ADS-B approval and flight-planning requirements.

Item 13 — Departure aerodrome and time

Enter the departure aerodrome's four-letter ICAO location indicator, followed by the estimated off-block time for a plan filed before departure. Off-block time means the planned time the aircraft starts leaving its parking position; it is not take-off time. If the aerodrome has no assigned indicator, use `ZZZZ` and identify it after `DEP/` in Item 18.

For a flight plan received from an aircraft already in flight, use `AFIL`. The time then refers to the actual or estimated time over the first point to which the plan applies. Item 18 identifies the ATS unit from which supplementary flight-plan information can be obtained.

Source: Doc 4444, Appendix 2 §2.2, Item 13.

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Item 15 — Cruising speed, level and route

Begin with the first cruising speed and level, then enter the route. Cruising speed here is true airspeed or the prescribed Mach format, not IAS. Use `N` plus four figures for knots, `K` plus four for kilometres per hour, or `M` plus three for Mach. A flight level uses `F` plus three figures; an altitude in hundreds of feet uses `A` plus three. Metric formats apply where prescribed.

> `N0120A045` means 120 kt TAS and an altitude of 4,500 ft. `N0450F330` means 450 kt TAS and FL330. These are format examples, not route or level clearances.

Describe the route using the applicable ATS-route designators and significant points. Use `DCT` for direct portions where the instructions require it. Identify planned changes of speed, level or flight rules at the relevant point. For a speed or level change, give both the speed and level after the point and slash, even if only one changes. An IFR/VFR change point must agree with Item 8.

Source: Doc 4444, Appendix 2 §2.2, Item 15(a)–(c), including (c)(3) and (4).

Item 16 — Destination, total EET and alternates

Enter the destination's four-letter ICAO indicator followed by total estimated elapsed time in hours and minutes. Enter up to two destination alternate aerodromes in the spaces provided. Where an indicator has not been assigned, enter `ZZZZ` and identify the aerodrome in Item 18 using `DEST/` or `ALTN/`, as appropriate.

Total EET is a duration, not an arrival clock time and not fuel endurance. For a flight plan received in flight, it is measured from the first point to which that plan applies.

Total estimated elapsed time — where it ends: For IFR, the CAR measures from take-off to the designated navigation-reference point where the instrument approach is intended to begin; if no navigation aid is associated with the destination aerodrome, it ends over that aerodrome. For VFR, it measures from take-off to arrival over the destination aerodrome. It is a duration, and does not automatically mean time to touchdown.

Source: DGCA CAR Section 9, Series C, Part I, Definitions, "Total estimated elapsed time", printed page 12.

Source: Doc 4444, Appendix 2 §2.2, Item 16.

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Item 18 — Other information and cross-references

Use the prescribed indicators followed by a slash. This field supplies detail that cannot be expressed fully in the earlier boxes. For example, `REG/` identifies registration when different from Item 7; `TYP/` expands ZZZZ in Item 9; `DEP/`, `DEST/` and `ALTN/` expand the relevant aerodrome entries; `EET/` gives accumulated elapsed times to specified points or FIR boundaries. `RMK/` carries required or necessary plain-language remarks.

AAI requires accumulated elapsed times to FIR boundary points where applicable, together with the pilot-in-command's telephone/mobile number, in Item 18. Additional authority details apply to the non-scheduled operations specified in ENR 1.10. Match Item 18 to the earlier fields; do not leave a ZZZZ entry unexplained.

Sources: Doc 4444, Appendix 2 §2.2, Item 18; AAI eAIP ENR 1.10 §§2.2.3, 2.3.4 and 2.4.4. This explanation covers the field's function and common cross-references; it is not the full current indicator catalogue.

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Item 19 — Supplementary information

These details support assistance and search and rescue. They are not transmitted in the basic FPL message shown on the form. Complete them as required by the appropriate ATS authority. On the paper form, cross out equipment indicators that do not apply; a remaining indicator describes equipment available.

Entry	What to enter or indicate
E/ — Endurance	Fuel endurance in four figures, hours and minutes.
P/ — Persons on board	Total passengers and crew. TBN means “to be notified” when the total is not known at filing.
R/ — Emergency radio	U: 243.0 MHz; V: 121.5 MHz; E: ELT. Cross out unavailable equipment.
S/ — Survival equipment	P: polar; D: desert; M: maritime; J: jungle. Cross out categories not carried.
J/ — Life jackets	L: lights; F: fluorescein; U/V: radio capability. Cross out unavailable features; cross out all if no jackets.
D/ — Dinghies	Number, combined person capacity, cover and colour. Cross out D and C if no dinghies; cross out C if not covered.
A/ — Aircraft appearance	Colour and significant markings to help identify the aircraft.
N/ — Remarks	Other survival equipment or relevant survival remarks; cross out N if none.
C/ — Pilot	Name of the pilot-in-command.

Source: Doc 4444, Appendix 2 §2.2, Item 19; Appendix 2 §3 for FPL-message transmission.

Filed by, acceptance and the remaining space

“Filed by” identifies the person, agency or unit submitting the plan. ATS acceptance is recorded in the manner prescribed by the appropriate authority. The final reserved space accommodates additional requirements. Filing, acceptance and an ATC clearance are separate steps.

Source: Doc 4444, Appendix 2 §§2.3–2.5; AAI eAIP ENR 1.10 §§2 and 4.

Indian filing and departure requirements

Except where necessary for operational or technical reasons, submit the flight plan to the appropriate ATS unit at least **180 minutes before estimated off-block time**. For RPA, the provision specifies at least 24 hours before estimated departure. These are Indian submission requirements, separate from the time formats on the form.

For a proposed departure delay of **15 minutes or more** after filing, the pilot-in-command or representative must notify ATC of the revised departure time. Amend the plan where applicable, or submit a new plan and cancel the old one.

At the engine-start or taxi request, pass total persons on board and report that security checks have been completed to the appropriate clearance-delivery, surface-movement or aerodrome-control unit. Where RNAV/RNP SIDs are in use, report non-RNAV status before pushback and start-up.

Source: AAI eAIP ENR 1.10 §§3.1–3.4.

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Flight Plans

Read the CAR baseline together with the Indian ATS requirement

CAR §3.3.1.4 specifies at least 60 minutes before departure for a flight receiving ATC or air traffic advisory service, **unless otherwise prescribed by the appropriate ATS authority**. AAI eAIP ENR 1.10 §3.1 prescribes at least 180 minutes before EOBT in India, except where necessary for operational or technical reasons. The authority qualification matters.

For submission during flight, CAR §3.3.1.4 requires receipt by the appropriate ATS unit at least 10 minutes before the estimated point of entry into a control or advisory area, or the point of crossing an airway or advisory route.

Sources: DGCA CAR Section 9, Series C, Part I, §3.3.1.4, printed page 25; AAI eAIP ENR 1.10 §3.1.

WHEN IS A FLIGHT PLAN MANDATORY?

A flight plan shall be submitted prior to operating (exceptions for nano/micro RPA below 50 ft/200 ft AGL in uncontrolled airspace/enclosed premises):

- a) Any flight or portion thereof to be provided with **air traffic control service**
- b) Any **IFR flight** within advisory airspace
- c) Any flight within or into **designated areas or along designated routes** when required by the ATS authority for flight information, alerting, and search & rescue services
- d) Any flight within or into designated areas or along designated routes requiring **co-ordination with military units or ATS units in adjacent States** (to avoid interception)
- e) Any flight **across International borders**

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FLIGHT PLAN SUBMISSION TIMES

Type	Submission Time
Standard intended flights	**At least 180 minutes before EOBT**, except where necessary for operational or technical reasons. Source: AAI eAIP ENR 1.10 §3.1.
RPA (Remotely Piloted Aircraft)	**At least 24 hours before estimated time of departure**
Delay of 15 minutes or more (previously filed FP)	PIC or representative must **notify ATC of revised ETD**; amend FP or file new FP and cancel old one

COMPLETION OF A FLIGHT PLAN

Whatever the purpose for which it is submitted, a flight plan shall contain information as applicable on relevant items up to and including **"Alternate aerodrome(s)"** regarding the whole route or the portion thereof for which the flight plan is submitted.

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RNP/RVSM/RCP/RSP REQUIREMENT

An operator shall, prior to departure, ensure that where the flight is intended to operate on a route or in an area where an **RNP/RVSM/RCP/RSP type specification is prescribed**, the aircraft has an appropriate approval and that all conditions applying to that approval will be satisfied.

Changes to a Flight Plan

Flight Type	When to Report Changes
IFR flight OR VFR flight operated as controlled flight	As soon as practicable to the appropriate ATS unit
Other VFR flights	Significant changes shall be reported as soon as practicable
Fuel endurance or number of persons on board (if incorrect at departure)	Constitutes a significant change — must be reported

Closing a Flight Plan

ARRIVAL REPORT

Unless otherwise prescribed by the appropriate ATS authority, a **report of arrival** shall be made in person, by radiotelephony or via data link at the **earliest possible moment** after landing, to the appropriate ATS unit at the arrival aerodrome, by any flight for which a flight plan has been submitted covering the **entire flight** or the **remaining portion** of a flight to the destination aerodrome.

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Closing the plan when normal arrival reporting is unavailable

- A plan covering only part of a flight, other than the remaining part to destination, is closed by an appropriate report to the relevant ATS unit when required (§3.3.5.2).
- If no ATS unit exists at the arrival aerodrome, report as soon as practicable after landing, using the quickest means available, to the nearest ATS unit or Flight Information Centre (§3.3.5.3).
- If arrival communications are known to be inadequate and there are no alternate ground-reporting arrangements, transmit a message comparable to an arrival report immediately before landing, if practicable and where a report is required. Normally send it to the aeronautical station serving the ATS unit responsible for the FIR (§3.3.5.4).
- The arrival report contains aircraft identification, departure aerodrome, destination aerodrome only for a diversionary landing, arrival aerodrome and arrival time (§3.3.5.5).

Closing the plan tells ATS that the flight has arrived. An omitted report can trigger unnecessary search-and-rescue action.

Source: DGCA CAR Section 9, Series C, Part I, §§3.3.5.2–3.3.5.5, printed pages 26–27.

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Time (UTC)

TIME STANDARD

- **Co-ordinated Universal Time (UTC)** shall be used and expressed in hours and minutes (and when required, seconds) of the **24-hour day beginning at midnight**
- A **time check** shall be obtained prior to operating a controlled flight and at such other times during the flight as may be necessary
- ATS unit clocks and time-recording devices shall be checked to ensure correct time to within **±30 seconds of UTC**
- Wherever time is utilised in the application of **data link communications**, it shall be accurate to within **1 second of UTC**

ATS clock accuracy: **±30 sec of UTC**
Data link time accuracy: **±1 sec of UTC**
Time format: **24-hour UTC**

Air Traffic Control Service

ATC CLEARANCE

An **ATC clearance shall be obtained prior to operating a controlled flight**, or a portion of a flight as a controlled flight. Such clearance shall be requested through the **submission of a flight plan** to an ATC unit.

ADHERENCE TO FLIGHT PLAN

An aircraft shall adhere to the **current flight plan** or the applicable portion unless a request for a change has been made and **clearance obtained** from the appropriate ATC unit.

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Deviations from the Current Flight Plan

Deviation Type	Required Action
Deviation from Track	Take action forthwith to **adjust heading to regain track as soon as practicable**
Deviation from ATC assigned Mach number/indicated airspeed	Inform appropriate ATS unit **immediately**
Deviation from Mach number / true airspeed — varies by ± Mach 0.02 or more, or ± 19 km/h (10 kt) TAS or more	Inform appropriate ATS unit
Change in Time Estimate — exceeds 2 minutes from notified time	Notify ATS unit as soon as possible (unless ADS-C is activated and serviceable in airspace where ADS-C services are provided)

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ADS-C AUTOMATIC REPORTING

When ADS-C services are provided and ADS-C is activated, the ATSU shall be informed **automatically via data link** whenever changes occur beyond the threshold values stipulated by the **ADS event contract.**

Change Requests — Required Information

Type of Change	Information to Include in Request
Change of cruising level	Aircraft identification; requested new cruising level and Mach/TAS at this level; revised time estimates at subsequent reporting points or FIR boundaries
Change of Mach/TAS	Aircraft identification; requested Mach/TAS
Change of route (destination unchanged)	Aircraft identification; flight rules; description of new route including related FP data beginning with position from which change is to commence; revised time estimates; any other pertinent info
Change of route (destination changed)	Aircraft identification; flight rules; description of revised route to revised destination including related FP data; alternate aerodrome(s); any other pertinent info

Weather Deterioration Below VMC (VFR Controlled Flight)

OPTIONS WHEN VMC CANNOT BE MAINTAINED (VFR CONTROLLED FLIGHT)

- a) Request an **amended clearance to continue in VMC** to destination or alternate, or to leave the airspace requiring ATC clearance
- b) If no clearance can be obtained, **continue in VMC and notify ATC** of the action being taken either to leave the airspace or to land at the nearest suitable aerodrome
- c) If operated within a control zone, request authorisation to operate as a **Special VFR flight**
- d) Request clearance to operate in accordance with **IFR**

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Position Reports & Communications

POSITION REPORTS — CONTROLLED FLIGHTS

Unless exempted, a controlled flight shall report to the ATS unit, as soon as possible, the **time and level** of passing each **designated compulsory reporting point**, together with any other required information. Position reports shall also be made in relation to additional points when requested.

COMMUNICATIONS — CONTROLLED FLIGHTS

An aircraft operated as a controlled flight shall maintain **continuous air-ground voice communication watch** on the appropriate channel and establish **two-way communication** as necessary with the appropriate ATC unit.

SELCAL or similar automatic signalling devices satisfy the requirement to maintain an air-ground voice communication watch.

The requirement for air-ground voice communication watch **remains in effect after CPDLC has been established.**

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Radio Communication Failure (RCF) Procedures

RCF DECISION FLOWCHART

IF IN VMC (RADIO FAILURE) — ACTIONS

- a) **Continue to fly in VMC**, land at the nearest suitable aerodrome, and report arrival by the most expeditious means to the appropriate ATC unit
- b) If considered advisable, complete an **IFR flight in accordance with the provisions below**

IF IN IMC (OR IFR PIC CONSIDERS INADVISABLE TO COMPLETE VMC PROVISIONS) — DETAILED PROCEDURE

Step	Airspace	Action Required
a)	**Non-radar airspace**	Unless otherwise prescribed (regional air navigation agreement) — maintain **last assigned speed and level, or minimum flight altitude if higher, for 20 minutes** following the aircraft's failure to report over a compulsory reporting point, then adjust level and speed per filed flight plan
b)	**Radar-provided airspace**	Maintain last assigned speed and level, or minimum flight altitude if higher, for **7 minutes** following: (1) time last assigned level/min altitude reached; OR (2) time transponder set to Code 7600; OR (3) aircraft's failure to report over compulsory reporting point — whichever is later. Then adjust per filed flight plan.
c)	Radar vectored / RNAV offset	Rejoin current flight plan route no later than the **next significant point**, taking into account applicable minimum flight altitude
d)	En-route	Proceed per current FP route to the **appropriate designated navigation aid or fix serving the destination aerodrome**, and hold over this aid until commencement of descent
e)	Approach	Commence descent from the nav aid/fix at **expected approach time last received and acknowledged**, or if no EAT received, at the **estimated time of arrival from the current FP**
f)	Approach	Complete a **normal instrument approach procedure** as specified for the designated nav aid or fix
g)	Landing	Land, if possible, within **30 minutes after the ETA** specified in step (e) or the last acknowledged EAT, whichever is later

RADIO COMMUNICATION FAILURE: TIME VALUES

Non-Radar: maintain for **20 minutes**

Radar: maintain for **7 minutes**

Land within **30 min** of ETA/EAT

Squawk: **7600**

Note: The provision of ATC service to other flights operating in the airspace concerned will be based on the premise that an aircraft experiencing communication failure will comply with these rules.

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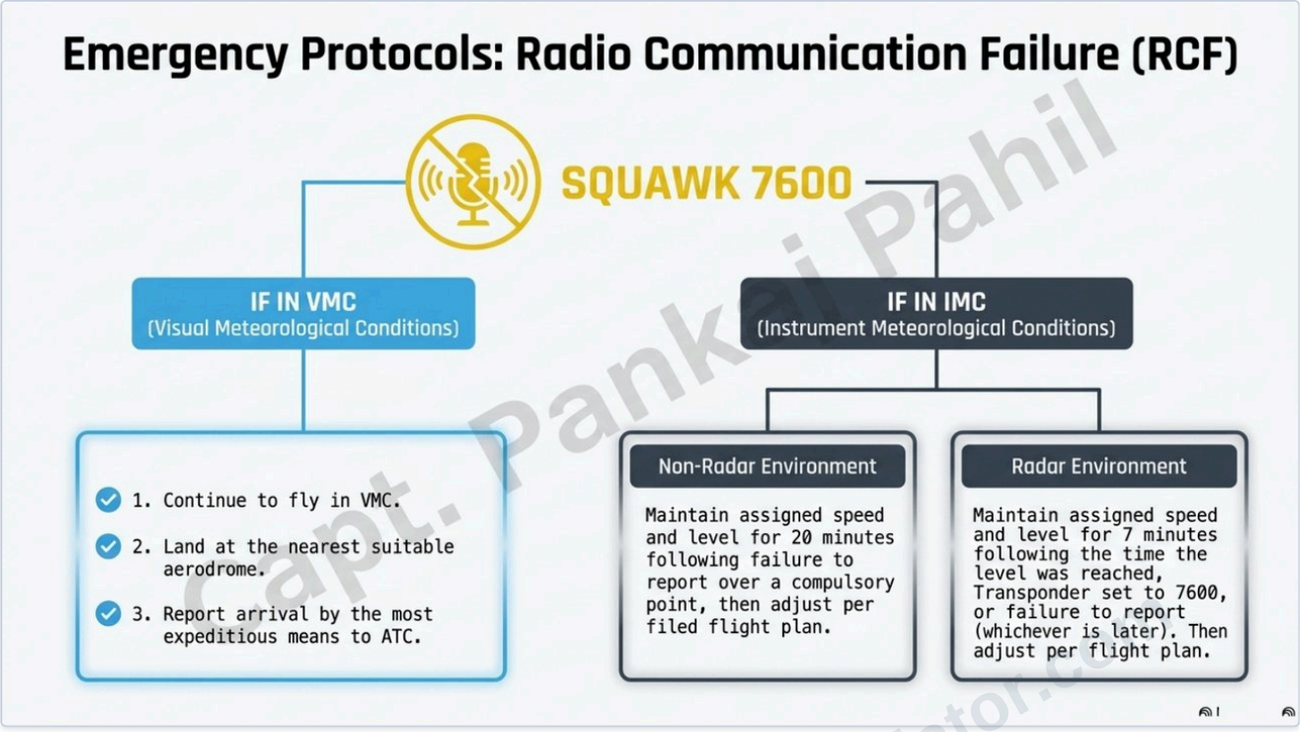

Figure 3.6 *Radio communication failure*

VMC Visibility and Distance from Cloud Minima (Table 3-1)

Source: DGCA CAR Section 9, Series C, Part I, §3.9, Table 3-1 and its notes, printed page 35. The lower-band cloud-clearance exception is for Classes F and G; use the higher of 3,000 ft AMSL and 1,000 ft above terrain to determine the lower-band boundary.

Altitude Band	Airspace Class	Flight Visibility	Distance from Cloud	
			Horizontal	Vertical
At and above 3,050 m (10,000 ft) AMSL	A*** / BCDEFG	8 km	1,500 m	300 m (1,000 ft)
Below 3,050 m (10,000 ft) AMSL and above 900 m (3,000 ft) AMSL, or above 300 m (1,000 ft) above terrain, whichever is higher	A*** / BCDEFG	5 km	1,500 m	300 m (1,000 ft)
At and below 900 m (3,000 ft) AMSL, or 300 m (1,000 ft) above terrain, whichever is higher	A***BCDE	5 km	1,500 m	300 m (1,000 ft)
	F, G	**5 km****	**Clear of cloud and with the surface in sight**	

*** Note on Transition Altitude:** When the height of the transition altitude is lower than 3,050 m (10,000 ft) AMSL, **FL 100 should be used in lieu of 10,000 ft.**

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**** Special Provisions for Airspace F & G (India — at/below 900m/3000ft AMSL or 300m/1000ft AGL):**

- Flight visibilities reduced to not less than **3,000 m** may be permitted for flights operating: (1) at speeds giving adequate opportunity to observe other traffic/obstacles, or (2) in areas where encounter probability with other traffic is normally low
- When so prescribed by the appropriate ATS authority: helicopters may operate up to **1,000 m flight visibility**

Authority qualification: These reduced-visibility permissions apply when prescribed by the appropriate ATS authority. The helicopter provision in the opened Indian CAR specifies 1,000 m, with speed allowing adequate opportunity to see and avoid traffic or obstacles.

***** Class A Airspace:** VMC minima in Class A airspace are included for guidance to pilots and **do not imply acceptance of VFR flights in Class A airspace.**

Visual Flight Rules (VFR)

VFR FLYING CONDITIONS

Except when operating as a **Special VFR** flight, VFR flights shall be conducted so that the aircraft is flown in conditions of **visibility and distance from clouds equal to or greater than those specified in the VMC Table.**

VFR — CONTROL ZONE ENTRY/DEPARTURE RESTRICTIONS

Except when a clearance is obtained from an ATC unit, VFR flights shall **NOT** take off or land at an aerodrome within a control zone, or enter the aerodrome traffic zone or traffic pattern:

- a) When the **ceiling is less than 450 m (1,500 ft)**, or
- b) When the **ground visibility is less than 5 km**

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VFR OPERATING HOURS

VFR flights shall be operated during the period from **20 minutes before sunrise to 20 minutes after sunset**, except when exempted by ATC for local flights and training flights of flying club aircraft cleared by ATC.

VFR ALTITUDE / AIRSPACE RESTRICTIONS (UNLESS ATS AUTHORITY AUTHORIZES)

- VFR flights shall **NOT** operate **above FL 150**
- VFR flights shall **NOT** operate at **transonic and supersonic speeds**
- VFR flights shall **NOT** operate more than **100 NM seaward** from the shoreline within controlled airspace

Authorization for VFR flights to operate above FL 290 shall NOT be granted in areas where a vertical separation minimum of **300 m (1,000 ft) is applied above FL 290.**

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WHEN VFR FLIGHTS COMPLY WITH ATC PROVISIONS

- a) When operated within **Class B, C, and D airspace**
- b) When forming part of aerodrome traffic at **controlled aerodromes**
- c) When operated as **Special VFR flights**

CHANGE FROM VFR TO IFR

An aircraft operating VFR that wishes to change to IFR compliance shall:

- a) If a FP was submitted, **communicate the necessary changes** to be effected to its current flight plan, or
- b) When required, submit a FP to the appropriate ATS unit and **obtain a clearance prior to proceeding IFR** when in controlled airspace.

Special VFR Flights

WHAT IS A SPECIAL VFR FLIGHT?

A VFR flight cleared by ATC to operate within a **control zone in meteorological conditions below VMC**, but following certain special provisions.

CONDITIONS FOR SPECIAL VFR AUTHORISATION

- a) When traffic conditions permit, Special VFR flights may be authorised subject to approval from the unit providing **approach control service**
- b) Requests shall be handled **individually**
- c) When ground visibility is **not less than 1,500 metres**, Special VFR flights may be authorised to enter a CTR for landing, take-off, depart, cross, or operate locally within a CTR
- d) Performance Class I and Class II helicopters may be authorised when ground visibility is **not less than 1,000 metres**
- e) Pilot is responsible for meeting criteria for performance class helicopters and shall state this in Field 18 of the flight plan and report on RTF to ATC
- f) **Special VFR flights shall be operated ONLY by pilots holding:**
  - Instrument Rating (IR), or
  - Assistant Flight Instructor (AFI) rating, or
  - Flight Instructor (FI) rating

Fixed-wing min visibility: **1,500 m**
Helicopter min visibility: **1,000 m**
Required rating: **IR / AFI / FI**

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Instrument Flight Rules (IFR)

IFR AIRCRAFT EQUIPMENT

Aircraft shall be equipped with **suitable instruments and with navigation equipment appropriate to the route to be flown.**

IFR MINIMUM LEVELS (WHERE NO STATE MINIMUM HAS BEEN ESTABLISHED)

Terrain Type	Minimum IFR Level	Obstacle Clearance Reference
High terrain or mountainous areas	600 m (2,000 ft) above highest obstacle	Within **8 km** of estimated position
All other areas	300 m (1,000 ft) above highest obstacle	Within **8 km** of estimated position

Mountains: **600m (2000ft)** above highest obstacle within 8km

Elsewhere: **300m (1000ft)** above highest obstacle within 8km

IFR — Rules Applicable to Controlled Airspace

CRUISING LEVELS IN CONTROLLED AIRSPACE

An IFR flight in cruising flight in controlled airspace shall be flown at a cruising level, or if authorised for cruise climb, between two levels or above a level, selected from:

- a) The **Tables of cruising levels**, or
- b) A modified table for flight above **FL 410**, except that the correlation of levels to track shall not apply whenever otherwise indicated in ATC clearances or specified by ATS authority in AIPs.

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IFR — Rules Applicable Outside Controlled Airspace

CRUISING LEVELS OUTSIDE CONTROLLED AIRSPACE

An IFR flight operating in level cruising outside controlled airspace shall be flown at a cruising level appropriate to its track per:

- a) Tables of cruising levels, except when otherwise specified by appropriate ATS authority for flight at or below 900 m (3,000 ft) AMSL
- b) Modified table for flight above FL 410

COMMUNICATIONS — IFR OUTSIDE CONTROLLED AIRSPACE

An IFR flight operating outside controlled airspace but within or into areas or along routes designated by ATS authority shall maintain **air-ground voice communication watch** and establish two-way communication with the air traffic services unit providing flight information service.

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Change from IFR to VFR

IFR → VFR CANCELLATION

An aircraft electing to change from IFR to VFR compliance shall, if a FP was submitted, **notify the appropriate ATS unit specifically that the IFR flight is cancelled** and communicate thereto the changes to be made to the current FP.

When an aircraft operating IFR encounters VMC, it shall **NOT cancel its IFR flight unless it is anticipated, and intended, that the flight will be continued for a reasonable period of time in uninterrupted VMC.**

Flight by Night (Definition)

NIGHT DEFINITION

A flight performed between **half an hour after sunset** and **half an hour before sunrise.**

> **CONTRAST WITH VFR DAY OPERATIONS**
>
> VFR flights are operated **20 minutes before sunrise to 20 minutes after sunset.**
> **Night definition for IFR/general purposes:** 30 min after sunset to 30 min before sunrise.

IFR — Position Reports Outside Controlled Airspace

IFR POSITION REPORTS (ADVISORY AIRSPACE USERS)

Aircraft electing to use the air traffic advisory service whilst operating IFR within specified advisory airspace are expected to comply with these provisions, **except that flight plan changes are not subjected to clearances** and that two-way communication will be maintained with the unit providing the air traffic advisory service.

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Distress, Urgency & Interception Signals

DISTRESS SIGNALS — MAYDAY (GRAVE AND IMMINENT DANGER)

The following signals, used **either together or separately**, mean that **grave and imminent danger threatens, and immediate assistance is requested:**

- a) Radiotelegraphy: **SOS (· · · — — — · · ·)** in Morse Code
- b) Radiotelephony: spoken word **MAYDAY**
- c) Distress message via **data link** transmitting intent of word MAYDAY
- d) **Rockets or shells** throwing red lights, fired one at a time at short intervals
- e) A **parachute flare showing a red light**

URGENCY SIGNALS — PAN PAN (DIFFICULTIES COMPELLING LANDING, NO IMMEDIATE ASSISTANCE)

Type 1 — Aircraft has difficulties compelling it to land without requiring immediate assistance:

- a) Repeated switching on and off of the **landing lights**
- b) Repeated switching on and off of the **navigation lights** in such manner as to be distinct from flashing nav lights

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Type 2 — Aircraft has a very urgent message concerning safety of a ship, aircraft, vehicle or person:

- a) Radiotelegraphy: **XXX (— · · — — · · — — · · —)**
- b) Radiotelephony: spoken words **PAN, PAN**
- c) Urgency message via **data link** transmitting intent of words PAN, PAN

DISTRESS VS URGENCY — KEY DISTINCTION

Signal	Words Used	Morse	Meaning
DISTRESS	MAYDAY MAYDAY MAYDAY	SOS · · · — — — · · ·	Grave & imminent danger; immediate assistance needed
URGENCY	PAN PAN PAN PAN	XXX — · · — — · · — — · · —	Very urgent message / difficulties compelling land

Interception — Actions of the Intercepted Aircraft

WHEN YOUR AIRCRAFT IS INTERCEPTED — IMMEDIATE ACTIONS

- a) Attempt to establish two-way communication via any available means, including **emergency radio frequency 121.5 MHz** (and if practicable, also on **243 MHz**)
- b) If equipped with SSR transponder: select **Mode A, Code 7700** (unless otherwise instructed by ATS unit)
- c) If equipped with ADS-B or ADS-C: select the **appropriate emergency functionality**, if available

> **Conflict of Instructions:** If any instructions received by radio from any sources **conflict** with those given by the intercepting aircraft by visual signals, the intercepted aircraft shall **request immediate clarification while continuing to comply with the visual instructions** given by the intercepting aircraft. If instructions by radio conflict with those from intercepting aircraft by radio, request clarification while complying with **radio instructions from the intercepting aircraft.**

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INTERCEPTION COMMUNICATION PHRASEOLOGY

PHRASES FOR USE BY INTERCEPTING AIRCRAFT			PHRASES FOR USE BY INTERCEPTED AIRCRAFT		
Phrase	Pronunciation	Meaning	Phrase	Pronunciation	Meaning
CALL SIGN	KOL SA-IN	What is your call sign	CALL SIGN	KOL SA-IN	My call sign is ...
FOLLOW	FOL-LO	Follow me	WILCO	VILL-KO	Understood, will comply
DESCEND	DEE-SEND	Descend for landing	CAN NOT	KANN NOTT	Unable to comply
YOU LAND	YOU LAAND	Land at this aerodrome	REPEAT	REE-PEET	Repeat your instruction
PROCEED	PRO-SEED	You may proceed	AM LOST	AM LOSST	Position unknown
			MAYDAY	MAYDAY	I am in distress
			HIJACK	HI-JACK	I have been hijacked

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			LAND (place)	LAAND	I request to land at (place)
			DESCEND	DEE-SEND	I require descent

Visual Signals — Interception Sequences

Series	Intercepting Aircraft Signal	Meaning	Intercepted Aircraft Response	Meaning
1	DAY/NIGHT — Rocking aircraft + flashing nav lights at irregular intervals from position slightly above and ahead of, to the left of intercepted aircraft (or right if helicopter); slow level turn to left on desired heading	You have been intercepted. Follow me.	DAY/NIGHT — Rocking aircraft + flashing nav lights at irregular intervals, and following	Understood, will comply
2	DAY/NIGHT — Abrupt breakaway climb of 90° or more without crossing the line of flight of intercepted aircraft	You may proceed.	DAY/NIGHT — Rocking the aircraft	Understood, will comply
3	DAY/NIGHT — Lowering landing gear + showing steady landing lights + overflying runway in use	Land at this aerodrome.	DAY/NIGHT — Lowering landing gear + steady landing lights + following and landing if safe	Understood, will comply
4	*(Response to Series 4 from intercepted aircraft)* Raises landing gear + Series 1 signals = follow to alternate aerodrome; or Series 2 signals = release intercepted aircraft		DAY/NIGHT — Raising landing gear + flashing landing lights while passing over runway at 300–600m, continuing to circle runway	Aerodrome you have designated is inadequate
5	Use Series 2 signals	Understood	DAY/NIGHT — Regular switching on/off all available lights but distinct from flashing lights	Cannot comply
6	Use Series 2 signals	Understood	DAY/NIGHT — Irregular flashing of all available lights	In distress

VISUAL SIGNAL — WARNING: RESTRICTED/PROHIBITED/DANGER AREA

By day and by night: a series of **projectiles discharged from the ground at intervals of 10 seconds**, each showing on bursting **red and green lights or stars**, indicate to an unauthorised aircraft that it is flying in or about to enter a restricted, prohibited, or danger area, and that it is to take such remedial action as may be necessary.

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Figure 3.7 *Warning to an aircraft entering a restricted, prohibited or danger area: projectiles bursting with red and green lights or stars.*

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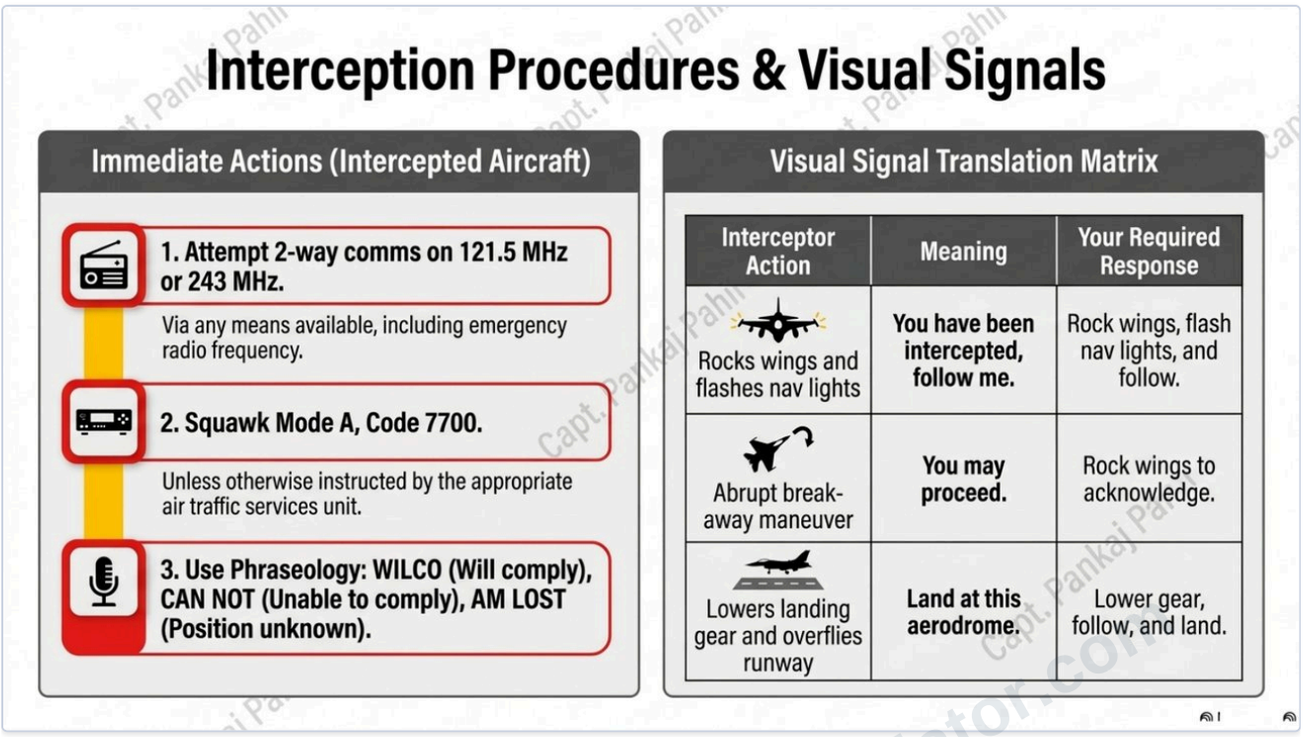

Figure 3.8 *Interception: what to do and what the signals mean*

ADIZ (Air Defense Identification Zone) & Air Defense Clearance (ADC)

ADIZ ZONES IN INDIA

Air Defense Identification Zones (ADIZ) have been established for air defense clearance:

- **ADIZ North, South, East, West, Central, and Sub-ADIZ South East**

WHEN IS ADC REQUIRED?

Except local flights within **5 NM radius** centred at ARP and vertical limits of **1,000 ft AGL** of an aerodrome: aircraft operating to, through, or within the ADIZ shall obtain **Air Defense Clearance (ADC) before take-off**, through the ATC concerned.

ADC Rule	Details
ADC validity *(the period during which a document is accepted)* — Scheduled/RPL/FPL	**(-) 15 minutes to (+) 45 minutes of EOBT**
ADC validity — Non-scheduled domestic	**(-) 15 minutes to (+) 3 hours of EOBT**
ADC route validity	Valid for the **entire route, irrespective of intermediate halts** for flights originating in one ADIZ/FIR and transiting through other ADIZ/FIR
All flights entering Indian airspace	Shall obtain ADC before entering ADIZ from respective FIC, **ten minutes prior to entering Indian airspace**
Communication difficulty at departure	Aircraft with radio may take off with instructions to obtain ADC immediately after airborne from FIC concerned

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Flying Club aircraft (no ATC at aerodrome)	May obtain ADC from nearest IAF ATC Unit; IAF ATC will advise FIC concerned
Same-day return flight	ADC can be issued for return flight also; however, a **fresh ADC is required if delay exceeds 30 minutes** from estimated departure time

Exemption radius: **5 NM / 1000 ft AGL**

Scheduled ADC validity: **-15 to +45 min EOBT**

Non-scheduled ADC validity: **-15 min to +3 hr EOBT**

Entry timing: **10 min before entering Indian airspace**

Visual Ground Signals

PROHIBITION OF LANDING	**Signal:** Horizontal RED square panel with YELLOW diagonals (X pattern) **Meaning:** Landings are prohibited and the prohibition is liable to be prolonged
SPECIAL PRECAUTIONS WHILE LANDING	**Signal:** Horizontal RED square panel with ONE yellow diagonal **Meaning:** Owing to bad state of the manoeuvring area or for any other reason, special precautions must be observed in approaching to land or in landing
USE RUNWAYS AND TAXIWAYS ONLY	**Signal:** Horizontal WHITE DUMBBELL **Meaning:** Aircraft are required to land, take off, and taxi on runways and taxiways only
RUNWAYS ONLY (Other manoeuvres not confined)	**Signal:** White dumbbell with a BLACK BAR perpendicular to shaft across each circular portion **Meaning:** Aircraft required to land and take off on runways only, but other manoeuvres need not be confined to runways and taxiways
CLOSED RUNWAYS OR TAXIWAYS	**Signal:** Crosses of a single contrasting colour — WHITE or YELLOW — displayed horizontally on runways and taxiways **Meaning:** Area unfit for movement of aircraft
DIRECTION FOR LANDING/TAKEOFF	**Signal:** Horizontal white or orange landing "T" **Meaning:** Indicates direction to be used by aircraft for landing and take-off (parallel to shaft, towards cross arm). At night: illuminated or outlined in white lights.
DIRECTION FOR TAKEOFF (Digits)	**Signal:** A set of two digits displayed vertically at/near aerodrome control tower — Yellow background **Meaning:** Indicates direction for takeoff, expressed in units of 10 degrees to the nearest 10 degrees of the magnetic compass *Example: "12" = 120° magnetic*
RIGHT HAND TRAFFIC	**Signal:** Right-hand arrow of conspicuous colour displayed in signal area or horizontally at end of runway/strip in use **Meaning:** Turns are to be made to the RIGHT before landing and after take-off

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ATS REPORTING OFFICE	**Signal:** Letter "C" displayed vertically in BLACK against a YELLOW background **Meaning:** Indicates the location of the air traffic services reporting office
GLIDER FLIGHTS IN OPERATION	**Signal:** Double white cross displayed horizontally in the signal area **Meaning:** Aerodrome is being used by gliders and glider flights are being performed

Figure 3.9 *Light signals from the control tower*

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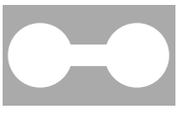
Signal	Meaning
	Landings prohibited; the prohibition is liable to be prolonged. Appendix 1, paragraph 4.2.1
	Special precautions are needed when approaching or landing. Appendix 1, paragraph 4.2.2
	Land, take off and taxi on runways and taxiways only. Appendix 1, paragraph 4.2.3.1
	Land and take off on runways only; other manoeuvres need not be confined to runways and taxiways. Appendix 1, paragraph 4.2.3.2
	A marked runway, taxiway or part is unfit for aircraft movement. The symbol is an X-shaped cross. Appendix 1, paragraph 4.2.4
	Land or take off parallel to the shaft of the T, towards its cross-arm. Appendix 1, paragraph 4.2.5.1

Figure 3.10 *Visual ground signals. Vector symbols matched to the official CAR figures. DGCA CAR9-C-I, Appendix1 §§4.2.1–4.2.5.1, pp45–46.*

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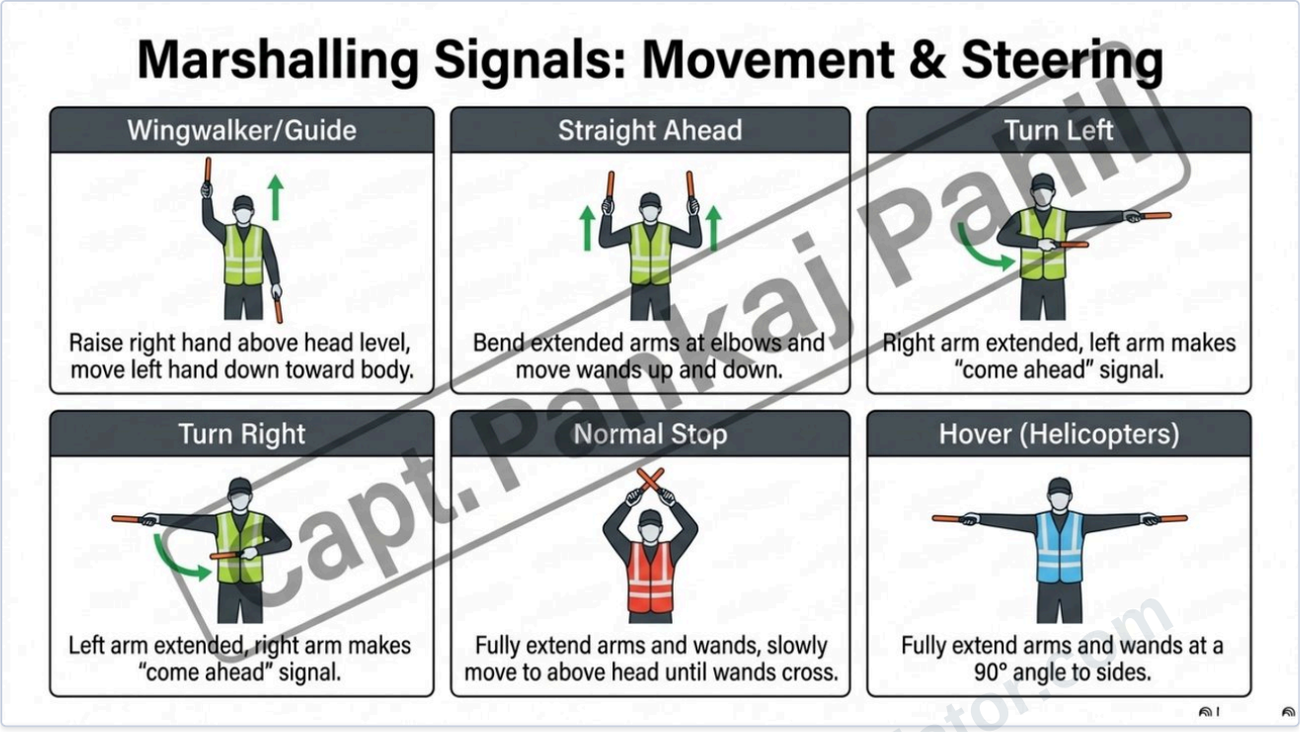

Figure 3.11 *Marshalling signals: movement and steering*

Table of Cruising Levels (RVSM & NON-RVSM)

CVSM VS RVSM

- **CVSM (Conventional VSM) / Non-RVSM:** General vertical separation minimum applied everywhere. Separation = 2,000 ft (same direction below FL290)
- **RVSM (Reduced VSM):** 300 m (1,000 ft) applied between **FL 290 and FL 410 inclusive**. RVSM is applied in **Indian airspace** per the table.
- In RVSM airspace (FL290–FL410): VFR flights are **NOT allowed.**
- Above RVSM airspace (>FL410): Separation doubles again.

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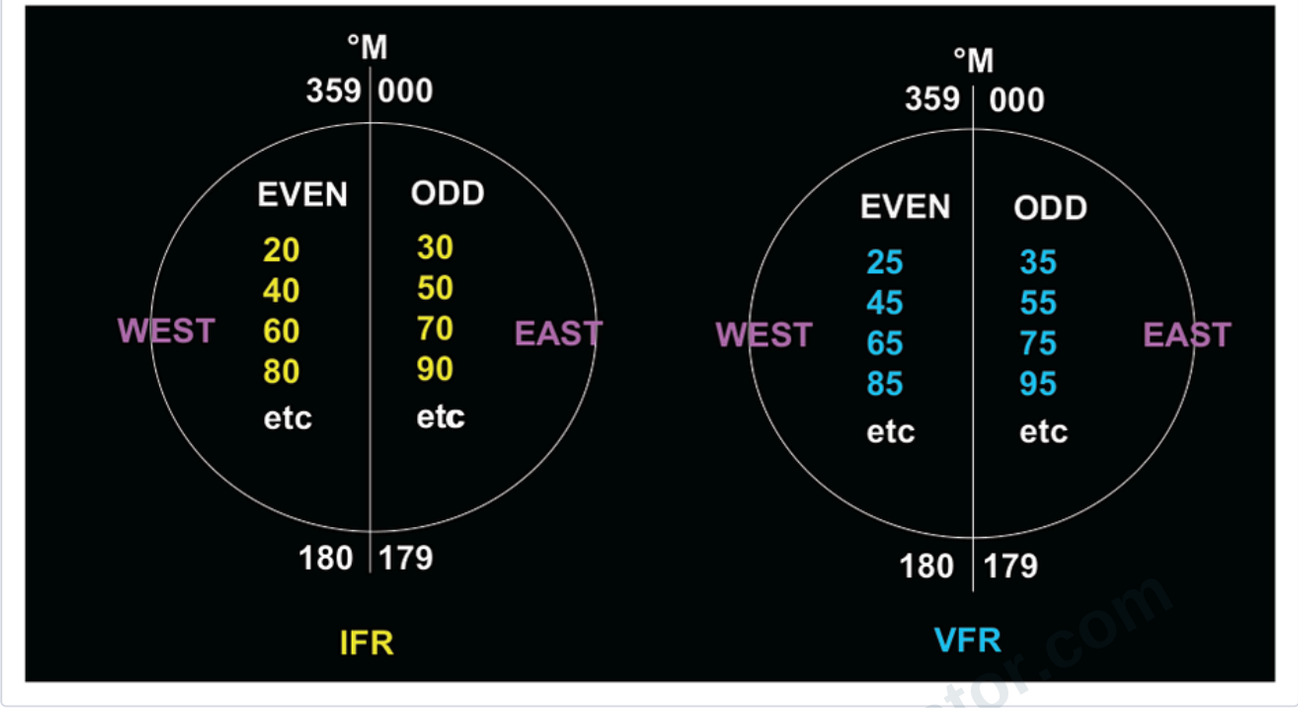

Figure 3.12 *Semi-circular rule up to FL290: eastbound (000°–179° magnetic) odd levels, westbound (180°–359°) even; VFR levels are 500 ft above the matching IFR levels.*

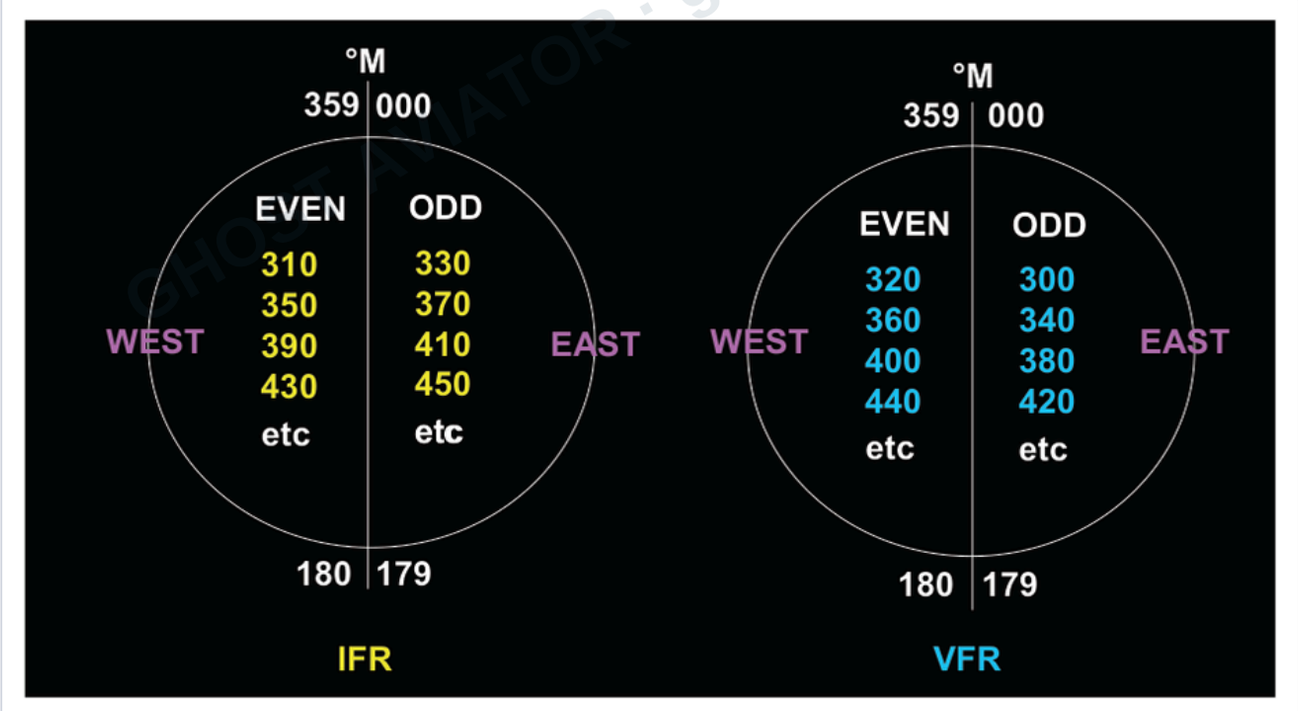

Figure 3.13 *Conventional (non-RVSM) levels above FL290: levels in the same direction are 4,000 ft apart.*

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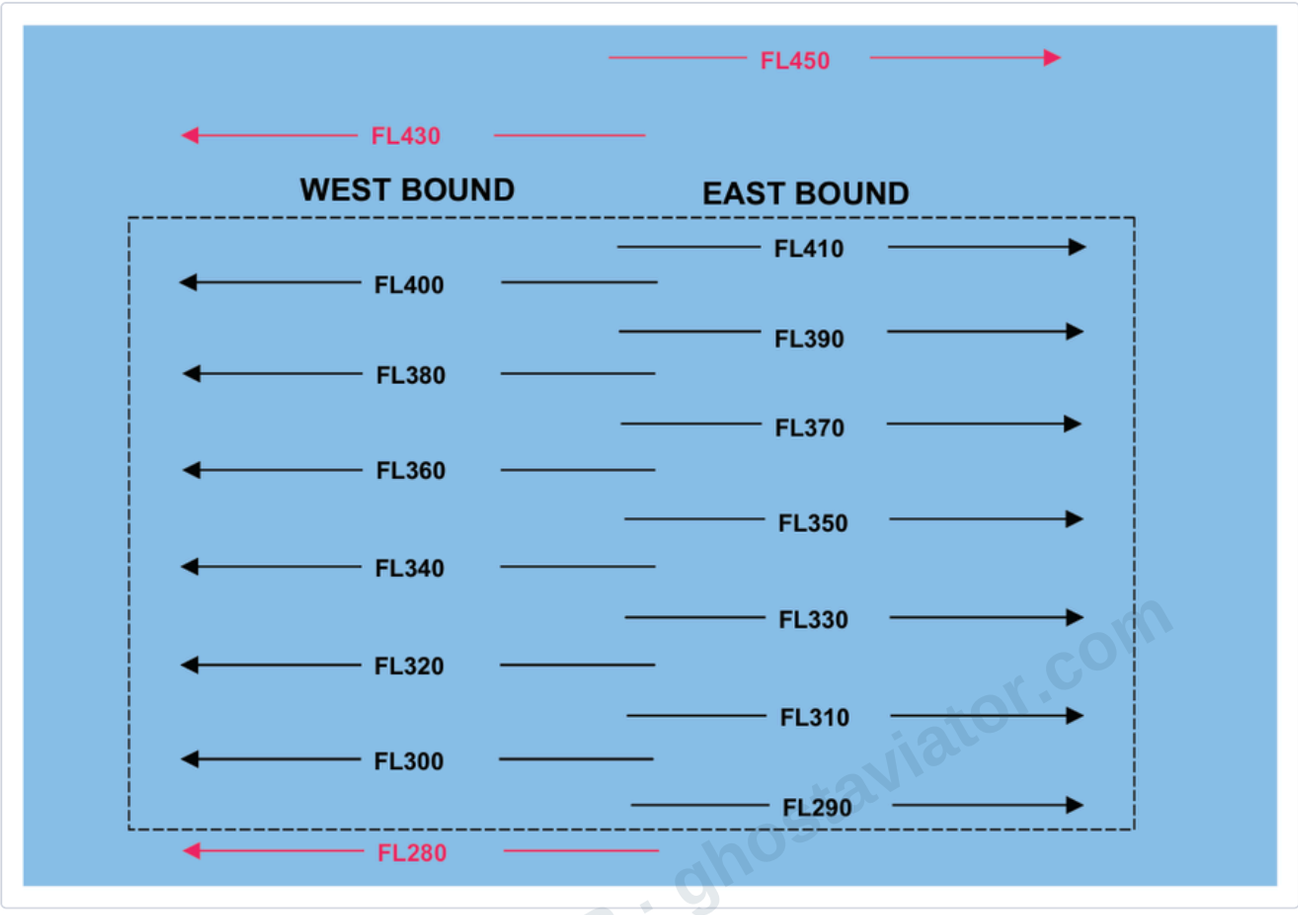

Figure 3.14 *RVSM, FL290–FL410: 1,000 ft spacing - eastbound odd levels (FL290 … FL410), westbound even levels (FL300 … FL400).*

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SEMI-CIRCULAR RULE — MEMORY TABLE

Airspace	Flight Level	Magnetic Track 000°–179°	Magnetic Track 180°–359°
Below FL290 (all areas)	FL010 – FL280	**IFR:** Odd (FL010, 030, 050 … FL270) **VFR:** Odd + 500 ft (FL035, 055, 075 … FL275)	**IFR:** Even (FL020, 040, 060 … FL280) **VFR:** Even + 500 ft (FL045, 065, 085 … FL285)
RVSM (1,000 ft VSM)	FL290 to FL410	**IFR:** Odd – FL290, 310, 330, 350, 370, 390, 410; then FL450, 490 … **VFR:** N/A	**IFR:** Even – FL300, 320, 340, 360, 380, 400; then FL430, 470 … **VFR:** N/A
Non-RVSM (conventional, 2,000 ft VSM)	FL290 and above	**IFR:** FL290, 330, 370, 410, 450 … **VFR (where permitted):** FL300, 340, 380, 420 …	**IFR:** FL310, 350, 390, 430, 470 … **VFR (where permitted):** FL320, 360, 400, 440 …

RVSM separation note: Minimum separation in airspaces below FL290 between IFR flights in the same direction is **2,000 ft** and in reciprocal direction **1,000 ft**. Minimum separation between IFR and VFR both ways is **500 ft**. Above FL290 in RVSM, separation is **2,000 ft same direction and 1,000 ft reciprocal direction**. Above RVSM airspace it doubles.

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Navigation Lights & Right of Way at Night

AEROPLANE NAVIGATION LIGHT GEOMETRY

Navigation Light	Coverage Angle	Position	Night Right-of-Way Rule
Red	110° projected L + R of horizontal plane	Port (left) wing	If the light is **NOT moving relative** to you = risk of collision
Green	110° projected L + R of horizontal plane	Starboard (right) wing	
White	140° rearward (70° each side)	Tail	

Right of Way at Night — What You See

What You See (Not Moving)	Other Aircraft Position	Action Required
RED + GREEN together	Coming directly HEAD-ON at you	BOTH alter heading RIGHT
RED + WHITE	Moving from right to left (you see their port side)	Give way (you are on their right — converging rule)
GREEN + WHITE	Moving from left to right (you see their starboard side)	YOU HAVE RIGHT OF WAY
WHITE only	Flying away from you	Monitor; you may be overtaking
GREEN to GREEN	Passing port to port (safe)	Safe passage
RED to RED	Passing starboard to starboard (safe)	Safe passage
GREEN to RED	Crossing paths — unsafe	Collision risk — take action

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Unlawful Interference

WHEN UNABLE TO PROCEED PER RULES (HIJACKING/UNLAWFUL INTERFERENCE)

If the PIC cannot proceed to an aerodrome in accordance with the rules, he/she should attempt to continue flying on the **assigned track** and at the **assigned cruising level** at least until able to notify an ATS unit or until within radar or ADS-B coverage.

When an aircraft subjected to unlawful interference must depart from its assigned track or level **without being able to make radiotelephony contact with ATS**, the PIC should, whenever possible:

- a) Attempt to broadcast warnings on the **VHF channel in use or the VHF emergency frequency**, and other appropriate channels; use transponders and data links when advantageous
- b) Proceed in accordance with applicable **special procedures for in-flight contingencies** (Doc 7030 Regional Supplementary Procedures)
- c) If no regional procedures, proceed at a level which differs from the cruising levels normally used for IFR flight by:
  - 1) **150 m (500 ft)** in an area where a vertical separation minimum of 300 m (1,000 ft) is applied
  - 2) **300 m (1,000 ft)** in an area where a vertical separation minimum of 600 m (2,000 ft) is applied

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In RVSM area: depart cruising level by **±500 ft (150m)**
In CVSM area: depart cruising level by **±1000 ft (300m)**
Squawk: **7500 (Hijack)**

Exam Tips and Key Numbers

EXAM TIPS FROM CAPT. PANKAJ PAHIL

- High-frequency exam topics: VMC minima table, cruising levels, RCF procedures, light gun signals, interception signals, right-of-way rules, marshalling signals
- Numbers to memorise: acrobatics min height (600m/2000ft), RVSM range (FL290–FL410), RCF non-radar time (20 min), RCF radar time (7 min), special VFR min visibility (1500m fixed-wing, 1000m helicopters), overtaking angle (<70°), FP delay notify ≥15 min
- Watch for trick questions: VFR ceiling/visibility in CTR = 1500ft (450m) ceiling OR <5km vis; Night definition = 30 min after SS to 30 min before SR; VFR hours = 20 min before SR to 20 min after SS

QUICK REFERENCE CARD — Critical Numbers to reference

Acrobatics min height: **600m / 2000ft AGL**
Acrobatics near aerodrome: **<2NM prohibited (unless >1800m/6000ft)**
Formation lateral: **≤1km / 0.5NM**
Formation vertical: **≤30m / 100ft**
VFR ceiling for CTR entry: **≥450m / 1500ft**
VFR visibility for CTR entry: **≥5km**
VFR max altitude: **FL150**
VFR operating hours: **-20min SR to +20min SS**
Night definition: **+30min SS to -30min SR**
VMC visibility above 10,000ft: **8km**
VMC visibility below 10,000ft: **5km**
VMC horizontal cloud distance: **1,500m**
VMC vertical cloud distance: **300m / 1,000ft**
Special VFR fixed-wing: **≥1500m vis**
Special VFR helicopter: **≥1000m vis**
IFR mountains: **600m / 2000ft within 8km**
IFR elsewhere: **300m / 1000ft within 8km**
FP submission: **180 min before EOBT**

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RPA FP: **24hr before ETD**
FP delay report: **≥15min delay → notify ATC**
UTC clock accuracy: **±30 sec**
Data link time: **±1 sec UTC**
Mach deviation notify: **±Mach 0.02 or ±10kt TAS**
ETA deviation notify: **>2 min**
RCF non-radar hold: **20 minutes**
RCF radar hold: **7 minutes**
RCF land within: **30 min of ETA/EAT**
Squawk RCF: **7600**
Squawk hijack: **7500**
Squawk emergency: **7700**
Emergency frequency: **121.5 MHz / 243 MHz**
RVSM range: **FL290–FL410**
RVSM separation: **1000ft same dir, 2000ft recip**
ADC scheduled: **-15 to +45 min EOBT**
ADC non-scheduled: **-15 min to +3hr EOBT**
ADIZ exemption: **5NM / 1000ft AGL**
Overtaking angle: **<70° plane of symmetry**
Nav lights: PORT RED 110° | STBD GREEN 110° | TAIL WHITE 140°
VFR congested area: **≥300m / 1000ft above highest obstacle**
VFR elsewhere: **≥150m / 500ft AGL**

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Official right-of-way provisions

DGCA CAR Section 9, Series C, PartI, §§3.2.2–3.2.2.7.3, pp 19–20. The explanations earlier in this chapter should be read with these provisions.

Paragraph 3.2.2

Right-of-way The aircraft that has the right-of-way shall maintain its heading and speed.

Paragraph 3.2.2.1

An aircraft that is obliged by the following rules to keep out of the way of another shall avoid passing over, under or in front of the other, unless it passes well clear and takes into account the effect of aircraft wake turbulence.

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Paragraph 3.2.2.2

Approaching head-on. When two aircraft are approaching head-on or approximately so and there is danger of collision, each shall alter its heading to the right.

Paragraph 3.2.2.3

Converging. When two aircraft are converging at approximately the same level, the aircraft that has the other on its right shall give way, except as follows: a) power-driven heavier-than-air aircraft shall give way to airships, gliders and balloons; b) airships shall give way to gliders and balloons; c) gliders shall give way to balloons; d) power-driven aircraft shall give way to aircraft which are seen to be towing other aircraft or objects.

Paragraph 3.2.2.4

Overtaking. An overtaking aircraft is an aircraft that approaches another from the rear on a line forming an angle of less than 70 degrees with the plane of symmetry of the latter, i.e. is in such a position with reference to the other aircraft that at night it should be unable to see either of the aircraft's left (port) or right (starboard) navigation lights. An aircraft that is being overtaken has the right-of- way and the overtaking aircraft, whether climbing, descending or in horizontal flight, shall keep out of the way of the other aircraft by altering its heading to the right, and no subsequent change in the relative positions of the two aircraft shall absolve the overtaking aircraft from this obligation until it is entirely past and clear.

Paragraph 3.2.2.5.1

An aircraft in flight, or operating on the ground or water, shall give way to aircraft landing or in the final stages of an approach to land.

Paragraph 3.2.2.5.2

When two or more heavier-than-air aircraft are approaching an aerodrome for the purpose of landing, aircraft at the higher level shall give way to aircraft at the lower level, but the latter shall not take advantage of this rule to cut in in front of another which is in the final stages of an approach to land, or to overtake that aircraft. Nevertheless, power-driven heavier-than-air aircraft shall give way to gliders.

Paragraph 3.2.2.5.3

Emergency landing. An aircraft that is aware that another is compelled to land shall give way to that aircraft.

Paragraph 3.2.2.6

Taking off. An aircraft taxiing on the manoeuvring area of an aerodrome shall give way to aircraft taking off or about to take off.

Paragraph 3.2.2.7.1

In case of danger of collision between two aircraft taxiing on the movement area of an aerodrome the following shall apply: a) when two aircraft are approaching head on, or approximately so, each shall stop or where practicable alter its course to the right so as to keep well clear; b) when two aircraft are on a converging course, the one which has the other on its right shall give way; c) an aircraft which is being overtaken by another aircraft shall have the right-of-way and the overtaking aircraft shall keep well clear of the other aircraft. Note.— For the description of an overtaking aircraft see 3.2.2.4.

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Paragraph 3.2.2.7.2

An aircraft taxiing on the manoeuvring area shall stop and hold at all runway-holding positions unless otherwise authorized by the aerodrome control tower. Note.— For runway-holding position markings and related signs, see CAR Section 4 Series B Part II, 5.2.10 and 5.4.2.

Paragraph 3.2.2.7.3

An aircraft taxiing on the manoeuvring area shall stop and hold at all lighted stop bars and may proceed further when the lights are switched off.

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CHAPTER 4

Air Traffic Services

04

Objectives of Air Traffic Services

CORE DEFINITION

The primary purpose of ATS is to maintain safe, orderly, and efficient air traffic operations across all phases of flight. Five objectives are prescribed *(laid down by the authority or the rules)* under ICAO Annex 11.

1. **Prevent collisions between aircraft.**
2. **Prevent collisions between aircraft on the manoeuvring area and obstructions on that area.**
3. **Expedite and maintain an orderly flow of air traffic.**
4. **Provide advice and information useful for the safe and efficient conduct of flights.**
5. **Notify appropriate organisations regarding aircraft in need of search and rescue aid, and assist such organisations as required.**

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Divisions of Air Traffic Services

ATS comprises **three services**:

Service	Objectives	Divisions
Air Traffic Control	a, b, c	Area · Approach · Aerodrome
Flight Information Service	d	—
Alerting Service	e	—

Figure 4.1 *The three divisions of Air Traffic Services and their objectives*

Air Traffic Control Service (ATC)

ATC accomplishes **objectives a, b, and c**. It is divided into three parts:

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A) AREA CONTROL SERVICE

Provision of ATC service for **controlled flights**, except for those parts described in approach and aerodrome control service. Accomplishes objectives **a and c**.
Provided by: Area Control Centre (ACC), or by the Approach Control unit where no ACC is established.

B) APPROACH CONTROL SERVICE

Provision of ATC service for **controlled flights associated with arrival or departure**. Accomplishes objectives **a and c**.
Provided by: Aerodrome Control Tower / ACC (combined), or by a separate Approach Control Unit.

C) AERODROME CONTROL SERVICE

Provision of ATC service for **aerodrome traffic**, except parts covered by Approach Control. Accomplishes objectives **a, b, and c**.
Provided by: Aerodrome Control Tower (ACT).

Flight Information Service (FIS)

Provided to accomplish **objective d** — advice and information for safe and efficient flight conduct.

Alerting Service

Provided to accomplish **objective e** — notifying SAR organisations and assisting them.

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Provisions of Control Service – AIP India

Radio Communication Establishment

PROCEDURE

Radio communication shall be established with the appropriate **Aerodrome / Approach Control Unit**:

- **Prior to taxiing for departure;** or
- **When intending to operate in a Class C and D airspace.**

CRITICAL RULE – COMMUNICATION

While operating in Class C and D airspace, **only direct controller-pilot communication is permitted**. RTF communication through an interpreter **shall NOT be permitted**.

Equipment Requirements – Class C & D

EQUIPMENT REQUIRED

For IFR or VFR operation in Class C and D airspace, aircraft shall be equipped with:

- **Appropriate two-way VHF radio apparatus**
- **A radio compass**

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Entering a CTR or ATZ

PROCEDURE

A pilot-in-command under IFR or VFR intending to **enter, cross, or operate within a CTR or ATZ** shall request a Clearance from the Aerodrome/Approach Control on the appropriate radio frequency.

The pilot shall:

- Pass the aircraft's **position, level, track and estimated time of crossing** the airspace.
- **Maintain a continuous listening watch** on that frequency while the aircraft is within the airspace.
- **Carry out any instructions** received from Aerodrome/Approach Control.

Classification & Designation of ATS Airspace in India

Designation of Airspace Portions

FLIGHT INFORMATION REGIONS (FIR)

Portions where **Flight Information Service and Alerting Service** will be provided are designated as **Flight Information Regions**.

CONTROL AREAS & CONTROL ZONES

Portions where **ATC service will be provided to IFR flights** are designated as **Control Areas or Control Zones**. Where ATC is also provided to VFR flights, these are designated as **Class B, C, or D airspace**.

CONTROLLED AERODROMES

Aerodromes where ATC service will be provided to aerodrome traffic are designated as **Controlled Aerodromes**.

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Airspace Classes in India

IMPLEMENTATION STATUS

Class A* and Class B* — Implementation under consideration. Not yet fully implemented in India.

CLASS A*

Applicable to designated airspaces in **TCA, CTA, and CTZ**. **IFR flights only** are permitted. All flights provided with ATC service and **separated from each other**.

CLASS B*

Applicable to designated airspaces in **TCA, CTA, and CTZ**. **IFR and VFR** flights permitted. All flights provided with ATC service and **separated from each other**.

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CLASS C

IFR and VFR flights permitted. All flights provided with ATC service. **IFR separated from IFR and from VFR**. VFR separated from IFR; VFR receives **traffic information in respect of other VFR flights**.

CLASS D

IFR and VFR flights permitted. All flights provided with ATC service. **IFR separated from IFR only**; IFR receives traffic info about VFR. **VFR receives traffic information on all other flights** (no separation from VFR).

CLASS E

Airspace within **ATS Route segment outside controlled airspace**. IFR and VFR flights permitted. IFR separated from IFR. All flights receive traffic info as far as practical. **Class E shall NOT be used for Control Zones.**

CLASS F

Airspace within **ATS Route segment outside controlled airspace**. IFR and VFR flights permitted. Participating IFR flights receive **Air Traffic Advisory Service**. All flights receive **Flight Information Service if requested**.

CLASS G

Airspace **outside ATS route segment and outside controlled airspace**. IFR and VFR flights permitted. Flights receive **Flight Information Service if requested**. No separation provided.

BOUNDARY NOTE – VERTICAL ADJACENCY

Where ATS airspaces adjoin vertically (one above the other), flights at a common level comply with requirements of **the less restrictive class of airspace**. Order: **A (most restrictive) → B → C → D → E → F → G (least restrictive)**.

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ATS Airspace Classes – Complete Reference Table

Class	Flight Type	Separation Provided	Service Provided	Speed Limit**	Radio Comm.	ATC Clearance
A*	IFR only	All aircraft	Air Traffic Control Service	Not applicable	Continuous two-way	Yes
B*	IFR	All aircraft	Air Traffic Control Service	Not applicable	Continuous two-way	Yes
	VFR	All aircraft	Air Traffic Control Service	Not applicable	Continuous two-way	Yes
C	IFR	IFR from IFR; IFR from VFR	Air Traffic Control Service	Not applicable	Continuous two-way	Yes
	VFR	VFR from IFR	1. ATC for separation from IFR; 2. VFR/VFR traffic info (avoidance advice on request)	**250 kt IAS** below 3050m (10,000 ft) AMSL	Continuous two-way	Yes

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Class	Flight Type	Separation Provided	Service Provided	Speed Limit**	Radio Comm.	ATC Clearance
D	IFR	IFR from IFR	ATC incl. traffic info about VFR (avoidance advice on request)	**250 kt IAS** below 3050m (10,000 ft) AMSL	Continuous two-way	Yes
	VFR	Nil	IFR/VFR & VFR/VFR traffic info (avoidance advice on request)	**250 kt IAS** below 3050m (10,000 ft) AMSL	Continuous two-way	Yes
E	IFR	IFR from IFR	ATC and, as far as practical, traffic info about VFR flights	**250 kt IAS** below 3050m (10,000 ft) AMSL	Continuous two-way	Yes
	VFR	Nil	Traffic information as far as practical	250 kt IAS below 3050m (10,000 ft) AMSL	**No**	**No**
F	IFR	IFR from IFR (as far as practicable)	Air Traffic Advisory Service; Flight Information Service	250 kt IAS below 3050m (10,000 ft) AMSL	Continuous two-way	Yes
	VFR	Nil	Flight Information Service	250 kt IAS below 3050m (10,000 ft) AMSL	No	No
G	IFR	Nil	Flight Information Service	250 kt IAS below 3050m (10,000 ft) AMSL	Continuous two-way	No
	VFR	Nil	Flight Information Service	250 kt IAS below 3050m (10,000 ft) AMSL	No	No

** When transition altitude is lower than 3600m (10,000 ft) AMSL, FL 100 should be used in lieu of 10,000 ft. * Implementation of Class A & Class B under consideration.

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Procedures in Class C Airspace (India-Specific)

WHERE CLASS C AND CLASS D APPLY

AIP India (ENR 1.4) says that, besides the other classes in the table, **Class C** is a designated part of controlled airspace and **Class D** is the rest of it. The height limits and the lateral limits of each Class C and Class D area are different at each airport and are published for each control area in AIP India **ENR 2.1**. Do not learn one single height for the whole country; learn the rules of each class.

- VFR flights must ensure they can meet Class C requirements **(MET/COM/SPEED)**.
- **Pre-departure clearance** from appropriate ATC shall be obtained by all VFR flights intending to enter Class C.
- Aircraft on SID/STAR shall follow **published speed restrictions**.
- All aircraft shall follow any **additional speed restrictions when instructed by ATC**.
- Responsibility of **terrain clearance remains with the flight crew of VFR flights at all times**.
- VFR flights unable to adhere to ATC clearance shall **obtain alternate clearance**.

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FIRs and Location Indicators

Flight Information Regions (FIRs)

Flight Information Centres are established to provide FIS and Alerting Service **outside controlled airspaces**. FIRs are delineated to cover the **whole air route structure** to be served.

KEY RULE

A FIR includes **all airspace within its lateral limits**, except as limited by an Upper FIR (UIR).

ICAO Location Indicators – Structure

4-LETTER LOCATION INDICATOR

ICAO has divided the world into **22 non-overlapping AFSRAs**. Each AFSRA has a unique identifying letter.

Structure of a 4-letter indicator:

`1st letter` → Identifies the **AFSRA**
`2nd letter` → Identifies the **State/Country**
`3rd & 4th letters` → Identify the **Place/Station**

India's AFSRA letter = V. The 2nd alphabets assigned to India are: I, E, A, O.

VIDF	Delhi FIR Northern India 2nd letter "I"
VABF	Bombay FIR Western India 2nd letter "A"
VECF	Calcutta FIR Eastern India 2nd letter "E"
VOMF	Madras FIR Southern India 2nd letter "O"
VEGF	Guwahati sub-FIR Eastern India 2nd letter "E"

FIR CODE VS AERODROME CODE

The four FIR codes above end in **F** (VIDF Delhi, VABF Mumbai, VECF Kolkata, VOMF Chennai). The aerodromes keep the same first two letters but end differently: VIDP Delhi, VABB Mumbai, VECC Kolkata, VOMM Chennai.

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EXAMPLES FROM TEXT

VIDP → V=AFSRA, I=India, DP=Delhi
OPCL → O=AFSRA, P=Pakistan, KC=Karachi
OAKB → O=AFSRA, A=Afghanistan, KB=Kabul

Aeronautical Chart Indicators

NDB	2 (sometimes 3) capital letters Examples: SP = Sarsawa NDB, JG = Dehradun NDB, DBR = Dibrugarh NDB
VOR	3 capital letters Examples: AAT = Agartala VOR, BBB = Mumbai VOR
ILS	4 capital letters, starting with "I" Examples: IGHT = Guwahati ILS RW 02, ILUC = Lucknow ILS 27
Aerodrome	4 capital letters — India: V + FIR letter + 2 letters Examples: VABB = Mumbai, VOMM = Madras. Amsterdam = EHAM, Lisbon = LPPT
Significant Point	5 capital letters forming a pronounceable name-code Examples: GOLEM, IDOLA
ATS Routes	≥1 alphabet + number 1–999 Examples: A791, B209. Domestic India routes start with W (e.g. W49). Route designators A,B,G,R,W = Amber, Blue, Green, Red, White.
SIDs & STARs	NDB/VOR/Significant Point + number [+ alphabet suffix] Examples: LETPU 1D = Departure, SSB 1A = Arrival

ATS Route Designator Letters – Selection

Letter Group	Route Type
A, B, G, R	Routes forming part of **regional networks of ATS routes** — NOT area navigation routes
L, M, N, P	Area navigation routes forming part of **regional networks**
H, J, V, W	Routes NOT forming part of regional networks — NOT area navigation routes
Q, T, Y, Z	Area navigation routes NOT forming part of regional networks

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Supplementary Prefix Letters

PREFIX RULES

- K — Low-level route established primarily for **helicopters**
- U — Route established in the **upper airspace**
- S — Route established exclusively for **supersonic aircraft** during acceleration, deceleration, and supersonic flight

Total characters in designator: **SHALL NOT exceed 6**; should be kept to **maximum 5** where possible.

Supplementary Suffix Letters

SUFFIX RULES

- F — **Advisory service only** provided on that route/portion
- G — **Flight information service only** provided on that route/portion

Special Airspace Areas – P, R, D, TSA, TRA

Identification System for P, R, D Areas

Identification = **Nationality letters** + **P/R/D** + **Unduplicated number**

Type	Letter	Definition	Example
Prohibited Area	P	Airspace above land/territorial waters of India within which flights are **NOT permitted at any time under any circumstances**	VA(P)-2 = Prohibited Area No. 2 in Mumbai FIR
Restricted Area	R	Airspace above land/territorial waters of India within which flight of aircraft **is restricted**	VO(R)-81 = Restricted Area No. 81 in Chennai FIR
Danger Area	D	Airspace within which activities **dangerous to flight** exist at specified times (notified by NOTAMs)	VE(D)-72 = Danger Area No. 72 in Calcutta FIR

Military Exercise and Training Areas

TSA – TEMPORARY SEGREGATED AREA

Airspace **temporarily segregated and allocated for the exclusive use of a particular user** during a determined period. **Other traffic will NOT be allowed to transit.**

Identifier: TSA + number (501–999) + space + [place name in square brackets].
For multiple sub-areas: number followed by capital letter in round brackets.
Example: TSA801(A) [Thanjavur]

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TRA – TEMPORARY RESERVED AREA

Airspace **temporarily reserved for specific use**, but through which **other traffic may be allowed to transit under ATC clearance**.

Identifier: TRA + number (501–999) + space + [place name in square brackets].
Example: `TRA601(A) [Panagarh]`

NUMBER ALLOCATION – TSA/TRA

A **single series (501–999)** is used for both TSA and TRA regardless of type. A number is **NOT duplicated** in any type.

Mumbai FIR	501 – 600
Kolkata FIR	601 – 700
Delhi FIR	701 – 800
Chennai FIR	801 – 900
Reserved for Future Use	901 – 999

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Control Areas and Control Zones

Control Areas (CTA)

Control areas (including airways and terminal control areas) shall be delineated to encompass sufficient airspace to contain flight paths of IFR flights.

LOWER LIMIT OF CONTROL AREA

A lower limit of a Control Area shall be established at a height above the ground or water of **not less than 200 m (700 ft)**.

Control Zones (CTR)

Control zone lateral limits encompass portions of airspace **not within control areas**, containing paths of IFR flights arriving at and departing from aerodromes under IMC.

LATERAL LIMIT OF CTR

Shall extend to at least **9.3 km (5 NM)** from the centre of the aerodrome(s) in the directions from which approaches may be made.

NOTES

- Aircraft holding in the vicinity of aerodromes are considered as **arriving aircraft**.
- A control zone may include **two or more aerodromes situated close together**.

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Local / Training Flights

PROCEDURE

- For local training/test flights by scheduled/non-scheduled operators: **flight plan submitted as for non-scheduled flights**.
- For training flights by flying clubs within ATZ: **flight plan may be submitted by Fax or telephone**.
- If no ATZ established and no reserved airspace: an area of **5 NM radius from ARP, up to 3000 ft AGL** = vicinity of aerodrome for VFR.
- For cross-country flights from airfields with no ATS reporting office: flight plan submitted to FIC by Fax. Must include **accumulated elapsed times to FIR boundary points** and **contact tel/mobile of PIC in Field 18**.

Identification of ATS Units and Airspaces

- An **Area Control Centre or FIC** → identified by name of a **nearby town/city or geographic feature**.
- An **Aerodrome Control Tower or Approach Control Unit** → identified by **name of the aerodrome**.
- A **Control Zone, CTA, or FIR** → identified by **name of the unit having jurisdiction** ***(legal power over a person, place or event)*** over that airspace.

Upper Airspace FIRs / Control Areas

To limit the number of FIRs or CTAs through which high-flying aircraft operate, a FIR or CTA may be delineated to include **upper airspace within the lateral limits of a number of lower FIRs or CTAs**.

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Air Traffic Control Service

Application – Where ATC is Provided

- To **all IFR flights** in airspace Classes A, B, C, D and E.
- To **all VFR flights** in airspace Classes B, C and D.
- To **all special VFR flights**.
- To **all aerodrome traffic at controlled aerodromes**.

Separation Provided by ATC Clearances

Clearances shall provide separation:

- Between **all flights** in airspace Classes **A and B**.
- Between **IFR flights** in airspace Classes **C, D and E**.
- Between **IFR and VFR flights** in airspace Class **C**.
- Between **IFR and Special VFR flights**.
- Between **Special VFR flights** when so prescribed by the appropriate ATS authority.

SEPARATION MINIMA

Selection of separation minima within a given airspace shall be as per provisions of **PANS-ATM** and **Regional Supplementary Procedures**.

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Read-back of Clearances and Safety-Related Information

MANDATORY READ-BACK ITEMS

The flight crew shall read back safety-related parts of ATC clearances/instructions. The following shall **ALWAYS** be read back:

1. **ATC route clearances**
2. Clearances and instructions to **enter, land on, take off from, hold short of, cross and backtrack** on any runway
3. **Runway-in-use, altimeter settings, SSR codes, level instructions, heading and speed instructions**, and whether issued by the controller or contained in ATIS broadcasts, **transition levels**

CONTROLLER OBLIGATION

The controller shall listen to the read-back to ascertain that the clearance has been correctly acknowledged and shall take **immediate action to correct any discrepancies**. Voice read-back of CPDLC messages shall **NOT be required**.

Service to Aircraft in Emergency

EMERGENCY PRIORITY

An aircraft known or believed to be in a state of emergency, including being subjected to **unlawful interference**, shall be given **maximum consideration, assistance and priority over other aircraft**.

Transponder Code	Meaning
`Mode A, 7700`	**General Emergency**
`Mode A, 7500`	**Unlawful Interference (Hijack)**
Also: activate ADS emergency/urgency capability OR transmit emergency message via CPDLC	

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Air Traffic Flow Management (ATFM)

ATFM is primarily meant to **balance demand against capacity** to achieve optimum utilisation of major resources — Airport, Airspace, and Aircraft — at every Indian airport where there is a capacity constraint.

INDIA'S C-ATFM SYSTEM

AAI has established a **Central Air Traffic Flow Management (C-ATFM)** system consisting of:

- **Central Command Center (CCC)** at Delhi
- **Flow Management Positions (FMPs)** at major ACCs and ATC towers
- Phase I: FMPs at **Delhi, Mumbai, Bengaluru, Chennai, Kolkata, Hyderabad** (handle 60% of Indian air traffic)
- Phase II: FMPs at additional 30 airports (phased manner)

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Flight Information Service (FIS)

Application

FIS shall be provided to all aircraft which are likely to be affected by the information and which are:

- Provided with air traffic control service; or
- Otherwise known to the relevant ATS units.

Scope of FIS – Information Included

FIS INCLUDES

- **SIGMET** and AIRMET information (*Note: AIRMET is NOT provided in India*).
- Information on **pre-eruption volcanic activity, eruptions, volcanic ash clouds**.
- Information on **release of radioactive materials or toxic chemicals** into atmosphere.
- Changes in **serviceability of navigation aids**.
- Changes in condition of **aerodromes and associated facilities** (including state of movement areas affected by snow, ice, or significant depth of water).
- Information on **unmanned free balloons** and any other information likely to affect safety.

FIS Provided to Flights

- **Weather conditions** at departure, destination and alternate aerodromes.
- **Collision hazards** for aircraft operating in airspace Classes **C, D, E, F and G**.
- For flights over water: radio call sign, position, track, speed of **surface vessels** (if requested and practicable).

FIS to VFR Flights

FIS provided to VFR flights shall include **traffic and weather conditions along the route likely to make operation under visual flight rules impracticable**.

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Automatic Terminal Information Service (ATIS)

Voice-ATIS Broadcasts

PURPOSE

Provided at aerodromes where there is a requirement to **reduce the communication load on the ATS VHF air-ground communication channels**.

CRITICAL RULES – VOICE-ATIS

- A **discrete VHF frequency** shall, whenever practicable, be used.
- If discrete freq unavailable, may use VOR voice channel — but **NOT on ILS voice channel**.
- Broadcast shall be **continuous and repetitive**.
- Message should, whenever practicable, **NOT exceed 30 seconds**.

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D-ATIS (Data Link ATIS)

Where D-ATIS supplements Voice-ATIS, the information shall be **identical in both content and format** to the Voice-ATIS broadcast.

ATIS Operational Rules

WHEN ATIS IS PROVIDED

- Information relates to a **single aerodrome**.
- Updated **immediately a significant change occurs**.
- Preparation and dissemination is the responsibility of **Air Traffic Services**.
- Each message identified by a **letter of the ICAO spelling alphabet** (designators in alphabetical order).
- Aircraft shall acknowledge receipt upon establishing communication with the ATS unit.
- ATS unit provides aircraft with **current altimeter setting** upon reply.
- Meteorological information extracted from the **local met routine or special report**.
- If rapidly changing conditions, ATIS shall indicate that **relevant weather info will be given on initial contact**.
- Information in a current ATIS (acknowledged by aircraft) need not be re-transmitted, **except the altimeter setting**.

ATIS Message Contents (Arriving and Departing)

Item	Content
a	Name of aerodrome
b	Arrival and/or departure indicator
c	Contract type (if D-ATIS)
d	Designator (ICAO alphabet letter)
e	Time of observation (if appropriate)
f	Type of approach(es) to be expected
g	Runway(s) in use; status of arresting system (if any)
h	Significant runway surface conditions; braking action
i	Holding delay (if appropriate)
j	Transition level (if applicable)
k	Other essential operational information
l	Surface wind direction and speed (incl. significant variations)
*m	Visibility and, when applicable, RVR
*n	Present weather
*o	Cloud below 1500m (5000 ft) or below highest MSA (whichever greater); cumulonimbus; vertical visibility if sky obscured
p	Air temperature

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Item	Content
q	Dew point temperature
r	Altimeter setting(s)
s	Significant meteorological phenomena in approach/climb-out areas; wind shear; recent weather
t	Trend forecast (when available)
u	Specific ATIS instructions

* Items m, n, o are replaced by **"CAVOK"** whenever visibility, cloud and present weather are better than prescribed values.

Runway Surface Condition Reporting Terms

RUNWAY SURFACE CONDITION TERMS

Standardised terms for runway surface conditions affecting braking:

- COMPACTED SNOW
- DRY
- DRY SNOW
- DRY SNOW ON TOP OF COMPACTED SNOW
- DRY SNOW ON TOP OF ICE
- FROST
- ICE
- SLUSH
- STANDING WATER
- WATER ON TOP OF COMPACTED SNOW
- WET
- WET ICE
- WET SNOW
- WET SNOW ON TOP OF COMPACTED SNOW
- WET SNOW ON TOP OF ICE

Friction measuring devices shall correlate in a manner acceptable to **DGCA**.

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Alerting Service

Application

Alerting service shall be provided:

- For **all aircraft provided with ATC service**.
- As far as practicable, to **all aircraft having filed a flight plan or otherwise known to ATS**.
- To **any aircraft known or believed to be subject of unlawful interference**.

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CENTRAL POINT FOR EMERGENCY INFO

Flight Information Centres or Area Control Centres serve as the **central point for collecting all information relevant to a state of emergency** and forwarding to the appropriate Rescue Coordination Centre.

The Three Emergency Phases

INCERFA	Uncertainty Phase • No communication for **30 minutes** after it was due, or from time of unsuccessful attempt • Aircraft fails to arrive within **30 minutes** of ETA (last notified/estimated), if no doubt about safety
ALERFA	Alert Phase • Following INCERFA, further attempts to contact aircraft and inquiries have **failed** • Aircraft cleared to land, **fails to land within 5 minutes** of ETA, comm not re-established • Operating efficiency impaired but **forced landing NOT likely** • Aircraft known/believed subject of **unlawful interference**
DETRESFA	Distress Phase • Following ALERFA, further attempts indicate aircraft **probably in distress** • Fuel on board considered **exhausted** or insufficient to reach safety • Operating efficiency impaired to extent that **forced landing is likely** • Information received that forced landing **is about to be or has been made**

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Emergency phase	Meaning
INCERFA — uncertainty	Uncertainty about the safety of an aircraft and its occupants.
ALERFA — alert	Apprehension about their safety.
DETRESFA — distress	Reasonable certainty of grave and imminent danger requiring immediate assistance.

Figure 4.2 *Emergency phases describe the situation. They do not require waiting through a fixed sequence. DGCA CAR Section9,SeriesS,PartI, definitions and §5.2.*

Emergency Information to Other Aircraft

UNLAWFUL INTERFERENCE – SPECIAL RULE

When ATS believes an aircraft is being subjected to unlawful interference, **no reference shall be made in ATS air-ground communications to the nature of the emergency** unless it has first been referred to in communications from the aircraft and it is certain such reference will NOT aggravate the situation.

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ATS Routes in India – Rules and Requirements

ATC CLEARANCE REQUIREMENTS

- Aircraft shall obtain ATC clearance from appropriate ATS unit at least **10 minutes prior to entering controlled airspace**.
- Aircraft shall NOT join or cross ATS routes without prior approval/ATC clearance:
  - At least **10 minutes prior** if in direct VHF contact.
  - At least **20 minutes prior** if contact is through en-route radio frequency.
- All aircraft transiting from one FIR to another shall forward **estimates for FIR boundary (including FL)** to the ATS unit of the FIR to be entered, at least **10 minutes prior**. *(Does NOT apply between Indian FIRs — only entering Indian FIR from a foreign FIR.)*

CROSSING ATS ROUTES

- Aircraft will join or cross ATS routes at or close to a **designated reporting point**.
- Crossing shall be done at **90° to the direction of the route** and at a level appropriate to the magnetic track.
- VFR flights to cross ATS routes outside controlled airspace shall only cross at an appropriate VFR level at **right angle to the direction** of the ATS route.

POSITION REPORTING

- On routes defined by **significant points**: report as soon as possible after passing each designated reporting point.
- On routes **NOT defined by significant points**: report after the **first half hour** of flight and at **hourly intervals** thereafter. Additional reports at shorter intervals may be requested by ATC.

INTERNATIONAL BORDER RULE

All aircraft are **forbidden to operate within 15 NM of the international border of India** unless specifically permitted, following an ATS route, or operating to/from an aerodrome situated within 15 NM of the border.

MFA OBSTACLE CLEARANCE

Minimum flight altitude on ATS routes is determined to ensure at least **1000 ft (300 m) vertical clearance above the highest obstacle within 10 KM on each side of the centre line** of the route.

FLYING OUTSIDE ATS ROUTES

Flying outside of ATS routes is **prohibited within Indian FIRs**, unless otherwise authorised by ATC. Scheduled international flights are permitted to flight plan using **domestic ATS route segments** to/from destination, departure and approved alternate airports not connected by international ATS Routes.

ATS Route Description in AIP (With effect 4/11/2021)

AIP Section	Old Description	New Description
ENR 3.1	Lower ATS Routes	**Conventional Navigation Routes**

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AIP Section	Old Description	New Description
ENR 3.2	Upper ATS Routes	**Air Navigation Routes**

Types of Services (at Civil Aerodromes)

SERVICES PROVIDED (EXCEPT AT MILITARY BASES)

1. Air Traffic Control Service
2. Air Traffic Advisory Services
3. Flight Information Service and Alerting Service
4. Surveillance Services

Note: There is **no distinction between upper and lower controlled airspace** in India.

Strayed Aircraft

DEFINITION

An aircraft that has **deviated significantly from its intended track or reports that it is lost**. An aircraft may be considered "strayed" by one unit and "unidentified" by another. ATS takes all necessary steps to assist and safeguard its flight.

Air Traffic Advisory Service

Provided to aircraft operating **outside controlled airspace on ATS routes classified "F"**.

ADVISORY SERVICE VS ATC

IFR flights using advisory service in Class F comply with same procedures as controlled flights **except** that the flight plan and changes are **NOT subject to clearance**. The unit only provides advice on essential traffic and suggests possible courses of action.

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Quick Reference – Key Numbers & Facts

Parameter	Value / Rule	Source
Lower limit of CTA	**≥ 200 m (700 ft) AGL**	ICAO Annex 11
Lateral limit of CTR	**≥ 9.3 km (5 NM)** from aerodrome centre in approach directions	ICAO Annex 11
Speed limit (all classes with limit)	**250 kt IAS** below 3050m (10,000 ft) AMSL	ICAO Annex 11
INCERFA trigger time (no comms)	**30 minutes**	ICAO Annex 11
INCERFA trigger (late arrival)	**30 minutes** after last notified ETA	ICAO Annex 11

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Parameter	Value / Rule	Source
ALERFA trigger (cleared, no land)	**5 minutes** after ETA + no comms	ICAO Annex 11
ATIS max duration	**30 seconds** (whenever practicable)	ICAO Annex 11
ATC clearance before controlled airspace	At least **10 minutes** prior (direct VHF)	AIP India
ATC clearance before ATS route entry (en-route radio)	At least **20 minutes** prior	AIP India
FIR boundary estimate	At least **10 minutes** prior to entry	AIP India
MFA obstacle clearance	**1000 ft (300 m)** above highest obstacle within **10 km** each side of route centre line	AIP India
International border buffer	**15 NM** — no operations unless specifically permitted	AIP India
ATC clearance crossing ATS route angle	**90°** to direction of route	AIP India
Transponder – General Emergency	**Mode A, 7700**	ICAO Annex 10
Transponder – Unlawful Interference	**Mode A, 7500**	ICAO Annex 10
TSA/TRA number series	**501–999** (single series)	AIP India
Delhi FIR TSA/TRA numbers	**701–800**	AIP India
Mumbai FIR TSA/TRA numbers	**501–600**	AIP India
Kolkata FIR TSA/TRA numbers	**601–700**	AIP India
Chennai FIR TSA/TRA numbers	**801–900**	AIP India
ATIS update trigger	**Immediately** a significant change occurs	ICAO Annex 11
ATIS acknowledgement — exception	Altimeter setting **must always** be re-transmitted	ICAO Annex 11
Annex for ATS standards	**Annex 11** to Chicago Convention	ICAO
Number of world AFSRAs	**22** non-overlapping	ICAO Doc 7910
ATC clocks accuracy	**± 30 seconds** UTC	ICAO Annex 11

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CHAPTER 5

Separation Methods and Minima

05

General Provisions – Separation of Controlled Traffic

Vertical or horizontal separation shall be provided in the following situations:

- **Between all flights** in **Class A** and **Class B** airspaces
- **Between IFR flights** in **Class C**, **D** and **E** airspaces
- **Between IFR and VFR flights** in **Class C** airspace
- **Between IFR and Special VFR flights**
- **Between Special VFR flights** — when prescribed *(laid down by the authority or the rules)* by the appropriate ATS authority — *except* for IFR flights in Class D & E, during daylight when flights have been cleared to climb/descend maintaining own separation and remaining in VMC

DEFINITION — COMPOSITE SEPARATION

A combination of **vertical separation** and **horizontal separation**, using minima for each which may be **lower than, but not less than half of**, those used for each element when applied individually.

Composite separation shall only be applied on the basis of **regional air navigation agreements**.

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Vertical Separation

Vertical Separation Minima

STANDARD — BELOW FL 290

Nominal **1 000 ft** below FL 290

Nominal **2 000 ft** at or above FL 290

(except as per RVSM rules)

RVSM — REDUCED VERTICAL

Nominal **1 000 ft** when:

- Both aircraft are **RVSM compliant**, AND
- Operating within **designated RVSM airspace**

CRUISE CLIMB — INDIA SPECIFIC

Cruise climb is NOT permitted in Indian FIRs.

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Vertical Separation during Climb & Descent

GENERAL RULE

An aircraft **may** be cleared to a level previously occupied by another aircraft *after the latter has reported vacating it*, **EXCEPT when:**

1. Severe turbulence is known to exist; OR
2. The aircraft concerned are established at the **same holding pattern**; OR
3. The difference in aircraft performance is such that **less than the applicable separation minimum may result** — in which case clearance shall be withheld until the vacating aircraft has reported at or passing another level separated by the required minimum.

PILOT–PILOT COMMUNICATION

Pilots in direct communication with each other may, with their concurrence, be cleared to **maintain a specified vertical separation between their aircraft during ascent or descent**.

VMC Climb and Descents

VMC OWN SEPARATION — CONDITIONS

When requested by an aircraft **and** agreed by the pilot of the other aircraft, an ATC unit may clear a controlled flight (including departing/arriving) operating in **Class D** or **Class E** airspace in **VMC during daylight hours** to fly subject to:

- Maintaining **own separation** to one other aircraft, AND
- Remaining in **VMC**

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When so cleared, the following shall apply:

- Clearances shall be for a specified portion of the flight at or below **10,000 ft** during climb and descent
- **Essential traffic information** shall be passed
- If VMC may become impracticable, an **IFR flight shall be provided alternative instructions** to be complied with in the event VMC cannot be maintained

Horizontal Separation

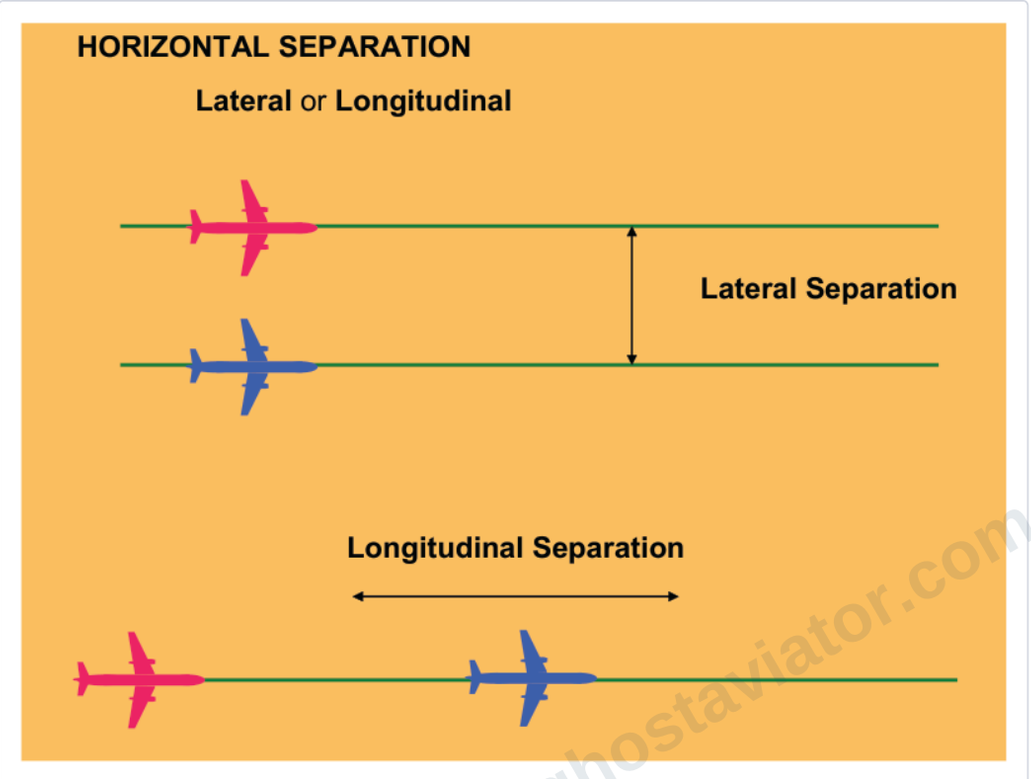

Figure 5.1 *Horizontal separation is either lateral (aircraft on different tracks) or longitudinal (aircraft on the same track, kept apart by time or distance).*

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THREE TYPES OF HORIZONTAL SEPARATION

A) LATERAL SEPARATION | B) LONGITUDINAL SEPARATION | C) ATS SURVEILLANCE-BASED

Lateral Separation

Lateral separation may be applied by the following means:

By Geographic Location (Position Reports)

GEOGRAPHIC SEPARATION

By position reports which positively indicate the aircraft are over **different geographic locations** as determined visually or by reference to a navigation aid.

By Use of NDB, VOR or GNSS on Intersecting Tracks / ATS Routes

Aircraft are required to fly on specified tracks separated by a minimum amount appropriate to the navigation aid employed.

Nav Aid	Angular Divergence Requirement	Distance Condition	Additional Rule
VOR	Radials diverging by at least **15°**	At least one aircraft is **15 NM** or more from the facility	—

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Nav Aid	Angular Divergence Requirement	Distance Condition	Additional Rule
NDB	Tracks diverging by at least **30°**	At least one aircraft is **15 NM** or more from the facility	Track of second aircraft must be at least **20°** from the radial of the first aircraft
GNSS / GNSS	Zero offset between two waypoints on track	At least one aircraft at minimum distance from common point (see table below)	Controller must confirm GNSS navigation and no lateral offsets
VOR / GNSS	VOR aircraft on radial; GNSS aircraft zero-offset between waypoints	At least one aircraft at minimum distance from common point (see table below)	Same as GNSS/GNSS

GNSS / VOR-GNSS Distance Table

Angular Difference at Common Point (°)	Aircraft: VOR or GNSS + GNSS · FL 010–FL 190 · Distance from common point	Aircraft: VOR or GNSS + GNSS · FL 200–FL 600 · Distance from common point
15 – 135°	**27.8 km (15 NM)**	**43 km (23 NM)**

IMPORTANT NOTE

The distances in the table are **ground distances**. States must account for **slant range** from DME source to receiving antenna when DME is used for range information.

GNSS TRACK SEPARATION — CONTROLLER CHECKLIST

Before applying GNSS-based track separation, the controller **shall confirm**:

- The aircraft is navigating using **GNSS**, AND
- In airspace where strategic lateral offsets are authorized, a **lateral offset is NOT being applied**

GNSS-based track separation shall NOT be applied in cases of pilot-reported RAIM outages.

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Different Navigation Aids or Methods

MIXED NAV AID LATERAL SEPARATION

Lateral separation between aircraft using different navigation aids (or when one aircraft uses RNAV equipment) shall be established by ensuring the **derived protected airspaces for the navigation aid(s) or RNP do not overlap**.

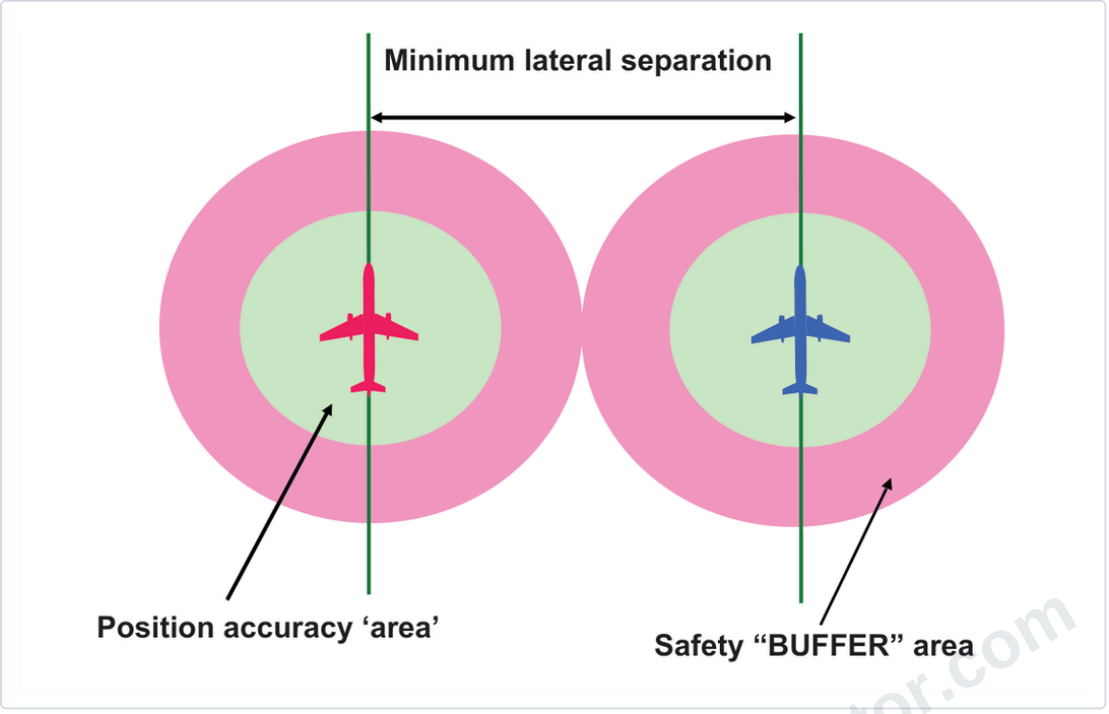

Figure 5.2 *Lateral separation: the protected areas of the two aircraft never overlap.*

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Longitudinal Separation

PRINCIPLE

Longitudinal separation shall be applied so that the spacing between the **estimated positions** of the aircraft being separated is **never less than a prescribed minimum**.

Track Definitions (Critical for Exam)

A) SAME TRACK

Same direction tracks and intersecting tracks or portions thereof, the angular difference of which is **less than 45° or more than 315°**, and whose protection areas overlap.

B) RECIPROCAL TRACKS

Opposite tracks and intersecting tracks or portions thereof, the angular difference of which is **more than 135° but less than 225°**, and whose protection areas overlap.

C) CROSSING TRACKS

Intersecting tracks or portions thereof **other than** those specified in Same Track (a) and Reciprocal Tracks (b) above — i.e. angular difference **45° to 135°** or **225° to 315°**.

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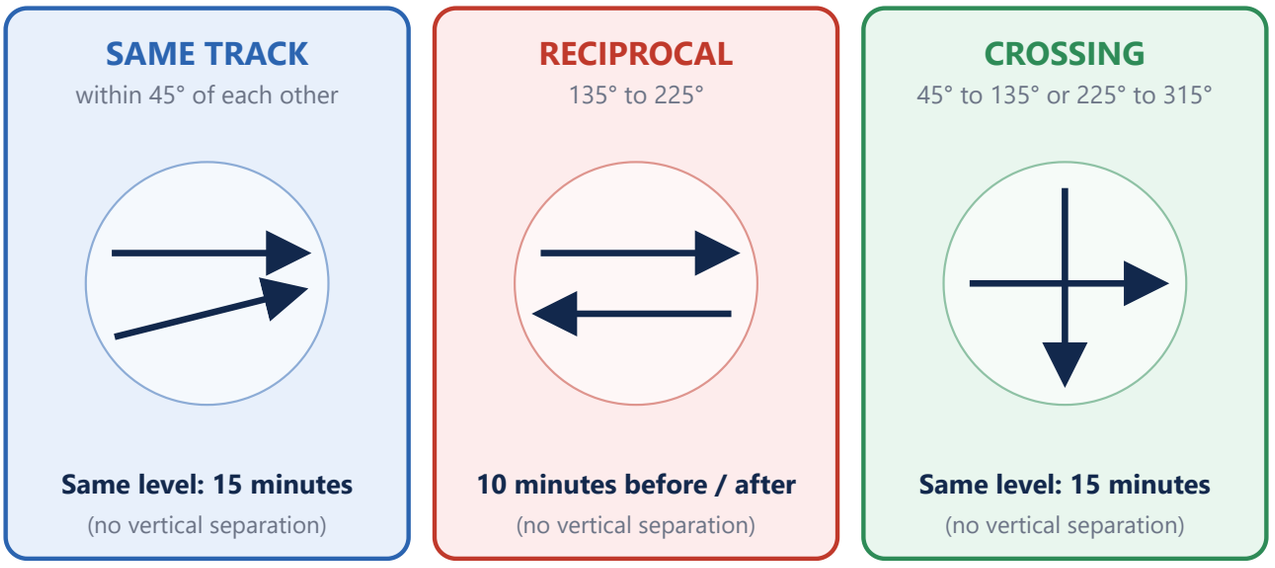

Figure 5.3 *SAME TRACK*

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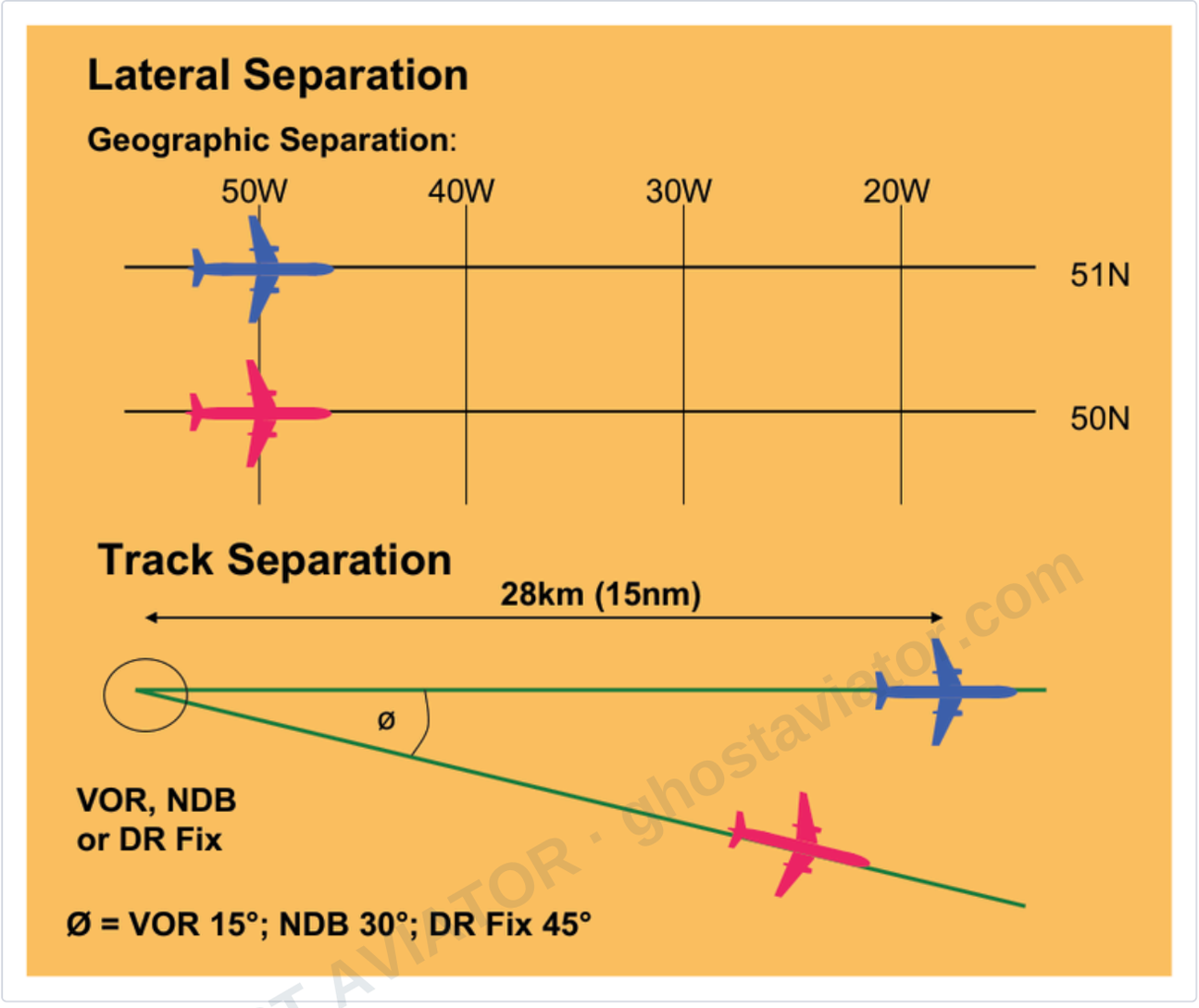

Figure 5.4 *Lateral separation: geographic separation by different positions, and track separation by angular divergence from a VOR (15°), NDB (30°) or DR fix.*

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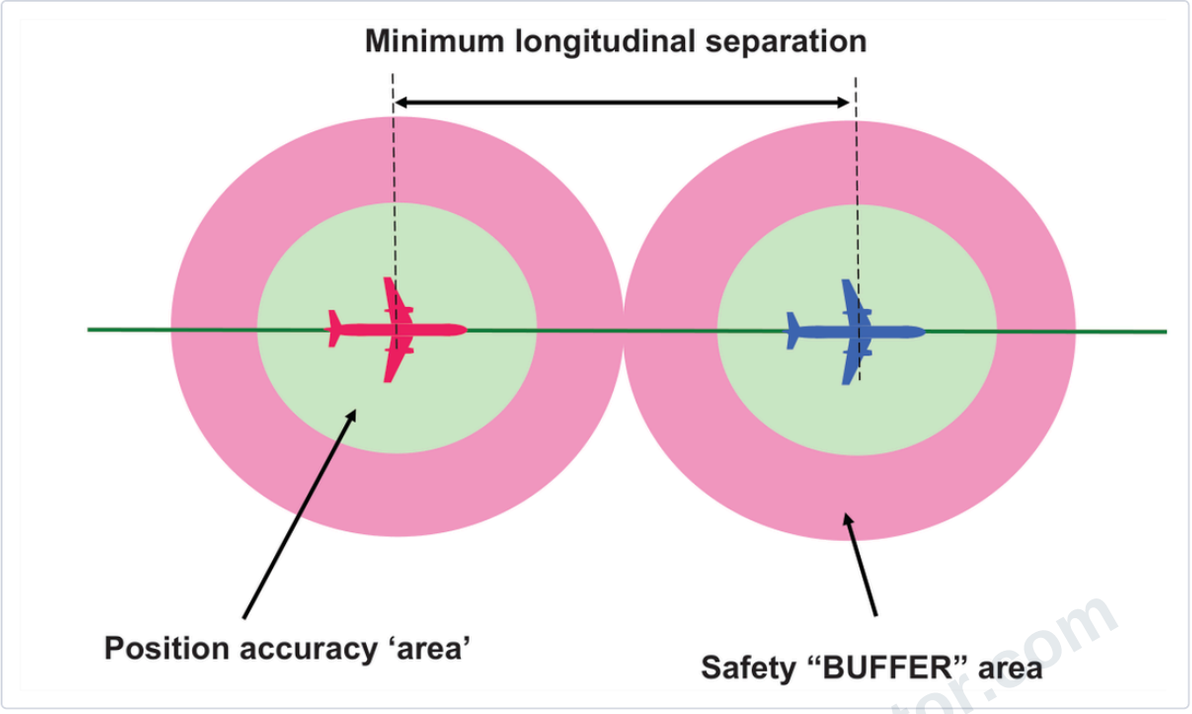

Figure 5.5 *Longitudinal separation: a buffer is kept between aircraft on the same track.*

Longitudinal Separation Minima — Time Based

CROSS-CHECK REQUIREMENT

Separation requirements must be **cross-checked** to ensure the integrity of calculations. The cross-check validates the initial calculation and confirms it is consistent with the traffic disposition.

Aircraft at the Same Cruising Level

Aircraft flying on the SAME TRACK

15	Minutes Standard — when nav aids do NOT permit frequent position/speed determination
10	Minutes When nav aids PERMIT frequent determination of position and speed
5	Minutes Preceding a/c ≥37 kmph (20 kt) FASTER. Requires: same aerodrome departure, OR same exact significant point, OR 5-min guaranteed at route join
3	Minutes Preceding a/c ≥74 kmph (40 kt) FASTER. Same conditions as 5-minute cases

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5-MINUTE SEPARATION — THREE QUALIFYING CASES

1. Between aircraft that have **departed from the same aerodrome**
2. Between en-route aircraft that have **reported over the same exact significant point**
3. Between departing and en-route aircraft after the en-route aircraft has reported over a fix so located as to ensure 5-min separation can be established at the **point the departing aircraft will join the route**

3-MINUTE SEPARATION — CONDITION

Same three cases as for 5-minute separation, BUT the preceding aircraft must be maintaining a true airspeed faster than the succeeding aircraft of **74 kmph (40 knots) or more**.

Aircraft flying on CROSSING TRACKS — Same Level

15	Minutes Standard — no frequent position/speed determination
10	Minutes Nav aids permit frequent position/speed determination
10	Minutes While vertical separation does NOT exist, with navigation aids permitting frequent position/speed determination
5	Minutes While vertical sep does not exist, level change commenced within 10 min of second a/c reporting over exact reporting point

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Aircraft Climbing or Descending — Same Track

RULE: PASSING THROUGH ANOTHER AIRCRAFT'S LEVEL (SAME TRACK)

When an aircraft will **pass through the level** of another aircraft on the same track, the following minimum longitudinal separation shall be provided:

- **15 minutes** while vertical separation does not exist

Aircraft Climbing or Descending — Crossing Tracks

Least 10 minutes prior to and after the time the aircraft are estimated to pass:

15	Minutes While vertical separation does not exist (standard)
10	Minutes While vertical sep does not exist — only where ground-based nav aids or GNSS permit frequent position/speed determination
5	Minutes While vert sep does not exist, provided: level change within 10 min of second a/c reporting over common point AND CPDLC restriction added if applicable

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Traffic on Reciprocal Tracks — Climbing or Descending

RECIPROCAL TRACK RULE

Where lateral separation is NOT provided, **vertical separation shall be provided for at least 10 minutes prior to and after the time the aircraft are estimated to pass, or are estimated to have passed**.

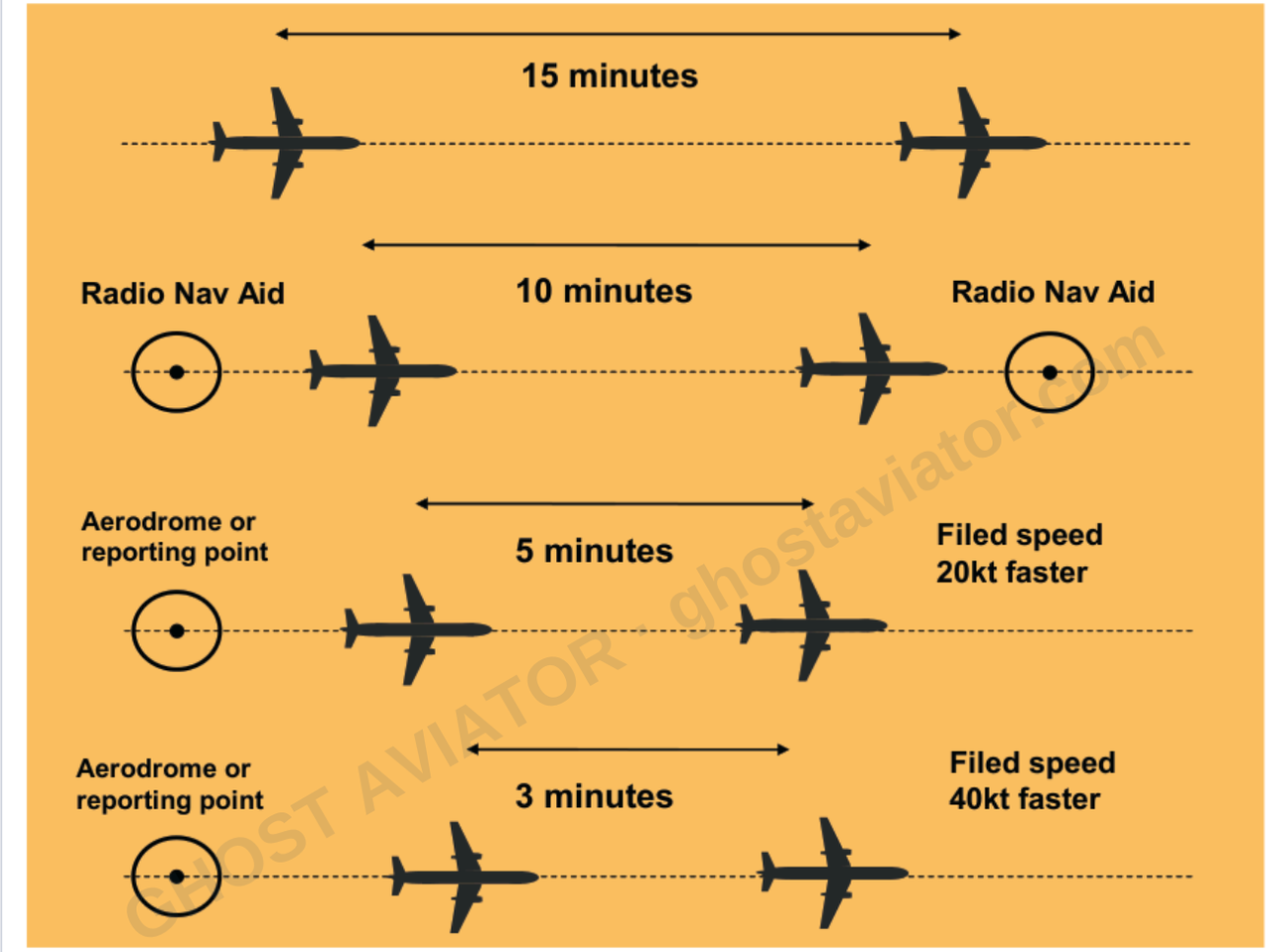

Figure 5.6 *Same-track longitudinal separation at the same level: 15 min standard; 10 min where navigation aids give frequent position/speed checks; 5 min when the preceding aircraft is 20 kt faster; 3 min when it is 40 kt faster.*

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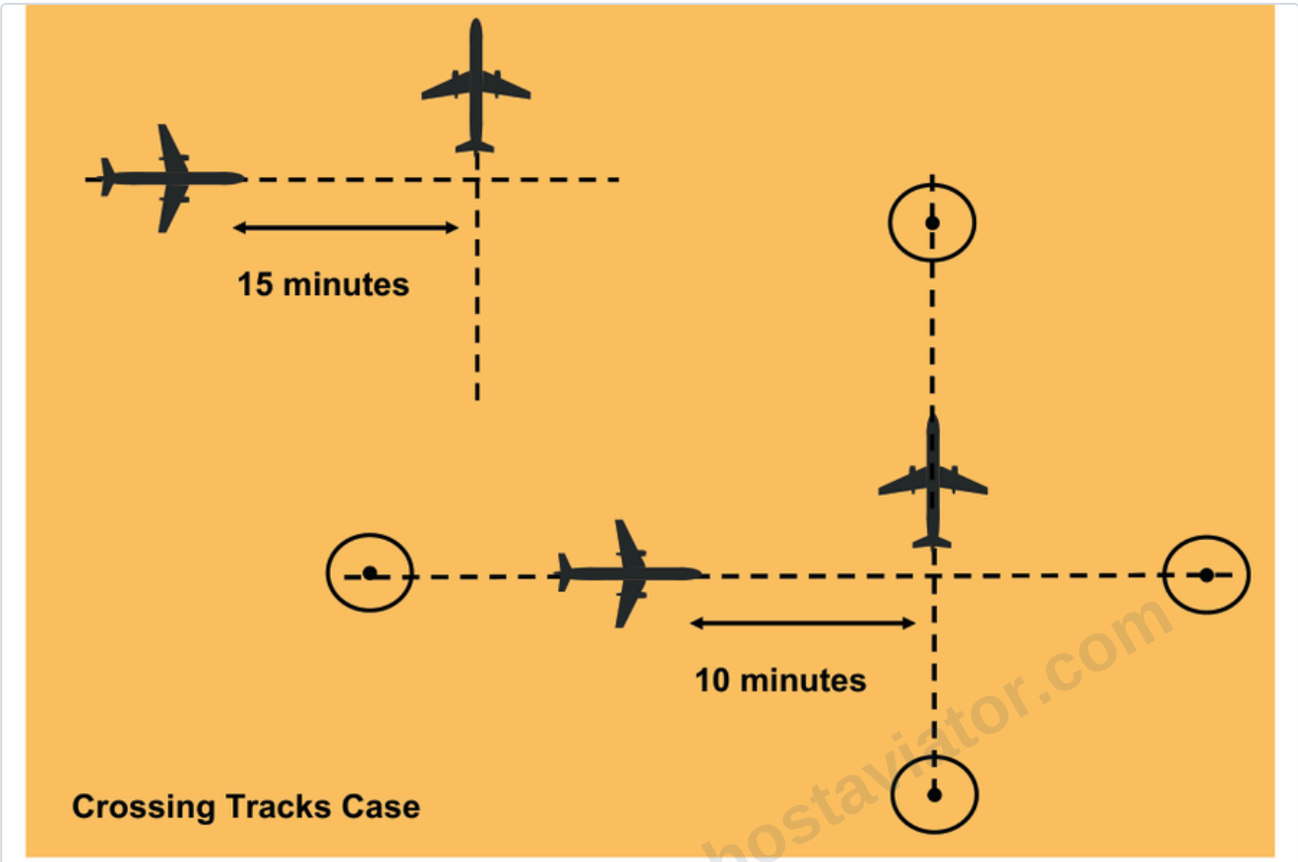

Figure 5.7 *Crossing tracks at the same level: 15 min standard, 10 min with frequent position/speed determination.*

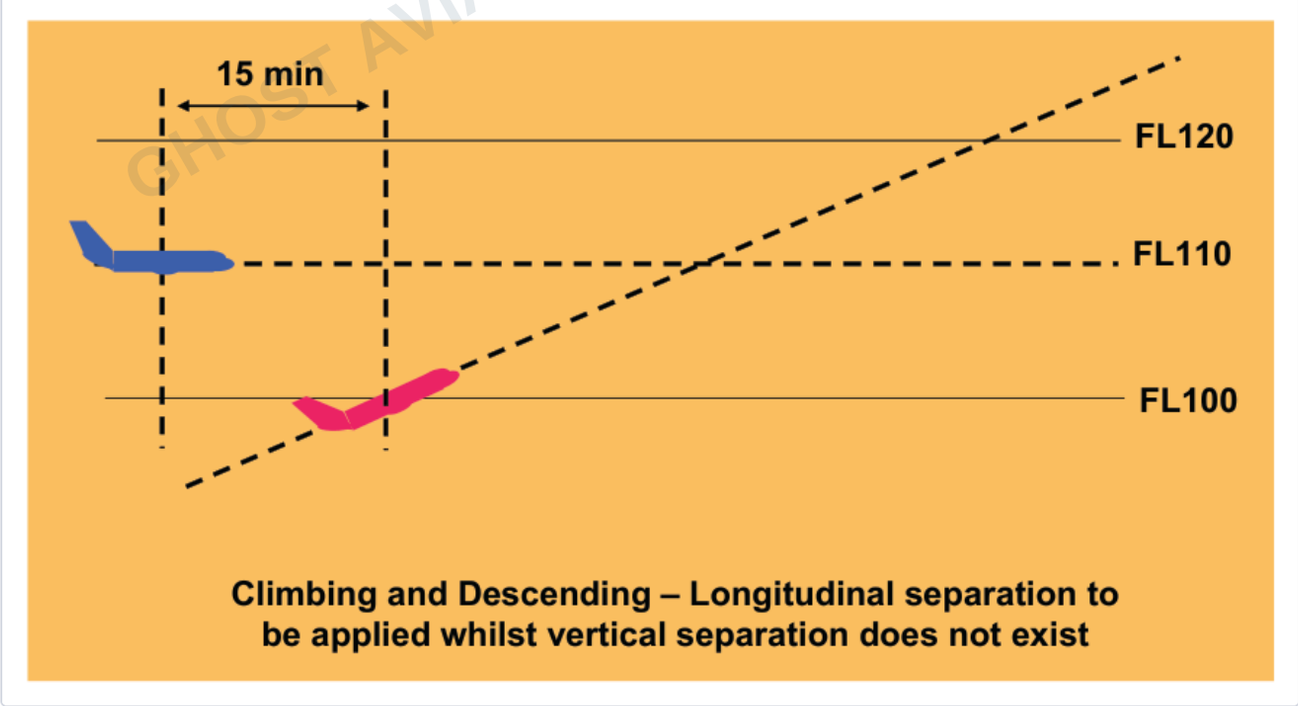

Figure 5.8 *Climbing/descending through another aircraft's level on the same track: 15 min while vertical separation does not exist.*

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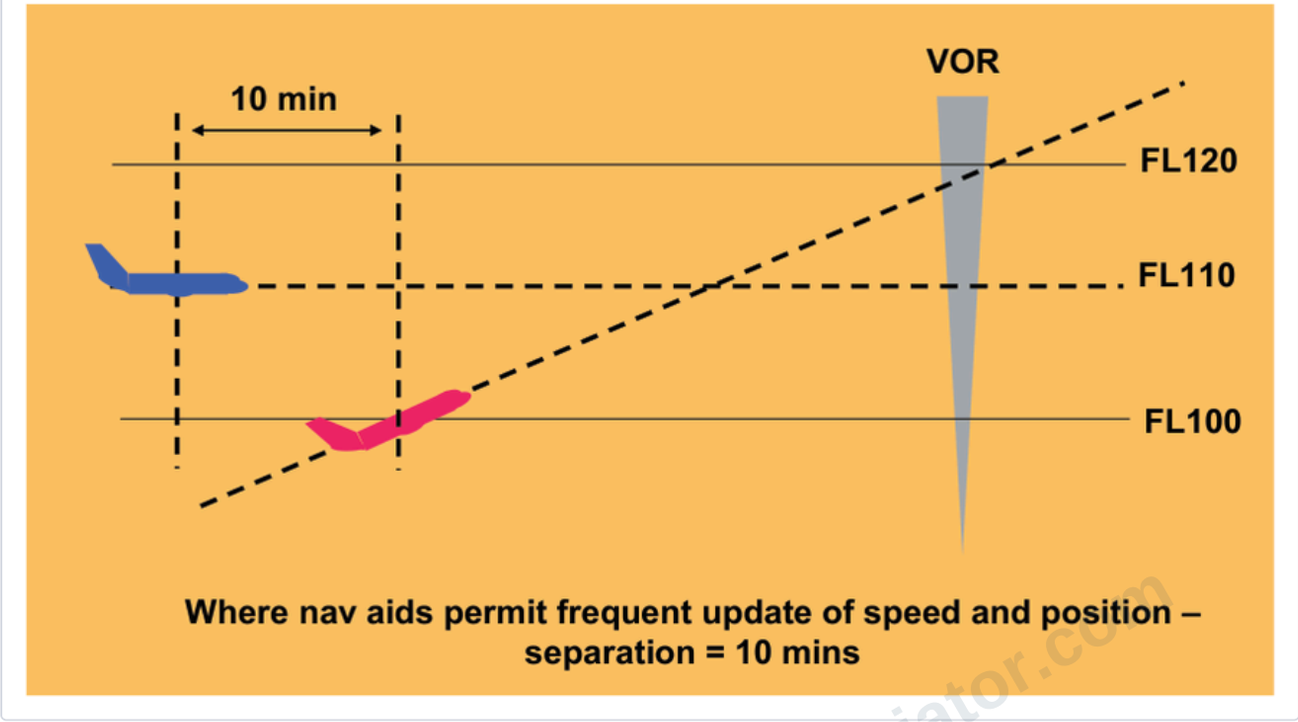

Figure 5.9 *Where navigation aids permit frequent position and speed checks: 10 min while vertical separation does not exist.*

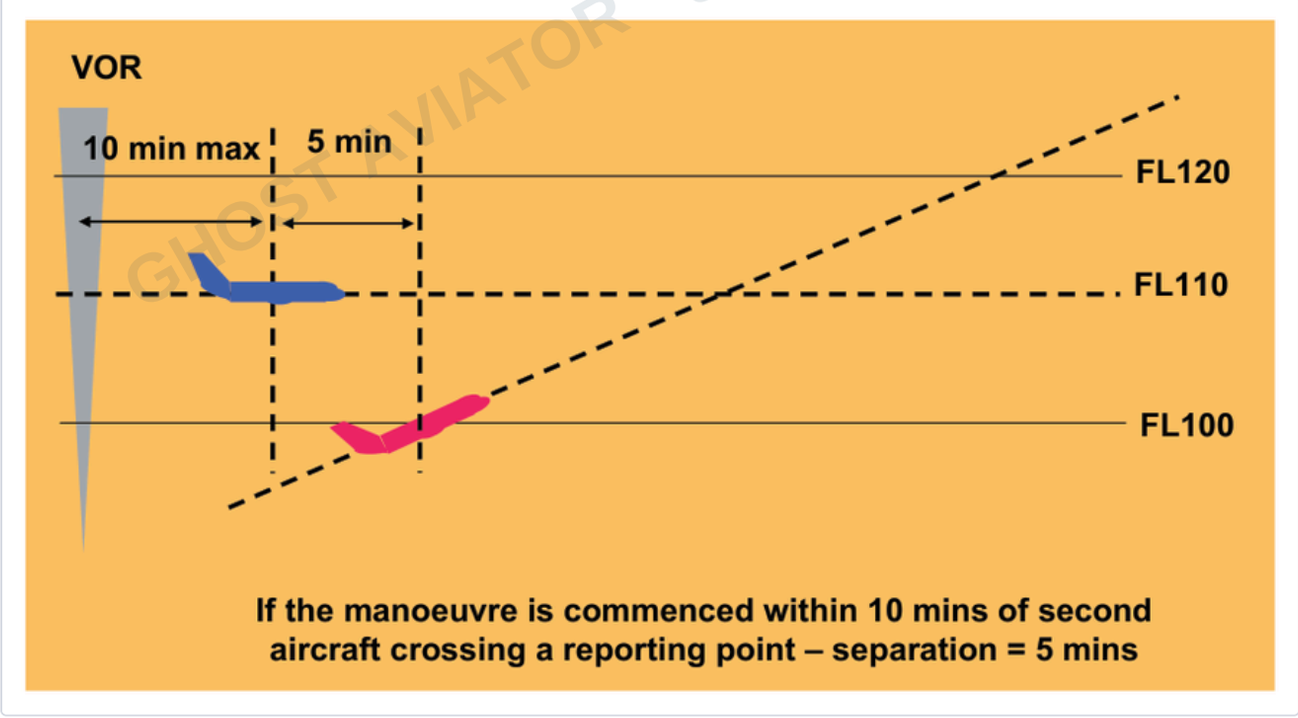

Figure 5.10 *5 min: level change commenced within 10 min of the second aircraft reporting over a common reporting point.*

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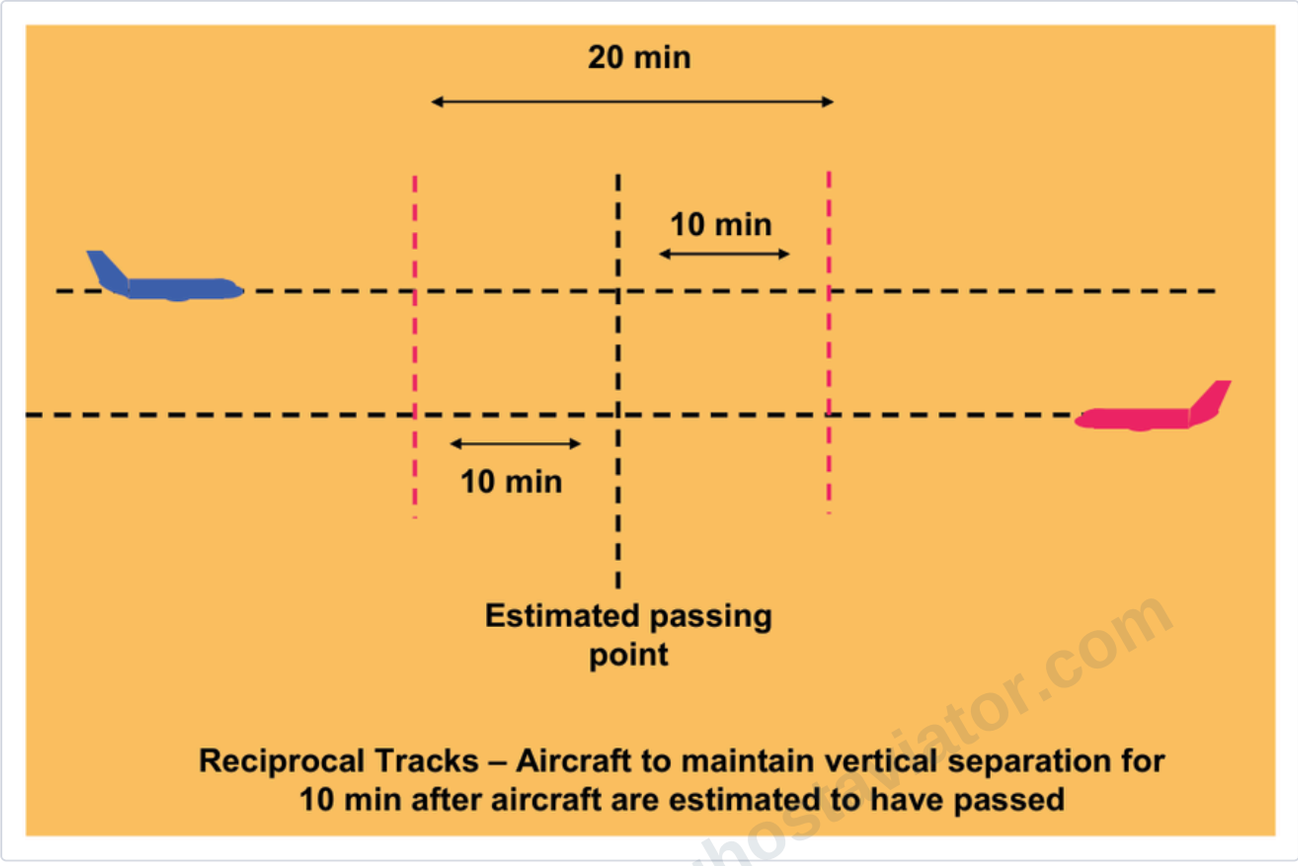

Figure 5.11 *Reciprocal tracks: provide vertical separation until 10 min after the aircraft are estimated to have passed.*

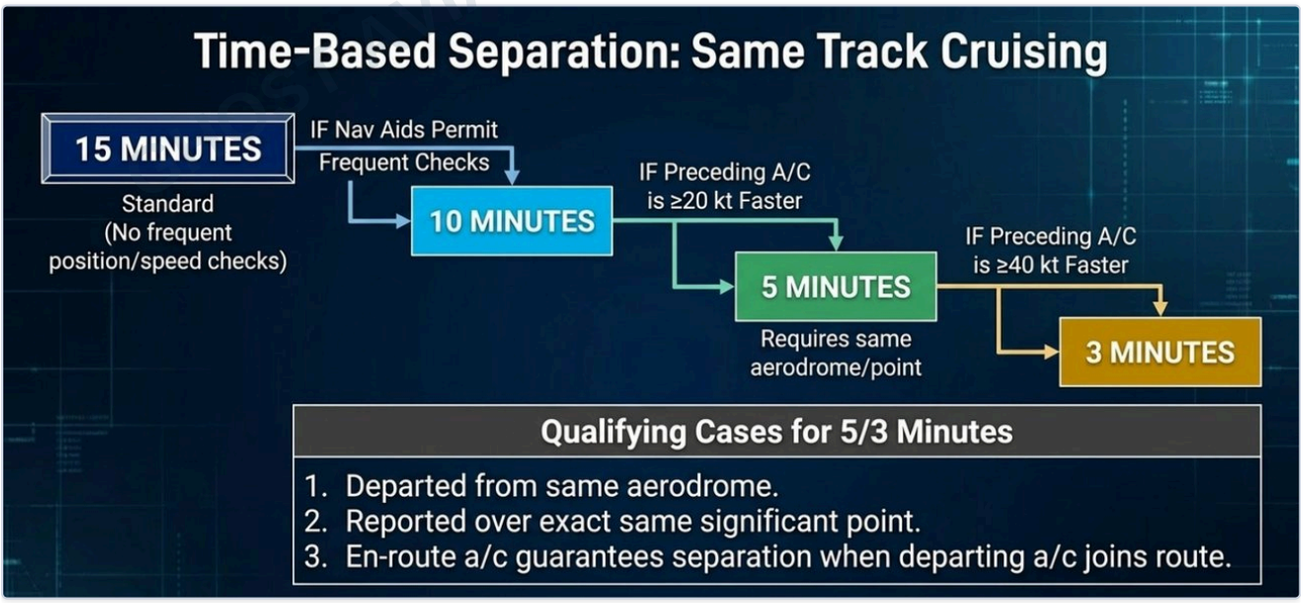

Figure 5.12 *Time-based separation on the same track*

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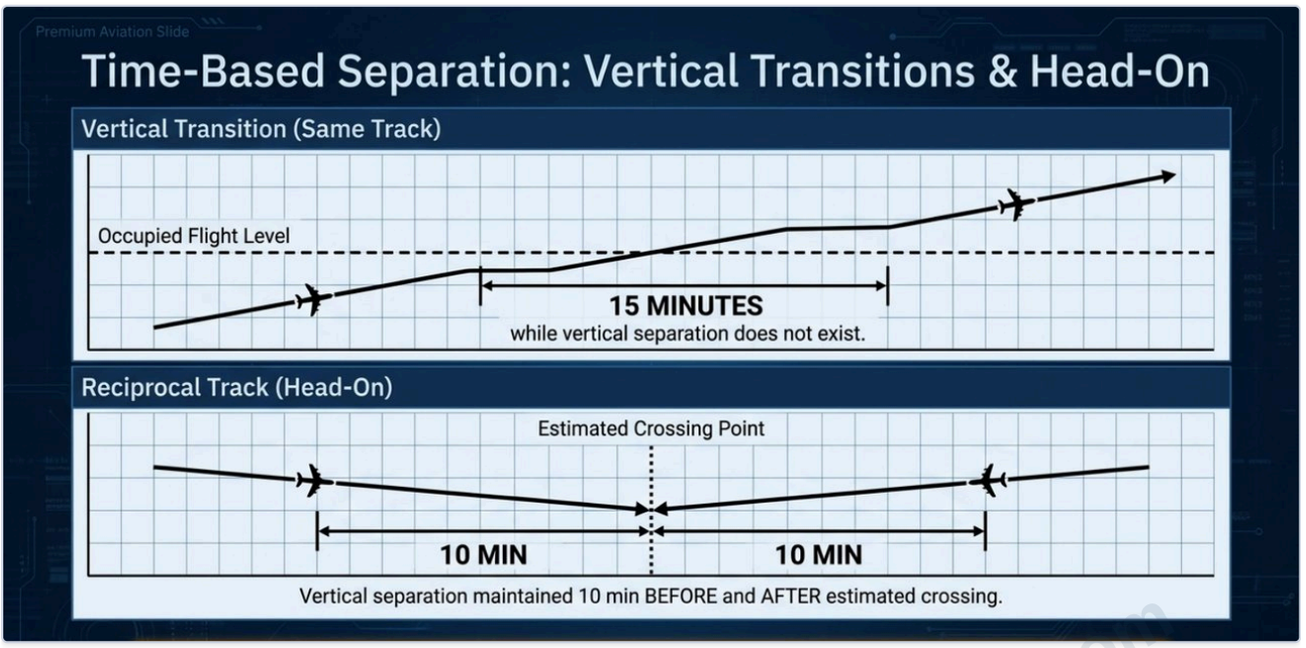

Figure 5.13 *Climbing or descending through a level, and reciprocal tracks*

Longitudinal Separation — Distance Based (DME / GNSS)

APPLICABILITY

Separation shall be established by maintaining not less than specified distances between aircraft positions as reported by reference to **DME in conjunction with other nav aids and/or GNSS**. Applied between:

- Two aircraft using **DME**
- Two aircraft using **GNSS**
- One aircraft using **DME** + one using **GNSS**

Direct controller-pilot VHF voice communication shall be maintained while this separation is used.

GNSS NOTE

For GNSS-based separation minimum, a distance derived from an **integrated navigation system incorporating GNSS input is regarded as equivalent to GNSS distance**.

All distance reports must be made with reference to the **same DME station**.

When applying these minima to any aircraft with area navigation capability, controllers shall specifically **request GNSS derived distance**.

WHEN LATERAL SEPARATION IS NOT PROVIDED

Vertical separation shall be provided for inadequate onboard equipment, on GNSS input into an integrated navigation system, or a **loss of GNSS integrity**.

Note: Traffic on reciprocal tracks may make it impossible for a pilot to provide GNSS distance information.

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Aircraft at Same Cruising Level — Same Track

A) 20 NM — BASELINE

Provided each aircraft utilizes:

- Same **"on track" DME station** (both on DME), OR
- "On track" DME + **collocated waypoint** (one DME, one GNSS), OR
- Same **waypoint** (both GNSS)

AND separation is checked by obtaining **simultaneous DME/GNSS readings** at frequent intervals.

B) 10 NM — REDUCED (SPEED CONDITION)

Same reference requirements as 20 NM, PLUS:

The **leading aircraft** maintains a true airspeed of **20 kt or more faster** than the succeeding aircraft.

Separation checked simultaneously at frequent intervals.

Aircraft at Same Level — Crossing Tracks

Minimum	Conditions	Speed Condition	Angle Condition
20 NM	Each a/c reports distance from DME/collocated waypoint/same waypoint at crossing point	None required	Relative angle between tracks **<90°**
10 NM	Same reference + simultaneous DME/GNSS checks	Leading a/c **≥20 kt faster**	Relative angle **<90°**

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Aircraft Climbing or Descending — Same Track

10 NM WHILE VERTICAL SEPARATION DOES NOT EXIST, PROVIDED

- Each aircraft utilizes the same "on track" DME / collocated waypoint / same waypoint
- One aircraft **maintains a level** while vertical separation does not exist
- Separation established by obtaining **simultaneous DME/GNSS readings**

Aircraft on Reciprocal Tracks — DME/GNSS

RECIPROCAL TRACKS — CLIMB/DESCENT

Aircraft utilizing on-track DME and/or collocated waypoint or same waypoint may be cleared to climb or descend to or through levels occupied by other aircraft utilizing same references, **provided it has been positively established that the aircraft have passed each other and are at least 10 NM apart.**

Separation Minima — ATS Surveillance Systems (Radar / ADS-B / MLAT)

System	Range / Condition	Horizontal Separation Minimum
Radar / ADS-B / MLAT	Within reliable surveillance coverage	5 NM standard; reductions only where authorized

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System	Range / Condition	Horizontal Separation Minimum
Radar (Special Authorization)	Specifically authorized conditions	3 NM

OFFICIAL STATEMENT

Five nautical miles (5 NM) within airspace under surveillance coverage.

AAI AIP India, ENR 1.6, paragraph 7.5.1; web eAIP has no fixed PDF page. Read alongside paragraphs 7.5.2–7.5.3. Official document

In plain English: The standard horizontal minimum in surveillance coverage is 5 NM. A smaller minimum needs a specific authorization. Other rules may require greater separation.

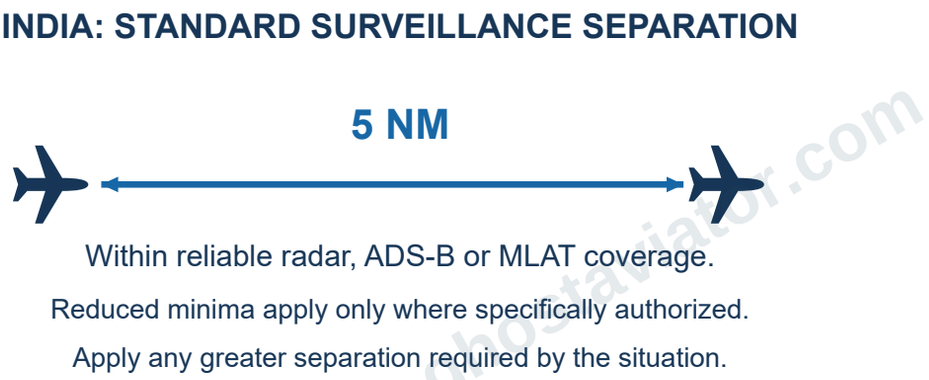

Figure 5.14 *AAI standard horizontal separation within surveillance coverage. A fixed 60 NM boundary does not create a blanket 10 NM minimum. Local authorizations and other applicable separation requirements must still be observed.*

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Master Summary — Longitudinal Separation

Time-Based Longitudinal Separation Summary

Separation	Nav Aid Condition	Track Type	Phase	Speed Condition
15 min	Nav aids do NOT permit frequent pos/speed determination	Same track, Crossing track	Same level, Climb/Descend	None
10 min	Nav aids permit frequent pos/speed determination	Same track, Crossing track	Same level, Climb/Descend	None
10 min	Any	Reciprocal	10 min before & after crossing	None
5 min	Common reference point	Same track	Same level, Climb/Descend	Preceding a/c ≥20 kt faster
5 min	Common ref (GNSS/ground)	Crossing	Climb/Descend (level change within 10 min)	None
3 min	Common reference point	Same track	Same level	Preceding a/c ≥40 kt faster

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DME/GNSS Distance-Based Separation Summary

Minimum	Track Type	Phase	Speed Condition	Angle Condition
20 NM	Same track	Same level	None	—
20 NM	Crossing	Same level	None	<90°
10 NM	Same track	Same level	Leading a/c ≥20 kt faster	—
10 NM	Crossing	Same level	Leading a/c ≥20 kt faster	<90°
10 NM	Same track	Climbing/Descending (one a/c maintains level)	None	—
10 NM	Reciprocal	Climb/Descend after confirmed passing	None	—

Longitudinal Separation — Mach Number Technique

WHEN MACH NUMBER TECHNIQUE IS APPLIED

Minimum longitudinal separation between turbojet aircraft on the **same track**, whether in level, climbing or descending flight, shall be:

- **10 minutes**, OR
- The prescribed minima based on application of **differential Mach number** on prescribed ATS routes

RNAV DISTANCE SEPARATION WITH MACH TECHNIQUE

- RNAV distance-based separation may be applied between **RNAV-equipped aircraft** on designated RNAV routes or ATS routes defined by VOR
- A **150 km (80 NM)** RNAV distance-based separation minimum with Mach number technique may be used on same-direction tracks **in lieu of a 10-minute longitudinal separation minimum**

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Performance-Based Concepts (Definitions)

RNP — REQUIRED NAVIGATION PERFORMANCE

A statement of navigation performance necessary for operation within a defined airspace. For en-route phase: **RNP 1, 4, 10, 12.6 and 20**.

PBN — PERFORMANCE-BASED NAVIGATION

The prescribed navigation specification shall be appropriate to the level of communications, navigation and ATS provided in the airspace. (Refer Doc 9613)

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RCP — REQUIRED COMMUNICATION PERFORMANCE

A set of requirements for ATS provision, aircraft capability, and operations to support performance-based communication. RCP number = **seconds for instruction to travel ground→aircraft→ground**. Examples: **RCP 10, 60, 120, 240, 400**.

RSP — REQUIRED SURVEILLANCE PERFORMANCE

Number of seconds for surveillance data from the CSP interface to arrive at the ATSU flight data processing system. **RSP 180, 240, 400** indicate time in seconds between flight crew/HMI and controller/HMI. HMI = Human Machine Interaction.

PBS — PERFORMANCE-BASED SURVEILLANCE

ATS surveillance services and capability based on performance requirements for ATS provision, aircraft flight operations along an ATS route, on an instrument approach procedure, or in a designated airspace.

Performance-Based Longitudinal Separation Minima (Table)

For aircraft cruising, climbing or descending on the **same track** OR crossing tracks (relative angle **<90°**):

Separation Minima	RNP	RCP	RSP	Max ADS-C Periodic Reporting Interval
93 km (50 NM)	10	240	180	27 minutes
	4	240	180	32 minutes
55.5 km (30 NM)	2 or 4	240	180	12 minutes
5 minutes	2 or 4 or 10	240	180	14 minutes

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FMS/RNAV IN CONVENTIONAL PROCEDURES

When authorised in the **Aircraft Flight Manual (AFM)**, FMS/RNAV may be used to fly conventional procedures provided the procedures are monitored using basic display associated with these procedures and **tolerances for using raw data on the basic display are complied with**.

Essential Traffic Information

DEFINITION — ESSENTIAL TRAFFIC

Essential traffic is that **controlled traffic to which the provision of separation by ATC is applicable**, but which, in relation to a particular controlled flight, is **not, or will not be, separated from other controlled traffic by the appropriate separation minimum**.

Essential traffic information shall be given to controlled flights concerned whenever they constitute essential traffic to each other.

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Quick Reference Card — All Key Values at a Glance

Parameter	Value	Condition / Note
Vertical Sep below FL 290	**1 000 ft**	Standard non-RVSM
Vertical Sep at/above FL 290	**2 000 ft**	Standard non-RVSM
Vertical Sep (RVSM)	**1 000 ft**	Both a/c RVSM compliant, in RVSM airspace
Cruise Climb	**NOT permitted**	Indian FIRs
VOR Lateral Sep	**15°, 15 NM**	Radials ≥15° diverging; at least one a/c ≥15 NM
NDB Lateral Sep	**30°, 15 NM**	Tracks ≥30° diverging; at least one a/c ≥15 NM; 2nd track ≥20° from first radial
GNSS/GNSS distance (FL 010–190)	**15 NM (27.8 km)**	Angular diff 15–135°
GNSS/GNSS distance (FL 200–600)	**23 NM (43 km)**	Angular diff 15–135°
Same Track (standard)	**15 min**	Angular diff <45° or >315°
Reciprocal Tracks	**135°–225°**	10 min before/after crossing
Crossing Tracks	**45°–135° or 225°–315°**	15 min standard
Long Sep with 20 kt speed adv	**5 min**	Same aerodrome/point; 37 kmph faster
Long Sep with 40 kt speed adv	**3 min**	Same aerodrome/point; 74 kmph faster
DME baseline (same track)	**20 NM**	Same level
DME reduced (20 kt faster)	**10 NM**	Leading a/c ≥20 kt faster
DME climbing/descending	**10 NM**	Same track; one a/c maintains level
Radar / ADS-B / MLAT	Within reliable surveillance coverage	5 NM standard; reductions only where authorized
Radar Sep (special auth)	**3 NM**	Specifically authorised only
RNAV/Mach technique	**80 NM (150 km)**	Same direction; lieu of 10-min sep
VMC own sep — altitude limit	**10 000 ft**	During climb/descent in Class D/E, daylight
RCP transaction time	**seconds (ground↔a/c)**	RCP 10/60/120/240/400

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CHAPTER 6

Separation Near Aerodromes

06

Procedures for Departing Aircraft — General

REGULATORY BASIS

All departure clearances are governed by **Doc 4444 (PANS-ATM)** and **AIP India**. The ATC clearance for a departing aircraft shall specify, when necessary for the separation of aircraft:

ELEMENTS OF A DEPARTURE CLEARANCE

- **Direction of take-off** and the **turn after take-off**
- **Heading or track** to be made good before taking up the cleared departure track
- **Level to maintain** before continuing climb to the assigned level
- **Time, point and/or rate** at which a level change shall be made
- Any **other necessary manoeuvre** consistent with the safe operation of the aircraft

SID REQUIREMENT

At aerodromes where **Standard Instrument Departures (SIDs)** have been established, departing aircraft should **normally be cleared to follow the appropriate SID**.

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Minimum Separation Between Departing Aircraft

Three distinct separation minima apply between departing aircraft, depending on track divergence, speed difference, and level relationship:

1	MINUTE **Condition:** Aircraft are to fly on tracks **diverging by at least 45°** immediately after take-off so that **lateral separation is provided**.
2	MINUTES **Condition:** The preceding aircraft is **40 kt or more faster** than the following aircraft AND **both propose to follow the same track**.
5	MINUTES **Condition:** A departing aircraft will be **flown through the level of a preceding departing aircraft** and both follow the **same track**. *Vertical separation does not exist during this phase.*

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CRITICAL RULE — 5-MINUTE SEPARATION

Action **must be taken** to ensure that the **5-minute separation will be maintained or increased** while vertical separation does not exist. This is a strict ATC obligation — not discretionary.

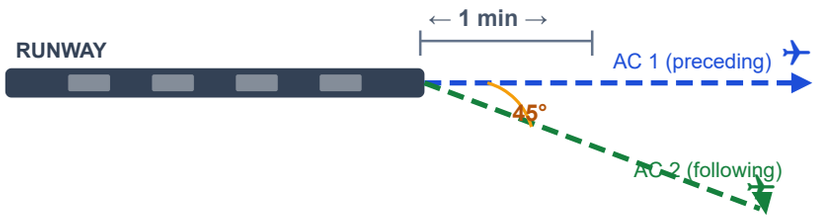

Figure 6.1 *1-Minute Separation (Diverging Tracks ≥ 45°)*

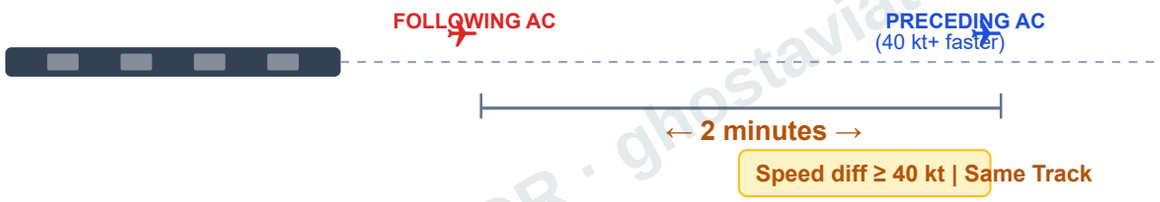

Figure 6.2 *2-Minute Separation (Preceding AC 40 kt or More Faster, Same Track)*

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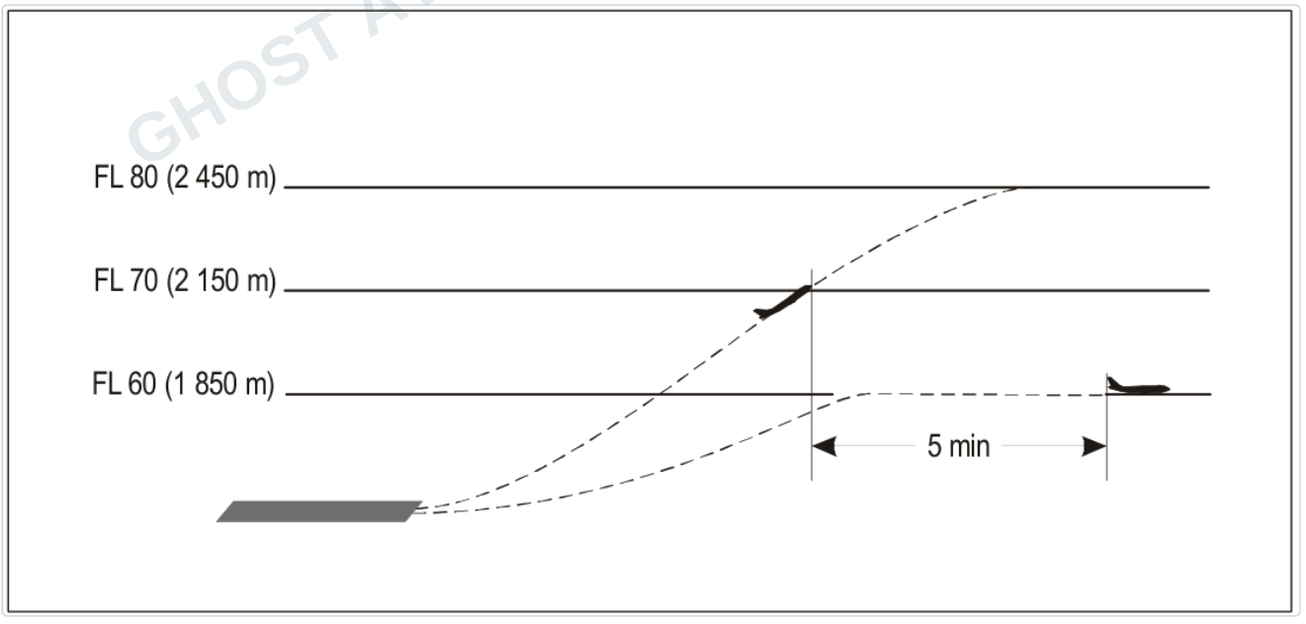

Figure 6.3 *Five-minute departure separation on the same track when a following departure will pass through the preceding departure's level. Separation must be maintained or increased while vertical separation does not exist.*
ICAO Doc 4444, 15 th ed, §5.6.3, Figure 5-37.

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Quick Reference — Separation Minima Between Departing Aircraft

Separation	Condition	Track Requirement	Additional Notes
1 Minute	Standard diverging departure	Tracks diverge by **≥ 45°** immediately after T/O	Lateral separation is provided
2 Minutes	Speed differential on same track	**Same track**	Preceding AC is **≥ 40 kt faster**
5 Minutes	Through-level conflict	**Same track**	Following AC passes through level of preceding AC; vertical separation absent — must be maintained or increased

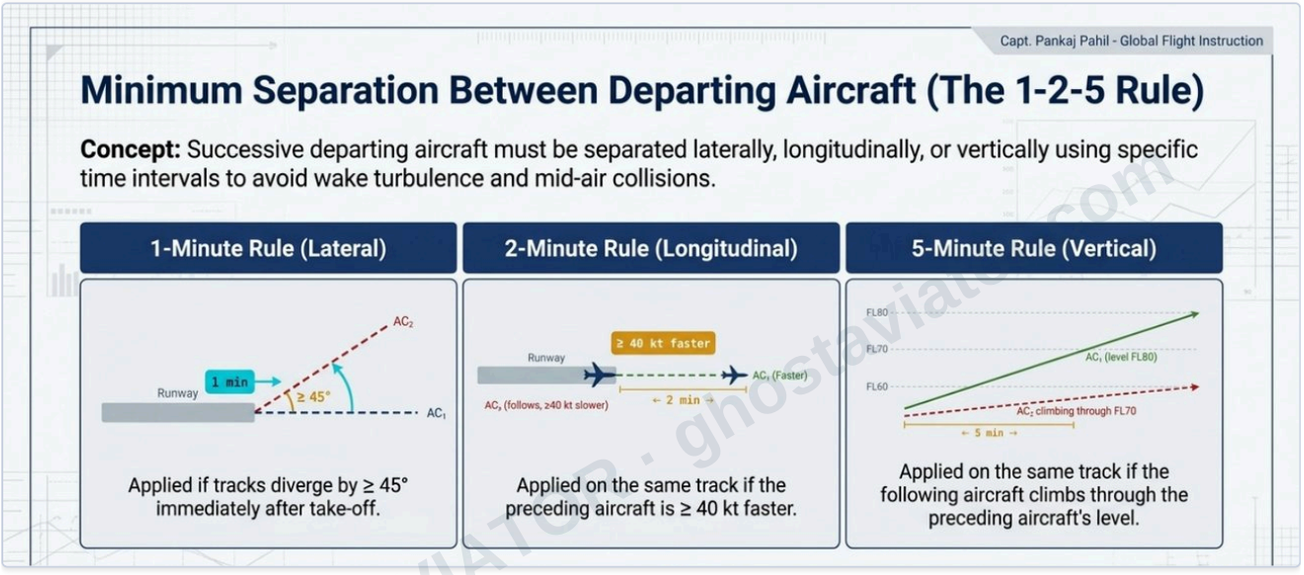

Figure 6.4 *Departing aircraft: the 1, 2 and 5 minute rules*

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Separation of Departing Aircraft from Arriving Aircraft

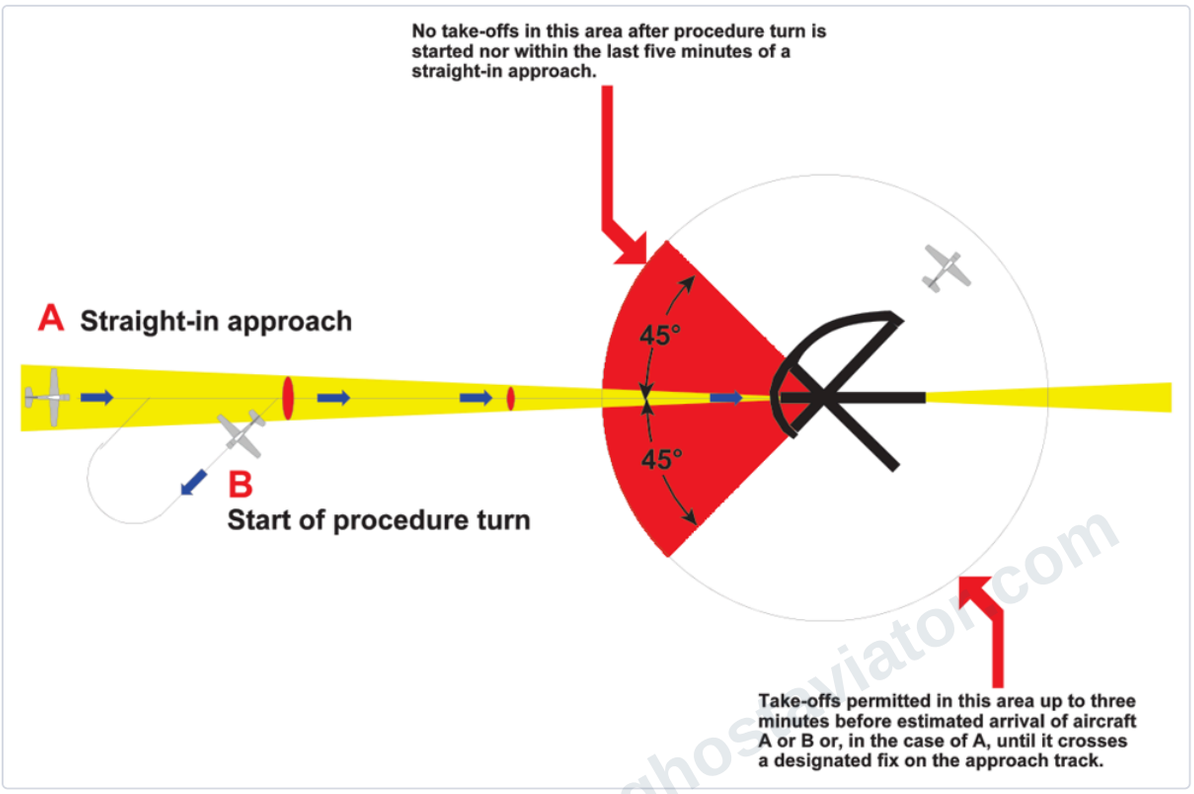

Figure 6.5 *Departure and arrival conflict area. In the red sector, 45 degrees either side of the approach track, no take-off is made after the arriving aircraft starts its procedure turn, nor in the last five minutes of a straight-in approach. Outside it, take-off is allowed up to three minutes before the arrival is estimated over the runway.*

Separation rules when take-off clearance is based on the position of an arriving aircraft. Two scenarios apply:

Scenario A — Arriving Aircraft Making a Complete Instrument Approach

A DEPARTING AIRCRAFT MAY TAKE OFF

a. **In any direction** — until the arriving aircraft has **started its procedure turn or base turn** leading to final approach.

b. **In a direction differing by at least 45° from the reciprocal of the approach direction** — after the arriving aircraft has started its procedure turn or base turn, provided:

- The take-off is made **at least 3 minutes before** the arriving aircraft is estimated to be over the beginning of the instrument runway.

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Scenario B — Arriving Aircraft Making a Straight-In Approach

A DEPARTING AIRCRAFT MAY TAKE OFF

a. **In any direction** — until **5 minutes before** the arriving aircraft is estimated to be over the instrument runway.

b. **In a direction differing by at least 45° from the reciprocal of the approach direction**:

i. Until **3 minutes before** the arriving aircraft is estimated to be over the beginning of the instrument runway, **OR**

ii. Before the arriving aircraft **crosses a designated fix on the approach track** (the location of such fix to be determined by the appropriate ATS authority after consultation with operators).

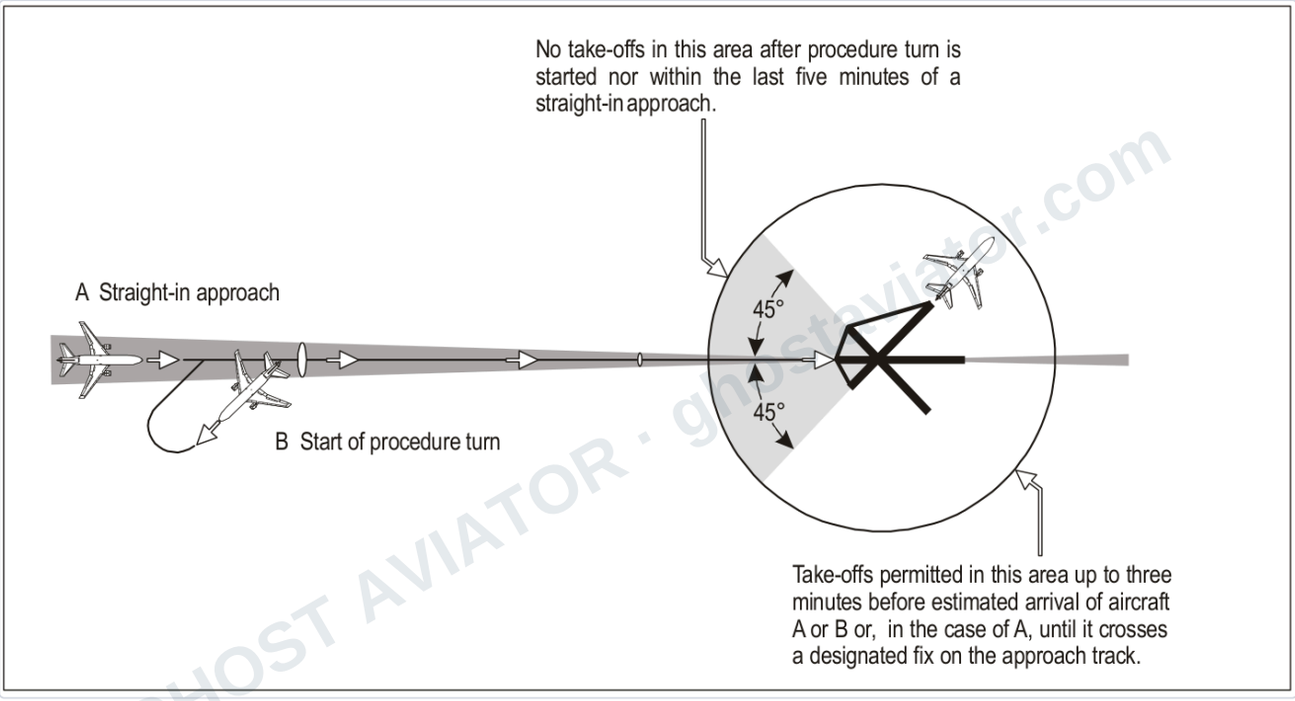

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Figure 6.6 *Departure directions relative to an arriving aircraft. Read each sector with the approach stage, timing and designated-fix conditions in the text.* *ICAO Doc 4444, 15 th ed, §§5.7.1.1–5.7.1.2, Figure 5-38.*

Approach Type	Departure Direction	Condition / Time Limit
Complete Instrument Approach	Any direction	Until arriving AC has **started procedure turn or base turn**
	≥ 45° from reciprocal of approach direction	After proc. turn starts, T/O at least **3 min before** arriving AC over runway THR
Straight-In Approach	Any direction	Until **5 minutes before** arriving AC estimated over instrument runway
	≥ 45° from reciprocal of approach direction	Until **3 minutes before** arriving AC over runway THR
	≥ 45° from reciprocal of approach direction	Before arriving AC crosses the **designated fix on approach track**

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Reduction in Separation Minima in the Vicinity of Aerodromes

Separation minima **in the vicinity of aerodromes may be reduced** under specific conditions. All three conditions are independent — any one is sufficient:

THREE CONDITIONS PERMITTING REDUCTION

a. **Controller Visual Observation:** Adequate separation can be provided by the aerodrome controller when **each aircraft is continuously visible to this controller**.

b. **Mutual Visual Contact — Both Crews:** Each aircraft is continuously visible to the **flight crews of the other aircraft concerned**, and the pilots report that they **can maintain their own separation**.

c. **Following Aircraft Reports Visual Contact:** In the case of one aircraft following another — the **flight crew of the succeeding aircraft reports that the other aircraft is in sight** and **separation can be maintained**.

CRITICAL DISTINCTION

Reduction is **not** at the sole discretion of the controller. Condition (a) requires *continuous visibility by the controller*. Conditions (b) and (c) require *explicit pilot reports* that they can maintain their own separation. No reduction is permissible without satisfying one of these conditions.

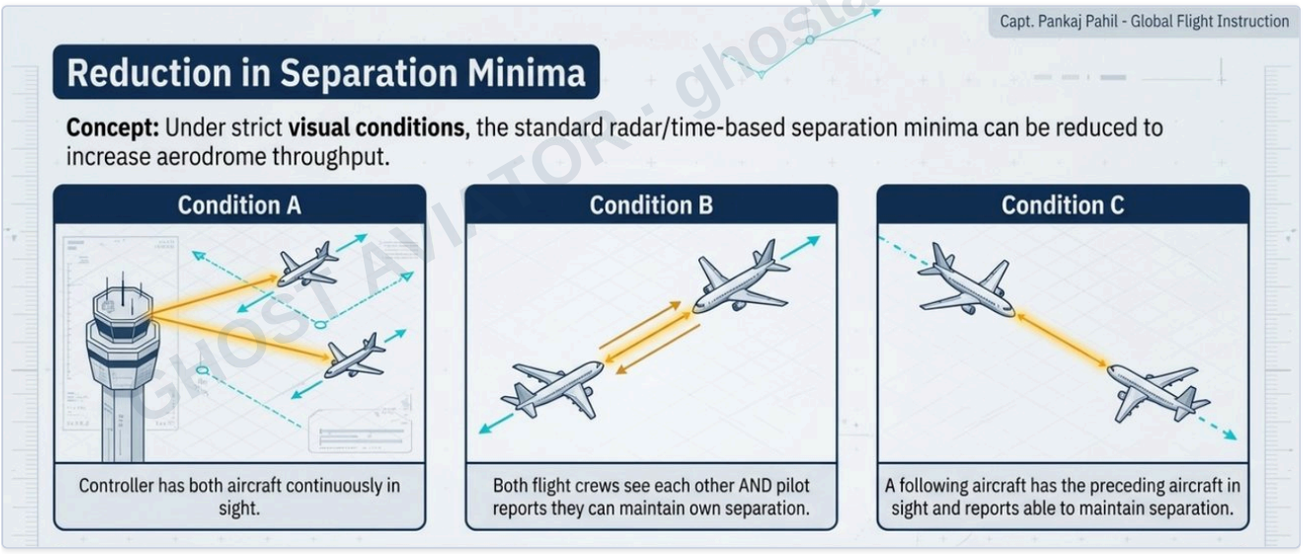

Figure 6.7 *When separation near an aerodrome may be reduced*

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Information for Departing Aircraft

A — Meteorological Conditions

RULE

Information regarding **significant changes in meteorological conditions** in the take-off or climb-out area, obtained by the unit providing approach control service after a departing aircraft has established communication, **shall be transmitted to the aircraft without delay**.

Exception: Transmission is not required when it is known that the aircraft already has received the information.

"SIGNIFICANT CHANGES" — DEFINED (REGULATORY NOTE)

Significant changes include those relating to:

- Surface wind direction or speed
- Visibility
- Runway visual range
- Air temperature *(for turbine-engine aircraft)*
- Occurrence of **thunderstorm or cumulonimbus**
- **Moderate or severe turbulence**
- **Wind shear**
- Hail
- **Moderate or severe icing**
- **Severe squall line**
- Freezing precipitation
- Severe mountain waves
- Sand storm, dust storm, blowing snow
- **Tornado or waterspout**

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B — Operational Status of Visual or Non-Visual Aids

TRANSMISSION REQUIREMENT

Information regarding **changes in the operational status of visual or non-visual aids essential for take-off and climb** shall be transmitted **without delay** to a departing aircraft, except when it is known that the aircraft already has received the information.

Procedures for Arriving Aircraft — General

STAR CLEARANCE

At aerodromes where **Standard Instrument Arrivals (STARs)** have been established, arriving aircraft should **normally be cleared to follow the appropriate STAR**.

The aircraft shall be informed of the **type of approach to expect** and **runway-in-use** as early as possible.

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DIRECT APPROACH BY ACC

After coordination with the approach control unit, the **ACC may clear the first arriving aircraft for approach** rather than to a holding fix.

Visual Approach — Definitions & Rules

DEFINITION — VISUAL APPROACH

A **visual approach** is an approach by an **IFR flight** when either part or all of an instrument approach procedure is **not completed** and the approach is executed in **visual reference to terrain**.

WHO CAN INITIATE?

- Clearance for an IFR flight to execute a visual approach **may be requested by the pilot** OR **initiated by the controller**.

CONTROLLER RESTRICTION — WHEN NOT TO INITIATE

The controller **shall NOT initiate** a visual approach when there is a reason to believe that the **flight crew concerned is not familiar with the aerodrome and its surrounding terrain**.

The controller shall also take into consideration **prevailing traffic and meteorological conditions** before initiating a visual approach.

DIRECT CLEARANCE

Aircraft **may be cleared for direct base leg/final** if there is **reasonable assurance that visual approach and landing can be completed**.

SEPARATION DURING VISUAL APPROACH

- **Separation shall be provided** between an aircraft cleared to execute a visual approach and other arriving and departing aircraft.
- When clearance to execute visual approach has been issued, it shall be the **responsibility of the pilot to maintain terrain clearance**.

Visual Approach — By Flight Crew Request

A flight crew **may request visual approach** if the pilot has the runway in sight and can maintain visual reference to terrain — subject to **all of the following conditions**:

Condition	Aircraft Category	Required Visibility / RVR
(a) Ground Visibility *Not below the higher of aerodrome operating minima for associated non-precision approach OR minimum visibility/RVR below:*	**Category A/B**	**2 800 m**
	Category C	**3 200 m**
	Category D	**3 600 m**

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Condition	Aircraft Category	Required Visibility / RVR
Circling Approach Only Runway (visual approach requested)		Ground visibility **not less than 5 km**
(b) Ceiling	Reported ceiling is at or **above the level of the beginning of the initial approach segment**	
(c) Pilot Report	Pilot reports at the level of the beginning of the initial approach segment (or at any time during instrument approach) that meteorological conditions are such that with **reasonable assurance a visual approach and landing can be completed**	

POSITION REPORT REQUIRED

The pilot at the time of requesting visual approach **should give a position report**.

PILOT SHALL ADVISE CONTROLLER IMMEDIATELY WHEN

- Weather has **deteriorated** and unable to keep the terrain in sight; or
- Unable to **continue flight following the preceding aircraft**; or
- **Additional spacing** is required from preceding aircraft.

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Instrument Approach

ATC RESPONSIBILITY

The **approach control unit shall specify the instrument approach procedure** to be used by arriving aircraft.

A flight crew **may request an alternative procedure** and, if circumstances permit, should be cleared accordingly.

Clearance to Fly Maintaining Own Separation While in VMC

DEFINITION — "OWN SEPARATION" CLEARANCE

When so requested by an aircraft and provided it is **agreed by the pilot of the other aircraft** and **authorised by the appropriate ATS authority**, an ATC unit may clear a controlled flight — including departing and arriving flights — operating in **Airspace Classes D and E in VMC** during hours of daylight to fly subject to **maintaining own separation to one other aircraft** and remaining in VMC.

CONDITIONS WHEN A CONTROLLED FLIGHT IS SO CLEARED

a. The clearance shall be for a **specified portion of the flight at or below 3 050 m (10 000 ft)**, during climb or descent and subject to further restrictions as prescribed *(laid down by the authority or the rules)* on the basis of regional air navigation agreements.

b. If there is a possibility that flight under VMC may become impracticable, an IFR flight shall be provided with **alternative instructions** to be complied with in the event that flight in VMC cannot be maintained for the term of the clearance.

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c. The **pilot of an IFR flight**, on observing that conditions are deteriorating and considering that operation in VMC will become impossible, shall:
   - Inform ATC **before entering IMC**
   - Proceed in accordance with the **alternative instructions given**
   - **Essential traffic information** shall be given to controlled flights concerned whenever they constitute essential traffic to each other.

KEY LIMITATIONS AT A GLANCE

- Valid only in **Class D and E airspace**
- Only during **hours of daylight**
- Only at or below **3 050 m / 10 000 ft**
- Only while the aircraft **remains in VMC**
- Requires **pilot request + other pilot's agreement + ATS authority authorization**
- Applicable to **one other aircraft only**

Expected Approach Time (EAT)

DEFINITION — EAT

The **Expected Approach Time (EAT)** is the time at which ATC expects that an arriving aircraft, following a delay, will leave the holding fix to commence its approach for a landing.

Rule	Detail
When EAT is Issued	When an arriving aircraft will be subjected to a delay of **10 minutes or more**
When to Transmit	As soon as practicable and preferably **not later than at the commencement of its initial descent from cruising level**
Revised EAT	Transmitted without delay whenever it differs from previously transmitted EAT by **5 minutes or more**, or such lesser period as agreed between ATS units
Most Expeditious Means	EAT shall be transmitted by the **most expeditious means** whenever the aircraft will be required to hold for **30 minutes or more**
Holding Fix ID	The holding fix to which an EAT relates shall be **identified together with the EAT** whenever circumstances are such that this would not otherwise be evident to the pilot

EAT NUMBERS TO REMEMBER FOR EXAM

- Delay ≥ **10 minutes** → EAT must be issued
- EAT revised if differs by ≥ **5 minutes**
- Hold ≥ **30 minutes** → EAT by most expeditious means

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Onward Clearance Time

DEFINITION — ONWARD CLEARANCE TIME

"*Onward clearance time*" is the **time at which an aircraft can expect to leave the fix at which it is being held**.

WHEN IS IT ISSUED?

In the event an aircraft is held **en route or at a location or aid other than the initial approach fix**, the aircraft concerned shall, as soon as practicable, be given an **expected onward clearance time from the holding fix**. The aircraft shall also be advised if **further holding at subsequent holding fix is expected**.

Information for Arriving Aircraft

A — Information on Establishing Communication

ELEMENTS TRANSMITTED (IN THIS ORDER)

As early as practicable after an aircraft has established communication with the approach control unit, the following shall be transmitted:

a. Type of approach and **runway-in-use**
b. **Meteorological information**
c. Current **runway surface conditions** (in case of precipitants or other temporary hazards)
d. Changes in the **operational status of visual and non-visual aids** essential for approach and landing

If it becomes necessary to use a different runway or instrument approach procedure, the **flight crew shall be advised without delay**.

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B — Information at Commencement of Final Approach

TRANSMITTED AT COMMENCEMENT OF FINAL APPROACH

a. Significant changes in the **mean surface wind direction and speed**
b. Latest information on **wind shear and/or turbulence** in the final approach area
c. The current **visibility representative of the direction of approach and landing**, or when provided, the **current RVR value(s) and the trend**, supplemented by slant visual range if provided

SIGNIFICANT WIND CHANGES (WHEN CONTROLLER HAS COMPONENT FORM)

Component	Significant Change Value
Mean Head-Wind Component	**10 kt**
Mean Tail-Wind Component	**2 kt**
Mean Cross-Wind Component	**5 kt**

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C — Information Transmitted Without Delay During Final Approach

TRANSMITTED WITHOUT DELAY DURING FINAL APPROACH

a. The **sudden occurrence of hazards** (e.g., unauthorized traffic on the runway)

b. Significant variations in the **current surface wind**, expressed in terms of **minimum and maximum values**

c. Significant changes in **runway surface conditions**

d. Changes in the **operational status of required visual or non-visual aids**

e. Changes in **observed RVR values** in accordance with the reported scale in use, or changes in the **visibility representative of the direction of approach and landing**

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CHAPTER 7

Aerodrome Control Service

07

Section 1 · Functions & Organisation

Functions of Aerodrome Control Towers

CORE PURPOSE (REGULATORY STATEMENT)

Aerodrome control towers shall issue **information and clearances** to aircraft under their control to achieve a **safe, orderly and expeditious flow of air traffic** on and in the vicinity of an aerodrome with the object of **preventing collision(s)**.

Collisions to be Prevented — Five Categories

Item	Description
a	Aircraft flying within the **designated area of responsibility** of the control tower, including the **aerodrome traffic circuits**
b	Aircraft operating on the **manoeuvring area**
c	Aircraft **landing and taking off**
d	**Aircraft and vehicles** operating on the manoeuvring area
e	Aircraft on the manoeuvring area and **obstructions on that area**

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CONTINUOUS WATCH OBLIGATION

Aerodrome controllers shall maintain a **continuous watch** on all flight operations on and in the vicinity of an aerodrome as well as vehicles and personnel on the manoeuvring area.

- Watch maintained by **visual observation**
- In low visibility — augmented by an **ATS surveillance system** (when available)
- Traffic controlled in accordance with procedures herein **and all applicable TLIs** (Temporary Local Instructions) of the concerned airport
- If other aerodromes exist within a control zone — **traffic circuits must be coordinated** to avoid conflict

Working Positions within an Aerodrome Control Tower

Position	Primary Responsibility
Aerodrome Controller	Operations on the **runway** and aircraft flying within the area of responsibility of the aerodrome control tower
Ground Controller	Traffic on the **manoeuvring area — excluding runways**

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Position	Primary Responsibility
Clearance Delivery Position	Delivery of **start-up and ATC clearance** to departing **IFR flights**

PARALLEL RUNWAY OPERATIONS

Where **parallel or near-parallel runways** are used for simultaneous operations, individual aerodrome controllers should be responsible for operations on **each of the runways**.

Alerting Service Provided by Aerodrome Control Towers

Aerodrome control towers are responsible for alerting the **rescue and fire fighting services** whenever:

a	An **aircraft accident** has occurred on or in the vicinity of the aerodrome
b	Information is received that the **safety of an aircraft** which is or will come under the jurisdiction *(legal power over a person, place or event)* of the aerodrome control tower **may have or has been impaired**
c	**Requested by the flight crew**
d	When otherwise **deemed necessary or desirable**

Section 2 · Departure Procedures

Procedure for Start-Up & Assignment of Flight Level to Departing Aircraft

Pre-Clearance Obligations on the Pilot

1	**Before** asking for startup or push-back clearance — pilot must ensure the **step ladder / Aerobridge has been removed** and **doors are closed**.
2	Pilot shall intimate **total number of persons on board** (including crew) and **security check completed** to aerodrome control tower when requesting start-up clearance.
3	**Sequence of departure** determined & intimated based on taxiing sequence and sequence at holding point — except where deviation is made to facilitate a **VIP aircraft** or change of order for **traffic reasons**.

REGULATORY DELAYS — KNOW THESE!

- **Delays may be expected** for the second aircraft to push-back when it is parked adjacent to another aircraft being pushed-back.
- Delay in take-off due to **restrictions in the ATC clearance** and over-flights or the appropriate ATS authority.
- Delay when proceeding on the **same track/level**, or climbing through the level, or to climb at a **higher rate of climb** behind a preceding traffic to establish the prescribed *(laid down by the authority or the rules)* separation.

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- A departing aircraft requesting the **same cruising level as an over-flying aircraft** may have to accept an **alternate level** or delay its departure to establish prescribed separation.

Take-Off and Landing

ABSOLUTE RULES — NO EXCEPTIONS

- The **pilot-in-command shall not take-off or land** without a clearance from the appropriate **ATC unit**.
- The pilot-in-command shall **not run-up on the runway in use** unless **authorised by Aerodrome Control**.

ENGINE RUN-UPS — WHERE THEY CAN BE DONE

Engine run-ups in the **holding pan** or **runway holding position** clear of the runway in use may be carried out — subject to **approval by Aerodrome Control**.

Minimum Runway Occupancy Procedures

To increase runway capacity it is essential to **minimise the runway occupancy time**. Follow these procedures:

1	**Cockpit checks should be completed prior to lineup.** Any checks requiring completion on the runway should be kept to the **minimum required**.
2	Pilots must be able to **commence the take-off run immediately** after take-off clearance is issued.
3	Pilots **not able to comply** must **notify ATC prior to commencement of taxi**.
4	Pilots of arriving aircraft are reminded that **rapid exits from the landing runway** enable ATC to apply minimum spacing on final approach — achieving **maximum runway utilisation**.

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INDIAN AIRSPACE-SPECIFIC PROHIBITIONS

- **Cruise Climb is NOT permitted** in Indian Flight Information Regions.
- **Transonic and Supersonic phases of flight are NOT permitted** over Indian airspace.

Section 3 · Control of Aerodrome Traffic

Designated Positions in the Aerodrome Traffic & Taxi Circuits

The positions of aircraft in the traffic and taxi circuits are the positions where the aircraft normally receive aerodrome control tower clearances. Where practicable, **all clearances should be issued without waiting for the aircraft to initiate the call**.

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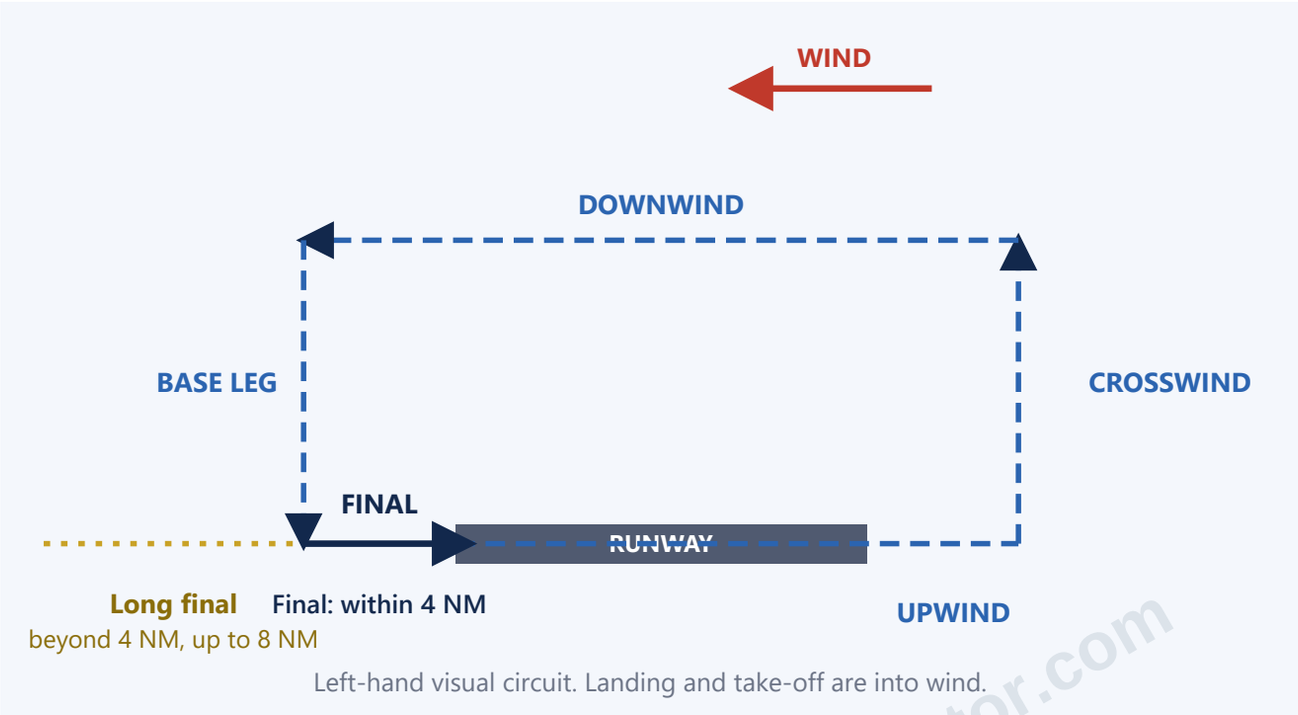

Figure 7.1 *RUNWAY*

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Taxi Clearance

WHAT TAXI CLEARANCES MUST CONTAIN

a. Concise instructions and adequate information to assist the flight crew to follow the **correct taxi routes**

b. To **avoid collision** with other aircraft or objects

c. To minimise the potential for the aircraft **inadvertently entering an active runway**

RUNWAY CROSSING RULE

When a taxi clearance contains a **taxi limit beyond a runway**, it shall contain an **explicit clearance to cross** or an **instruction to hold short** of that runway.

Taxiing on a Runway-in-Use

a	Aircraft may be permitted to taxi on the runway-in-use for the purpose of **expediting air traffic**, provided **no delay or risk** to other aircraft will result.
b	If the control tower is unable to determine (visually or via ATS surveillance) that a vacating/crossing aircraft has cleared the runway — aircraft shall be **requested to report when it has vacated**. Report made when the **entire aircraft is beyond the relevant runway-holding position**.
c	Pilot who requires to **back-track the runway for departure** must notify ATC **prior to commencement of taxi**.

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d	Unless prior permission has been obtained from ATC, the **Pilot-in-Command shall not hold on the runway-in-use**.
e	**Only one aircraft** will be cleared to land on the runway-in-use at any one time — **except formation flight by military aircraft**.

Use of Runway-Holding Positions

HOLDING DISTANCE RULES

a. Except as provided in (b) below, aircraft shall not be held **closer to a runway-in-use than at a runway-holding position**.

b. Aircraft shall **NOT be permitted to line up and hold on the approach end** of a runway-in-use whenever another aircraft is effecting a landing — until the landing aircraft has **passed the point of intended holding**.

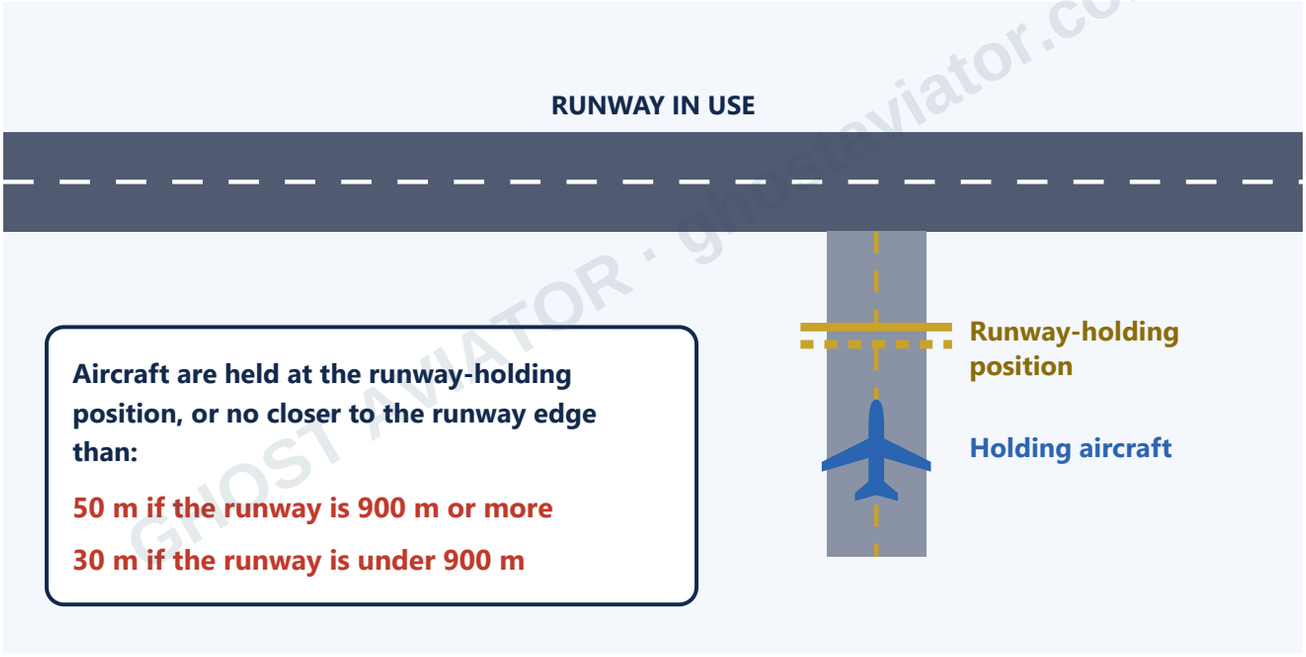

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Figure 7.2 *RUNWAY IN USE*

Runway Length	Minimum Distance from Runway Edge
≥ 900 m	**50 metres** from runway edge
< 900 m	**30 metres** from runway edge

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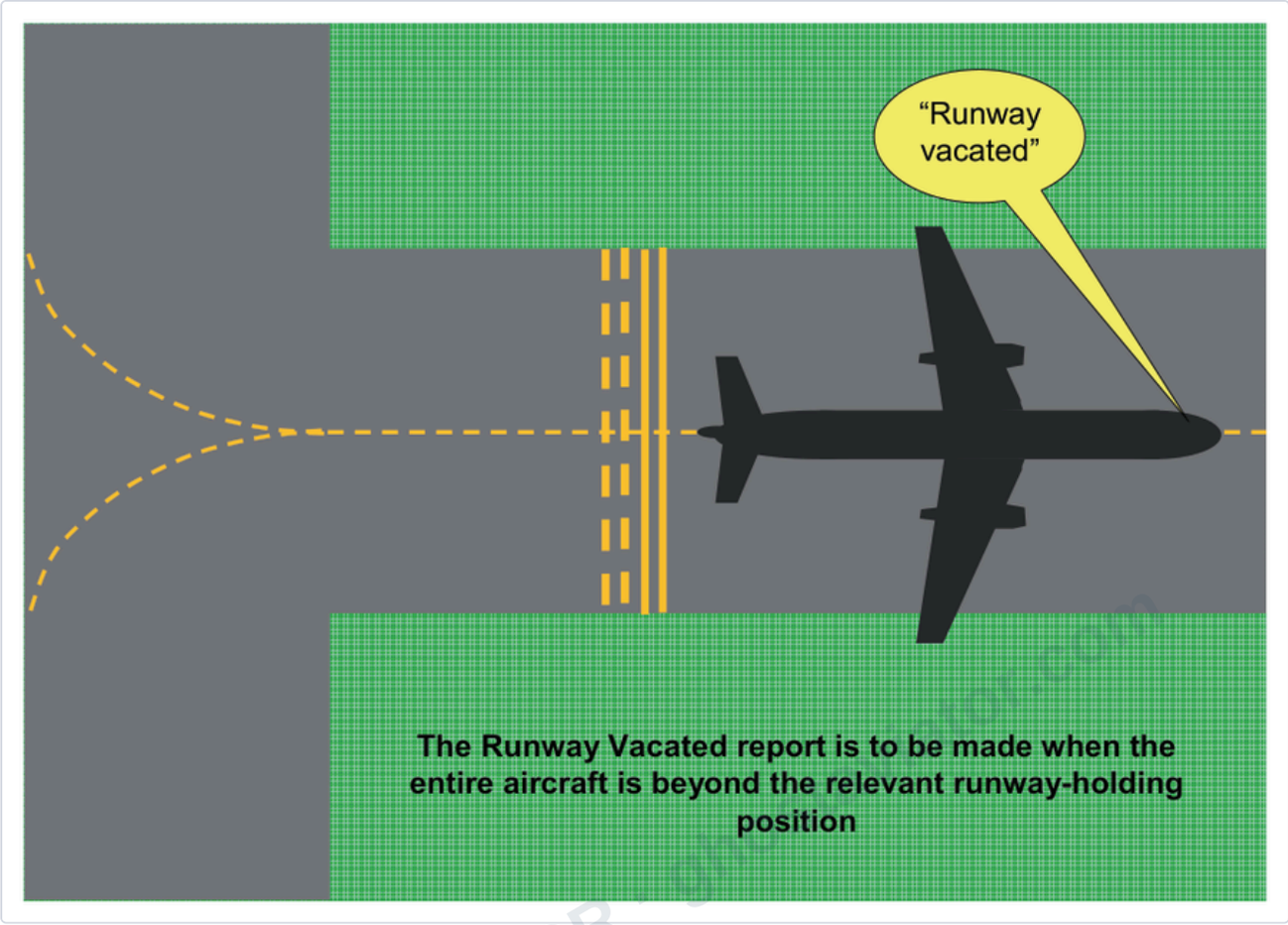

Figure 7.3 *Runway vacated: the report is made when the whole aircraft is beyond the relevant runway-holding position.*

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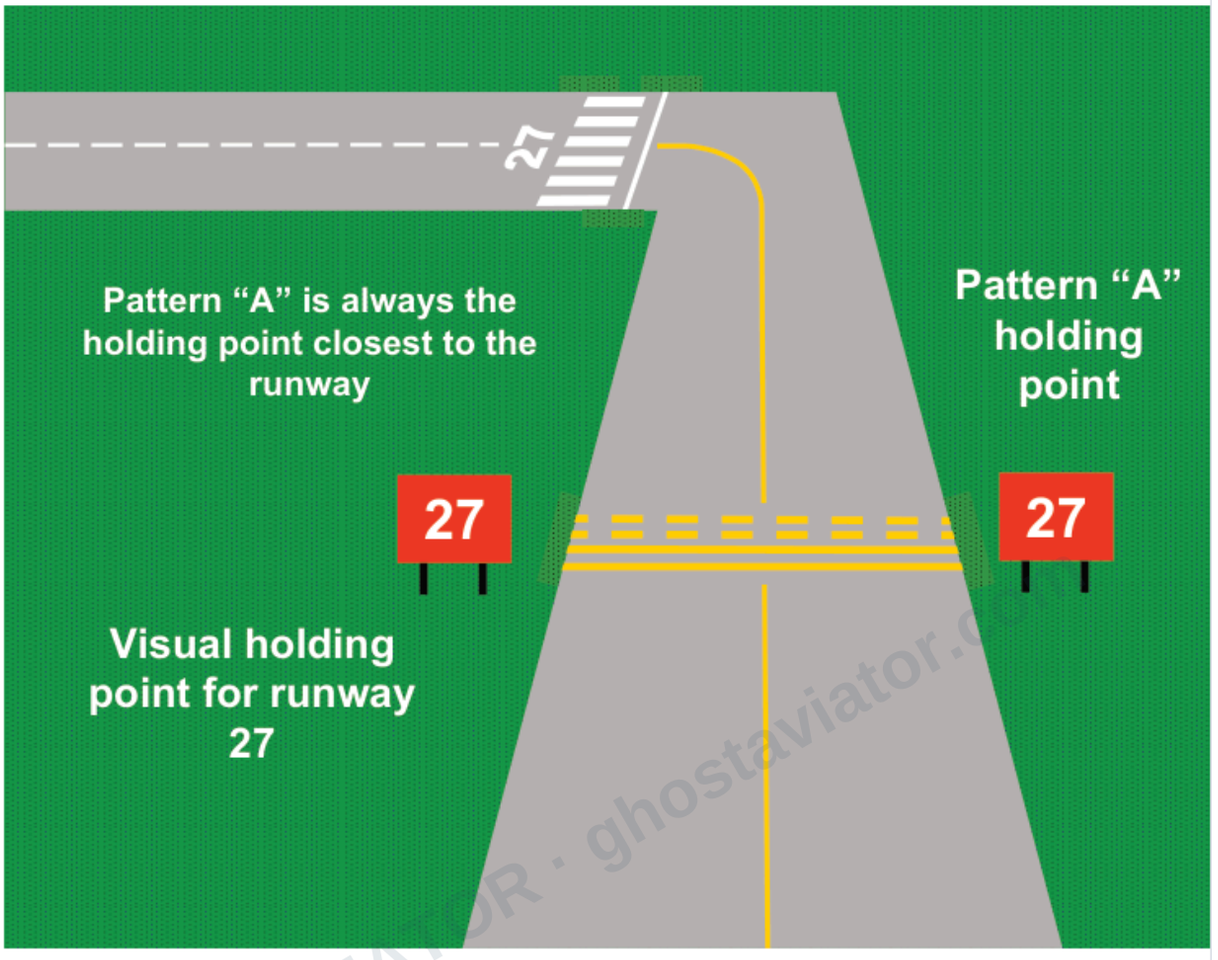

Figure 7.4 *Runway-holding position (pattern A): aircraft hold at the holding position, never closer to the runway-in-use.*

Section 4 · Information to Aircraft by Aerodrome Control Towers

Aerodrome & Meteorological Information (Pre-Taxi)

WHEN IS THIS INFORMATION PROVIDED?

Prior to **taxiing for take-off**, aircraft shall be advised of the following elements of information — in the order listed — with the exception of elements already known to the aircraft.

#	Element of Information
a	The **runway to be used**
b	The **surface wind direction and speed**, including significant variations
c	The **QNH altimeter setting** and, either on a regular basis or if requested — the **QFE** altimeter setting
d	The **air temperature for the runway** — in the case of **turbine-engine aircraft**
e	The **visibility** representative of the direction of take-off and initial climb, if less than **10 km**, or — when applicable — the **RVR value(s)** for the runway to be used

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#	Element of Information
f	The **correct time**

Information Prior to Take-Off

Prior to take-off, aircraft shall be advised of:

a	Any **significant changes** in the surface wind direction and speed, the air temperature, and the visibility or RVR value(s)
b	**Significant meteorological conditions** in the take-off and climb-out area — except when it is known the aircraft has already received the information

NOTE — SIGNIFICANT MET CONDITIONS INCLUDE
Cumulonimbus or thunderstorm, moderate or severe turbulence, wind shear, hail, moderate or severe icing, severe squall line, freezing precipitation, severe mountain waves, sand storm, dust storm, blowing snow, tornado or waterspout — in the take-off and climb-out area.

Pre-Traffic-Circuit / Approach Information

Prior to entering the traffic circuit or commencing approach to land, aircraft shall be provided:

#	Element
a	Runway to be used
b	Surface wind direction and speed, including significant variations
c	QNH altimeter setting and, if requested, QFE altimeter setting

Runway Incursion or Obstructed Runway

If the aerodrome controller observes — after a take-off or landing clearance has been issued — any obstruction on the runway likely to impair safety (runway incursion by aircraft/vehicle, animals, birds), the following actions shall be taken:

a	**In ALL cases:** Inform the aircraft concerned of the obstruction and its **location on the runway**
b	**Cancel the take-off clearance** for an aircraft which has **not started to roll**
c	**Instruct a landing aircraft to go around**

ESSENTIAL LOCAL TRAFFIC INFORMATION
Essential local traffic shall be considered to consist of any **aircraft, vehicle or personnel on or near the manoeuvring area** or traffic operating in the vicinity of the aerodrome, which **may constitute a hazard** to the aircraft concerned.

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Uncertainty of Position on the Manoeuvring Area

PILOT ACTION — POSITION UNCERTAINTY (NOT ON RUNWAY)

A pilot in doubt as to the position of the aircraft with respect to the manoeuvring area shall immediately:

a. **Stop the aircraft**

b. Simultaneously **notify the appropriate ATS unit** of the circumstances (including the last known position)

PILOT ACTION — RECOGNISES AIRCRAFT IS ON A RUNWAY

If a pilot recognises the aircraft is on a runway (while uncertain of position), the pilot shall immediately:

a. **Notify the appropriate ATS unit** of circumstances (including last known position)

b. If able to locate a nearby suitable taxiway — **vacate the runway as expeditiously as possible**, unless otherwise instructed by the ATS unit

c. **Stop the aircraft**

VEHICLE DRIVER — POSITION UNCERTAINTY

A vehicle driver in doubt as to position on the manoeuvring area shall immediately:

a. **Notify the appropriate ATS unit** of circumstances (including last known position)

b. Unless otherwise instructed — **vacate the landing area, taxiway, or other part of the manoeuvring area** to a safe distance as expeditiously as possible

c. **Stop the vehicle**

CONTROLLER ACTION — LOST/UNCERTAIN AIRCRAFT OR VEHICLE

In the event the aerodrome controller becomes aware of an aircraft or vehicle that is lost or uncertain of its position, appropriate action shall be taken **immediately** to safeguard operations and assist the aircraft or vehicle to determine its position.

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Wake Turbulence and Jet Blast Hazard

CONTROLLER OBLIGATIONS

- Aerodrome controllers shall, when applicable, **apply the wake turbulence separation minima specified**.
- Whenever the responsibility for wake turbulence avoidance rests with the pilot-in-command — aerodrome controllers shall, to the extent practicable, **advise aircraft of the expected occurrence of hazards** caused by turbulent wake.

Condition of the Movement Area and Related Facilities

Information on the condition of the movement area and operational status of related facilities shall be provided to arriving and departing aircraft particularly in respect of:

#	Condition Type
a	Construction or maintenance work on, or immediately adjacent to the movement area
b	Rough or broken surfaces on a runway, taxiway or apron — whether marked or not
c	Water, snow, slush, ice or frost on a runway, taxiway or apron

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#	Condition Type
d	Anti-icing or de-icing liquid chemicals or other contaminant on a runway, taxiway or apron
e	Snow banks or drifts adjacent to a runway, taxiway or apron
f	Other temporary hazards — including parked aircraft
g	Failure or irregular operation of part or all of the aerodrome visual aids
h	Failure of normal or secondary power supply

Section 5 · Priority & Special Traffic

Control of Other Than Aircraft Traffic

Entry to the Manoeuvring Area

AUTHORITY

The movement of persons or vehicles (including towed aircraft) on the manoeuvring area of an aerodrome shall be **controlled by the aerodrome control tower** as necessary to avoid hazard to them or to aircraft landing, taxiing or taking off.

LOW VISIBILITY PROCEDURES — RESTRICTIONS ON MANOEUVRING AREA

a. Persons and vehicles operating on the manoeuvring area shall be **restricted to the essential minimum**, and particular regard shall be given to the requirements to **protect the ILS sensitive area(s)** when Category II or Category IIIA precision instrument operations are in progress.

b. Vehicles shall remain at **safe distance from taxiing aircraft**.

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Priority on the Manoeuvring Area

EMERGENCY PRIORITY

Emergency vehicles proceeding to the assistance of an aircraft in distress shall be afforded **priority over all other surface movement traffic**.

LIGHT SIGNAL — EMERGENCY / NO RADIO CONDITIONS

In emergency conditions or if radio signals are not observed, the signal given hereunder shall be used for runways or taxiways equipped with a lighting system:

Flashing runway lights → Meaning: **Vacate the runway or taxiway and observe the tower for light signal**

Priority Rules for Aircraft Operations

The following general rules determine the **order of priority of aircraft operations**:

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1	**Emergency Landing (Compelling)** Aircraft known/believed to be in emergency compelling it to land without delay — e.g. engine failure, acute fuel shortage, total communication failure
2	**Military Fighter Flights** Invariably experience fuel shortage if not per planned ops. Ground time limitations — departures accommodated without delay
3	**Live Scramble** An aircraft engaged in a live scramble
4	**Medical / MEDEVAC** Aircraft carrying sick/injured persons requiring urgent medical attention or carrying out approved medical evacuation
5	**VVIP Aircraft** Aircraft carrying VVIP as laid down in procedures for their handling
6	**Search & Rescue** Aircraft proceeding on Search and Rescue missions
7	**Military Large Formation** Military flight carrying out large integrated formations in military exercises
8	**Urgent Military Ops (RPA)** Aircraft/RPA in approved urgent operational military commitment coordinated with appropriate ATC Centres
9	**Large Scale Weather Deviations** Aircraft on large scale weather deviations
10	**Adverse Condition Diversions** Aircraft landing at alternate aerodromes after diversion due adverse operational or weather conditions at intended destination

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Suspension of Visual Flight Rules Operations

WHO CAN SUSPEND VFR?

Any or all VFR operations on and in the vicinity of an aerodrome may be suspended by any of the following — whenever safety requires such action:

a. The **approach control unit or the appropriate ACC**

b. The **aerodrome control tower**

c. The **appropriate ATS authority**

All such suspensions of VFR operations shall be accomplished through or notified to the **aerodrome control tower**.

Procedures When VFR Operations Are Suspended

The following procedures shall be observed by the aerodrome control tower:

a	**Hold all VFR departures**

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b	**Recall all local flights** operating under VFR or obtain approval for **special VFR operations**
c	**Notify the approach control unit or ACC** as appropriate of the action taken
d	**Notify all operators** or their designated representatives of the reason for taking such action — if necessary or requested

Section 6 · Traffic Circuit & Clearances

Control of Traffic in the Traffic Circuit

Entry of Traffic Circuit

WHEN IS ENTRY CLEARANCE ISSUED?

The clearance to enter the traffic circuit should be issued to an aircraft whenever it is desired that the aircraft approach the landing area in accordance with current traffic circuits but traffic conditions do **not yet allow a landing clearance to be issued**.

Depending on circumstances and traffic conditions, an aircraft may be cleared to **join at any position** in the traffic circuit.

An arriving aircraft executing an **instrument approach** shall normally be cleared to **land straight in** unless visual manoeuvring to the landing runway is required.

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Order of Priority for Arriving and Departing Aircraft

FUNDAMENTAL PRIORITY RULE

An aircraft **landing or in the final stages of an approach to land** shall normally have **priority** over an aircraft intending to depart from the same or an intersecting runway.

Control of Departing Aircraft

Departure Sequence

DEPARTURE SEQUENCING RULE

Departures shall normally be cleared in the **order in which they are ready for take-off**, except that deviations may be made from this order to **facilitate the maximum number of departures with the least average delay**.

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Separation of Departing Aircraft

WHEN MAY NEXT DEPARTURE COMMENCE?

A departing aircraft will **not normally** be permitted to commence take-off until:

- The **preceding departing aircraft has crossed the end of the runway-in-use**, OR
- The preceding departing aircraft has **started a turn**, OR
- All **preceding landing aircraft are clear of the runway-in-use**

Take-Off Clearance

WHEN CAN TAKE-OFF CLEARANCE BE ISSUED?

Take-off clearance may be issued to an aircraft when there is **reasonable assurance that the separation will exist** when the aircraft commences take-off.

IMMEDIATE TAKE-OFF CLEARANCE

In the interest of expediting traffic, a clearance for **immediate take-off** may be issued to an aircraft **before it enters the runway**. On acceptance of such clearance — the aircraft shall **taxi out to the runway and take off in one continuous movement**.

TAKE-OFF CLEARANCE — RUNWAY DESIGNATOR

To reduce the potential for misunderstanding — the take-off clearance **shall include the designator of the departure runway**.

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Control of Arriving Aircraft

Separation on the Same Runway

WHEN MAY LANDING AIRCRAFT CROSS THRESHOLD?

A landing aircraft will **not normally be permitted to cross the runway threshold** on its final approach until:

- The **preceding departing aircraft has crossed the end of the runway-in-use**, OR
- Has **started a turn**, OR
- All **preceding landing aircraft are clear of the runway-in-use**

Clearance to Land

WHEN IS LANDING CLEARANCE ISSUED?

- An aircraft may be cleared to land when there is **reasonable assurance that the separation will exist** when the aircraft crosses the runway threshold.
- A clearance to land shall **not be issued** until a preceding landing aircraft has **crossed the runway threshold**.
- The landing clearance shall include the **designator of the landing runway** — to reduce the potential for misunderstanding.

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Procedures for Low Visibility Operations

CONTROL OF SURFACE TRAFFIC IN LOW VISIBILITY

In conditions where **low visibility procedures are in operation**:

- Persons and vehicles operating on the manoeuvring area shall be **restricted to the essential minimum**
- Particular regard shall be given to protecting the **ILS sensitive area(s)** when **Category II or Category IIIA** precision instrument operations are in progress

Aeronautical Ground Lights

WHAT IS RUNWAY LIGHTING?

Runway lighting includes: **edge lights, threshold lights, centre line lights, end lights, touchdown zone lights, and wing bar lights**.

When Aeronautical Ground Lights Shall Be Operated

#	Condition
a	During the time from **sunset to sunrise**
b	From sunrise to sunset when **visibility is 5,000 m or less in India**
c	When **requested by pilot**
d	At any other time when their use — based on meteorological conditions — is considered **desirable for the safety of air traffic**

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LIGHTS MAY BE TURNED OFF — WHEN?

Lights on and in the vicinity of aerodromes not intended for en-route navigation purposes may be turned off, if:

- No likelihood of either **regular or emergency operation** exists, AND
- They can be **again brought into operation at least one hour before** the expected arrival of an aircraft

Variable Intensity Lights

At aerodromes equipped with lights of variable intensity — a **table of intensity settings** based on conditions of visibility and ambient light should be provided for ATC guidance. When requested by an aircraft, further adjustment of intensity **shall be made whenever possible**.

VISUAL APPROACH SLOPE INDICATOR (VASI) LIGHTS

The lights of a visual approach slope indicator system shall be operated during the **hours of daylight as well as of darkness and irrespective of the visibility conditions** when the associated runway is being used.

Runway Lighting Rules

RUNWAY LIGHTING SHALL NOT BE OPERATED UNLESS

Runway lighting shall not be operated if that runway is **not in use for landing, take-off or taxiing purposes**, unless required for **runway inspections or maintenance**.

Lighting After Take-Off — Not Operated Continuously

Aerodrome Type	Rule
With ATC service + centrally controlled lights	Lights of **one runway** shall remain lighted after take-off as long as considered necessary for the **return of the aircraft due to an emergency occurring during or immediately after take-off**
Without ATC service OR without centrally controlled lights	Lights of **one runway** shall remain lighted until such time as would normally be required to reactivate the lights in likelihood of the departing aircraft returning for an emergency landing, and in any case **not less than fifteen minutes after take-off**

Wind Shear Advisories

DEFINITION OF WIND SHEAR

Wind shear is a **sustained change in the wind velocity along the aircraft flight path**, which occurs significantly faster than the aircraft can accelerate or decelerate. It can occur at any level, but it is **'low level wind shear'** occurring from the surface to a height of approximately **1,500 feet**, which can cause problems of sufficient magnitude to affect the control of aircraft in **departure or final approach phases of flight**.

WIND SHEAR REPORTING PROCEDURE

- Whenever a pilot reports wind shear conditions to ATC — the information shall be **passed to subsequent arriving and departing aircraft** until either:
  - Confirmation is received that the condition **no longer exists**, OR
  - Wind shear information has been **included in ATIS** and the flight crew reports receipt of the appropriate ATIS designator
- Reports on wind shear from aircraft should be **passed to the Meteorological office**.

Designation of Hot Spot

HOT SPOT

The aerodrome operator shall whenever necessary designate a **location or several locations on the movement area** of the aerodrome as **hot spot(s)**. The hot spot(s) shall be charted in accordance with **Annex 4**.

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Coefficient of Friction

CONTEXT

The table below was developed from friction data collected only in **compacted snow and ice** — and should not be taken as absolute values applicable in all conditions. Surface condition information (including estimated surface friction) must be provided for **each third of a runway**. The "first part" always means the **first third of the runway as seen in the direction of landing**.

Measured Coefficient (μ)	Estimated Surface Friction	Code
0.40 and above	GOOD	**5**
0.39 to 0.36	MEDIUM TO GOOD	**4**
0.35 to 0.30	MEDIUM	**3**
0.29 to 0.26	MEDIUM TO POOR	**2**
0.25 and below	POOR	**1**

KEY EXAM POINT — RUNWAY SEGMENTS

The runway is divided into **thirds (1/3 segments)** for reporting surface condition. When giving landing information, sections are referred to as **first, second, or third part of the runway**. The **first part always means the first third of the runway as seen in the direction of landing**.

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CHAPTER 8

Air Traffic Surveillance Systems

08

Introduction — ATS Surveillance Systems

WHAT IS AN ATS SURVEILLANCE SYSTEM?

ATS Surveillance System is a *generic term* covering any ground-based system that enables the identification of aircraft to provide Air Traffic Services. It encompasses:

- **ADS-B** — Automatic Dependent Surveillance – Broadcast
- **PSR** — Primary Surveillance Radar
- **SSR** — Secondary Surveillance Radar
- **MLAT** — Multilateration
- Any comparable **ground-based** system enabling aircraft identification

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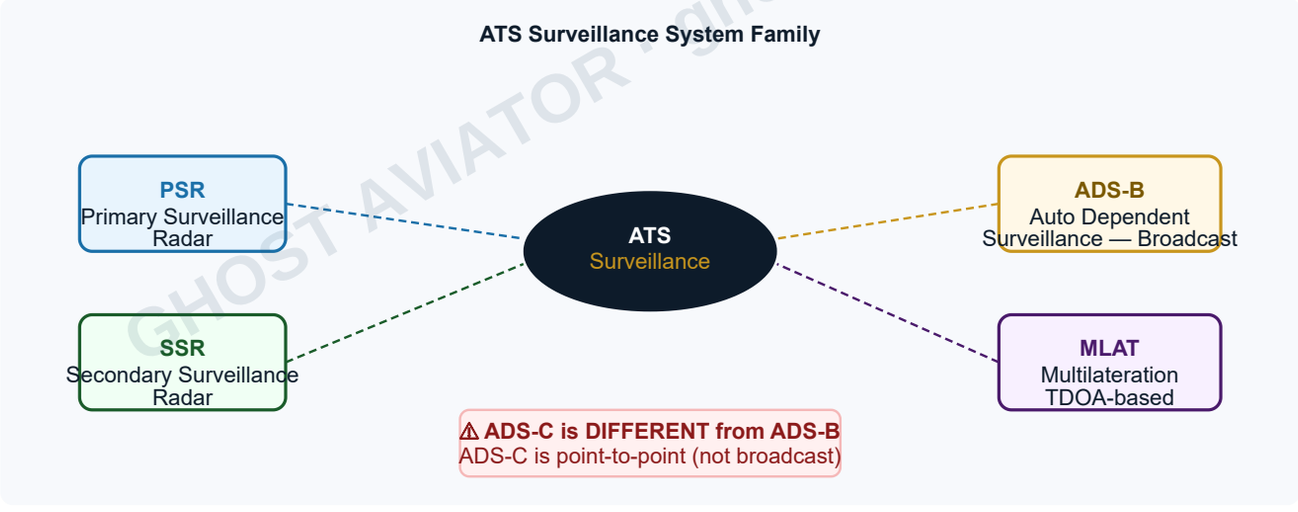

Figure 8.1 *ATS Surveillance System Family*

When is PSR Used?

PSR USAGE RULE

PSR systems should be used in circumstances where *other* ATS surveillance systems **alone** would **not meet** the air traffic services requirements.

PSR is the supplementary backstop — not the primary tool in modern ATC.

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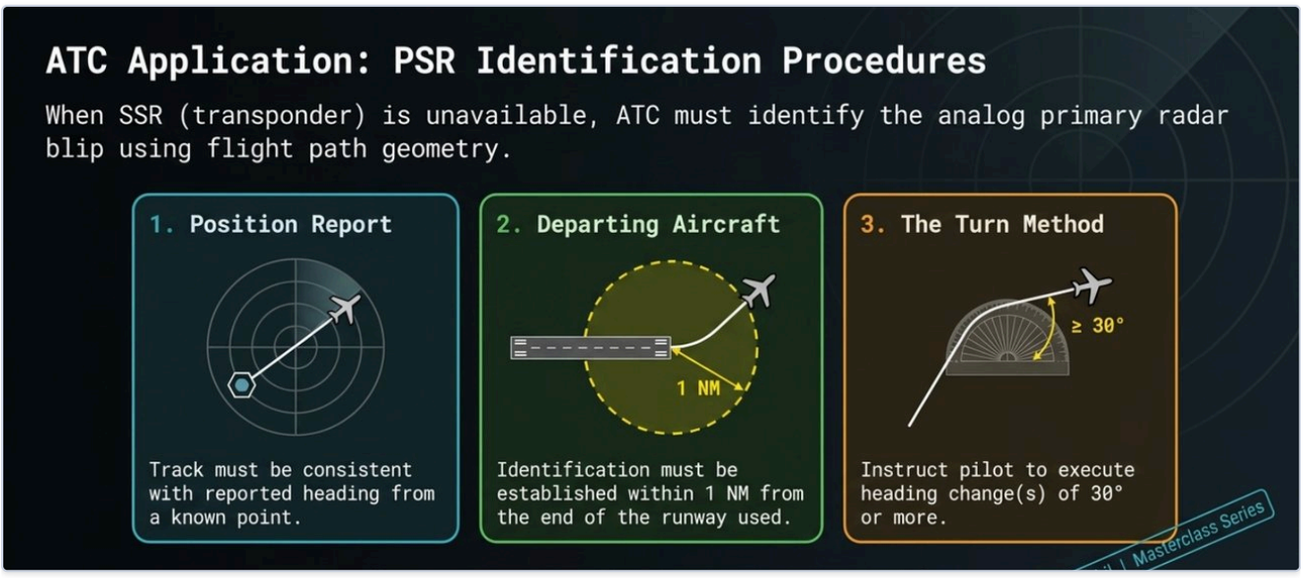

Figure 8.2 *Identifying an aircraft with primary radar*

When Can SSR Be Used Alone?

SSR ALONE — CONDITIONS FOR SEPARATION

SSR systems (especially those with **monopulse techniques** or **Mode S capability**), or **MLAT**, may be used alone — including for **separation between aircraft** — provided *both* conditions are met:

- The carriage of SSR transponders is **mandatory** within the area
- Identification is **established and maintained**

ADS-B — Automatic Dependent Surveillance – Broadcast

DEFINITION — ADS-B

ADS-B, like PSR and SSR, is an **ATS surveillance system**. It allows ATC to **automatically and repeatedly access data** from all suitably equipped aircraft and both *use* and *re-broadcast* it to suitably equipped other aircraft within range.

KEY POINTS — ADS-B

- **Dependent:** Relies on the aircraft's own navigation systems (GPS/GNSS)
- **Automatic:** No pilot input required — broadcasts continuously
- **Surveillance:** Provides traffic picture to ATC and other aircraft
- **Broadcast:** Signal goes to ALL receivers in range (unlike ADS-C which is point-to-point)
- Moves from **ground radar** and navigational aids to **precise satellite-based tracking**
- Forms the **foundation for NextGen** (Next Generation Air Transportation System)

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ADS-C — Automatic Dependent Surveillance – Contract

DEFINITION — ADS-C

ADS-C uses the **same on-board systems** as ADS-B to transmit similar information (position, altitude, speed, navigational intent, meteorological data) — but *only* to **one or more specific** Air Traffic Services Units (ATSU) or Aeronautical Operational Control (AOC) facilities — for **surveillance and/or route conformance monitoring.**

ADS-B (Broadcast)	• Signal goes to **ALL** receivers within range • Used by ATC *and* other aircraft (TCAS/traffic display) • No specific "contract" required • Suitable for busy, high-density airspace • Ground stations re-broadcast to other aircraft
ADS-C (Contract)	• Signal goes to **specific** ATSUs / AOC only • Based on an **ADS contract** between aircraft & ground • Contract defines *type of data* and *reporting conditions* • Used for oceanic / remote area surveillance • Aircraft can also send *unsolicited* emergency reports

CRITICAL EXAM DISTINCTION

Although the names are similar, **ADS-C and ADS-B are two different applications.**

Data provision under ADS-C is *only* generated in response to a **request within the terms of an ADS contract** held by the ground system. The contract identifies the types of information and the conditions under which reports are sent.

Some information types are included in *every* report; others are provided *only if specified* in the contract.

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Advantages of ADS-B

ADVANTAGES — ADS-B (THE FULL LIST)

- **Environmentally friendly technology** — enhances safety and efficiency
- **Pilots see what controllers see:** displays showing other aircraft in the sky (first time in aviation history)
- **Cockpit displays** also pinpoint **hazardous weather and terrain**, and give important flight information such as temporary flight restrictions
- **Reduces runway incursions:** cockpit and controller displays show the location of aircraft *and equipped ground vehicles* on airport surfaces — even at night or in heavy rainfall
- **Collision alerts:** ADS-B applications give pilots indications or alerts of potential collisions
- **Greater coverage:** ground stations are much easier to place than radar — remote areas without radar coverage now have surveillance with ADS-B
- **Satellite-based routing:** aircraft fly more directly from Point A to Point B, saving time, money, and reducing fuel burn and emissions

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- **Reduces separation minima:** improved accuracy of satellite signals over radar allows controllers to eventually reduce minimum separation and increase airspace capacity
- **Foundation for NextGen:** moves from ground radar and navigational aids to precise tracking using satellite signals

MLAT — Multilateration Applications

DEFINITION — MLAT

Multilateration (MLAT) is a technology to accurately locate aircraft using **Time Difference of Arrival (TDOA)**. It employs a number of **ground stations** placed in strategic locations around an airport.

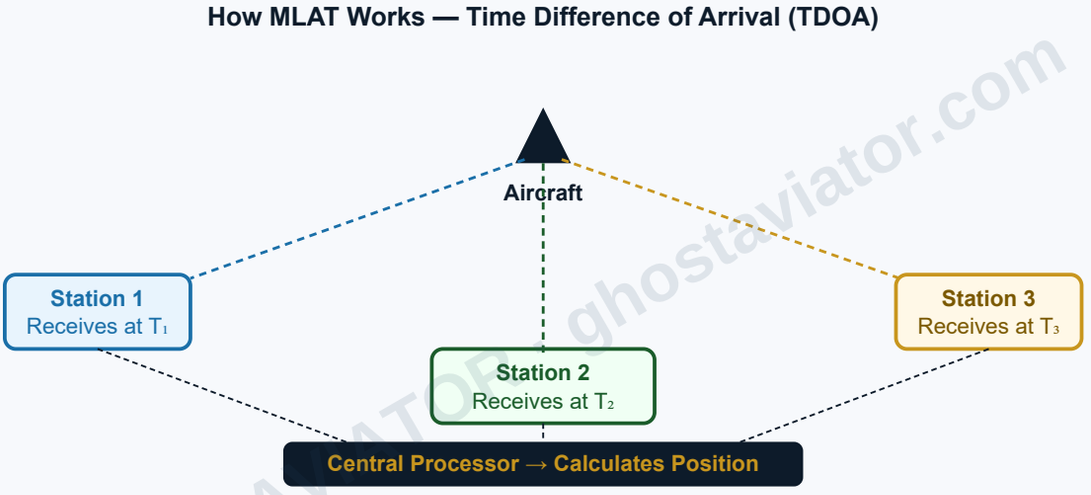

Figure 8.3 *How MLAT Works — Time Difference of Arrival (TDOA)*

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MLAT KEY POINTS

- **No additional avionics required** — uses replies from **Mode A, C and S transponders**, as well as **military IFF** and **ADS-B transponders**
- Individual aircraft will be at **different distances** from each ground station — their replies are received at **fractionally different times**
- Advanced computer processing techniques use these individual time differences to **precisely calculate** aircraft position
- MLAT is particularly useful in **surface movement** (airport ground) and **terminal area** applications

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Radar Applications

RADAR VECTORS — HEADINGS

Radar vectors given to an aircraft will be related to **magnetic headings only.**

Radar control procedures will be used by ATC *in preference* to non-radar control procedures whenever ATS or the aircraft served will gain **operational advantage.**

Types of Radar Services

(i) Radar Control Service	For aircraft operating within **controlled airspaces**
(ii) Radar Advisory Service	For aircraft operating within **Class F airspace**
(iii) Radar Flight Information Service	For **identified** aircraft operating in **any part of FIR**

TWO-WAY COMMUNICATION REQUIREMENT

The provision of any of the above types of radar service requires that aircraft remain in **direct two-way communication** with the unit providing the service.

Exception: Radar separation may be provided between two radar-identified aircraft even when *only one* of the aircraft is in direct communication with the radar unit.

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Radar in Approach Control Service

APPROACH CONTROL — RADAR USES

Surveillance Radar is extensively used in approach control to:

- Sequence the traffic
- Vector aircraft to position for pilot-interpreted approach aids
- Conduct **surveillance radar approaches**

Radar in Aerodrome Control Service

AERODROME CONTROL — RADAR USES

- Surveillance radar is used as an **approach monitor aid**
- **Surface Movement Radar (SMR)** is used to control traffic on the ground in the movement area

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Position Indication Display Formats

HOW POSITION MAY BE DISPLAYED

Position indications may be displayed as:

- **(a)** Individual position symbols — e.g. PSR, SSR, ADS-B or MLAT symbols, or combined symbols
- **(b)** PSR blips
- **(c)** SSR responses

Use of ATS Surveillance Service in Air Traffic Control Service

PRIMARY PURPOSES OF ATS SURVEILLANCE SERVICE

Provide radar service to:

- (i) Improve airspace utilisation | (ii) Reduce delays | (iii) Enhance safety

Provide radar vectoring to:

- (i) Departing aircraft for expeditious and efficient departure flow and expediting climb to cruising level
- (ii) Arriving aircraft for expediting descent from cruising level and establishing an expeditious and efficient approach sequence
- (iii) Aircraft for the purpose of resolving potential conflict
- (iv) Assist pilot in their navigation

Provide separation and maintain normal flow when an aircraft experiencing **communication failure** is within area of ATS surveillance service coverage.

Monitor the progress of air traffic in order to:

- (i) Obtain improved position information regarding aircraft under control
- (ii) Obtain supplementary information regarding other traffic
- (iii) Detect significant deviations by aircraft from their assigned routings or level

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Use of SSR Without Primary Radar

SSR ALONE — SEPARATION PROVISION

SSR information may be used **alone** in the provision of **separation between aircraft** provided:

- Aircraft identification is **established and maintained** by use of **discrete SSR codes**

NON-TRANSPONDER AIRCRAFT — NON-RADAR SEPARATION

Non-radar separation will be applied between:

- A transponder-equipped aircraft, AND
- An aircraft *without* SSR transponder, OR an aircraft with a **non-functioning** SSR transponder

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TRANSPONDER FAILURE PROCEDURES

1. **Failure detected during flight:** The aircraft (for whom carriage of transponder is mandatory) will *normally be permitted to continue to operate to the next point of landing.*
2. **Failure detected before departure:** The aircraft may be specifically *authorised by ATC* to operate without a serviceable transponder — provided a **request is included in the flight plan.**

ATS Surveillance Service Identification Procedures

MANDATORY RULE — IDENTIFICATION BEFORE SERVICE

Before providing ATS surveillance service to an aircraft, **radar identification shall be established** by one of the methods in *Doc 4444-PANS-ATM*, and the **pilot so informed.**

If identification is subsequently *lost*, the pilot shall be **informed accordingly** and instructions will be issued so as to restore **non-ATS surveillance service separation.**

ADS-B Identification Procedures

Where ADS-B is used for identification, aircraft may be identified by:

ADS-B — IDENTIFICATION METHODS (3 METHODS)

- **(a)** Direct recognition of the aircraft identification in an **ADS-B label**
- **(b)** Transfer of ADS-B identification
- **(c)** Observation of compliance with an instruction to **TRANSMIT ADS-B IDENT**

SSR and/or MLAT Identification Procedures

Where SSR and/or MLAT is used for identification, aircraft may be identified by:

SSR/MLAT — IDENTIFICATION METHODS (6 METHODS)

- **(a)** Recognition of the aircraft identification in a SSR and/or MLAT label
- **(b)** Recognition of an **assigned discrete code**, the setting of which has been verified, in a SSR and/or MLAT label
- **(c)** Direct recognition of the aircraft identification of a **Mode S-equipped** aircraft in a SSR and/or MLAT label
- **(d)** By **transfer** of identification
- **(e)** Observation of compliance with an instruction to **set a specific code**
- **(f)** Observation of compliance with an instruction to **squawk IDENT**

PSR Identification Procedures

When SSR is not available, PSR identification may be achieved by the following methods:

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Method 1 — Position Report Method

POSITION REPORT METHOD

By **correlating** a particular radar position indication with an aircraft reporting:

- Its position **over** or as **bearing and distance from** a point shown on the situation display; **AND**
- By **ascertaining** that the track of the particular radar position is consistent with the aircraft path or reported heading

Method 2 — Departing Aircraft Method

DEPARTING AIRCRAFT METHOD

By correlating an observed radar position indication with an aircraft **known to have just departed**, provided that *(an exception or condition added to a rule)*:

- Identification is established **within 1 NM from the end of the runway used**
- Particular care should be taken to avoid confusion with aircraft:
  - Holding over or overflying the aerodrome
  - Departing from or making a missed approach over *adjacent runways*

Method 3 — The Turn Method

THE TURN METHOD

(a) An aircraft may be identified by ascertaining the aircraft heading, if circumstances require, and following a period of track observation by:

- ▶ Instructing the pilot to execute **one or more changes of heading of 30 degrees or more** and correlating the movements of one particular radar position indication with the aircraft's acknowledged execution of the instructions given; **OR**
- ▶ Correlating the movements of a particular radar position indication with manoeuvres currently executed by an aircraft having so reported

(b) When using these methods, the radar controller shall:

- (i) Verify that the movements of **not more than one** radar position indication correspond with those of the aircraft
- (ii) Ensure that the manoeuvre(s) will **not carry the aircraft outside** the coverage of the radar or the situation display

PSR ID Method	Key Requirement	Critical Limit
Position Report	Aircraft reports position over / bearing & distance from a known point	Track must be consistent with reported heading/path
Departing Aircraft	Aircraft known to have just departed a runway	Within 1 NM from end of runway used
Turn Method	Aircraft executes an instructed heading change	30° or more heading change

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Separation Minima

SEPARATION MINIMA — CORE RULE

The separation minima specified shall **only be applied** between *identified aircraft* when there is **reasonable assurance that identification will be maintained.**

When is Procedural (Non-Radar) Separation Applied?

PROCEDURAL SEPARATION MINIMA WILL BE APPLIED IN

- **(i)** In the event of **ATS surveillance service failure**
- **(ii)** In the area **outside** the ATS surveillance service coverage
- **(iii)** To aircraft **leaving** ATS surveillance service coverage or entering an **adjacent FIR** — except where ATS surveillance service transfer is effected

ICAO Separation Minima — Radar / ADS-B / MLAT

Horizontal Separation Minima — Visual Scale

Condition	Separation Minimum	Authority	Notes
Standard (Default) — Radar / ADS-B / MLAT	9.3 km (5.0 NM)	ICAO	Unless otherwise prescribed *(laid down by the authority or the rules)*
Reduced — where capabilities at a given location so permit	5.6 km (3.0 NM)	ICAO	May be reduced but *not below* this minimum (unless category (b) applies)
Further Reduced — Succeeding aircraft on same final approach track within 18.5 km (10 NM) of runway end threshold	4.6 km (2.5 NM)	ICAO	Provided local factors permit such reduction

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INDIAN SURVEILLANCE MINIMUM AND THE CONDITIONAL ICAO REDUCTION

AAI ENR 1.6 §7.5.1: "Five nautical miles (5 NM) within airspace under surveillance coverage." Section 7.5.2 specifies the authorised airspace in which 3 NM applies. Apply the location, level, coverage and equipment conditions published for that airspace.

ICAO's 2.5 NM provision is conditional. Doc 4444 §8.7.3.2(b) concerns succeeding aircraft established on the same final approach track within 10 NM of the runway end. It requires prescription by the appropriate ATS authority and all the following conditions:

1. Average landing runway-occupancy time is demonstrated not to exceed 50 seconds.
2. Braking action is good; contamination does not adversely affect occupancy time.
3. Radar has suitable resolution and an update rate of 5 seconds or less, with suitable displays.
4. The aerodrome controller can observe the runway and associated entry/exit taxiways visually or through the specified surface systems.
5. The applicable distance-based wake-turbulence restrictions do not apply.
6. The controller closely monitors and, when necessary, adjusts approach speeds to protect the minimum.

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7. Operators and pilots know the need to vacate the runway promptly.
8. The reduced-minimum procedures are published in the AIP.

Being established on the same localizer alone does not meet all these conditions. Use the applicable published Indian procedure; 2.5 NM is not the general minimum stated in AAI ENR 1.6 §7.5.

Sources: AAI eAIP ENR 1.6 §§7.5.1–7.5.2; ICAO Doc 4444 §8.7.3.2(b)(i)–(viii).

CRITICAL EXAM VALUES — MEMORISE THESE

Minimum	NM	km	Condition
Standard Horizontal	5.0 NM	9.3 km	Default — Radar / ADS-B / MLAT
Reduced	3.0 NM	5.6 km	Where ATS authority permits; local capabilities
Final Approach (same track)	2.5 NM	4.6 km	Within 10 NM (18.5 km) of runway threshold; local factors permit
Departing Aircraft ID	1 NM	—	PSR ID must be established within 1 NM of runway end
Turn Method heading change	—	—	30° or more

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India — Separation Minima

DGCA / AIP INDIA — SEPARATION MINIMA

DGCA For separation minima based on ATS surveillance systems in **India**, please refer to **Chapter Six** of the Air Regulations text.

The ICAO standard of **5.0 NM / 9.3 km** applies unless otherwise prescribed by the appropriate ATS authority.

MLAT and PSR Blip Separation Application

SEPARATION USING MLAT SYMBOLS / PSR BLIPS

Separation based on the use of MLAT position symbols and PSR blips shall be applied so that:

The distance between the **centres** of the position symbols and PSR blips, representing the positions of the aircraft concerned, is **never less than a prescribed minimum.**

Speed Control: Identify the Applicable Procedure

First establish which procedure applies: non-ATS-surveillance speed control, or surveillance-based approach control. The following limits and the arrival schedule belong to different provisions of AAI eAIP.

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Non-ATS-surveillance procedures

250
kt IAS maximum

Below 10,000 ft — all aircraft
Applies to arrivals and departures under the non-ATS-surveillance procedure.

220
kt IAS maximum

Arrivals below 10,000 ft and within 15 NM of the serving VOR/DME
All three conditions matter: arriving aircraft, below the stated level, and within the stated radius. The radius is measured from the VOR/DME serving the aerodrome.

ATC may impose additional restrictions on arriving and en-route aircraft when traffic requires. When traffic permits, ATC may suspend speed control using **"NO SPEED RESTRICTIONS"**.

Source: AAI eAIP ENR 1.1 §§2.5.1–2.5.4. Illustration is conceptual and not to scale.

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Surveillance-based approach control: arrival schedule

AAI publishes a separate schedule for terminal areas where surveillance-based approach control is established. Distinguish distances in NM from touchdown from the DME distances specified in the source.

Phase	Turboprop	Turbojet
En-route and initial descent up to FL290	Not applicable	250 kt or actual speed, whichever is higher
Below FL290 and up to FL150	250 kt or actual speed, whichever is lower	250 kt or actual speed, whichever is higher
Below FL150; within 25D to 20 NM (30D to 20 NM for straight-in), or on downwind	220 kt or actual speed, whichever is lower	220 kt or minimum clean speed, whichever is higher
Within 20 NM from touchdown	180 kt	180 kt

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Phase	Turboprop	Turbojet
Intercept leg, or 12 NM from touchdown for straight-in	180–160 kt	180–160 kt
10–5 NM from touchdown	160–150 kt	160 kt
Within 5 NM from touchdown	Speed control not applicable	Speed control not applicable

The first two rows are optional, according to ATC requirements; speed below 250 kt requires pilot concurrence. Below FL210, ATC may reduce speed to 240 kt with pilot concurrence. The terminal rows from below FL150 to 5 NM are mandatory, subject to the specified operational constraints and exceptions.

Reduce to 160 kt during the intercept leg. ATC may further reduce 180 kt to 170 kt. Below 10,000 ft AMSL, ATC may reduce speed to 210 kt with pilot concurrence. A turboprop unable to maintain the specified speed must tell ATC as early as possible, preferably during interception or at 12 NM from touchdown. Approach clearance does not cancel a speed restriction: it remains applicable unless ATC withdraws it.

The pilot may refuse a speed restriction considered excessive or contrary to aircraft operating specifications. Speed control is not applicable to aircraft entering or established in a hold, encountering turbulent weather, conducting CAT II/III operations within 20 NM from touchdown, within 5 NM from touchdown, flying the published instrument approach until intercepting the final approach track, carrying VVIP, or conducting priority/emergency landing.

Source: AAI eAIP ENR 1.6 §§8.1, 8.7, 8.9 and Annexure I. D denotes DME distance from the VOR; NM distances in this schedule are measured from touchdown.

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Termination of ATS Surveillance Service

TERMINATION — MANDATORY NOTIFICATION

An aircraft that has been informed it is provided with ATS surveillance service **should be informed immediately** when for any reason ATS surveillance service is **interrupted or terminated.**

AUTOMATIC TERMINATION — ARRIVING AIRCRAFT

ATS surveillance service is **automatically terminated** when an arriving aircraft receiving ATS surveillance service has been **instructed to contact tower frequency.**

Important: Position of aircraft from *touchdown* should be given to the aircraft **before** changing over the aircraft to tower.

SSR Transponder — Operating Procedures (India)

TRANSPONDER OPERATION — MANDATORY RULE (INDIA)

DGCA All aircraft carrying a serviceable transponder shall **operate the transponder at all times during flight** within **Chennai, Delhi, Guwahati, Kolkata and Mumbai FIR** — regardless of whether the aircraft is within or outside airspace where SSR is used for ATS.

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Operating Procedures — Pilots

TRANSPONDER MODE & CODE PROCEDURES

i. **Departing aircraft** from aerodromes in Chennai, Delhi, Guwahati, Kolkata and Mumbai FIR shall be assigned an *appropriate SSR code on departure*. This SSR code setting shall continue **unless instructed otherwise.**

ii. **International flights entering** Chennai, Delhi, Guwahati, Kolkata and Mumbai FIR shall *continue to maintain the SSR code* being squawked in the adjacent FIR.

iii. **Domestic flights** shall operate the transponder on the *last assigned code*.

iv. **Aircraft not assigned a SSR code** shall operate transponder on *Mode A3 code **2000*** before entry into Chennai, Delhi, Guwahati, Kolkata and Mumbai FIR and maintain that code setting until otherwise instructed.

v. To **avoid interference on radar display**, the pilot shall *NOT operate the transponder when the aircraft is on ground* except:

- When **entering the runway for take-off**, OR
- Until **vacating the runway after landing**

EMERGENCY / SPECIAL TRANSPONDER CODES — MEMORISE

Code	Condition	Meaning
7700	**Emergency**	General Emergency (distress)
7600	**Communication Failure (Radio Failure)**	NORDO — Lost communication
7500	**Unlawful Interference**	Hijack / Unlawful seizure of aircraft
2000	**No SSR code assigned**	Default code before entry into Indian FIR

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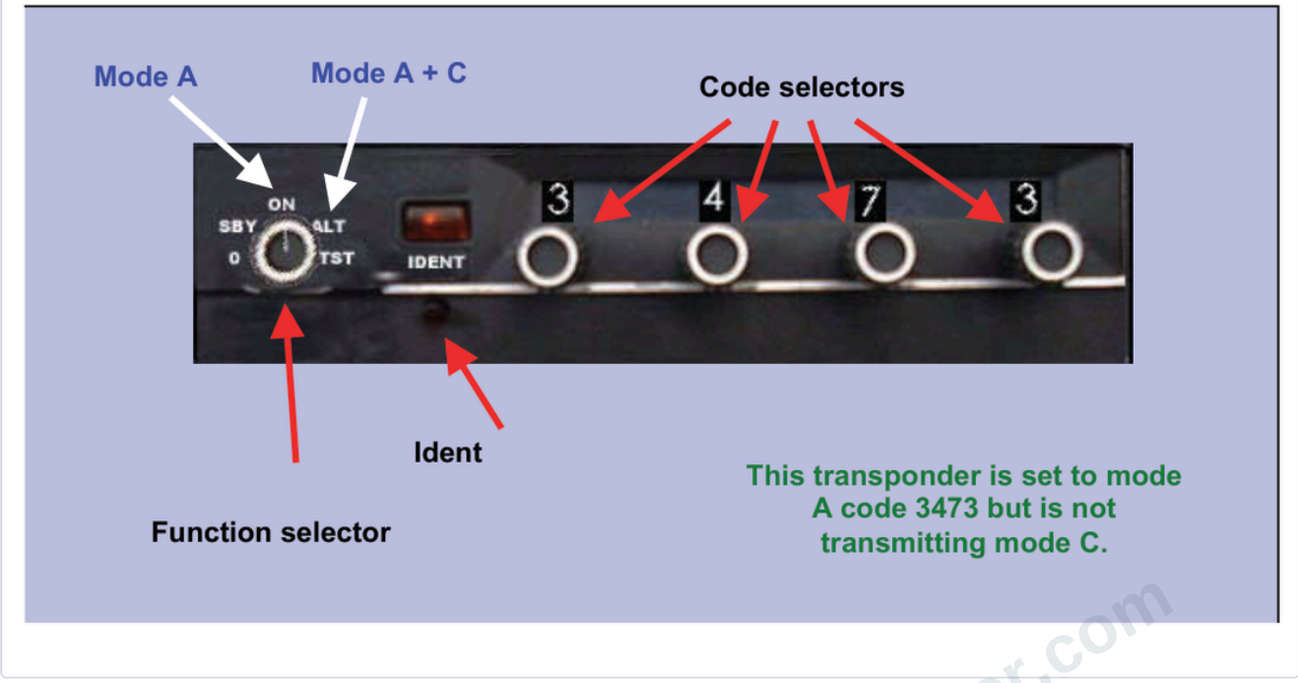

Figure 8.4 *A transponder control panel: function selector, code selectors and the ident button.*

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Mode C / ADS-B Altitude Verification

ALTITUDE TOLERANCE VALUES — EXACT DGCA/ICAO VALUES

Airspace Type	Tolerance (metres)	Tolerance (feet)
RVSM Airspace	±60 m	±200 ft
Other (non-RVSM) Airspace	±90 m	±300 ft
ATS Authority may specify smaller criterion	But **not less than ±60 m (±200 ft)**	

GEOMETRIC HEIGHT — SEPARATION RULE

Geometric height information shall NOT be used for separation.

Only pressure-altitude-derived level information and Mode C / ADS-B altitude data transmission are used for verification and separation.

Verification Procedure — Step by Step

MODE C / ADS-B ALTITUDE VERIFICATION PROCEDURE

1. All aircraft must **report the level/altitude maintaining/passing** on *first contact* with a radar unit — to facilitate verification of Mode C altitude information / ADS-B altitude data transmission.
2. Verification shall be effected **at least once** by each suitably equipped ATC unit on *initial contact* with the aircraft — or as soon as possible thereafter. Verification is by **simultaneous comparison** with *altimeter derived level information* received from the aircraft by radio telephony.

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3. If Mode C / ADS-B altitude data transmission is **within the approved tolerance** → pilot is **NOT** advised of verification.
4. If the displayed information is **NOT within approved tolerance**, or if a discrepancy *in excess* of the approved tolerance is detected subsequently: Pilot is **advised and requested to check his pressure setting and confirm his level.**
5. If, following confirmation of level and correct pressure setting, the **discrepancy continues**: Controller may request the pilot to **stop his Mode C transmission.**

ATC PHRASEOLOGY — STOP MODE C

If the Mode C discrepancy continues after confirmation:
"STOP SQUAWK CHARLIE. WRONG INDICATION" "STOP ADS-B ALTITUDE DATA TRANSMISSION (WRONG)"

ATS Identification — Decision Flowchart

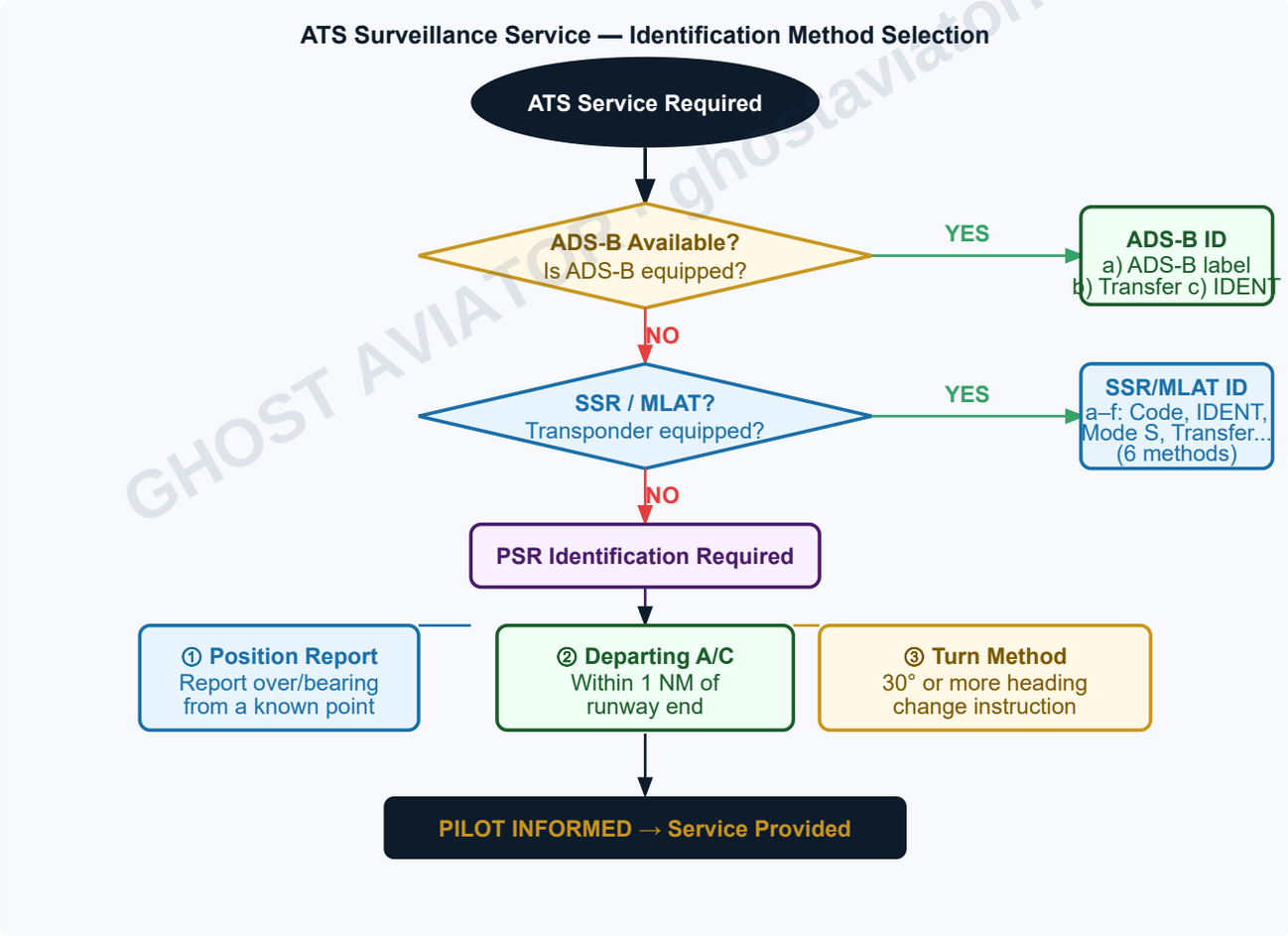

Figure 8.5 *ATS Surveillance Service — Identification Method Selection*

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EXAM PRACTICE — ALL 21 QUESTIONS FROM CHAPTER 9

Quick Reference — All Critical Values

Parameter	Value	Condition / Notes
Standard Horizontal Separation (ICAO)	5.0 NM / 9.3 km	Radar / ADS-B / MLAT — default
Reduced Horizontal Separation	3.0 NM / 5.6 km	Where ATS authority permits; local capability
Final Approach Separation (same track)	2.5 NM / 4.6 km	Within 10 NM (18.5 km) of runway threshold; local factors permit
Departing Aircraft PSR ID — limit	1 NM	Must identify within 1 NM from end of runway
Turn Method — minimum heading change	30° or more	For PSR identification when SSR unavailable
RVSM altitude tolerance (Mode C / ADS-B)	±200 ft / ±60 m	RVSM airspace — hard limit
Non-RVSM altitude tolerance (Mode C / ADS-B)	±300 ft / ±90 m	Non-RVSM airspace; ATS may specify smaller
Speed limit — below 10,000 ft (all flights)	250 KT IAS	AIP India — arrivals and departures
Speed limit — arrivals within 15 NM of VOR/DME	220 KT IAS	AIP India — below 10,000 ft
Emergency transponder code	7700	Distress / General Emergency
Radio failure transponder code	7600	Communication failure (NORDO)
Unlawful interference transponder code	7500	Hijack / Unlawful seizure
Default code (no SSR assigned) — India	Mode A3 — 2000	Before entry into Chennai/Delhi/Guwahati/Kolkata/Mumbai FIR
Transponder on ground — when to operate?	Only on/off runway	Enter runway for T/O; till vacating after landing
Radar vectors — heading reference	Magnetic only	All radar vectoring uses magnetic headings

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PART III

Information, Aerodromes and Rescue

How information reaches pilots, how aerodromes are marked and lit, and what happens when something goes wrong.

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CHAPTER 9

09

Aeronautical Information Services

General — AIS Overview & Responsibilities

What is AIS?

DEFINITION

The **Aeronautical Information Service (AIS)** is the body responsible for ensuring the continuous flow of **aeronautical data and aeronautical information** that is necessary for the safety, regularity, economy, and efficiency of global **Air Traffic Management (ATM)** — in an environmentally sustainable manner.

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Why Has AIS Become More Critical?

The role and importance of aeronautical data changed significantly with the implementation of modern navigation technologies:

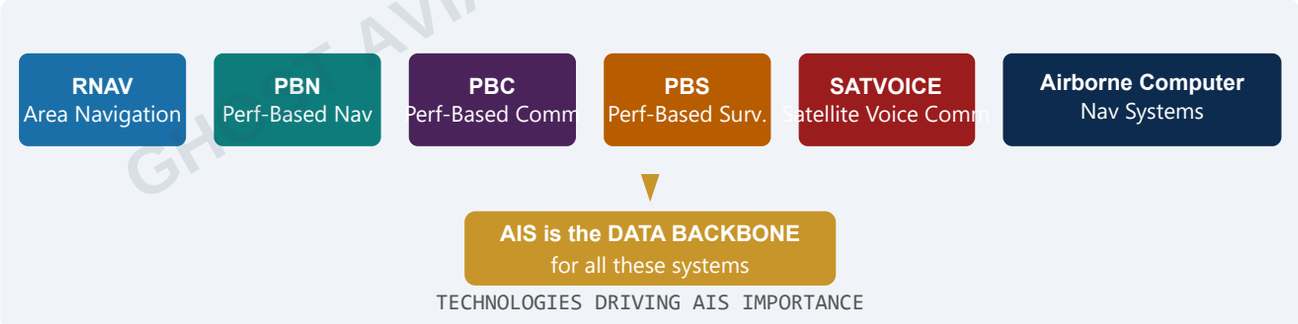

Figure 9.1 *RNAV*

What Does AIS Publish?

Aeronautical information shall be provided in a **standardized presentation** and shall include:

THE AIS PRODUCT SUITE

- **AIP** — Aeronautical Information Publication
- **AIP Amendments**
- **AIP Supplements**

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- **AIC** — Aeronautical Information Circulars
- **NOTAM** — Notice to Airmen
- **Aeronautical Charts**

Responsibilities in India

Publication / Service	Responsible Authority
AIS (all publications except AIC)	**Airports Authority of India (AAI)**
Aeronautical Information Circulars (AIC)	**DGCA** (Directorate General of Civil Aviation)

EXAM ALERT

AAI provides all AIS *except* AIC. **AIC is published by DGCA**. This distinction is a common exam question!

Aeronautical Information Publications (AIP)

AIP DEFINITION

AIP is intended primarily to satisfy **international requirements** for the exchange of aeronautical information of a **lasting character essential to air navigation**. When practicable, the form of presentation is designed to facilitate their use in flight.

AIP constitutes the **basic information source** for **permanent information** and **long-duration temporary changes**.

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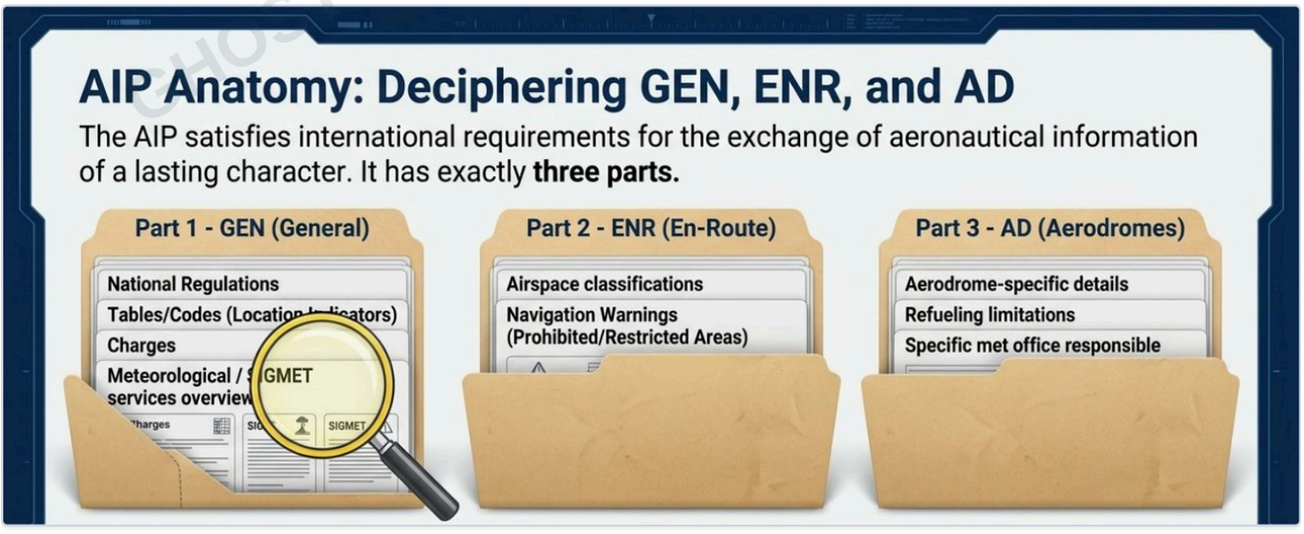

Figure 9.2 *The three parts of the AIP*

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AIP Contents

An AIP shall contain information in **three parts**, with sections and subsections uniformly referenced to allow for standardized electronic data storage and retrieval, arranged under subjects enumerated in Annex 15.

AIP Part	Code	Contents
Part 1 — General	GEN	National regulations, tables & codes, services, ATS, comms, met, SAR, charges
Part 2 — En-Route	ENR	General rules, FIR/CTA/TMA, radio nav aids, navigation warnings, en-route charts
Part 3 — Aerodromes	AD	Aerodromes/heliports, rescue & firefighting, licensing, aerodrome details

AIP General Specifications

- A **checklist** giving the current date of each page shall be reissued frequently. The page number/chart title and date of the checklist shall appear on the checklist itself.
- Operationally significant changes shall be published in accordance with **AIRAC procedures** and clearly identified by the acronym **AIRAC**.
- AIP shall be amended or reissued at regular intervals to keep them up to date. The normal method of amendment shall be by means of **replacement sheets**.
- Recourse to hand amendments or annotations shall be kept to the **minimum**.
- The regular interval shall be specified in the **AIP Part 1 — GEN**.

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Specifications for AIP Amendments

AIP AMENDMENT RULES

- **Permanent changes** to the AIP shall be published as AIP Amendments.
- Each AIP Amendment shall be allocated a **serial number** which shall be consecutive.
- Each AIRAC AIP Amendment page (including the cover sheet) shall display an **effective date**.
- When an AIP Amendment will not be published at the established interval or publication date, a **NIL notification** shall be originated and distributed by the monthly printed plain-language list of valid NOTAM.

Specifications for AIP Supplements

WHAT GOES INTO AN AIP SUPPLEMENT?

Temporary changes of long duration (three months or longer) AND information of short duration which contains **extensive text and/or graphics** shall be published as AIP Supplements.

3	months or longer = "Long Duration"
≤1 mo	Checklist of valid AIP SUPs issued interval

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Yellow	Preferred page color for AIP SUPs

- Each AIP Supplement shall be allocated a **serial number** which shall be consecutive and based on the **calendar year**.
- AIP Supplement pages shall be kept in the AIP as long as all or some of their contents remain valid.
- When an AIP Supplement is sent in replacement of a NOTAM, it shall include a reference to the **serial number of the NOTAM**.
- A checklist of valid AIP Supplements shall be issued at intervals of **not more than one month**. This information shall be issued through the monthly printed plain language list of valid NOTAM.
- AIP Supplement pages shall be **coloured** in order to be conspicuous, preferably in **yellow**.
- AIP Supplement pages shall be kept as the **first item** in the AIP parts.
- When an error occurs in an AIP Supplement or when the period of validity *(the period during which a document is accepted)* is changed, a **new AIP Supplement** shall be published as a replacement.

REGULATORY NOTE

The requirements for NOTAM apply when **time constraints do not allow sufficient time** for the distribution of an AIP Supplement.

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Electronic AIP (eAIP)

The AIP, AIP Amendment, AIP Supplement, and AIC should also be published in a format that allows for:

- Displaying on a computer screen
- Printing on paper

NOTE ON EAIP

This composite electronic document is named **"Electronic AIP" (eAIP)** and may be based on a format that allows for **digital data exchange**.

NOTAM — Notice to Airmen

DEFINITION

A **NOTAM** is a notice distributed by means of telecommunication containing information concerning the establishment, condition or change in any **aeronautical facility, service, procedure or hazard**, the timely knowledge of which is essential to personnel concerned with flight operations.

NOTAM Types

Type	Suffix	Meaning
NOTAM (N)	N	NOTAM **New** — Original issuance of a new NOTAM
NOTAM (R)	R	NOTAM **Replacement** — Replaces a previously issued NOTAM

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Type	Suffix	Meaning
NOTAM (C)	C	NOTAM **Cancellation** — Cancels a previously issued NOTAM

Trigger NOTAM

WHAT IS A TRIGGER NOTAM?

The intent of a **Trigger NOTAM** is to serve as a **reminder in the pre-flight information bulletin (PIB)** for the forthcoming operationally significant change in the AIP (via AIP Supplement), if applicable.

When an AIP Supplement is published in accordance with AIRAC procedures, a trigger NOTAM must be originated giving:

- A **brief description** of the contents
- The **effective date/time**
- The **serial number** of the AIP Supplement

CRITICAL RULE

The Trigger NOTAM **must come into force on the same date** as the supplement to which it refers. It is included in the PIB to ensure pilots and operators are reminded that changes will take place as of a given effective date.

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NOTAM Origination Rules

WHEN SHALL A NOTAM BE ORIGINATED?

- Information is of a **temporary nature** and of **short duration**, OR
- **Operationally significant permanent changes**, OR
- **Temporary changes of long duration** made at **short notice**, except for extensive text and/or graphics.

The need for origination shall be considered in any circumstance which may affect the operations of aircraft.

Information NOT Notified by NOTAM (9 Exceptions)

THE FOLLOWING SHALL NOT BE NOTIFIED BY NOTAM

#	Situation / Condition
a)	Routine maintenance work on aprons and taxiways which does **not affect** the safe movement of aircraft
b)	Runway marking work, when aircraft operations can safely be conducted on other available runways, or the equipment used can be removed when necessary
c)	Temporary obstructions in the vicinity of aerodromes/heliports that do **not affect** the safe operation of aircraft
d)	Partial failure of aerodrome/heliport lighting facilities where such failure does **not directly affect** aircraft operations

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#	Situation / Condition
e)	Partial temporary failure of air-ground communications when **suitable alternative frequencies are known to be available and operative**
f)	The lack of apron marshalling services and road traffic control
g)	The unserviceability of location, destination or other instruction signs on the aerodrome movement area
h)	Parachuting when in uncontrolled airspace under VFR, when controlled at promulgated sites, or within danger or prohibited areas
i)	Other information of a similar temporary nature

Advance Notice Requirements

7 days	Min. advance notice for activation of danger/restricted/prohibited areas or temporary airspace restrictions (non-emergency)
24 hrs	Desirable advance notice whenever possible for airspace utilization planning

NOTAM notifying unserviceability of aids to air navigation, facilities, or communication services shall give an **estimate of the period of unserviceability** or the time at which restoration of service is expected.

SNOWTAM

SNOWTAM DEFINITION

A **SNOWTAM** is a special series NOTAM given in a standard format providing a **surface condition report** notifying the presence or cessation of hazardous conditions due to:

- Snow, ice, slush, frost
- Standing water
- Water associated with snow, slush, ice or frost on the movement area

Runway Condition Code (RWYCC)

RWYCC — SURFACE CONDITION ASSESSMENT AND REPORTING

The **Runway Condition Code (RWYCC)** is used for reporting assessed runway information.

- The **aerodrome operator** assesses the runway surface conditions whenever **water, snow, slush, ice or frost** are present on an operational runway.
- From this assessment, a RWYCC and a description of the runway surface are reported which can be used by the flight crew for **aeroplane performance calculations**.
- This format, based on the type, depth, and coverage of contaminants, is the best assessment of the runway surface condition by the aerodrome operator.
- Changes in conditions shall be reported **without delay**.
- The RWYCC shall be reported for each **one third of the runway** to be assessed.

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REFERENCE DOCUMENTS FOR RWYCC

For details refer **Doc 9981 PANS-Aerodrome** and **Doc 10066 PANS-AIM**.

ASHTAM

ASHTAM — VOLCANIC ACTIVITY NOTAM

Information concerning an operationally significant change in volcanic activity, a **volcanic eruption and/or volcanic ash cloud** shall, when reported by means of an ASHTAM, contain the information in the order shown in the **ASHTAM Format**.

NOTAM General Specifications

- The NOTAM originator shall allocate to each NOTAM a **series identified by a letter** and a **four-digit number** followed by a stroke and a **two-digit number for the year**. The four-digit number shall be consecutive and based on the **calendar year**.
  Example: A0047/26 = Series A, NOTAM #0047 of year 2026
- Each NOTAM shall deal with **only one subject and one condition** of the subject.
- A checklist of valid NOTAM shall be issued as a NOTAM over the **Aeronautical Fixed Service (AFS)** at intervals of **not more than one month** using the NOTAM Format specified. **One NOTAM shall be issued for each series.**
- A checklist of NOTAM shall refer to the latest AIP Amendments, AIP Supplements, and at least the internationally distributed AIC.
- A monthly printed plain-language list of valid NOTAM (including indications of the latest AIP Amendments, AIC issued and a checklist of AIP Supplements) shall be prepared with a minimum of delay and forwarded by the most expeditious means to recipients of the Integrated Aeronautical Information Package.

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NOTAM Distribution

- The originator shall select the NOTAM that are to be given **international distribution**.
- **Selective distribution lists** should be used when practicable.

AIRAC — Aeronautical Information Regulation and Control

AIRAC DEFINITION

AIRAC (Aeronautical Information Regulation and Control) is a system aimed at advance notification based on **common effective dates**, of circumstances necessitating significant changes in operating procedures. It uses a 28-day amendment cycle.

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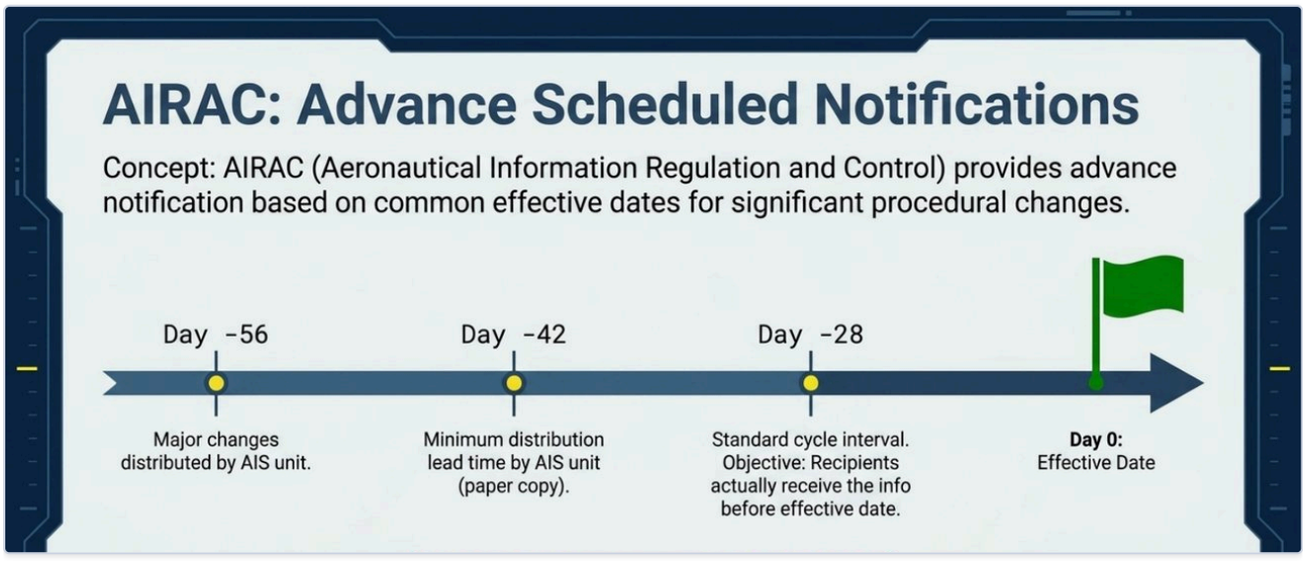

Figure 9.3 *AIRAC: advance notice on fixed dates*

AIRAC General Specifications

AIRAC CYCLE RULES

- Information concerning establishment, withdrawal or significant changes upon a series of **common effective dates at intervals of 28 days**, including 29 January 1998.
- The information notified therein shall not be changed further for at least **another 28 days after the effective date**, unless the circumstance notified is of a temporary nature and would not persist for the full period.
- The regulated system (AIRAC) shall also be used for the provision of information relating to the establishment and withdrawal of, and **premeditated significant changes**.

28	Day AIRAC Cycle (standard interval)
42 days	Min. distribution lead time by AIS unit (paper copy)
28 days	Objective — recipients receive info before effective date
56 days	For MAJOR changes — distribution by AIS unit in advance

Information to be Notified by AIRAC

Establishment, Withdrawal, and Premeditated Significant Changes to

Category	Specifics
Limits & Procedures for Airspace	a) Flight Information Regions b) Control Areas c) Control Zones d) Advisory Areas e) ATS Routes f) Permanent Danger/Prohibited/Restricted Areas & ADIZ g) Areas/Routes where interception is possible

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Category	Specifics
Radio Navigation & Comms	Positions, frequencies, call signs, known irregularities and maintenance periods of radio navigation aids and communication facilities
Procedures	Holding and approach procedures, arrival and departure procedures, noise abatement procedures and any other pertinent ATS procedures
Met & Infrastructure	Meteorological facilities (including broadcasts) and procedures · Runways and stopways

Establishment and Withdrawal, and Premeditated Significant Changes to

- Position, height and lighting of **navigational obstacles**
- Taxiways and aprons
- Hours of service: aerodromes, facilities and services
- Customs, immigration and health services
- Temporary danger, prohibited and restricted areas and navigational hazards, military exercises and mass movements of aircraft
- Temporary areas or routes or portions thereof where the possibility of interception exists

Provision of Information in Paper Copy Form

56 days before	For **major changes** where advance notice is desirable and practicable — information in paper copy should be distributed by the AIS unit.
42 days before	For **all AIRAC information** in all instances — must be distributed by AIS unit to achieve 28-day recipient receipt objective.
28 days before	The **objective for recipients** to have received paper copy before the effective date.
Effective Date	AIRAC change **comes into force**. No further change for at least 28 days unless temporary and non-persistent.

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KEY NUMBERS TO MEMORIZE

AIRAC Cycle = 28 days | AIS distributes paper copy at least 42 days ahead | Recipients to receive 28 days ahead | Major changes: AIS distributes 56 days ahead

Aeronautical Information Circulars (AIC)

AIC — WHEN IS IT ORIGINATED?

An AIC shall be originated when it is **necessary to promulgate aeronautical information** which does not qualify for inclusion in an AIP *or* for the origination of a NOTAM.

AIC Origination Criteria

An AIC shall be originated whenever it is desirable to promulgate:

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- A **long-term forecast** of any major change in legislation, regulations, procedures or facilities
- Information of a **purely explanatory or advisory nature** liable to affect flight safety
- Information or notification of an explanatory or advisory nature concerning **technical, legislative or purely administrative matters**

AIC General Specifications

Specification	Detail
Format	AIC shall be issued in **printed form**. Both text and diagrams may be included.
Numbering	Each AIC shall be allocated a **serial number** which shall be consecutive and based on the **calendar year**.
Multiple Series	When AIC are distributed in more than one series, each series shall be separately identified by a **letter**.
International Distribution	Originator shall select the AIC that are to be given international distribution.
Checklist	A checklist of AIC currently in force shall be issued at least **once a year**, with distribution as for the AIC.
Distribution Rule	AIC selected for international distribution shall be the **same as for the AIP**.
India Responsibility	Published by **DGCA** (not AAI)

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Aeronautical Charts

- Aeronautical charts shall, when available for designated international aerodromes/heliports, **form part of the AIP**, or be provided separately to recipients of the AIP.
- Electronic aeronautical charts should be provided based on **digital databases and the use of geographic information systems**.
- The chart resolution of aeronautical data shall be that as specified for a particular chart.

Pre-Flight & Post-Flight Information / PIB

Pre-Flight Information Requirements

AT INTERNATIONAL AERODROMES

At any aerodrome/heliport normally used for international air operations, aeronautical information essential for the **safety, regularity and efficiency** of air navigation and relative to the route stages originating at the aerodrome/heliport shall be made available to:

- Flight operations personnel
- Flight crews
- Services responsible for pre-flight information

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Aeronautical information provided for pre-flight planning purposes at the aerodromes/heliports shall include relevant:

- Elements of the **Integrated Aeronautical Information Package**
- **Maps and charts**

The documentation may be limited to national publications and when practicable, those of immediately adjacent States, provided a **complete library** of aeronautical information is available at a central location and means of **direct communications** are available between the aerodrome AIS unit and that library.

Pre-Flight Information Bulletin (PIB)

PIB DEFINITION

A **recapitulation of current NOTAM and other information of urgent character** shall be made available to flight crews in the form of **plain-language pre-flight information bulletins (PIB)**.

PIB Availability in India (Four Locations)

#	Aerodrome
1	**Kolkata**
2	**Chennai**
3	**Delhi**
4	**Mumbai**

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The information provided at each aerodrome includes: information on departure aerodrome, elements of the Integrated Aeronautical Information Package and maps and charts.

Automatic Self Briefing System (ASBS)

ASBS

The **Automatic Self Briefing System (ASBS)** provides an automatic method of receiving, storing and retrieving the data essentially required for pilot's pre-flight briefing. It has the capability to include information pertaining to the stations and routes concerned.

Post-Flight Information

POST-FLIGHT INFORMATION PURPOSE

The purpose of post-flight information is to ensure that:

- Malfunctioning/unserviceability of **visual or non-visual aids** or any other facility essential to the safety of flight operations, **observed by the Pilot during the flight**, are reported without delay to the authorities responsible for those facilities.

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- Presence of **birds and wildlife hazards** on or around the airport constituting a potential hazard to aircraft operations, observed by a Pilot, should be reported without delay.

This ensures information is made available for distribution as circumstances necessitate.

Telecommunication Requirements & Digital Data Sets

Telecom Requirements

- NOTAM offices shall be connected to the **Aeronautical Fixed Service (AFS)**.
- The connections shall provide for **printed communications**.
- Each NOTAM office shall be connected, through the AFS, to the following points within the territory for which it provides service:
  - Area control centres and flight information centres
  - Aerodromes/heliports at which an information service is established

Digital Data Sets

Digital data shall be in the form of the following data sets:

#	Data Set
a)	**AIP data set**
b)	**Terrain data sets**
c)	**Obstacle data sets**
d)	**Aerodrome mapping data sets**
e)	**Instrument flight procedure data sets**

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Aeronautical Information Products

AIS PRODUCTS

Aeronautical data and aeronautical information provided either as **digital data sets** or as a **standardized presentation in paper or electronic media**. Products include:

- AIP, AIC, NOTAMS
- Aeronautical charts
- Digital data sets

These products are intended primarily to satisfy **international requirements** for the exchange of aeronautical information.

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National Provisions — India (AAI, NOF, NOTAM Series)

AIS Infrastructure in India

The **Aeronautical Information Service** which forms part of the **Airports Authority of India (AAI)** ensures the flow of information necessary for the safety, regularity and efficiency of International and National Air Navigation within the areas of its responsibility. It consists of:

Component	Location / Detail
AIS Headquarters (AIS HQ)	**Safdarjung Airport, New Delhi – 110 003**
International NOTAM Office (NOF)	4 offices: **Mumbai, Calcutta (Kolkata), Delhi, Chennai**
AIS Units	At various aerodromes throughout India

EXAM TRAP: NOF = INTERNATIONAL NOTAM OFFICE

NOF = International NOTAM Office. Four NOFs established at **Mumbai, Kolkata, Delhi, and Chennai ONLY** — NOT at all major aerodromes, NOT at all international airports.

Area of Responsibility

The Aeronautical Information Service is responsible for the collection and dissemination of information for:

- The **entire territory of India**
- Airspace over the **high seas under the jurisdiction** ***(legal power over a person, place or event)*** **of India** for air traffic control purposes

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NOTAM Series in India

NOTAMs are originated by the International NOTAM Offices at Mumbai, Kolkata, Delhi and Chennai. NOTAMs are distributed in **five series** identified by letters **A, B, C, D and G**:

A	International — Long Changes/unserviceability of aeronautical facilities likely to last **more than 2 hours** in respect of locations of direct importance to **international aircraft operations**, requiring **General International Distribution**.
B	International — Short Changes/unserviceability likely to last **more than 30 minutes but less than 2 hours** for international locations, requiring **limited distribution to adjacent stations only**.
C	Domestic Only Changes/unserviceability at locations used by **domestic flights only** for which no separate Series A International Distribution is given.

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D	Military Airfields Changes/unserviceability at **military controlled airfields** used by domestic flights. Issued by **AHQ through a VVO broadcast**.
G	General (Lasting) Information of **general and lasting character** affecting aircraft operations in general. Operated **only by the International NOTAM Office, Delhi**.

NOTAM SERIES NUMBERING

NOTAM of each series are allocated a serial number by the respective NOTAM offices, commencing with **No. 0001** preceded by the designated letter of series A, B, C, D & G as the case may be, at **0000 UTC on 1st January** every year.

Checklist and Summary of NOTAM

- A checklist of all NOTAM current on the **1st of each month** is originated by the respective NOTAM office and transmitted over the AFTN to all addresses on the distribution list.
- In addition, a monthly checklist and summary of NOTAM in force at the **end of the month** is compiled and posted.
- The **complete summary** is prepared once a year on the **1st January**.
- Subsequent monthly summaries contain the text of **only those NOTAMs promulgated during the month of issue** and remaining current on the date of compilation.

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AIP India — Structure

AIP India has been published in **two volumes**:

Volume	Contents	Parts
Volume I	Part 1 — General (GEN) & Part 2 — En-route (ENR)	GEN ENR
Volume II	Part 3 — Aerodromes (AD)	AD

The AIP is published in **loose leaf form in English** for use in International and Domestic Operations.

India — Amendment and Supplement Schedule

Document	India Schedule
AIP Amendment (AMDT)	Made **as required**. Operationally significant changes are issued as AIRAC amendments on the common 28-day AIRAC effective dates (AIP India is now an electronic AIP with an AMDT tab); an older printed schedule of one amendment a year in April is no longer the rule.
AIP Supplement (SUP)	Checklist of AIP Supplements currently in force issued in the **month of January** of each year

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AIP Content Detail — GEN / ENR / AD (India)

GENERAL (GEN)

National Regulation and Requirements

- Designated authorities
- Entry, transit and departure of aircraft; passengers and crew; cargo
- Aircraft instruments, equipment and flight documents
- Summaries of Indian Regulations and International Agreements/Conventions
- Differences from ICAO Standards, Recommended Practices and Procedures

Tables and Codes

- Measuring systems, aircraft marking, holidays, units of measurement, time system, geodetic reference
- Aircraft nationality and registration, public holidays
- Abbreviation used in AIS publication, chart symbols, location indicators
- List of radio navigation aids, conversion tables, sunrise/sunset tables

Services

- Aeronautical information services: responsible services, area of responsibility, aeronautical publication, AIRAC system, pre-flight information service at aerodromes
- Aeronautical charts: responsible service(s), maintenance of charts, purchase arrangements, chart series available, list of aeronautical charts available

Air Traffic Services

- Responsible service, area of responsibility, types of services, minimum flight altitude, ATS units address

Communication Services

- Responsible service, area of responsibility, types of service, requirements and conditions

Meteorological Services

- Responsible service, area of responsibility, meteorological observations and reports, types of services
- Notification required from operators, aircraft reports, VOLMET service, **SIGMET service**, other automated meteorological services

Search and Rescue

- Responsible service(s), area of responsibility, types of services, SAR agreements, conditions of availability, procedures and signals

Charges

- Aerodrome/Heliport Charges, Air Navigation Services Charges

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EXAM POINT

Charges for aerodromes/heliports and Air Navigation Services are in **Part 1 — GEN**.

Location indicators list is in **Part 1 — GEN**.

SIGMET service information is in **Part 1 — GEN (Meteorological services subsection)**.

EN-ROUTE (ENR)

General Rules and Procedures

- Visual Flight Rules (VFR), Instrument Flight Rules (IFR)
- ATS Airspace Classification
- Approach and Departure Procedures
- ATS Surveillance Services, Altimeter Setting
- Regional Supplementary Procedures, Air Traffic Flow and Capacity Management Service
- Flight Plans (PLN), Addressing of Flight Plan Messages
- Interception of Civil Aircraft Procedures, Unlawful Interference, Air Traffic Incidents

FIR / CTA / TMA, Other Regulated Airspace

- ATS Routes, RNAV Routes, Helicopter Routes, Other Routes, EN-ROUTE Holding

Radio Navigation Aids/systems

- Radio Navigation Aids & Systems
- Special Navigation Systems
- Global Navigation Satellite System (GNSS)

Navigation Warnings

- **Prohibited, Restricted and Danger Areas**
- Military Exercise and Training Areas
- Other Activities of Dangerous Nature
- Air Navigation Obstacles Enroute
- Aerial Sporting and Recreational Activities

EXAM POINT

Prohibited, restricted, and dangerous areas are in **Part 2 — ENR** (Navigation Warnings subsection). Holding, approach and departure procedures are in **Part 2 — ENR**.

AERODROME (AD)

- Aerodromes / Heliports — Introduction
- Aerodrome / Heliport Availability

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- Rescue and Fire Fighting Services and Snow Plan
- List of Aerodromes, Heliports, Water Aerodromes
- Grouping of Aerodromes / Heliports
- Status of Aerodrome Licensing
- Aerodromes (individual aerodrome entries)

EXAM POINT

Refuelling facilities and limitations on refuelling services are in **Part 3 — AD**. Detailed meteorological information at the aerodrome and which meteorological office is responsible is in **Part 3 — AD**.

Integrated Aeronautical Information Package (IAIP)

IAIP ELEMENTS (INDIA)

The Aeronautical Information is provided in the form of the **Integrated Aeronautical Information Package** consisting of the following elements:

#	Element	Code	Distribution Method (India)
a.	Aeronautical Information Publication	`AIP`	Distributed by mail
b.	Amendment service to the AIP	`AIP AMDT`	Distributed by mail
c.	Supplement to the AIP	`AIP SUP`	Distributed by mail
d.	NOTAM and pre-flight information bulletins	`NOTAM / PIB`	NOTAMS via AFS; PIB at ATS Reporting Office
e.	Aeronautical information circulars	`AIC`	Distributed by mail
f.	Checklists and summaries	—	NOTAMS via AFS

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COMPLETE IAIP

The correct answer for "elements of the Integrated Aeronautical Information Package" is: **AIP (including amendment service) + supplements to AIP + NOTAM and pre-flight information bulletins (PIB) + AIC + checklists and summaries** — i.e., Option B in the exam questions.

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AIS Products Flowchart

Information	Publication
Permanent information or permanent change	AIP / AIP Amendment. Apply AIRAC when required for the information concerned.
Temporary information lasting three months or longer, or extensive text/graphics	AIP Supplement.
Short-duration temporary operational information, or operationally significant information requiring timely notice	NOTAM, subject to the publication criteria.
Explanatory, advisory or administrative information outside AIP/NOTAM criteria	Aeronautical Information Circular (AIC).

Figure 9.4 *Choose the publication by the information's purpose, duration and format. An AIC is a separate category, not a stage after a NOTAM.* *Chapter 9 source content; DGCA CAR 9-I-I AIS publication categories.*

Critical Numbers & Facts at a Glance

Topic	Critical Value / Fact
AIP Supplement — Long Duration	≥ 3 months
AIP SUP Checklist Interval	≤ 1 month
AIP SUP Page Colour	Yellow (preferred)
AIRAC Cycle	28 days
AIRAC — AIS distributes paper copy by	42 days before effective date
AIRAC — Recipients should receive by	28 days before effective date
AIRAC — Major changes advance distribution	56 days before effective date
NOTAM Checklist Interval (AFS)	≤ 1 month (one per series)
NOTAM Series in India	A, B, C, D, G
Series A duration	> 2 hours, international importance
Series B duration	> 30 min but < 2 hours
Series G operator (India)	International NOTAM Office Delhi ONLY
AIS HQ India	Safdarjung Airport, New Delhi – 110 003
NOF in India	Mumbai, Kolkata, Delhi, Chennai (4 only)
AIC published by (India)	DGCA (NOT AAI)
AIS published by (India)	AAI (Airports Authority of India)
AIC Checklist Interval	At least once a year
RWYCC assessed per	Each 1/3 of the runway

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Topic	Critical Value / Fact
Advance notice for danger/restricted area activation	≥ 7 days
Desirable advance notice (general)	24 hours
PIB available at (India)	Kolkata, Chennai, Delhi, Mumbai
AIP India — annual amendment month	April
AIP India — SUP checklist month	January
NOTAM numbering restarts	0001 at 0000 UTC on 1st January each year

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CHAPTER 10

Search and Rescue

10

SAR Services — International Obligations

Annex 12, ICAO · AIP India Chapter 11

Core Regulatory Obligation — Contracting States	Contracting States shall, individually or in cooperation with other States, **arrange for the establishment and prompt provision** of search and rescue services within their territories to ensure that **assistance is rendered to persons in distress**. Such services shall be provided on a **24-hour basis**.

HIGH SEAS / UNDETERMINED SOVEREIGNTY AREAS

Those portions of the **high seas or areas of undetermined sovereignty** ***(complete power of a State over its own territory)*** for which SAR services will be established shall be determined on the basis of **Regional Air Navigation Agreements**.

24	Hour SAR Service Coverage
365	Days Per Year — No Exceptions
ICAO	Annex 12 Authority

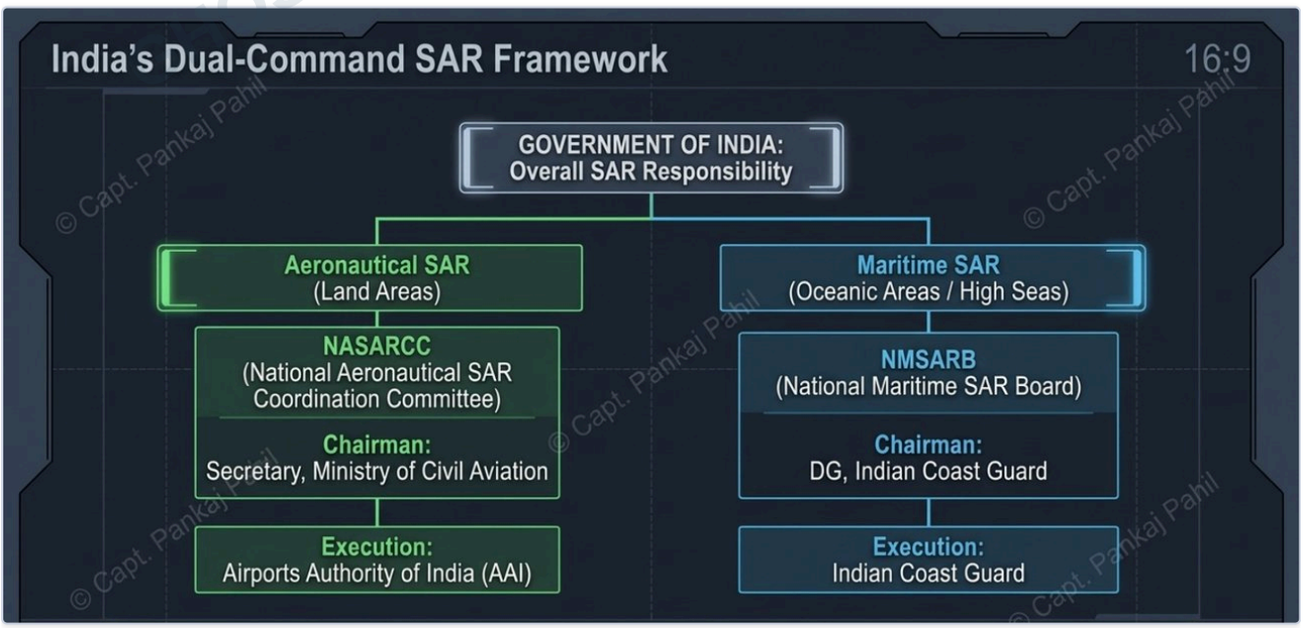

Figure 10.1 *Who is responsible for search and rescue in India*

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Search & Rescue Regions (SRR)

Annex 12 · National Provisions

DEFINITION — SEARCH & RESCUE REGION (SRR)

An area of defined dimensions associated with a Rescue Coordination Centre within which SAR services are provided.

STRICT RULES FOR SAR REGIONS

- Contracting States **shall delineate** the search and rescue regions within which they will provide services.
- SAR regions **shall NOT overlap**.
- Neighboring SAR regions **shall be contiguous** (no gaps allowed).

INDIA'S SAR REGION (SRR) — KEY FACT

The boundary of the **Indian Aeronautical SRR coincides with the Indian FIR boundary**, *excluding* the portion of FIR over the sovereign territory of the **Kingdom of Bhutan**.

Rescue Coordination Centres & Sub-Centres

Annex 12 · Chapter 11

RESCUE COORDINATION CENTRE (RCC)

A unit responsible for **promoting efficient organization** of search and rescue services and for **coordinating the conduct of SAR operations** within a SAR region.

RESCUE SUB-CENTRE (RSC)

A unit subordinate to an RCC, established to **complement the latter** according to particular needs of the SAR authority.

INDIA'S FOUR AERONAUTICAL RCCS (AAI)

Airports Authority of India (AAI) coordinates SAR services through **four Rescue Coordination Centres** established at:

Delhi	Delhi FIR
Mumbai	Mumbai FIR
Kolkata	Kolkata FIR
Chennai	Chennai FIR

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National Provisions — India

AIP India · CAR Section 9 · Series 'S', Part I Issue II

Government of India — SAR Responsibility	Indian Government has assumed the responsibility of providing **Search and Rescue (SAR) cover in Indian Search & Rescue Regions (SRR)**.

Organisation	Responsibilities	Coordination centres / chair
Government of India	Overall SAR responsibility	—
National Aeronautical Search & Rescue Coordination Committee (NASARCC)	Land areas; aeronautical SAR coordination	Secretary, Ministry of Civil Aviation
National Maritime SAR Board (NMSARB)	Oceanic areas; maritime SAR	Director General, Indian Coast Guard
Airports Authority of India	Coordinates SAR excluding oceanic areas	RCCs: Delhi · Mumbai · Kolkata · Chennai
Indian Coast Guard	Maritime areas — high seas; oceanic areas of Kolkata, Mumbai and Chennai FIRs	MRCCs: Mumbai · Chennai · Port Blair

Figure 10.2 *Indian SAR Organizational Structure*

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ADDITIONAL SAR RESOURCES AVAILABLE

The following departments and organisations are also available for SAR missions when required:

- Railways, Post & Telegraph, All India Radio
- Police and District Collectors / Magistrates
- Municipal and Local bodies
- Airline operators, Flying clubs, Professional Pilots
- Mercantile Marine, Port Trust
- **Armed Forces**

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SAR COVERAGE — TERRITORIAL EXTENT

- SAR services cover the entire **Indian Territory including territorial waters**.
- Also covers portions of high seas / areas of undetermined sovereignty where responsibility has been **delegated to India**.
- Indian Aeronautical SRR boundary **= Indian FIR boundary**, EXCLUDING sovereign territory of Kingdom of Bhutan.

SAR Equipment Requirements

Annex 12 · Chapter 11

EQUIPMENT — SAR UNITS

Search and rescue units shall be provided with equipment for:

- **Locating promptly** the scene of an accident
- **Providing adequate assistance** at the scene of an accident

SAR Aircraft Equipment Requirements

Requirement	Detail
Communications	Equipped to communicate on the **aeronautical distress** and **on-scene frequencies**, and on such other frequencies as may be prescribed *(laid down by the authority or the rules)*.
Homing Device	Equipped with a device for **homing on distress frequencies**.
Maritime Communication	When used over maritime areas — equipped to **communicate with vessels**.
International Code of Signals	When used over maritime areas — must carry a **copy of the International Code of Signals** to overcome language difficulties with ships.

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VESSEL COMMUNICATION FREQUENCIES

Many vessels can communicate with aircraft on:

`2182 KHZ` `4125 KHZ` `121.5 MHZ`

NOTE: These frequencies, and in particular **121.5 MHz**, may **NOT be routinely monitored** by vessels.

SURVIVAL EQUIPMENT — COLOUR CODE FOR DROPPED CONTAINERS

Survival equipment dropped by SAR containing **food and water** will be packed in containers and indicated by streamers of **Blue colour**.

Operating Procedures — Pilot-in-Command

Annex 12 · Operating Procedures

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PIC at the Scene of an Accident

WHEN PIC OBSERVES ANOTHER AIRCRAFT OR SURFACE CRAFT IN DISTRESS — SHALL IF POSSIBLE AND UNLESS UNREASONABLE/UNNECESSARY

a. Keep the craft in distress **in sight** until compelled to leave the scene or advised by the RCC that it is no longer necessary.

b. **Determine the position** of the craft in distress.

c. As appropriate, **report to the RCC or ATS unit** as much information as possible.

COMMAND RESPONSIBILITY — NON-SAR AIRCRAFT FIRST ON SCENE

If the **first aircraft to reach the scene of an accident is NOT a search and rescue aircraft**, it shall **take charge of on-scene activities** of all other aircraft subsequently arriving until the **first SAR aircraft reaches the scene** of the accident.

Situation	On-scene command
First aircraft is a SAR aircraft	SAR aircraft takes command
First aircraft is not a SAR aircraft	First non-SAR aircraft takes command of subsequent aircraft until the first SAR aircraft arrives
SAR aircraft arrives	SAR aircraft takes over

Figure 10.3 *On-Scene Command — Who's In Charge?*

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PIC Intercepting a Distress Transmission

WHENEVER A DISTRESS TRANSMISSION IS INTERCEPTED — PIC SHALL IF FEASIBLE

a. **Acknowledge** the distress transmission.

b. **Record the position** of the craft in distress *if given*.

c. **Take a bearing** on the transmission.

d. **Inform the appropriate RCC or ATS unit** of the distress transmission, giving all available information.

e. At the pilot's discretion, while awaiting instructions, **proceed to the position given in the transmission**.

EXAM TIP — DISTRESS TRANSMISSION STEPS

The key qualifier is "*if feasible*" — the pilot must do **all five actions if feasible**, but is not obligated if they genuinely cannot.

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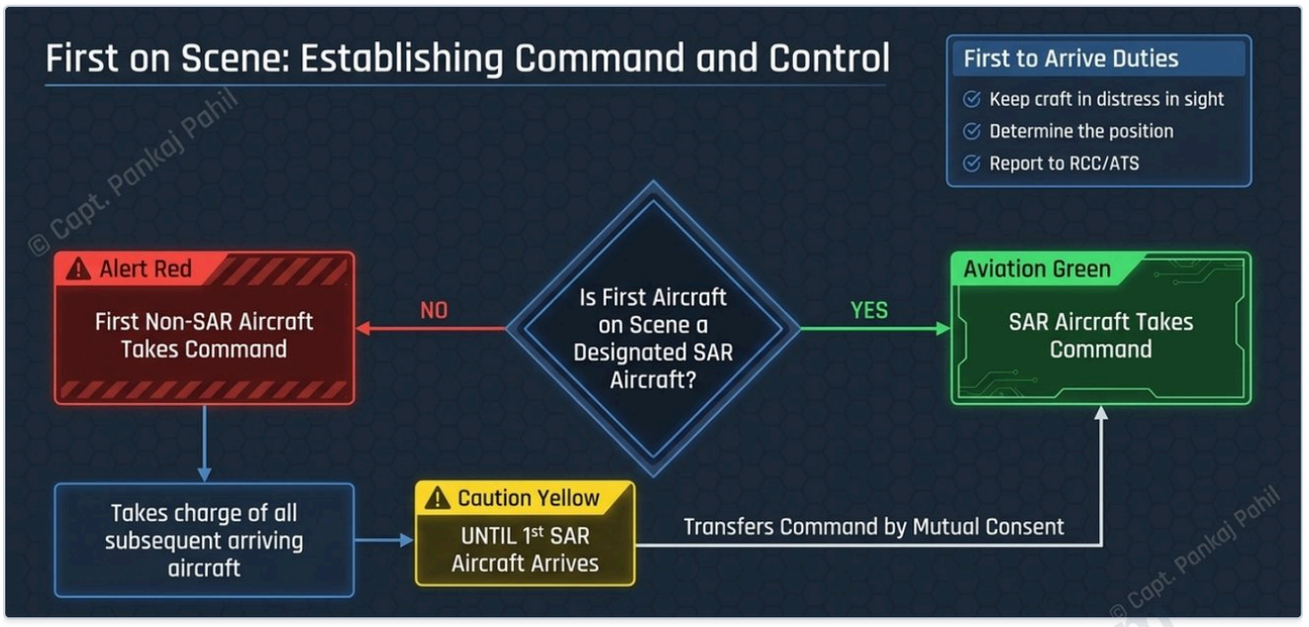

Figure 10.4 *First aircraft on the scene*

SAR Signals — Aircraft to Surface Craft

Annex 12 · Search and Rescue Signals

Aircraft Directing Surface Craft to a Distress Location

Signal Sequence — Meaning: "Follow Me to distress location"	The following manoeuvres performed **in sequence** by an aircraft mean that it wishes to direct a surface craft towards an aircraft or a surface craft in distress:

a. **Circling** the surface craft at least once.

b. **Crossing the projected course** of the surface craft close ahead at *low altitude*, and one of:

- **(1) Rocking the wings** — primary visual signal
- **(2) Opening and closing the throttle** — sound signal (alternative)
- **(3) Changing the propeller pitch** — sound signal (alternative)

c. **Heading in the direction** in which the surface craft is to be directed.

> **IMPORTANT NOTE ON SOUND SIGNALS**
>
> Due to high noise levels on board surface craft, the sound signals in (2) and (3) may be **less effective** than the visual signal in (1) and are regarded as **alternative means of attracting attention**.
>
> **Repetition of such manoeuvres has the same meaning.**

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Aircraft Signal: "Assistance No Longer Required"

SIGNAL MEANING ASSISTANCE IS NO LONGER REQUIRED

Crossing the wake of the surface craft close astern at a low altitude, and one of:

- Rocking the wings
- Opening and closing the throttle
- Changing the propeller pitch

Surface Craft Replies to Aircraft Signals

For Acknowledging Receipt of Signals (Meaning Understood)	
1	Hoisting the **"code pennant"** (vertical red and white stripes) **close up** — meaning: understood
2	Flashing a succession of **"T's"** by signal lamp in **Morse code**
3	**Changing heading to follow** the aircraft

For Indicating Inability to Comply	
1	Hoisting the international flag **"N"** (a blue and white checkered square)
2	Flashing a succession of **"N's"** in Morse code

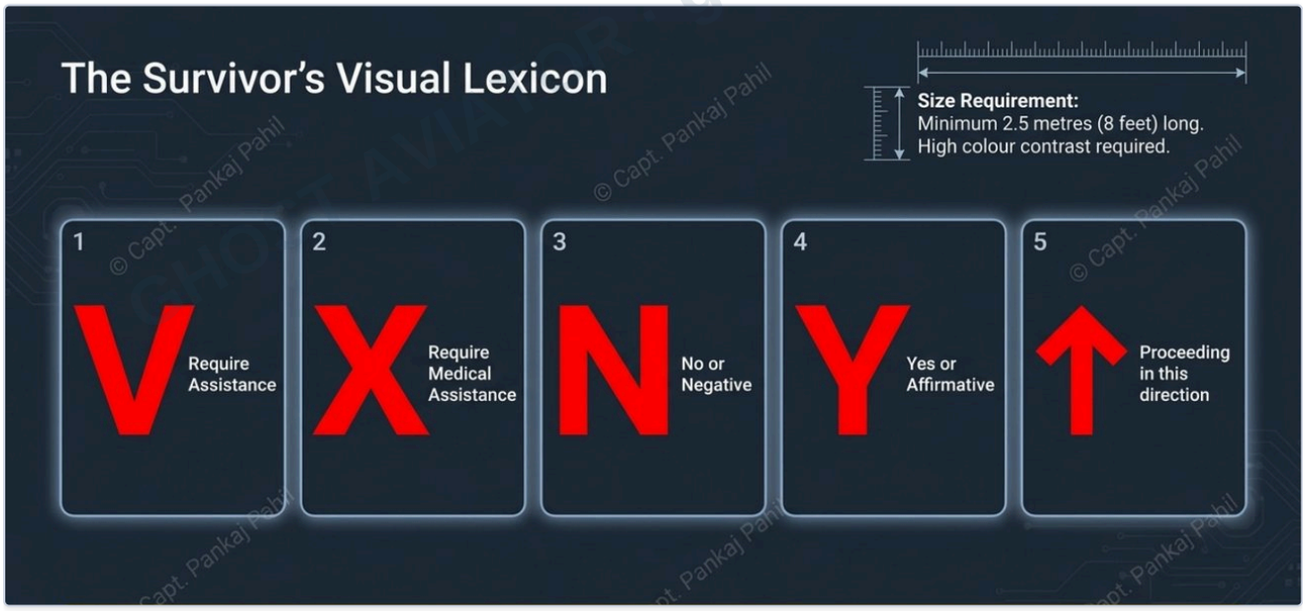

Figure 10.5 *Ground-air signals used by survivors*

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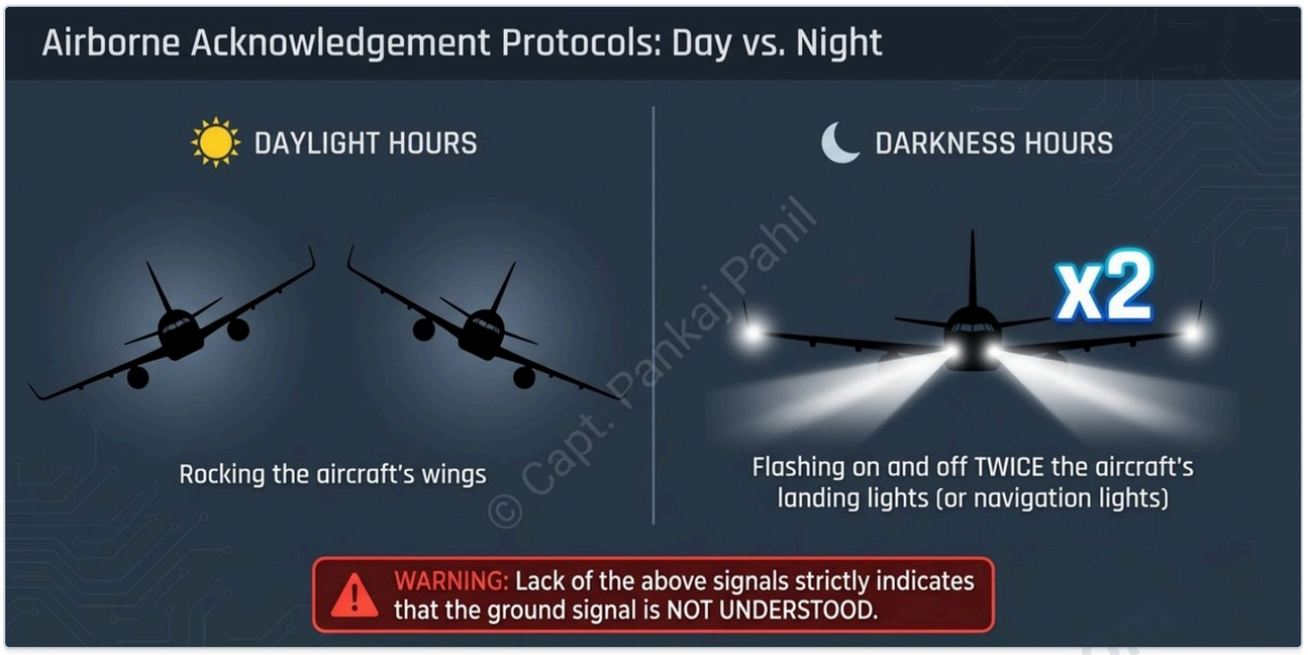

Figure 10.6 *How an aircraft shows it has understood*

Ground-Air Visual Signal Codes

Annex 12 · SAR Signals

Code for Use by SURVIVORS

No.	Message	Code Symbol
1	Require Assistance	V
2	Require Medical Assistance	X
3	No or Negative	N
4	Yes or Affirmative	Y
5	Proceeding in This Direction	↑

SYMBOL SIZE REQUIREMENT

Symbols shall be at least **2.5 metres (8 feet) long** and shall be made as conspicuous as possible. Provide as much colour contrast as possible between signals and background.

HOW SYMBOLS MAY BE FORMED

- Strips of fabric, parachute material, pieces of wood, stones or similar material
- Marking the surface by trampling or staining with oil
- Attention may also be attracted by: **radio, flares, smoke, and reflected light**

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Code for Use by RESCUE UNITS

No.	Message	Code Symbol
1	Operation Completed	LLL
2	We have found ALL personnel	LL
3	We have found ONLY SOME personnel	++
4	We are not able to continue. Returning to base.	XX
5	Have divided into two groups. Each proceeding in direction indicated.	⇄
6	Information received that aircraft is in this direction.	→ →
7	Nothing found. Will continue to search.	NN

Air-to-Ground Signals

Annex 12 · SAR Signals

Aircraft Acknowledging Ground Signals	The following signals by aircraft mean that the **ground signals have been understood**:

DURING DAYLIGHT HOURS

Rocking the aircraft's wings

A clear, visible acknowledgement that ground signals are understood.

DURING DARKNESS HOURS

Flashing on and off TWICE the aircraft's landing lights, or if not so equipped, by **switching on and off TWICE** its navigation lights.

LACK OF SIGNAL = NOT UNDERSTOOD

Lack of the above signals indicates that the **ground signal is NOT understood**.

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Satellite-Aided SAR — COSPAS-SARSAT

AIP India · Chapter 11 · Satellite-aided SAR

WHAT IS COSPAS-SARSAT?

India has evolved a **Satellite-aided Search and Rescue programme** by participating in the **COSPAS/SARSAT system**. It is a **SAR satellite-aided tracking system** that helps locate distress beacons.

406	MHz — Operating Frequency
5 km	Normal Location Accuracy

2	Local User Terminals (LUTs)

COSPAS-SARSAT INFRASTRUCTURE IN INDIA

- **2 Local User Terminals (LUTs)** established at: **Bangalore** and **Lucknow**
- **Indian Mission Control Centre (INMCC)** located at **Bangalore**
- INMCC is responsible for coordination with RCCs and other International Mission Control Centres
- INMCC at Bangalore is connected with RCCs at **Mumbai, Delhi, Kolkata and Chennai** through **AFS network**
- Any distress alert received from covered areas is **automatically transmitted** to the RCC

COUNTRIES COVERED BY INDIAN COSPAS-SARSAT DETECTION

The system detects transmissions throughout the Indian SRR and also SRR of:

Bangladesh Myanmar Bhutan Indonesia Kenya Malaysia Maldives Mauritius Nepal Seychelles Singapore Somalia Sri Lanka Thailand Tanzania

SAR Agreements

AIP India · Chapter 11

INDIA'S SAR ARRANGEMENT

- India currently has SAR Arrangement with **Royal Government of Bhutan**.
- India can seek SAR assistance from adjoining RCCs of other nations in accordance with **bi-lateral agreement**.
- For the purpose of SAR, authorities of other States who wish their SAR units to enter the territory of India shall **transmit a request, giving full details of the projected missions and the need for it**.

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INSTRUCTIONS

ANSWER: C

Recognised international distress frequencies include 121.5 MHz (civil), 243.0 MHz (military), 2182 kHz (maritime), and 406 MHz (COSPAS-SARSAT). 2430 kHz is NOT an internationally designated distress frequency.

ANSWER: C

Per the procedure for intercepting a distress transmission (step e): "At the pilot's discretion, while awaiting instructions, proceed to the position given in the transmission."

Quick Reference — Key Facts Summary

Examination Ready Recall Sheet

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Topic	Key Fact
SAR Service Hours	**24 hours** — no exception
SAR Regions	Must NOT overlap · Must be contiguous
India Aeronautical RCCs	Delhi · Mumbai · Kolkata · Chennai (4 total — AAI)
India Maritime MRCCs	Mumbai · Chennai · Port Blair (3 total — Indian Coast Guard)
Indian Aeronautical SRR	Coincides with Indian FIR boundary (excluding Bhutan)
Land SAR Authority	NASARCC — Secy, Ministry of Civil Aviation as Chairman
Maritime SAR Authority	NMSARB — DG, Indian Coast Guard as Chairman
COSPAS-SARSAT Frequency	**406 MHz**
COSPAS-SARSAT Accuracy	Normally within **5 km**
LUTs (Local User Terminals)	Bangalore and Lucknow
INMCC Location	Bangalore
SAR Dropped Container Colour	**Blue** streamers — food and water
Survivor "V"	Require Assistance
Survivor "X"	Require Medical Assistance
Survivor "N"	No / Negative
Survivor "Y"	Yes / Affirmative
Rescue Unit "LLL"	Operation Completed
Rescue Unit "LL"	Found ALL personnel
Rescue Unit "++"	Found ONLY SOME personnel
Rescue Unit "XX"	Cannot continue — returning to base
Rescue Unit "NN"	Nothing found — will continue to search
Air-to-Ground (Day)	Rock wings = ground signal understood
Air-to-Ground (Night)	Flash landing/nav lights TWICE = understood
Symbol Minimum Size	**2.5 metres (8 feet) long**
SAR Agreement	India has arrangement with Royal Government of Bhutan
Vessel Comms Frequencies	2182 kHz · 4125 kHz · 121.5 MHz

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CHAPTER 11

Visual Aids and Aerodrome Services

11

Indicators & Signalling Devices

Location sign: inscription in yellow on a black background — stand-alone signs shall have a yellow border.

Wind Direction Indicator (WDI)

> **APPLICATION — MANDATORY**
>
> An aerodrome **shall** be equipped with at least **one wind direction indicator**.

Location

A WDI shall be located so as to be:

- Visible from aircraft **in flight** or on the **movement area**
- Free from the effects of **air disturbances** caused by nearby objects

Characteristics

> **RECOMMENDATION**
>
> Provision should be made for **illuminating at least one wind indicator** at an aerodrome intended for use at night.

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Landing Direction Indicator (LDI)

Location

Where provided, a landing direction indicator shall be located in a **conspicuous place** on the aerodrome.

Characteristics

> **RECOMMENDATION**
>
> The landing direction indicator should be in the form of a **"T"** (Tee shape).

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Signaling Lamp

MANDATORY RULE

A signaling lamp **shall** be provided at a **controlled aerodrome** in the **aerodrome control tower**.

Characteristics

RECOMMENDATION

A signaling lamp should be capable of producing **red, green, and white** signals.

Signal Panels & Signal Area

Location of Signal Area

RECOMMENDATION

The signal area should be located so as to be visible from **all angles of azimuth above an angle of 10° above the horizontal** when viewed from a height of **300 m**.

Characteristics of Signal Area

DGCA RULE

- The signal area shall be an **even horizontal surface** at least **9 m square**.

RECOMMENDATIONS

- The colour of the signal area should contrast with the signal panels used
- It should be surrounded by a **white border** not less than **0.3 m wide**

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Runway Markings

General — Marking Colours & Conspicuity

COLOUR RULES — MANDATORY

- **Runway markings** — shall be **WHITE**
- **Taxiway markings, runway turn pad markings, and aircraft stand markings** — shall be **YELLOW**
- **Apron safety lines** — shall be of a conspicuous colour contrasting with aircraft stand markings.

Runway — White
Taxiway / Stand — Yellow

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NOTES ON RUNWAY MARKINGS

1. On **light-coloured** runway surfaces, white markings may be improved by **outlining them in black**.
2. It is preferable to use paints with **reduced friction variation** to minimise uneven friction characteristics.
3. Markings may consist of solid areas or **longitudinal stripes** providing an equivalent effect.

RECOMMENDATION — NIGHT OPERATIONS

At aerodromes where operations take place at night, pavement markings should be made with **reflective materials** to enhance visibility.

Interruption of Runway Markings at Intersections

RULE — INTERSECTION PRIORITY

At an intersection of two or more runways, the markings of the **more important runway** shall be displayed and those of the other runway(s) **interrupted**. Runway side stripe markings may be continued or interrupted.

ORDER OF IMPORTANCE AT INTERSECTIONS

1. **1st:** Precision approach runway
2. **2nd:** Non-precision approach runway
3. **3rd:** Non-instrument runway

RUNWAY × TAXIWAY INTERSECTIONS

At a runway–taxiway intersection: **runway markings shall be displayed; taxiway markings interrupted** (runway side stripe markings may be interrupted).

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Runway Designation Marking

APPLICATION — MANDATORY

A runway designation marking **shall** be provided at the **thresholds of a paved runway**.

RECOMMENDATION

A runway designation marking *should* be provided, so far as practicable, at the thresholds of an **unpaved runway**.

Location

The runway designation marking shall be located at a **threshold**, as per the appropriate figure.

NOTE

If the runway threshold is displaced from the extremity of the runway, a sign showing the designation may be provided for aeroplanes taking off.

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Characteristics — Designation Numbers & Letters

HOW RUNWAY NUMBERS ARE ASSIGNED

- Consists of a **two-digit number** (nearest one-tenth of the magnetic North — approach direction)
- If a single-digit result would occur, it shall be **preceded by a zero** (e.g., 09 not 9)
- On **parallel runways**, each number is supplemented by a letter:

Number of Parallel Runways	Letters Used (L to R when viewed on approach)
2 parallel	L, R
3 parallel	L, C, R
4 parallel	L, R, L, R
5 parallel	L, C, R, L, R or L, R, L, C, R
6 parallel	L, C, R, L, C, R

Runway Centre Line Marking

APPLICATION — MANDATORY

A runway centre line marking **shall** be provided on a **paved runway**.

Location

Located along the **centre line of the runway** between the runway designation markings, except when interrupted at the intersection of two or more runways.

Characteristics — Stripe & Gap Dimensions

Parameter	Specification
Length of stripe + gap	**Not less than 50 m**, not more than **75 m**
Length of each stripe	At least equal to the gap or **30 m**, whichever is greater
Width — CAT II & III precision approach runways	**≥ 0.90 m**
Width — Non-precision approach (code 3 or 4) & CAT I	**≥ 0.45 m**
Width — Non-precision approach (code 1 or 2) & Non-instrument	**≥ 0.30 m**

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Threshold Marking

APPLICATION — MANDATORY CONDITIONS

A threshold marking **shall** be provided at the threshold of:

- A **paved instrument runway**
- A **paved non-instrument runway** where code number is **3 or 4** AND intended for use by **international commercial air transport**

RECOMMENDATIONS

1. Should be provided at non-instrument runway (code 3 or 4) for *other than* international commercial air transport.
2. Should be provided, so far as practicable, at the thresholds of an **unpaved runway**.

Location

CRITICAL VALUE

Stripes of the threshold marking shall commence **6 m** from the threshold.

Characteristics — Number of Stripes vs. Runway Width

Runway Width	Number of Stripes (TOTAL, both sides of the centre line)
18 m	4
23 m	6
30 m	8
45 m	12
60 m	16

Exception: On non-precision approach and non-instrument runways ≥ 45 m wide, the optional pattern (Fig. 1C) may be used.

Transverse Stripe

RECOMMENDATION

Where a threshold is **displaced** from the extremity of the runway, or where the extremity is **not square** with the runway centre line, a **transverse stripe** should be added to the threshold marking.

RULE

A transverse stripe shall be **not less than 1.80 m wide**.

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Displaced Threshold — Arrows

MANDATORY RULES FOR DISPLACED THRESHOLDS

- **Permanently displaced:** Arrows conforming to Fig. 2(B) shall be provided on the runway portion *before* the displaced threshold.
- **Temporarily displaced:** Marked as per Fig. 2(A) or 2(B). All markings prior to the displaced threshold shall be **obscured** except the runway centre line marking, which shall be converted to **arrows**.

Aiming Point Marking

APPLICATION — MANDATORY

An aiming point marking **shall** be provided at each approach end of a paved **instrument runway where the code number is 2, 3 or 4**.

RECOMMENDATIONS — PROVIDE ALSO AT

- Paved **non-instrument** runway where code number is 3 or 4
- Paved **instrument** runway where code number is 1, when additional conspicuity is desirable

Location — Distance from Threshold

TABLE 1 — AIMING POINT MARKING: LOCATION & DIMENSIONS

Parameter	LDA < 800 m	800–1200 m (excl.)	1200–2400 m (excl.)	≥ 2400 m
Distance: Threshold → start of marking	150 m	250 m	300 m	400 m
Length of stripe*	30–45 m	30–45 m	45–60 m	45–60 m
Width of stripe	4 m	6 m	6–10 m**	6–10 m**
Lateral spacing (inner sides)	6 m***	9 m***	18–22.5 m	18–22.5 m

*Length shall be at least that of the gap. **Width shall be at least that of TDZ barrettes. ***Or the width of the runway if less.

SPECIAL RULE — VISUAL APPROACH SLOPE INDICATOR

On a runway equipped with a **VASIS/PAPI**, the beginning of the aiming point marking shall be **coincident with the visual approach slope origin**.

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Touchdown Zone (TDZ) Marking

APPLICATION — MANDATORY

A TDZ marking **shall** be provided in the touchdown zone of a paved **precision approach runway** where the code number is **2, 3 or 4**.

RECOMMENDATION

Should also be provided on a paved **non-precision approach or non-instrument runway** (code 3 or 4) where additional conspicuity is desirable.

Number of TDZ Marking Pairs vs. LDA / Distance Between Thresholds

Landing Distance Available (or Distance Between Thresholds)	Pairs of Markings
Less than 900 m	1
900 m up to but not including 1 200 m	2
1 200 m up to but not including 1 500 m	3
1 500 m up to but not including 2 400 m	4
2 400 m or more	6

Characteristics

A TDZ marking consists of **pairs of rectangular markings** symmetrically disposed about the runway centre line. Where marking is displayed at both approach directions, the number is based on the **distance between thresholds**.

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Runway Side Stripe Marking

APPLICATION

Shall be provided between the thresholds of a paved runway where there is a **lack of contrast** between the runway edges and the shoulders or surrounding terrain.

RECOMMENDATION

Should be provided on a **precision approach runway irrespective of contrast**.

- Should consist of **two stripes**, one along each edge, with the outer edge of each stripe approximately on the edge of the runway
- Where runway is > 60 m wide, stripes should be located **30 m from the runway centre line**
- Width: at least **0.9 m** on runways ≥ 30 m wide; at least **0.45 m** on narrower runways

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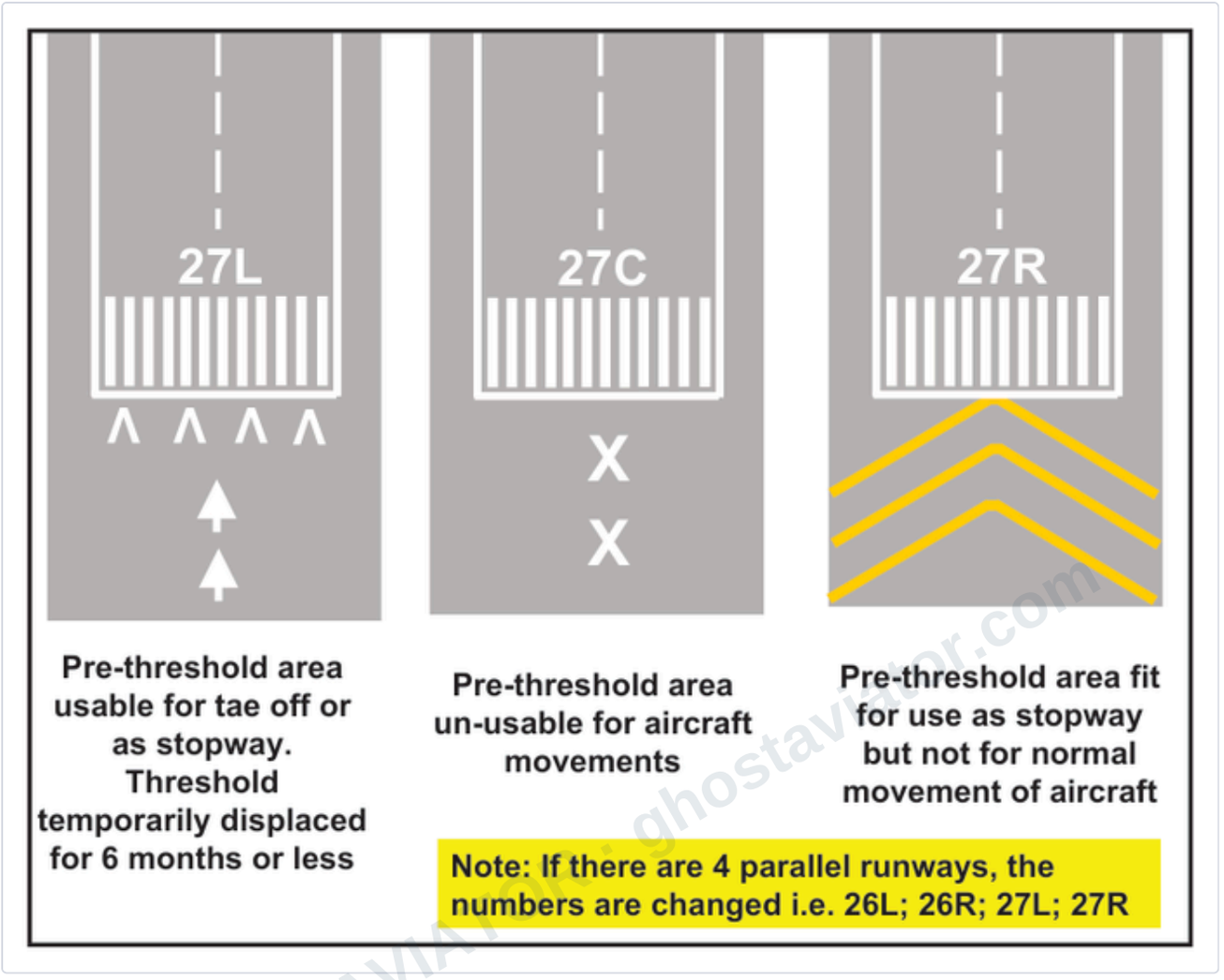

Figure 11.1 *Runway designation markings on parallel runways (L, C, R) with the pre-threshold area; with more parallel runways the numbers themselves are changed.*

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Taxiway Markings

Taxiway Centre Line Marking

MANDATORY APPLICATION — CODE 3 OR 4

Taxiway centre line marking **shall** be provided on a paved taxiway, de-icing/anti-icing facility and apron where the **code number is 3 or 4**, providing continuous guidance between the runway centre line and aircraft stands.

RECOMMENDATION — CODE 1 OR 2

Should be provided on a paved taxiway, de-icing/anti-icing facility and apron where the **code number is 1 or 2** for continuous guidance.

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MANDATORY — ON RUNWAYS AS PART OF TAXI-ROUTE

Taxiway centre line marking shall be provided on a paved runway when the runway is part of a standard taxi-route and:

- (a) there is **no runway centre line marking**, or
- (b) where the taxiway centre line is **not coincident** with the runway centre line

Characteristics

WIDTH REQUIREMENT

A taxiway centre line marking shall be at least **15 cm** in width and **continuous in length** except where it intersects with a runway-holding position marking or intermediate holding position marking.

Enhanced Taxiway Centre Line Marking

RECOMMENDATIONS — ENHANCED MARKING

1. Where it is necessary to denote proximity of a **runway-holding position**, enhanced taxiway centre line marking should be provided. It shall extend from the runway-holding position Pattern A to a distance of up to **47 m** in the direction of travel away from the runway.
2. On a straight section of taxiway, the marking should be located along the taxiway centre line. On a curve, it should continue from the straight portion at a **constant distance from the outside edge** of the curve.
3. At an intersection of a taxiway with a runway where the taxiway serves as an exit from the runway, the taxiway centre line marking should be **curved into the runway centre line marking**.

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Isolated Aircraft Parking Position

MANDATORY RULE

An isolated aircraft parking position **shall** be designated, or the aerodrome control tower shall be advised of an area or areas suitable for parking of an aircraft which is known or believed to be the subject of **unlawful interference**, or which for other reasons needs isolation from normal aerodrome activities.

RECOMMENDATION

The isolated parking position should be located at the **maximum distance practicable** and in any case **never less than 100 m** from other parking positions, buildings or public areas. Care should be taken to ensure no underground utilities (gas, aviation fuel, electrical, communication cables) are beneath.

Runway-Holding Position Marking

MANDATORY RULE

A runway-holding position marking **shall** be displayed along a runway-holding position.

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Characteristics — Pattern A and Pattern B

PATTERN SELECTION RULES

- At intersection of taxiway and **non-instrument, non-precision approach or take-off runway**: **Pattern A**
- Where a single holding position exists at intersection of a taxiway and a **precision approach CAT I, II or III runway**: **Pattern A**
- Where two or three holding positions exist at such an intersection — those **closer to the runway: Pattern A**; those **further: Pattern B**
- At a position established to avoid infringement of obstacle limitation surface or interference with radio/navaids: **Pattern A**

PATTERN A AND PATTERN B

The marking nearest the runway is **Pattern A** and a marking farther from the runway is **Pattern B** (CAR Section 4, Series B, Part I, para 5.2.10 and Figure 5-6).

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PATTERN SPECIFICATIONS SUMMARY

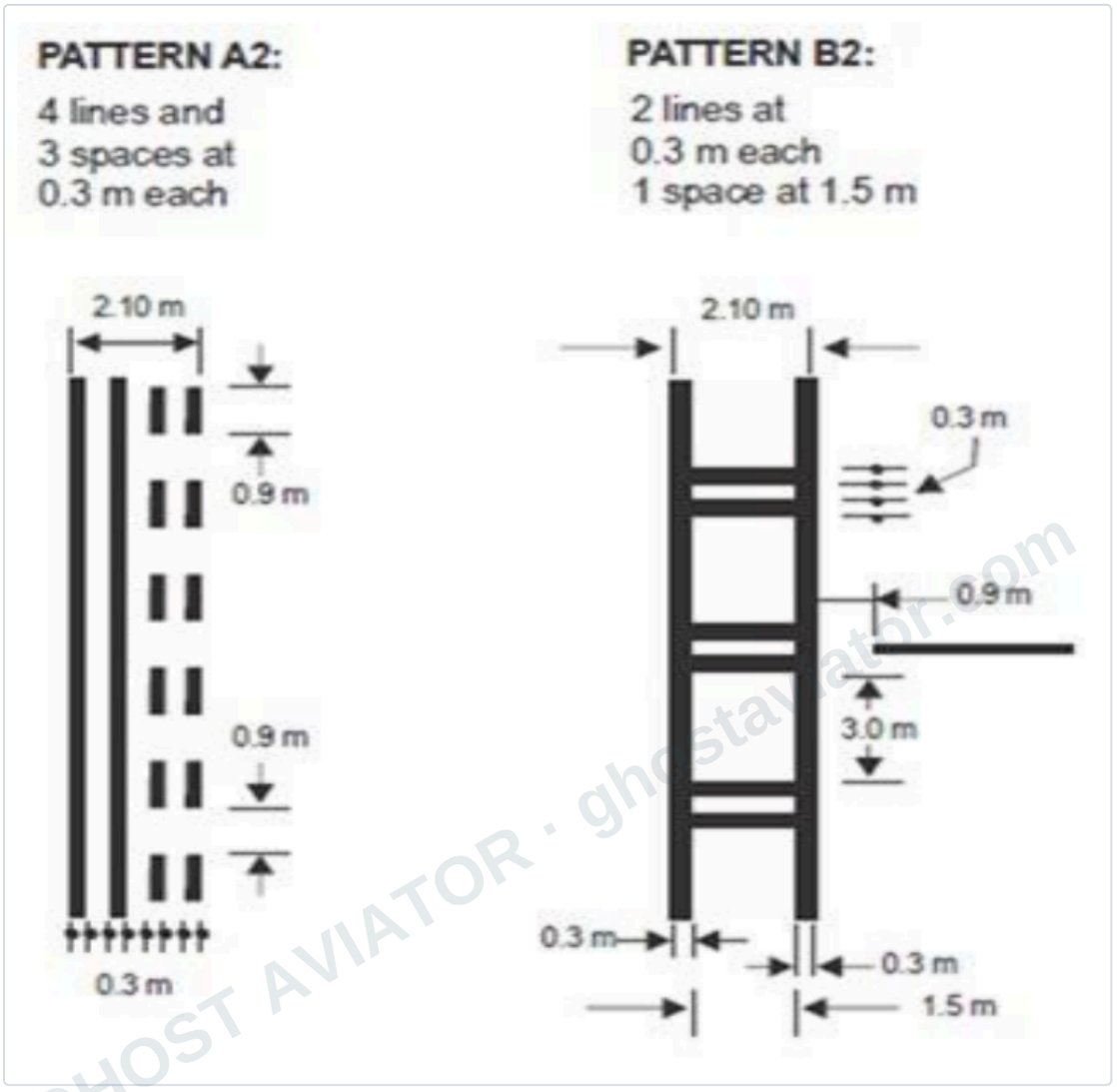

Figure 11.2 *Runway-holding position marking dimensions: PatternsA 2 and B2. DGCA CAR 4-B-I, §5.2.10.5 and Figure 5-8, p 124.*

Intermediate Holding Position (IHP) Marking

RECOMMENDATIONS — APPLICATION & LOCATION

1. An IHP marking should be displayed along an **intermediate holding position**.
2. Should be displayed at the **exit boundary of a remote de-icing/anti-icing facility** adjoining a taxiway.

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VOR Aerodrome Checkpoint Marking

CHARACTERISTICS — MANDATORY

A VOR aerodrome checkpoint marking shall consist of:

- A circle **6 m in diameter** with a line width of **15 cm**

RECOMMENDATIONS

1. When aircraft alignment in a specific direction is preferred, a **direction line** should be provided — passing through the centre of the circle, extending **6 m** outside the circle in the desired heading direction, terminating in an **arrowhead**. Line width: **15 cm**.
2. The marking should preferably be **white**, differing from taxiway marking colour.

Aircraft Stand Markings

RECOMMENDATIONS

- Should be provided for designated parking positions on a **paved apron** and on a **de-icing/anti-icing facility**.
- Should include elements such as: **stand identification, lead-in line, turn bar, turning line, alignment bar, stop line, lead-out line**, as required.
- Stand identification (letter and/or number) should be included in the lead-in line, in a short distance after the beginning. Height should be adequate to be readable from the **cockpit** of aircraft using the stand.
- Lead-in, turning, and lead-out lines should be **continuous** and have a width of **not less than 15 cm**.

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Apron Safety Lines

LOCATION — MANDATORY

Apron safety lines shall be located so as to define the areas intended for use by **ground vehicles and other aircraft servicing equipment** to provide safe separation from aircraft.

RECOMMENDATIONS

- Should include wingtip clearance lines and service road boundary lines as required.
- Should be **continuous** in length and at least **10 cm in width**.

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Roadholding Position Marking

MANDATORY RULES

- **Application:** A roadholding position marking **shall** be provided at **all road entrances to a runway**.
- **Location:** Shall be located across the road **at the holding position**.
- **Characteristics:** Shall be in accordance with **local road traffic regulations**.

Mandatory Instruction Marking

APPLICATION — MANDATORY

Where it is **impracticable to install a mandatory instruction sign**, a mandatory instruction marking **shall** be provided on the surface of the pavement.

RECOMMENDATION — WHEN TO SUPPLEMENT

Where operationally required, such as on taxiways exceeding **60 m in width**, a mandatory instruction sign should be supplemented by a mandatory instruction marking.

Location

RULES

- Located on the **left-hand side** of the taxiway centre line marking
- On the **holding side** of the runway-holding position marking
- Distance between the nearest edge of the marking and the runway-holding position marking or taxiway centre line marking shall be **not less than 1 m**

RECOMMENDATION

A mandatory instruction marking should **not** be located on a runway (except where operationally required).

Characteristics

APPEARANCE RULES

- Inscription in **white on a red background**
- Except for a **NO ENTRY** marking: inscription identical to the associated mandatory instruction sign
- A **NO ENTRY** marking shall consist of the inscription "NO ENTRY" in white on a red background
- Where there is insufficient contrast between marking and pavement, an appropriate border (preferably **white or black**) shall be included

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RECOMMENDATIONS

1. Character height should be **4 m**
2. Background should be rectangular and extend a maximum of **0.5 m** laterally and vertically beyond the extremities of the inscription

Information Marking

APPLICATION — MANDATORY

Where an information sign would normally be installed and it is physically impossible to install a sign, an information marking **shall** be displayed on the surface of the pavement.

RECOMMENDATION ON LOCATION

The information marking should be displayed across the surface of the taxiway or apron where necessary, and positioned so as to be **legible from the cockpit** of an approaching aircraft.

Characteristics

INFORMATION MARKING COLOURS

- (a) Inscription in **yellow**, when it replaces or supplements a **location sign**
- (b) Inscription in **black**, when it replaces or supplements a **direction or destination sign**
- Where insufficient contrast — (a) black background with yellow inscriptions; (b) yellow background with black inscriptions

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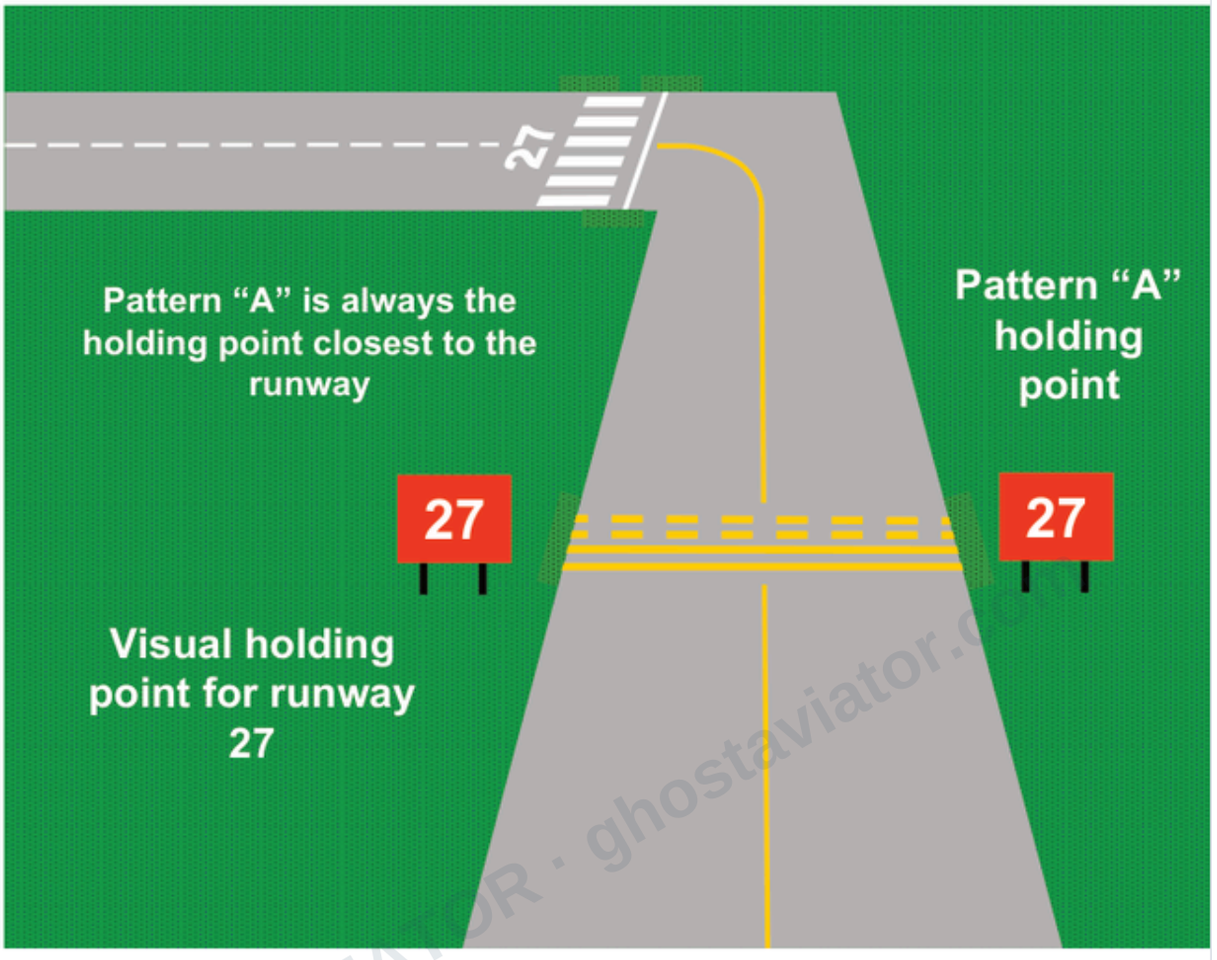

Figure 11.3 *Runway-holding position, Pattern A: the holding position closest to the runway (with its sign).*

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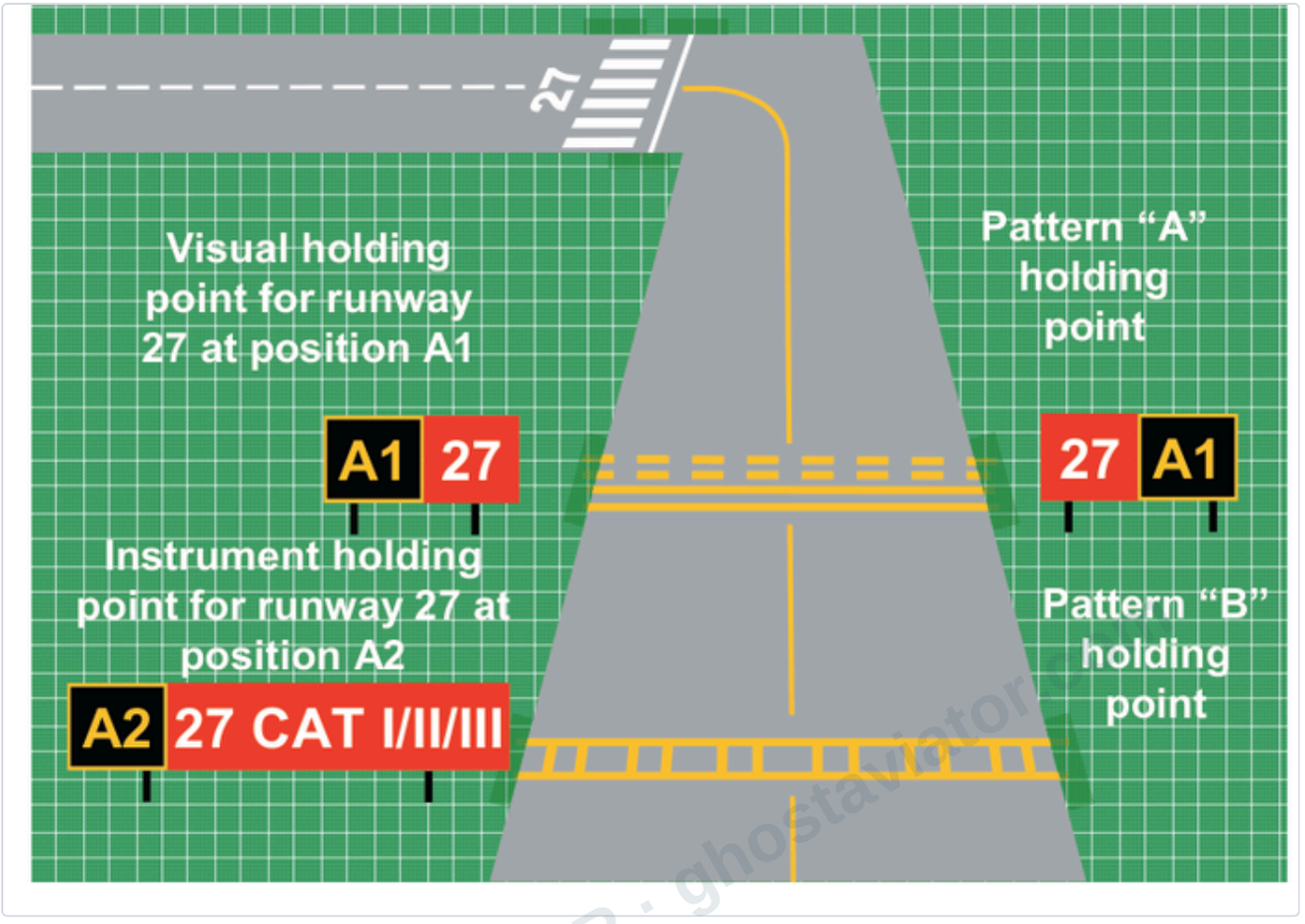

Figure 11.4 *Pattern B: the holding position further from the runway; Pattern A stays closest.*

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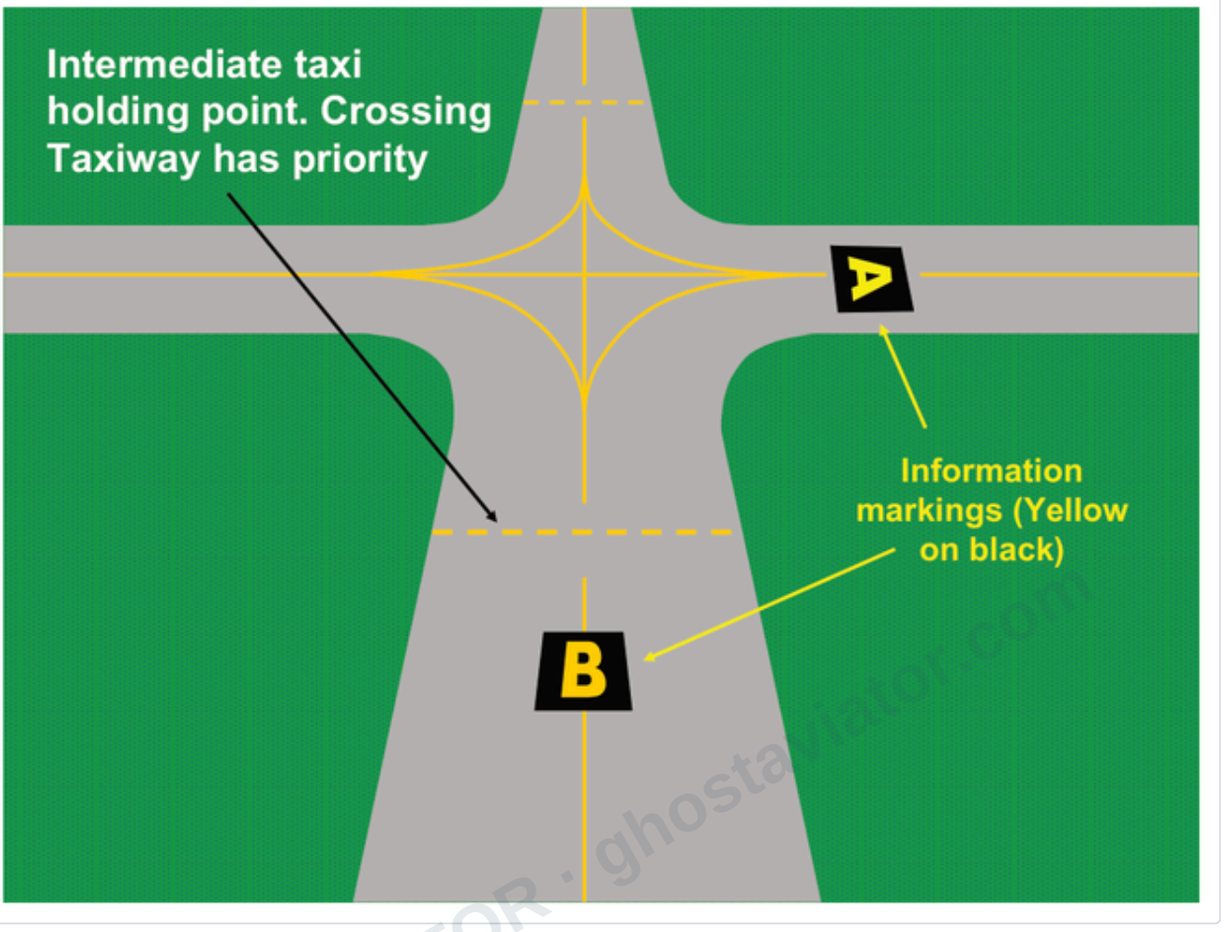

Figure 11.5 *Intermediate holding position marking at a taxiway intersection.*

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Lights: General & Beacons

Laser Emissions — Endanger Safety of Aircraft

LASER PROTECTED ZONES

To protect aircraft safety against hazardous effects of laser emitters, the following protected zones should be established around aerodromes:

- **LFFZ** — Laser-Beam Free Flight Zone
- **LCFZ** — Laser-Beam Critical Flight Zone
- **LSFZ** — Laser-Beam Sensitive Flight Zone

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Non-Aeronautical Lights Near Aerodrome

SAFETY RULE

A non-aeronautical ground light near an aerodrome which might endanger the safety of aircraft **shall** be extinguished, screened or otherwise modified to eliminate the source of danger.

LIGHTS WHICH MAY CAUSE CONFUSION — ZONES OF ATTENTION

Particular attention should be directed to non-aeronautical ground lights visible from the air within these areas:

- **(a) Instrument runway code 4:** Before the threshold and beyond the runway end, extending at least **4500 m in length** from threshold and runway end, and **750 m either side** of the extended runway centre line
- **(b) Instrument runway code 2 or 3:** As (a), except length should be at least **3000 m**
- **(c) Instrument runway code 1 and non-instrument runway:** Within the approach area

Emergency Lighting

RECOMMENDATION — APPLICATION

At an aerodrome provided with runway lighting and **without a secondary power supply**, sufficient emergency lights should be conveniently available for installation on at least the **primary runway** in the event of failure of the normal lighting system.

Note: Emergency lighting may also be useful to mark obstacles or delineate taxiways and apron areas.

LOCATION

When installed on a runway, the emergency lights should, as a minimum, conform to the configuration required for a **non-instrument runway**.

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Aerodrome Beacon

APPLICATION — WHEN REQUIRED (NIGHT OPERATIONS)

An aerodrome beacon **shall** be provided at an aerodrome intended for use at night if one or more of the following conditions exist:

- (a) Aircraft navigate predominantly by visual means
- (b) Reduced visibilities are frequent
- (c) It is difficult to locate the aerodrome from the air due to surrounding lights or terrain

LOCATION

The aerodrome beacon shall be located on or adjacent to the aerodrome in an area of **low ambient background lighting**.

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LOCATION RECOMMENDATION

The beacon should not be shielded by objects in significant directions and should **not dazzle a pilot approaching to land**.

Characteristics — Flash Colours & Frequency

MANDATORY SPECIFICATIONS

- Shall show either **coloured flashes alternating with white flashes**, or **white flashes only**
- Frequency of total flashes: **20 to 30 per minute**
- **Land aerodrome beacons:** Coloured flashes shall be **green**
- **Water aerodrome beacons:** Coloured flashes shall be **yellow**
- **Combined water-and-land aerodrome:** Colour characteristics of whichever section is the principal facility

Identification Beacon

CHARACTERISTICS

- **Land aerodrome:** Shall show **flashing green**
- **Water aerodrome:** Shall show **flashing yellow**
- Characters transmitted in **International Morse Code**

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Approach Lighting Systems

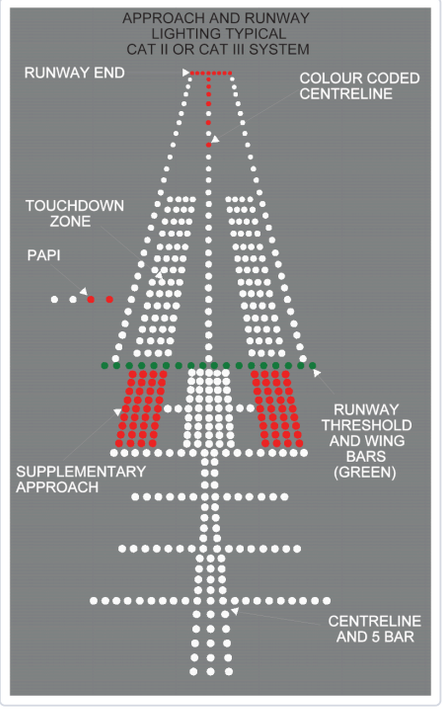

Figure 11.6 *Approach and runway lighting for a Category II or III runway.*

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Application — When to Provide Approach Lighting

APPROACH LIGHTING REQUIREMENTS BY RUNWAY TYPE

Runway Type	Requirement	System Type
Non-instrument runway (code 3/4, night, not good visibility)	RECOMMENDATION	Simple Approach Lighting System
Non-precision approach runway	MANDATORY (where physically practicable)	Simple Approach Lighting System
Precision approach runway CAT I	MANDATORY (where physically practicable)	CAT I Lighting System
Precision approach runway CAT II & III	MANDATORY	CAT II & III Lighting System

Simple Approach Lighting System (SALS)

Location & Configuration

KEY DIMENSIONS

- Row of lights on the **extended centre line of the runway**
- Total length: at least **420 m** from the threshold
- Crossbar: **18 m or 30 m** in length, located at **300 m** from the threshold
- Gaps in a 30 m crossbar shall be kept to a minimum, not exceed **6 m**
- Centre line lights at longitudinal intervals of **60 m** (may be reduced to **30 m** to improve guidance)
- Innermost light located either **60 m or 30 m** from the threshold

ILS/MLS ANTENNA — OBSTACLE RULE

No object other than an **ILS or MLS azimuth antenna** shall protrude through the plane of the approach lights within a distance of **60 m** from the centre line. Any protruding antenna shall be treated as an obstacle and marked/lighted accordingly.

Characteristics

LIGHT TYPE

- Fixed lights
- Colour: shall ensure system is **readily distinguishable from other aeronautical ground lights and extraneous lighting**
- Each centre line light shall consist of either: (a) a **signal source**, or (b) a **barrette at least 3 m in length**

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Precision Approach CAT I Lighting System

Location & Configuration

KEY DIMENSIONS — CAT I

- Row of lights on extended centre line extending over a distance of **900 m** from the runway threshold
- Crossbar: **30 m in length** at **300 m** from the runway threshold
- Centre line lights at longitudinal intervals of **30 m**, innermost light at **30 m** from threshold
- Crossbar spacing: **1 m to 4 m** (gaps on each side of centre line may improve directional guidance)

OPERATIONAL LIMITATION NOTE

Installation of an approach lighting system of **less than 900 m in length may result in operational limitations** on the use of the runway.

Characteristics — CAT I

LIGHT TYPE & CONFIGURATION

- Centre line and crossbar lights: **fixed lights showing variable white**
- Each centre line light position shall consist of either:
  - (a) Signal light sources — 1 source in innermost 300 m, 2 sources in central 300 m, 3 sources in outer 300 m (distance information)
  - (b) A **barrette**
- If barrettes are used, each shall be supplemented by a **flashing light** (unless unnecessary given system characteristics and meteorological conditions)
- Flashing lights: flashed **twice a second in sequence**, beginning with the **outermost light progressing toward the threshold** (innermost light)

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Precision Approach CAT II & III Lighting System

Location & Configuration

KEY DIMENSIONS — CAT II & III

- Row of lights on extended centre line extending over a distance of **900 m** from the runway threshold
- Plus two **side rows of lights** extending **270 m** from the threshold
- Two crossbars: one at **150 m** and one at **300 m** from the threshold
- Centre line and side row lights at longitudinal intervals of **30 m**, with innermost lights at **30 m** from threshold
- Crossbar at **150 m** fills in the gaps between centre line and side row lights
- Crossbar at **300 m** extends on both sides to a distance of **15 m** from the centre line

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Characteristics — CAT II & III

CENTRE LINE & SIDE ROW SPECIFICATIONS

- **Centre line (inner 300 m from threshold):** Barrettes showing **variable white** (where threshold is displaced ≥ 300 m, signal light sources showing variable white may be used)
- **Centre line (beyond 300 m):** Either a barrette as used in inner 300 m, OR two/three light sources showing variable white
- **Side rows:** Barrettes showing **red**
- Side row barrette length and spacing = equal to those of touchdown zone light barrettes
- Each barrette supplemented by a **flashing light** (unless unnecessary)
- Flashing lights: flashed **twice a second in sequence**, from outermost to innermost

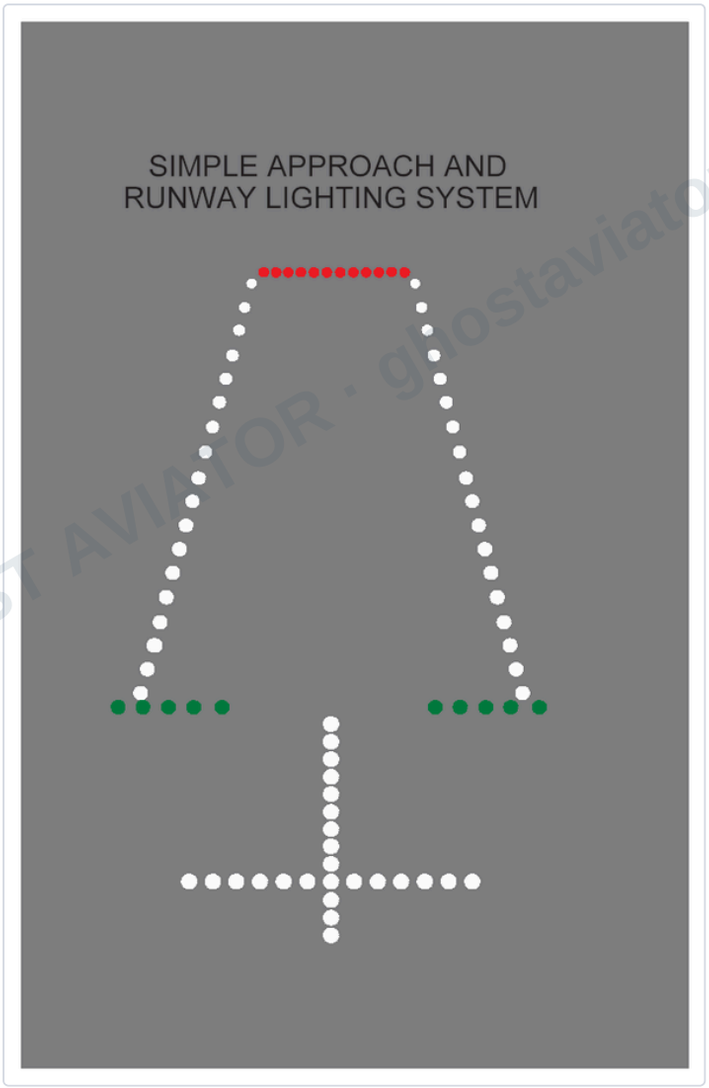

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Figure 11.7 *Simple approach lighting: a centre line row with a crossbar, threshold lights in green, and runway end lights in red.*

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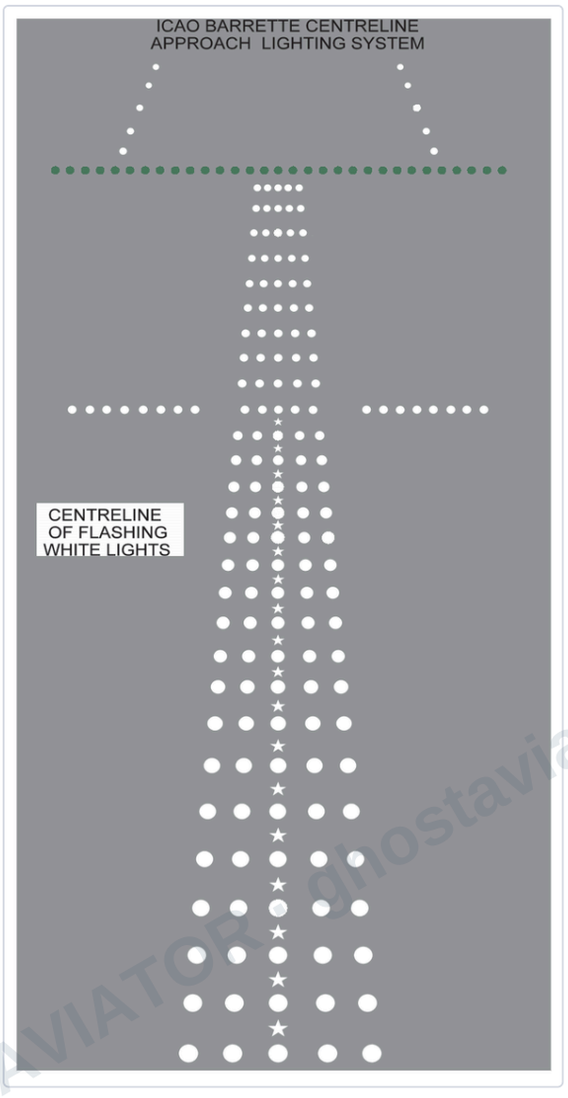

Figure 11.8 *Precision approach (barrette centre line) lighting for CAT I, with a crossbar and green threshold bar.*

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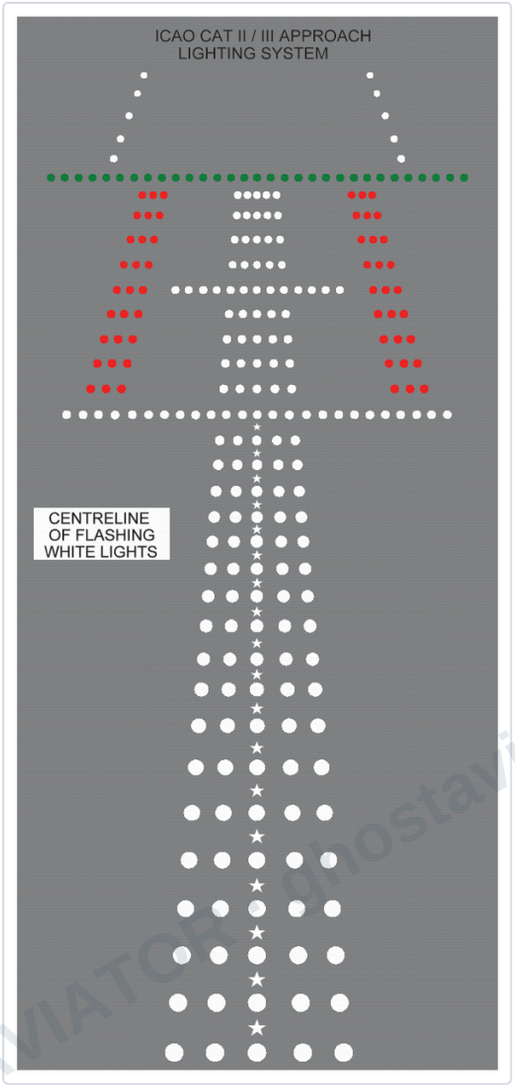

Figure 11.9 *CAT II / III approach lighting: red side rows near the threshold and extra crossbars.*

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Visual Approach Slope Indicator Systems (PAPI & APAPI)

When VASIS Must Be Provided

APPLICATION — VASIS REQUIRED WHEN

- (a) The runway is used by **turbojet or other aeroplanes with similar approach guidance requirements**
- (b) A pilot of any type of aeroplane may have difficulty judging the approach due to:
  - (i) Inadequate visual guidance over water/featureless terrain by day, or absence of sufficient extraneous lights at night
  - (ii) Misleading information from deceptive surrounding terrain or runway slopes
- (c) Presence of objects in the approach area may involve serious hazard if an aeroplane descends below the normal approach path

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- (d) Physical conditions at either end of the runway present a serious hazard in the event of an aeroplane **undershooting or overrunning** the runway
- (e) Terrain or meteorological conditions are such that the aeroplane may be subjected to **unusual turbulence during approach**

STANDARD SYSTEMS

The standard visual approach slope indicator systems shall consist of **PAPI** and **APAPI** systems.

PAPI — Precision Approach Path Indicator

DESCRIPTION

- Consists of a **wing bar of FOUR sharp transition multi-lamp (or paired single lamp) units** equally spaced
- Located on the **left side of the runway** unless physically impracticable

*Note: Where a runway is used by aircraft requiring visual roll guidance not provided by other external means, a second wing bar may be provided on the **opposite side**.*

PAPI — What the Pilot Sees

W W W W	TOO HIGH
R W W W	SLIGHTLY HIGH
R R W W	ON GLIDE PATH
R R R W	SLIGHTLY LOW
R R R R	TOO LOW

Light order above: unit nearest runway (leftmost) to unit farthest from runway (rightmost)

PAPI — FORMAL RULES

- **On or close to approach slope:** Two units nearest runway = **red**; two units farthest = **white**
- **Above approach slope:** One unit nearest runway = red; three units farthest = white. Further above: **all white**
- **Below approach slope:** Three units nearest runway = red; one unit farthest = white. Further below: **all red**

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APAPI — Abbreviated Precision Approach Path Indicator

DESCRIPTION

- Consists of a **wing bar of TWO sharp transition multi-lamp (or paired single lamp) units**
- Located on the **left side of the runway** unless physically impracticable

Note: Second wing bar on opposite side may be provided if visual roll guidance is required.

APAPI — WHAT THE PILOT SEES

- **On or close to approach slope:** Unit nearer the runway = **red**; unit farther from runway = **white**
- **Above approach slope:** Both units = **white**
- **Below approach slope:** Both units = **red**

LIGHT UNIT CHARACTERISTICS

The PAPI and APAPI systems shall be suitable for both **day and night operations**.

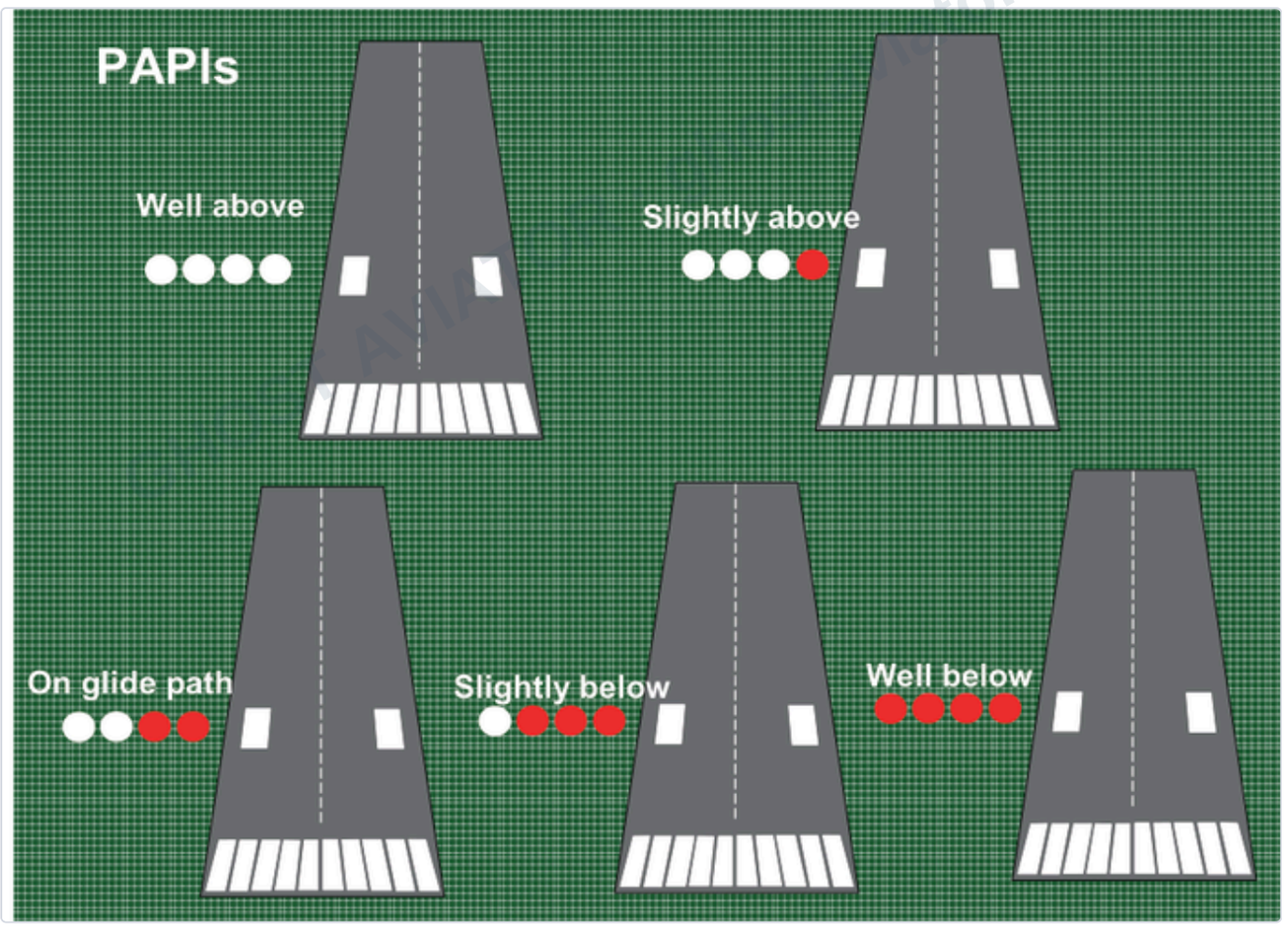

Figure 11.10 *PAPI indications from well above to well below the glide path (on slope: two white, two red).*

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Runway Lighting

Circling Guidance Lights

RULES

- Shall be provided when existing approach and runway lighting systems do **not satisfactorily permit identification of the runway and/or approach area** to a circling aircraft in the conditions for which it is intended.
- Shall be **fixed or flashing** lights of an intensity and beam spread adequate for the visibility and ambient light conditions.
- Flashing lights shall be **white**; steady lights either **white or gaseous discharge lights**.

Runway Lead-in Lighting System

RECOMMENDATION

Should be provided where it is desired to provide visual guidance along a specific approach path (e.g., avoiding hazardous terrain, noise abatement).

- Each group: at least **three flashing lights** in a linear or cluster configuration
- Flashing lights and steady burning light shall be **white**
- Where practicable, flashing lights in each group shall **flash in sequence towards the runway**

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Runway Threshold Identification Lights (RTIL)

RECOMMENDATION — INSTALL WHEN

- (a) At the threshold of a **non-precision approach runway** when additional threshold conspicuity is necessary or where it is not practicable to provide other approach lighting aids
- (b) Where a runway threshold is **permanently or temporarily displaced** and additional threshold conspicuity is necessary

LOCATION & CHARACTERISTICS

- Located **symmetrically about the runway centre line**, in line with the threshold and approximately **10 m outside** each line of runway edge lights
- Should be **flashing white lights** with a flash frequency between **60 and 120 per minute**
- Lights shall be visible **only in the direction of approach** to the runway

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Runway Edge Lights

APPLICATION — MANDATORY

- Shall be provided for a runway intended for **use at night**
- Shall be provided for a **precision approach runway** intended for use by day or night

RECOMMENDATION

Should be provided on a runway intended for takeoff with an operating minimum below an **RVR of the order of 800 m by day**.

Location

POSITIONING RULES

- Along the **full length of the runway**, in two parallel rows equidistant from the centre line
- Along the edges of the area declared for use as the runway or **outside the edges at a distance of not more than 3 m**
- Uniformly spaced at intervals of **not more than 60 m** for an instrument runway
- Uniformly spaced at intervals of **not more than 100 m** for a non-instrument runway

WHERE RUNWAY WIDTH > 60 M

The distance between the rows of lights should be determined taking into account the nature of the operations, the light distribution characteristics, and other visual aids serving the runway.

Characteristics — Colours

RUNWAY EDGE LIGHT COLOURS

Fixed lights showing **variable white**, EXCEPT:

- (a) In the case of a **displaced threshold**, the lights between the beginning of the runway and the displaced threshold shall show **red** in the approach direction
- (b) A section of lights **600 m** or **one-third of the runway length**, whichever is less, at the remote end of the runway (from the end where takeoff run is started), **may show yellow**

Variable White (Normal)
Red (Displaced threshold — approach dir.)
Yellow (Last 600 m or 1/3 — takeoff end)

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Runway Threshold & Wing Bar Lights

Application of Threshold Lights

MANDATORY

Runway threshold lights **shall** be provided for a runway equipped with runway edge lights, except on a **non-instrument or non-precision approach runway where the threshold is displaced** and wing bar lights are provided.

Location of Threshold Lights

RULES

- **Threshold at extremity of runway:** Lights placed in a row at right angles to the runway axis, as near to the extremity as possible and in any case **not more than 3 m outside the extremity**
- **Displaced threshold:** Lights placed at right angles to the runway axis at the displaced threshold
- **Number — Non-instrument or Non-precision approach (Cat. I incl.):**
  - At least **six lights**, equally spaced between the rows of runway edge lights, or symmetrically disposed in two groups with a gap equal to the TDZ marking/lighting gauge or no more than half the distance between edge light rows
- **Number — CAT II or III:** Uniformly spaced between the rows of runway edge lights at intervals of **not more than 3 m**

Wing Bar Lights

RECOMMENDATION — WING BAR LIGHTS

Should be provided on a **precision approach runway when additional conspicuity is desirable**. Shall be provided on a non-instrument or non-precision approach runway where the threshold is displaced and runway threshold lights are required but are not provided.

WING BAR LIGHTS — LOCATION

- Symmetrically disposed about the runway centre line at the threshold in two groups (wing bars)
- Each wing bar formed by **at least five lights** extending at least **10 m outward** from and at right angles to the line of runway edge lights
- Innermost light of each wing bar in the line of runway edge lights

Characteristics — Threshold & Wing Bar Lights

COLOUR

Fixed **unidirectional lights showing green** in the direction of approach to the runway.

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Runway End Lights

APPLICATION & LOCATION

- Shall be provided for a runway equipped with runway edge lights
- *Note: When threshold is at the runway extremity, fittings serving as threshold lights may be used as runway end lights.*
- Placed on a line at right angles to the runway axis as near to the end of the runway as possible, in any case **not more than 3 m outside the end**

NUMBER — RECOMMENDATION

At least **six lights**, either equally spaced between the rows of runway edge lights, or symmetrically disposed in two groups with a gap of not more than half the distance between edge light rows.

For **CAT III runway**: Spacing between runway end lights, except between two innermost lights if a gap is used, should **not exceed 6 m**.

Characteristics

COLOUR

Fixed **unidirectional lights showing red** in the direction of the runway.

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Runway Centre Line Lights

APPLICATION — MANDATORY

- Shall be provided on a **precision approach runway category II or III**
- Shall be provided on a runway intended for use for takeoff with an operating minimum below an **RVR of the order of 400 m**

RECOMMENDATIONS — PROVIDE ALSO ON

- Precision approach runway category I, particularly when used by aircraft with **high landing speeds** or where width between edge lights > 50 m
- Runway for takeoff with RVR minimum **400 m or higher** when used by aeroplanes with **very high takeoff speed**, particularly where width between edge lights > 50 m

Location

SPACING & OFFSET RULES

- Along the centre line of the runway, from the threshold to the end
- May be uniformly offset to the same side of the runway centre line by **not more than 60 cm** where not practicable to locate them on the marking
- Longitudinal spacing: **approximately 15 m**
- *Note: Existing centre line lighting spaced at 7.5 m need not be replaced.*

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Characteristics — Runway Centre Line Light Colours

COLOUR CODING — CRITICAL FOR EXAM!

Section	Colour
From threshold to point **900 m from the runway end**	Variable White
From **900 m to 300 m from the runway end**	Alternate Red & Variable White
Last **300 m to runway end**	Red
Exception: For runways less than 1800 m in length, the alternate red and variable white lights shall extend from the midpoint of the runway usable for landing to 300 m from the runway end.	

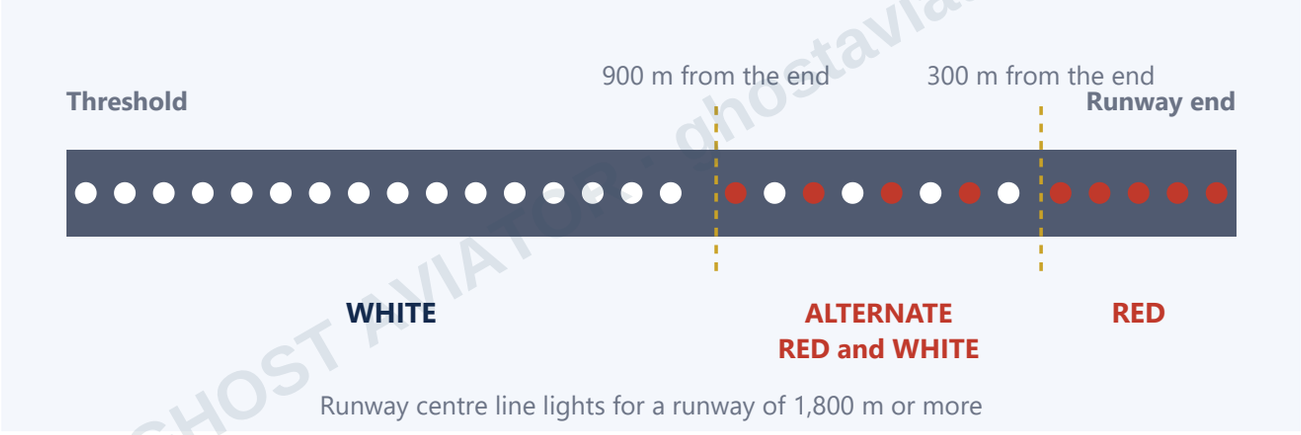

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Figure 11.11 *Runway Centre Line Light Colour Coding (Colour zones for a runway of 1800 m or more)*

Runway Touchdown Zone (TDZ) Lights

APPLICATION

Shall be provided in the touchdown zone of a **precision approach runway category II or III**.

Location

EXTENT & PATTERN

- Extend from the threshold for a longitudinal distance of **900 m**
- Exception: On runways less than 1800 m in length, shortened so it **does not extend beyond the midpoint** of the runway
- Formed by **pairs of barrettes** symmetrically disposed about the runway centre line

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- Lateral spacing between innermost lights of a pair = equal to the lateral spacing for TDZ marking
- Longitudinal spacing between pairs of barrettes: **30 m or 60 m**
- *Note: For operations at lower visibility minima it may be advisable to use a 30 m spacing.*

Characteristics

KEY TECHNICAL SPECS

- A barrette shall be composed of **at least three lights** with a spacing between lights of **not more than 1.5 m**
- TDZ lights shall be **fixed unidirectional lights showing variable white**

BARRETTE LENGTH RECOMMENDATION

A barrette should be **not less than 3 m** nor more than **4.5 m in length**.

Simple Touchdown Zone Lights (STZL)

PURPOSE

Provides pilots with enhanced situational awareness in all visibility conditions and helps enable pilots to decide whether to commence a go-around if the aircraft has not landed by a certain point on the runway.

WHEN STZL SHOULD BE PROVIDED

Except where TDZ lights are provided at an aerodrome where:

- The approach angle is greater than **3.5 degrees**, and/or
- The Landing Distance Available combined with other factors increases the risk of an overrun

Simple TDZ Lights should be provided — a pair of fixed unidirectional lights showing variable white, located on each side of the runway centreline, visible to the pilot in the direction of approach.

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Stopway Lights

APPLICATION, LOCATION & CHARACTERISTICS

- **Application:** Shall be provided for a stopway intended for **use at night**
- **Location:** Along full length of stopway in two parallel rows equidistant from centre line; also across the end of the stopway at right angles to stopway axis, not more than **3 m outside the end**
- **Colour:** Fixed unidirectional lights showing **red** in the direction of the runway

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Taxiway Lighting & Aerodrome Signs

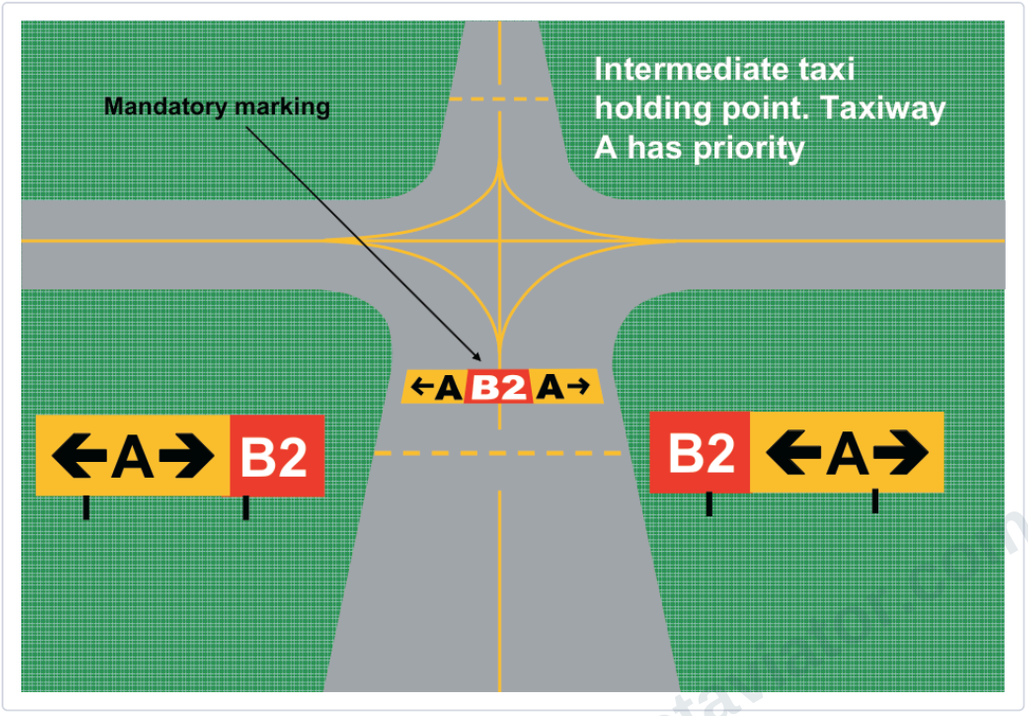

Figure 11.12 *Signs at a taxiway intersection: yellow information signs and a red mandatory sign.*

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Taxiway Centre Line Lights

APPLICATION — MANDATORY

Taxiway centre line lights **shall** be provided on an exit taxiway, taxiway, de-icing/anti-icing facility and apron intended for use in **runway visual range conditions less than 350 m** to provide continuous guidance between the runway centre line and aircraft stands, except that these lights need not be provided where:

- Traffic density is light, and
- Taxiway edge lights and centre line marking provide adequate guidance

RECOMMENDATIONS

- Should be provided on a taxiway intended for use at night in RVR conditions of **350 m or greater**, and particularly on complex taxiway intersections and exit taxiways
- Should also be provided as components of an **Advanced Surface Movement Guidance and Control System (ASMGCS)**

Characteristics — Exit Taxiway vs. Regular Taxiway

TAXIWAY CL LIGHTS COLOUR RULES

- **Regular taxiway (not exit taxiway) and runway part of standard taxi-route:** Fixed lights showing **green**, beam dimensions visible only from aeroplanes on or in the vicinity of the taxiway

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- **Exit taxiway:** Fixed lights — alternate taxiway centre line lights shall show **green and yellow**:
  - From beginning near runway centre line to the perimeter of the ILS/MLS critical/sensitive area or the lower edge of the inner transitional surface (whichever is farthest from runway)
  - Thereafter: all lights show **green**
  - Light nearest to the perimeter shall always be **green**
- Where aircraft may follow the same centre line in both directions: all centre line lights shall show **green to aircraft approaching the runway**

Location — Taxiway CL Lights on Taxiways

SPACING RECOMMENDATIONS

- Straight section: longitudinal intervals of **not more than 30 m**, except:
  - (a) Larger intervals up to **60 m** where meteorological conditions allow adequate guidance
  - (b) Intervals less than 30 m on short straight sections
  - (c) For use in **RVR < 350 m**: longitudinal spacing shall **not exceed 15 m**
- On a taxiway curve: continue from straight portion at a constant distance from the outside edge of the curve, spaced so as to give a clear indication of the curve

Taxiway Edge Lights

APPLICATION

- Shall be provided at the edges of a holding bay, de-icing/anti-icing facility, apron etc., intended for use at night, and on a taxiway not provided with taxiway centre line lights and intended for use at night
- Shall be provided on a runway forming part of a standard taxi-route and intended for taxiing at night where the runway is not provided with taxiway centre line lights

LOCATION — SPACING

- Straight section and runway forming part of standard taxi-route: uniform longitudinal intervals of **not more than 60 m**
- Lights on a curve: intervals **less than 60 m** so that a clear indication of the curve is provided
- Holding bay, de-icing/anti-icing facility, apron: uniform longitudinal intervals of **not more than 60 m**

COLOUR

Taxiway edge lights shall be **fixed lights showing blue**.

Taxiway CL — Green (or Green/Yellow on exit)
Taxiway Edge — Blue
Runway Edge — Variable White
Threshold / Wing Bar — Green

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Runway End / Stopway — Red

Aerodrome Signs

General Rules for Aerodrome Signs

GENERAL RULES — ALL SIGNS

- Signs shall be **frangible**. Those near a runway or taxiway shall be **sufficiently low to preserve clearance** for propellers and engine pods of jet aircraft. The installed height shall not exceed the dimension specified with the **longer side horizontal**.
- The **only signs** on the movement area utilizing **red** shall be **mandatory instruction signs**.
- Signs shall be **retro reflective and/or illuminated** when intended for use at night in association with non-instrument runways where the code number is 1 or 2.

Mandatory Instruction Signs

APPLICATION

A mandatory instruction sign shall be provided to identify a location beyond which an **aircraft taxiing or vehicle shall not proceed unless authorized by the aerodrome control tower**.

TYPES OF MANDATORY INSTRUCTION SIGNS

- Runway designation signs
- Category I, II or III holding position signs
- Runway-holding position signs
- Road-holding position signs
- **NO ENTRY** signs

A NO ENTRY sign shall be provided when entry into an area is prohibited.

APPEARANCE — MANDATORY INSTRUCTION SIGNS

- Inscription in **white on a red background**
- Where environmental factors require enhanced conspicuity — the outside edge of the white inscription shall be supplemented by a **black outline**:
  - **10 mm wide** for runway code numbers 1 and 2
  - **20 mm wide** for runway code numbers 3 and 4

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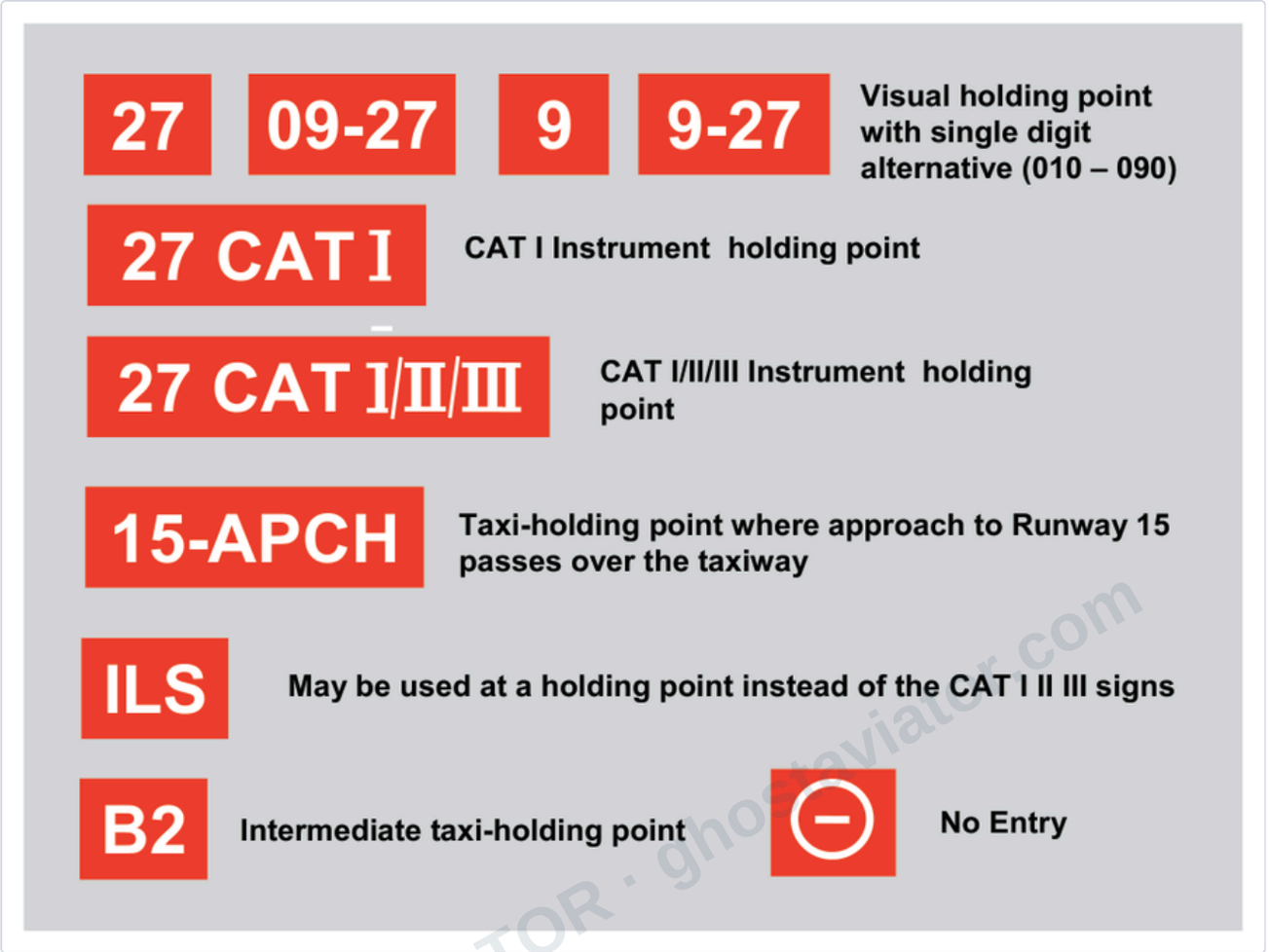

Figure 11.13 *Aerodrome sign characteristics: mandatory instruction signs (white on red), location signs (yellow on black), direction signs (black on yellow).*

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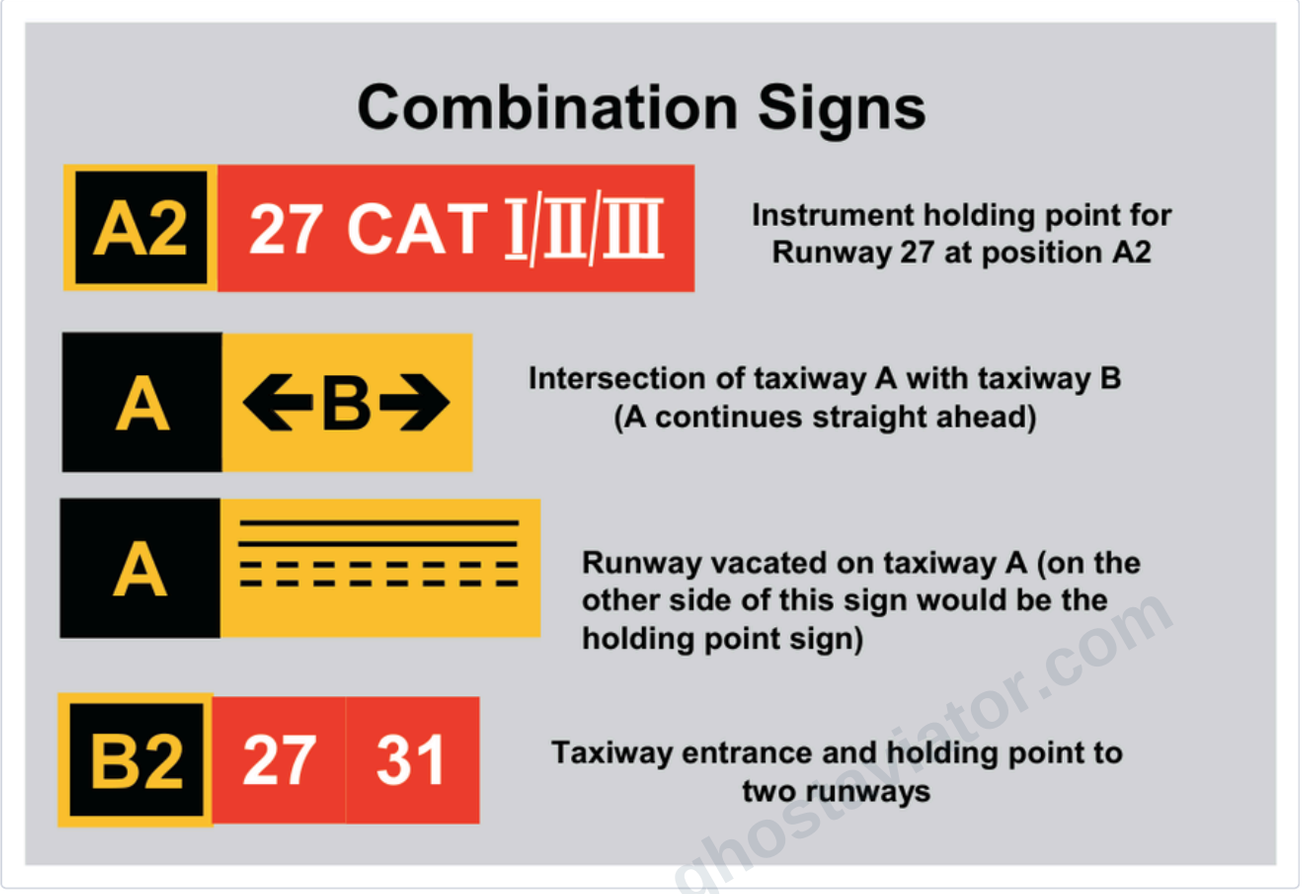

Figure 11.14 *Combination signs: location sign (yellow on black) next to direction signs (black on yellow) and holding position signs.*

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Visual Aids for Denoting Obstacles

Objects to be Marked and/or Lighted

MANDATORY MARKING & LIGHTING REQUIREMENTS

- A fixed obstacle that extends above a **take-off climb surface within 3,000 m of the inner edge** shall be marked and, if runway is used at night, lighted.
- A fixed object (other than an obstacle) **adjacent to a take-off climb surface** shall be marked and, if used at night, lighted.
- A fixed obstacle above a **horizontal surface** shall be marked and, if aerodrome used at night, lighted.
- A fixed object that extends above an **obstacle protection surface** shall be marked and, if runway used at night, lighted.
- **Vehicles and mobile objects** (excluding aircraft) on the movement area of an aerodrome are obstacles and shall be marked and, if used at night or low visibility, lighted (except aircraft servicing equipment and vehicles used only on aprons may be exempt).
- **Elevated aeronautical ground lights** within the movement area shall be marked to be conspicuous by day. Obstacle lights shall **not** be installed on elevated ground lights or signs in the movement area.

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- All obstacles within the distance specified from the centre line of a taxiway, apron taxiway or aircraft stand taxi lane shall be marked and, if taxiway used at night, lighted.

Marking of Objects

COLOUR & PATTERN RULES

- All fixed objects to be marked shall, whenever practicable, be **coloured**; if not practicable, **markers or flags** shall be displayed on or above them (objects conspicuous by shape, size, or colour need not be otherwise marked).
- All mobile objects to be marked shall be coloured or display flags.
- An object shall be coloured to show a **chequered pattern** if it has essentially unbroken surfaces and its projection on any vertical plane equals or exceeds **4.5 m in both dimensions**.
- The pattern: rectangles of **not less than 1.5 m and not more than 3 m on a side**, with corners in the darker colour, colours contrasting each with the other and with the background.
- **Colours: Orange & White** or alternatively **Red & White**, except where such colours merge with the background.

MARKING VS. LIGHTING TRADE-OFF

Obstacles should be both marked **and** lighted, except that the **marking may be omitted** when high-intensity obstacle lights by day light the obstacle.

Reference: ICAO Annex 14 · CAR Section-4, Series 'B', Part I Issue II

Capt. Pankaj Pahil

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Signs, Rescue and Fire-Fighting, and Declared Distances

Aerodrome Signs (Complete)

SIGN COLOUR SUMMARY — THE MASTER RULE

- The **only signs on the movement area using red** shall be **mandatory instruction signs**.
- Information signs other than a location sign: inscription in **black on a yellow background**.

Sign Colour/Style Quick Reference

27	MANDATORY INSTRUCTION White on Red
NO ENTRY	NO ENTRY SIGN White on Red
A	LOCATION SIGN Yellow on Black (Yellow Border)

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← G	DIRECTION SIGN Black on Yellow
← APRON	DESTINATION SIGN Black on Yellow
VOR	VOR CHECKPOINT Black on Yellow
AERODROME NAME	AERODROME ID SIGN Conspicuous colour Char height ≥ 3 m
2500m→	INTERSECTION TAKE-OFF SIGN Black on Yellow

Information Signs — Detailed Rules

WHEN EACH INFORMATION SIGN IS MANDATORY

- **Runway exit sign:** Shall be provided where there is an operational need to identify a runway exit.
- **Runway vacated sign:** Shall be provided where the exit taxiway is not provided with taxiway centre line lights and there is need to indicate to a pilot leaving a runway the perimeter of the ILS/MLS critical/sensitive area or the lower edge of the inner transitional surface (whichever is farther from the runway centre line).
- **Intersection take-off sign:** Shall be provided when there is an operational need to indicate the remaining take-off run available (**TORA**) for intersection take-offs.
- **Direction sign:** Shall be provided when there is an operational need to identify the designation and direction of taxiways at an intersection.
- **Combined location and direction sign:** Shall be provided when it is intended to indicate routing information prior to a taxiway intersection.

DESTINATION SIGN RECOMMENDATION

A destination sign should be provided to indicate the direction to a specific destination on the aerodrome, such as **cargo area, general aviation**, etc.

Location Sign — Rules

MANDATORY RULES

- Shall be provided **in conjunction with a runway designation sign** except at a runway/runway intersection.
- Shall be provided **in conjunction with a direction sign**, except it may be omitted where an aeronautical study indicates it is not needed.
- Where a taxiway ends at a "T" intersection and identification is necessary, a **barricade, direction sign and/or other appropriate visual aid shall be used**.

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APPEARANCE

- Inscription in **yellow on a black background**
- Stand-alone location sign shall have a **yellow border**
- Memory: **"Black Square — You Are There"**

Taxiway Designators — Rules

RULES FOR NAMING TAXIWAYS

- A taxiway shall be identified by a designator used **only once** on an aerodrome — comprising a single letter, two letters, or a combination of a letter or letters followed by a number.
- The use of words such as **"inner" and "outer"** shall be avoided wherever possible.
- The letters **I, O, and X** shall **not** be used (to avoid confusion with numerals 1, 0 and the closed-marking).
- The use of numbers alone is reserved for **runway designation**.
- **Apron stand designators shall not be the same as taxiway designators.**

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Aerodrome Identification Sign

RULES

- Shall be provided at an aerodrome where there is **insufficient alternative means of visual identification of the aerodrome**.
- Shall be placed on the aerodrome so as to be legible, in so far as is practicable, at **all angles above the horizontal**.
- Shall consist of the **name of the aerodrome**.
- Colour shall give **adequate conspicuity** when viewed against its background.
- Characters shall have a height of **not less than 3 m**.

Aircraft Stand Identification Signs

APPEARANCE

An aircraft stand identification sign shall consist of an inscription in **black on a yellow background**. An aircraft stand identification marking shall be supplemented with an aircraft stand identification sign where feasible.

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Road-Holding Position Sign

RULES

- Shall be provided at **all road entrances to a runway**.
- Located **1.5 m** from one edge of the road (left or right as appropriate to local traffic regulations) at the holding position.
- Consists of an inscription in **white on a red background**.

VOR Aerodrome Checkpoint Sign

RULES

- When a VOR aerodrome check-point is established, it shall be indicated by a **VOR aerodrome check-point marking AND sign**.
- Sign consists of an inscription in **black on a yellow background**.

ARIWS — Advanced Runway Incursion Warning System

DEFINITION & PURPOSE

ARIWS (Advanced Runway Incursion Warning System) is a system used for **warning a pilot about runway incursion in near real time**. It is a safety system designed to detect and warn pilots when there is an unauthorised entry onto an active runway, helping prevent runway incursions and collisions.

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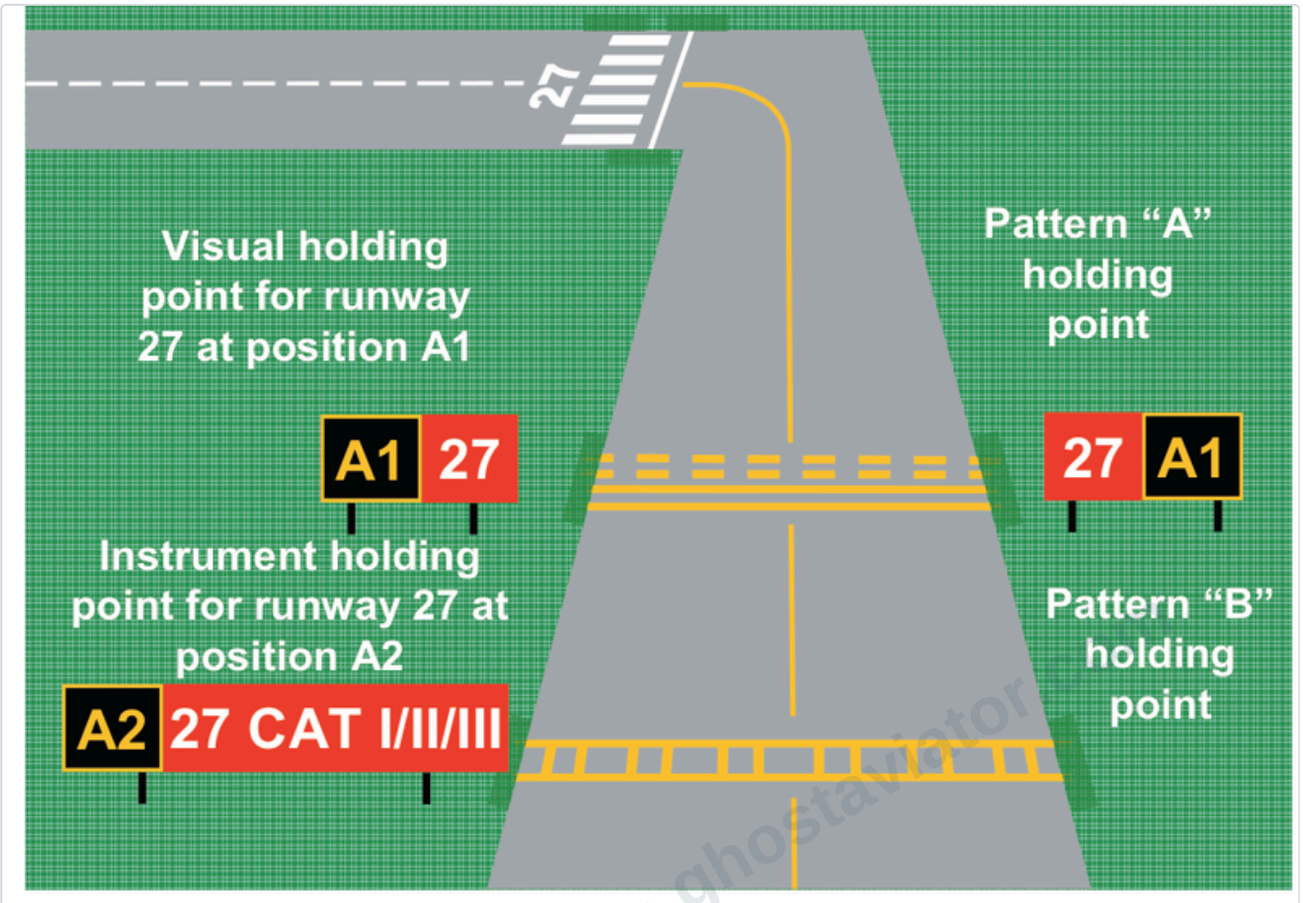

Figure 11.15 *Information signs: a location sign (yellow on black) with direction signs (black on yellow) and runway-holding position signs (white on red).*

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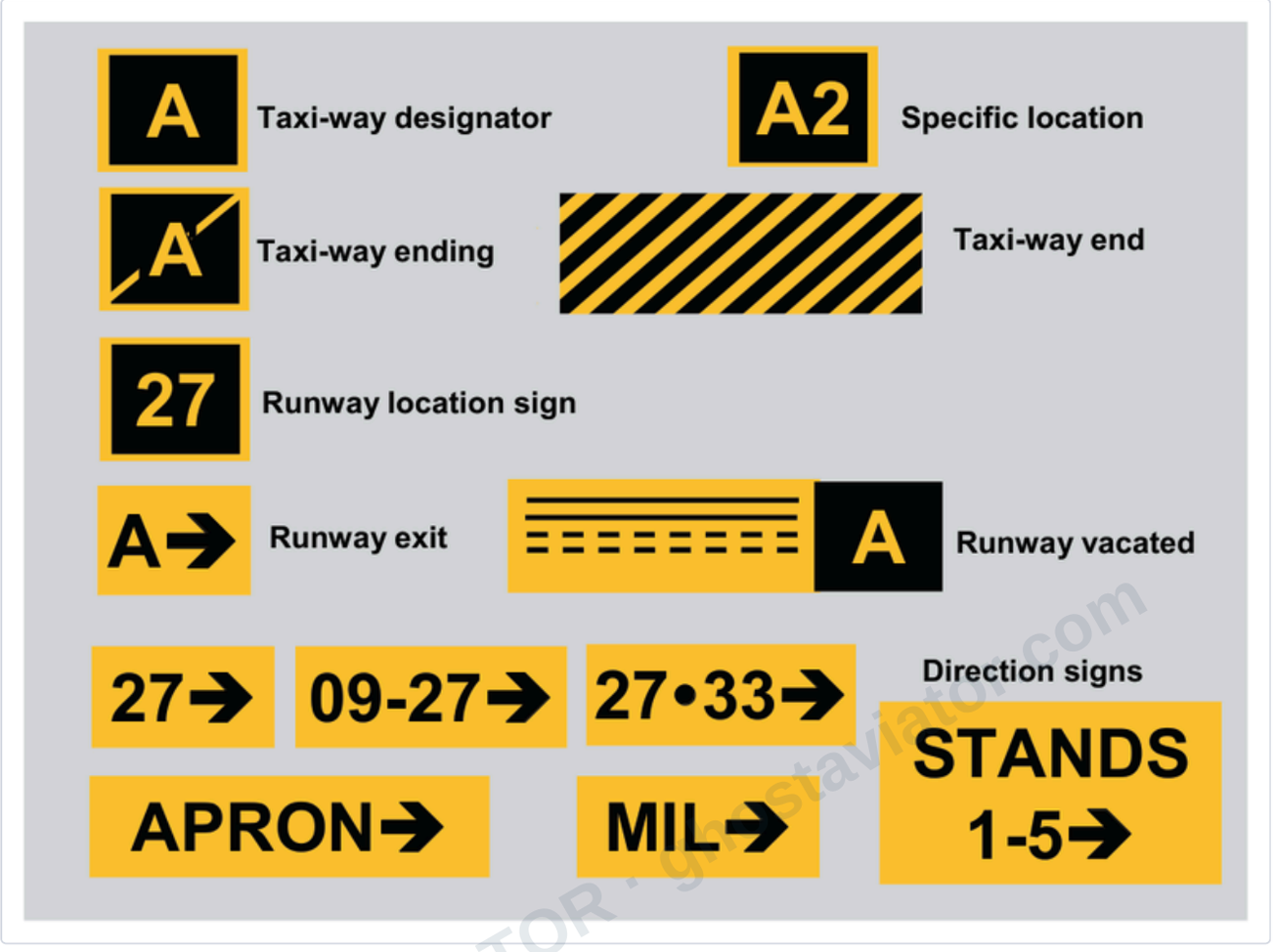

Figure 11.16 *Sign types: taxiway designator, taxiway ending, runway location, runway exit, runway vacated, direction and destination signs.*

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Markers

GENERAL RULE — ALL MARKERS

Markers shall be **frangible**. Those located near a runway or taxiway shall be sufficiently low to preserve clearance for propellers and for the engine pods of jet aircraft.

Unpaved Runway Edge Markers

APPLICATION

Shall be provided when the extent of an unpaved runway is **not clearly indicated** by the appearance of its surface compared with the surrounding ground.

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Stopway Edge Markers

RECOMMENDATION

Should be provided when the extent of a stopway is not clearly indicated by its appearance compared with the surrounding ground. Shall be **sufficiently different from runway edge markers** to ensure the two types cannot be confused.

Edge Markers for Snow-Covered Runways

RECOMMENDATION

Should be used to indicate the usable limits of a snow-covered runway when the limits are not otherwise indicated.

Taxiway Edge Markers

RECOMMENDATION

Should be provided on a taxiway where the code number is **1 or 2** and **taxiway centre line or edge lights** or taxiway centre line markers are not provided.

Unpaved Taxiway Edge Markers

RECOMMENDATION

Where the extent of an unpaved taxiway is not clearly indicated by its appearance compared with the surrounding ground, markers should be provided.

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Boundary Markers

APPLICATION & COLOUR

- Shall be provided at an aerodrome where the **landing area has no runway**.
- Shall be coloured to contrast with the background: a single colour (**orange or red**), or two contrasting colours (**orange and white**, or alternatively **red and white**), except where such colours merge with the background.

Obstacle Markers

RULES

- Shall be located in **conspicuous positions** so as to retain the general definition of the object and shall be recognizable in clear weather from:

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- A distance of at least **1000 m** for an object to be viewed **from the air**
- **300 m** for an object to be viewed from the ground in all directions in which an aircraft is likely to approach the object

- The shape of markers shall be **distinctive** to ensure they are not mistaken for markers employed to convey other information.
- The hazard presented by the object shall not be increased.

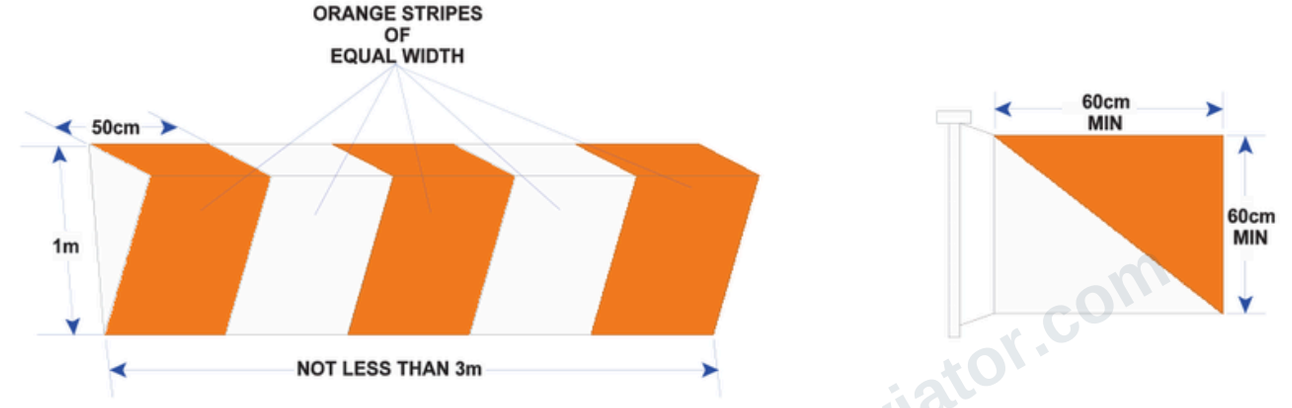

Figure 11.17 *Boundary markers: orange and white stripes of equal width.*

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Obstacle Marking & Lighting (Detailed)

Colour Rules for Object Marking

WHEN OBJECTS ARE SMALL (BOTH DIMENSIONS < 1.5 M)

An object shall be coloured in a **single conspicuous colour** if its projection on any vertical plane has both dimensions less than **1.5 m**. Colour: **orange or red**, except where such colours merge with background.

WHEN OBJECTS ARE LARGE (BOTH DIMENSIONS ≥ 4.5 M)

An object shall be coloured to show a **chequered pattern** if it has essentially unbroken surfaces and its projection on any vertical plane equals or exceeds **4.5 m in both dimensions**. Pattern: rectangles of **1.5 m to 3 m on a side**, corners in the darker colour. Colours: **orange & white** or **red & white**.

MOBILE OBJECT MARKINGS

- Emergency vehicles on the manoeuvring area: a single conspicuous colour, preferably **red** (or yellowish green)
- Other service vehicles: **yellow** (or orange)
- Follow-me vehicles: **yellow** (flashing low-intensity lights)

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Lighting of Objects — Use of Obstacle Lights

RULES

- The presence of objects which must be lighted shall be indicated by **low-, medium- or high-intensity obstacle lights**, or a combination.
- One or more low-, medium- or high-intensity obstacle lights shall be located. Lights shall be arranged to **depict the outline of the object**.

Low-Intensity Obstacle Lights

TYPES AND COLOURS

- **Type A (fixed obstacle): Fixed Red**
- **Type B (fixed obstacle): Fixed Red**
- **Type E: Flashing Red**
- Spaced at longitudinal intervals **not exceeding 45 m**
- **Vehicles — Emergency/Security: Flashing Blue**
- **Vehicles — Other service: Flashing Yellow**
- **Follow-me vehicles: Flashing Yellow**

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Medium-Intensity Obstacle Lights

TYPES AND SPACING

- Spaced at longitudinal intervals **not exceeding 900 m**
- **Type A: Flashing White**
- **Type B: Flashing Red**
- **Type C: Fixed Red**

High-Intensity Obstacle Lights

TYPE

- **High-intensity obstacle lights** shall be **Flashing White**

Wind Turbines

RULE

A wind turbine shall be **marked and/or lighted** if it is determined to be an obstacle. When lighting is deemed necessary, **medium intensity obstacle lights shall be used**.

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Characteristics of Obstacle Lights — Complete Table

Light Type	Colour	Signal Type / Flash Rate	Peak Intensity (cd) — Day (>500 cd/m²)	Peak Intensity (cd) — Twilight (50–500 cd/m²)	Peak Intensity (cd) — Night (<50 cd/m²)
Low Intensity, Type A (fixed obstacle)	Red	Fixed	N/A	N/A	10
Low Intensity, Type B (fixed obstacle)	Red	Fixed	N/A	N/A	32
Low Intensity, Type C (Mobile obstacle)	Yellow / Blue	Flashing (60–90 fpm)	N/A	40	40
Low Intensity, Type D (Follow Me Vehicle)	Yellow	Flashing (60–90 fpm)	N/A	200	200
Low Intensity, Type E	Red	Flashing	N/A	N/A	32
Medium Intensity, Type A	White	Flashing (20–60 fpm)	20,000	20,000	2,000
Medium Intensity, Type B	Red	Flashing (20–60 fpm)	N/A	N/A	2,000
Medium Intensity, Type C	Red	Fixed	N/A	N/A	2,000
High Intensity, Type A	White	Flashing (40–60 fpm)	200,000	20,000	2,000
High Intensity, Type B	White	Flashing (40–60 fpm)	100,000	20,000	2,000

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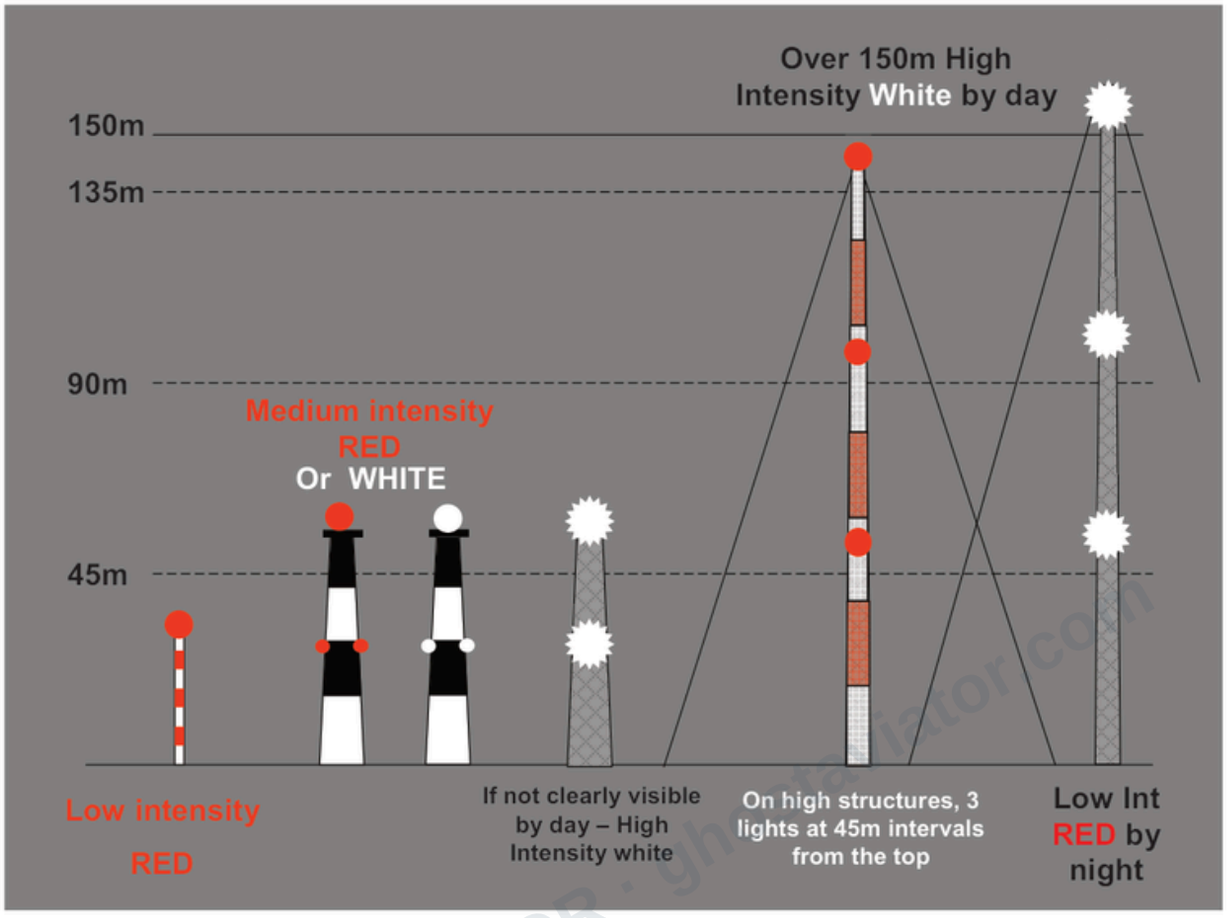

Figure 11.18 *Obstacle lighting by height: low-intensity red lights on low objects, medium-intensity red or white on higher structures, high-intensity white by day where needed.*

Visual Aids for Denoting Restricted Use Areas

Closed Runways and Taxiways, or Parts Thereof

MANDATORY RULES

- A **closed marking shall be displayed** on a runway or taxiway, or portion thereof, which is **permanently closed** to use by all aircraft.
- When permanently closed: all normal runway and taxiway markings shall be **obliterated**.
- Lighting on a closed runway or taxiway or portion thereof **shall not be operated**, except as required for maintenance purposes.

RECOMMENDATION — TEMPORARILY CLOSED

A closed marking should be displayed on a temporarily closed runway or taxiway or portion thereof, except that such marking may be omitted when the closing is of short duration and adequate warning by air traffic services is provided.

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Pre-Threshold Area

RULES

- When the surface before a threshold is paved, **exceeds 60 m in length**, and is **not suitable for normal use by aircraft**, the entire length before the threshold shall be marked with a **chevron marking**.
- A chevron marking shall **point in the direction of the runway** and be placed as shown in Fig. 17.
- A chevron marking shall be of **yellow colour**.
- It shall have an **overall width of 0.9 m**.

Unserviceable Areas

MANDATORY RULES

- **Unserviceability markers** shall be displayed wherever any portion of a taxiway, apron or holding bay is **unfit for the movement of aircraft** but it is still possible for aircraft to bypass the area safely.
- On a movement area used at night, **unserviceability lights** shall be used.
- Markers and lights shall be placed at intervals **sufficiently close so as to delineate the unserviceable area**.

Characteristics

MARKERS & LIGHTS

- **Unserviceability markers** shall consist of conspicuous upstanding devices such as **flags, cones, or marker boards**.
- **Unserviceability light:** A **red fixed light**. The intensity shall be sufficient to ensure conspicuity considering the intensity of adjacent lights and the general level of illumination.

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Rescue and Fire Fighting (RFF)

PRINCIPAL OBJECTIVE

The principal objective of a rescue and fire fighting service is to **save lives** in the event of an aircraft accident or incident occurring at, or in the immediate vicinity of, an aerodrome. Rescue and fire fighting equipment and services shall be provided at an aerodrome.

Level of Protection & Aerodrome Category

HOW THE AERODROME RFF CATEGORY IS DETERMINED

- Based on the **overall length** of the longest aeroplane normally using the aerodrome **AND** its **maximum fuselage width**.
- The level of protection shall be **appropriate to the aerodrome category**.

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- Exception: Where the number of movements of the aeroplanes in the highest category normally using the aerodrome is **less than 700 in the busiest consecutive three months**, the level of protection provided shall be **not less than one category below the determined category**.
- *Note: Either a take-off or a landing constitutes a movement.*

Aerodrome Category for RFF — Complete Table

Aerodrome Category	Aeroplane Overall Length	Maximum Fuselage Width
1	0 m up to but not including 9 m	2 m
2	9 m up to but not including 12 m	2 m
3	12 m up to but not including 18 m	3 m
4	18 m up to but not including 24 m	4 m
5	24 m up to but not including 28 m	4 m
6	28 m up to but not including 39 m	5 m
7	39 m up to but not including 49 m	5 m
8	49 m up to but not including 61 m	7 m
9	61 m up to but not including 76 m	7 m
10	76 m up to but not including 90 m	8 m

1–3	Light / GA aircraft <18 m
4–5	Regional jets 18–28 m
6–7	Narrow-body jets 28–49 m
8–9	Wide-body jets 49–76 m
10	A380 class 76–90 m MAX 8 m FW

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Extinguishing Agents

RULES

- Both **principal and complementary agents** shall be provided at an aerodrome.
- The **principal extinguishing agent** should be **foam**.
- The **complementary extinguishing agent** should be a **dry chemical powder** suitable for extinguishing hydrocarbon fires.
- Complementary agents shall comply with the appropriate specifications of the **Bureau of India Standards (BIS) / ISO**.

Response Times

CRITICAL VALUES

- A response time **not exceeding 3 minutes** shall be maintained by RFF services for any point of each operational runway and for any other part of the movement area.
- The **operational objective** is to achieve a response time **not exceeding 2 minutes** to any point of each operational runway, in **optimum visibility and surface conditions**.

3 min	maximum RFF Response Time (movement area)
2 min	operational objective RFF Response Time (optimum conditions)
700	movements / 3 months Threshold for category reduction allowance
1,000 m	from threshold Emergency access road provision target

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Emergency Access Roads

RECOMMENDATION

Emergency access roads should be provided on an aerodrome where terrain conditions permit, so as to facilitate achieving minimum response times. Particular attention should be given to provision of ready access to approach areas up to **1,000 m from the threshold**, or at least within the aerodrome boundary.

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Fire Stations

RECOMMENDATIONS

- All RFF vehicles should normally be housed in a **fire station**.
- Satellite fire stations should be provided whenever the response time cannot be achieved from a single fire station.
- The fire station should be located so that access into the runway area is **direct and clear**, requiring a **minimum number of turns**.

Communication and Alerting Systems

RECOMMENDATIONS

- A **discrete communication system** should be provided linking a fire station with the control tower, any other fire station on the aerodrome and the RFF vehicles.
- An **alerting system** for RFF personnel, capable of being operated from that station, shall be provided at a fire station, any other fire station on the aerodrome and the **aerodrome control tower**.

Radio Altimeter Operating Area

MANDATORY

A radio altimeter operating area **shall** be established in the pre-threshold area of a **precision approach category II & III runway**.

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Declared Distances, Clearway & Stopway

Declared Distances — Definitions

FOUR DECLARED DISTANCES

- **TORA — Takeoff Run Available:** The length of runway declared available and suitable for the **ground run of an aeroplane taking off**.
- **TODA — Takeoff Distance Available:** The length of the takeoff run available **plus the length of the clearway**, if provided.
- **ASDA — Accelerate-Stop Distance Available:** The length of the takeoff run available **plus the length of the stopway**, if provided.
- **LDA — Landing Distance Available:** The length of runway which is declared available and suitable for the **ground run of an aeroplane landing**.

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Clearway (CWY)

DEFINITION

A defined **rectangular area on the ground or water** under the control of the appropriate authority, selected or prepared as a suitable area over which an aeroplane may make a portion of its initial climb to a specified height.

- Should extend laterally to a distance of at least **75 m** on each side of the **extended centre line of the runway**
- The ground in a clearway should **not project over a plane having an upward slope of 1.25%**
- An object situated on a clearway which may endanger aeroplanes in the air should be regarded as an obstacle and should be **removed**

Stopway (SWY)

DEFINITION

A defined **rectangular area on the ground at the end of takeoff run available** prepared as a suitable area in which an aircraft can be stopped in the case of an **abandoned takeoff**.

- Shall have the **same width as the runway** with which it is associated
- Should be capable, in the event of an abandoned takeoff, of **supporting the aeroplane without inducing structural damage** to the aeroplane

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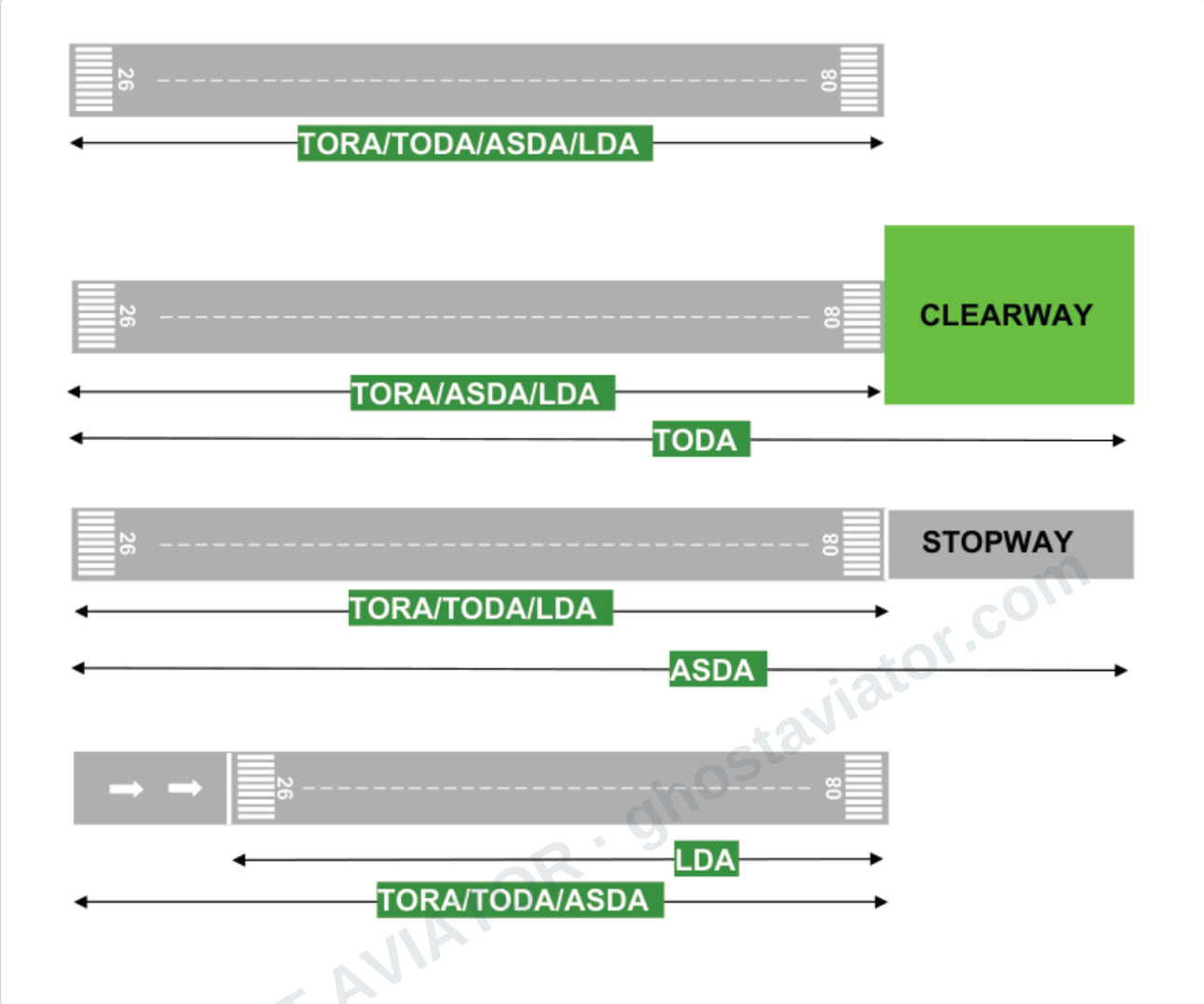

Figure 11.19 *Declared distances: TORA, TODA, ASDA and LDA with clearway, stopway and displaced threshold.*

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Wildlife Hazard Management & FOD Control

Wildlife Hazard Management

HAZARD STATEMENT

The presence of **wildlife (birds and other animals)** on and within the aerodrome vicinity may pose a **serious hazard to aircraft operational safety**. Therefore, to reduce the risk to aviation safety, **active assessments, reporting and management of wildlife** are necessary.

WILDLIFE HAZARD MANAGEMENT PROGRAMME (WHMP)

A **WHMP** is a method for aerodrome operators to adopt reasonable wildlife risk control measures, in order to **prevent wildlife from colliding with aircraft**.

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Foreign Object Debris (FOD) Control

DEFINITION OF FOD

FOD is defined as an **inanimate object within the movement area** which has no operational or aeronautical functions, and which has the potential to be a hazard to aircraft operations.

SOURCES OF FOD

FOD can be generated from:

- **Personnel**
- **Aerodrome infrastructure** (pavements, lights, signs)
- **The environment** (wildlife, snow, ice)
- **Equipment** operating at the aerodrome (aircraft, vehicles, maintenance equipment, fueling trucks, aircraft servicing equipment, construction equipment)

HAZARD STATEMENT & ACTION

The presence of FOD on the movement area may pose a **significant hazard** to the safety of aircraft operations — FOD has the potential to **damage aircraft during critical phases of flight**, which can lead to **catastrophic loss of life and airframe**, and increased maintenance and operating costs.

FOD hazards can be reduced through the implementation of an **FOD control program**, which would normally include **FOD prevention, detection, removal and evaluation**.

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Final Master Summary — Chapter 12 Key Flashpoints

THE COMPLETE COLOUR CODE — COMMIT TO MEMORY

Item	Colour	Type
Runway markings	White	Painted surface
Taxiway / Aircraft stand / Turn pad markings	Yellow	Painted surface
Apron safety lines (India)	Red	Painted surface
Chevron marking (pre-threshold)	Yellow	Painted surface
Runway edge lights	Variable White (Red near displaced thr; Yellow last 600 m / 1/3)	Fixed
Runway threshold / wing bar lights	Green (unidirectional)	Fixed
Runway end lights	Red	Fixed unidirectional
Runway CL lights — normal section	Variable White	Fixed
Runway CL lights — 900→300 m from end	Alt. Red & Variable White	Fixed

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Item	Colour	Type
Runway CL lights — last 300 m	Red	Fixed
TDZ lights	Variable White	Fixed unidirectional
Stopway lights	Red	Fixed unidirectional
Taxiway CL lights (regular)	Green	Fixed
Taxiway CL lights (exit taxiway)	Alternate Green & Yellow → Green	Fixed
Taxiway edge lights	Blue	Fixed
Mandatory instruction sign	White on Red	Sign
Location sign	Yellow on Black (+ yellow border)	Sign
Direction / destination sign	Black on Yellow	Sign
Aerodrome beacon (land)	Green/White flashing	Light
Aerodrome beacon (water)	Yellow/White flashing	Light
PAPI on glide path	2 Red + 2 White	Approach aid
Low intensity obstacle (fixed)	Red (fixed)	Obstacle light
Medium intensity obstacle (Type A)	White (flashing 20–60 fpm)	Obstacle light
High intensity obstacle	White (flashing 40–60 fpm)	Obstacle light

Reference: ICAO Annex 14 · CAR Section-4, Series 'B', Part I Issue II

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CHAPTER 12

12

PANS-OPS: Instrument Flight Procedures

WHAT IS PANS-OPS?

PANS-OPS stands for **Procedures for Air Navigation Services – Aircraft Operations**. It is ICAO Document 8168 and consists of two volumes:

- **Volume I – Flight Procedures:** Operational procedures recommended for flight operations personnel and flight crew. Outlines the parameters on which Volume II criteria are based.
- **Volume II – Construction of Visual and Instrument Flight Procedures:** For procedures specialists. Describes areas, obstacle clearance requirements, and guidelines for constructing instrument flight charts.

BASELINE ASSUMPTIONS FOR ALL CALCULATIONS

- All engines operating (unless contingency procedures are being developed — that is the operator's responsibility).
- All procedures depict **tracks**. Pilots apply corrections to heading for known wind.
- Calculations are based on an altitude of **600 m (2,000 ft) above MSL** and **ISA +15°C** unless otherwise stated.

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Obstacle Clearance — Areas

AREA STRUCTURE

Where track guidance is provided, each segment comprises a specified volume of airspace. The vertical cross-section is an **area located symmetrically about the centre line** of each segment.

- Vertical cross-section = divided into **Primary** and **Secondary** areas.
- Full obstacle clearances are applied over the **primary area**.
- Clearance reduces to **zero at the outer edges** of the secondary areas.

KEY AREA RULES

- On **straight segments**: Primary area width = **½ of total width**. Each secondary area = **¼ of total width**.
- Where **no track guidance is provided** during a turn: the entire total width is considered **primary area**.
- MOC is provided at full value across the primary area. In secondary areas, MOC reduces from inner edge to **zero at outer edge**.

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MOC Values — Departure Phase

MINIMUM OBSTACLE CLEARANCE VALUES

At the Departure End of Runway (DER):
MOC = **ZERO**

Increases by:
0.8% of the horizontal distance from DER (assuming max turn of 15°).

In the Turn Initiation Area and Turn Area:
MOC = **90 m (295 ft)**

Mountainous/precipitous terrain:
Procedures designer may **increase MOC**.

DESIGN CONSIDERATIONS

- Design of an IDP is dictated by the **terrain surrounding the aerodrome**.
- May also be required for **ATC requirements** in the case of SID routes.
- These factors influence the type and siting of navaids and the departure route.
- Airspace restrictions may also affect routing and siting of navaids.

AERODROME OPERATING MINIMA

Where obstacles cannot be cleared by the appropriate margin when flown on instruments, aerodrome operating minima are established to permit **visual flight clear of obstacles**.

Wherever possible, a **straight departure aligned with the runway centre line** is specified.

TURNING DEPARTURE TRIGGER

When a departure route requires a turn of more than **15°** to avoid an obstacle, a **turning departure** is constructed.

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Establishment of a Departure Procedure

A departure procedure is established for each runway where instrument departures are expected. It will include procedures for the **various categories of aircraft**.

WIND EFFECT — KEY RULE

- When **radar vectored**: Pilots will **NOT compensate** for wind effects.
- When flying departure routes expressed as **tracks to be made good**: Pilots **WILL compensate** for known or estimated wind.

DEFINITION OF PDG

The PDG is an aid to the procedures designer, who adjusts the route to **minimise the PDG** consistent with other constraints.

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STANDARD PDG VALUES — CRITICAL

Unless otherwise published:
PDG = **3.3 per cent**

For helicopters (Cat H):
PDG = **5.0 per cent**

Basis of PDG:
OIS gradient = **2.5%**
+ Additional margin = **0.8%**
= PDG = **3.3%**

Gradient Specification — How it Works

PDG SPECIFICATION RULES

- Published gradients are specified to an **altitude/height after which the minimum gradient of 3.3%** is considered to prevail.
- The final PDG continues until obstacle clearance is ensured for the next phase (en-route, holding, or approach).
- At that point, the departure procedure **ends** and is marked by a significant point.

Close-In Obstacles Rule

CLOSE-IN OBSTACLE RULE

Gradients to a height of **60 m (200 ft) or less**, caused by close-in obstacles, are **NOT specified**. However, a **note will be published** stating that close-in obstacles exist.

Fixes as Aid in Obstacle Avoidance

DME / RNAV FIXES

Where a suitably located DME exists, additional specific height/distance information for obstacle avoidance may be published. RNAV waypoints or other suitable fixes may be used to provide a means of **monitoring climb performance**.

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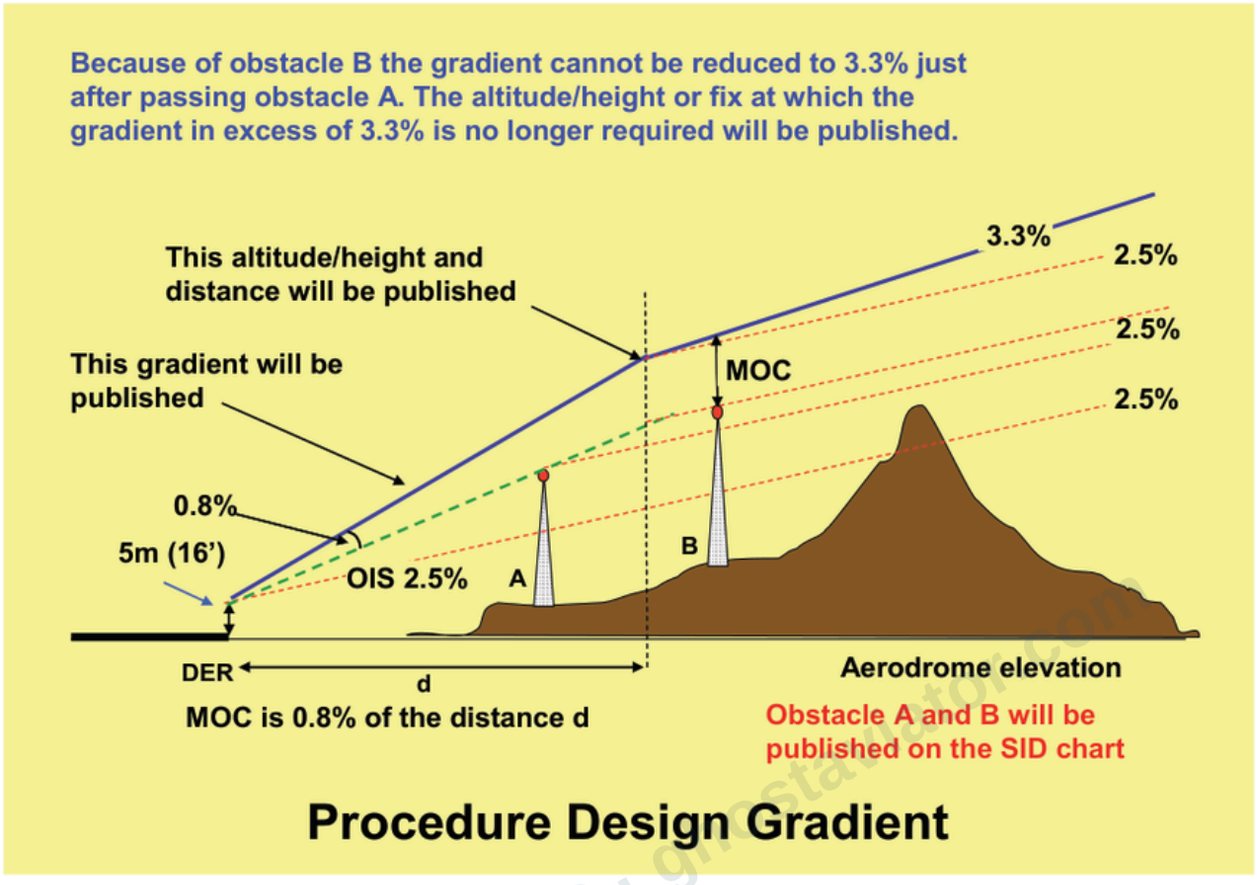

Figure 12.1 *Procedure design gradient: the OIS climbs at 2.5 %, the MOC grows by 0.8 % of the distance from the departure end of the runway (MOC = 0.8 % of d), giving the published PDG of 3.3 %.*

WHEN TO ACCEPT RADAR VECTORS

Pilots should **NOT accept radar vectors during departure UNLESS** one of the following applies:

1. They are **above the minimum altitudes/heights** required to maintain obstacle clearance in the event of engine failure. (This relates to engine failure between V1 and minimum sector altitude, or end of contingency procedure.)
2. The departure route is **non-critical with respect to obstacle clearance**.

SID TERMINATION

The SID terminates at the **first fix/facility/waypoint of the en-route phase** following the departure procedure.

Straight Departures

ALIGNMENT RULE

A straight departure is one in which the initial departure track is **within 15° of the alignment of the runway centre line**.

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STRAIGHT DEPARTURE — OBSTACLE RULES

- If obstacles exist that affect the departure route, PDGs **greater than 3.3%** may be specified.
- When such a gradient is specified, the altitude/height to which it extends **shall be promulgated**. After this point, PDG of 3.3% (Cat H: 5.0%) resumes.
- Gradients to a height of **60 m (200 ft)** or less due to close-in obstacles are **not specified**; a note will be published instead.

Turning Departures

TURNING DEPARTURE — DEFINITION & ALTITUDE RULES

- Turn of **more than 15°** from runway C/L = **Turning Departure**.
- Straight flight is assumed until reaching an altitude/height of at least **120 m (394 ft)** for fixed-wing aircraft.
- For helicopters: **90 m (295 ft)**.

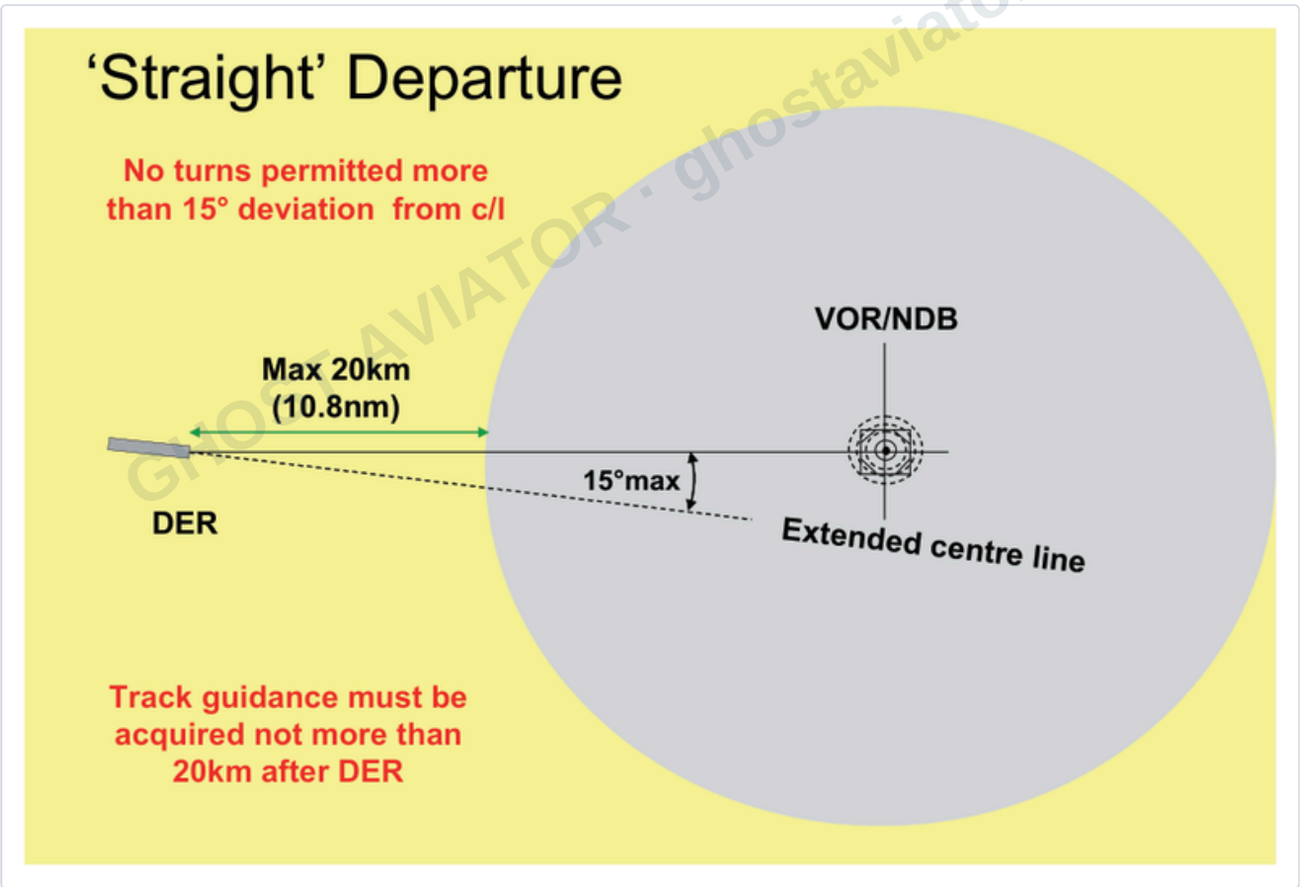

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Figure 12.2 *Straight departure: initial track within 15° of the runway centre line; track guidance acquired within 20 km of the DER.*

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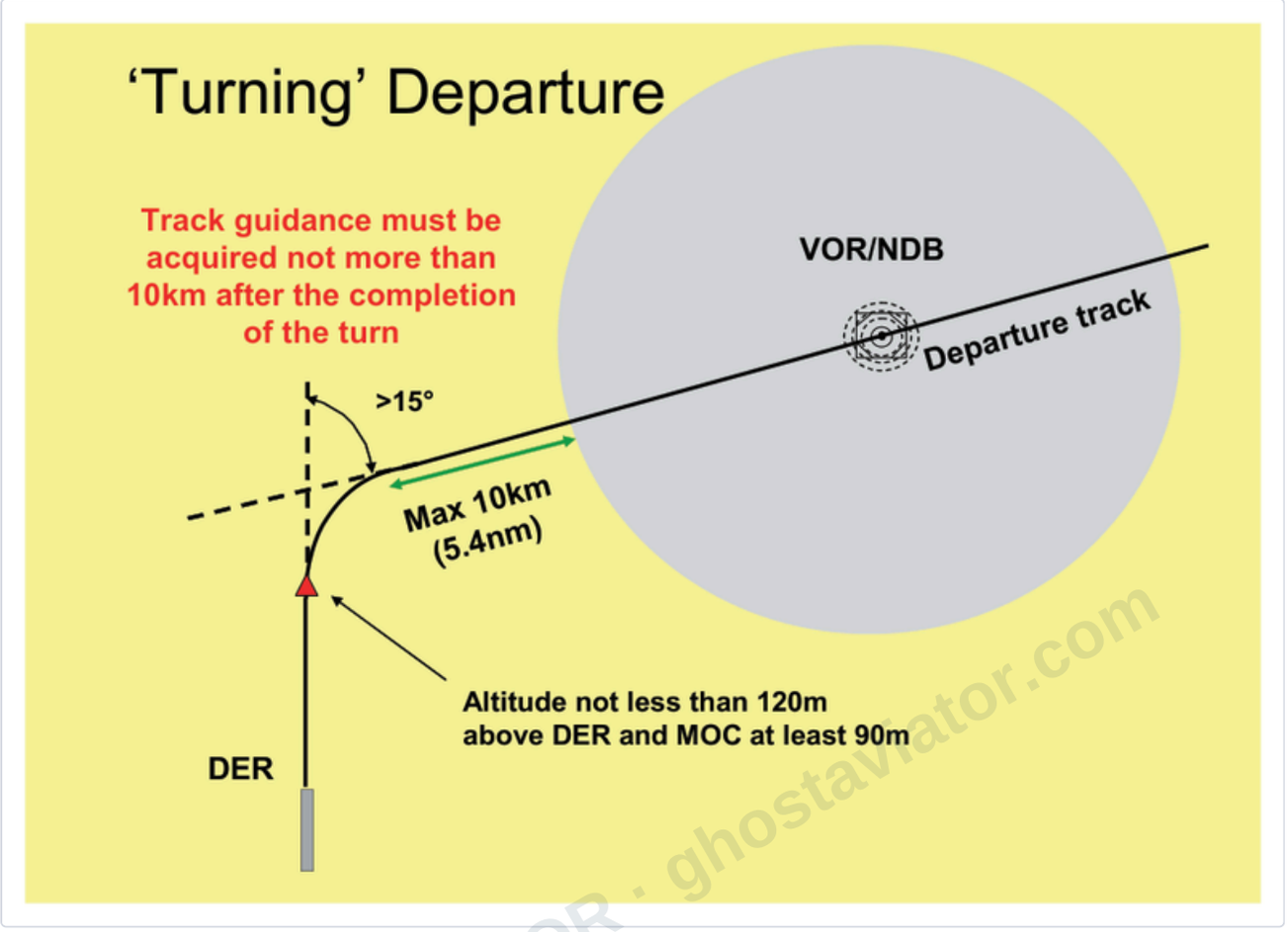

Figure 12.3 *Turning departure: climb straight ahead to at least 120 m above the DER (MOC at least 90 m) before the turn.*

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GENERAL

Where **no track guidance** is provided, departure procedures are designed using the **omnidirectional method**.

Where obstacles do not permit omnidirectional procedures, one of the following is necessary:

- Fly a Standard Instrument Departure (SID) route; or
- Ensure ceiling and visibility permit obstacles to be avoided by visual means.

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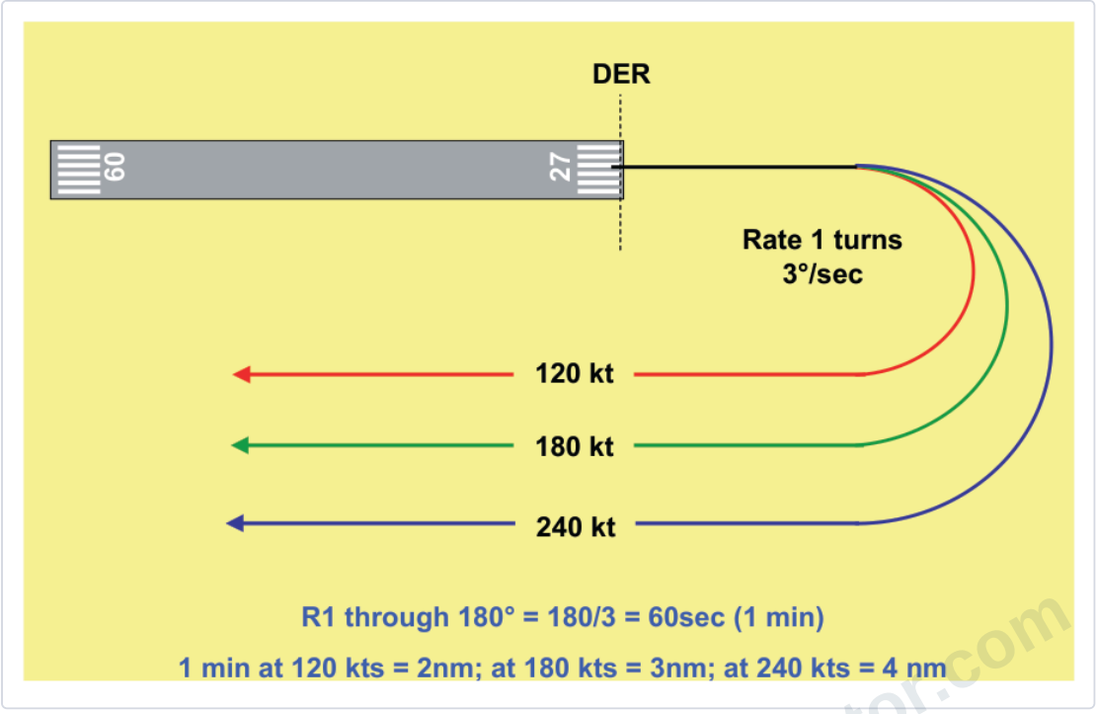

Figure 12.4 *Turn radius grows with speed: a rate 1 turn through 180 degrees takes one minute.*

Beginning of Departure

DER & TURN INITIATION RULES

- Departure procedure begins at the **Departure End of Runway (DER)** = end of area declared suitable for take-off.
- Since point of lift-off varies, a turn at **120 m (394 ft)** above aerodrome elevation is **not initiated sooner than 600 m from the beginning of the runway**.
- Procedures are normally designed/optimised for turns at **600 m** from the beginning of the runway.
- Turns may NOT be initiated before the DER (or specified point); this will be noted on the departure chart.
- For **Category H**: procedure turns can be initiated at **90 m (295 ft)** if the DER and earliest initiation point are at the beginning of the runway/FATO.

PDG for Omnidirectional Departures

PDG RULES

- Default PDG = **3.3%** (helicopters: 5%) + straight climb on extended runway C/L until **120 m (394 ft)** above aerodrome elevation (helicopters: 90 m).
- The basic procedure ensures:
  (a) Aircraft climbs on extended C/L to **120 m (394 ft)** before turns can be specified.
  (b) At least **90 m (295 ft)** of obstacle clearance is provided before turns greater than 15° are specified.

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Four Design Options for Omnidirectional Procedures

(A) STANDARD CASE

No obstacles penetrate the **2.5% OIS** and 90 m (295 ft) clearance prevails. A 3.3% climb to 120 m satisfies requirements for a **turn in any direction**.

(B) SPECIFIED TURN ALT/HEIGHT

Where obstacles preclude omnidirectional turns at 120 m, the procedure specifies a **3.3% climb to a higher alt/height** where omnidirectional turns can be made.

(C) SPECIFIED PDG

Where obstacles exist, the procedure defines a minimum gradient **more than 3.3%** to a specified altitude/height before turns are permitted.

(D) SECTOR DEPARTURES

The procedure identifies **sectors** for which either a minimum gradient or minimum turn alt/height is specified. E.g., "climb straight ahead to alt X before turning east; climb to alt Y before turning west."

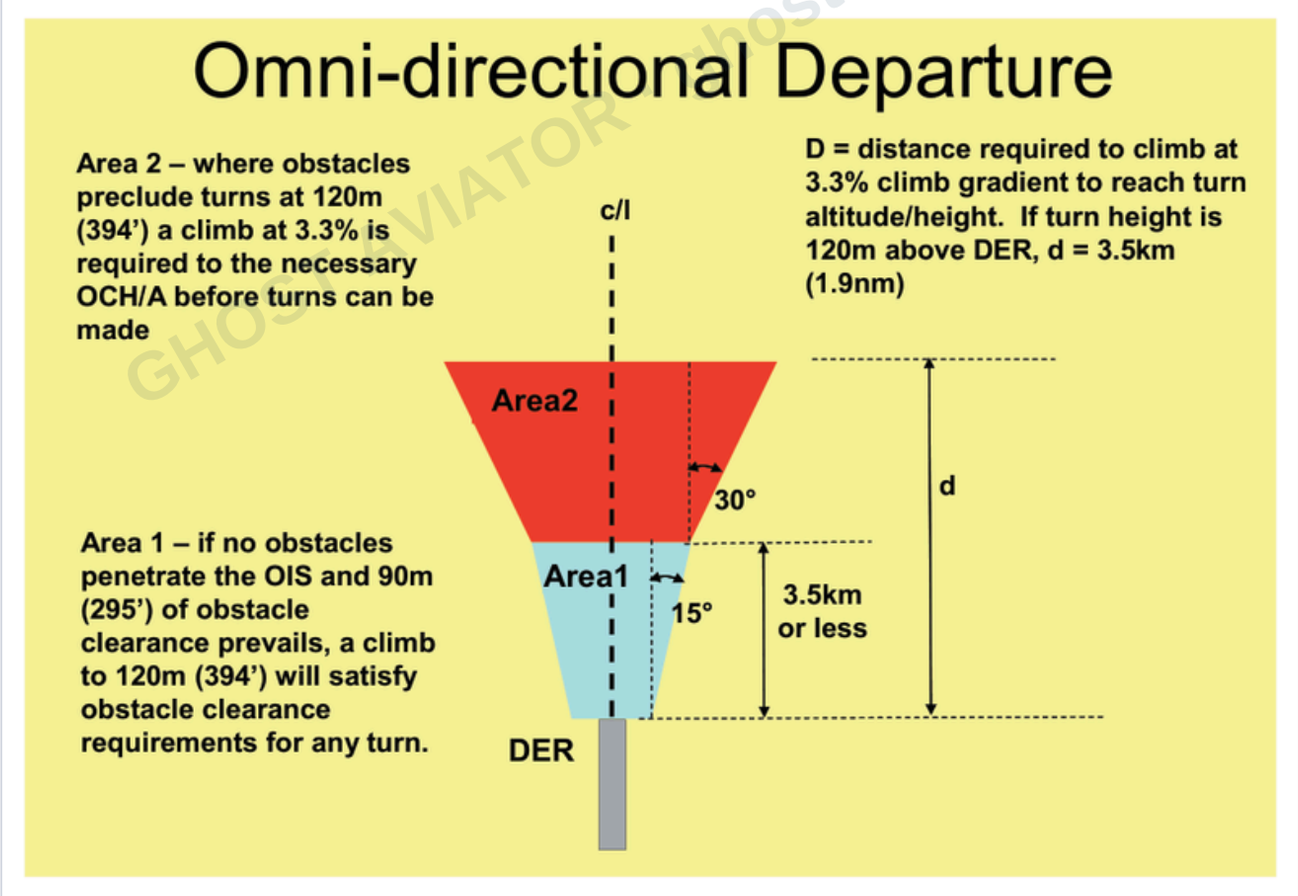

Figure 12.5 *Omnidirectional departure: Area 1 and Area 2 turn restrictions, and the distance d needed to climb at 3.3 % to the turn height.*

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Key Definitions

GNSS

Global Navigation Satellite System. A worldwide position and time determination system including one or more satellite constellations, aircraft receivers, and system integrity monitoring, augmented as necessary.

GBAS

Ground Based Augmented System. User receives augmentation information directly from a **ground-based transmitter**.

SBAS

Satellite Based Augmentation System. User receives augmentation information from a **satellite-based transmitter**. (WAAS, EGNOS, GAGAN, MSAS)

ABAS

Aircraft Based Augmentation System. Augments and/or integrates GNSS information with information available **on board the aircraft**.

GLS

GBAS Landing System. For approach and landing using GNSS augmented by GBAS as the primary navigational reference.

PBN

Performance Based Navigation. Area navigation based on performance requirements for aircraft operating along an ATS route, instrument departure/approach procedure, or in designated airspace.

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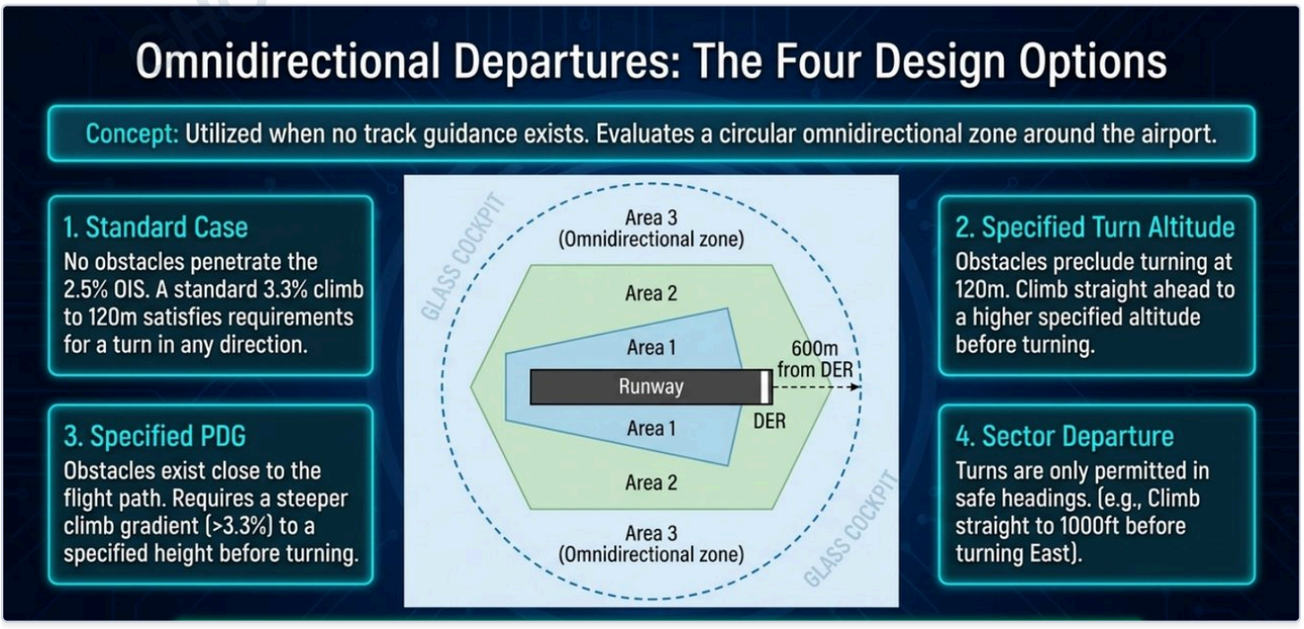

Figure 12.6 *Omnidirectional departures: the four design options*

Four Kinds of Turns in RNAV

RNAV TURN TYPES

1. **Turn at a fly-by waypoint** — Turn anticipation always provided.
2. **Turn at a flyover waypoint** — No turn anticipation.
3. **Turn at an altitude/height** — Some systems cannot code this; must be executed manually.
4. **Fixed radius turn** — Generally associated with RNP procedures.

SBAS — Straight & Turning Departures

SBAS KEY VALUES

- From DER to turn initiation point: SBAS receiver provides nominal **full-scale deflection (FSD) of 0.3 NM**.
- Larger FSDs may be acceptable with augmentations (e.g., autopilot).
- SBAS receiver will NOT transition to en-route integrity performance until the **final waypoint** in the departure procedure is sequenced.

GBAS NOTE

No departure criteria specifically designed for GBAS exists. Departure operations based on basic GNSS or SBAS may be flown by aircraft with a GBAS receiver using the optional GBAS positioning service.

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Accuracy of Fixes

FIX TOLERANCE AREA

Because all navigation facilities and waypoints have accuracy limitations, the geographic point identified is not precise but may be **anywhere within an area called the fix tolerance area** which surrounds its plotted point of intersection.

Fix Tolerances for Specific Navigation Systems

Fix Tolerances by Navigation System

Navigation System	Fix Tolerance	Notes
TAR (Terminal Area Radar)	±1.5 km (0.8 NM)	Within 37 km (20 NM)
RSR (En-route Surveillance Radar)	±3.1 km (1.7 NM)	Within 74 km (40 NM)
DME	±0.46 km (0.25 NM) + 1.25% of distance	To the antenna
75 MHz Marker Beacon	Used for ILS & "z" markers	Instrument approach

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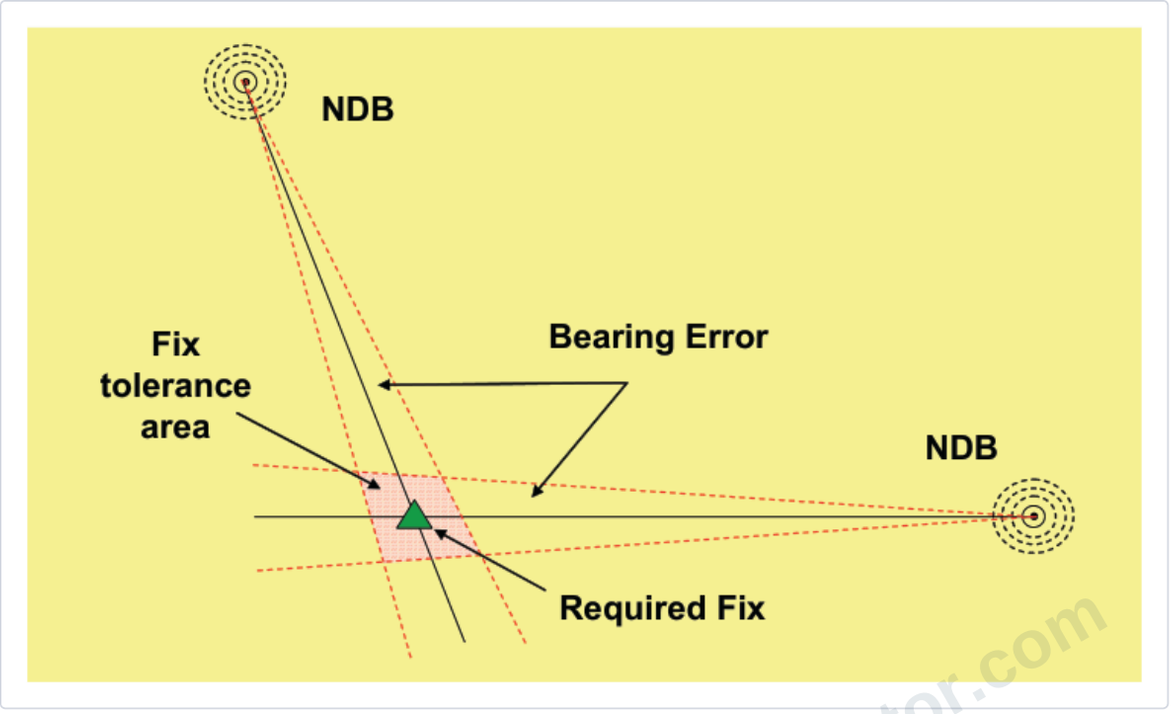

Figure 12.7 *Fix tolerance area: where the two bearing errors overlap.*

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Types of Approach

STRAIGHT-IN APPROACH

Wherever possible, a straight-in approach aligned with the runway centre line is specified.

For **non-precision approaches**, a straight-in approach is acceptable if the angle between the final approach track and runway C/L is **30° or less**.

CIRCLING APPROACH

Specified when terrain or other constraints cause the final approach track alignment or descent gradient to fall outside the criteria for a straight-in approach.

The final approach track is in most cases aligned to pass over some portion of the **usable landing surface**.

BASIS OF CATEGORISATION

Five categories based on the **indicated airspeed at threshold (V_{at})** which equals the stall speed V_{SO} × 1.3 or stall speed V_{S1g} × 1.23 (whichever is higher) in landing configuration at maximum certificated landing mass.

A	V_{at} < 91 kt
B	91 – 120 kt
C	121 – 140 kt
D	141 – 165 kt

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E	166 – 210 kt
H	Helicopters

Speeds for Procedure Calculations (kt) — DOC 8168 Table 1-4-1-1

Cat	V_{at} (kt)	Initial Approach (kt)	Final Approach (kt)	Circling Max (kt)	Missed App Intermediate (kt)	Missed App Final (kt)
A	<91	90 / 150 (110*)	70 / 100	100	100	110
B	91–120	120 / 180 (140*)	85 / 130	135	130	150
C	121–140	160 / 240	115 / 160	180	160	240
D	141–165	185 / 250	130 / 185	205	185	265
E	166–210	185 / 250	155 / 230	240	230	275
H	N/A	70 / 120**	60 / 90***	N/A	90	90

* Maximum speed for reversal and racetrack procedures. ** Max speed for racetrack: ≤6000 ft = 100 kt; >6000 ft = 110 kt. *** GNSS point-in-space: 120 KIAS initial/intermediate, 90 KIAS final/MA (or 90/70 KIAS based on operational need).

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Descent Gradient (Final Approach)

DESCENT GRADIENT / ANGLE RULES — FINAL APPROACH

Minimum/Optimum (with FAF):
5.2% / 3.0° (52 m/km = 318 ft/NM)

Maximum for Cat A & B:
6.5% / 3.7° (65 m/km = 395 ft/NM)

Maximum for Cat C, D, E:
6.1% / 3.5° (61 m/km = 370 ft/NM)

Maximum for Cat H:
10% / 5.7°

CDFA — CONTINUOUS DESCENT FINAL APPROACH

- Requires a **continuous descent without level-offs**, flown with VNAV guidance or manual calculation.
- Rate of descent adjusted to achieve continuous descent to a point approximately **15 m (50 ft) above the landing runway threshold**.
- Descent shall pass at or above the minimum altitude at any stepdown fix.
- CDFA with advisory VNAV = **3D instrument approach operation**.
- CDFA with manual calculation = **2D instrument approach operation**.

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Arrival Segment (STAR)

STAR — STANDARD TERMINAL ARRIVAL ROUTE

Permits transition from the **en-route phase to the approach phase**. Arrival route normally ends at the IAF.

Protection area convergence: Begins at **46 km (25 NM)** before the IAF, with a maximum convergence angle of **30° each side** of the axis.

MSA / TAA — CRITICAL VALUES

- **MSA (Minimum Sector Altitude)** provides at least **300 m (1,000 ft)** obstacle clearance within **46 km (25 NM)** of the navigation aid, IAF, or IF.
- **TAA (Terminal Arrival Altitude)**: Associated with PBN procedures based on the "T" or "Y" arrangement with three IAFs around the IF. TAAs replace the 46 km (25 NM) MSA where published.

Initial Approach Segment

INITIAL APPROACH — KEY FACTS

- Begins at the **IAF** and ends at the **IF**.
- Aircraft has left the en-route structure and is manoeuvring to enter the intermediate segment.
- Speed and configuration depend on distance from aerodrome and descent required.
- Max angle of interception of initial segment to IF: 90° for Precision 120° for Non-Precision
- MOC: **300 m (1,000 ft)** in primary area, reducing to zero at outer edge of secondary area.

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Types of Manoeuvres in Initial Approach

REVERSAL PROCEDURES

Used where no suitable IAF or IF is available. Entry is restricted to a specific direction or sector — a **base turn or procedure turn** is prescribed *(laid down by the authority or the rules)*. Directions and timing specified must be strictly followed. A racetrack or holding manoeuvre cannot be conducted unless so specified.

Three manoeuvre types:

1. **45°/180° Procedure Turn** (1 to 3 minutes outbound timing)
2. **80°/260° Procedure Turn** (1 to 3 minutes outbound timing)
3. **Base Turn** (1, 2, or 3 minutes — end of outbound leg by radial or DME distance)

RACETRACK PROCEDURE

- 180° turn overhead facility/fix onto outbound track for 1, 2, or 3 minutes.
- Followed by a 180° turn in the same direction to return to inbound track.
- Outbound leg may alternatively be limited by a **DME distance or intersecting radial/bearing**.

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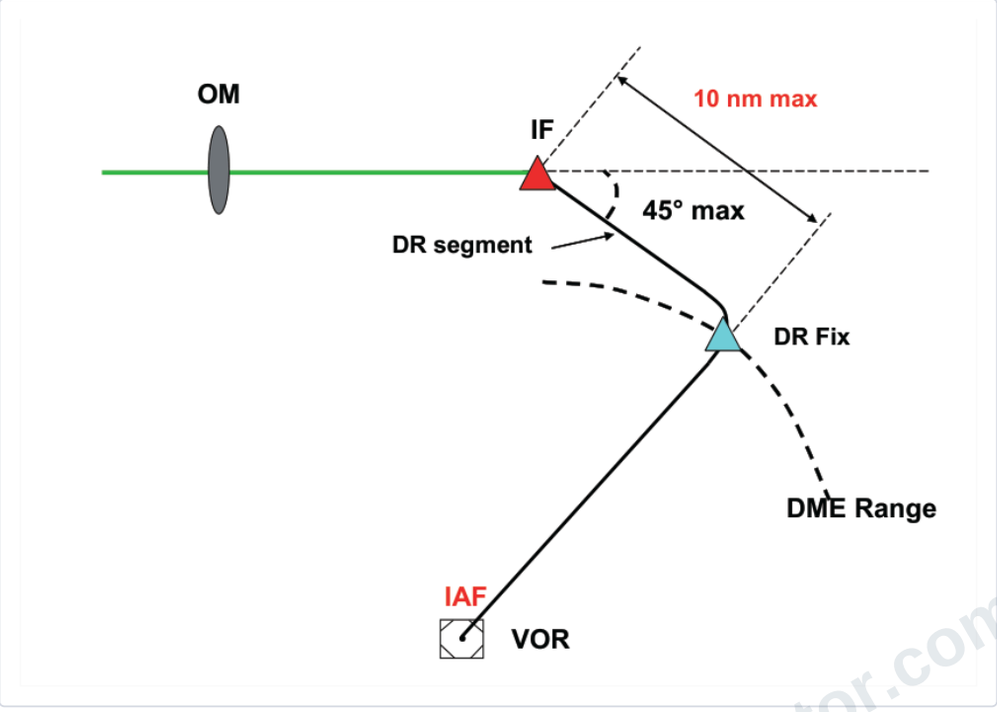

Figure 12.8 *A dead-reckoning segment: track not more than 45 degrees from the final track, not longer than 10 NM.*

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Intermediate Approach Segment

INTERMEDIATE APPROACH — KEY FACTS

- Aircraft speed and configuration adjusted to prepare for final approach. **Descent gradient kept as shallow as possible**.
- MOC: Reduces from **300 m (984 ft)** to **150 m (492 ft)** in the primary area, reducing to zero at outer edge of secondary area.
- Begins: On inbound track of procedure turn / base turn / final inbound leg of racetrack.
- Ends: At the FAF or Final Approach Point (FAP).
- *Note: Where no FAF is specified, the inbound track IS the final approach segment.*

Final Approach Segment

TYPES OF FINAL APPROACH

1. **NPA with FAF** — Non-precision approach with final approach fix
2. **NPA without FAF** — Non-precision approach; facility is both IAF and MAPt
3. **APV** — Approach with vertical guidance
4. **PA** — Precision approach

NPA WITH FAF

- Begins at **FAF**, ends at **MAPt**.

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- FAF is sited on the final approach track.
- **Optimum distance FAF to threshold: 9.3 km (5.0 NM)**
- **Maximum length:** normally not greater than **19 km (10 NM)**
- MOC: **75 m** with FAF
- Stepdown fixes may be incorporated. Two OCA/H values published (higher for primary, lower if stepdown fix is positively identified).

NPA WITHOUT FAF

- Single facility on or near aerodrome = both **IAF and MAPt**.
- Indicates: (a) minimum alt/height for reversal/racetrack; and (b) OCA/H for final approach.
- In absence of FAF, descent to MDA/H starts once established inbound on final approach track.
- MOC: **90 m** without FAF
- Final approach track **cannot normally be aligned on runway C/L**.

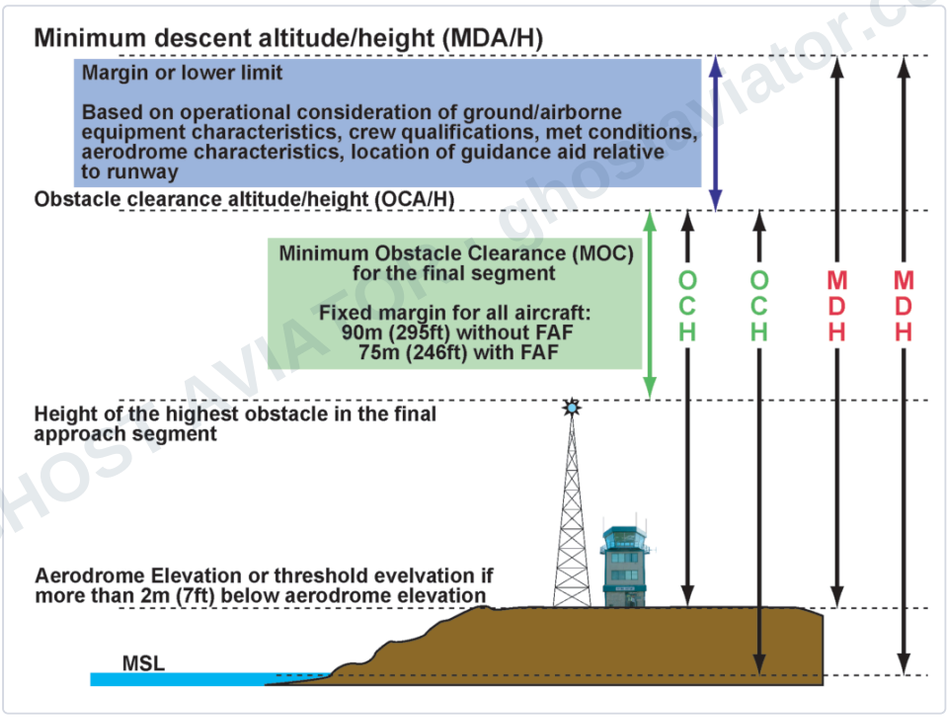

Figure 12.9 *Non-precision approach: obstacle clearance height and minimum descent height. Fixed margin 75 m with a final approach fix, 90 m without.*

Precision Approach — FAP

FINAL APPROACH POINT (FAP) — PRECISION

- FAP = point in space on final approach track where **intermediate approach altitude/height intercepts the nominal glide path / MLS elevation angle**.
- Intermediate altitude generally intercepts glidepath at heights of **300 m (1,000 ft) to 900 m (3,000 ft)** above runway elevation.

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- For a 3° glide path, interception occurs between **6 km (3 NM) and 19 km (10 NM)** from the threshold.
- **Outer Marker/DME fix**: Permits verification of glidepath/altimeter relationship. Descent below fix crossing altitude should NOT be made prior to crossing the fix.
- *Note: FAF is associated with NPA; FAP is associated with PA.*

OCA/H, MDA/H, DA/H — KEY DEFINITIONS

- **OCA (Obstacle Clearance Altitude)** — Referenced to MSL.
- **OCH (Obstacle Clearance Height)** — Referenced to threshold elevation (or aerodrome elevation if threshold >2 m/7 ft below aerodrome elevation).
- **MDA (Minimum Descent Altitude)** — NPA/circling: altitude below which descent must NOT be made without required visual reference. Referenced to MSL.
- **MDH (Minimum Descent Height)** — Referenced to aerodrome elevation or threshold elevation.
- **DA (Decision Altitude)** — PA/APV: altitude at which a missed approach must be initiated if required visual reference not established. Referenced to MSL.
- **DH (Decision Height)** — Referenced to threshold elevation.

VSS — VISUAL SEGMENT SURFACE

Constructed per ICAO Doc 8168/PANS-OPS Volume II. Identifies obstacles that may affect execution of an IAP (except circling). Refers to the final approach leg between MDA/DA of the procedure and the runway threshold.

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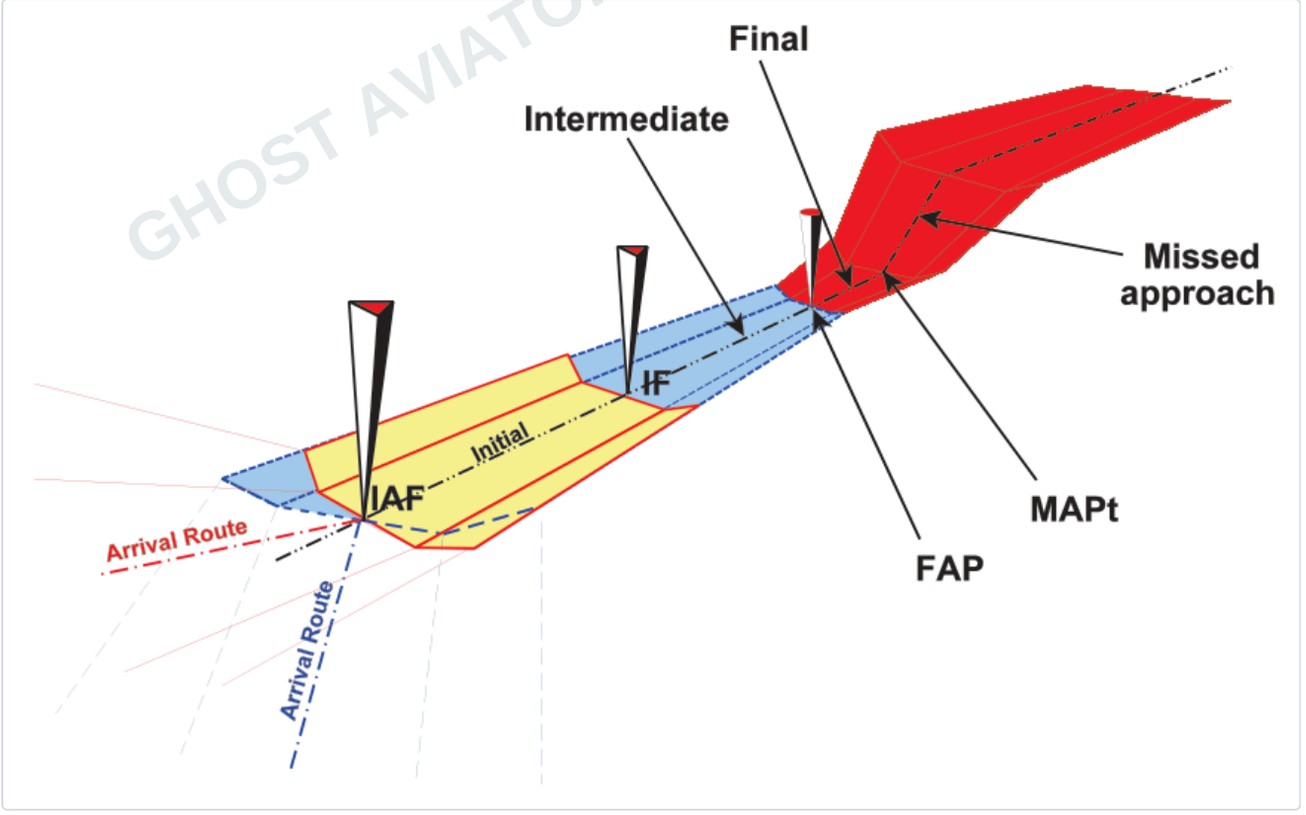

Figure 12.10 *Instrument approach segments: arrival, initial, intermediate, final and missed approach.*

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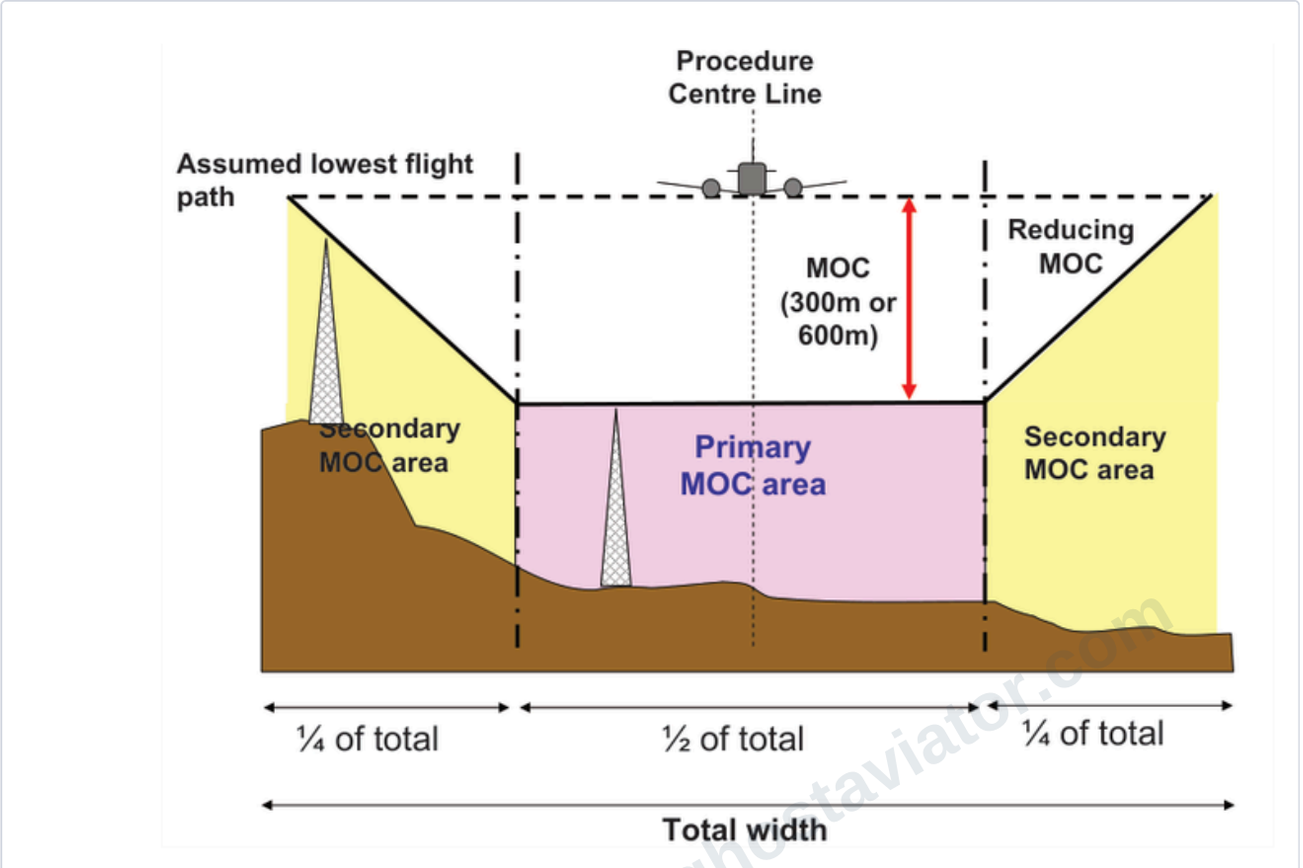

Figure 12.11 *Cross-section of a segment: primary area (half of the total width) with full MOC, and a secondary area (a quarter of the width each side) where MOC reduces to zero.*

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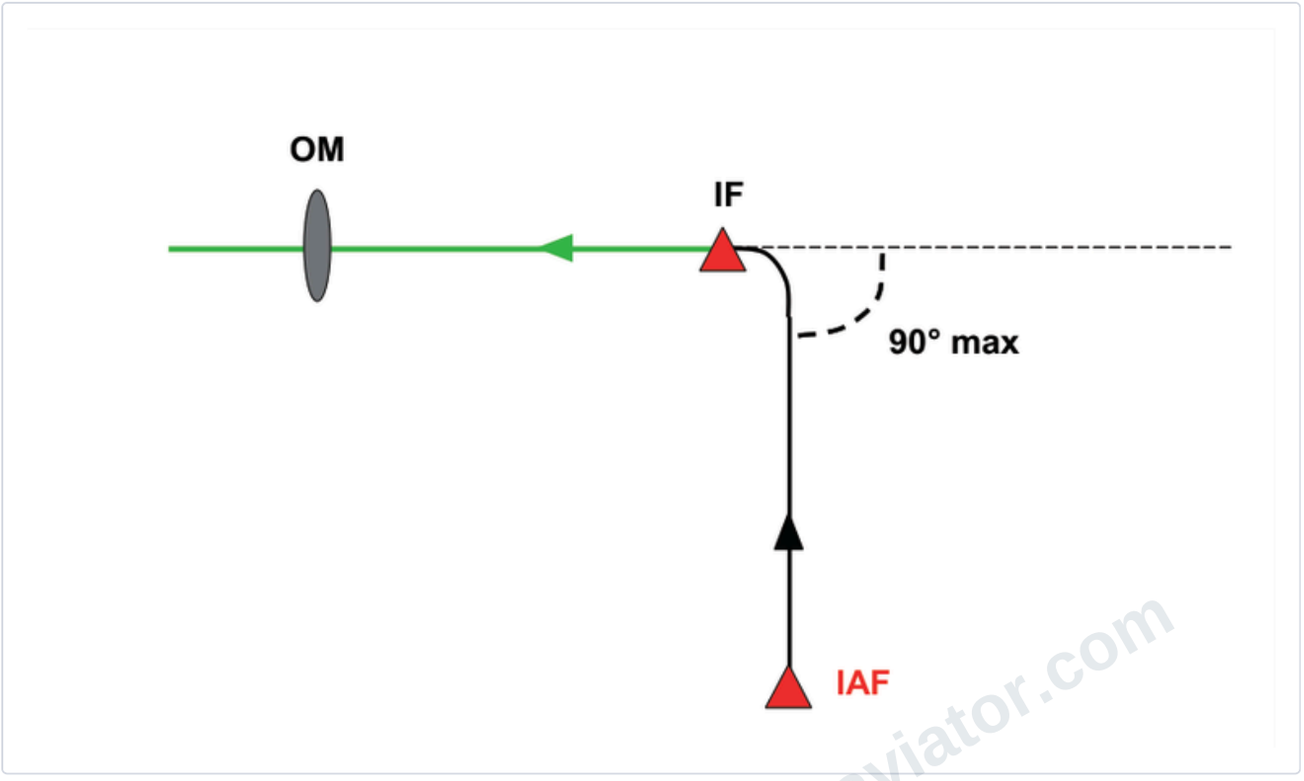

Figure 12.12 *Initial-to-intermediate approach: maximum interception angle at the IF is 90° for a precision approach.*

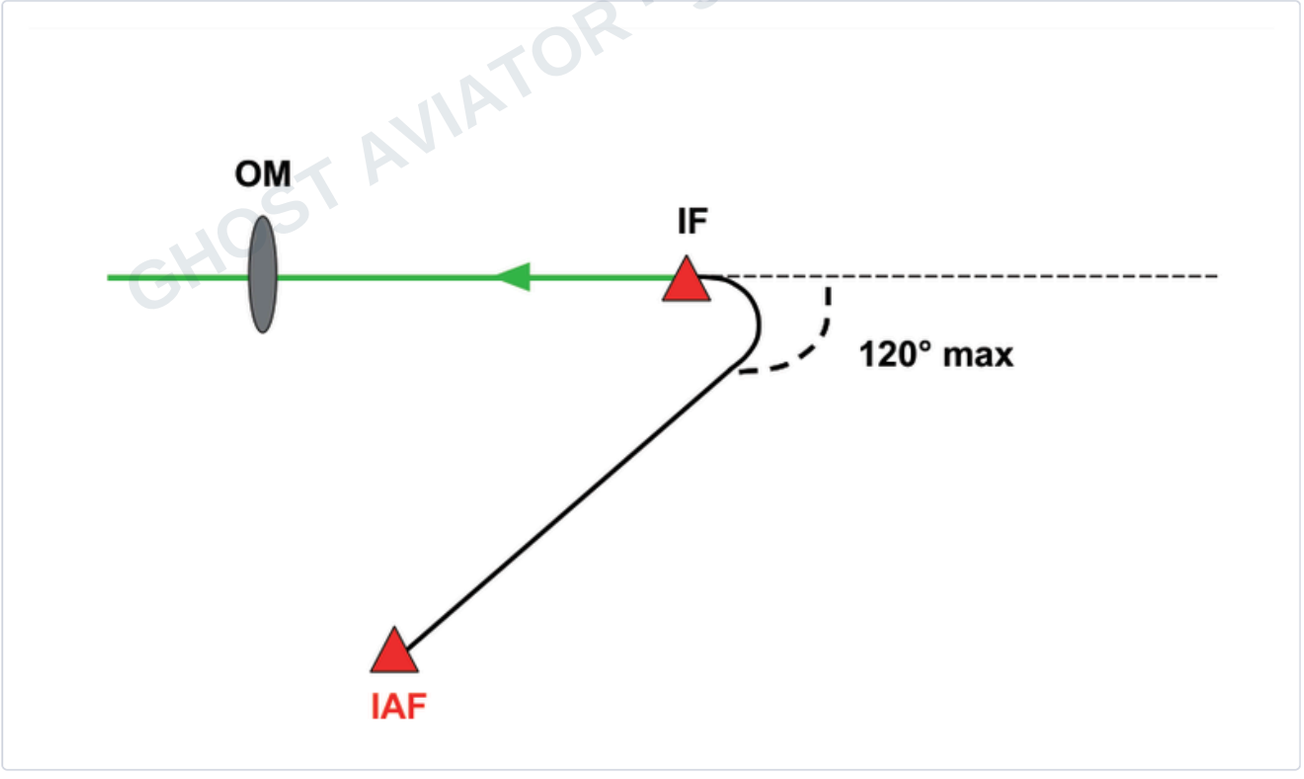

Figure 12.13 *Maximum interception angle at the IF for a non-precision approach is 120°.*

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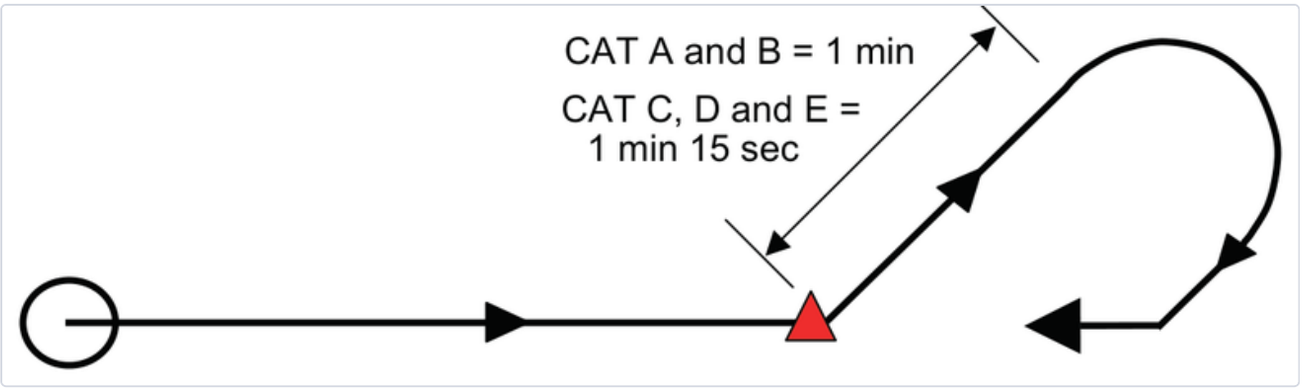

Figure 12.14 *45°/180° procedure turn (outbound timing 1 min for categories A and B, 1 min 15 s for C, D and E in the book diagram).*

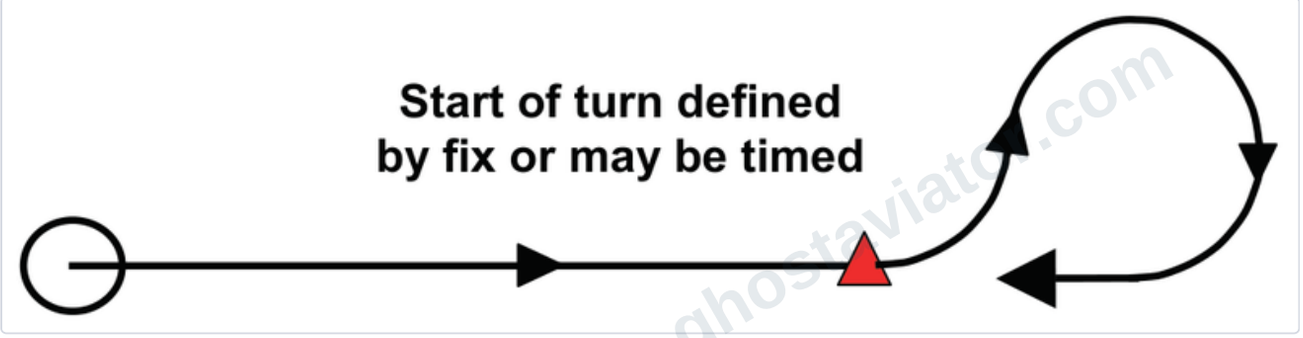

Figure 12.15 *80°/260° procedure turn.*

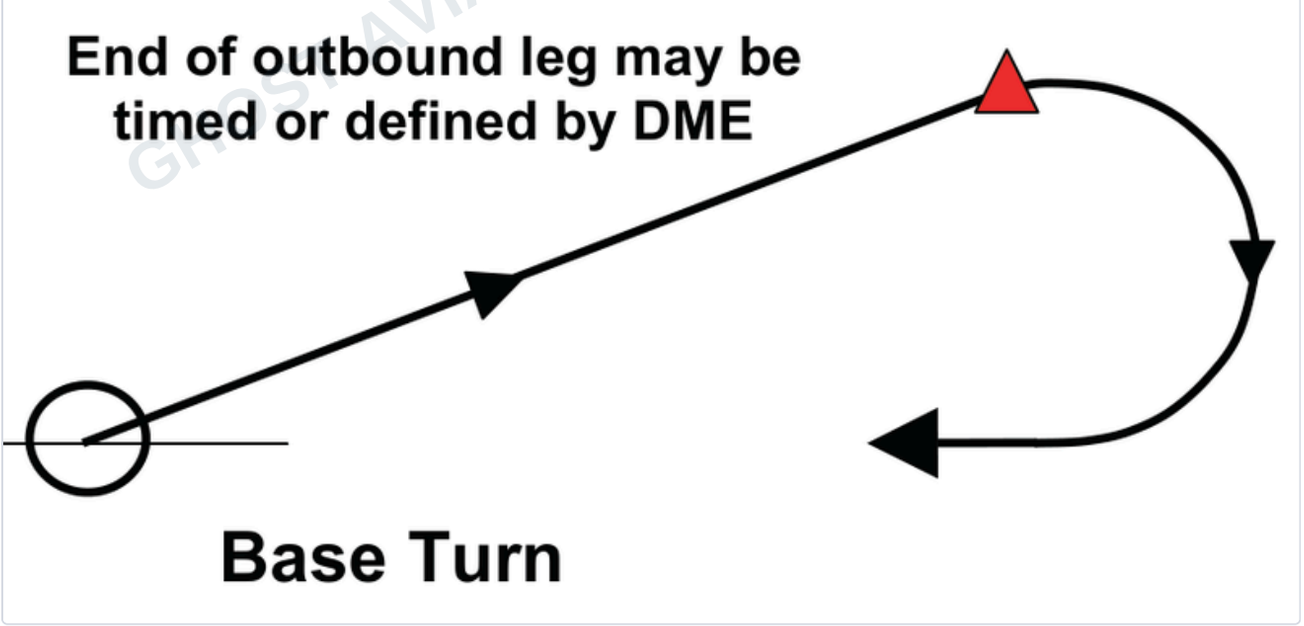

Figure 12.16 *Base turn: the end of the outbound leg may be timed or defined by DME.*

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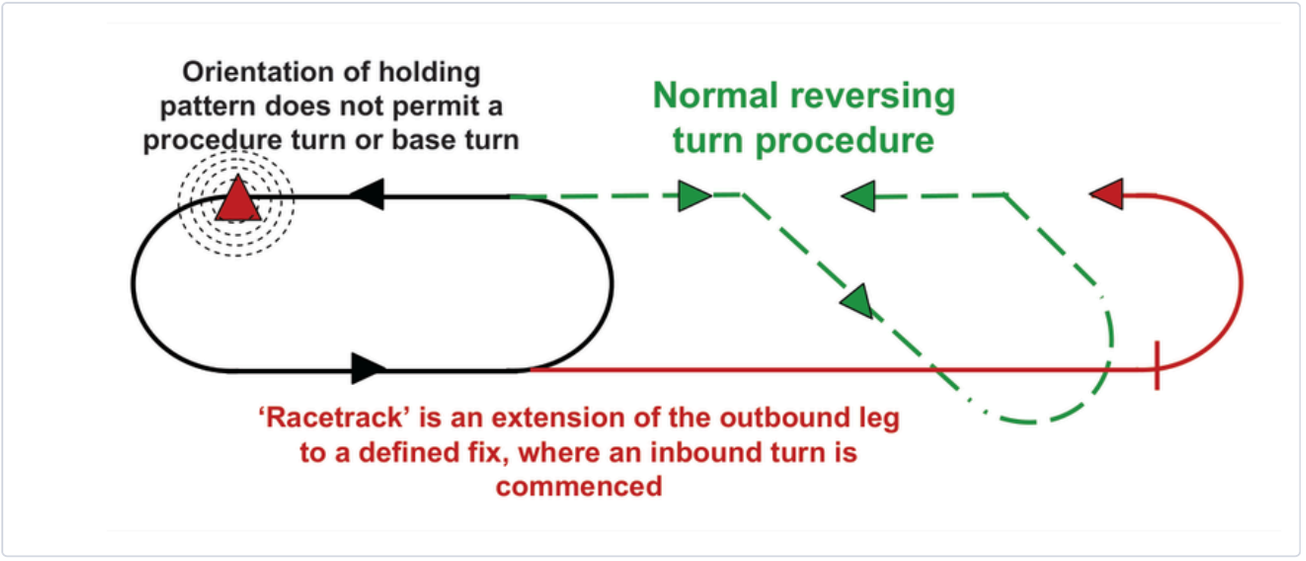

Figure 12.17 *Racetrack procedure: an extension of the outbound leg to a defined fix where the inbound turn is commenced.*

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MISSED APPROACH — GENERAL

- Only **one** missed approach procedure per instrument approach procedure.
- Three phases: **Initial, Intermediate, and Final**.
- Specifies a point where MA begins and a point or alt/height where it ends.
- MA should be initiated not lower than **DA/H** in precision approaches, or at the MAPt not lower than **MDA/H** in non-precision.

MAPt Definition

FOR PA/APV

MAPt = point of intersection of the electronic glide path with the applicable DA/H.

FOR NPA

MAPt = a navigation facility, a fix, or a specified distance from the FAF. Upon reaching the MAPt if required visual reference is not established, missed approach must be initiated at once.

MISSED APPROACH GRADIENT

Normally based on a minimum missed approach **climb gradient of 2.5 per cent**.

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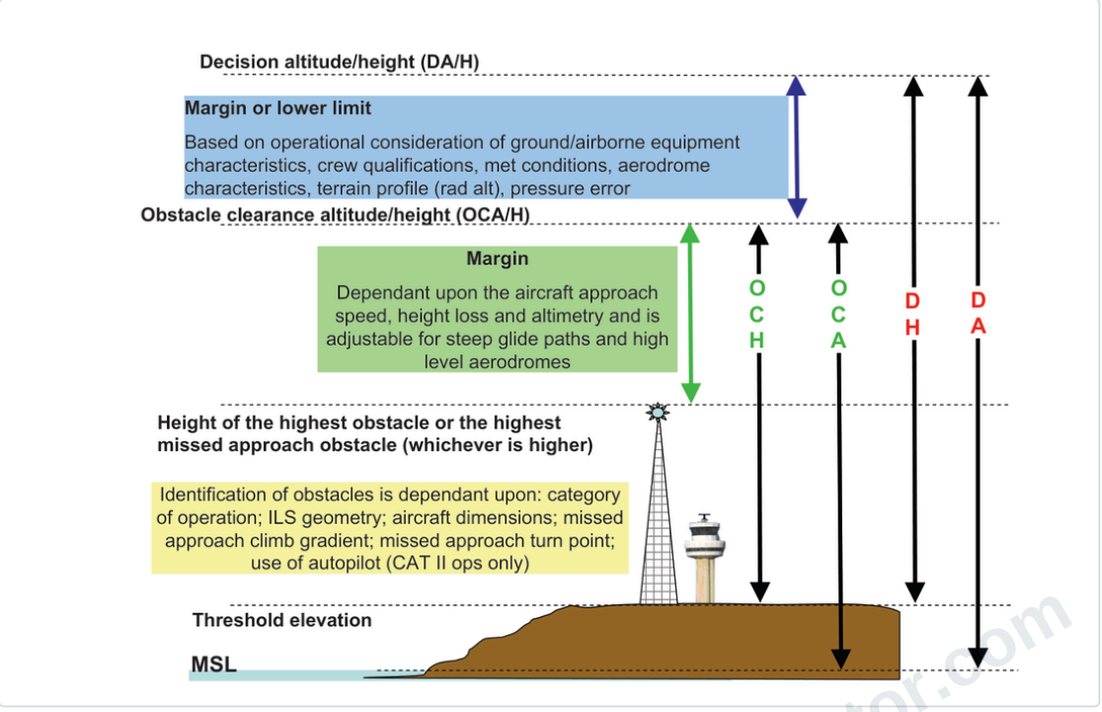

Figure 12.18 *Precision approach: how obstacle clearance height and decision height relate.*

Instrument approach classification

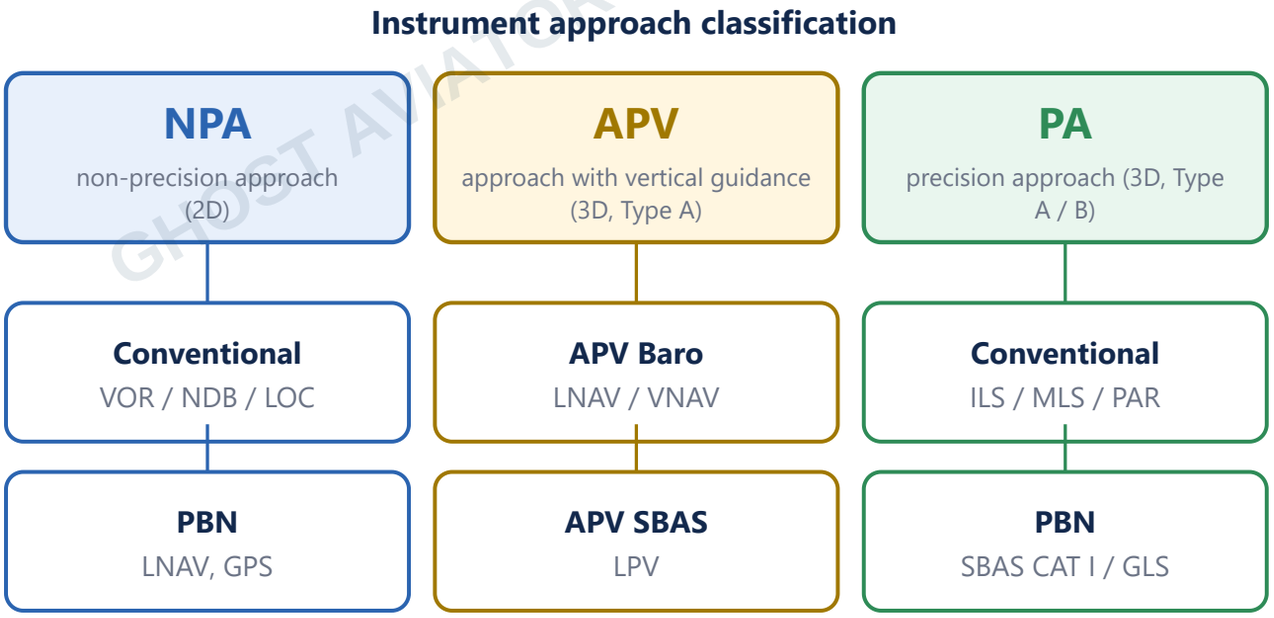

Figure 12.19 *Instrument approach classification*

Three Phases of Missed Approach

Missed Approach Phases

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Phase	Begins At	Ends At	Key Characteristics
Initial	MAPt	SOC (Start of Climb)	Concentrated attention on establishing climb. Guidance equipment not extensively utilised. **No turns specified** in this phase.
Intermediate	SOC	First point where 50 m (164 ft) obstacle clearance is obtained	Climb continued straight ahead. Track may be changed by max **15°**. Aircraft begins track corrections.
Final	50 m (164 ft) clearance point [Cat H: 40 m (131 ft)]	Point where new approach/holding/en-route flight is initiated	**Turns may be prescribed** in this phase. Airspace for turns based on *final missed approach speeds*.

TURNING MISSED APPROACH

Turns in a missed approach procedure are only prescribed where **terrain or other factors make a turn necessary**. The protected airspace for turns is based on speeds for the *final missed approach*.

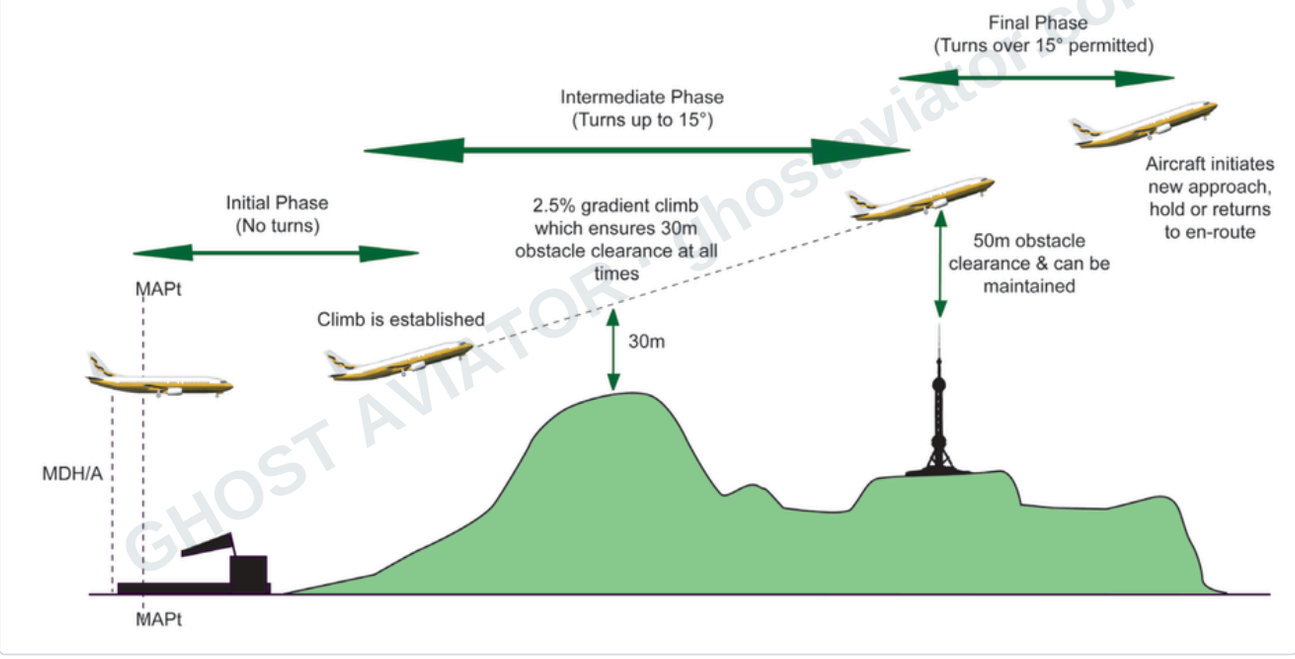

Figure 12.20 *Three phases of the missed approach: initial (no turns), intermediate (track change up to 15°) and final (turns permitted); 2.5 % gradient gives 30 m obstacle clearance.*

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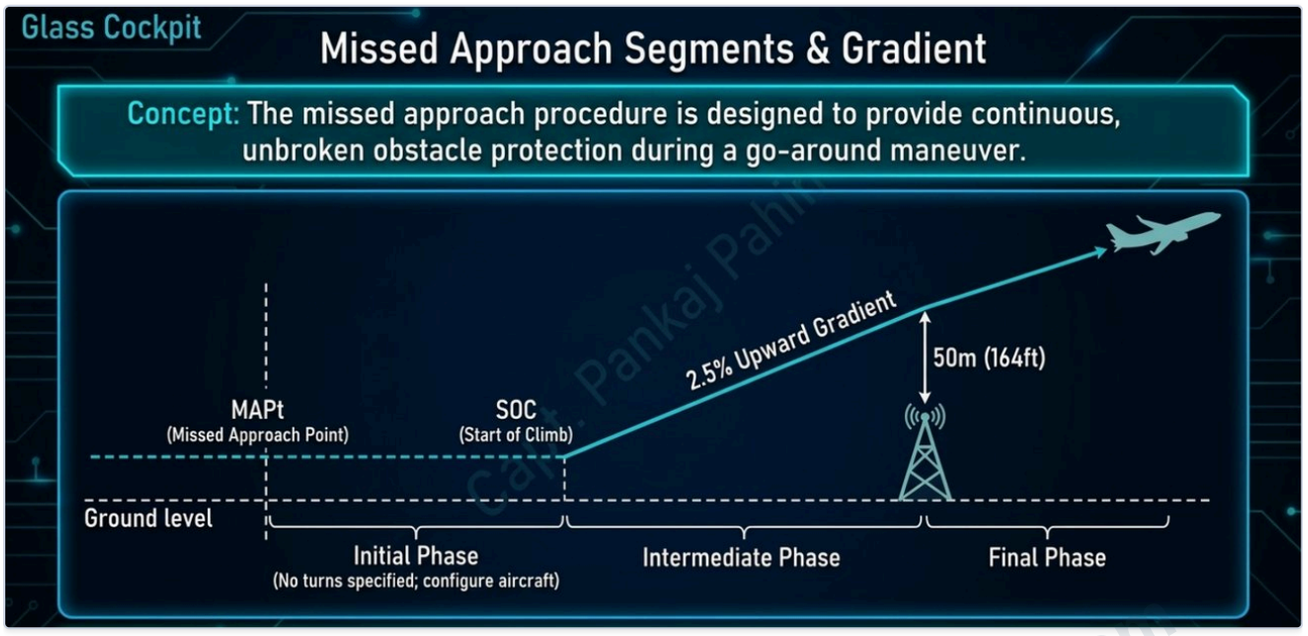

Figure 12.21 *Missed approach: the three phases and the 2.5% climb gradient*

Approach Operation Types

TYPE A VS TYPE B — CRITICAL FOR EXAMS

Type A: Minimum descent height or decision height at or above **75 m (250 ft)**.

Includes: NPA, APV, PA Cat I with DH ≥75 m.

Type B: Decision height **below 75 m (250 ft)**. Categorised as Precision Approach Runway:

- **CAT I:** DH ≥ 60 m (200 ft); RVR ≥ 800 m or visibility ≥ 550 m
- **CAT II:** DH 30–60 m (100–200 ft); RVR ≥ 300 m
- **CAT III:** DH < 30 m (100 ft) OR no DH; RVR < 300 m OR no RVR limitation

CAT II and CAT III operations shall NOT be authorized unless RVR information is provided.

Performance Requirements in Support of Instrument Approach Operations

Type of Approach	Annex 10 System Performance	Annex 6 Approach Operation Category
Non-Precision (NPA)	—	2D – Type A
APV (Vertical Guidance)	—	3D – Type A
Precision Approach (PA)	CAT I, DH ≥ 75 m (250 ft)	3D – Type A
	CAT I, DH ≥ 60 m (200 ft) and < 75 m (250 ft)	3D – Type B — CAT I
	Category II	3D – Type B — CAT II
	Category III	3D – Type B — CAT III

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TERMINAL INSTRUMENT FLIGHT PROCEDURE TYPES

- **NPA Conventional:** Ground-based (VOR/DME, NDB, Localiser). Runway visibility ≥ 1,000 m.
- **NPA PBN (RNP APCH):** Uses GNSS. Charted as RNAV(GNSS) or RNAV(GPS). Types: RNP App, NPA GPS, LNAV.
- **APV (RNP AR APCH):** 3D guidance but lower performance than PA. Types: APV BaroVNAV, APV SBAS (LPV), LNAV/VNAV.
- **PA Conventional:** ILS, MLS, PAR. 3D approach ops Type A or B.
- **PA PBN:** SBAS Cat I, GLS (GBAS Landing System).

DEFINITION

Visual manoeuvring (circling) is the phase of flight after an instrument approach has been completed, which brings the aircraft into position for landing on a runway that is **not suitably located for straight-in approach** (i.e., alignment or descent gradient criteria cannot be met).

The Visual Manoeuvring Area

AREA CONSTRUCTION

Arcs centred on each runway threshold, joined by tangent lines. Radius depends on:

- **Aircraft category**
- **Speed** for each category
- **Wind speed: 46 km/h (25 kt)** throughout the turn
- **Bank angle: 20° average** or **3° per second**, whichever requires less bank

DESCENT BELOW MDA/H RULES

Descent below MDA/H should NOT be made until ALL three of the following are met:

1. Visual reference has been established and **can be maintained**.
2. The pilot has the **landing threshold in sight**.
3. The required obstacle clearance can be maintained and the aircraft is **in a position to carry out a landing**.

MISSED APPROACH WHILE CIRCLING

If visual reference is lost while circling, the **specified missed approach procedure must be followed**. The pilot will make an initial climbing turn toward the landing runway and overhead the aerodrome, then establish on the missed approach track.

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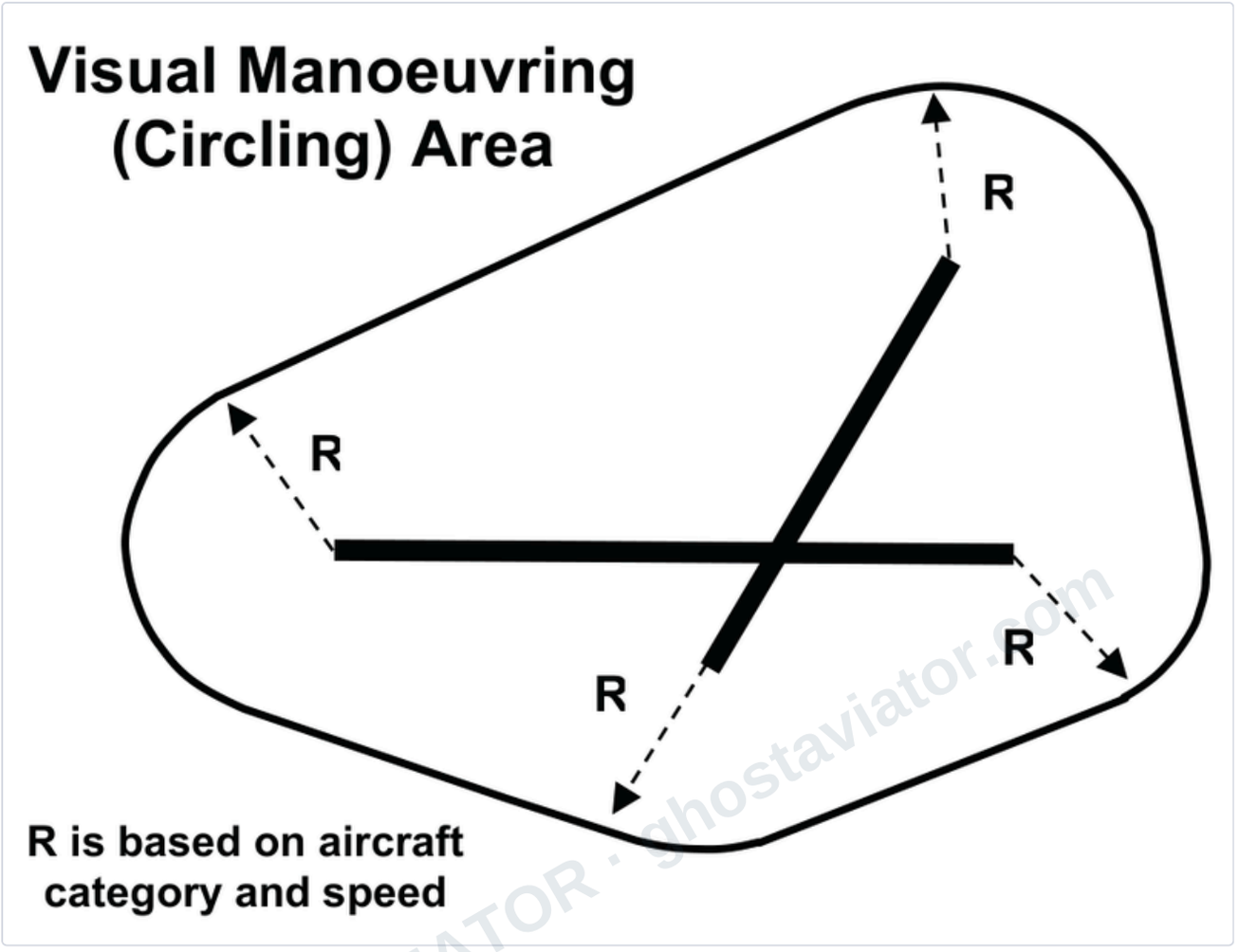

Figure 12.22 *Visual manoeuvring (circling) area: arcs of radius R centred on each runway threshold joined by tangents; R depends on aircraft category and speed.*

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General Rules

BANK ANGLE / RATE OF TURN

All turns to be made at a **bank angle of 25°** or at a **rate of 3° per second**, whichever requires the **lesser bank**.

WIND CORRECTION

Pilots should attempt to maintain the track by applying allowance for known wind — both heading and timing corrections. This applies during **entry and while flying in the holding pattern**.

OUTBOUND TIMING

Timing begins **over or abeam the fix**, whichever occurs later. If abeam position cannot be determined, start timing when the turn to outbound is completed.

DME OUTBOUND LEG

If outbound leg is based on a DME distance, it terminates as soon as the limiting DME distance is reached.

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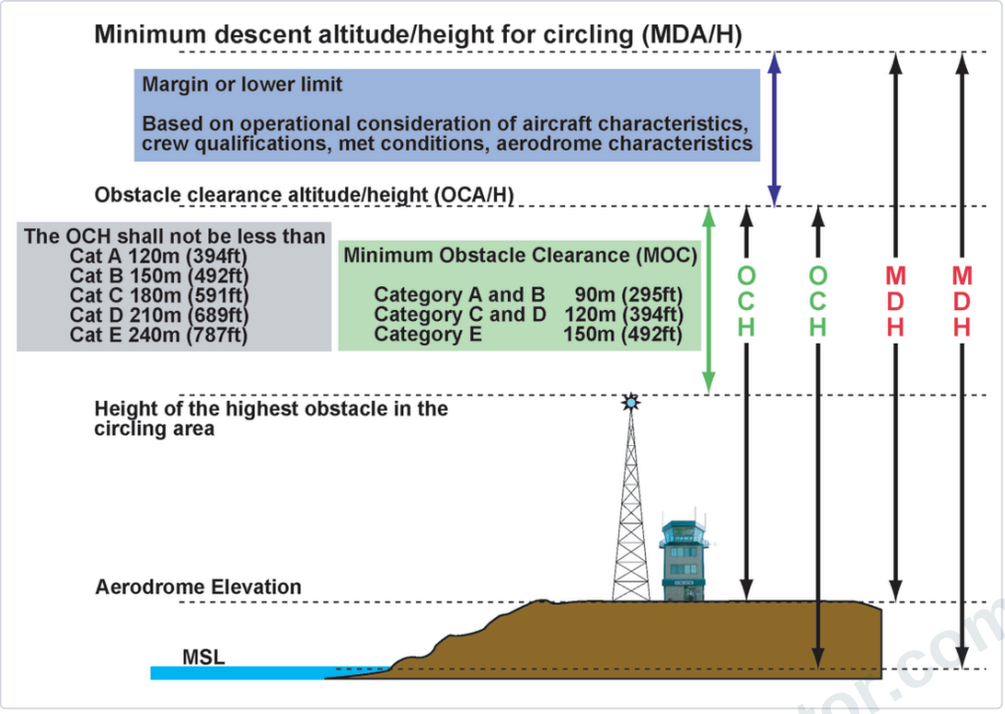

Figure 12.23 *Circling: obstacle clearance and minimum descent height for each aircraft category.*

Holding Entry — Three Sectors

Holding Entry Procedures

Sector	Entry Type	Procedure
Sector 1	Parallel Entry	At the fix, turn LEFT onto outbound heading for the appropriate period of time; then turn LEFT to intercept inbound track or return to fix; on second arrival, turn RIGHT to follow holding pattern.
Sector 2	Offset Entry (30°)	At the fix, turn to make good a track at **30°** from the reciprocal of the inbound track on the holding side; fly outbound for the appropriate time; turn RIGHT to intercept inbound track; on second arrival over fix, turn RIGHT to follow holding pattern.
Sector 3	Direct Entry	Having reached the fix, turn RIGHT to follow the holding pattern.

VOR/DME FIX — ENTRY TRACK LIMITATIONS

- The entry track to a VOR/DME fix is limited to: (a) the VOR radial; (b) the DME arc; or (c) the entry radial to a VOR/DME fix at the end of the outbound leg as published.
- *Note: DME arc entry is specified only when there is a specific operational difficulty making other entry procedures impossible.*

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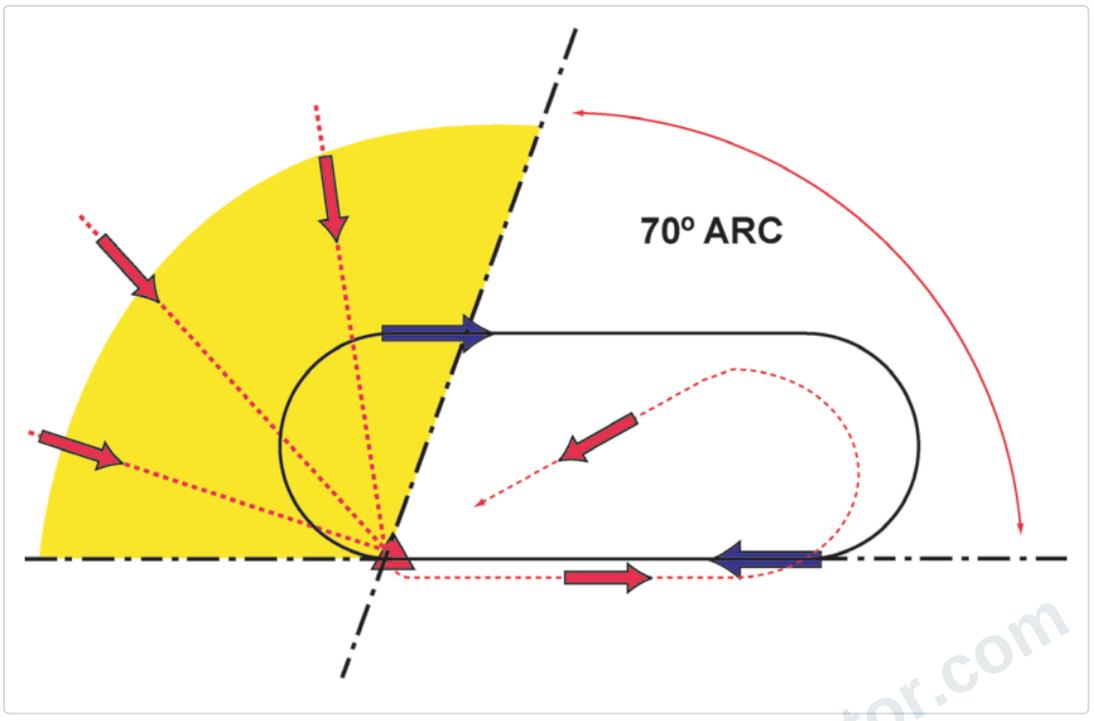

Figure 12.24 *Sector 1: parallel entry.*

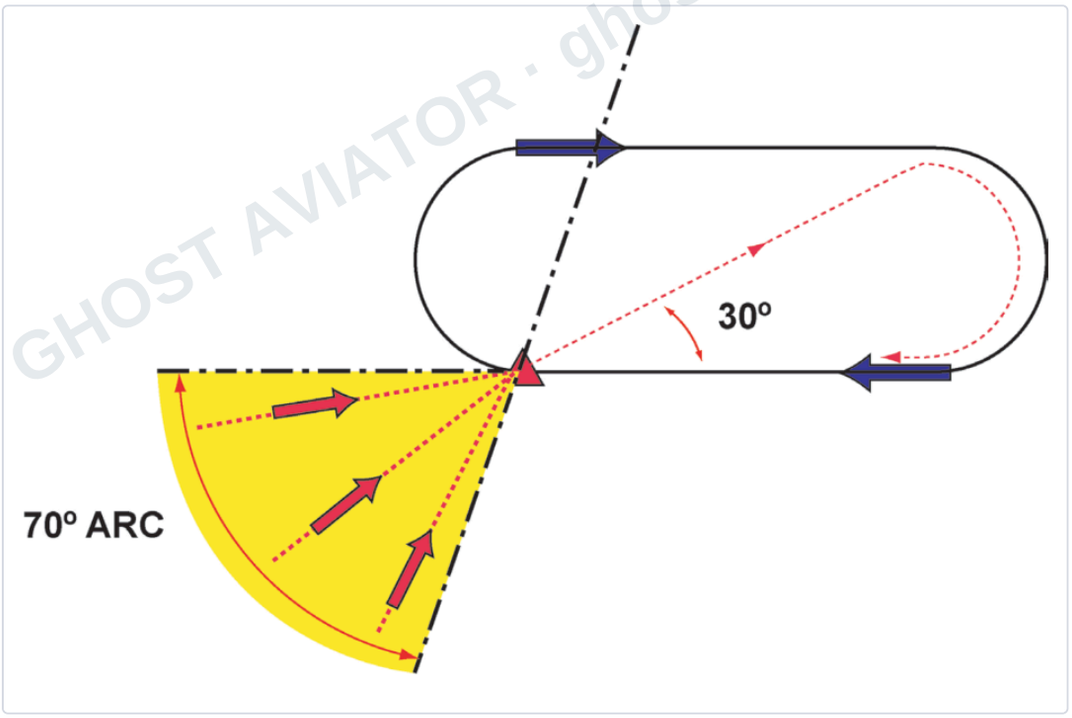

Figure 12.25 *Sector 2: offset (teardrop) entry, 30 degrees from the reciprocal of the inbound track.*

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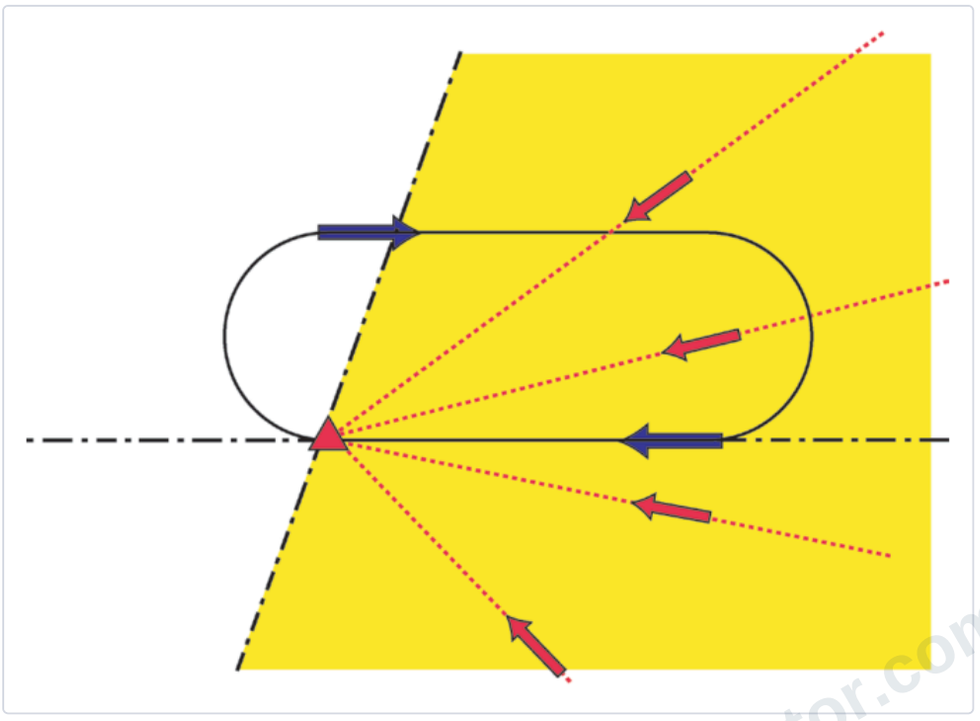

Figure 12.26 *Sector 3: direct entry.*

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Holding Speeds

Holding Speeds — Categories A through E

Level / Altitude	Normal Conditions	Turbulence Conditions
Up to 4,250 m (14,000 ft)	425 km/h (230 kt) — Cats C,D,E 315 km/h (170 kt) — Cats A,B	520 km/h (280 kt) [or 0.8 Mach] 315 km/h (170 kt) — Cats A,B
4,250 m (14,000 ft) to 6,100 m (20,000 ft) inclusive	445 km/h (240 kt)	520 km/h (280 kt) or 0.8 Mach — lesser
6,100 m (20,000 ft) to 10,350 m (34,000 ft) inclusive	490 km/h (265 kt)	520 km/h (280 kt) or 0.8 Mach — lesser
Above 10,350 m (34,000 ft)	0.83 Mach	0.83 Mach

Notes: Speed of 520 km/h (280 kt) for turbulence requires prior ATC clearance, unless publications indicate the holding area can accommodate it. 315 km/h (170 kt) applies to Cat A and B only.

DME ARC ENTRY — OUTBOUND TIME LIMITS

- At or below **4,250 m (14,000 ft)**: still air time for outbound entry heading must not exceed **1 minute**.
- Above **4,250 m (14,000 ft)**: must not exceed **1.5 minutes**.

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MINIMUM HOLDING LEVEL

- Provides clearance of at least **300 m (984 ft)** above obstacles in the holding area.
- Minimum holding altitude published shall be rounded **up** to the nearest 50 m or 100 ft.
- Over high terrain/mountainous areas: additional clearance up to **600 m (1,969 ft)** is provided to accommodate turbulence, down-drafts, and altimeter errors.

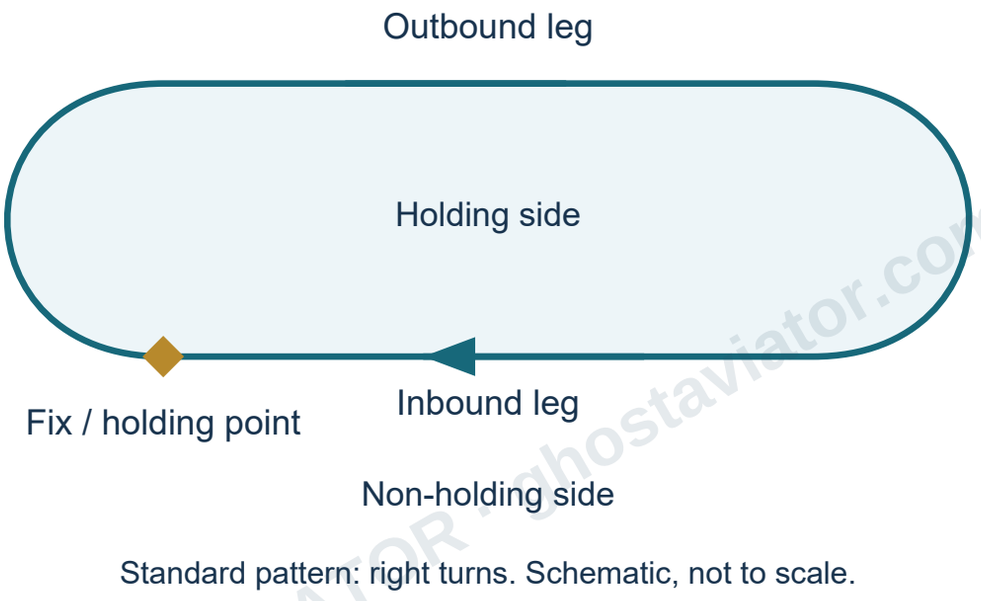

Figure 12.27 *Holding-pattern terminology. The source specifies turns at25°bank or3°per second, whichever requires less bank; use any published or instructed procedure that applies.*

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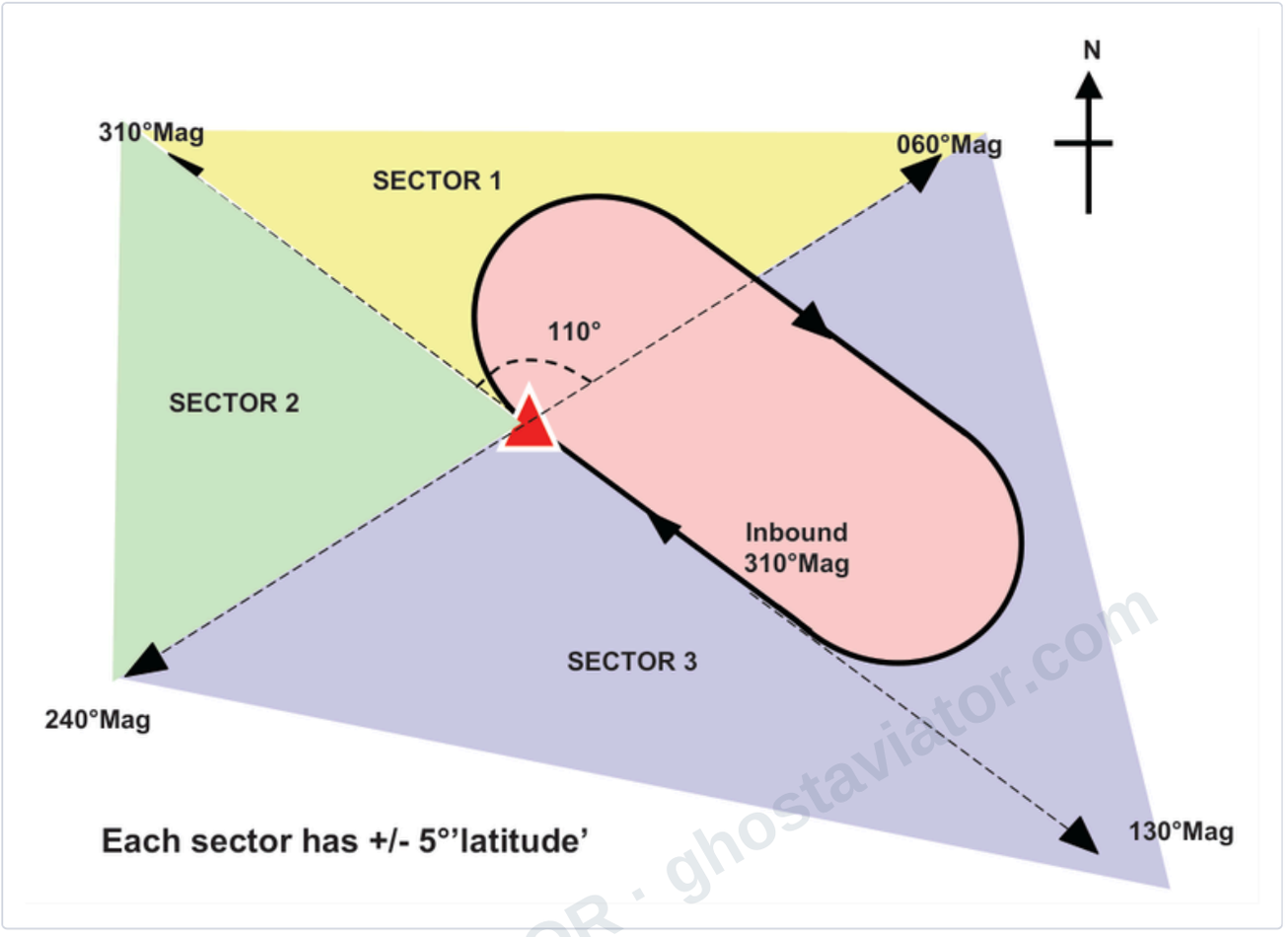

Figure 12.28 *Holding entry sectors: parallel, offset and direct; each sector boundary has a ±5° zone of flexibility.*

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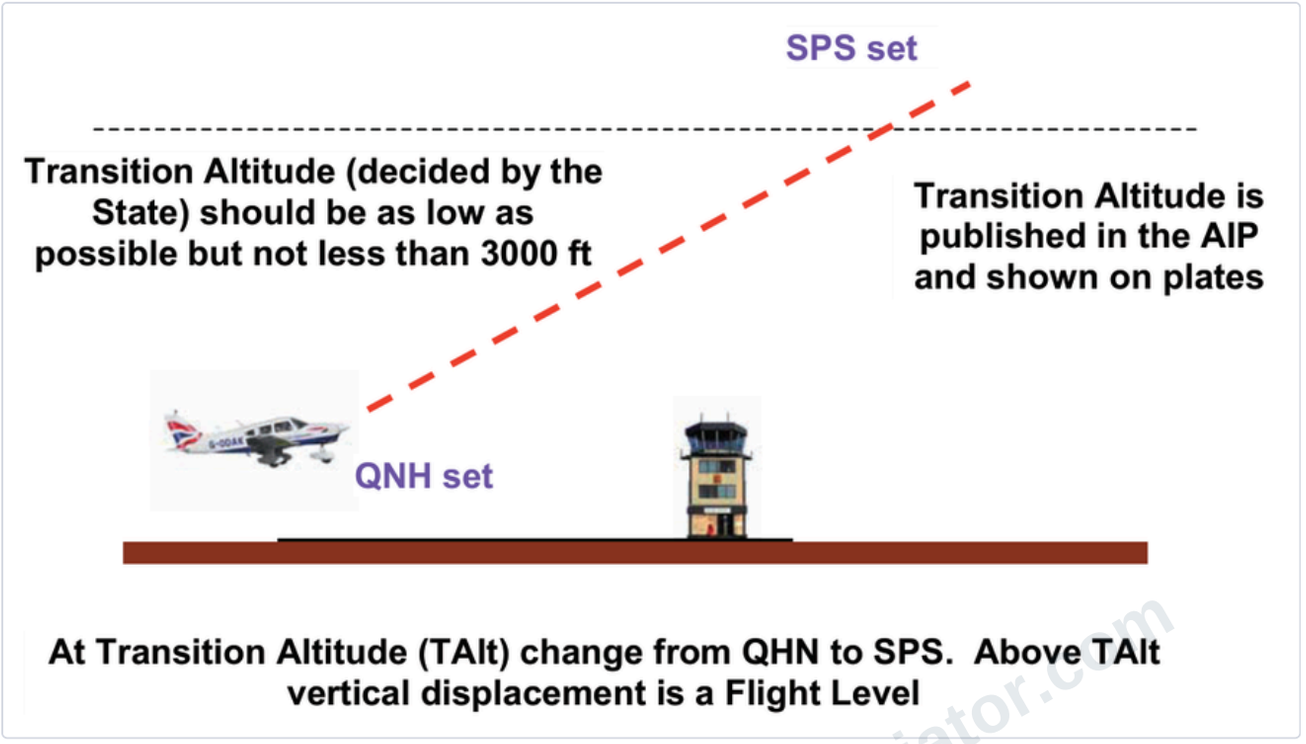

Figure 12.29 *Transition altitude: on climb the altimeter changes from QNH to the standard setting at the transition altitude; above it, vertical position is a flight level.*

Modes of Operation for Parallel Instrument Runways

Mode	Name	Description	Runway Separation
Mode 1	Independent Parallel Approaches	Radar separation minima between aircraft using adjacent ILS/MLS are **not prescribed**.	≥ 1,035 m between C/Ls
Mode 2	Dependent Parallel Approaches	Radar separation minima between aircraft using adjacent ILS/MLS **are prescribed**.	≥ 915 m between C/Ls
Mode 3	Independent Parallel Departures	Aircraft departing in same direction from parallel runways simultaneously. (Less than specified C/L distance for wake turbulence = single runway.)	≥ 760 m between C/Ls
Mode 4	Segregated Parallel Ops	One runway for approaches, one for departures.	≥ 760 m between C/Ls; decrease by 30 m for each 150 m the arrival runway is staggered toward the arriving aircraft, to a minimum of 300 m (increase by 30 m for each 150 m staggered away)

VOR/DME RNAV APPROACH

Based on one reference facility composed of a **VOR and collocated DME**. The reference facility will be indicated. The VOR/DME RNAV approach is a **non-precision approach procedure**.

RNAV VOR/DME — Key Values by Segment

Segment	FTT Value	Key Rules

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Segment	FTT Value	Key Rules
STARs (Arrival)	3.7 km (2.0 NM) until 46 km (25 NM) from IAF; 1.9 km (1.0 NM) after	Based on RNP 1 or better; or specific RNAV criteria
Final Approach	MOC = 75 m (246 ft) in primary area	Generally aligned with runway. FAF = fly-by waypoint. Flyover waypoint also provided at runway threshold.
Missed Approach	Area splays at 15° each side from earliest MAPt	MAHF (Missed Approach Holding Fix) defines end of missed approach segment.

PURPOSE

Provide adequate vertical separation between aircraft and adequate terrain clearance during all phases of flight. Method is based on the following basic principles:

Core Principles

ALTITUDE VS FLIGHT LEVEL

- At or below **transition altitude**: vertical position expressed as **ALTITUDE** (altimeter set to QNH).
- Above **transition altitude**: vertical position expressed as **FLIGHT LEVELS** (altimeter set to 1,013.2 hPa).
- Change from altitude → flight level: at **transition altitude** (on climb).
- Change from flight level → altitude: at **transition level** (on descent).
- **Transition layer** = airspace between transition level and transition altitude.

TRANSITION ALTITUDE — CRITICAL VALUES

- Normally specified for each aerodrome by the State.
- **Minimum transition altitude in India = 4,000 ft**
- Height above aerodrome: as low as possible but **normally not less than 900 m (3,000 ft)**.
- Calculated height rounded **up** to the next full 300 m (1,000 ft).
- Transition level located **300 m (1,000 ft)** above the transition altitude (to permit concurrent use in cruising flight with vertical separation ensured).
- **Flight Level Zero = 1,013.2 hPa**. Consecutive FLs separated by a pressure interval corresponding to at least **500 ft (152.4 m)** in the standard atmosphere.

Pre-Flight Altimeter Test

QNH / QFE SETTING PRE-FLIGHT OPERATIONAL TEST

1. With aircraft at a known elevation on the aerodrome, set the altimeter to the current QNH/QFE setting.
2. Vibrate the instrument by tapping unless mechanical vibration is provided.

A serviceable altimeter indicates the elevation/height within a tolerance of:

- Test range 0 to 9,000 m (0 to 30,000 ft): **±20 m or 60 ft**
- Test range 0 to 15,000 m (0 to 50,000 ft): **±25 m or 80 ft**

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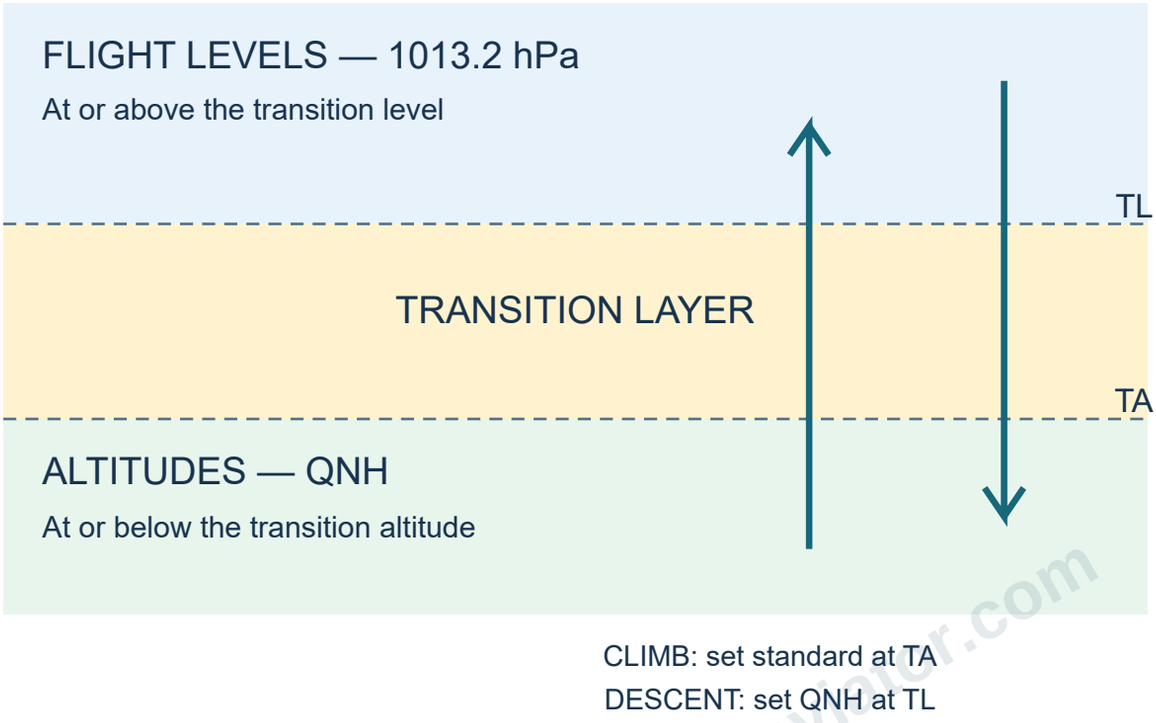

Figure 12.30 *Altimeter setting through the transition layer. On climb, use flight levels in the layer; on descent, use altitudes. The layer is not a cruising band.* *AAI MATS Part 1, Ed 6.2, §4.16.1.1 and §4.16.2.2.*

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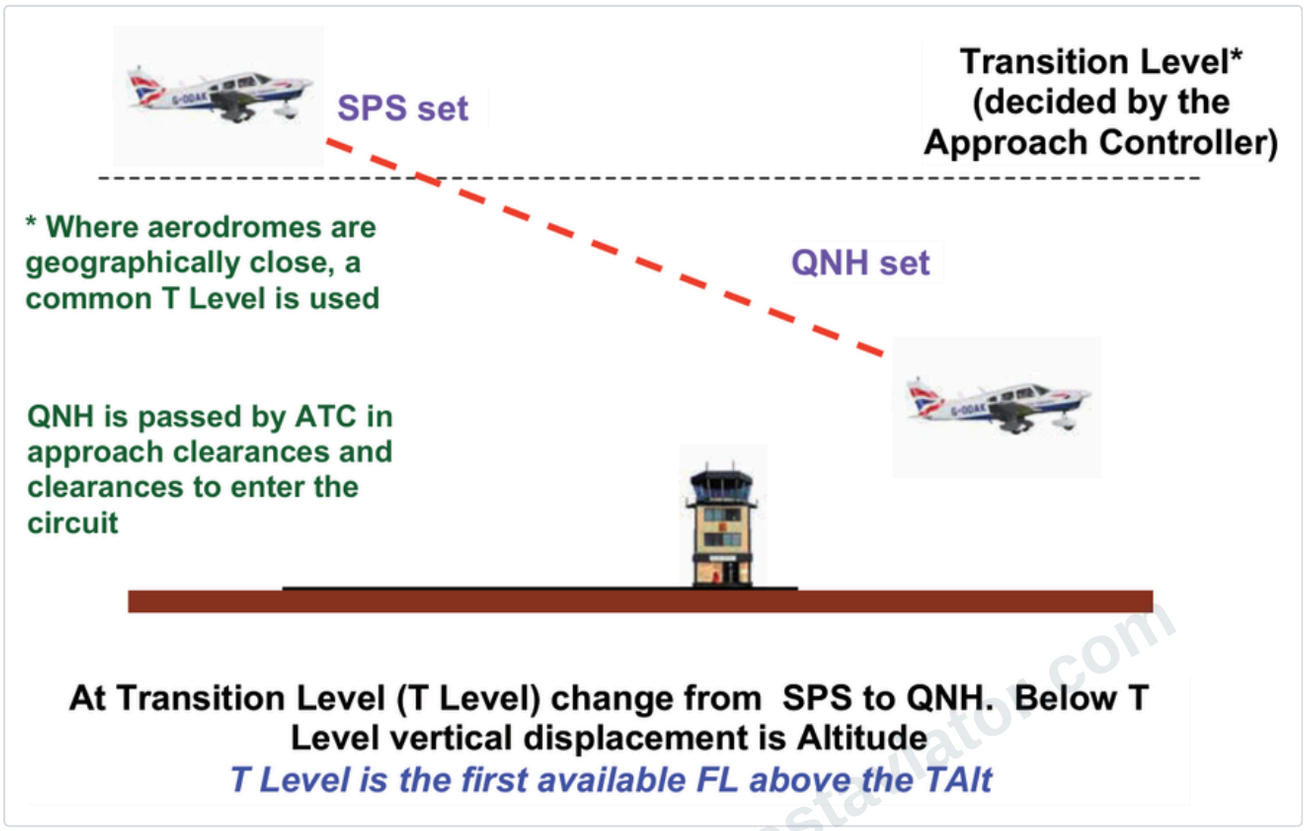

Figure 12.31 *Transition level: on descent the altimeter changes from the standard setting to QNH at the transition level; below it, vertical position is an altitude.*

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Key Definitions

SSR

Secondary Surveillance Radar. Uses **transmitters/receivers (interrogators) and transponders**.

ACAS (DEFINITION)

Aircraft system based on SSR transponder signals operating **independently of ground-based equipment** to provide advice to the pilot on potential conflicting aircraft equipped with SSR transponders.

MODE A

To elicit transponder replies for **identity and surveillance**.

MODE C

To elicit transponder replies for **automatic pressure-altitude transmission and surveillance**.

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Transponder Operating Procedures

GENERAL OPERATION RULE

When an aircraft carries a serviceable transponder, the pilot shall operate it **at all times during flight**, regardless of whether SSR is used for ATS purposes in that airspace.

Squawk codes are four-digit octal numbers. Dials read 0–7 inclusive. Lowest squawk = **0000**. Highest = **7777**.

CODE SELECTION PRIORITY

1. Operate on Mode A codes **as directed by the ATC unit** with which contact is being made.
2. Operate on Mode A codes as prescribed by **regional air navigation agreements**.
3. In absence of any ATC directions or regional agreements: operate on **Mode A Code 2000**.

CONFIRM SQUAWK PROCEDURE

When ATC requests "CONFIRM SQUAWK (code)", the pilot shall:

1. Verify the Mode A code setting on the transponder.
2. Reselect the assigned code if necessary.
3. Confirm to ATC the setting displayed on the transponder controls.

Pilots shall NOT SQUAWK IDENT unless requested by ATC.

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Emergency Squawk Codes

7700	GENERAL EMERGENCY
7600	COMMUNICATION FAILURE
7500	UNLAWFUL INTERFERENCE (HIJACK)

EMERGENCY TRANSPONDER RULES

- **7700 (Emergency):** Set unless ATC has previously directed to operate on a specified code. However, pilot may select 7700 whenever there is specific reason to believe it is the best course of action.
- **7600 (Comm Failure):** Set when losing two-way communications. Controller will determine extent of failure by instructing pilot to SQUAWK IDENT or change code.
- **7500 (Unlawful Interference):** If circumstances warrant, Code 7700 should be used instead. If ATC requests pilot to confirm 7500, pilot shall — according to circumstances — **either confirm or not reply at all**. Non-reply = confirmation that 7500 is not an inadvertent selection.

TRANSPONDER FAILURE AFTER DEPARTURE

ATC units shall attempt to provide for continuation of the flight to the destination aerodrome in accordance with the flight plan. Pilots may, however, expect to comply with specific restrictions.

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PURPOSE OF ACAS

ACAS indications shall be used by pilots in the avoidance of potential collisions, enhancement of situational awareness, and the active search for, and visual acquisition of, conflicting traffic. Nothing in ACAS procedures prevents pilots-in-command from exercising their best judgment and full authority in the choice of the best course of action.

ACAS Operations — Surveillance

SURVEILLANCE RANGE

- Nominal surveillance range: **26 km (14 NM)**
- Minimum guaranteed range when airborne: **8.5 km (4.5 NM)**
- Range can be reduced in geographic areas with large numbers of ground interrogators and/or ACAS-equipped aircraft.

TA vs RA — Critical Distinction

TRAFFIC ADVISORY (TA)

- Intended to **alert pilots** to the possibility of an RA, to enhance situational awareness, and to assist in visual acquisition of conflicting traffic.
- Pilots shall **NOT manoeuvre** in response to TAs only.
- On receipt of a TA, use all available information to **prepare for appropriate action** if an RA occurs.
- TAs can be issued against any Mode C transponder-equipped aircraft.

RESOLUTION ADVISORY (RA) — MANDATORY ACTIONS

1. **Respond IMMEDIATELY** by following the RA as indicated, unless doing so would jeopardize safety.
2. Follow the RA even if there is a **conflict between RA and ATC instruction** to manoeuvre.
3. **Do NOT manoeuvre in the opposite sense** to the RA.
4. As soon as possible, **notify the appropriate ATC unit** of the RA, including the direction of any deviation.
5. Promptly comply with any **modified RAs**.
6. Limit alterations to the **minimum extent** necessary.
7. Promptly return to ATC clearance when conflict is resolved.
8. Notify ATC when returning to the current clearance.

ACAS COLLISION AVOIDANCE — KEY RULES

- TAs can be issued against any transponder-equipped aircraft responding to Mode C interrogations, even without altitude-reporting capability.
- RAs can be issued **only against aircraft that are reporting altitude** and in the vertical plane only.
- RAs against ACAS-equipped intruders are coordinated to ensure complementary RAs are issued.

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- Failure to respond to an RA deprives the aircraft of collision protection AND restricts the other aircraft's ACAS choices.
- Manoeuvring in the opposite direction to an RA is **likely to result in further reduction in separation**.

ACAS ALERT PRIORITY ORDER

The following alerts take precedence over ACAS:

- **Stall Warning**
- **Wind Shear Warning**
- **Ground Proximity Warning System (GPWS) alerts**

Exam Tips and Key Numbers

PANS-OPS Chapter 13 — Critical Numerical Values

Topic	Value	Unit	Context
Standard PDG (fixed-wing)	3.3%	gradient	Unless otherwise published
Standard PDG (helicopters)	5.0%	gradient	Cat H procedures
OIS gradient (basis of PDG)	2.5%	gradient	Obstacle Identification Surface
MOC increase rate from DER	0.8%	of horizontal dist	Departure, max turn 15°
MOC in turn initiation area	90 m (295 ft)	m / ft	Departure turns
Straight departure track alignment	±15°	degrees	From runway C/L
Min altitude before turn (fixed-wing)	120 m (394 ft)	m / ft	Above aerodrome elevation
Min altitude before turn (Cat H)	90 m (295 ft)	m / ft	Helicopter turns
Turn initiation distance from runway start	600 m	m	Normal design optimisation
Close-in obstacle gradient not published	≤60 m (200 ft)	m / ft	A note published instead
MSA / TAA obstacle clearance	300 m (1,000 ft)	m / ft	Within 46 km (25 NM)
Initial approach MOC	300 m (1,000 ft)	m / ft	Primary area
Intermediate approach MOC (start)	300 m (984 ft)	m / ft	Reduces to 150 m
Intermediate approach MOC (end)	150 m (492 ft)	m / ft	Primary area
NPA MOC with FAF	75 m	m	Final approach
NPA MOC without FAF	90 m	m	Final approach
FAF optimal distance from threshold	9.3 km (5.0 NM)	km / NM	NPA with FAF
FAF maximum length from threshold	19 km (10 NM)	km / NM	NPA with FAF
FA interception angle (precision)	90°	degrees	Max angle at IF
FA interception angle (non-precision)	120°	degrees	Max angle at IF
Straight-in approach alignment	≤30°	degrees	NPA acceptable angle to C/L

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Topic	Value	Unit	Context
Missed approach gradient (standard)	2.5%	gradient	Minimum climb
MA intermediate phase turn	Max 15°	degrees	From initial MA track
MA final phase MOC (fixed-wing)	50 m (164 ft)	m / ft	Start of final MA phase
MA final phase MOC (Cat H)	40 m (131 ft)	m / ft	Cat H final MA
CAT I DH minimum	60 m (200 ft)	m / ft	Precision approach
CAT I RVR minimum	800 m	m	Or visibility 550 m
CAT II DH range	30–60 m (100–200 ft)	m / ft	RVR ≥ 300 m
CAT III DH	<30 m (100 ft)	m / ft	Or no DH; RVR < 300 m
Type A minimum DH/MDH	≥75 m (250 ft)	m / ft	2D/3D Type A operations
Holding turn bank / rate	25° or 3°/sec	°/°/sec	Whichever requires lesser bank
Holding entry zone of flexibility	±5°	degrees	Either side of sector boundaries
Holding speed ≤ 14,000 ft (Cat C,D,E)	230 kt	kt	Normal; 280 kt turbulence
Holding speed ≤ 14,000 ft (Cat A,B)	170 kt	kt	Normal and turbulence
Holding outbound time ≤ 14,000 ft	1 minute	min	Outbound leg timing
Holding outbound time > 14,000 ft	1.5 minutes	min	Above 4,250 m
Min holding clearance (normal)	300 m (984 ft)	m / ft	Above obstacles
Min holding clearance (mountainous)	600 m (1,969 ft)	m / ft	High terrain areas
India minimum transition altitude	4,000 ft	ft	INDIA specific
Altimeter test tolerance (0–30,000 ft)	±20 m / 60 ft	m / ft	Pre-flight test
Altimeter test tolerance (0–50,000 ft)	±25 m / 80 ft	m / ft	Pre-flight test
ACAS nominal surveillance range	26 km (14 NM)	km / NM	Nominal
ACAS minimum guaranteed range	8.5 km (4.5 NM)	km / NM	When airborne
Circling wind speed assumption	46 km/h (25 kt)	kt	Throughout the turn
Circling bank angle assumption	20° or 3°/sec	° / °/sec	Whichever requires less bank
SBAS straight departure FSD	0.3 NM	NM	From DER to turn initiation
Mode 1 parallel runway C/L separation	≥1,035 m	m	Independent parallel approaches
Mode 2 parallel runway C/L separation	≥915 m	m	Dependent parallel approaches

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Capt. Pankaj Pahil

PART IV

National Law, Licensing and Operations

The Indian Act and Rules, the licences you will hold, and the rules for operating an aircraft.

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CHAPTER 13

National Law

13

CURRENT STATUTORY FRAMEWORK

The Bharatiya Vayuyan Adhiniyam,2024 replaces the Aircraft Act,1934. Sections 3,5 and 7 continue the statutory bodies DGCA, BCAS and AAIB. Section 43(2) saves earlier rules and actions insofar as they are consistent with the new Act.

Bharatiya Vayuyan Adhiniyam 2024, §§3,5,7,32 and 43.

The older Act's section numbers below are historical references. Read current legal duties against the new Act and applicable rules.

The Aircraft Act, 1934 Central Legislation

WHAT IS THE AIRCRAFT ACT, 1934?

The Aircraft Act, 1934 is an **Act of Parliament** for the control of manufacture, possession, use, operation, sale, import and export of aircraft in India. Rules made under this Act are called the **Aircraft Rules, 1937**.

Extent & Applicability — The Aircraft Act Extends To

The Act extends to the whole of India and also applies to:

Clause	Applies To
(a)	Citizens of India **wherever they may be** (even outside India)
(b)	Persons on aircraft **registered in India**, wherever they may be
(c)	Aircraft registered **outside India** but for the time being **in or over India**
(d)	Aircraft operated by a non-citizen of India whose **principal place of business or permanent residence is in India**

Sec 3 — Power of Central Government to Exempt Certain Aircraft

The Central Government may, by notification in the **Official Gazette**, exempt any aircraft or class of aircraft and any person or class of persons from all or any provisions of this Act, or may direct that such provisions apply with modifications.

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Sec 5(qa) — Prohibition Near Aerodromes

CRITICAL LIMIT — AERODROME PROTECTION ZONE

The following are prohibited within a radius of **10 km** from the **aerodrome reference point**:

- Slaughtering and flaying of animals
- Depositing rubbish, filth and other polluted and obnoxious matter

Sec 8 — Power to Detain Aircraft

Any authority **authorised by the Central Government** may detain any aircraft if, in their opinion:

- **(a)** The intended flight would involve **danger to persons in the aircraft or to other persons or property**
- **(b)** Detention is necessary to **secure compliance** with any provision of the Act, or to prevent a contravention *(breaking a rule)* of any rule made under clause (h) or (i) of sub-section (2) of section 5, or to implement any court order

HOW DETENTION IS EFFECTED (RULE 18)

- By issuing a written direction to the pilot or person in charge
- By use of force, denial of access, removal of parts, defuelling
- If aircraft is kept at a Government aerodrome — **housing/picketing charges are payable** for the entire detention period

Sec 9 — Wreck and Salvage

The provisions of **Part XIII of the Merchant Shipping Act, 1958** relating to Wreck and Salvage apply to aircraft on or over the sea or tidal waters. The owner of an aircraft is entitled to a reasonable reward for salvage services, like the owner of a ship.

Sec 10 — Penalties for Contravention

Sub-section	Offence	Punishment
(1)	Contravention of rules under clause (l) — carriage of arms/explosives or giving **false information** about dangerous goods. Owner also liable unless offence was without owner's knowledge/consent.	**Imprisonment up to 2 years + Fine up to ₹1 Crore**
(1A)	Contravention of rules under clause (qa) — slaughter/flaying of animals, depositing rubbish within **10 km** of aerodrome reference point	**Imprisonment up to 3 years OR Fine up to ₹1 Crore, or both**
(1B)	Offence under sub-section (1A)	**Cognizable offence** (police can arrest without warrant)
(2)	Breach of any other rule under sections 4, 7, 8, 8A, 8B	**Imprisonment up to 2 years OR Fine up to ₹1 Crore, or both**

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Sec 10A — Additional Penalty Powers

The Central Government may, in making rules under sections 4, 5, 7, 8, 8A or 8B, provide for imposition of a penalty **not exceeding ₹1 Crore** for contravention of any rule for which no other punishment is provided elsewhere in the Act.

The Central Government may also appoint designated officers (not below the rank of **Deputy Secretary**) to adjudge penalties.

Sec 10B — Suspension or Cancellation of Licences

If any person **contravenes any provision** of the Act or rules thereunder, the licence, certificate or approval issued to that person under this Act may be **suspended or cancelled** in such manner as the Central Government may make rules.

Sec 11A — Penalty — Failure to Comply with Directions Under Section 5A

PENALTY — SECTION 5A NON-COMPLIANCE

Willful failure to comply with any direction issued under **Section 5A**:
Imprisonment up to 2 years OR Fine up to ₹1 Crore, or both

Sec 11B — Penalty — Failure to Comply with Directions Under Section 9A

PENALTY — SECTION 9A NON-COMPLIANCE

(1) Willful failure to comply with any direction in a notification issued under Section 9A:
Imprisonment up to 2 years OR Fine up to ₹1 Crore, or both

(2) Failure to demolish building/structure or cut trees within specified period → authorised officer may demolish the structure or cut the tree on behalf of the government.

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Sec 17 — Bar of Certain Suits

PROTECTION FOR AIRCRAFT OPERATORS

No suit shall be brought in any **Civil Court** in respect of trespass or nuisance merely because an aircraft has flown over any property at a **reasonable height** (having regard to wind, weather and all circumstances), or by reason of the **ordinary incidents of such flight**.

Sec 18 — Saving for Acts Done in Good Faith

No suit, prosecution, or other legal proceeding shall lie against any person for anything in **good faith done or intended to be done** under this Act.

The Aircraft Rules, 1937 Operational Rules

WHAT ARE THE AIRCRAFT RULES, 1937?

These are rules made by the **Government of India under the Aircraft Act, 1934** to regulate civil aviation activity in India. These are found in **Vol II, Air Regulations (Aircraft Manual India)**.

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Rule 1 — Short Title and Extent

These rules are called the **Aircraft Rules, 1937**. They extend to the whole of India and apply (unless contrary intention appears) to:

- **(a)** Aircraft registered in India and persons thereon, *wherever they may be* (except cases under sub-rule 4)
- **(b)** All aircraft for the time being *in or over India*

IMPORTANT PROVISO — FOREIGN-REGISTERED AIRCRAFT

For aircraft registered in a country other than India, the **regulations of that country** relating to registration, licence of personnel, airworthiness, and log books apply **in place of Parts IV, V, VI and IX** of these Rules — BUT only if that country's regulations are based on standards at least equal to the minimum standards of the **Chicago Convention (7 Dec 1944)**.

SUB-RULE 2A — ART. 83 AGREEMENTS (LEASED FOREIGN AIRCRAFT)

For aircraft registered in a contracting State (other than India), operated under a **lease, charter, or interchange agreement** by an operator whose principal place of business is in another contracting State — the regulations of that other State apply, provided an **Article 83 agreement** has been reached and officially notified to India / ICAO.

SUB-RULE 3 & 4 — CONVERSE SITUATIONS

- **Sub-rule 3:** Indian Rules apply to foreign-registered aircraft operated by India-based operators (Art. 83 agreement in place).
- **Sub-rule 4:** Indian Rules do **NOT** apply to Indian-registered aircraft leased/chartered to foreign operators (Art. 83 agreement in place). Such aircraft are treated as *not registered in India*.

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Rule 2 — Nationality of Aircraft

An aircraft shall be deemed to possess the **nationality of the State on the register** of which it is entered.

Rule 4 — Use and Operation of Aircraft

PROHIBITION

No person shall use, operate, or assist in using or operating an aircraft **except in accordance with these rules**.

Rule 5 — Registration, Nationality and Registration Marks

Subject to Rule 33, no person shall fly or assist in flying any aircraft unless:

- **(a)** It has been **registered**
- **(b)** It bears its **nationality and registration marks** and the **name and residence of the owner** affixed or painted thereon (per Rule 37, or per regulations of the State of registration for foreign aircraft)

EXCEPTION

This prohibition does NOT apply to aircraft flown with **special written permission of the Central Government**, subject to conditions and limitations specified therein.

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Rule 5A — Prohibited Flight

Except under, and in accordance with, a **permit issued by the Director-General of Civil Aviation (DGCA)**:

PROHIBITED WITHOUT DGCA PERMIT

- **(1)** No aircraft **registered in India** shall leave India for a flight to a place **outside India**
- **(2)** No aircraft shall fly to any **territory declared prohibited** by the Central Government via Official Gazette notification

Note: For this rule — foreign aircraft under sub-rule 3 of Rule 1 are treated as Indian-registered aircraft; Indian aircraft under sub-rule 4 of Rule 1 are treated as not registered in India.

Rule 6 — Licensing of Personnel

Every aircraft shall carry and be operated by the **personnel prescribed** ***(laid down by the authority or the rules)*** **in Part V**. Such personnel shall be licensed in the manner prescribed in Part V and in **Schedule II**.

For non-Indian registered aircraft, personnel shall be licensed per the regulations of the **State of registration**.

Rule 6A — Type of Aircraft to Be Included in Rating

TYPE RATING REQUIREMENT

No person shall fly as pilot of an aircraft which is **not included or entered in the aircraft rating of the licence** except as provided in Rules 6B and 6C.

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Rule 6B — Flights to Qualify for Extension of Licence

The holder of a pilot's licence may fly within the **Indian territory** as pilot of an aircraft of a type **not included in the aircraft rating** of his licence, for the purpose of **qualifying for the inclusion of such type**.

EXAM POINT — RULE 6B FLIGHTS

These are flights for type rating endorsement *(an official entry added to a licence or certificate)*. They can be conducted **within Indian territory only** (Rule 6B). Expected answer: Within Local Flying Area of the aerodrome.

Rule 6C — Flights for Testing and Non-Revenue Special Purposes

The Director-General may authorise the holder of a licence to fly an aircraft **not entered in the aircraft rating** for testing or non-passenger carrying flight. The authorisation is **limited in validity** ***(the period during which a document is accepted)*** **to the time needed to complete the testing or the specific flight**.

Rule 7 — Documents to Be Carried on Aircraft

- **(1)** No person shall fly an aircraft unless **valid documents** as required by the law of the country of registration are carried on board
- **(2)** Aircraft registered in India shall carry on board valid documents as required by these rules

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Rule 7A — Prohibition of Carriage of Persons Without Passport

- **(1)** No person-in-charge of any aircraft shall allow it to enter India from outside India unless all persons on board are in possession of **valid passports** as required under the **Indian Passport Act, 1920**
- **(2)** If an aircraft is brought into India in contravention of (1), an empowered authority may direct the owner or person-in-charge to take persons without passports back on board and remove them from India immediately

Rule 7B — Carriage of Cockpit Checklists

MANDATORY CARRIAGE

Every aircraft registered in India shall carry **Cockpit Check Lists and Emergency Check Lists** as specified by the Director-General for that particular type of aircraft. Such lists shall be carried in the cockpit **readily accessible to the pilot in flight**.

Rule 8 — Carriage of Arms, Ammunition, Explosives, Military Stores

ABSOLUTE PROHIBITION UNLESS PERMITTED

No person shall carry or cause or permit to be carried in any aircraft to, from, within or over India, any **arms, ammunition, munitions of war, implements of war, explosives and military stores**, except with the **written permission of the Central Government** and subject to the terms and conditions of such permission.

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Rule 8A — Security Check of Persons Boarding Aircraft at Aerodrome

For securing safety of aircraft operations, every person boarding an aircraft at an aerodrome and his **hand-baggage** shall be liable to be **searched at the aerodrome** by an officer authorised by the Central Government, before such person proceeds to the aircraft for embarkation.

Rule 9 — Radio-Telegraph Apparatus

No person shall operate radio transmitting apparatus in any aircraft registered in India unless he holds a **licence of the type required** by Part V in respect of aircraft required to carry radio-telegraph or radio-telephone apparatus, issued in accordance with those provisions.

Rule 10 — Mails

No person shall carry mails or allow mails to be carried in any aircraft except with the **consent in writing of the Director-General of Posts and Telegraphs**.

Rule 11 — Aerodromes

No pilot or person-in-charge of any aircraft **carrying passengers for hire or reward** shall use any place for a series of landings and departures. No pilot/person-in-charge of any aircraft employed on a **scheduled air transport service** shall use any place as a **regular place of landing or departure** other than an **aerodrome licensed or approved** for that purpose per Part XI.

Prohibited Areas — Aircraft Rules, Schedule I Rule 12

Rule 12 — Prohibited Areas

- **(1)** No person shall fly or assist in flying an aircraft over any of the areas specified in **Schedule I** except in accordance with conditions specified in that schedule.
- **(2)** A pilot who finds himself above a prohibited area shall — as soon as aware — **give the distress signal** and land as soon as possible outside the prohibited area at one of the **nearest aerodromes in India**.
- Provided: shall NOT fly further into the prohibited area nor commence descent while still above it (unless compelled by weather or unavoidable cause).
- **(3)** When signals warning aircraft about a restricted/prohibited/danger area are given, the pilot shall **immediately change course and fly away** from the prohibited area.
- **(4)** When signals from an aerodrome control tower are given, the pilot shall immediately give the distress signal and land in accordance with sub-rule (2).

Situation	Action
Pilot realises the aircraft is over a prohibited area	Immediately give a distress signal
Leaving the area	Do not fly deeper into the area, unless weather requires it
Landing	Land at the nearest aerodrome outside the area

Figure 13.1 *Action When Inadvertently Over Prohibited Area*

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Schedule I — Areas Over Which Flight Is ABSOLUTELY PROHIBITED

#	Location	Extent of Prohibition
1	Within a radius of **1 mile** from the **Towers of Silence, Malabar Hills, Bombay**	Absolute — all altitudes
2	Area near **Baroda (Vadodara) aerodrome**	Vertically from **ground level to unlimited upper level**
3	Area around **Rashtrapati Bhavan**	Vertically from **ground level to unlimited upper level**
4	Within a radius of **10 km** from **Mathura Refineries**	Vertically from **ground level to unlimited upper level**
5	Area near **Bhubaneshwar**	Vertically from ground level up to **50,000 ft**
6	Within a radius of **10 km** from **Kalpakkam Nuclear facility**	Vertically from ground level up to **10,000 ft**

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EXAM FAVOURITE — MATHURA REFINERY CLASSIFICATION

Mathura Refinery is a **PROHIBITED AREA** (not restricted, not danger). It is within 10 km of the aerodrome reference point.

Aircraft Rules 1937 — Operations, Safety & Conduct Rules 13–29D

Rule 13 — Photography at Aerodromes or from Aircraft in Flight

No person shall take, cause or permit to be taken, at a **Government aerodrome** or from an aircraft in flight, any photograph except in accordance with and subject to the terms and conditions of a **permission in writing** granted by:

- The Director-General
- A Joint Director General
- A Deputy Director-General
- The Director of Regulations and Information of the Civil Aviation Department

EXAM POINT

Photography at an aerodrome requires prior written permission from **DGCA (or equivalent DG-level officer)**.

Rule 14 — Aerial Work and Public Transport Reserved for Certain Aircraft

No aircraft registered in Category B (Rule 30) shall be used as an aerial work aircraft or as a public transport aircraft.

The Central Government may permit and impose restrictions on aerial work or public transport by aircraft other than those registered in Category A; such restrictions are effective when notified in writing to the owner/operator or via Official Gazette.

Rule 15 — Conditions to Be Complied with by Aircraft in Flight

No aircraft shall be flown unless the following conditions are complied with:

Condition	Detail
(i)	The aircraft possesses a **valid Certificate of Airworthiness (C of A)** or **Special C of A** issued by the Director-General
(ii)	The aircraft is **certified as airworthy** and maintained per **Part VI** (or per State of registration regulations for foreign aircraft)
(iii)	**All terms and conditions** on which the C of A was granted are duly complied with
(iv)	The aircraft carries on board its **C of A (or special C of A)** and any other certificate prescribed by Part VI

EXCEPTION — SPECIAL FLIGHT PERMIT

Aircraft not in compliance with the above may be flown under a **Special Flight Permit** issued by the DG under **Rule 55A**, subject to conditions specified therein.

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Rule 15A — Operation of Remotely Piloted Aircraft System (RPAS)

RPAS / DRONE REGULATIONS — KEY LIMITS

- No remotely piloted aircraft of **all-up weight more than 250 grams** shall be flown unless it has been allotted a **Unique Identification Number (UIN)** by the Director-General.
- An RPAS with all-up weight exceeding a limit specified by the DG from time to time shall not be flown by the operator except as per the conditions of a **permit issued by the DG**.

Rule 16 — Rules of the Air

Every person shall comply with the **Rules of the Air** issued by the Director-General in accordance with **Annex 2 to the Convention** as may be applicable to that person.

Rule 17 — Production of Licences

Any licence, certificate, authorisation and approval, log book or document granted or required to be maintained under these rules shall, **on demand for inspection**, be produced by the licensee or the owner/hirer/person-in-charge. The demand may be made by:

- Any magistrate
- Any police officer above the rank of sub-inspector
- Any Customs officer
- Any commissioned officer of the Naval, Military or Air Force
- Any gazetted officer of the Civil Aviation Department
- Any other person authorised by the Central Government by special or general order in writing

PROVISO — DOCUMENTS NOT REQUIRED TO BE CARRIED ON BOARD

Any document not required by these rules to be carried in the aircraft shall be produced within **7 days** of the demand being made.

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Rule 19 — Cancellation, Suspension or Endorsement of Licences, Certificates, Authorisation and Approval

Where any person is convicted of a contravention of, or failure to comply with, these rules or any direction issued under Rule 133A in respect of any aircraft, the Central Government may:

- **Cancel or suspend** any certificate of registration granted under these rules relating to that aircraft
- **Cancel or suspend** any certificate relating to airworthiness of an aircraft or Type Certificate of an aircraft component, if a **reasonable doubt exists** as to the safety of the aircraft or the airworthiness of the aircraft

Rule 19A — Restrictions on Licence, Certificate, Authorisation or Approval

The Director-General may impose restrictions on any licence, certificate, authorisation or approval:

- **(a)** In the event of non-compliance with any condition imposed on said licence/certificate/authorisation/approval
- **(b)** If any safety concern that emerged during an inspection has remained **unresolved beyond the period specified by the DG**

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Rule 20 — Certain Rules Not Applicable to Gliders, Kites and RPAS

Aircraft Type	Rules Not Applicable
Gliders	Rule 7 shall not apply
Kites	Rules 5, 6, 7, 12, 15, 17, 19 and all rules in Part III (except Rule 26)
RPAS	Rules 5, 6, 15 in Part II; all rules in Part III (except Rules 21, 21A, 26); ALL rules in Part IV, Part V, Part VI and Part VII

Rule 21 — Dangerous Flying

PROHIBITION — DANGEROUS FLYING

No person shall fly any aircraft in such circumstances as — by reason of **low altitude or proximity to persons or dwellings or for other reason** — to cause **unnecessary danger** to any person or property.

Refer Rule 21A for General Safety, Rule 21B for Aircraft in Distress.

EXAM POINT

A pilot flying low due to **thundershowers for safety reasons** — this is an **unavoidable cause**. Rule 161(2) provides a defence where contravention is due to accident, stress of weather or other unavoidable cause.

Rule 22 — Assault and Other Acts of Interference Against a Crew Member

No person shall, on board an aircraft:

- **(a)** Assault, intimidate or threaten (physically or verbally) a crew member in a way that may **interfere with the performance of duties** or lessen ability to perform those duties
- **(b)** Refuse to follow a **lawful instruction** given by the Pilot-in-Command (or on behalf of PIC by a crew member) for the safety of the aircraft or passengers or for maintaining good order and discipline on board

Rule 23 — Assault and Other Acts Endangering Safety or Jeopardizing Good Order and Discipline

No person shall, on board an aircraft:

- **(a)** Assault, intimidate or threaten (physically or verbally) *any person*
- **(b)** Intentionally cause damage to or destroy any property
- **(c)** Consume alcoholic beverages or drugs which is likely to **endanger the safety of the aircraft** or any person or **jeopardize the good order and discipline** on board

Rule 24 — Prohibition on Consumption of Intoxicating and Psychoactive Substances

CRITICAL LIMIT — 12 HOUR RULE

No person acting as or carried in aircraft for the purpose of acting as **pilot, commander, navigator, engineer, cabin crew or other operating member of the crew** thereof shall have taken or used any alcoholic drink, sedative, narcotic or stimulant drug or preparation within **12 hours** of the commencement of the flight or take or use any such preparation in the course of the flight.

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No such person shall, while so acting or carried, be in a state of **intoxication or have detectable blood alcohol** whatsoever in his breath, urine or blood, or be in a state where his capacity is **impaired** due to any alcoholic, sedative, narcotic or stimulant drug.

No other person while in a state of intoxication shall enter or be in aircraft.

DOMESTIC FLIGHTS — ALCOHOL PROHIBITION

No operator operating a **domestic air transport service** in India shall serve any alcoholic drink on board, and no passenger on such service shall consume any alcoholic drink while on board.

LICENCE HOLDERS — ADDITIONAL OBLIGATIONS

- Licence holders shall NOT exercise licence privileges *(what a licence allows its holder to do)* while under the influence of any **psychoactive substance** that renders them unable to safely exercise those privileges.
- Licence holders shall not engage in **problematic use of substances**.

EXAM POINT

The member of operating crew shall not have taken alcoholic drinks within **12 hours** before commencement of flight.

Rule 24A — Carriage of Persons Suffering from Mental Disorders or Epilepsy

GENERAL PROHIBITION

No person shall knowingly carry or permit to be carried a person suffering from any **mental disorder or epilepsy** in any aircraft.

EXCEPTION — WHEN CARRIAGE IS PERMITTED

Carriage is permitted if the person is **certified by a registered medical practitioner** to be fit to travel by air without being a risk to other passengers or the aircraft, AND:

- **(a)** Has not taken any alcoholic drink or preparation within **12 hours** of commencement of flight
- **(b)** Is kept under proper sedative, if in a state of excitement, during the flight and stops en route
- **(c)** Is accompanied by an attendant. However, if the person has been in a state of excitement requiring sedation within the **two weeks** preceding the date of commencement of flight, he shall be accompanied by a **registered medical practitioner and adequate escort**, who shall collectively ensure no alcohol is taken and person is kept suitably sedated during the flight and all stops

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Rule 24B — Carriage of Prisoners in Aircraft

REQUIRES WRITTEN PERMIT

No prisoner shall be taken aboard or carried on an aircraft except under a **permit in writing** issued by the **Director-General, Deputy Director-General, Director of Regulations and Information**, or any other officer of the Civil Aviation Department authorised by the Central Government.

Definition — "Prisoner": A person who is confined in any prison AND includes a person who is arrested under any law for the time being in force.

EXAM POINT

Prisoners on board an aircraft can be carried with the permission of **DGCA**.

Rule 24C — Carriage of Animals, Birds and Reptiles in Aircraft

No animal, bird or reptile shall be taken aboard or carried on any aircraft to, from and within India, except under a **general or special permit in writing issued by the Director-General**, subject to such conditions as may be specified therein.

Rule 25 — Smoking in Aircraft

- **(1)** The owner/operator and the PIC of every Indian-registered aircraft shall exhibit in prominent places notices stating where and to what extent smoking is prohibited or permitted.
- **(2)** A notice permitting smoking may be exhibited only if smoking is permitted by the **certificate of airworthiness** or by direction of the Central Government, and only in accordance with conditions in such C of A or direction.

EXAM POINT

Smoking on a private aircraft is permitted only if the **certificate of airworthiness of the aircraft permits**.

Rule 25A — Fuelling of Aircraft

FUELLING RESTRICTIONS — CRITICAL DISTANCES

Item	Regulation
Fuelling location	Outdoors and **not less than 15 m** from any building
No Smoking notice	Must be **prominently displayed**
Naked flame / spark-producing appliance	Not permitted within **30 m** of aircraft or fuelling equipment
Aircraft engines	Must be **shut down**; ignition switches in **"OFF" position**
Electrical/radar/radio systems	**Not to be operated**; switches in "OFF" position (Exceptions: power/light for fuelling, minimum cabin lighting, steady parking lights)
Non-authorised persons	Not permitted within **15 m** of aircraft
Jet aircraft refuelling cessation	When rear jet outlets come within **43 m** of fuelling equipment or aircraft

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Item	Regulation
Bonding & earthing	Fuelling equipment and aircraft shall be **bonded to each other and earthed**
Flame-proof equipment distance	If not flame-proof: **≥15 m** for gasoline/wide-cut fuels, **≥6 m** for straight kerosene
Freight/baggage handling	Shall NOT proceed simultaneously with fuelling unless **adequate precautions** against fire risk have been taken
Aircraft maintenance during fuelling	**No maintenance** that may provide a source of ignition
Fire extinguishers	Of adequate capacity and suitable type, approved by DG — available for **immediate use near aircraft**
Fuel spill	Stop fuelling, stop ground power engine, clean spilled fuel — fuel must NOT be washed into sewers or drains

PASSENGER PRESENCE DURING FUELLING — CONDITIONS

- **(i)** An attendant shall remain on duty in the cabin at all times; shall ensure no smoking and shall assist in removal of passengers in case of fire
- **(ii)** Passenger loading ramp shall be correctly positioned at cabin exit door; adequate provision for equilibrium of aircraft if all passengers attempt to leave by one exit
- **(iii)** For marine aircraft: adequate means of water transport stationed at cabin exit door

EXAM POINTS — FUELLING

- Refuelling must stop when jet aircraft comes within **43 m**
- No naked light within **30 m** during refuelling
- No unauthorised person within **15 m** of aircraft during refuelling

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Rule 25B — Housing of Aircraft

- **(1)** No aircraft containing **dangerous petroleum** in bulk in any of its tanks may be housed in a hangar unless the hangar is constructed of **uninflammable material** and is effectively and safely **ventilated to the open air**.
- **(2)** Every such hangar shall be in the charge of a **competent person** responsible for taking all proper precautions against fire and for preventing unauthorised access.

Rule 26 — Dropping of Articles and Descent by Parachutes

GENERAL RULE

No person shall drop or project or cause or permit to be dropped or projected from an aircraft in motion anything except **ballast in the form of fine sand or water**.

PERMITTED EXCEPTIONS

- **(a)** In an emergency: dropping of **liquid fuel**

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- **(b)** In an emergency: dropping of **cargo** over areas where hazard to persons/property is not created
- **(c)** Dropping of **message bags, smoke-producing apparatus, or materials for navigation/communication** — subject to precautions to avoid injuring persons
- **(d)** Dropping of **printed matter or flower petals** if:
  - Prior written permission of **District Magistrate or Commissioner of Police** is obtained
  - Aircraft is suitable for dropping these articles
  - Minimum safe heights specified in rules are observed
  - Necessary precautions taken to avoid injury/damage
- **(e)** Dropping of **ropes used for towing aircraft**

(2) Parachute Descents: No person shall, except in an emergency, descend by parachute from an aircraft, and no article shall be dropped from an aircraft in flight (whether attached to a parachute or not), unless the descent/drop is made in accordance with the **general or special order of the Central Government in writing**.

EXAM POINT

Dropping of paper leaflets requires the permission of the **Local District Magistrate**.

Rule 27 — Carriage of Persons in Unauthorised Parts of Aircraft

No person shall at any time be carried on the wings, undercarriage, or any part of aircraft not designed for accommodation of personnel or passengers, or on anything attached externally to the aircraft.

PERMITTED EXCEPTIONS

- **(a)(i)** Temporary access for executing repairs or adjusting machinery/equipment, or for safety of aircraft/persons/goods carried
- **(a)(ii)** Access to a part where goods/stores are being carried and to which proper means of access is provided
- **(b)** A person may be carried anywhere on/in the aircraft *with the permission in writing of the Central Government* and subject to conditions specified

Rule 28 — Minimum Age for Sole Control of Aircraft

MINIMUM AGE LIMIT

No person **under the age of 16 years** shall have sole control of an aircraft in motion. No person shall cause or permit any other person to have sole control unless he knows or has reasonable cause to believe such other person has attained the age of **16 years**.

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Rule 28A — Maximum Age Limit for Professional Pilots

MAXIMUM AGE LIMITS — COMMERCIAL OPERATIONS

Age Reached	Restriction	Exception
65 years	Cannot act as PIC or Co-pilot of any aircraft engaged in **commercial air transport operations**	None
60 years	Cannot act as PIC or Co-pilot in commercial air transport unless in a **multi-crew environment** and the other pilot is **less than 60 years**	Does NOT apply to aircraft **certified for single pilot operations** not exceeding **5700 kg MCTOW** engaged in commercial air transport within India territory in multi-crew environment

Rule 29 — Acts Likely to Imperil the Safety of Aircraft

ABSOLUTE PROHIBITION

No person shall **interfere with the pilot** or with a member of the operating crew, or **tamper with the aircraft or its equipment**, or conduct himself in a **disorderly manner** in an aircraft, or **commit any act likely to imperil** the safety of an aircraft or its passengers or crew.

Rule 29A — Prohibition of Operating Civil Aircraft Causing Sonic Boom

SUPERSONIC FLIGHT RESTRICTION

No person shall operate a civil aircraft at a **true flight Mach number greater than ONE (Mach 1)** over the territory of India or over the high seas in a manner which may cause or is likely to cause **sonic boom over the territory of India**.

Rule 29B — Prohibition on the Use of Portable Electronic Devices (PEDs)

- **(1)** No person shall operate, nor shall the operator or PIC allow the operation of, any **portable electronic device on board an aircraft in flight**.
- **(2)** Exceptions — Sub-rule (1) does NOT apply to:
  - Portable voice recorders
  - Hearing aids
  - Heart pacemakers
  - Electric shavers
  - Other PEDs which, in the opinion of the operator, do not cause interference with navigation or communication systems — and for which the operator has obtained the **approval of the Director-General**
- **(3)** Notwithstanding *(in spite of)* sub-rule (1), the **PIC may permit access to internet services** by passengers in flight through **Wi-Fi on board**, when laptop, smartphone, tablet, smartwatch, e-reader or point of sale device is used in **flight mode or airplane mode**. Provided that *(an exception or condition added to a rule)* the **DG has certified the aircraft** for internet service through Wi-Fi on board.

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DEFINITION — "IN FLIGHT"

For purposes of Rule 29B, an aircraft shall be deemed to be in flight when all its **external doors are closed following embarkation** until the moment when any such door is opened for disembarkation.

MOBILE PHONES — SPECIAL PROVISION

The PIC may permit use of **cellular telephone by passengers** after the aircraft has **landed and cleared active runway**, except when the landing takes place in **low visibility conditions** as may be determined by the DG from time to time.

EXAM POINT

PIC may permit internet via Wi-Fi provided **the Director-General has certified the aircraft for internet service in flight through Wi-Fi on board**.

Rule 29C — Adoption of the Convention and Annexes

- The DG may lay down standards and procedures not inconsistent with the Aircraft Act, 1934 to carry out the Convention and any Annex thereto.
- The DG shall formulate the **State Safety Programme (SSP)** and oversee its implementation.

Rule 29D — Safety Management Systems

Every organisation engaged in the operation of aircraft and aerodromes, provision of air traffic services, training of personnel, maintenance, design and manufacture of aeronautical products shall:

- **(a)** Establish and maintain **Safety Management Systems (SMS)**
- **(b)** Prepare a **Safety Management Systems Manual** in the form and manner specified by the DG and submit the same to the DG for approval

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Aircraft Registration & Change of Ownership Rules 30, 33

Rule 30 — Certificate of Registration — Categories

An aircraft may be registered in India in either of the following categories:

Category	Ownership Requirement
Category A	Aircraft *wholly owned* by: (i) Citizens of India; or (ii) A company/corporation registered and having principal place of business *within India*; or (iii) The Central/State Government, or any company/corporation owned or controlled by either Government; or (iv) A company/corporation registered *outside India*, provided such company has **leased** the aircraft to a person mentioned in (i), (ii), or (iii)
Category B	Aircraft *wholly owned* by: (i) Persons resident in or carrying on business in India who are **not citizens** of India; or (ii) A company/corporation registered *outside India* and carrying on business in India

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KEY DISTINCTION — CATEGORY A VS B

Only **Category A aircraft** may be used as **aerial work aircraft or public transport aircraft**. **Category B aircraft cannot** be used for these purposes (Rule 14).

Rule 33 — Change in Ownership

In the event of any change in ownership or if a registered aircraft ceases to meet the conditions of Rule 30:

- **(a)** The registered owner shall **forthwith notify** the DG of such change
- **(b)** The new owner shall forthwith inform the DG in writing of his ownership and may apply for a new certificate of registration
- **(c)** The existing registration and certificate shall remain valid **until cancelled by the DG**

Licensing Authority, Operating Crew & Cabin Crew Rules 38, 38A, 38B

Rule 38 — Licensing Authority

The authority by which licences and ratings may be granted, renewed or varied shall be the **Central Government**, which may withhold the grant or renewal of a licence or rating if for any reason it considers it desirable to do so.

Rule 38A — Carriage of Operating Crew

Every aircraft registered in India shall comply with the following requirements in respect of the personnel it carries and by which it is operated:

(1) Pilot Requirements

Category of Aircraft	Minimum Pilot Requirement
Private Aircraft	A person holding a valid **Private Pilot's Licence (PPL)** issued per Schedule II. *However:* • NOT to be flown by a holder of a **Student Pilot Licence** • NOT to be flown by PPL holder **for remuneration or hire of any kind** • NOT to be flown by PPL holder at **night without a valid Instrument Rating**
Public Transport or Aerial Work Aircraft (other than microlight, glider, balloon)	A person holding an appropriate **professional pilot's licence** — i.e., a **Commercial Pilot's Licence (CPL) or Airline Transport Pilot's Licence (ATPL)** issued per Schedule II

(2) Flight Instructor or Assistant Flight Instructor

- **(a)** Every aircraft used for dual instruction shall carry a person holding a CPL or ATPL (or microlight/glider/balloon licence) with an appropriate **Flight Instructor's or Assistant Flight Instructor's rating** per Schedule II, or an **authorisation in writing by the DG**
- **(b)** No person other than one having an FI or AFI rating shall impart instruction in piloting an aircraft, unless **specifically authorised in writing by the DG**

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(3) Flight Navigator

Every public transport aircraft engaged on a flight without landing over a great circle distance of more than **600 NM** and **not equipped** with the navigational equipment capable of providing instant and continuous ground position shall carry a **Flight Navigator licensed per Schedule II**, if the total distance between any two consecutive radio navigational fixing aids located within **30 NM** of the route and capable of being used by the aircraft is more than **600 NM**.

(4) Flight Engineer

Where a Flight Engineer is required to be carried on board as a flight crew member (per sub-rule 7), he shall hold the appropriate licence per Schedule II.

(6) Flight Radio Telephone Operator

Aircraft equipped with radio apparatus per Rule 63 and communicating by radio telephony shall carry a person holding a **Flight Radio Telephone Operator's Licence or Flight Radio Telephone Operator's Licence (Restricted)** per Schedule II.

(7) Minimum Crew for Any Flight

The number and description of flight crew members operating any flight of an aircraft registered in India shall be **as per the certificate of airworthiness**.

> **EXAM POINT**
>
> Minimum crew required on a private aircraft = **as specified in the Certificate of Airworthiness**.

Rule 38B — Carriage of Cabin Crew

No aircraft registered in India shall be operated for public transport of passengers unless the following minimum number of cabin crew are on board:

Aircraft Type	Seating Capacity	Minimum Cabin Crew
Aeroplane	**10 – 50** passengers	**1** cabin crew
Helicopter	**20 – 50** passengers	**1** cabin crew
Aeroplane or Helicopter	More than **50** passengers	**2 cabin crew** PLUS **1 additional cabin crew for each unit (or part of a unit) of 50 passenger seats** above a seating capacity of **99 passengers**

> **EXAM POINTS — CABIN CREW**
>
> - No. of cabin crew is based on **No. of seats**.
> - Aircraft with seating capacity of 150 — Rule 38B: two cabin crew **plus one for each unit (or part of a unit) of 50 passenger seats above a seating capacity of 99**. 150 − 99 = 51 seats above 99 → 2 units (51 ÷ 50 rounds up) → 2 + 2 = **4 cabin crew**. (The threshold is 99, not 100: a 250-seater needs 2 + 4 = 6.)

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Disqualification, Medical Standards & Licence Management Rules 39A–42, 44, 45

Rule 39A — Disqualification from Holding or Obtaining a Licence

Where the licensing authority is satisfied, after giving the person an opportunity of being heard, that any person:

Ground for Disqualification
(a) Is habitually intemperate in the use of alcohol, or is an addict of narcotics, drugs and the like
(b) Is using, has used or is about to use an aircraft in the commission of a cognizable offence or in contravention of these rules
(c) Has, by his previous conduct as crew member, shown that he is **irresponsible** in the discharge of his duties or is likely to endanger the safety of the aircraft or any person or thing carried therein, or of other aircraft or persons or things on the ground
(d) Is a habitual criminal or has been convicted by a Court in India for an offence involving **moral turpitude** or an offence amounting to **heinous crime**
(e) Has obtained the licence, rating, aircraft type rating, or extension thereof, or renewal of any of them, by **suppression of material information or on the basis of wrong information**
(f) Has **unauthorisedly varied or tampered** with the particulars entered in a licence or rating

The licensing authority may, for reasons to be recorded in writing, make an order **disqualifying that person for a specified period** from holding or obtaining a licence.

> **PERMANENT DEBARMENT**
>
> The Central Government may **permanently or temporarily debar** a person from holding any licence or rating mentioned in Rule 38, if it is in the **public interest** to do so.

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Rule 39B — Medical Standards

No licence or rating referred to in Rule 38 shall be **issued or renewed** unless the applicant:

- Undergoes a **medical examination with an approved medical authority**
- Satisfies the **medical standards as notified by the Director-General**

> **RULE 42(2) — LICENCE NOT TO BE EXERCISED AFTER SICKNESS/INJURY**
>
> The holder of a licence shall NOT exercise the privileges of his licence without being declared fit after a fresh medical examination if:
>
> - **(a)** A sickness or injury involving incapacity for a period of **15 days or more** for the work for which he is licensed
> - **(b)** An injury sustained in any accident occurring during the exercise of his licence's privileges or otherwise, which is likely to **cause incapacity or impair his efficiency**
>
> The licence holder or his employer shall **immediately notify** all relevant details of the sickness/injury to the Director-General.

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Rule 40 — Signature of Licence Holder

On the issue of a licence to an applicant, he shall **forthwith sign his name** on the licence as the holder thereof with his ordinary signature.

Rule 41 — Proof of Competency

Applicants for licences and ratings shall produce proof of having acquired the **flying experience** and having passed satisfactorily the **tests and examinations specified in Schedule II**.

> **DEFENCE FORCES EXEMPTION**
>
> A qualified pilot from the **Indian Air Force, Indian Navy, Indian Army or Indian Coast Guard** who produces satisfactory evidence of possessing the necessary flying experience, competency, and physical fitness standards may be exempted by the DG from all or any flying tests and medical or technical examinations required for the issue of licences.

Rule 41A — Checks, Tests and Examinations

- The DG may conduct examinations specified in Schedule II, fix examination centres, appoint invigilators, and lay down the procedure for conducting examinations.
- The DG may appoint **Examiners** for carrying out flying tests and technical examinations required under Schedule II.
- The DG may appoint a **Board** to conduct oral examinations when necessary.

Rule 42 — Licences and Their Renewal

(1) Licences and ratings mentioned in Rule 38 may be issued or renewed for any period not exceeding the period specified in Rule 39C and Schedule II.

If on the date of application for renewal, the licence or rating has **expired** for the following periods, the applicant may be required to qualify in:

Expiry Period	Requirement
Exceeding **2 years** but not exceeding **3 years**	Test of skill and Air Regulations
Exceeding **3 years**	All examinations and tests required for the **fresh issue** of the licence or rating

Rule 44 — Aircraft Not Registered in India

An aircraft not registered in India shall carry the **personnel prescribed by the laws of the State in which it is registered** and such personnel shall be licensed in accordance with the laws of that State.

Rule 45 — Validation of Foreign Licences

When a licence has been granted by the **duly competent authority in any foreign State** and is in force, the Central Government may, subject to such conditions and limitations and for such periods as it shall think fit, **render such licence valid** by an authorisation for flying aircraft registered in India. A validated licence is subject to Rule 19, and validation ceases if the licence is revoked *(cancel for good)* or suspended.

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Certificate of Airworthiness, Modifications & Log Books Rules 50, 52, 65, 67A, 67B

Rule 50 — Certificate of Airworthiness

- The owner or operator of an aircraft may apply to the DG for **issue or renewal of a C of A**, or special C of A, or for **validation of a C of A issued elsewhere**.
- The DG may **validate a C of A** in respect of any aircraft that may be imported.
- Subject to these rules, a C of A shall remain in force for such period as may be specified in the certificate and may from time to time be **renewed by the DG**.

Rule 52 — Modification and Repairs

- **(1)** A person shall not carry out any modification or repair affecting safety of any aircraft with a valid C of A unless he has been required to do so under these rules or has obtained the **prior approval of the DG**.
- **(2)(a)** Modifications issued by the manufacturer which have been issued a **Type Certificate** by the DG or elsewhere may be deemed as approved modifications, unless otherwise specified by the DG.
- **(2)(b)** Repair schemes issued by the manufacturer of an aircraft with a Type Certificate and other repairs in accordance with **standard aeronautical engineering practice** may be deemed as approved repairs, unless otherwise specified by the DG.

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Rule 65 — Aeronautical Beacon and Aeronautical Ground Lights

- **(1)** No aeronautical beacon or aeronautical ground light shall be established or maintained within India, nor shall the character of the light be altered, except with the **approval in writing of the Central Government**.
- **(2)** No person shall wilfully or negligently endanger or interfere with any aeronautical beacon or aeronautical ground light established or maintained by or with the approval of the Central Government, or any light exhibited therefrom.

Rule 67A — Log Books of Flight Crew Personnel and Logging of Flight Time

LOG BOOK REGULATIONS

Requirement	Detail
Maintenance	Every licensed flight crew member shall maintain a **personal log book** in the form prescribed by the DG. All flight times shall be logged therein.
Entries	All entries in log books shall be made in **ink**
Preservation	Log books shall be preserved for **not less than 5 years** after the date of the **last entry**
Monthly certification	Each crew member shall certify the accuracy of entries in his log book with respect to flight time at least at the end of each **calendar month**
Quarterly certification	At the end of every quarter (end of March, June, September, December), log books shall be certified for correctness by: **Operations Manager/Deputy OM/Chief Pilot** (professional pilots), **Chief Flying Instructor** (FTO members), **DG-designated Officers** (all others)
Dual instruction entries	Flight time under dual instruction shall be entered as **"dual"**. The FI shall make entries in the trainee's log book showing the nature of instruction given

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Logging Rules for Different Crew Members

Licence Type	How to Log PIC Time
Student Pilot	As PIC only when he is the **sole occupant** of the aircraft
Private Pilot (PPL)	As PIC only during which he **acts as PIC**
CPL / ATPL Holder	As PIC — during which he **acts as PIC**; as Co-pilot — during which he **acts as Co-pilot**
Flight Instructor	As PIC — during which he acts as Instructor; **log entries shall indicate in remarks column** that flight time was flown as Instructor
Examiner	As PIC — during which he acts as Examiner, if entitled and authorised to fly in command of that type
Supervised PIC duties	May be logged as PIC if person is **entitled and authorised** to fly in command of that type by virtue of ratings and privileges. Otherwise logged as **Co-pilot with appropriate indication in remarks column**
Instrument time	Logged only when aircraft is flown **solely by reference to instruments**, under actual or simulated instrument conditions. **Over-the-top flying shall NOT be logged as instrument flying time**
Instrument ground time	Logged in full while flying solely by reference to instruments in any **recognised synthetic device** simulating instrument flight conditions

EXAM POINTS — LOG BOOKS

- Log book certified for correctness by competent authority at end of **every quarter**.
- Pilot's log book preserved for **5 years from date of last entry**.

Rule 67B — Prohibition on False Entries

ABSOLUTE PROHIBITION

No person shall destroy, mutilate, alter or render illegible any entry made, or wilfully make or procure or assist in the making of any **false or fraudulent entry** in or omission from any log book referred to in Rules 67 and 67A.

Aerodromes, Scheduled Services & Operations Manual Rules 79, 80, 133A, 134, 140–161

Rule 79 — Qualifications of Licensee (Aerodrome)

A licence for an aerodrome shall not be granted to any person other than:

- **(a)** A citizen of India
- **(b)** A Company or body corporate registered and having its principal place of business in India; meeting the equity holding criteria specified by the Central Government; or
- **(c)** The Central/State Government or any company/corporation owned or controlled by either Government; or

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- **(d)** A society registered under the Societies Registration Act, 1860

Rule 80 — Procedure for Grant of Aerodrome Licence

- **(1)** Application made to the DG along with the **Aerodrome Manual**
- **(2)** Application shall be in such form and contain such particulars as may be specified by the DG
- **(3)** DG may require the applicant to furnish additional information
- **(4)** DG may also require the applicant to produce evidence in support of any information furnished

Rule 133A — Directions by Director-General

The DG may, through:

- **Notices to Airmen (NOTAMs)**
- **Aeronautical Information Publication (AIP)**
- **Aeronautical Information Circulars (AICs)**
- **Notices to Aircraft Owners and Maintenance Engineers**
- **Civil Aviation Requirements (CARs)**

...issue special directions not inconsistent with the Aircraft Act, 1934, relating to the operation, use, possession, maintenance or navigation of aircraft flying in or over India or registered in India.

> **PUBLIC CONSULTATION REQUIREMENT**
>
> Civil Aviation Requirements (CARs) shall be issued after placing the draft on the **website of the DGCA for 30 days** for inviting objections and suggestions from all persons likely to be affected.

Rule 134 — Scheduled Air Transport Services

No person shall operate any Scheduled air transport service from, to, in, or across India except with the **permission of the Central Government**, granted in accordance with the provisions contained in **Schedule XI**.

The Central Government may permit any air transport undertaking of another country to operate scheduled services from/to/across India in accordance with the terms of any bilateral agreement, or where there is no such agreement, of a **temporary authorisation** by the Government of India.

Rule 140 — Minimum Requirements to Be Complied with by Operators

All aircraft owners and operators shall comply with the **engineering, inspection and manual requirements** contained in Part XIII-A and with the **safety requirements** in respect of air routes, aircraft and aircrew, as may be specified by the DG.

Rule 140A — DG Sanction for New Routes or Alterations to Existing Routes

Before operating a scheduled air transport service on a new route or making a **substantial alteration** in, or effecting the **discontinuance** of, any existing route, or introducing a new time-table, the Corporation shall:

- Obtain the **concurrence of the DG**
- Give at least **7 days'** previous notice to the DG before the date proposed for the operation of the new route, substantial alteration, new time-table, or discontinuance

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Rule 140B — Operations Manual

- **(1)** An Operations Manual in the form approved by the DG shall be maintained by the Corporation.
- **(2)** The Operations Manual shall contain:

Item	Content Required
(a)	Instructions outlining responsibilities of operations personnel for the conduct of flight operations
(b)	The flight crew for each stage of all routes to be flown, including the **designation of the succession of command**
(c)	In-flight procedure
(d)	Emergency flight procedure
(e)	The **minimum safe flight altitude** for each route to be flown
(f)	The circumstances in which a **radio listening watch** is to be maintained
(g)	A list of the **navigational equipment** to be carried

- **(3)** A copy of the Operations Manual, or such part as prescribed by the DG, shall be **carried in all aircraft engaged in scheduled air transport services**.

> **EXAM POINTS — OPERATIONS MANUAL**
>
> - Succession of command is given in the **Operations Manual**.
> - Operational Manual is prepared by the **operator and authorised by the DGCA**.

Rule 140C — Route Guide

All aircraft of the Corporation engaged in scheduled air transport services shall carry a Route Guide containing:

- **(a)** Communication facilities, navigation aids, and a list of aerodromes available on the route
- **(b)** Instrument 'let down' procedure for aerodromes on the route or those likely to be used as alternates
- **(c)** Meteorological minima for each aerodrome on the route (regular and alternate)
- **(d)** Specific instructions for computation of quantities of fuel and oil to be carried on each route, having regard to all circumstances including the **possibility of failure of one or more engines**

Rule 161 — Penalties

- **(1)** Any person who has contravened or failed to comply with any of these rules or any direction issued under Rule 133A shall, where no punishment is provided for such contravention in the Aircraft Act, 1934, be punishable to the extent laid down in **Schedule VI**.
- **(2)** It shall be a defence to any proceedings for contravention if the contravention or failure is proved to have been due to **accident, stress of weather or other unavoidable cause**. Also, it is a defence for the owner, hirer, operator, pilot or commander that the alleged contravention took place without his **actual fault or privity**.

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The Indian Aircraft Rules, 1920 Customs Rules

PURPOSE

These are **custom rules governing aircraft arriving in or departing from India**. They are found in the Aircraft Manual (India) Vol I.

Rule 53 — Customs Aerodromes and Customs Officers

The Central Government may:

- **(a)** By notification in the Official Gazette, **declare any aerodrome in India to be a customs aerodrome**
- **(b)** Appoint persons to be **Chief Customs Officers and Customs Collectors**, and define areas within which each shall exercise the powers and perform the duties conferred

Rule 54 — Arrival and Departure

- **(1)** No person in-charge of an aircraft entering India shall cause it to be landed in India for the first time in any journey except at a **customs aerodrome**, unless compelled to land by accident, stress of weather or other unavoidable cause.
- **(2)** No person in-charge of an aircraft shall fly it to a place outside India except from a **customs aerodrome** (unless compelled to land elsewhere).

EXAM POINT

An international flight from India can take off only from a **customs aerodrome**.

Rule 55 — Import and Export of Goods

- **(1)** No person in any aircraft entering or departing from India shall carry or allow to be carried any goods of which the import or export by sea or by land is **prohibited** by any law in force.
- **(2)** No person in any aircraft entering India shall **break or alter any seal** placed upon any part of the aircraft or upon any goods therein by an officer of customs at the aerodrome at which such aircraft departed for India.

Rule 62 — Examination

- **(1)** The person in-charge of any aircraft shall permit any **Customs Collector or other authorised officer** to board and examine the aircraft and any goods laden thereon, at any time.
- **(2)** The importer or exporter of any goods shall produce such goods to the **Customs Collector at the customs aerodrome** and permit him to examine such goods.

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Aircraft Manual (India) — Vol I & Vol II

VOL I

Contains the following **Acts and Rules**:

- **Aircraft Act, 1934** — Control of manufacture, possession, use, operation, sale, import/export of aircraft
- **Aircraft Rules, 1937** — Regulating civil aviation activity in India (Vol II Air Regulations)
- **Indian Aircraft Rules, 1920** — Customs rules governing aircraft arriving in/departing from India

VOL II

- **Part I**: National Conventions/Rules/Acts
- **Part II**: International Conventions

Indian Aircraft (Public Health) Rules, 1954 (Amended 1969)

PURPOSE & AUTHORITY

Framed by the Government of India under the power given in the Aircraft Act, 1934. These rules govern public health measures at airports and on aircraft.

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Key Definitions

Term	Definition
Health officer	A person appointed by the Central Government to the health office of the airport
Infected aircraft	An aircraft which has any person or animal who carries an infectious disease
Infected area	Any area outside India declared as such in relation to a quarantinable or other infectious disease by the Central Government via notification in the Official Gazette
Isolation	The separation of that person/group from other persons (except health staff on duty) so as to prevent the spread of infection

Periods of Incubation for Quarantinable Diseases

INCUBATION PERIODS

6	days — Yellow Fever
6	days — Plague
5	days — Cholera
14	days — Small Pox
14	days — Typhus

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8	days — Relapsing Fever

COMMANDER'S OBLIGATION — HEALTH REPORT BEFORE LANDING

The Commander of an aircraft on its way to India from any place outside India shall send to the officer in-charge of the airport where he proposes to land a **health report at least 2 hours in advance** stating:

- **(a)** Whether any person on board is suffering from any illness, and if so, what its signs and symptoms are (giving if possible the name of the illness)
- **(b)** Whether at any time during the voyage there has occurred on board any case or suspected case of a quarantinable or other infectious disease

MEASURES TAKEN BY HEALTH OFFICER ON ARRIVAL

- When an infected/suspected aircraft comes from a place outside India, the health officer may take appropriate measures as prescribed by the rules
- May consider medical examination/inspection of any passenger or crew, particularly when coming from a **yellow fever infected area**
- If aircraft comes from an area where **malaria or mosquito-borne disease** could develop from imported vectors, the aircraft may be **disinfected**

PILGRIM AIRCRAFT — SPECIAL REQUIREMENT

For pilgrim aircraft, all persons should produce valid certificates of vaccination against **cholera and yellow fever**.

HUMAN REMAINS — RESTRICTIONS

No person shall bring into India any dead body or human remains of persons who may have died of **yellow fever, plague, anthrax, ganders** or such other diseases notified by the Central Government.

- If cremated: ashes shall be placed in an urn or casket having an outer packing of suitable material
- If not cremated: the corpse should be closed in a **shell of zinc or other equally suitable metal** with all joints soldered to seal hermetically, enclosed in a stoutly built teak or hard wood coffin, enclosed in a zinc or tin-lined wooden packing case filled with sawdust impregnated with carbolic powder

PENALTY — PUBLIC HEALTH RULES

Whoever contravenes any provision of these rules may be punished for a term not exceeding **3 months** or with fine up to **₹1,000** or with both.

EXAM POINTS — PUBLIC HEALTH

- Passenger from yellow fever infected area without certificate — Referral of suspect travellers to designated quarantine facility by **Airport Health Organisation**.
- Bringing human remains — subject to **public health clearance of dead bodies/human remains**.
- If passenger falls sick on board with suspected infectious disease — PIC's action = **To inform health officer of destination aerodrome**.

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International Health Regulations (IHR), 2005

AIRPORT HEALTH ORGANIZATION (APHO)

APHO works under the **Directorate General of Health Services (DGHS), Ministry of Health & Family Welfare, Government of India** to co-ordinate with all service providers and stakeholders to ensure minimum core facilities requirements as per IHR 2005.

Major Functions of APHO at Airports

- **Airport Quarantine:** Inspection of international travel documents for yellow fever disease; referral of suspect travellers to designated quarantine facility
- Disinfection, disinsection and deratting of aircraft including inspection and supervision of control measures
- Supervision of sanitation, quality of drinking water supply, anti-mosquito and anti-rodent work inside airports
- Public health clearance of dead bodies/human remains
- Administration of yellow fever vaccine and issue of yellow fever vaccination certificate
- Arrangements for quarantine of yellow-fever suspects at designated quarantine facilities
- Attend aircraft medical emergencies
- Food inspection for VVIP; Inspection of food stuff, catering establishments inside airport under FSSAI, 2011
- Assistance in sampling of imported food items and forwarding the lab analysis report
- Inspection and Licensing of eating establishments within local limits of airports as designated officer under FSSAI, 2006
- Activities during **Public Health Emergencies of International Concern (PHEIC)** like passenger screening, isolation & quarantine, contact listing, aircraft disinsection, transportation
- Effective surveillance activities on international travellers, aircrafts, cargos, conveyances, goods, postal parcels & human remains
- Ensure a safe environment for travellers at points of entry (potable water, eating establishments, flight catering facilities, public washrooms, waste disposal)

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EXAM POINT

Responsibility for implementing IHR 2005 at Indian Airports is that of **Director General Health Services**.

Other Applicable Rules Brief Summaries

Rule/Act	Year	Coverage
Aircraft (Demolition of Obstructions Caused by Building and Trees Etc.) Rules	1994	Procedure for demolition of obstructions around aerodromes
Aircraft (Carriage of Dangerous Goods) Rules	2003	Procedures for carriage of dangerous goods on aircraft
Aircraft (Security) Rules	2011	Deals with airport security
Aircraft (Investigation of Accidents and Incidents) Rules	2017	Deals with accidents and incidents investigations

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Rule/Act	Year	Coverage
The Unmanned Aircraft System (UAS) Rules	2021	See below

The Unmanned Aircraft System (UAS) Rules, 2021

These rules apply to:

- **(a)** Unmanned Aircraft System (UAS) registered in India, wherever they may be
- **(b)** A person owning, possessing, or engaged in exporting, importing, manufacturing, trading, leasing, operating, transferring or maintaining an Unmanned Aircraft System **in India**
- **(c)** All Unmanned Aircraft Systems for the time being **in or over India**

RELATIONSHIP WITH AIRCRAFT RULES, 1937

The provisions contained in the **Aircraft Rules, 1937 shall NOT apply** to the Unmanned Aircraft System and matters connected therewith except for those provisions whose application on UAS is specifically provided in these rules. These rules apply to UAS for **civil aviation purposes only**.

EXAM POINT

The provisions contained in the Aircraft Rules, 1937 shall **not apply to UAS except for those specifically provided in these rules**.

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Quick Reference — Critical Numbers & Limits

Parameter	Value	Rule Reference
No-fly radius from aerodrome (animal slaughter/rubbish)	**10 km**	Sec 5(qa)
Alcohol abstinence before flight (operating crew)	**12 hours**	Rule 24
Minimum age for sole control of aircraft	**16 years**	Rule 28
Maximum age for PIC/Co-pilot (commercial)	**65 years**	Rule 28A
Age limit requiring multi-crew & younger co-pilot	**60 years**	Rule 28A
Maximum Mach for civil aircraft (sonic boom)	**Mach 1**	Rule 29A
RPAS requiring UIN (all-up weight)	**>250 g**	Rule 15A
Fuelling — minimum distance from building	**15 m**	Rule 25A
Fuelling — naked flame exclusion zone	**30 m**	Rule 25A
Fuelling — unauthorised persons exclusion	**15 m**	Rule 25A
Fuelling — jet aircraft rear outlet distance (stop fuelling)	**43 m**	Rule 25A
Equipment distance (gasoline/wide-cut fuels)	**≥15 m**	Rule 25A
Equipment distance (straight kerosene)	**≥6 m**	Rule 25A
Documents production after demand (not on board)	**7 days**	Rule 17
Flight Navigator required: circle distance	**>600 NM**	Rule 38A(3)

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Parameter	Value	Rule Reference
Flight Navigator required: fix-to-fix distance (within 30 NM of route)	**>600 NM**	Rule 38A(3)
Log book preservation period	**5 years (from last entry)**	Rule 67A(3)
Log book quarterly certification period	**End of March, June, September, December**	Rule 67A(4)
Sickness/injury period requiring fresh medical before flying	**≥15 days**	Rule 42(2)
Advance notice for new route / route change	**7 days**	Rule 140A
CAR draft on DGCA website for public comment	**30 days**	Rule 133A
Towers of Silence (Malabar Hills) — prohibited radius	**1 mile**	Schedule I
Mathura Refineries — prohibited radius	**10 km**	Schedule I
Kalpakkam Nuclear — prohibited radius	**10 km**	Schedule I
Bhubaneshwar — prohibited upper limit	**50,000 ft**	Schedule I
Kalpakkam — prohibited upper limit	**10,000 ft**	Schedule I
Penalty — dangerous goods false info / arms carriage	**2 yrs + ₹1 Cr**	Sec 10(1)
Penalty — animal slaughter / rubbish near aerodrome	**3 yrs + ₹1 Cr**	Sec 10(1A)
Commander's health report before landing (India)	**≥2 hours in advance**	Public Health Rules
Mental disorder epilepsy — alcohol restriction before flight	**12 hours**	Rule 24A
Mental disorder — if excited in previous 2 weeks	**Requires registered medical practitioner + escort**	Rule 24A
Single pilot commercial: max weight (age 60-65, multi-crew exemption)	**5700 kg MTOW**	Rule 28A

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Capt. Pankaj Pahil

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CHAPTER 14

Personnel Licensing

14

Introduction & ICAO Background

WHAT IS "PERSONNEL LICENSING"? Personnel Licensing is the body of **Standards and Recommended Practices (SARPs)** for licensing aviation personnel. It ensures that everyone with safety-critical duties is trained, tested and certified to a common world standard.

Who is covered by Annex 1?

The SARPs for the licensing of the following personnel are provided by **Annex 1** to the Convention on International Civil Aviation (the Chicago Convention):

- **Flight crew members** — pilots, flight engineers and flight navigators
- Air traffic controllers
- Aeronautical station operators
- Aircraft maintenance technicians
- Flight dispatchers (flight operations officers)

Role of the Training Manuals

Related ICAO training manuals guide States on the **scope and depth** of training curricula, so that the confidence in safe air navigation intended by the Convention and Annex 1 is maintained. These manuals also guide the training of other aviation personnel such as **aerodrome emergency crews, flight operations officers and radio operators**.

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PURPOSE OF THE MEDICAL STANDARDS The medical standards of the Annex require **periodic health examinations**. They serve as an **early warning** for possible incapacitating medical conditions and contribute to the general health of flight crews and controllers.

Remember: the ICAO Annex that contains the Standards and Recommended Practices for **Personnel Licensing is Annex 1**. This is the single most repeated fact from this chapter (it answers two separate questions in the DGCA bank).

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National Provisions — General Rules

Licensing Authority

WHO ISSUES YOUR LICENCE? The authority by which licences and ratings may be **granted, renewed or varied** shall be the **Central Government**.

DISCRETIONARY POWER The Central Government **may withhold** the grant or renewal of a licence or a rating **if for any reason it considers it desirable to do so**. There is no automatic entitlement to a licence.

Educational Qualification

The minimum schooling required depends on the category of licence applied for. DGCA splits this into two groups:

Table 2.1 — Minimum Educational Qualification by Licence

Minimum Qualification	Licences Requiring It
Pass in Class 10 from a recognised Board (or equivalent examination)	Student Pilot's Licence (Aeroplanes / Helicopters / Gliders / Balloons / Microlight aircraft); Private Pilot's Licence (Aeroplanes); Private Pilot's Licence (Helicopters); Pilot's Licence (Gliders / Balloons / Microlight aircraft); Flight Radio Telephone Operator's Licence.
Pass in Class 10 + 2 with **Physics & Mathematics**, from a recognised Board / University (or equivalent)	Commercial Pilot's Licence (Aeroplanes); Commercial Pilot's Licence (Helicopters); Airline Transport Pilot's Licence (Aeroplanes); Airline Transport Pilot's Licence (Helicopters); Student Navigator's Licence; Flight Navigator's Licence; Student Flight Engineer's Licence; Flight Engineer's Licence; and Flight Radio Telephone Operator's Licence.

Exam Tip: For a **CPL/ATPL** you must have **10+2 with Physics & Maths**. For an **SPL or PPL**, a simple **Class 10 pass** is enough. The FRTOL appears in *both* lists.

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Cross-Country Flight — Definitions

DEFINITION (A) — GENERAL A cross-country flight, unless otherwise stated, means a flight to a point **beyond a radius of 100 nautical miles** from the aerodrome of departure, **whether over land or sea**.

DEFINITION (B) — PPL & MICROLIGHT For the purpose of the **PPL and Pilot Licence (Microlight)**, a cross-country flight means a flight to a point **beyond a radius of 50 nautical miles** from the aerodrome of departure.

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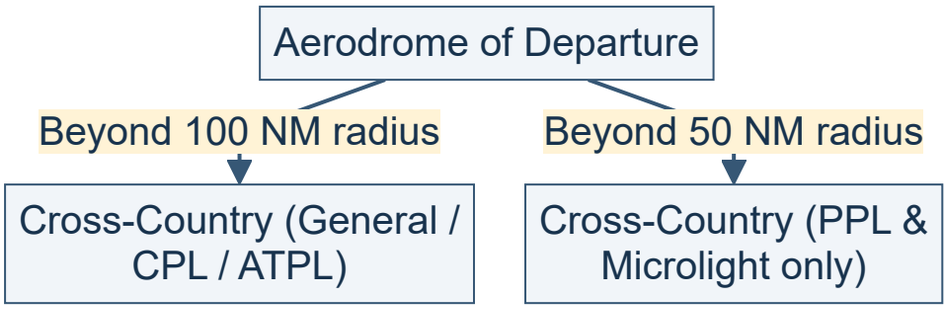

Figure 14.1 *Cross-Country Radius at a Glance*

Language Proficiency

THE ENGLISH REQUIREMENT An applicant for the **PPL, CPL, ATPL, Flight Navigator, Flight Engineer, ATC and Aeronautical Station Operator's** licences shall have the ability to **speak and understand the English language** used for radiotelephony communications, to the level of proficiency specified by the Director General.

Table 2.2 — ICAO English Language Proficiency Levels & Re-evaluation

Level	Description	Acceptable?	Re-evaluation Period
6	Expert	Yes	**Re-evaluation NOT required**
5	Extended	Yes	Every **6 years**
4	Operational	Yes	Every **3 years**
3	Pre-operational	No	**Further training required**
2	Elementary	No	
1	Pre-elementary	No	

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CRITICAL LIMIT Levels **1, 2 and 3 are NOT acceptable** — the holder must undergo further training. Only **Levels 4, 5 and 6** permit operation, and only Level 6 escapes periodic re-evaluation.

Memory hook — "6 = forever, 5 = six, 4 = three": Level 6 needs no re-evaluation, Level 5 every 6 years, Level 4 every 3 years.

Validity of Medical Fitness

DGCA medical assessments come in **three classes**. The class you need depends on the licence you hold or apply for.

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Which Class for Which Licence?

CLASS 1 MEDICAL ASSESSMENT — REQUIRED FOR APPLICANTS & HOLDERS OF:

1. Commercial Pilot's Licence (Aeroplane & Helicopter)
2. Airline Transport Pilot's Licence (Aeroplane & Helicopter)
3. Private Pilot's Licence (Aeroplane & Helicopter) **where Instrument Rating (Aeroplane & Helicopter) privileges** ***(what a licence allows its holder to do)*** **are required**

CLASS 2 MEDICAL ASSESSMENT — REQUIRED FOR APPLICANTS & HOLDERS OF:

1. Private Pilot's Licence (Aeroplane & Helicopter)
2. Pilot's Licence (Microlight)
3. Student Pilot's Licence (Aeroplane)
4. Student Pilot's Licence (Helicopter)
5. Student Pilot's Licence (Glider)
6. Student Pilot's Licence (Balloons)
7. Student Pilot's Licence (Microlight)
8. Flight Radio Telephone Operator's Licence (Restricted)

CLASS 3 MEDICAL ASSESSMENT — REQUIRED FOR APPLICANTS & HOLDERS OF:

1. Student ATCO Licence
2. ATCO Licence

Exam Tip: A normal **PPL needs Class 2** — but the moment **Instrument Rating privileges** are wanted, the PPL holder must move up to a **Class 1** medical. Class 3 is exclusively for **ATCOs**.

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Validity of the Medical Assessment

Table 3.1 — Medical Validity *(the period during which a document is accepted)* Periods

Class	Validity
Class 1	**One year** up to the age of 60 years; **six-monthly** thereafter. **For single-crew operations**, validity is **six months above 40 years of age**.
Class 2	**2 years** up to the age of 50 years; **One year** thereafter.
Class 3	Up to 40 years — **4 years**; Above 40 years up to 50 years — **2 years**; Above 50 years — **One year**.

Medical class	Age / operating condition	Medical validity
Class 1: CPL / ATPL / PPL + IR	Up to 60 years	1 year

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Medical class	Age / operating condition	Medical validity
Class 1	Above 60 years	6 months
Class 1	Single-crew and above 40 years	6 months
Class 2: PPL / SPL / Microlight / FRTOL(R)	Up to 50 years	2 years
Class 2	Above 50 years	1 year
Class 3: Student ATCO / ATCO	Up to 40 years	4 years
Class 3	Above 40, up to 50 years	2 years
Class 3	Above 50 years	1 year

Figure 14.2 *Medical assessment validity — DGCA CAR Section 7, Series C, Part I, paragraphs 2.4 and 5.1.*

Single-crew trap: A CPL holder flying a **single-pilot helicopter (e.g. charters) above 40 years of age** gets only a **6-month** medical validity — *not* 12 months.

Conditions Under Which a Licence is Deemed Invalid

NO FLYING WITHOUT A FRESH "FIT" DECLARATION The holder of a licence **shall not exercise the privileges of his licence** without being declared fit after a **fresh medical examination** in the event of his having:

1. a **sickness or injury involving incapacity** for a period of **15 days or more** for the work for which he is licensed; **or**
2. an **injury sustained in any accident** occurring during the exercise of the privileges of his licence or otherwise, which is likely to cause incapacity or impair his efficiency in the discharge of his duties.

REPORTING PROCEDURE & FURTHER CONDITIONS

- The licence holder **or his employer** shall **immediately notify** all relevant details of the sickness or injury to the **Director-General**.
- The licence of a person **disqualified** shall be deemed to be invalid **until the holder passes a fresh medical examination**.
- The holder shall **not exercise the privileges** of his licence during any period when he is aware that his **physical condition has deteriorated below the standard** required for that category of licence.

Exam Tip: The threshold for informing the Authority of an illness causing incapacity is **15 days** or more — this is directly tested.

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Licence Validity, Special Medical Exam & Age Limits

Validity of Licences

STANDARD VALIDITY & RENEWAL PROCEDURE

- ATPL, CPL and PPL are valid for **10 years** since **10 Oct 2023** (the Student Pilot's Licence is still 5 years, renewed in 12-month steps - see 5.3).
- Licences are valid **from the date of issue**.
- **Renewal** will be done **from the date of expiry**, provided the application is made **within one month prior** to the date of expiry.

Special Medical Examination

WHEN A "SPECIAL" MEDICAL APPLIES A Special Medical Examination is for **medical ailments resulting in unfitness for flying for 15 days or more, but less than or equal to 60 days**.

WHO CONDUCTS IT & REPORTING OF INCAPACITATION

- It will be undertaken by a **DGCA-empanelled Class 1 Medical Examiner**, on a **case-to-case basis**, only as decided by **DGCA (Medical Directorate)**.
- All cases of **in-flight incapacitation** of aircrew shall be reported by the **concerned operator** to DGCA **at the earliest**.

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Minimum Age

SOLE CONTROL OF AN AIRCRAFT IN MOTION No person **under 16 years of age** shall have **sole control** of an aircraft in motion, and no person shall cause or permit any other person to have sole control of an aircraft in motion unless he knows, or has reasonable cause to believe, that the other person has attained the age of 16 years.

Table 4.1 — Minimum Age for Each Licence

Licence	Minimum Age
Student Pilot's Licence (SPL)	**16 years**
Private Pilot's Licence (PPL)	**17 years**
Commercial Pilot's Licence (CPL)	**18 years**
Airline Transport Pilot's Licence (ATPL)	**21 years**

Maximum Age Limit for Professional Pilots

RULE (1) — THE HARD CEILING No person holding a pilot's licence, having attained the age of **sixty-five (65) years**, shall act as **Pilot-in-Command or Co-pilot** of an aircraft engaged in **commercial air transport operations**.

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RULE (2) — THE 60-YEAR MULTI-CREW RULE No person holding a pilot's licence, having attained the age of **sixty (60) years**, shall act as Pilot-in-Command or Co-pilot of an aircraft engaged in commercial air transport operations **unless it is operated in a multi-crew environment and the other pilot is less than sixty years of age**.

EXCEPTION TO RULE (2) The provisions of sub-rule (2) shall **not apply** in respect of aircraft **certified for single-pilot operations and not exceeding an all-up weight of 5700 kg**, engaged in commercial air transport operations within the territory of India, and while operating in a **multi-crew environment**.

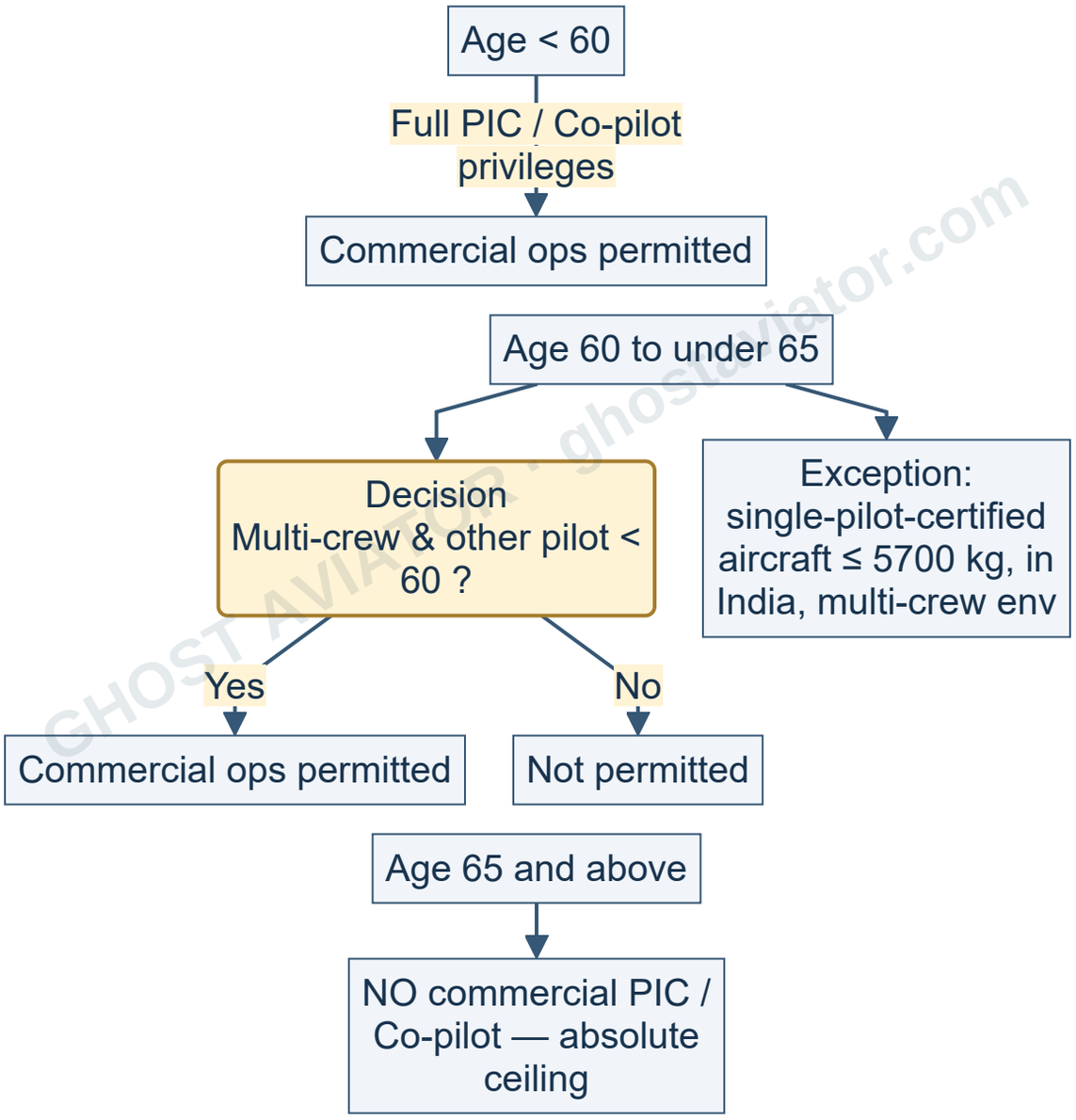

Figure 14.3 *Professional Pilot Age Ladder*

Exam Tip: With an ATPL, the operating window for a public-transport flight is **minimum 21 years — maximum 65 years**.

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Student Pilot's Licence (SPL) · Aeroplanes / Helicopters / Gliders

Knowledge

EXAMINATIONS REQUIRED The applicant shall pass an **oral examination** in **Air Regulations, Air Navigation, Aviation Meteorology, and Aircraft & Engines**, as per the syllabus prescribed *(laid down by the authority or the rules)* by the Director-General — **unless** he previously held a pilot's licence of a higher category, or is able to produce evidence (in the prescribed manner) that he is well-versed with the aforesaid subjects.

Security Clearance

PROCEDURE Security clearance for the Student Pilot / Pilot will be obtained from the concerned **State Police authorities** by the **Flying Club / Government Flying Training School**, **before the commencement of flying training**, and furnished to the Director-General.

Renewal

SPL RENEWAL LIMITS The licence may be renewed for a period **not exceeding twelve (12) months** from the date of fresh medical examination, **subject to the total validity of the licence not exceeding twenty-four (24) months** from the date of issue.

Aircraft Rating

WHAT THE RATING SHOWS The licence shall indicate the **class and the types** of aeroplanes, helicopters or gliders the holder is entitled to fly. **Only those types** of aircraft may be entered in the licence in respect of which the candidate has **passed the examination in Aircraft & Engines**.

Privileges

WHAT AN SPL HOLDER MAY DO The privileges are to fly **within Indian territory only**, as **Pilot-in-Command** of any aeroplane, helicopter or glider entered in the aircraft rating of his licence — **provided that** ***(an exception or condition added to a rule)***:

THE FOUR CONDITIONS (A)–(D) ON SPL PRIVILEGES

a. he shall fly **at all times under the authority and supervision** of a Flight Instructor or an Approved Examiner;

b. he shall fly under **Visual Flight Rules (VFR) only**;

c. he shall **not carry passengers, animals or goods**, nor fly for **hire, reward or remuneration** of any kind;

d. he shall **not undertake cross-country flights** unless he has a minimum of **ten (10) hours of solo flight time** and has passed the examinations in **Air Navigation and Aviation Meteorology**.

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ISSUING BODY The Student Pilot's Licence shall be issued by a **Flying Club / Government Flying Training School** specifically authorised in this regard, and subject to the conditions laid down by the Director-General.

Memory hook for SPL: "**VFR · Supervised · No pax/cargo · 10 hrs solo before XC**." Renewal: **12-month steps, 24-month total cap**.

Private Pilot's Licence — PPL (Aeroplanes)

Knowledge

WRITTEN EXAMINATION The applicant shall pass a **written examination** in **Air Regulations, Air Navigation, Aviation Meteorology, and Aircraft & Engines**, as per the syllabus prescribed by the Director-General.

CONCESSION FOR HELICOPTER-LICENCE HOLDERS An applicant in possession of a valid **PPL (Helicopters)** or a **Commercial / Airline Transport Pilot's Licence (Helicopters)** shall pass an examination in **Aircraft & Engines only**.

Experience

MINIMUM FLYING EXPERIENCE FOR A PPL(A) The applicant shall produce evidence of having satisfactorily completed, as a pilot of an aeroplane, **not less than 40 hours of flight time**, which shall include:

1. not less than **20 hours of solo flight time**;
2. not less than **5 hours of cross-country flight time** as the **sole occupant** of an aeroplane, including a flight of **not less than 150 nautical miles** in the course of which **full-stop landings at two different aerodromes** shall have been made;
3. not less than **10 hours of solo flight time** completed within a period of **twelve months** immediately preceding the date of application for the issue of the licence;
4. **50% of solo flying experience on microlight aircraft** acquired during the preceding 24 months may be credited towards the total experience, **subject to a maximum of 10 hours**;
5. **50% of solo gliding experience** shall count towards total flying experience, **subject to a maximum of 10 hours** towards total flight time.

Flying Training & Skill

FLYING TRAINING The applicant shall have completed flying training in accordance with the syllabus prescribed by the Director-General.

SKILL TEST The applicant shall have demonstrated his competency to perform as a Pilot-in-Command or a Co-pilot of an aeroplane — the procedures and manoeuvres prescribed in the syllabus — to the satisfaction of an Examiner, on the type of aeroplane to which the application relates, **within a period of six months** immediately preceding the date of application.

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Renewal

EVIDENCE REQUIRED FOR PPL RENEWAL The licence may be renewed on receipt of satisfactory evidence of the applicant:

a. having undergone a **medical examination**; **and**

b. having satisfactorily completed **not less than 5 hours of flight time as Pilot-in-Command** of an aeroplane within a period of **twelve months** immediately preceding the date of application for renewal — **or, in lieu thereof**, having satisfactorily completed the **flying test within a period of six months** immediately preceding the date of application.

Ratings on a PPL

(a) Aircraft Rating & the "Open Rating"

OPEN RATING — PPL The licence indicates the class and types of aeroplanes the holder may fly. An **open rating** for **all single-piston-engine types** of aeroplanes having an all-up weight **not exceeding 1500 kg** may be granted if the holder has completed **not less than 250 hours as Pilot-in-Command** and has at least **4 different types** of aeroplanes entered in the aircraft rating.

CONDITION ON THE OPEN RATING Open-rating privileges are exercised **only after** a **ground and flight familiarisation** with a Flight Instructor or an Approved Examiner for the type of aircraft, and a **certificate to this effect** shall be recorded by the Instructor / Examiner in the Pilot's **Log Book** before the pilot is released to exercise the open-rating privileges.

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(b) Night Rating

WHAT IT ALLOWS A Night Rating entitles the holder of the licence to **carry passengers at night**.

CONDITIONS FOR ISSUE OF A NIGHT RATING

1. He must have completed **not less than 50 hours of flight time as Pilot-in-Command** and as **sole manipulator** of the controls, including **not less than 5 hours by night**, which must include a minimum of **5 take-offs and 5 landings** carried out within the preceding 6 months of the date of application;
2. He must have completed a **dual cross-country flight by night of at least 100 nautical miles** before he can be permitted to undertake **sole cross-country flights by night**;
3. He must have completed **not less than 5 hours of dual instruction in instrument flying**, which may include **not more than 2½ hours on an approved synthetic flight trainer**.

(c) Instrument Rating

WHAT IT ALLOWS An Instrument Rating entitles the holder of the licence to fly **under the Instrument Flight Rules (IFR)**.

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Extension of Aircraft Rating

EVIDENCE REQUIRED TO ADD A NEW TYPE For extension of the Aircraft Rating to include an additional type of aeroplane, the applicant shall produce evidence of —

a. having passed the **written examination in Aircraft & Engines**;

b. having undergone adequate **dual instruction and solo flying** to gain competency on the type;

c. having satisfactorily completed the **flight test on the type** within a period of **six months** immediately preceding the date of application for the extension of the Aircraft Rating.

Privileges of a PPL

CORE PRIVILEGE The privileges of a PPL holder are to act, **but not for remuneration**, as **Pilot-in-Command or Co-pilot** of any aeroplane entered in the Aircraft Rating of his licence, and to **carry passengers** therein.

THE THREE "PROVIDED THAT" CONDITIONS

1. **No flight** is undertaken for **hire or remuneration** of any kind whatsoever;
2. Passengers are carried **by night only** when the holder is in possession of a **valid Night Rating** and has carried out **not less than 5 take-offs and 5 landings by night as Pilot-in-Command** within a period of 6 months immediately preceding the date of the intended flight;
3. **No flight** shall be undertaken under the **Instrument Flight Rules** without being in possession of a **valid Instrument Rating**.

Exam Tip: A PPL holder may carry passengers **without remuneration** — never for hire/reward. The headline PPL number is **40 hours total**, of which **20 hours** solo.

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Commercial Pilot's Licence — CPL (Aeroplanes)

Knowledge

EXAMINATIONS REQUIRED The applicant shall pass a **written examination** in **Air Regulations, Air Navigation, Meteorology, and Aircraft & Engines**, and a **Signals (practical) examination** for the interpretation of **aural and visual signals**, as per the syllabus prescribed by the Director-General.

CONCESSION FOR CPL(H) HOLDERS The holder of a current **Commercial Pilot's Licence (Helicopters)** shall be required to pass an examination in **Aircraft & Engines only**.

Experience

MINIMUM EXPERIENCE FOR A CPL(A) The applicant shall produce evidence of having satisfactorily completed, as a pilot of an aeroplane, on the date of application, **not less than 200 hours of flight time**, which shall include:

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1. not less than **100 hours as Pilot-in-Command**, of which not less than **15 hours** shall have been completed within a period of **six months** immediately preceding the date of application;
2. not less than **20 hours of cross-country flight time as Pilot-in-Command**, including a cross-country flight of **not less than 300 nautical miles** in the course of which **full-stop landings at two different aerodromes** shall be made;
3. not less than **10 hours of instrument time**, of which **not more than 5 hours** may be on an **approved simulator**; and
4. not less than **5 hours of flight time by night**, including a minimum of **10 take-offs and 10 landings as Pilot-in-Command** (sole manipulator of controls) carried out within six months immediately preceding the date of application.

REDUCED REQUIREMENT FOR EXPERIENCED HELICOPTER PILOTS For an applicant who holds a **CPL(H) / ATPL(H)** and has satisfactorily completed **not less than 1000 hours of flight time as Pilot-in-Command of a helicopter**, the requirement of 200 hours as pilot of an aeroplane is **reduced to 100 hours**, which shall include —

1. not less than **75 hours as Pilot-in-Command**, including a minimum of **25 hours cross-country** and **10 hours instrument time** (of which not less than 5 hours may be on an approved simulator);
2. not less than **5 hours by night**, including **10 take-offs and landing patterns**; and
3. not less than **10 hours on an aeroplane** within a period of six months immediately preceding the date of application.

Flying Training & Other Requirements

FLYING TRAINING Completed in accordance with the syllabus prescribed by the Director-General.

OTHER REQUIREMENT — FRTOL The applicant shall be in possession of a **current Flight Radio Telephone Operator's Licence** for the operation of radio-telephone apparatus on board an aircraft, issued by the Director-General.

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Skill

DEMONSTRATION OF COMPETENCY The applicant shall demonstrate competency within **six months** immediately preceding the date of application. The Director-General may, however, allow the Skill Test (or part thereof) to be carried out on an aircraft / approved **Zero Flight Time Training (ZFTT) simulator Level 'D'** for the type. Competency shall be demonstrated in:

i. general flying test **by day**;
ii. general flying test **by night**;
iii. a **cross-country flight test by day** consisting of a flight of **not less than 250 nautical miles**, in the course of which **at least one full-stop landing** at an aerodrome other than the aerodrome of departure shall be made; and
iv. a **cross-country flying test by night** consisting of a flight of **not less than 120 nautical miles**, returning to the place of departure **without landing elsewhere**.

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Renewal

EVIDENCE REQUIRED FOR CPL RENEWAL

a. having undergone a **medical examination**;

b. having satisfactorily completed **not less than 10 hours of flight time as Pilot-in-Command** (**50% of flight time as Co-pilot may be counted** towards the PIC requirement) within a period of **six months** immediately preceding the date of application for renewal — or, in lieu thereof, having completed the **general flying test by day and night** within the same period;

c. having a **current Flight Radio Telephone Operator's Licence** for operation of radio-telephone apparatus on board an aircraft, issued by the Director-General.

Aircraft Rating

OPEN RATING — CPL The licence indicates the types of aeroplane the holder may fly. An **open rating** for all **single-piston-engine** types of aeroplane having an all-up weight **not exceeding 1500 kg** may be granted if he has completed **not less than 1000 hours of flight time on such types**, including **not less than 500 hours as Pilot-in-Command**, and has at least **4 different types** entered in the aircraft rating. The open rating is exercised only after ground & flight familiarisation, with a certificate recorded in the pilot's log book.

- **Instructor's Rating** — entitles the holder to **impart flying instructions**.
- **Instrument Rating** — entitles the holder to fly **under the Instrument Flight Rules**.

Extension of Aircraft Rating

REQUIREMENTS TO ADD A TYPE

i. Having **passed a written examination in Aircraft & Engines** and having gained, under appropriate supervision, experience in flying the aircraft of such type (or on an approved flight simulator) in respect of:

- (a) normal flight procedures and manoeuvres during all phases of flight;
- (b) abnormal and emergency procedures and manoeuvres in the event of failures and malfunctions of equipment such as power plant, systems and airframe;
- (c) where applicable, instrument procedures — including instrument approach, missed approach and landing procedures under normal, abnormal and emergency conditions, including simulated engine failure;
- (d) procedures for crew incapacitation and crew coordination, including allocation of pilot tasks, crew cooperation and use of checklists; and

ii. having satisfactorily completed the **general flying tests by day and night** in respect of the type, within a period of **six months** immediately preceding the date of application.

Proficiency Check

WHEN A PROFICIENCY CHECK IS MANDATORY To act as a **co-pilot of transport aeroplanes having an all-up weight exceeding 5700 kg**, the licence holder shall undergo an appropriate proficiency check as specified by the Director-General, in respect of the type to be flown.

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VALIDITY & RENEWAL OF THE PROFICIENCY CHECK

- The proficiency check is valid for **six months** from the date of the check, and is renewed for a further period of **six months** at a time.
- On renewal, the period of validity shall **commence from the date of expiry** of the previous validity, provided the check has been carried out **within two months preceding** the date of expiry.

Privileges of a CPL

WHAT A CPL HOLDER MAY DO

a. to exercise **all the privileges of a Private Pilot's Licence**;

b. to act as **Pilot-in-Command of any aeroplane having an all-up weight not exceeding 5700 kg** which is entered in the aircraft rating — provided that, when passengers are carried at night, he shall have carried out **not less than 10 take-offs and 10 landings by night as Pilot-in-Command** within a period of six months immediately preceding the intended flight;

c. to act as **Co-pilot** of any aeroplane where a Co-pilot is required and which is entered in the aircraft rating — provided that, for all flights under the Instrument Flight Rules (as PIC or Co-pilot), he shall have a **current Instrument Rating**.

AGE RESTRICTION ON CPL PRIVILEGES On attainment of the age of **sixty-five (65) years**, such privileges shall be **restricted to that of a Private Pilot's Licence (Aeroplanes)**.

Exam Tip: CPL headline figures — **200 hrs total**, **100 hrs PIC**, XC flight **300 NM** with two full-stop landings. A CPL holder is PIC of aircraft **up to 5700 kg**.

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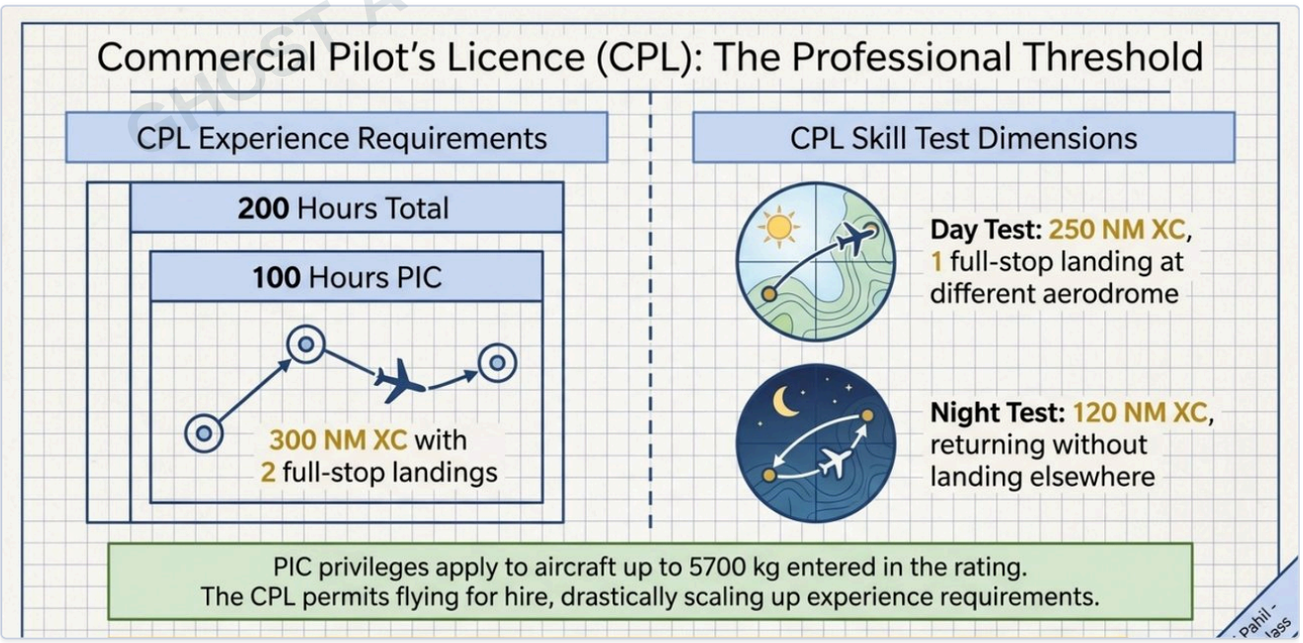

Figure 14.4 *Commercial Pilot Licence: experience and skill-test figures*

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CPL (Aeroplanes) with Instrument Rating

WHAT MAKES THIS LICENCE DIFFERENT This is a Commercial Pilot's Licence in which the privileges of an Instrument Rating are **built in** — there is **no separate Instrument Rating** issued. The experience requirements (especially instrument time) are correspondingly higher.

Knowledge

EXAMINATIONS REQUIRED A **written examination** in **Air Regulations, Air Navigation, Meteorology, and Aircraft & Engines**, and a **Signals (practical)** examination for interpretation of aural and visual signals, as per the prescribed syllabus. A holder of a current **CPL(H)** shall pass an examination in **Aircraft & Engines and Instrument Rating only**.

Experience

MINIMUM EXPERIENCE — CPL(A) WITH IR Within a period of **five years** immediately preceding the date of application, the applicant shall have completed, as a pilot of an aeroplane, **not less than 200 hours of flight time**, which shall include:

i. not less than **100 hours as Pilot-in-Command**, out of which not less than **30 hours as Student Pilot-in-Command** — which shall include **not more than 20 hours of cross-country** and **not more than 10 hours circuits-and-landings with a minimum of 10 landings**;

ii. not less than **15 hours as Pilot-in-Command** within a period of **six months** immediately preceding the date of application;

iii. not less than **50 hours of cross-country flight time as Pilot-in-Command**, including a cross-country flight of **not less than 300 nautical miles** with full-stop landings at two different aerodromes;

iv. not less than **50 hours of instrument time**, of which **not more than 20 hours** may be on an approved simulator, and out of which a **minimum of 5 hours** shall have been completed within six months immediately preceding the date of application for the Instrument Rating; and

v. not less than **5 hours by night**, including a minimum of **10 take-offs and 10 landings as Pilot-in-Command** (sole manipulator of controls) carried out within six months immediately preceding the date of application.

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REDUCTION FOR EXPERIENCED HELICOPTER PILOTS For an applicant in possession of a **CPL (Helicopters)** who has satisfactorily completed **not less than 1000 hours as Pilot-in-Command of a helicopter**, the requirement of 200 hours as pilot of an aeroplane is **reduced to 100 hours**.

Important Note on logging: The **Student-Pilot-in-Command flight time shall NOT be logged by the instructor** in his own log book. The student log book shall indicate Student-Pilot-in-Command flight time in the **remarks column as "SPIC"**, with the name of the instructor.

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Flying Training, Other Requirements & Skill

FLYING TRAINING & FRTOL Flying training completed per the prescribed syllabus. The applicant shall be in possession of a **current Flight Radio Telephone Operator's Licence**.

SKILL — DEMONSTRATED WITHIN SIX MONTHS Competency shall be demonstrated in:

i. general flying test **by day**;

ii. general flying test **by night**;

iii. a **cross-country flight test by day** of **not less than 250 NM**, with at least one full-stop landing at an aerodrome other than the aerodrome of departure;

iv. a **cross-country flying test by night** of **not less than 120 NM**, returning to the place of departure without landing elsewhere; and

v. **ability to fly an aeroplane solely with the aid of instruments**, by undergoing an **instrument flying test** within six months immediately preceding the date of application for the rating. The Director-General may allow such tests (or part) on an approved simulator for the type.

Renewal

EVIDENCE REQUIRED

a. having undergone a **medical examination**;

b. having satisfactorily completed **not less than 10 hours as Pilot-in-Command** (50% of Co-pilot flight time may be counted) within **six months** immediately preceding the date of application for renewal, or in lieu thereof the **general flying test by day and night** within the same period; and

c. having a **current Flight Radio Telephone Operator's Licence**.

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Rating

OPEN RATING & THE BUILT-IN INSTRUMENT RATING

- **Open Rating** — for all single-piston-engine types not exceeding **1500 kg**, granted on **not less than 1000 hours** including **not less than 500 hours as PIC** and at least **4 different types** entered in the rating (exercised only after ground & flight familiarisation, certificate in the log book).
- **Instructor's Rating** — entitles the holder to impart flying instructions.
- **Instrument Rating** — **No separate Instrument Rating is provided** in the licence. The privileges of the instrument rating are **included in the privileges of this licence**, provided the instrument rating flying tests have been carried out as per the conditions laid down by the Director-General.

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Extension of Aircraft Rating & Proficiency Check

EXTENSION OF AIRCRAFT RATING To add a type, the applicant shall produce evidence of having **passed a written examination in Aircraft & Engines** and of having gained, under appropriate supervision, experience in flying the type (or on an approved flight simulator) covering: (a) normal flight procedures and manoeuvres; (b) abnormal and emergency procedures and manoeuvres; (c) where applicable, instrument procedures including instrument approach, missed approach and landing under normal, abnormal and emergency conditions, including simulated engine failure; and (d) procedures for crew incapacitation and crew coordination — followed by satisfactorily completed **general flying tests by day and night** within **six months** immediately preceding the date of application.

PROFICIENCY CHECK To act as a co-pilot of transport aeroplanes exceeding **5700 kg**, the holder shall undergo a proficiency check. It is **valid for six months** and renewed for a further six months at a time; on renewal, validity **commences from the date of expiry** of the previous validity, provided the check is carried out **within two months preceding** the date of expiry.

Privileges

PRIVILEGES OF A CPL(A) WITH IR

a. to exercise **all the privileges of a Private Pilot's Licence (Aeroplanes)**;

b. to act as **Pilot-in-Command of any aeroplane having an all-up weight not exceeding 5700 kg** entered in the aircraft rating — provided that, when passengers are carried at night, he shall have carried out **not less than 10 take-offs and 10 landings by night as Pilot-in-Command** within six months immediately preceding the intended flight;

c. to act as **Co-pilot** of any aeroplane where a Co-pilot is required and which is entered in the aircraft rating — provided that, for all flights under the Instrument Flight Rules (as PIC or Co-pilot), he shall have **demonstrated his competency**.

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Airline Transport Pilot's Licence — ATPL (Aeroplanes)

Knowledge

EXAMINATIONS REQUIRED The applicant shall pass a **written and oral examination** in **Air Regulations, Air Navigation, Avionics (Radio Aids & Instruments), Aviation Meteorology, and Aircraft & Engines**, and a **Signals (practical)** examination for interpretation of aural and visual signals, as per the prescribed syllabus.

Important Note: A holder of a **current Commercial Pilot's Licence** shall **not be required to pass the examination in Air Regulations**; and a holder of a **Senior Commercial Pilot's Licence** shall **not be required to pass any of the examinations**.

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Experience

MINIMUM EXPERIENCE FOR AN ATPL(A) The applicant shall produce evidence of having satisfactorily completed, as a pilot of an aeroplane, **not less than 1500 hours of flight time**, and his total flying experience shall include:

i. not less than **250 hours as Pilot-in-Command or as Co-pilot** performing, under the supervision of a pilot who fulfils the flying-experience requirements of a check pilot, the duties and functions of a pilot-in-command — provided that **at least 100 hours** out of these shall be **cross-country** flight time, including **not less than 50 hours of flight time by night**;

ii. not less than **500 hours of total cross-country flight time**;

iii. not less than **75 hours of flight time by night**;

iv. not less than **100 hours of instrument time** under actual or simulated instrument conditions, of which **not less than 50 hours shall be in actual flight**; and

v. not less than **20 hours of flight time** completed within a period of **six months** immediately preceding the date of application for the licence.

NOTE — CO-PILOT FLIGHT-TIME CREDIT Where an aeroplane is required to be operated with a co-pilot in accordance with the provisions of the Flight Manual of the aeroplane: **not more than 50%** of the co-pilot flight time shall be credited towards the **total flight time** required for the issue of the licence; **but not more than 50%** of the co-pilot flight time under supervision shall be credited towards **pilot-in-command flight experience**; and **full credit** for such flight time shall be given towards the **total flight time** required.

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Other Requirements

THREE PREREQUISITES

i. He shall be the holder of a **Commercial or a Senior Commercial Pilot's Licence**. However, this will **not be applicable** for the issue of an ATPL to a pilot from the **Armed Forces** who otherwise meets the requirements;

ii. He shall have a **current Instrument Rating on multi-engine aircraft**;

iii. He shall be in possession of a **current Flight Radio Telephone Operator's Licence**.

Skill

DEMONSTRATION OF COMPETENCY The applicant shall have demonstrated competency, **by day and by night**, on the type of **multi-engine aeroplane** to which the application relates, within a period of **six months** immediately preceding the date of application. The Director-General may allow skill tests (or part) on an aircraft / approved **ZFTT simulator Level 'D'** for the type.

Note: A holder of a current **CPL(A) / Senior CPL(A)** issued by the Director-General shall **not be required to undergo general flying tests** for the issue of an ATPL(A), provided the licence is issued on the multi-engine type(s) included in the aircraft rating of the CPL/SCPL **and** the pilot possesses **10 hours as Pilot-in-Command or 20 hours as Co-pilot** on at least one such multi-engine type within the preceding six months.

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Renewal

EVIDENCE REQUIRED FOR ATPL RENEWAL

a. having undergone a **medical examination**;

b. having satisfactorily completed **not less than 10 hours as Pilot-in-Command** (50% of Co-pilot flight time may be counted) within **six months** immediately preceding the date of application for renewal, or in lieu thereof having completed the **flying tests by day and by night**;

c. having a **current Flight Radio Telephone Operator's Licence**;

d. having a **current Instrument Rating on a multi-engine aircraft**.

Ratings

ATPL AIRCRAFT RATING & OPEN RATING The licence indicates the class and types of aeroplanes the holder may fly. An **open rating** for **all types of aeroplanes** having an all-up weight **not exceeding 5700 kg** may be granted if he has completed **not less than 1000 hours as Pilot-in-Command on any aeroplane having an all-up weight of 14000 kg or above**.

- **Instructor's Rating** — entitles the holder to impart flying instructions.
- **Instrument Rating** — **No separate Instrument Rating is provided** in the licence; the privileges of the instrument rating are **included**, provided the instrument rating flying tests have been carried out as per the conditions laid down by the Director-General.

Extension of Aircraft Rating & Proficiency Check

EXTENSION OF AIRCRAFT RATING To add a type, the applicant shall have **passed a written examination in Aircraft & Engines** and shall have acquired, under appropriate supervision, experience in flying the type (or on an approved flight simulator) in accordance with the syllabus prescribed by the Director-General, followed by a **skill test** as laid down by the Director-General.

PROFICIENCY CHECK To act as a **Pilot-in-Command or Co-pilot** of transport aeroplanes exceeding **5700 kg**, the holder shall undergo a proficiency check. It is **valid for six months** and renewed for a further six months at a time; on renewal, validity **commences from the date of expiry** of the previous validity, provided the check is carried out **within two months preceding** the date of expiry.

Privileges of an ATPL

WHAT AN ATPL HOLDER MAY DO

a. to exercise the privileges of a **Private and a Commercial Pilot's Licence**;

b. to act as **Pilot-in-Command of an aeroplane having an all-up weight exceeding 5700 kg** entered in the aircraft rating, subject to compliance with the conditions below.

CONDITIONS (I)–(III) FOR PIC OF > 5700 KG AIRCRAFT

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i. He shall have completed, on that type of aeroplane, **not less than 100 hours of flight time as a Co-pilot**, followed by **10 consecutive satisfactory route checks**, of which **not less than 5 shall be by night**, under the supervision of a **Check Pilot** performing the duties and responsibilities of a Pilot-in-Command, and demonstrated his competency to fly as a Pilot-in-Command to the satisfaction of the Director-General.

> **CONCESSION** For a pilot having previous experience of **at least 100 hours as Pilot-in-Command of transport aeroplanes exceeding 5700 kg**, the required number of route checks may be **reduced to 6**, of which **at least 3 shall be by night**.

ii. He shall have undergone satisfactorily, within the preceding **six months** of the intended flight, appropriate **proficiency checks** in respect of that type of aircraft as required by the Director-General;

iii. He shall be in possession of a **current Instrument Rating** before operating any IFR flight.

AGE RESTRICTION ON ATPL PRIVILEGES On attainment of the age of **sixty-five (65) years**, such privileges shall be **restricted to that of a Private Pilot's Licence (Aeroplanes)**.

Exam Tip: ATPL headline figures — **1500 hrs total**, **500 hrs total cross-country**, **75 hrs night**, **100 hrs instrument** (≥50 hrs in actual flight), and **250 hrs cross-country with 50 hrs night** within the PIC/supervised block.

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Instrument Rating (Aeroplanes)

Knowledge

EXAMINATIONS REQUIRED The applicant shall pass a **written and oral examination** in **Air Regulations, Air Navigation, Aviation Meteorology and Instrument Rating**, as per the syllabus prescribed by the Director-General for the issue of a Commercial Pilot's Licence. He shall also pass a **practical test on the interpretation of aural and visual signals** as per the prescribed syllabus.

Experience

MINIMUM EXPERIENCE FOR AN INSTRUMENT RATING(A) The applicant shall produce evidence of having satisfactorily completed, as a pilot of an aeroplane —

i. not less than **100 hours of flight time as Pilot-in-Command**, including **not less than 50 hours of cross-country** flight time;

ii. not less than **40 hours of instrument time**, of which **not more than 20 hours shall be instrument ground time**. A minimum of **5 hours of instrument time** shall have been completed within a period of **six months** immediately preceding the date of application for the Instrument Rating.

CONCESSION FOR IR(HELICOPTERS) HOLDERS In the case of a pilot who holds a current **Instrument Rating (Helicopters)**, he shall have **not less than 100 hours as Pilot-in-Command of an aeroplane**, including not less than **25 hours of cross-country** flight time and not less than **20 hours of instrument time**, of which **not more than 10 hours** may be on an approved simulator.

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Other Requirements, Flying Training & Skill

OTHER REQUIREMENTS The applicant shall be —

i. holder of a **current Pilot's Licence (Aeroplanes)**;

ii. holder of a **current Flight Radio Telephone Operator's Licence** for the operation of radio-telephone apparatus on board an aircraft, issued by the Director-General.

FLYING TRAINING & SKILL Flying training shall be completed in accordance with the prescribed syllabus. For the **Skill**, the applicant shall demonstrate, to the satisfaction of the Examiner, his competency to fly an aeroplane **solely with the aid of instruments**, by undergoing an **instrument flying test** within a period of **six months** immediately preceding the date of application for the rating. The Director-General may allow such tests (or part) to be carried out on an **approved simulator** for the type.

Validity

VALIDITY OF THE INSTRUMENT RATING

a. The rating shall be valid for a period of **twelve months** from the date of the instrument rating flying test;

b. It shall be renewed for a further period of **twelve months** at a time from the date of expiry, provided the instrument rating flying test has been carried out **within two months preceding** the date of expiry and all other requirements for renewal are met;

c. In other cases, the validity of the renewal of the rating shall **commence from the date of the test**.

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Renewal & Extension

RENEWAL The Instrument Rating may be renewed on receipt of satisfactory evidence of the applicant: (a) having satisfactorily completed the **Instrument Rating Flight Test**; and (b) having a **valid Flight Radio Telephone Operator's Licence** issued by the Director-General for the operation of radio-telephone apparatus on board an aircraft.

EXTENSION OF INSTRUMENT RATING To include an additional type of aeroplane, the applicant shall produce evidence of having satisfactorily completed the **flight test** (in accordance with the skill requirement) in respect of the type for which the extension is desired, within a period of **six months** immediately preceding the date of application for the extension.

Privileges

CORE PRIVILEGE Subject to the validity of the Instrument Rating, the privileges of the holder shall be to fly under the **Instrument Flight Rules**, on the types of aeroplanes on which he has demonstrated his competency.

HOW THE PRIVILEGE SCALES BY AIRCRAFT CATEGORY

- A pilot who demonstrated competency by an IR Flight Test on a **single-engine** aeroplane **not exceeding 5700 kg** shall exercise the privileges on **all single-engine aeroplanes** within that weight category.

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- A pilot who demonstrated competency by an IR Flight Test on a **multi-engine** aeroplane **not exceeding 5700 kg** shall exercise the privileges on **all multi-engine aeroplanes** within that weight category.
- The holder of an IR on any **transport aeroplane** shall exercise the privileges **only on the type** on which the IR Flight Test was satisfactorily carried out.
- The holder of an IR on aeroplanes **exceeding 5700 kg** shall exercise the privileges **only on the type** on which the IR Flight Test was satisfactorily carried out.
- Notwithstanding *(in spite of)* the above, the holder of an IR on any **multi-engine** aeroplane **may exercise** the privileges of the Instrument Rating on **any single-engine aeroplane** entered in the aircraft rating of his licence.

Exam Tip: The Instrument Rating is valid **12 months**; for an IR an applicant needs **50 hours of cross-country as PIC** within the 100-hr PIC block.

Quick-Revision Comparison Charts Synthesised from

Table 11.1 — Licence Snapshot: Age, Schooling, Medical & Headline Hours

Licence	Min. Age	Education	Medical Class	Headline Experience
SPL	16 yrs	Class 10	Class 2	10 hrs solo before cross-country
PPL(A)	17 yrs	Class 10	Class 2 (Class 1 if IR privileges)	40 hrs total, incl. 20 hrs solo
CPL(A)	18 yrs	10+2 Physics & Maths	Class 1	200 hrs total, incl. 100 hrs PIC
CPL(A) + IR	18 yrs	10+2 Physics & Maths	Class 1	200 hrs total, incl. 50 hrs instrument
ATPL(A)	21 yrs	10+2 Physics & Maths	Class 1	1500 hrs total, incl. 500 hrs XC
Instrument Rating(A)	—	—	Class 1	100 hrs PIC + 40 hrs instrument

Table 11.2 — Renewal at a Glance

Licence / Rating	Renewal Requirement (key figures)
SPL	Renewed in 12-month steps from fresh medical; total validity capped at 24 months from issue.
PPL(A)	Medical + 5 hrs PIC in preceding 12 months, or flying test in preceding 6 months.
CPL(A) / CPL+IR / ATPL(A)	Medical + 10 hrs PIC in preceding 6 months (50% co-pilot time counts), or flying test by day & night + current FRTOL (ATPL also needs current multi-engine IR).
Instrument Rating	Valid 12 months; renewed 12 months at a time if flight test done within 2 months before expiry.
Proficiency Check (>5700 kg)	Valid 6 months; renewed 6 months at a time if check done within 2 months before expiry.

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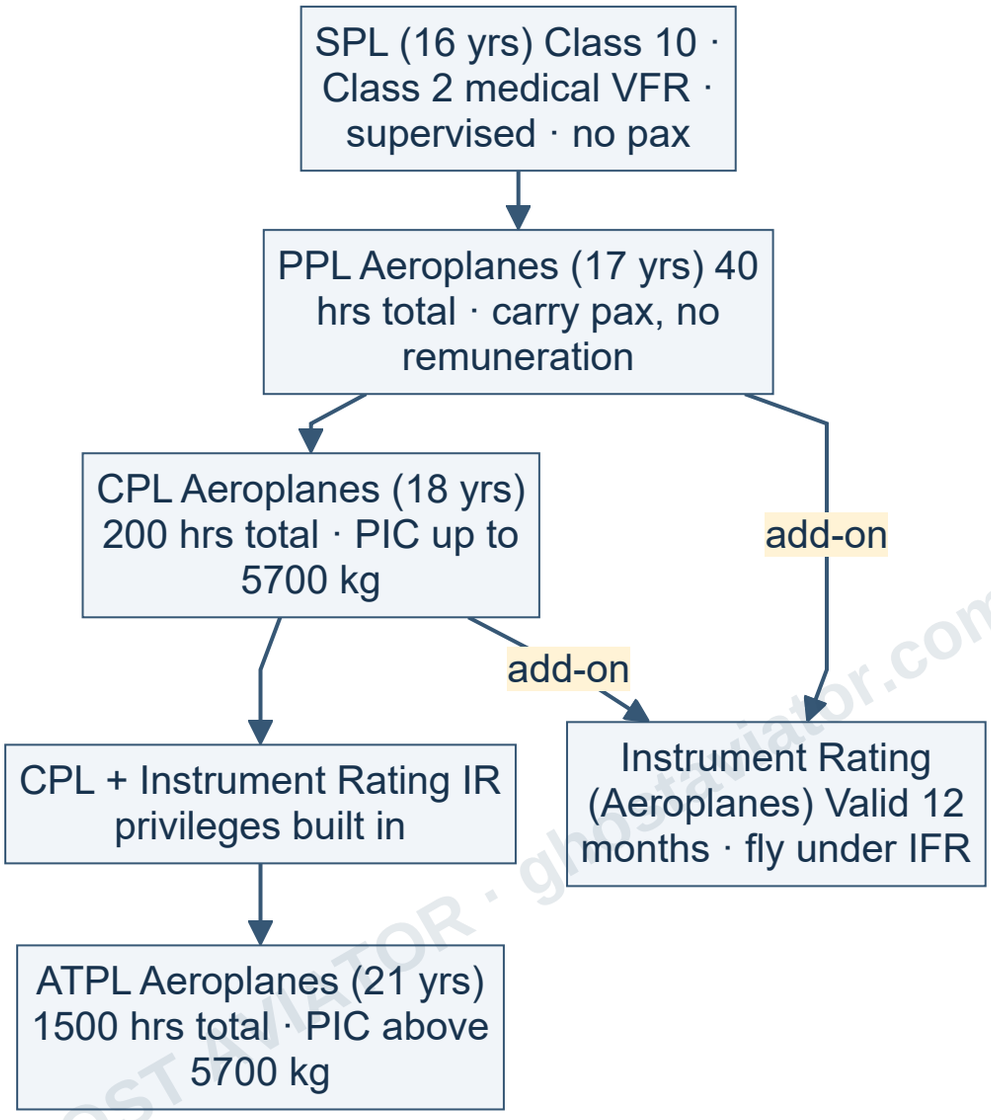

Figure 14.5 *The Pilot Licensing Career Ladder*

PPL Experience XC flight: 150 NM 2 full-stop landings

CPL Experience XC flight: 300 NM 2 full-stop landings

CPL Skill — XC by Day 250 NM · 1 full-stop landing

CPL Skill — XC by Night 120 NM · return, no landing elsewhere

Figure 14.6 *Cross-Country Test Distances*

Capt. Pankaj Pahil

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CHAPTER 15

Airworthiness of Aircraft

15

8	ICAO Annex — Airworthiness
6 (I)	Complementary Annex
5 700 kg	MCTOM Applicability
4	Parts of Annex 8

These notes break down every clause of the chapter into plain instructional language for student pilots — while keeping **every exact regulatory value, limit and rule** unchanged. Colour-coded boxes are used throughout: Blue = definitions/concepts, Green = procedures & allowances, Red = strict limitations & prohibitions, Amber = instructor & exam tips.

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Introduction — Airworthiness & the Certificate of Airworthiness

The whole chapter sits on a single safety idea: an aircraft is only allowed to fly if it has been **designed, constructed and operated** in line with the proper airworthiness requirements. The authority that sets and enforces those requirements for a particular aircraft is the **State of Registry** of that aircraft.

KEY DEFINITIONS

- **Airworthiness** — the condition of an aircraft being suitable and fit to fly safely.
- **Certificate of Airworthiness (C of A)** — the document issued to an aircraft **declaring that the aircraft is fit to fly**. It is the formal "fitness certificate" of the machine.
- **State of Registry** — the State on whose register the aircraft is entered; it owns the responsibility for that aircraft's airworthiness requirements.

In simple terms: **compliance with airworthiness requirements ⟶ the aircraft earns a C of A ⟶ the C of A declares it fit to fly.** Without that declaration, the machine is not permitted in the air.

INSTRUCTOR'S TIP

Remember the logic chain — *Design & Build correctly → Operate per requirements → Issue C of A → "Fit to fly"*. The C of A is about the **individual aircraft**; keep it separate in your mind from the **Type Certificate** (covered in Section 6), which is about the **design of the whole type**.

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Compatibility — Annex 8 vs Annex 6, Part I

Airworthiness does not live in isolation. The Standards of **Annex 6, Part I** — which deal with **aeroplane performance operating limitations** — are described as **complementary** to the airworthiness Standards of **Annex 8**. Both Annexes state **broad objectives**, but they look at the aircraft from two different angles.

> **TWO ANGLES ON THE SAME AIRCRAFT**
>
> **Annex 8** deals with airworthiness from the **engineering point of view**. **Annex 6 (Part I)** covers the subject from the **operational and safety point of view**. Together they form one complete picture.

Aspect	ICAO Annex 8	ICAO Annex 6, Part I
Subject	Airworthiness of Aircraft	Operation of Aircraft (International Commercial Air Transport — Aeroplanes)
Point of view	Engineering point of view	Operational & safety point of view
Core content	Airworthiness design & construction Standards	Aeroplane performance operating limitations
Nature	Both state **broad objectives** • the two are **complementary** to each other	

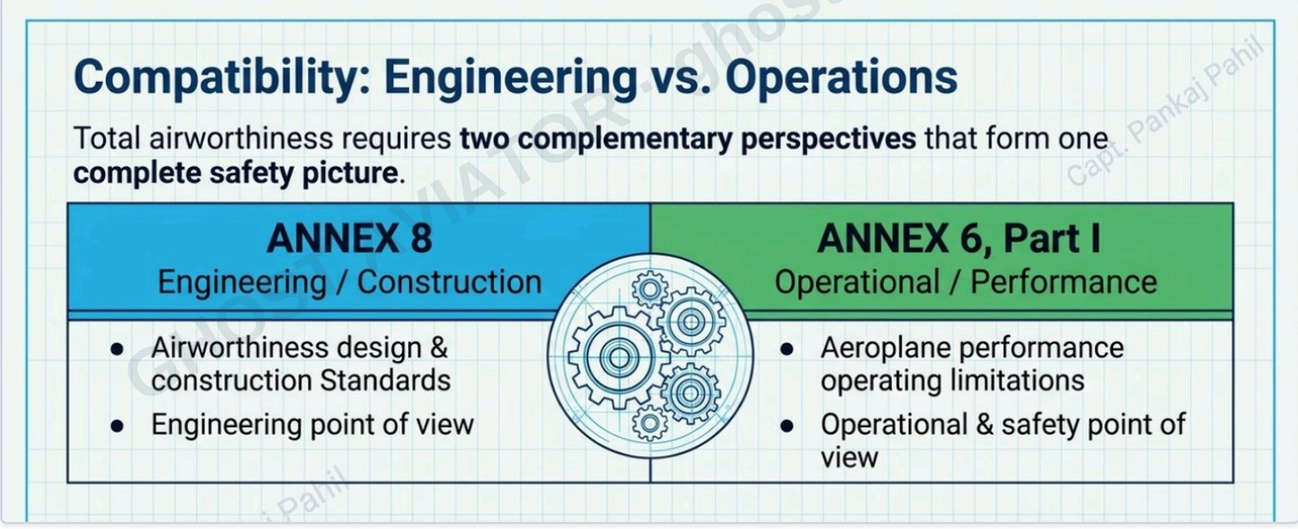

Figure 15.1 *Annex 8 looks at engineering; Annex 6 Part I looks at operations*

International Obligations & Recognition of a C of A

Protecting Visiting Aeroplanes

The ICAO **Council has urged Contracting States** not to impose on **visiting aeroplanes** operational requirements other than those established by the **State of Registry** — **provided** those State-of-Registry requirements are **not lower than the Standards of Annex 6, Part I**.

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WHAT THIS MEANS IN PRACTICE

A foreign aeroplane visiting another State should be judged by **its own State of Registry's rules** — not by a fresh set of local rules — as long as the State of Registry's rules at least meet the Annex 6, Part I floor. This avoids burdening international operations with duplicate requirements.

Recognising a Foreign Airworthiness Certificate

To make it easy to **import, export, lease, charter or interchange** aircraft and to support international air navigation, ICAO places a duty on the **State of Registry to recognise and render valid** an airworthiness certificate issued by another Contracting State.

THE STRICT CONDITION FOR RECOGNITION

A foreign C of A is recognised **only on the condition** that the airworthiness requirements under which it was issued or rendered valid are **equal to or above the minimum standards** that ICAO may establish from time to time. If the standards fall **below** the ICAO minimum, recognition does not follow automatically.

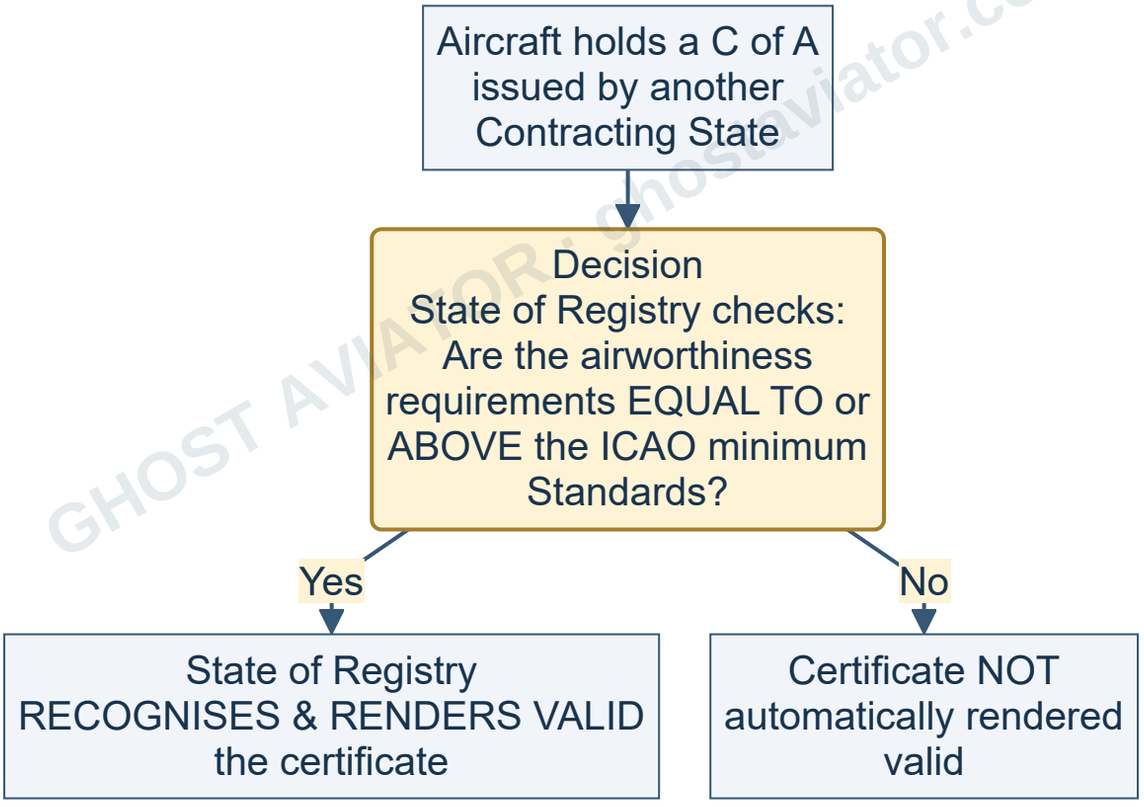

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Figure 15.2 *Recognition of a Foreign Certificate of Airworthiness*

National Codes vs ICAO Standards

ICAO is realistic about its own limits. It is **recognised that ICAO Standards would not replace national regulations.** ICAO Standards set the broad framework; the fine, detailed work of certifying individual aircraft needs a fuller, more detailed national code.

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WHY NATIONAL CODES ARE NEEDED

National codes of airworthiness — containing the **full scope and extent of detail** that an individual State considers necessary — are required as the **basis for the certification of individual aircraft**. ICAO's broad Standards alone are not detailed enough to certify a specific machine.

EACH STATE'S FREEDOM OF CHOICE

Every State is free to either —

1. **Develop** its own comprehensive and detailed code of airworthiness; **or**
2. **Select, adopt or accept** a comprehensive and detailed code already established by another Contracting State.

Scope of Annex 8 & the Airworthiness Technical Manual

Annex 8 includes **broad Standards** which define — for application by the **national airworthiness authorities** — the **minimum basis** for the recognition by States of Certificates of Airworthiness, so that aircraft of other States can fly **into and over their territories**.

PURPOSE OF ANNEX 8 STANDARDS

By setting that minimum basis, the Standards achieve — among other things — the **protection of other aircraft, third parties and property**.

WHERE THE DETAILED GUIDANCE LIVES

The **level of airworthiness** to be maintained by a national code is indicated by the broad Standards of Annex 8, **supplemented where necessary** by guidance material in ICAO's **Airworthiness Technical Manual — Doc 9760**.

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EXAM TIP — LOCK THE NUMBERS

Two reference labels are frequently tested: **Annex 8** = Airworthiness of Aircraft, and **Doc 9760** = Airworthiness Technical Manual. Memorise the document number exactly.

Type Certificate

While the C of A applies to an individual aircraft, the **Type Certificate** applies to the **design of a whole type** — and it is issued by a different authority.

TYPE CERTIFICATE — DEFINITION & ISSUING AUTHORITY

The **State of Design**, upon receipt of **satisfactory evidence** that the aircraft type (or **engine type** or **propeller type**, if certificated separately) complies with the design aspects of the appropriate airworthiness requirements, shall issue a **Type Certificate**. The Type Certificate:

- **Defines the type design**, and
- **Signifies its approval.**

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Certificate	Issued / Acted on by	What it does
Type Certificate	State of Design	Defines the type design of the aircraft / engine / propeller and signifies its approval — applies to the **design of the type**.
Certificate of Airworthiness	State of Registry (Director-General)	Declares an **individual aircraft** fit to fly / in a condition for safe operation.
Special Flight Permit	Director-General	May be issued, on application, to permit a flight while the C of A is suspended (see Section 9).

INSTRUCTOR'S TIP — DON'T CONFUSE THE "STATES"

State of Design ⟶ issues the **Type Certificate**. **State of Registry** ⟶ responsible for the **C of A** and continuing airworthiness. A common exam trap is swapping these two.

Applicability & Structure of Annex 8

Applicability of the Technical Standards

STRICT APPLICABILITY LIMIT — MEMORISE EXACTLY

At present, the technical Standards dealing with the **certification of aeroplanes** are **limited** to:

Multi-engined aeroplanes of OVER 5 700 kg maximum certificated take-off mass.

Note the precise wording: **"over"** 5 700 kg (not "up to"), **"multi-engined"**, and the mass is the **maximum certificated take-off mass**.

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These technical Standards include requirements relating to:

- **Performance** and **flying qualities**
- **Structural design and construction**
- **Engine and propeller design and installation**
- **Systems and equipment design and installation**, and operating requirements

The Four Parts of Annex 8

Annex 8 is divided into **four parts**:

Part	Content
Part I	Definitions
Part II	Procedures for **certification** and **continuing airworthiness** of aircraft
Part III	Technical requirements for the certification of **new large aeroplane designs**
Part IV	Helicopters

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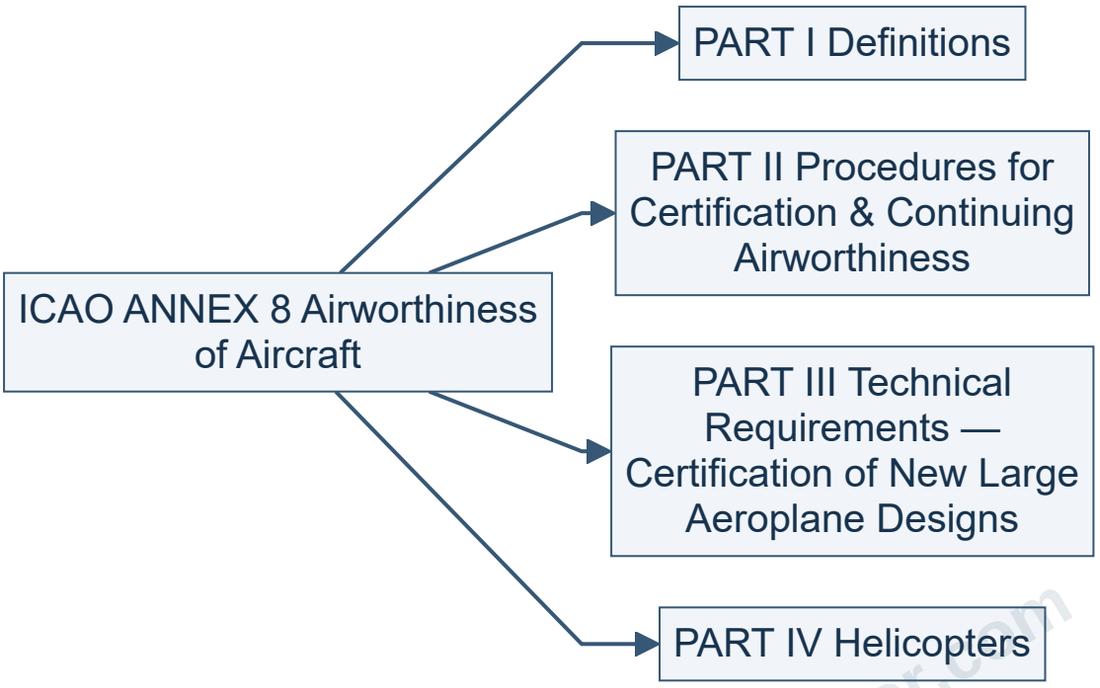

Figure 15.3 *Structure of ICAO Annex 8*

Special Security Features

Following the events of **hijacking and terrorist acts** on board aircraft, **special security features** have been built into aircraft design to improve the protection of the aircraft.

SECURITY FEATURES BUILT INTO DESIGN

These features include:

- Special features in **aircraft systems**;
- Identification of a **least-risk bomb location** on the aircraft;
- **Strengthening of the cockpit door**, and of the **ceilings and floors** of the cabin crew compartment.

National Provisions (DGCA / Indian Aircraft Rules)

This part of the chapter moves from the **international** ICAO framework to the **national** rules administered by the **Director-General** (DGCA). It covers how a C of A is issued, modified, suspended, cancelled and revalidated.

Issue of C of A / Special C of A / Airworthiness Review Certificate

WHO APPLIES, AND TO WHOM

The **owner or operator** of an aircraft may apply to the **Director-General** for —

- the issue of a **certificate of airworthiness** or a **special certificate of airworthiness** for the aircraft; **or**
- the **validation** of a certificate of airworthiness issued **elsewhere** for that aircraft.

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THE TWO CONDITIONS FOR ISSUE

The Director-General may issue a C of A or special C of A in respect of an aircraft **when**:

- **(a)** the applicant **furnishes the documents or other evidence** relating to the airworthiness of the aircraft, **as may be specified by the Director-General**; **and**
- **(b)** the Director-General is **satisfied** that the aircraft is **airworthy or in a condition for safe operation**.

Both conditions (a) *and* (b) must be met — they are joined by "and", not "or".

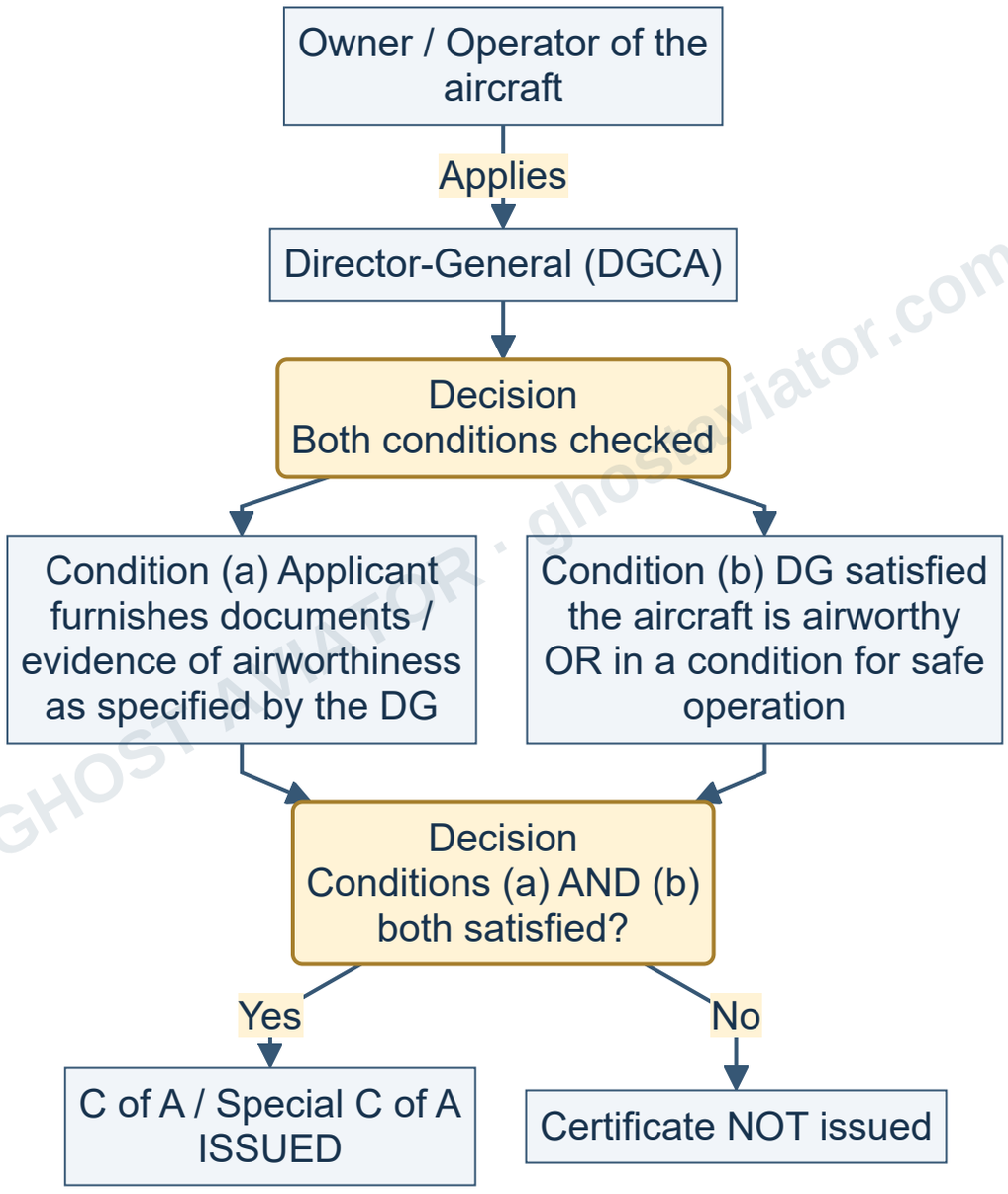

Figure 15.4 *Process for Issue of a Certificate of Airworthiness*

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Modification and Repairs

PROHIBITION — MODIFICATION / REPAIR AFFECTING SAFETY

A person **shall NOT carry out any modification or repair affecting the safety** of any aircraft that holds a valid certificate of airworthiness — **unless**:

- he has been **required to do so** in pursuance of these rules; **or**
- he has obtained the **prior approval of the Director-General**.

INSTRUCTOR'S TIP

The key trigger phrase is **"affecting safety"**. Such a modification or repair is permitted in only two situations — when the rules *require* it, or when the DG has given *prior* approval. "Prior" means **before** the work, not after.

Suspension or Cancellation — When a C of A is "Deemed Suspended"

A C of A or special C of A is **deemed to be suspended** automatically when an aircraft falls into any one of the following four situations:

FOUR GROUNDS FOR DEEMED SUSPENSION

- **(a) Ceases or fails to conform** with the requirement of the rules — in respect of **operation, maintenance, modification, repair, replacement, overhaul, process or inspection** applicable to that aircraft; **or**
- **(b)** Is **modified or repaired otherwise than in accordance** with the provisions of the rules; **or**
- **(c) Suffers major damage**; **or**
- **(d) Develops a major defect** which would affect the safety of the aircraft or its occupants in **subsequent flights**.

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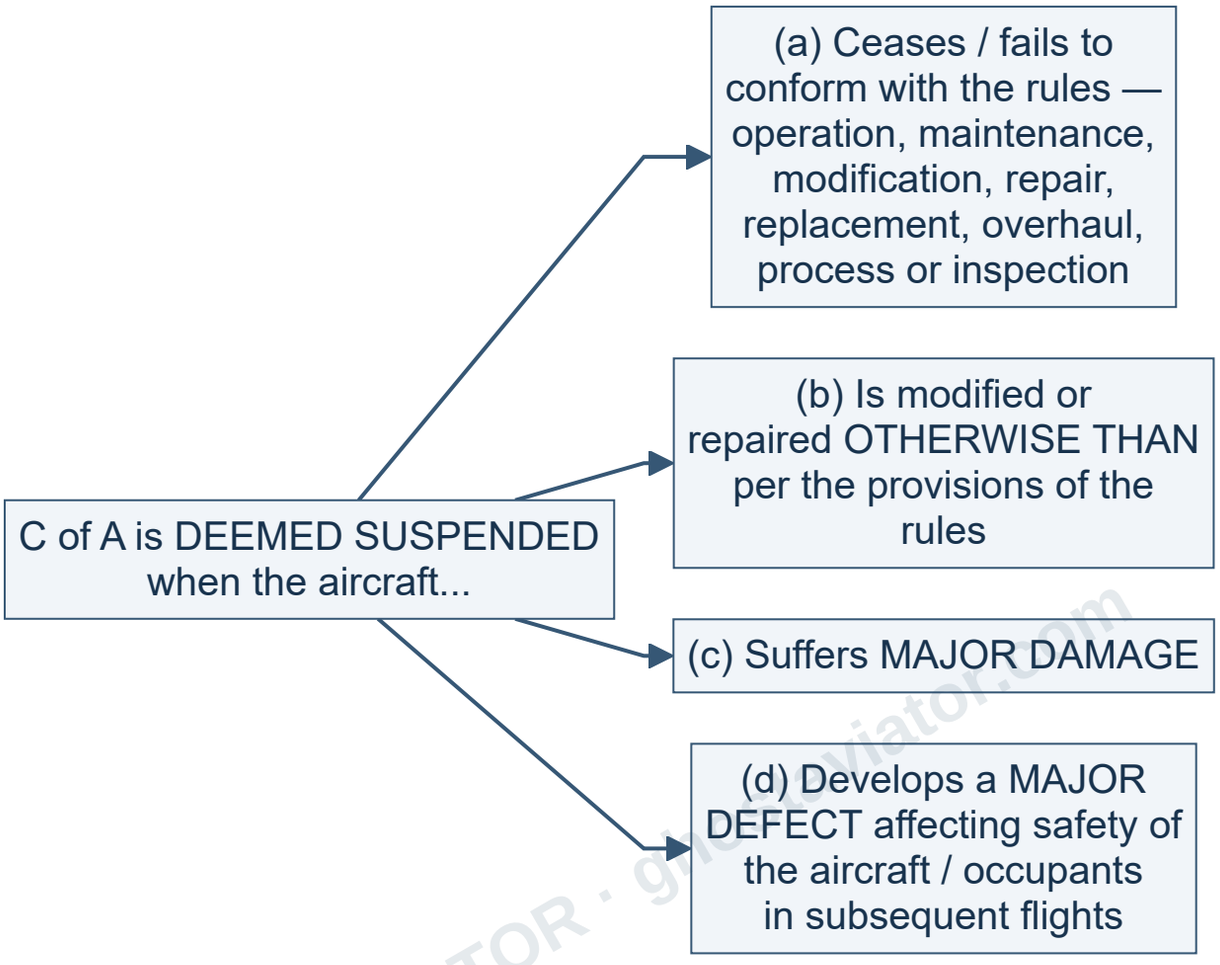

Figure 15.5 *Four Grounds on Which a C of A is Deemed Suspended*

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Director-General's Powers on "Reasonable Doubt"

If, **at any time**, the Director-General is satisfied that **reasonable doubt exists** as to the safety of an aircraft, or as to the safety of the **type** to which that aircraft belongs, he may take either of the following actions:

THE DG'S TWO OPTIONS

- **(a) Suspend or cancel** the C of A or special C of A in respect of the aircraft; **or**
- **(b) Require** the aircraft, an aircraft component, or an item of equipment to undergo **modification, repair, replacement, overhaul, inspection** — including **flight tests and examination** — under the supervision of an **approved person** as the DG may specify, as a **condition of the C of A remaining in force**.

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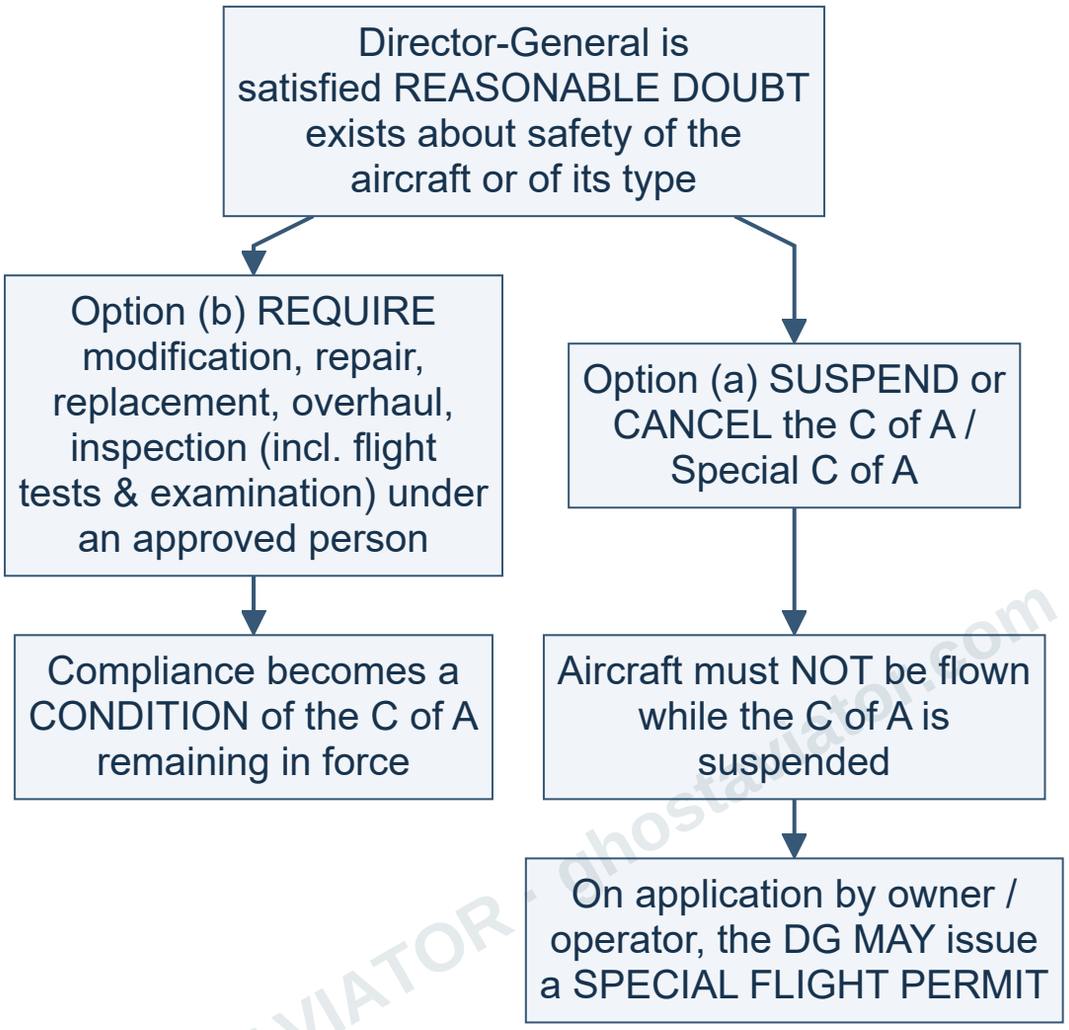

Figure 15.6 *DG's Action on Reasonable Doubt & the Special Flight Permit*

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Prohibition on Flying & the Special Flight Permit

ABSOLUTE PROHIBITION

An aircraft **shall NOT be flown** during any period for which its C of A or special C of A is **suspended or deemed to be suspended**.

THE EXCEPTION — SPECIAL FLIGHT PERMIT

Where the C of A or special C of A is suspended (or deemed suspended), the Director-General **may**, **upon an application by the owner or operator**, issue a **special flight permit**. This is the lawful route to move/fly the aircraft despite the suspension.

EXAM TIP — "MAY" VS "SHALL"

The aircraft **"shall not be flown"** — that is absolute. But the DG **"may"** issue a special flight permit — that is discretionary, and only on the owner/operator's **application**. Watch these modal words carefully in exam questions.

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Flight Manual

ENDORSEMENT OF THE C OF A

Where a **flight manual** is required to be kept for an aircraft in accordance with the provisions of these rules, the **Director-General shall endorse** ***(an official entry added to a licence or certificate)*** the certificate of airworthiness of the aircraft **accordingly**.

Instruments and Equipment

FITMENT REQUIREMENT

Every aircraft **shall be fitted and equipped** with the **instruments and equipment** — **including radio apparatus and special equipment** — as may be specified **according to the use and the circumstances** under which the flight is to be conducted.

INSTRUCTOR'S TIP

Note that the required instruments/equipment are **not fixed** — they depend on the **type of operation** and the **circumstances of the flight** (e.g. day/night, VFR/IFR). The fitment list scales with how and where the aircraft is used.

Self-Test All five chapter questions, the correct option, and the reasoning behind each answer.

Rapid Revision Snapshot

NUMBERS & LABELS TO MEMORISE

- **Annex 8** — Airworthiness of Aircraft
- **Annex 6, Part I** — complementary; operating limitations
- **Doc 9760** — Airworthiness Technical Manual
- **5 700 kg** — "over" this MCTOM, multi-engined aeroplanes
- **4 Parts** — Definitions / Certification & Continuing Airworthiness / New Large Aeroplane Designs / Helicopters

WHO DOES WHAT

- **State of Design** → issues the **Type Certificate**
- **State of Registry** → C of A & continuing airworthiness
- **Owner / Operator** → applies for C of A / special C of A / validation / special flight permit
- **Director-General** → issues, endorses, suspends, cancels; may issue a special flight permit

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HARD LIMITS & PROHIBITIONS — NEVER FORGET

- No modification/repair **affecting safety** on a C of A aircraft — unless **required by rules** or with **prior DG approval**.
- Aircraft **shall NOT be flown** while its C of A is suspended or deemed suspended.
- Foreign C of A recognised **only if** standards are **equal to or above** the ICAO minimum.
- C of A **deemed suspended** on 4 grounds: non-conformity • improper modification/repair • **major damage** • **major defect** affecting subsequent flights.

Starting point / condition	Branch	Next step / result
Design & build per airworthiness requirements	→	Type Certificate (State of Design)
Type Certificate (State of Design)	→	Owner/Operator applies to the Director-General
Owner/Operator applies to the Director-General	→	C of A ISSUED — aircraft 'fit to fly'
C of A ISSUED — aircraft 'fit to fly'	→	Aircraft stays in compliance?
Aircraft stays in compliance?	Yes	C of A remains valid (continuing airworthiness)
Aircraft stays in compliance?	No / damage / defect / doubt	C of A suspended, deemed suspended or cancelled
C of A suspended, deemed suspended or cancelled	→	Aircraft must not fly — DG may issue a Special Flight Permit

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Figure 15.7 *The Life-Cycle Logic of a Certificate of Airworthiness*

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CHAPTER 16

Operational Procedures

16

This chapter explains how an airline operation is run safely and legally — who may sit where, what paperwork must fly with you, how much fuel and oxygen you must carry, what survival gear is mandatory, and how long a crew may legally work before fatigue makes them unsafe. Every number here is examinable: learn the exact figures.

Section 01

ICAO Annex 6 & Applicability

ICAO **Annex 6** is titled "*Operation of Aircraft*". It contains the international **Standards and Recommended Practices (SARPs)** that tell operators how aircraft must be flown safely. It is split into three Parts, and each Part became applicable on a different date.

DEFINITION — THE THREE PARTS OF ANNEX 6

- **Part I — International Commercial Air Transport — Aeroplanes.** Applies to the operation of aeroplanes by operators authorised to conduct **international commercial air transport operations**.
- **Part II — International General Aviation — Aeroplanes.** Deals exclusively with international general aviation. Became applicable in **September 1969**.
- **Part III — International Operations — Helicopters.** Deals with all international helicopter operations. Became applicable in **November 1986**.

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Figure 16.1 *Structure of ICAO Annex 6*

Section 02

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General Requirements & Carriage of Persons

Quality System

Every operator must run a structured, self-checking organisation rather than relying on chance.

STANDARD REQUIREMENT

An operator shall establish **one quality system** and designate **one quality manager** to monitor compliance with — and the adequacy of — procedures required to ensure safe operational practices and airworthy aeroplanes.

Compliance monitoring must include a **feed-back system to the accountable manager** to ensure corrective action is taken as necessary.

IN PLAIN LANGUAGE

"One system, one manager, one feedback loop." The quality manager *watches* the operation; the accountable manager has the *authority and money* to fix what the quality manager finds. Without the feedback loop, problems would never reach the person who can correct them.

Methods of Carriage of Persons

People must not be carried in unsafe parts of the aircraft.

STRICT LIMITATION

An operator shall take all reasonable measures to ensure that **no person is in any part of an aeroplane in flight which is not designed for the accommodation of persons**, *unless temporary access has been granted by the commander* to any part of the aeroplane in the following cases:

- For the purpose of taking action necessary for the **safety** of the aeroplane, or of any person, animal or goods therein; **or**
- A part in which **cargo or stores are carried**, being a part designed to enable a person to have access thereto while the aeroplane is in flight.

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Admission to the Flight Deck

The flight deck is a restricted area. Only specific categories of person may be admitted.

STRICT LIMITATION — FLIGHT DECK ACCESS

An operator must ensure that **no person, other than a flight crew member assigned to a flight**, is admitted to or carried in the flight deck unless that person is:

- An **operating crew member**;
- A **representative of the Authority** responsible for certification, licensing or inspection, if this is required for the performance of his official duties; **or**
- Permitted by, and carried in accordance with, instructions contained in the **Operations Manual**.

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STANDARD PROCEDURE — WHO DECIDES

The **final decision** regarding admission to the flight deck shall be the responsibility of the **commander**.

Unauthorized Carriage

STRICT LIMITATION

An operator shall take all reasonable measures to ensure that **no person secretes himself, or secretes cargo, on board an aeroplane** (i.e. stowaways and hidden/undeclared cargo are prohibited).

Portable Electronic Devices (PEDs)

STRICT LIMITATION

An operator shall **not permit any person to use**, and shall take all reasonable measures to ensure that no person does use, on board an aeroplane, a **portable electronic device that can adversely affect the performance of the aeroplane's systems and equipment**.

WHY IT MATTERS

The rule does not ban every device outright — it bans devices that *can adversely affect* aircraft systems. The concern is electromagnetic interference with navigation and communication equipment, especially during critical phases of flight.

Section 03

Documents to be Carried on Board

No person in charge of any aircraft shall allow that aircraft to be flown unless the following valid documents — as applicable — are carried on board, in **original or attested copies**.

DEFINITION — THE MANDATORY DOCUMENT SET

1. **Certificate of Registration**
2. **Certificate of Airworthiness (C of A)**
3. **Airworthiness Review Certificate (ARC)**
4. A document attesting **Noise Certification** of the aeroplane / helicopter
5. **Air Operator's Permit (AOP)**
6. **Appropriate Licences** for each member of the flight crew
7. **Aeromobile Radio operation Licence** for the Radio Communication apparatus
8. **Journey Log Book** or equivalent documents approved by the DGCA
9. **Operations Manual**
10. **Minimum Equipment List (MEL)**
11. **Flight Manual**
12. **Cabin Crew Manual**
13. **Cockpit and Emergency Check List** — unless these form part of the Flight Manual carried on board

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14. **Aeroplane / Helicopter search procedure checklist**
15. **Maintenance Release / Certificate to Release to Service (CRS)**
16. **LOPA** — Layout of Passenger Arrangement
17. **Emergency and Safety Equipment Layout**
18. **Route guides**
19. **Current and suitable navigation charts / maps** for the planned route and all other routes along which the flight may reasonably be expected to be diverted
20. **Weight Schedule**
21. **Load and Trim Sheet**
22. If carrying passengers — a **list of passenger names** and their places of embarkation and destination
23. If carrying cargo — a **cargo manifest** and detailed declarations of the cargo
24. If carrying dangerous goods — a **list of such goods** (this list must be specifically brought to the notice of the Pilot-in-Command before the flight)
25. **Current comprehensive insurance policy** covering passengers and their baggage, crew, third party risks and hull loss

STANDARD PROCEDURE — CHECKLIST DISPLAY

Checklists for the **take-off, cruise and landing phases** shall be **displayed in the cockpit**, unless the lists form a part of the Flight Manual carried on board.

STANDARD PROCEDURE — PASSENGER SAFETY BRIEFING CARD

All operators of passenger aircraft shall provide a **passenger safety information briefing card at every passenger seat location**. Such a card shall make use of **symbology** to convey the instructions in a clear and concise manner.

EXAM FOCUS

Two favourite exam traps: (1) the **dangerous goods list** must be *personally shown to the PIC before flight* — not merely carried; (2) the Flight Manual is carried by **all aircraft** only *if the C of A states so* — the C of A determines the requirement.

Section 04

Information Retained on Ground & Power to Inspect

Information Retained on Ground

So that an investigation is always possible, key flight information must be safely preserved off the aircraft.

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STANDARD REQUIREMENT

An operator shall ensure that — **at least for the duration of each flight or series of flights**:

- Information relevant to the flight and appropriate for the type of operation is **preserved on the ground**; and
- The information is **retained until it has been duplicated** at the place at which it will be stored in accordance with the rules; **or**, if this is impracticable,
- The same information is carried in a **fireproof container** in the aeroplane.

IN PLAIN LANGUAGE

The aircraft can be lost — so a ground copy must exist. Until that ground copy is safely duplicated at the storage location, you must keep a protected copy. If keeping a ground copy is genuinely impracticable, the on-board copy must be inside a **fireproof container** so it can survive an accident.

Power to Inspect

STANDARD REQUIREMENT

An operator shall ensure that any person **authorised by the Authority** is permitted **at any time** to board and fly in any aeroplane operated under an AOC issued by that Authority, and to **enter and remain on the flight deck**.

THE COMMANDER'S OVERRIDE

The commander **may refuse access to the flight deck** if, in his opinion, the safety of the aeroplane would thereby be endangered. Safety always outranks the inspection right.

Section 05

Operator Certification (AOC)

Before an airline may sell tickets, it must prove to the State that it is capable of operating safely. That proof is the **Air Operator Certificate**.

DEFINITION — AIR OPERATOR PERMIT / CERTIFICATE (AOC)

A **Permit / Certificate authorising an operator to carry out specified commercial air transport operations**.

DEFINITION — STATE OF THE OPERATOR

The State in which the operator's **principal place of business** is located or, if there is no such place of business, the operator's **permanent residence**.

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Mutual Recognition

STANDARD REQUIREMENT

Contracting States shall **recognise as valid an AOC issued by another Contracting State**, provided that *(an exception or condition added to a rule)* the requirements under which the certificate was issued are **at least equal to the applicable Standards specified in Annex 6**.

Issue of an AOC

STANDARD REQUIREMENT — CONDITIONS OF ISSUE

The issuance of an AOC is **dependent upon the operator demonstrating** to the State that the following are **adequate** for the nature and extent of operations to be conducted:

- Its **organisation**;
- Its **training policy and programmes**;
- Its **flight operations**;
- Its **ground handling** arrangements; and
- Its **maintenance** arrangements.

The certification process involves the State's evaluation of each operator and a determination that the operator is capable of conducting **safe operations** before initial issue of an AOC, or before the addition of any subsequent authorizations to an AOC.

Validity of an AOC

CONTINUING OBLIGATION

The validity *(the period during which a document is accepted)* of an AOC shall depend upon the operator **maintaining the original certification standards** under the supervision of the State of the Operator. This supervision requires that a system of **continued surveillance** be established to ensure the required standards of operations are maintained.

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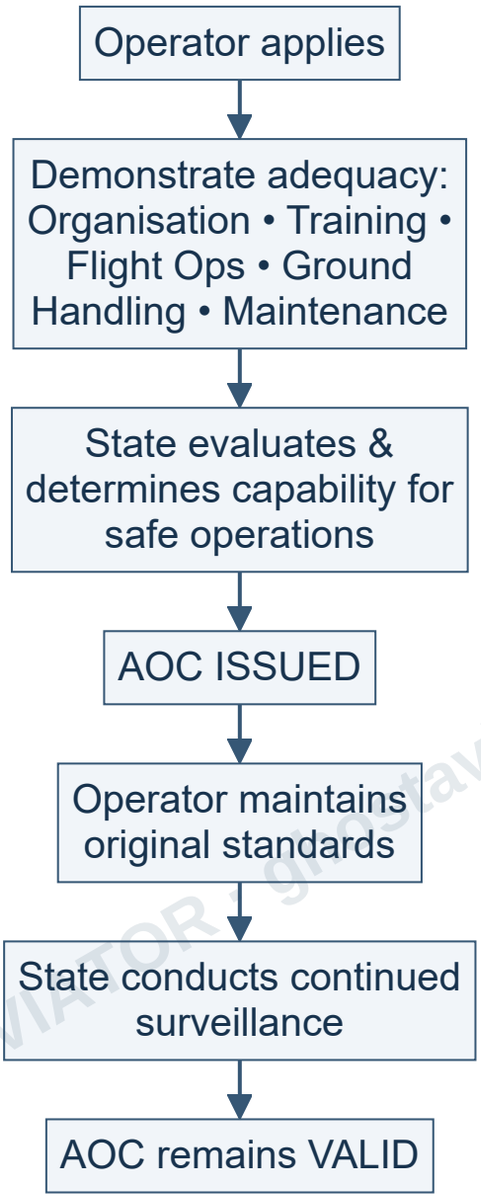

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Figure 16.2 *AOC — From Application to Continued Validity*

Section 06

Aerodrome Operating Minima & Approach Ban

Restricted Aerodrome Operating Minima (AOM)

When an aerodrome's normal minima are made stricter, "additives" are applied to the published Normal AOM.

STRICT LIMITATION — THE ADDITIVE FORMULA

Restricted Aerodrome Operating Minima shall be based on additives applied to the Normal AOM as below:

Restricted AOM = Normal AOM DA(H)/MDA(H) **+ 100 ft** **AND** Normal AOM Visibility/RVR **+ 400 m**

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Commencement & Continuation of Approach — The "Approach Ban Policy"

STRICT LIMITATION — COMMENCING AN APPROACH

The PIC shall **not commence an instrument approach** if the reported RVR/Visibility is **below the applicable minimum**.

STRICT LIMITATION — DISCONTINUING AN APPROACH

If, *after commencing* an instrument approach, the reported RVR/Visibility falls below the applicable minimum, the approach **shall not be continued**:

- (a) below **1 000 ft** above the aerodrome; **or**
- (b) into the **final approach segment (FAS)**.

STANDARD PROCEDURE — ONCE PAST THE GATE

Where RVR is not available, RVR values may be derived by **converting the reported visibility**.

If, **after entering the final approach segment** or descending below **1 000 ft** above aerodrome elevation, the reported RVR/visibility falls below the applicable minimum — the approach **may be continued to DA/H or MDA/H**.

The approach may be continued **below DA/H or MDA/H**, and the landing may be completed, **provided that the required visual reference is established at the DA/H or MDA/H and is maintained**.

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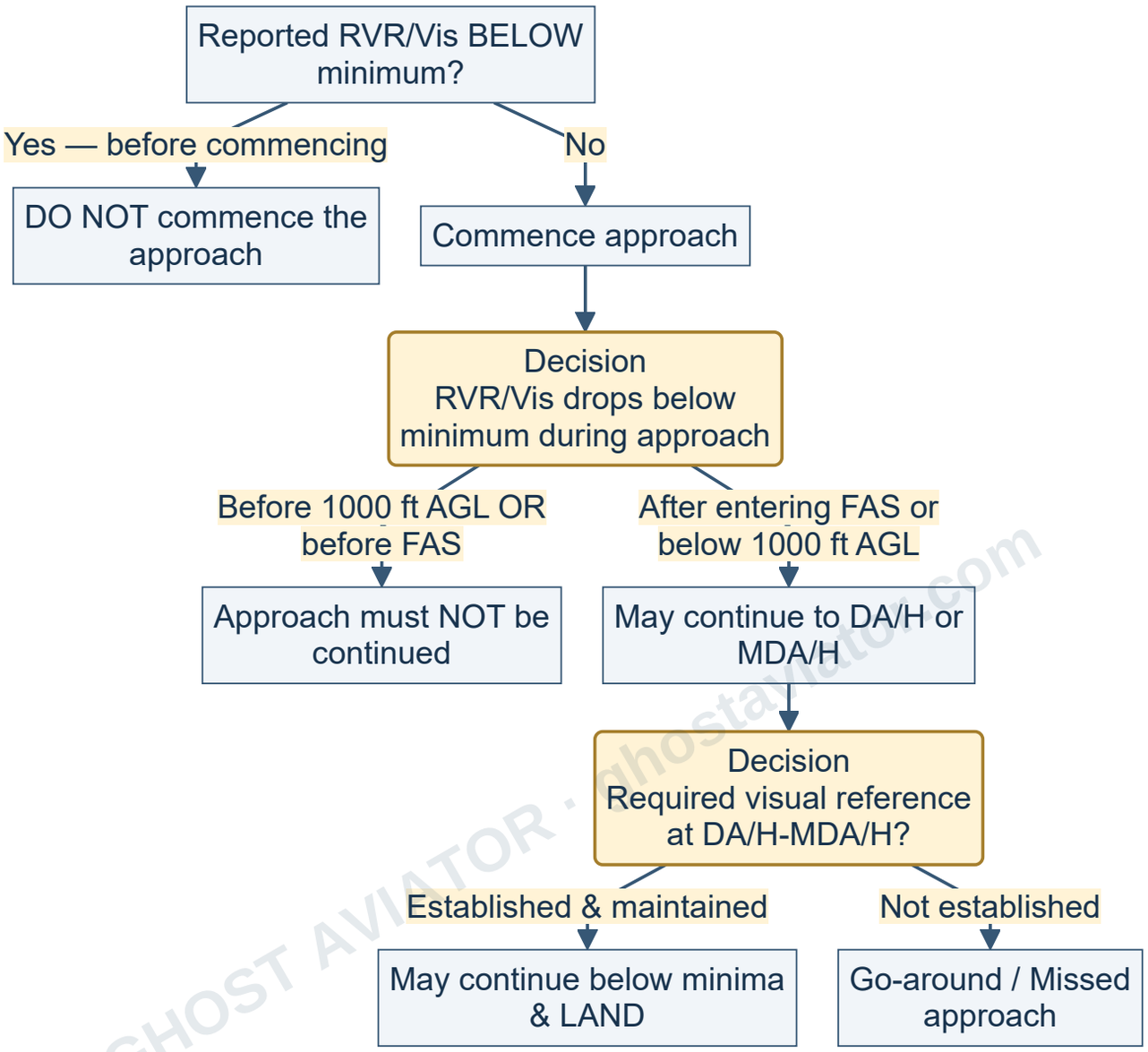

Figure 16.3 *Approach Ban Decision Logic*

Controlling RVR — Touch-down, Mid-point & Stop-end

DEFINITION — WHICH RVR CONTROLS?

The **touch-down zone (TDZ) RVR is always controlling.** If reported and relevant, the **mid-point and stop-end RVR are also controlling**.

Table 6.1 — Minimum RVR Values by Runway Segment

Runway Segment	Minimum RVR Value	Special Case
Touch-down zone (TDZ)	Always controlling	The primary reference
Mid-point	125 m	...or the RVR required for the TDZ if that is less. With a stop-end (roll-out) guidance or control system, the mid-point minimum is **50 m**.

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Runway Segment	Minimum RVR Value	Special Case
Stop-end	50 m	Controlling when reported and relevant

DEFINITION — "RELEVANT"

In this context, **"relevant"** means that part of the runway used during the **high-speed phase of the landing down to a speed of approximately 60 knots**.

WORKED EXAMPLE

Filed minima 300 m; TDZ RVR 400 m; mid-point RVR 100 m; stop-end RVR 300 m. The mid-point reads only 100 m, which is *below the 125 m minimum* — so the runway is **below minima**. A single segment failing the minimum makes the whole runway unusable.

VFR Operating Minima

STRICT LIMITATION

An operator is to ensure that VFR flights are conducted in accordance with the **Visual Flight Rules and in VMC**. **Special VFR flights are NOT permitted** for commercial air transport aeroplanes.

Section 07

Instrument & Safety Equipment

Circuit Protection Devices

STRICT LIMITATION — SPARE FUSES

An operator shall not operate an aeroplane in which **fuses are used** unless spare fuses are available for use in flight equal to **at least 10% of the number of fuses of each rating**, **OR three of each rating — whichever is the greater**.

Windshield Wipers

STRICT LIMITATION

An operator shall not operate an aeroplane with a maximum certificated take-off mass of **more than 5,700 kg** unless it is equipped at **each pilot station** with a windshield wiper or equivalent means to maintain a clear portion of the windshield during precipitation.

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Airborne Weather Radar Equipment

STANDARD REQUIREMENT

Pressurised aeroplanes carrying passengers should be equipped with **operative weather radar** whenever such aeroplanes are operated in areas where thunderstorms or other potentially hazardous weather conditions — regarded as detectable with airborne weather radar — may be expected to exist along the route, either **at night** or under **Instrument Meteorological Conditions (IMC)**.

Pressure Altitude Data Source

STRICT REQUIREMENT

All aeroplanes shall be equipped with a data source that provides **pressure altitude information with a resolution of 7.62 m (25 ft) or better**.

Microphones

STANDARD PROCEDURE

All flight crew members required to be on flight deck duty shall communicate through **boom or throat microphones below the transition level / altitude**.

Wind Shear Warning System & GPWS (Turbo-jet Aeroplanes)

STANDARD REQUIREMENT

All **turbo-jet aeroplanes** of a maximum certificated take-off mass **in excess of 5,700 kg**, **OR authorized to carry more than 9 passengers**, should be equipped with:

- A **forward-looking wind shear warning system**; and
- A **Ground Proximity Warning System (GPWS)** which has a forward-looking terrain avoidance function.

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ACAS II

STRICT REQUIREMENT

All **turbine-engined aeroplanes** of a maximum certificated take-off mass **in excess of 5,700 kg**, **OR authorized to carry more than 19 passengers**, shall be equipped with an **Airborne Collision Avoidance System (ACAS II)**.

EXAM TRAP — DON'T CONFUSE THE TWO

Both thresholds use **5,700 kg** — but the passenger figure differs. **Wind shear/GPWS** = turbo-jet, >**9** pax. **ACAS II** = turbine-engined, >**19** pax. Remember: *collision avoidance needs more seats (19) before it bites.*

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Security of the Flight Crew Compartment

STRICT REQUIREMENT

All **passenger-carrying aeroplanes** of a maximum certificated take-off mass **in excess of 45,500 kg**, **OR with a passenger seating capacity greater than 60**, shall be equipped with an approved **flight crew compartment door** that is designed to:

- Resist penetration by **small arms fire and grenade shrapnel**; and
- Resist **forcible intrusions** by unauthorized persons.

This door shall be capable of being **locked and unlocked from either pilot's station**.

First Aid Kits

First-aid kits are carried for use by **cabin crew** in managing incidents of ill health. The number required scales with passengers carried.

Table 7.1 — First Aid Kits Required

Passengers	First Aid Kits
0 – 100	1
101 – 199	2
200 – 299	3
300 or more	4

EXAM TRAP

An aircraft carrying **250 passengers** falls in the 200–299 band — answer is **3 kits**. Watch the band boundaries carefully.

Emergency Medical Kit

STANDARD REQUIREMENT — MEDICAL SUPPLIES

Universal Precaution Kit: For aeroplanes required to carry cabin crew as part of the operating crew — **one universal precaution kit** (**two** for aeroplanes authorized to carry more than **250 passengers**) for use by cabin crew in managing incidents of ill health associated with a suspected communicable disease, or illness involving contact with body fluids.

Medical Kit: For aeroplanes authorized to carry **more than 100 passengers**, on a **sector length of more than 2 hours** — a medical kit for use by medical doctors or other qualified persons in treating in-flight medical emergencies.

Section 08

Flight Recorders

Flight recorders capture what the aircraft and crew did. They fall into two families — **crash-protected** recorders (heavy, survivable) and **lightweight** recorders.

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DEFINITION — CRASH-PROTECTED FLIGHT RECORDERS

Comprise **one or more** of the following systems:

- **FDR** — Flight Data Recorder
- **CVR** — Cockpit Voice Recorder
- **AIR** — Airborne Image Recorder
- **DLR** — Data Link Recorder

Image and data link information may be recorded on **either the CVR or the FDR**.

DEFINITION — LIGHTWEIGHT FLIGHT RECORDERS

Comprise **one or more** of the following systems:

- **ADRS** — Aircraft Data Recording System
- **CARS** — Cockpit Audio Recording System
- **AIRS** — Airborne Image Recording System
- **DLRS** — Data Link Recording System

Image and data link information may be recorded on **either the CARS or the ADRS**.

Combination Recorders

STRICT REQUIREMENT

All aeroplanes of a maximum certificated take-off mass of **over 15,000 kg**, which are required to be equipped with **both a CVR and an FDR**, shall be equipped with **two combination recorders (FDR/CVR)**.

One recorder shall be located **as close to the cockpit as practicable**, and the other recorder located **as far aft as practicable**.

WHY TWO, FAR APART?

Putting one recorder forward and one aft maximises the chance that at least one survives an accident — a fire or impact that destroys one location may spare the other.

Recording Duration Requirements

Table 8.1 — Minimum Recording Retention Periods

Recorder	Minimum Retention	Applicability
FDR (all)	Last 25 hours	All FDRs
CVR & **Data Link Recorders**	Last 2 hours	General requirement
CVRs / CARS	Last 2 hours	All CVRs/CARS retain at least last 2 hours
CVR (large new aircraft)	Last 25 hours	Aeroplanes of MCTOM **over 27,000 kg** for which the individual C of A is first issued **on or after 1 January 2022**

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STANDARD PROCEDURE — CVR / CARS OPERATION

The CVR or a Cockpit Audio Recording System (CARS) shall **start recording before the aircraft moves under its own power**, and shall **continuously record until the aircraft is no longer capable of moving under its own power**.

CVR/CARS shall **NOT use magnetic tape or wire** for recording.

EXAM FOCUS — THE 25 / 2 / 25 PATTERN

FDR = **25 h**. Standard CVR & data link = **2 h**. But a new CVR on a heavy aircraft (>27,000 kg, C of A on/after 1 Jan 2022) = **25 h**. A common question pairs "FDR 25, CVR 2, data link 2" — that is the correct combination.

Section 09

Oxygen Supply & Protective Breathing Equipment

As an aeroplane climbs, the air thins. Oxygen rules are written around **absolute pressure values**, which correspond to standard-atmosphere altitudes.

Table 9.1 — Pressure / Altitude Equivalents (Standard Atmosphere)

Absolute Pressure	Metres	Feet
700 hPa	3 000 m	10 000 ft
620 hPa	4 000 m	13 000 ft
376 hPa	7 600 m	25 000 ft

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Oxygen Supply — Flights at Altitude

STRICT LIMITATION — PRESSURE BELOW 700 HPA

A flight to be operated at a flight altitude at which the atmospheric pressure in personnel compartments will be **less than 700 hPa** shall **not be commenced** unless sufficient stored breathing oxygen is carried to supply:

- **(a) All crew members and 10% of passengers** — for any period in excess of **30 minutes** that the pressure in the compartment occupied by them will be **between 700 hPa and 620 hPa**; and
- **(b) The crew and passengers** — for any period that the atmospheric pressure in compartments occupied by them will be **less than 620 hPa**.

Oxygen Supply — Pressurised Aeroplanes

STRICT LIMITATION — LOSS OF PRESSURISATION

A flight to be operated with a **pressurised aeroplane** shall **not be commenced** unless a sufficient quantity of stored breathing oxygen is carried to supply all crew members and passengers — as appropriate to the circumstances of the flight — in the event of **loss of pressurisation**, for any period that the atmospheric pressure in any compartment occupied by them would be **less than 700 hPa**.

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In addition, when an aeroplane is operated at flight altitudes at which the atmospheric pressure is **less than 376 hPa** — OR which, if operated at altitudes where the pressure is more than 376 hPa, **cannot descend safely within 4 minutes** to a flight altitude where the atmospheric pressure equals 620 hPa — there shall be **no less than a 10-minute supply** for the occupants of the passenger compartment.

STANDARD REQUIREMENT — PRESSURISATION WARNING DEVICE

Pressurised aeroplanes intended to be operated at flight altitudes where the atmospheric pressure would be **less than 376 hPa** shall be equipped with a device to provide **positive warning to the flight crew of any dangerous loss of pressurisation**.

Crew Protective Breathing Equipment (PBE)

STRICT REQUIREMENT

An operator shall not operate a **pressurised aeroplane**, OR an **unpressurised aeroplane with a MCTOM exceeding 5,700 kg or having a maximum approved seating configuration of more than 19 seats**, unless:

1. It has equipment to protect the **eyes, nose and mouth** of each flight crew member while on flight deck duty, and to provide oxygen for a period of **not less than 15 minutes**. The supply for PBE may be provided by the supplemental oxygen required. In addition, when the flight crew is **more than one and a cabin crew member is not carried**, portable PBE must be carried to protect the eyes, nose and mouth of **one member of the flight crew** and to provide breathing gas for not less than **15 minutes**; **and**
2. It has **sufficient portable PBE** to protect the eyes, nose and mouth of **all required cabin crew members** and to provide breathing gas for a period of **not less than 15 minutes**.

Section 10

Emergency & Survival Equipment

Hand Fire Extinguishers

STANDARD REQUIREMENT

Portable fire extinguishers provided shall be of a type which, when discharged, **will not cause dangerous contamination of the air** within the aeroplane. At least one shall be located in:

- The **pilot's compartment**; and
- **Each passenger compartment** that is separate from the pilot's compartment and that is not readily accessible to the flight crew.

Any portable fire extinguisher fitted in accordance with the Certificate of Airworthiness of the aeroplane may count as one prescribed *(laid down by the authority or the rules)*.

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Crash Axes and Crowbars

STRICT REQUIREMENT

An operator shall not operate an aeroplane with a MCTOM **exceeding 5,700 kg**, OR having a maximum approved passenger seating configuration of **more than 9 seats**, unless it is equipped with **at least one crash axe or crowbar located on the flight deck**.

If the maximum approved passenger seating configuration is **more than 200**, an **additional** crash axe or crowbar must be carried and located in or near the **most rearward galley area**.

Marking of Break-in Points

STANDARD REQUIREMENT

Areas of the fuselage suitable for break-in by rescue crews in an emergency are marked on an aeroplane. The colour of the markings shall be **red or yellow**, and if necessary they shall be **outlined in white** to contrast with the background.

Means for Emergency Evacuation

STANDARD REQUIREMENT

An operator shall, **for each type of aeroplane**, assign to all flight crew members the necessary functions they are to perform in an emergency or in a situation requiring emergency evacuation. **Annual training** in accomplishing these functions shall be contained in the operator's training programme and shall include instruction in the use of all emergency and lifesaving equipment required to be carried, and drills in the emergency evacuation of the aeroplane.

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Megaphones

STRICT REQUIREMENT

An operator shall not operate an aeroplane with a maximum approved passenger seating configuration of **more than 60** and carrying one or more passengers, unless it is equipped with portable battery-powered megaphones readily accessible for use by crew members during an emergency evacuation, to the following scales:

Table 10.1 — Megaphones Required

Passenger Seating Configuration	Number of Megaphones Required
61 to 99	1
100 or more	2

Emergency Lighting

STRICT REQUIREMENT

An operator shall not operate a passenger-carrying aeroplane which has a maximum approved passenger seating configuration of **more than 9** unless it is provided with an **emergency lighting system having an independent power supply** to facilitate the evacuation of the aeroplane.

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Automatic Emergency Locator Transmitter (ELT)

STRICT REQUIREMENT

All aeroplanes shall carry an **automatic ELT**.

All aeroplanes authorized to carry **more than 19 passengers** shall be equipped with **at least two ELTs**, **one of which shall be automatic**.

Life Jackets — Passenger Familiarisation

STANDARD REQUIREMENT

An operator shall ensure that passengers are made **familiar with the location and use of**:

- **Seat belts**;
- **Emergency exits**;
- **Life jackets**, if the carriage of life jackets is prescribed;
- **Oxygen dispensing equipment**, if the provision of oxygen for passengers is prescribed; and
- Other emergency equipment provided for individual use, including the **passenger emergency briefing card**.

Life Rafts & Survival ELTs for Extended Over-Water Flights

STRICT REQUIREMENT

On over-water flights, an operator shall not operate an aeroplane at a distance away from land suitable for making an emergency landing without **sufficient life-rafts to carry all persons on board**.

Unless excess rafts of enough capacity are provided, the buoyancy and seating capacity beyond the rated capacity of the rafts must accommodate **all occupants of the aeroplane in the event of a loss of one raft of the largest rated capacity**. The life-rafts shall be equipped with:

- A **survivor locator light**; and
- **Life-saving equipment**, including means of sustaining life as appropriate to the flight, and **at least two survival Emergency Locator Transmitters [ELT(S)]** capable of transmitting on the distress frequencies.

Survival Equipment — Information for Rescue Coordination Centres

STANDARD REQUIREMENT

Operators shall at all times have available for **immediate communication to rescue coordination centres** lists containing information on the emergency and survival equipment carried on board any of their aeroplanes engaged in international air navigation. The information shall include, as applicable:

- The **number, colour and type** of life rafts and pyrotechnics;
- Details of **emergency medical supplies** and **water supplies**; and
- The **type and frequencies** of the emergency portable radio equipment.

Section 11

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Communication & Navigation Equipment

Communication Equipment

STANDARD REQUIREMENT

An aeroplane shall be provided with **radio communication equipment capable of**:

- Conducting **two-way communication for aerodrome control** purposes;
- **Receiving meteorological information** at any time during flight; and
- Conducting **two-way communication at any time during flight** with at least one aeronautical station, and with such other aeronautical stations and on such frequencies as may be prescribed by the appropriate authority.

The radio communication equipment shall provide for communications on the **aeronautical emergency frequency 121.5 MHz**.

STANDARD REQUIREMENT — RCP-TYPE AIRSPACE

For flights in defined portions of airspace, or on routes where an **RCP (Required Communication Performance) type** has been prescribed, an aeroplane shall:

- Be provided with communication equipment which will enable it to operate in accordance with the prescribed RCP type(s); and
- Be **authorized by the State of the Operator** for operations in such airspace.

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Navigation Equipment

STANDARD REQUIREMENT

An aeroplane shall be provided with navigation equipment which will enable it to proceed:

- In accordance with its **operational flight plan**; and
- In accordance with the **requirements of air traffic services** — except when navigation for flights under the visual flight rules is accomplished by visual reference to landmarks.

STANDARD REQUIREMENT — PBN OPERATIONS

For operations where a **navigation specification for PBN (Performance-Based Navigation)** has been prescribed, an aeroplane shall:

- Be provided with navigation equipment which will enable it to operate in accordance with the prescribed navigation specification(s); and
- Be **authorized by DGCA** for such operations.

DEFINITION — NAT HLA (FORMERLY MNPSA)

The airspace formerly known as the **"North Atlantic Minimum Navigational Specifications Airspace" (MNPSA)** — **FL285–420 inclusive** — is re-designated as the **"North Atlantic High Level Airspace" (NAT HLA)**. The re-designation from MNPS to HLA is in keeping with the transition from capability-based navigation (MNPS) to **Performance-Based Navigation (PBN)** in this segment of the NAT airspace.

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Radio Equipment (Survival Radio)

STRICT REQUIREMENT

An aeroplane operated across land areas designated by AAI as areas in which search and rescue would be especially difficult shall be equipped with at least **one survival radio equipment**, stowed so as to facilitate its ready use in an emergency, which:

- Operates on **VHF**;
- Is **portable**;
- Is **not dependent for operation upon the aircraft power supply**; and
- Is **capable of being operated by unskilled persons**.

The aeroplane shall also be equipped with such **signaling devices and life-saving equipment** (including means of sustaining life) as may be appropriate to the area overflown.

Audio Selector Panel

STRICT REQUIREMENT

An operator shall not operate an aeroplane **under IFR** unless it is equipped with an **audio selector panel accessible to each required flight crew member**.

Section 12

Minimum Fuel Requirements

For all aeroplanes (except helicopters), fuel planning is built up from defined "blocks". You must understand each block and the exact rules attached to it.

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Basis of Fuel Calculation

STANDARD REQUIREMENT

The amount of usable fuel carried shall, as a minimum, be based on:

- Current aeroplane-specific data derived from a **fuel consumption monitoring system**, if available; or, if not available, data provided by the **aeroplane manufacturer**; and
- The **operating conditions** for the planned flight — including anticipated aeroplane mass, NOTAMs, current meteorological reports/forecasts, ATS procedures, restrictions and anticipated delays, and the effects of deferred maintenance items and/or configuration deviations.

The Fuel Blocks

DEFINITION — (A) TAXI FUEL

The amount of fuel expected to be consumed **before take-off**, taking into account local conditions at the departure aerodrome and **APU fuel consumption**.

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DEFINITION — (B) TRIP FUEL

The amount of fuel required to enable the aeroplane to fly from take-off (or the point of in-flight re-planning) **until landing at the destination aerodrome**, taking into account the operating conditions.

DEFINITION — (C) CONTINGENCY FUEL

The amount of fuel required to **compensate for unforeseen factors**. It shall be **5% of the planned trip fuel** (or of the fuel required from the point of in-flight re-planning, based on the consumption rate used to plan the trip fuel) — but in **no case** shall it be lower than the amount required to fly for **5 minutes at holding speed at 450 m (1 500 ft) above the destination aerodrome in standard conditions**.

NOTE — UNFORESEEN FACTORS

"Unforeseen factors" are those which could influence fuel consumption to the destination — deviations of an individual aeroplane from expected fuel data, deviations from forecast weather, extended delays, and deviations from planned routings and/or cruising levels.

(d) Destination Alternate Fuel

DEFINITION — (D) DESTINATION ALTERNATE FUEL

1) Where a destination alternate aerodrome is required — the amount of fuel required to enable the aeroplane to:

- i) perform a **missed approach** at the destination aerodrome;
- ii) **climb** to the expected cruising altitude;
- iii) **fly** the expected routing;
- iv) **descend** to the point where the expected approach is initiated; and
- v) conduct the **approach and landing** at the destination alternate aerodrome.

2) Where two destination alternate aerodromes are required — the amount of fuel required to enable the aeroplane to proceed to the destination alternate aerodrome which requires the **greater amount of alternate fuel**.

3) Where a flight is operated without a destination alternate aerodrome — the amount of fuel required to fly for **15 minutes at holding speed at 450 m (1 500 ft)** above destination aerodrome elevation in standard conditions.

4) Where the aerodrome of intended landing is an isolated aerodrome:

- vi) for a **reciprocating engine aeroplane** — fuel to fly for **45 minutes plus 15% of the flight time** planned to be spent at cruising level (including final reserve fuel), **or two hours, whichever is less**;
- vii) for a **turbine-engine aeroplane** — fuel to fly for **two hours at normal cruise consumption** above the destination aerodrome (including final reserve fuel).

(e) Final Reserve Fuel

DEFINITION — (E) FINAL RESERVE FUEL

The amount of fuel calculated using the estimated mass on arrival at the destination alternate aerodrome (or the destination aerodrome when no destination alternate is required):

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- For a **reciprocating engine aeroplane** — fuel to fly for **45 minutes**, under speed and altitude conditions specified by the State of the Operator;
- For a **turbine-engine aeroplane** — fuel to fly for **30 minutes at holding speed at 450 m (1 500 ft)** above aerodrome elevation in standard conditions.

(f) Additional Fuel

DEFINITION — (F) ADDITIONAL FUEL

The supplementary amount of fuel required if the minimum fuel calculated is not sufficient to:

- Allow the aeroplane to descend as necessary and proceed to an alternate aerodrome in the event of **engine failure or loss of pressurisation** — whichever requires the greater amount of fuel — based on the assumption that such a failure occurs at the **most critical point along the route**; then
- Fly for **15 minutes at holding speed at 450 m (1 500 ft)** above aerodrome elevation in standard conditions; and
- Make an approach and landing.

It must also allow an aeroplane engaged in **EDTO** to comply with the EDTO critical fuel scenario as established by the State of the Operator, and meet additional requirements not covered above.

DEFINITION — EXTENDED DIVERSION TIME OPERATIONS (EDTO)

Any operation by an aeroplane with **two or more turbine engines** where the diversion time to an en-route alternate aerodrome is **greater than the threshold time** established by the State of the Operator.

(g) Discretionary Fuel

DEFINITION — (G) DISCRETIONARY FUEL

The **extra amount of fuel** to be carried **at the discretion of the pilot-in-command**.

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Starting point / condition	Branch	Next step / result
TAXI FUEL before take-off + APU	→	TRIP FUEL take-off to destination
TRIP FUEL take-off to destination	→	CONTINGENCY FUEL 5% of trip fuel — min 5 min holding at 1500 ft
CONTINGENCY FUEL 5% of trip fuel — min 5 min holding at 1500 ft	→	DESTINATION ALTERNATE FUEL missed approach + climb + cruise + descent + approach & land
DESTINATION ALTERNATE FUEL missed approach + climb + cruise + descent + approach & land	→	FINAL RESERVE FUEL Turbine: 30 min holding at 1500 ft Recip: 45 min

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Starting point / condition	Branch	Next step / result
FINAL RESERVE FUEL Turbine: 30 min holding at 1500 ft Recip: 45 min	→	ADDITIONAL FUEL engine-out / depressurisation + EDTO critical scenario
ADDITIONAL FUEL engine-out / depressurisation + EDTO critical scenario	→	DISCRETIONARY FUEL extra — PIC's discretion

Figure 16.4 *Build-Up of Usable Fuel Required*

EXAM FOCUS — KEY FUEL NUMBERS

Contingency = **5%** of trip fuel (floor: 5 min holding). Final reserve — **turbine 30 min**, **recip 45 min**. No-alternate destination fuel = **15 min holding**. Holding height in every case = **450 m / 1 500 ft**.

Section 13

Flight Crew & Flight Duty Time Limitations (FDTL)

Fatigue is a flight-safety hazard. FDTL rules set the maximum a crew may legally work and the minimum rest they must receive. Every figure below is examinable.

The Governing Regulation

DEFINITION — CAR SECTION 7, SERIES J

Civil Aviation Requirement **Section 7 — Flight Crew Standards, Training and Licensing, Series J** (in three parts) prescribes the **maximum limits of Flight Time, Flight Duty Period, Duty Period** and the **minimum Rest Period** applicable to all flight crew and cabin crew of **Scheduled, Non-Scheduled, General Aviation and Scheduled Commuter** Air Transport Services.

RULE 42A OF THE AIRCRAFT RULES, 1937 — FATIGUE MANAGEMENT

1. Taking into account the likely impact of fatigue on alertness, the **maximum limits of flight time, flight duty period and duty period** of flight and cabin crew, and the **minimum rest periods**, **may be specified by the Director-General**.
2. **No flight or cabin crew member** of a flying machine registered in India, or wet-leased by an Indian operator, **shall contravene** ***(breaking a rule)*** **the limitations stipulated by the Director-General**.
3. Every Indian operator shall establish a scheme of flight and duty time limitations and minimum rest periods in accordance with the stipulations made by the Director-General, submit it for approval, and — after approval — such scheme shall form part of the **Operations Manual** of the operator.

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REFERENCE NOTE

The figures that follow are highlights of **CAR Series J, Part III, Issue III**, for flight crew engaged in Scheduled Air Transport Operations. For other operations, refer to Series J Parts I, II and IV.

Crew Member Categories

DEFINITION — CABIN CREW MEMBER

Crew who perform duties in the interest of the safety of passengers, and duties assigned by the operator and the pilot-in-command — but who **shall not act as a flight crew member**.

DEFINITION — FLIGHT CREW MEMBER

A **licensed** crew member charged with duties **essential to the operation of an aircraft** during a flight duty period.

DEFINITION — SCHEDULED COMMUTER OPERATORS

Operators permitted to operate aircraft weighing **up to 40,000 kg**. They are permitted to operate commercial flights between two or more local destinations, **except the Category I cities**, which include: Mumbai Int'l, Kolkata Int'l, Delhi Int'l, Bangalore Int'l, Hyderabad Int'l, Thiruvananthapuram and Chennai.

FDTL Definitions — Essential Vocabulary

DEFINITION — ACCLIMATIZED

A state in which a crew member's **circadian biological clock is synchronised** to the time zone where the crew member is. A crew member is considered acclimatized to a **3-hour-wide time zone** surrounding the local time at the point of departure. When the local time at the place where a flight duty commences differs by **more than 3 hours** from the local time where the next duty starts, the crew member — for calculating the maximum daily FDP — is considered acclimatized to the **departure time zone for the first 48 hours**. After 48 hours, the crew member is considered acclimatized to the local time where they start the next duty.

DEFINITION — ACCOMMODATION

For the purpose of standby and split duty — a **quiet and comfortable place not open to the public**, with the ability to control light and temperature, equipped with adequate furniture providing a crew member with the possibility to sleep, with enough capacity to accommodate all crew members present, and with access to food and drink.

DEFINITION — SUITABLE ACCOMMODATION

For the purpose of standby, split duty and rest — a **separate room for each crew member** located in a quiet environment, equipped with a **bed**, sufficiently ventilated, with a device for regulating temperature and light intensity, and with access to food and drink.

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DEFINITION — AUGMENTED FLIGHT CREW

A flight crew which comprises **more than the minimum number required** to operate the aircraft, allowing each flight crew member to leave the assigned post for the purpose of **in-flight rest** and to be replaced by another flight crew member of **equal or higher qualification**.

DEFINITION — DUTY

Any task that flight crew members are required by the operator to perform — including flight duty, administrative work, training, positioning and standby when it is likely to induce fatigue.

DEFINITION — DUTY PERIOD

A period which **starts when a flight crew member is required by an operator to report for, or to commence, a duty** and **ends when that person is free from all duties**.

DEFINITION — FLIGHT DUTY PERIOD (FDP)

A period intended to cover a continuous period of duty that **always includes a flight or series of flights**. It **commences when a flight crew member is required to report for duty** and **finishes at engine(s) off at the end of the last flight** on which he/she is a flight crew member.

DEFINITION — FLIGHT TIME

The total time from the moment an aeroplane **first moves for the purpose of taking off** until the moment it **finally comes to rest at the end of the flight**. *Note:* "Flight time" is synonymous with **"block-to-block"** or **"chocks-to-chocks"** time.

DEFINITION — MIXED DUTY

When a crew member is required to report for a duty **in advance of the stipulated reporting time**, for a scheduled flight or series of flights, to carry out a duty at the behest of the operator — the time spent on that duty shall be **part of the subsequent FDP**.

DEFINITION — FATIGUE

A **physiological state of reduced mental or physical performance capability** resulting from sleep loss or extended wakefulness, circadian phase, or workload, that can impair a crew member's alertness and ability to safely operate an aircraft or perform safety-related duties.

DEFINITION — HOME BASE

The location nominated by the operator, having a **high degree of permanence**, from which the crew member normally starts and ends a flight duty or a series of flight duty periods.

DEFINITION — TEMPORARY HOME BASE

The place of posting by the operator where the crew operates and starts a series of Duty Cycles, and which is **not the same as Home Base**.

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DEFINITION — LOCAL DAY

A **24-hour period** commencing at **00:00 local time**.

DEFINITION — LOCAL NIGHT

A period of **8 hours** falling between **2200 hours and 0800 hours** local time.

DEFINITION — NIGHT DUTY

Any Duty Period encroaching upon any portion of the time period between **0000 hrs and 0600 hrs** in the time zone to which the crew is acclimatized.

DEFINITION — OPERATING CREW MEMBER

A crew member **carrying out duties in an aircraft during a sector**.

DEFINITION — POSITIONING

The transferring of a **non-operating** flight crew member from place to place at the behest of the operator. *Note:* "Positioning" is synonymous with **"Deadheading"**.

DEFINITION — REPORTING TIME

The time at which a flight crew member is required by an operator to **report for duty**.

DEFINITION — REST PERIOD

A **continuous, uninterrupted and defined period of time**, subsequent to or prior to duty, during which a flight crew member is **free from all duties, standby and reserve**.

DEFINITION — REST FACILITY

A **bunk or seat with leg and foot support** suitable for crew members' sleeping on board an aircraft.

DEFINITION — ROSTER

A list provided by an operator of the times when a flight crew member is required to undertake duties. *Note:* "Roster" is synonymous with **"Schedule", "Line of Time", "Pattern"** and **"Rotation"**.

DEFINITION — STANDBY

A defined period of time during which a flight crew member is required by the operator to be **available to receive an assignment** for a specific duty **without an intervening rest period**.

DEFINITION — SECTOR

The segment of a Flight Duty Period between an aircraft **first moving for the purpose of taking off until it comes to rest after landing** on the designated parking position.

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DEFINITION — TIME ZONE

A defined region of the earth with a uniform local time which differs by **one hour, or part of one hour**, from the uniform local time of an adjoining region — irrespective of season.

DEFINITION — TRAINING FLIGHTS

Flights for the purpose of **Base Training, Familiarization, and Conducting Aircraft Training and Checks** (Skill Test / IR / PPC).

DEFINITION — UNFORESEEN OPERATIONAL CIRCUMSTANCE

An **unplanned event** — such as unforeseen weather, natural calamity, medical emergency, national requirement for relief and rescue operations (mission oriented), equipment malfunction, runway closure, passenger or aircraft security, etc. — that is **beyond the control of the operator**.

DEFINITION — ULTRA LONG RANGE (ULR) OPERATIONS

Continuous non-stop flights between a specific city pair, involving any sector having a **flight time of over 14 hours** — and flight duty periods of up to **22 hours** — at any time during a calendar year, taking into account the mean and seasonal wind changes. ULR operations apply to **both sectors** of a city pair.

DEFINITION — IN-FLIGHT REST PERIOD

A period of time within a flight duty time which is given to a crew member to **rest before recommencing duty**.

DEFINITION — WINDOW OF CIRCADIAN LOW (WOCL)

A period between **0200h and 0600h** in the time zone to which a crew member is acclimatized. This window is calculated from scientific data on the circadian low of performance, alertness (peak fatigue) and body temperature.

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Operator's Responsibility

STANDARD REQUIREMENT

- Every operator shall establish a **scheme** for complying with the limitations for Flight Time, FDP, Duty Periods and Rest Periods specified in the CAR.
- Each operator shall prepare the Flight Crew **Roster sufficiently in advance**. The roster should be published **at least for a 7-day period**, and the **weekly rest shall be printed** on the published roster.
- Every operator shall assign a **Home Base / Temporary Home Base** to each flight crew member — **minimum duration 7 days, maximum recommended 28 days**.

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Responsibility of the Flight Crew Member

CREW MEMBER'S DUTY

- No flight crew member may **accept an assignment which exceeds the prescribed limitation**.
- Flight crew members shall make best use of the facilities and opportunities provided for **rest and meals**, and shall plan and use rest periods to ensure they are fully rested.
- A flight crew member **shall not perform flight crew duties** on a flight if it is known or suspected that the crew member is **fatigued to the extent that the safety of flight may be adversely affected**.

Flight Time & FDP Limits — Two-Pilot Operation

The maximum flight time and maximum flight duty period during any 24 hours for a two-pilot crew:

Table 13.1 — FT & FDP Limits: Two-Pilot Operation

Maximum Flight Time	Maximum Flight Duty Period (Hours)	Maximum Number of Landings
8 Hrs	11:00	6
	11:30	5
	12:00	4
	12:30	3
9 Hrs	13:00	2
10 Hrs	13:00	1
For Operations Encroaching Upon Night		
8 Hrs	10:00	2

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READING THE TABLE

For a 2-crew international scheduled flight, the basic entry is **FDP 11:00 hrs, Flight Time 8 hrs, 6 landings**. As you trade FDP for more flying, landings drop — more flying hours allowed means fewer landings permitted. Operations encroaching on night are stricter: FDP capped at **10:00 hrs** with only **2 landings**.

Flight Time & FDP Limits — Augmented Crew

An augmented crew carries extra pilots so each can take in-flight rest. This permits longer limits:

Table 13.2 — FT & FDP Limits: Augmented Crew (any 24 hrs)

Crew	Max Flight Time	Max FDP — Bunk	Max FDP — Isolated Rest Seat	Max FDP — Basic Rest Seat	Max Landings
3 Crew Operations	12 Hrs	16 Hrs	15 Hrs	14 Hrs	2
4 Crew Operations	14 Hrs	18 Hrs	16 Hrs	NA	1
Above 14 hrs (ULR) — 4 Crew Operations	17 Hrs	21 Hrs	NA	NA	1

Maximum Cumulative Flight Time & Duty Period Limitations

Table 13.3 — Cumulative FT & Duty Period Limits

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Period	Maximum Flight Time (Hours)	Maximum Cumulative Duty Period
In **7** consecutive days	35	60
In **14** consecutive days	65	100
In **28** consecutive days	100	190
In **90** consecutive days	300	600
In **365** consecutive days	1000	1800

EXAM FOCUS — LOCK THESE IN

Flight Time: **7 days → 35 h**, **28 days → 100 h**, **90 days → 300 h**, **365 days → 1000 h**. A pilot can fly **100 hrs in 28 consecutive days** and **1000 hrs in 365 consecutive days** — both are classic MCQ answers.

Unforeseen Operational Circumstances — Extensions

"Flight Time", "Flight Duty Period", "Landing" and "Consecutive Night" may be extended due to unforeseen operational circumstances as follows:

STRICT LIMITATION — PERMITTED EXTENSIONS

- **(a)** Flight Time — by a maximum of **01 hour**.
- **(b)** FDP — by a maximum of **02 hours**.
- **(c)** Only **one extra landing** may be carried out in the event of a diversion to complete the flight, including a consecutive night for completion of the flight.
- **(d)** The above is subject to a cumulative limit of maximum **04 hours** (Flight Time) and maximum **08 hours** (FDP) during any period of **28 consecutive days**.
- **(e)** In the event of FDP extension up to 01 hour, OR Flight Time extension up to 30 minutes — the **rest shall increase by 02 hours**.
- **(f)** In the event of FDP extension beyond 01 hour, OR Flight Time extension beyond 30 minutes — the **rest shall increase by 04 hours**.
- **(g)** Extension of the maximum basic FDP **shall not be combined with split duty** in the same duty period.
- **(h)** If no flight time is utilized after commencement of FDP (reporting), the crew can be utilized after a minimum of **twice the time period spent after reporting**, subject to a minimum limit of **12 hours of rest**.

STANDARD PROCEDURE — CONSENT & REPORTING

The PIC, in consultation with the other flight crew members, will convey their willingness or consent to the Head of Operations for operating the flight. The Head of Operations will submit a **quarterly report to the DGCA** on all extensions granted. Cumulative Flight Time and Duty Time Limitations can be increased — as per (a) and (b) above — for the 7, 14, 28 and 90 consecutive-day maximums specified.

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Records

STRICT REQUIREMENT

Records shall be kept for **18 months** of the duties performed and the rest periods provided, so as to facilitate inspection by the operator's authorized personnel and surveillance/audit by DGCA officers. The records may be made available to flight crew on request.

Capt. Pankaj Pahil

Split duty, rest, standby and positioning

DGCA CAR Section 7, Series J, PartIII, paragraphs 9–12. These are the baseline scheduled-air-transport provisions in the cited CAR. Operator-specific variations must be applied only within their approved scope.

Split duty — a break within an FDP

A split duty divides a flight duty period into duty portions separated by a break. The break remains part of the FDP; it is not a normal rest period.

Consecutive break	Maximum FDP extension
Less than 3 hours	None.
Between 3 and 10 hours	Half the consecutive break taken.
More than 10 hours	No extension.

Paragraph 9 excludes pre-flight duties, post-flight duties and transport from the break. A break longer than 6 consecutive hours, or encroaching on the WOCL, requires suitable accommodation. For a break shorter than 6 hours, accommodation must be provided. The CAR states that the FDP portions before and after the break shall not exceed 10 hours. The whole split duty counts as FDP. No extension is permitted if the FDP encroaches on the night-duty period.

Minimum rest before flight

Under 10.1, provide at least the length of the preceding duty or the applicable minimum below, whichever is greater.

Situation	Minimum used in the comparison
Normal case	12 hours.
Crossing more than 3 and up to 7 time zones	18 hours.
Crossing more than 7 time zones	36 hours.

For positioning within the acclimatised zone, before an operating flight and outside paragraph 12’s special case, the note allows 12 hours after positioning provided the preceding 10.1 rest was taken before positioning. A crew proceeding farther after crossing 3–7 time zones receives 10.1 rest at the layover before the next FDP. Further time-zone difference is measured from the place of acclimatisation. On return from leave, the cited note requires 10 hours minimum rest from 0000 hours; leave is treated as local days.

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Paragraph 10.2 requires 10 hours for other duties apart from flight/mixed duty, with the operator defining the arrangement in its FDTL scheme. If preceding duty, including positioning, exceeds 18 hours, ensuing rest includes a local night. Airport transport counts as neither duty nor rest. Where transport exceeds 30 minutes, rest increases by twice the excess above 30 minutes.

Weekly rest and return to home base

Paragraph 10.6 requires 48 continuous hours including two local nights, at home base or temporary home base. No more than 168 hours may separate the end of one weekly rest and the start of the next. Under 10.6.1, weekly rest increases to 60 hours including two local nights when more than three duties encroached on the night-duty period in the preceding 168 hours.

Time-zone difference on return home	Minimum rest	Local nights
More than 3 and up to 7	36 hours	2
More than 7	72 hours	3

Standby — available for assignment

Standby must appear in regular rosters and the crew must be notified. Under 11.2.1, total standby does not exceed 12 hours; airport standby, with or without sleeping quarters, does not exceed 8 hours.

Standby situation	How it counts
Airport standby becomes flight duty	100% counts towards FDP and cumulative duty.
Airport standby does not become flight duty	100% counts towards cumulative duty.
Home/hotel standby becomes duty within its first 6 hours	No standby time counts towards FDP or cumulative duty under 11.2.3(a).
Home/hotel standby becomes duty at 6 hours or later	FDP is reduced by 50% of the standby time under 11.2.3(b).
Home/hotel standby does not become duty	25% counts towards cumulative duty.

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After standby finishes,12 hours rest is required by 10.3.1. If standby becomes duty, calculate subsequent rest using the duty plus the percentage of standby counted as duty.

Positioning — travelling at the operator's direction

Positioning time counts as duty. It forms part of FDP when it immediately precedes operating duty without intervening rest. Positioning after an FDP without intervening rest counts as duty when determining rest. Positioning normally is not an operating sector when calculating FDP. Under 12.4, it counts as a landing where, after positioning, the crew spends less than the minimum rest in suitable accommodation and then uses split duty to extend FDP.

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CHAPTER 17

Special Operational Procedures and Hazards

17

HOW TO USE THESE NOTES

Every box is colour-coded so your eye learns the rule before your brain does:

- **Blue boxes** — Definitions & core concepts. Learn the language first.
- **Green boxes** — Standard Operating Procedures & allowances. What you are permitted / expected to do.
- **Red boxes** — Strict limitations, warnings & prohibitions. Cross these and you bust a regulation.

Every exact DGCA figure is shown like this — **240 m (800 ft)** — so the precise value is never lost. Examiners test the number, not the paraphrase.

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Minimum Equipment List (MEL & MMEL)

Aircraft are certified with a great deal of redundant equipment. The MEL is the document that tells the crew whether the aeroplane may legally and safely fly with a particular item unserviceable — turning a "go / no-go" judgement into a documented decision.

DEFINITION — MEL

The operator **shall include in the Operations Manual a Minimum Equipment List (MEL)**, approved by the DGCA, which enables the **pilot-in-command** to determine whether a flight may be **commenced or continued from any intermediate stop** should any instrument, equipment or system become inoperative.

DEFINITION — MMEL

The **Master Minimum Equipment List (MMEL)** defines the equipment on which certain in-flight failures can be allowed, and the conditions under which this allowance can be accepted. The **MMEL is drawn up by the manufacturer and approved by the DGCA**.

The relationship is a chain of authority. The manufacturer writes the master list (MMEL); the operator tailors it into its own MEL for its specific fleet and operations; and the DGCA approves both. The MEL can never be *less* restrictive than the MMEL.

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Who Does What — The Approval Chain

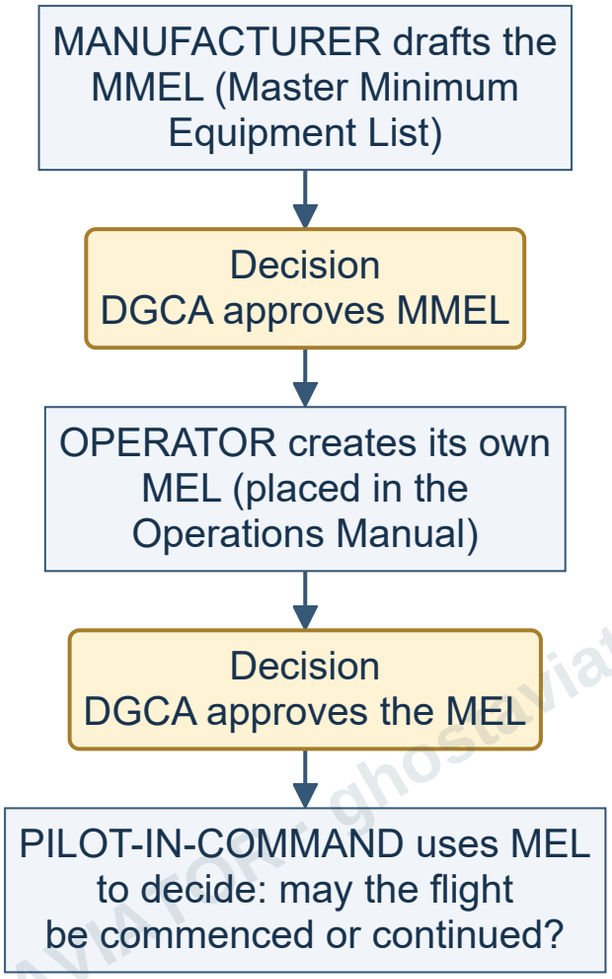

Figure 17.1 *Flowchart — From MMEL to the Crew's Go/No-Go Decision*

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SOP — DAY/NIGHT & IMC TAILORING

The DGCA may require an operator's approved MEL to **specify the operating equipment required for night and/or IMC operations, and separately for day/VMC operations**. The same aircraft may therefore be despatchable for a day VMC sector but grounded for a night IMC sector with the identical defect.

EXAM TRAP

MMEL = Manufacturer. MEL = found in the Operations Manual. The reference document a crew opens first when equipment fails while still parked is the **MEL** — not the Flight Manual, and not the "Abnormal & Emergency Procedures" chapter.

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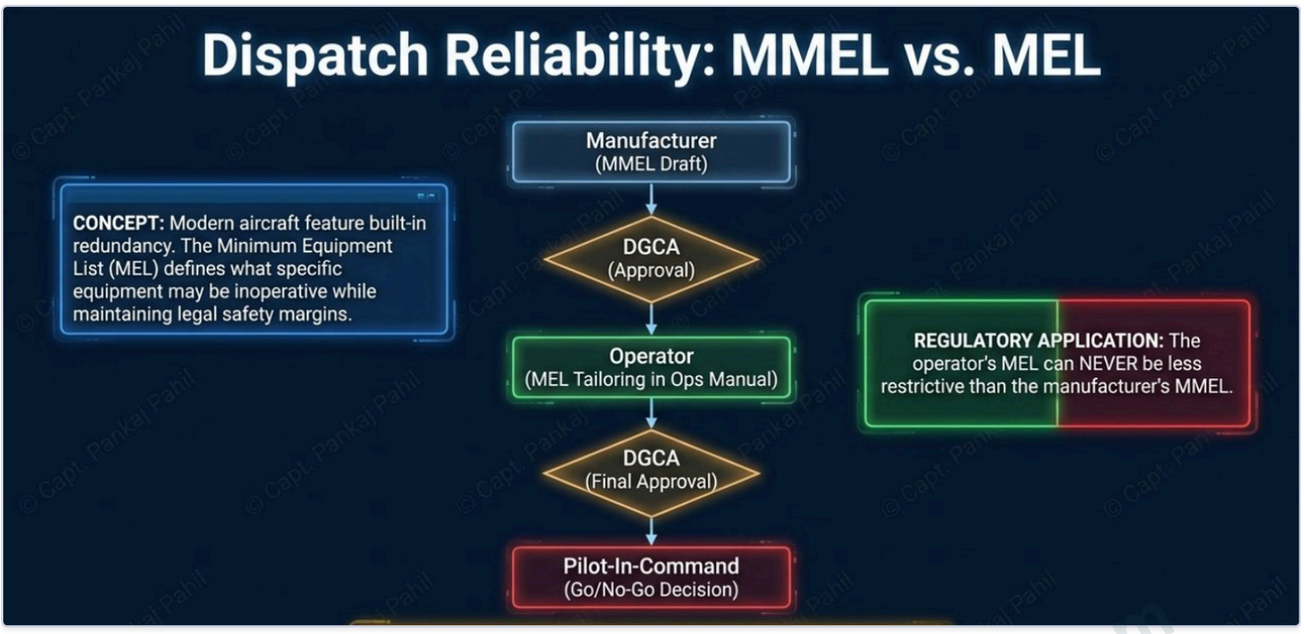

Figure 17.2 *From the master list to the operator list to the go or no-go decision*

Aeroplane Flight Manual (AFM)

CORE RULE

Aeroplanes **shall be operated in accordance with the provisions of the Flight Manual approved by the State of Design**.

LIMITATION

An aeroplane **shall be operated in compliance with the terms of its Certificate of Airworthiness** and **within the approved operating limitations contained in its Flight Manual**. The AFM limitations are not advisory — they are legally binding boundaries.

Factors That Must Be Accounted For

When applying AFM performance data, the crew/operator **shall take account of all factors that significantly affect the performance of the aeroplane**, including but not limited to:

PERFORMANCE FACTORS

- The **mass** of the aeroplane;
- The **operating procedures**;
- The **pressure-altitude** appropriate to the elevation of the aerodrome;
- The **ambient temperature**;
- The **wind**;
- The **runway slope**; and
- The **surface condition of the runway** — i.e. presence of snow, slush, water and/or ice for landplanes; water surface condition for seaplanes.

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Ground De-icing & Aircraft Icing

Ice is the silent destroyer of lift. This section covers when an aircraft may take off, how contamination is detected, the two distinct fluid procedures, and why even an invisible film of ice matters.

The Ground De-icing Rule

STRICT LIMITATION

A flight to be planned or expected to operate in **suspected or known ground icing conditions shall NOT take off** unless the aeroplane has been **inspected for icing** and, if necessary, has been given appropriate **de-icing / anti-icing treatment**. Accumulation of ice or other naturally occurring contaminants **shall be removed** so that the aeroplane is kept in an **airworthy condition prior to take-off**.

Definition & Recognition — On Ground / In-Flight

OPERATING PROHIBITIONS

(a) An operator **shall not operate an aeroplane in expected or actual icing conditions unless it is certificated and equipped to operate in icing conditions**.

(b) An operator **shall not operate an aeroplane in expected or actual icing conditions at night** unless it is equipped with a means to **illuminate or detect the formation of ice**. Any illumination used must be of a type that **will not cause glare or reflection** that would handicap crew members in the performance of their duties.

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De-icing, Anti-icing & Types of Fluid

The two most common techniques for protecting an aircraft's critical surfaces are **de-icing** and **anti-icing** — they sound similar but do opposite jobs.

Table 3.1 — De-icing vs. Anti-icing

Aspect	De-icing	Anti-icing
Purpose	Procedure by which existing frozen contamination (**frost, ice, snow or slush**) is **removed** from the aircraft.	Procedure that **protects a clean surface** against the accumulation of frozen contaminants for a limited period.
Fluid used	Heated **Aircraft Deicing Fluid (ADF)** — provides clean surfaces.	**Aircraft Anti-Icing Fluid (AAF)** — applied to a surface already free of frozen contaminants.
Surface state before	Surface is contaminated.	Surface is already free of frozen contamination.
Effect	Restores a clean surface.	Buys a limited "holdover" period of protection.

CONCEPT — WHEN ARE FLUIDS NEEDED?

De/anti-icing fluids are **only required until the aircraft becomes airborne**, after which the on-board de/anti-icing system then operates. The ground fluids are a bridge to get you safely into the air; the airframe system takes over thereafter.

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Performance Deterioration — On Ground / In-Flight

THE CRITICAL NUMBER

A very small amount of roughness — in thickness as low as **0.40 mm (1/64 in)** — caused by ice, snow or frost, **disrupts the airflow** over the lift and control surfaces. The consequence is **severe lift loss, increased drag and impaired manoeuvrability**, particularly during the take-off and initial climb phases.

WHAT ICE DOES TO THE AIRCRAFT

- **Interferes with movement** of control surfaces.
- **Adds significantly to aircraft weight.**
- **Severely affects instruments** by blocking static and pitot vents, or angle-of-attack vanes.
- **Obscures wind screens**; obstructs the undercarriage; forms skin friction.

Remember the instructor's rule: "There is no such thing as an insignificant amount of ice."

AERODYNAMIC EFFECT

Accumulation of snow or ice on an aircraft **in flight** causes an **increase in the stalling speed**. It does *not* raise the stall angle of attack — in fact contamination makes the wing stall *earlier* (at a lower angle).

Cold-Soaking & Clear Ice

DEFINITION — COLD-SOAKING

Ice can form **even when the Outside Air Temperature (OAT) is well above 0°C (32°F)**. An aircraft with wing fuel tanks may carry fuel cold enough to lower the wing-skin temperature **below freezing**. This is **cold-soaking**. It also occurs after cruising at high altitude for a period followed by a quick descent into a humid environment — liquid water contacting the below-freezing wing then freezes onto it.

Cold-soaking can also be caused by **fuelling an aircraft with cold fuel**. With rain or high humidity present, ice can form on the cold-soaked wing and accumulate over time. This ice can be **invisible to the eye** and is often referred to as **clear ice**.

DETECTING CLEAR ICE

Because clear ice is hard to see, it is detected by:

- Performing a **tactile (hands-on) inspection**; or
- Using specially designed ice-detecting systems such as a **Ground Ice Detection System (GIDS)**.

WHY CLEAR ICE IS DANGEROUS

Sheets of clear ice dislodged from the wing or fuselage during take-off or climb can be **ingested by aft-fuselage-mounted engines**, causing a **flameout or damage**. Dislodged sheets can also cause **impact damage to critical surfaces** such as the horizontal stabiliser.

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Planning a Flight in Icing Conditions

Starting point / condition	Branch	Next step / result
Flight expected in suspected / known ground icing conditions	→	Aircraft certificated & equipped for icing?
Aircraft certificated & equipped for icing?	No	Do NOT operate
Aircraft certificated & equipped for icing?	Yes	Operation at NIGHT?
Operation at NIGHT?	Yes	Means to illuminate / detect ice formation required (no handicapping glare)
Means to illuminate / detect ice formation required (no handicapping glare)	→	Inspect aircraft for icing
Operation at NIGHT?	No	Inspect aircraft for icing
Inspect aircraft for icing	→	Contamination present?
Contamination present?	Yes	Apply appropriate de-icing / anti-icing treatment
Apply appropriate de-icing / anti-icing treatment	→	Aircraft in airworthy condition → CLEARED to take off
Contamination present?	No	Aircraft in airworthy condition → CLEARED to take off

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Figure 17.3 *Decision Flow — Pre-Take-off in Suspected/Known Icing*

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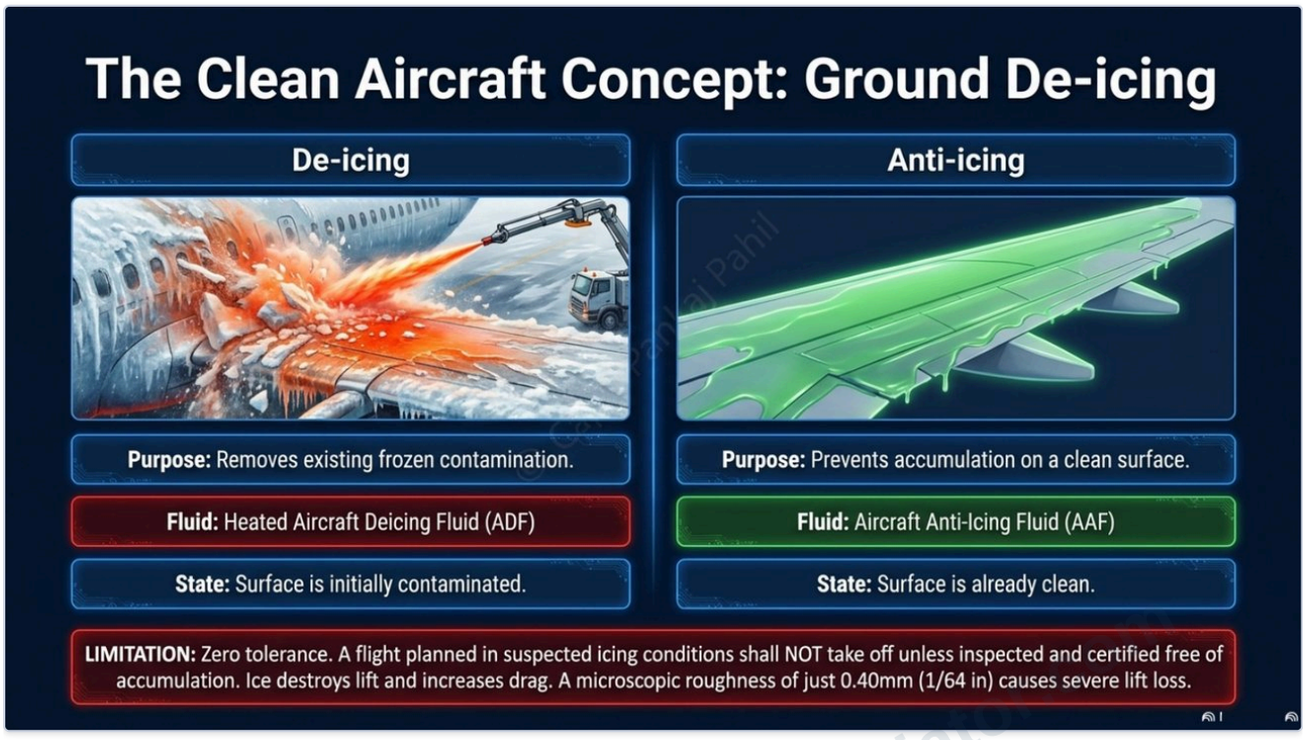

Figure 17.4 *De-icing removes; anti-icing prevents*

Bird Strike Risk & Avoidance

CORE CONCEPT

There is **no direct way to influence bird activity aloft**. Therefore the only chance to minimise bird-strike risk in the air is to **avoid flying through high bird concentrations**. Away from the airfield, bird-strike prevention is most effectively conducted by **warning procedures**.

Hazardous bird concentrations are well known during **migration periods** on a large temporal and spatial scale, but also occur occasionally throughout the year on a local or regional scale, **mainly governed by the diurnal cycle**. The major aspects of handling the problem are **monitoring, modelling, warning, predicting and forecasting** — producing advice passed to the aviation community.

Shell crackers	Most effective method for scaring birds
90%	of bird strikes occur under 500 m
135 kts	Avg. take-off speed in the study
~2 seconds	Birds fly away beforehand

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KEY FINDINGS

- The **most effective method for scaring birds is shell crackers**.
- **90% of bird strikes occur under 500 m.**
- Studies of birds on the ground ahead of an aircraft that has reached an average speed of **135 kts** show that **birds fly away about two seconds beforehand**.

CREW ACTION

If, after take-off, you (as captain) notice a flock of birds presenting a strike hazard, you must **immediately inform the appropriate ground station** — not merely file a report on arrival, and not "within a reasonable period."

Noise Abatement — Departure Climb & Approach

Noise abatement balances community quiet against flight safety. The golden principle: **safety always overrides noise.** Responsibility for establishing operating procedures for noise abatement during instrument flight (in compliance with ICAO PANS-OPS) rests with the **operator** — and the procedure specified for any one aeroplane type should be the same for all aerodromes.

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Departure Climb

Operating procedures for the departure climb must ensure flight safety is maintained while minimising noise exposure on the ground. The following requirements must be satisfied:

DEPARTURE CLIMB LIMITATIONS

- Noise-abatement procedures **shall not be executed below a height of 240 m (800 ft) above aerodrome elevation**.
- The noise-abatement procedure specified by an operator for any one aeroplane type **should be the same for all aerodromes**.
- Noise-abatement climb procedures are **not to be used where wind-shear warnings exist**, or where wind shear / downburst activity is suspected.
- The **maximum acceptable body angle** specified for an aeroplane type **shall not be exceeded**.

Aeroplane Operating Procedures — Approach

APPROACH NOISE PROCEDURES

- Subject to safety requirements, the use of **reverse thrust over reverse idle may be prohibited** during a specified period, especially during night hours.
- During a noise-abatement approach, the aeroplane is to be in the **final landing configuration** at any point **after passing the outer marker**, or at a point **5 NM** from the threshold of the landing runway — **whichever is earlier**.
- **Noise preferential routes** are established so departing and arriving aeroplanes avoid over-flying noise-sensitive areas near the aerodrome as far as practicable.

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- In establishing noise preferential routes, **no turns should be required coincident with a reduction of power** associated with a noise-abatement procedure.

WHEN NOISE ABATEMENT MUST NOT DECIDE THE RUNWAY

Noise abatement **shall not be the determining factor** in the designation of a runway under the following circumstances:

- (a) when the **crosswind component, including gust, exceeds 15 kt**;
- (b) when the **tailwind component, including gust, exceeds 5 kt**;
- (c) when the **runway is not clear or dry**.

ILS & VISUAL GUIDANCE

No ILS or visual guidance precludes a runway being used for noise-abatement procedures **if landing in VMC**. In other words, the absence of ILS/visual aids is what rules a runway out for noise abatement when landing visually.

Fire & Smoke

Fire-fighting strategy flows directly from one idea: a fire needs three things at once. Take any one away and the fire dies.

The Fire Triangle

DEFINITION — COMBUSTION

Combustion consists of **three elements: oxygen, heat and fuel**. Together these create a chemical chain reaction and result in a fire. The goal of fire-fighting is to **eliminate at least one element** from the fire in order to extinguish it.

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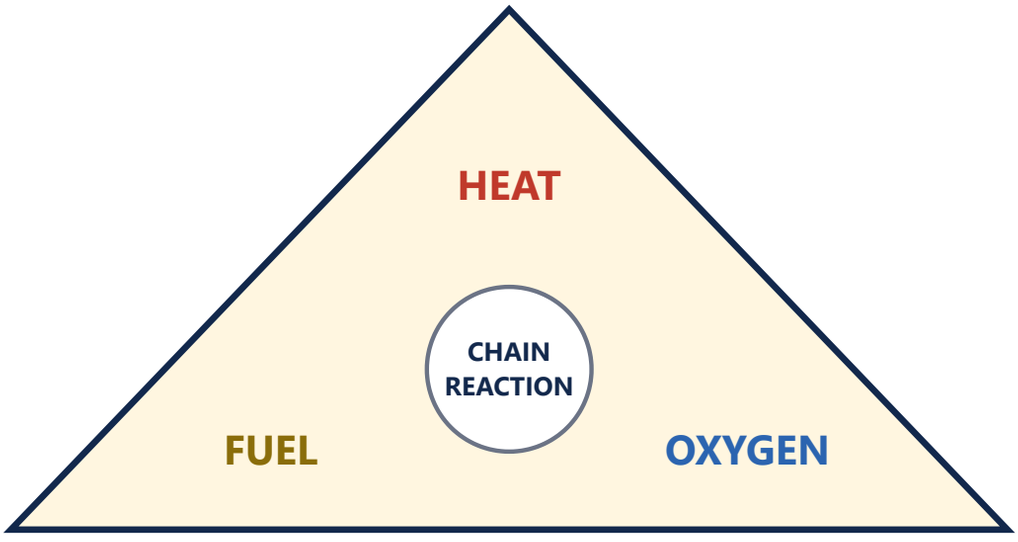

Fire needs heat, fuel and oxygen; the chain reaction keeps it burning.

Figure 17.5 *HEAT*

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A FIRE WILL CONTINUE, UNLESS

- The **fuel supply has been cut off**; or
- There is **no more oxygen available**; or
- The **temperature has been cooled below the flammability temperature**.

Classes of Fire & Their Smoke

Table 6.1 — Fire Classes, Materials, Extinguishing & Smoke Signature

Class	Materials Involved	How To Extinguish	Smoke Characteristics
Class A	Wood, paper, cloth or plastic — solids, generally of organic nature.	Fires need to be **cooled**. Water extinguisher, or liquid with a large percentage of water (coffee, tea, juice) works. **Water/glycol extinguishers are the most effective** for Class A. **Do NOT use liquid containing alcohol.**	Usually **gray/brown**; can be quite thick depending on quantity of fuel.
Class B	Flammable liquid, hydraulic fluid, oil, tar or aircraft fuel — liquid or liquefiable solid.	**Cannot be extinguished with water.** Use **Foam or Halon** extinguishers.	Usually **black**; very thick, with a distinct oil/petrol-like odour.
Class C	Electrical equipment.	Must be extinguished with a **non-conducting mixture** to avoid electrocution and damage to circuitry. **Halon** extinguishers are effective.	Usually **light grey or white**, with a bluish tinge; very fine, disperses rapidly; distinct acrid odour.
Class D	Flammable metals — sodium, magnesium, lithium, potassium.	**Special powder extinguishers** only (avoids dangerous chemical reaction). **Never use Halon on Class D.**	—

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HARD RULES — FIRE CLASSES

- **Never use Halon fire extinguishers on Class D fires.**
- **Do not use liquid containing alcohol on Class A fires.**
- **Class A and C fires are the most commonly encountered fires onboard aircraft.**

Fire Extinguishers

DISCHARGE TIME

Hand-held fire extinguishers discharge an extinguishing agent for **8 to 25 seconds**, depending on type and capacity. Because of this short period, it is **essential to select and use the appropriate extinguisher immediately**.

Table 6.2 — Extinguisher Types, Colour Codes & Suitable Fire Classes

Extinguisher Type	Colour Code	Suitable For
Halon / BCF (bromochlorodifluoromethane)	Green	Class **A, B and C** fires
Carbon dioxide (CO_2)	Black	Class **B and C** fires
Dry powder (DP)	Blue	Class **D** fires
Dry chemicals	Blue	Class **A, B and C** fires
Water solution (H_2O)	Red	Class **A** fires

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HALON FACTS & PRECAUTIONS

- A Halon extinguisher is **three times as effective as a CO_2 extinguisher** containing the same amount of agent.
- When a Halon extinguisher has been used on a **Class A fire**, the fire and surrounding area **must be cooled down with a non-alcoholic liquid**.
- When Halon is used in **crew compartments or confined areas**, **Portable Breathing Equipment (PBE) should be used**.

EXTINGUISHER LOCATION & SUPPORTING EQUIPMENT

At least one **Halon 1211 type extinguisher** should be located on the **flight deck**. Galleys, personnel and cargo compartments should be equipped with proper extinguishers. Beneath fire extinguishers, the supporting fire-fighting equipment carried on board includes **crash axes or crowbars**.

HALON ENVIRONMENTAL PHASE-OUT

ICAO Standards mandate the use of an **alternative agent to Halon** due to its impact on the environment — it **depletes the ozone layer**. As of now, Halon use is **permitted only in cargo compartments till 28 Nov 2024**.

Hand Fire Extinguisher Requirements by Seating

Table 6.3 — Hand Fire Extinguishers Required in the Passenger Compartment

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Maximum Approved Passenger Seating Configuration	Hand Extinguishers Required
61 to 200 passengers	**3** conveniently located in the passenger compartment
401 to 500 passengers	**6** conveniently located in the passenger compartment

ONBOARD FIRE-FIGHTING SOPS

- **Engine nozzle / jet-pipe fire on the ground:** carry out a **dry cranking (dry motoring) cycle**.
- **Engine fire warning:** switches on the relevant fire shut-off handle; the warning is switched off **when the fire is no longer detected**.
- **Fire in an air-conditioned cargo hold: turn off the cargo-hold ventilation and extinguish the fire**.
- **Belly compartment fire:** the system generally switched off is the **ventilation of the cargo compartment**.
- **Smoke in the air conditioning:** first action — **put on the mask and goggles**.
- **Fire-fighting in toilets:** use **all available extinguishers simultaneously**.
- **Burning cargo in cargo aircraft:** usually extinguished using **carbon dioxide**.
- **Wheel fire** (heated brakes): the safest extinguishant is **dry powder**; fight a brake-heat fire with a dry-powder extinguisher and a water-spray atomiser.

Overheated Brakes — After Aborted Take-off & Landing

The kinetic energy lost in slowing an aircraft is converted by friction into heat. After a rejected take-off or a heavy landing, that heat is concentrated in the brakes and tyres — and it bites.

CONSEQUENCES OF OVERHEATED BRAKES

Overheated brakes can result in:

- **Loss of braking performance**;
- **Fire**;
- **Tyre deflation / tyre burst**.

Brake failure — with subsequent poor directional control and deceleration — could in turn result in **runway excursion**, **uncommanded ground movements / taxiway excursion**, and **collision with objects on the ground or other aircraft**.

PRACTICAL PROBLEMS FOLLOWING BRAKE ISSUES

- **High level of stress & increased workload** — caused by directional-control and deceleration problems from brake failure during landing or a high-speed rejected take-off (RTO).
- **Lack of awareness** — the crew might not be aware of fire, tyre burst or deflation that could result from heavy braking on landing, RTO, or from smoke from the undercarriage.

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CREW ACTIONS

- **Request fire and rescue services** — hot-brake incidents may be considered by the crew a reason to request attendance of fire and rescue.
- **Decision for emergency evacuation** — the cockpit crew could take the decision for emergency evacuation if fire is detected following a high-energy brake application.

HOW FIREGUARDS MUST APPROACH HOT WHEELS

After a landing in overweight / overspeed conditions, with tyres and brakes extremely hot, the fireguards **should approach the landing-gear tyres ONLY from front or rear side** — **never from the sides**. A tyre that bursts releases its energy *sideways*; standing fore or aft keeps personnel out of the burst line. (Likewise: release the parking brake and approach the wheels either from aft or fore.)

EXAM HOOK

Following a **heavy-mass landing on a short runway**, the first thing you should check is the **temperature of the brakes**.

Decompression of a Pressurised Cabin

Pressurisation keeps a survivable atmosphere around the cabin at cruising altitude. Losing it is one of the most time-critical emergencies in aviation — because the real enemy is hypoxia *(too little oxygen reaching body tissues)*.

DEFINITION

Cabin pressurisation is the pumping of compressed air into an aircraft cabin to maintain a safe and comfortable environment for crew and passengers when flying at altitude. **Loss of pressurisation is a serious emergency** in an aircraft flying at the high cruising altitudes typical of passenger aircraft.

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Causes of Decompression

THREE CAUSES

- **Structural Failure** — failure of a window, door or pressure bulkhead, or an in-flight explosion (which may be due to a system failure, dangerous cargo, or a malicious act such as an explosive device carried on board by a terrorist).
- **Pressurisation system failure** — failure of some part of the pressurisation system, such as an outflow valve.
- **Deliberate Act** — a drastic measure an aircraft captain might consider, for example as a way of clearing the cabin of smoke.

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Rapid / Explosive vs. Slow Decompression

RAPID / EXPLOSIVE DECOMPRESSION

Decompression occurring rapidly — at a rate **greater than the rate the lungs can decompress** — will cause **lung damage**. The likelihood of reaching a lung-damaging level, for any given pressure-hull breach size, is **increased by the size of the pressure hull overall**.

A decompression taking **less than 0.5 seconds** is considered by most authorities to be **"explosive."** The cabin air may fill with **dust and debris**, and **fog** forms from the associated drop in temperature and change in relative humidity. The crew may be momentarily dazed or shocked — especially if unexpected — and may therefore be slow to fit oxygen masks.

SLOW DECOMPRESSION

Slow decompression is **similar to rapid decompression**. The only difference is in **severity and availability of reaction time**. In **both cases the actions by the crew remain the same**. A slow decompression may be caused by a **cracked window**, a **bad-functioning pressurisation system**, or a **minor leak in the fuselage** — not by the loss of a whole door or window.

RECOGNITION CUES

Fast decompression is recognised by: mist in the cabin, a blast towards the exterior of the aircraft, and expansion of body gases. **"Mist in the cabin, pressure and temperature drop"** characterise a **fast depressurisation**. During a slow depressurisation at cruise, the cabin rate-of-climb indicator shows **a rate of climb**.

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Crew & Passenger Incapacitation

THE REAL DANGER — HYPOXIA

The great danger of depressurisation is **crew incapacitation due to hypoxia**. Depending on aircraft altitude when depressurisation occurs, loss of pressurisation can **very quickly incapacitate** crew and passengers unless they receive supplementary oxygen. The **Time of Useful Consciousness is reduced by the explosive nature** of the decompression.

Cabin Altitude Trigger Points

Table 8.1 — Decompression Warning & Oxygen Mask Deployment

Event	Cabin Altitude Trigger
Depressurization **warning** is required	When cabin altitude **exceeds 10,000 ft**
Passenger oxygen masks **drop**	When cabin altitude rises to between **13,200 ft** and **14,000 ft** — but **no higher than 15,000 ft**
Supplemental oxygen for **pilots** after emergency descent	Available for the entire flight time the cabin pressure altitude exceeds a minimum of **13,000 ft**

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Crew Actions

ACTIONS ON LOSS OF PRESSURISATION

- **Oxygen** — the crew must don oxygen equipment **as soon as possible**. In the case of a **deliberate de-pressurisation**, the crew should be **on oxygen before the de-pressurisation commences**.
- **Emergency Descent** — in an emergency depressurisation the crew will want to **descend immediately** to an altitude at which they and the passengers can breathe without supplementary oxygen — conventionally **10,000 ft**.

STRUCTURAL-FAILURE HAZARD

If the decompression is caused by structural failure (e.g. failure of a window), there is a risk of crew or passengers being **buffeted by strong winds, hit by debris, exposed to extreme cold**, or even **being sucked out of the aircraft** — another reason for **wearing a seat belt or harness when seated**.

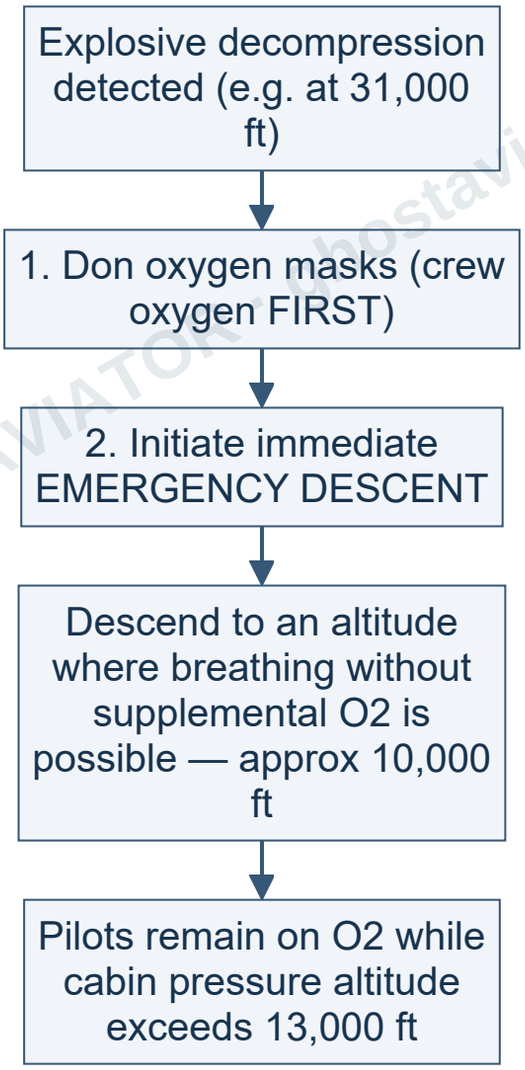

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Figure 17.6 *Explosive decompression: use the aircraft's approved emergency procedure and required crew oxygen. Any descent must respect the safe altitude for terrain and obstacles; 10,000ft is not permission to descend below that altitude.*

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EXAM HOOK

On an **explosive decompression at 31,000 ft**, the **initial action** of the operating crew is **to put on oxygen masks** — before seatbelt signs and before the MAYDAY call. Note also: using the **passenger oxygen system in severe cabin smoke is useless**, because the drop-down masks are not sealed — the **toxic cabin smoke mixes with the breathing oxygen**. Supplemental oxygen's defined purpose is to **provide people on board with oxygen during a cabin depressurisation**.

Wind Shear & Microburst

Wind shear is a sudden change in wind — and because lift depends on airspeed, a sudden change in wind is a sudden change in performance. Close to the ground, in the approach, landing or take-off phase, that is a hazard with very little margin to recover.

Wind Shear & Crew Reporting

DEFINITION

Wind shear may be described as a **change in wind direction and/or speed in space**, including up-draughts and down-draughts. Intense down-draughts — typically associated with **thunderstorms** — produce strong vertical and horizontal wind-shear components hazardous to aircraft in the approach, landing or take-off phase. One of its main characteristics is that it **can occur at any altitude, in both the vertical and horizontal planes**.

HOW TO REPORT WIND SHEAR

- Aircrew capable of reporting wind and altitude **above and below the shear layer** from the **Flight Management System (FMS)** are to do so.
- Pilots **without this equipment** should report wind shear by stating the **loss or gain of airspeed and the altitude** at which it was encountered.
- Pilots **unable to report in these specific terms** should report it in terms of its **general effect on the aircraft**.

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Significant Non-Convective Wind Shear — Thresholds

Table 9.1 — Guidelines To Establish Significant Non-Convective Wind Shear

Criterion	Threshold
Vector magnitude	Exceeding **25 kts** within **500 ft AGL**
Vector magnitude	Exceeding **40 kts** within **1 000 ft AGL**
Vector magnitude	Exceeding **50 kts** within **1 500 ft AGL**
Pilot report of IAS loss or gain	**20 kts** or more within **1 500 ft AGL**

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Microburst

DEFINITION

A microburst is a **small-scale, intense downdraft** which, on reaching the surface, **spreads outward from the down-flow centre**. This causes both vertical and horizontal wind shear that can be **extremely hazardous to all types and categories of aircraft**, particularly within **1 000 ft AGL** during approach to landing and in the take-off phases.

THE TAKE-OFF TRAP

On take-off through a microburst, the aircraft first encounters a **headwind (performance increasing)**, followed by a **downdraft**, then a **tailwind (both performance decreasing)**. The initial headwind can fool the crew into a false sense of climb performance — moments before the tailwind/downdraft robs it.

Characteristics of Microbursts

~1 NM	Diameter at 2 000 ft AGL (2–2½ NM at surface)
6 000 ft/min	Vertical winds; horizontal up to 45 KT (90 KT shear)
< 15 min	Life-cycle; peak winds last ~2–4 min
Up to 1 hr	If concentrated into a line structure

Table 9.2 — Microburst Characteristics

Characteristic	Detail
Size	Approximately **1 NM** in diameter at **2 000 ft AGL**, with a horizontal extent at the surface of approximately **2 to 2½ NM**.
Intensity	Vertical winds as high as **6 000 ft/min**. Horizontal winds giving as much as **45 KT** at the surface (i.e. **90 KT** shear).
Types	Normally accompanied by heavy rain in humid areas. In drier areas, falling raindrops may evaporate before reaching the ground — this is known as **VIRGA**.
Duration	Life-cycle seldom longer than **15 minutes**; maximum-intensity winds last approximately **2–4 minutes**. Concentrated into a line structure, activity may continue for as long as **1 hour**.

MULTIPLE MICROBURSTS

Once microburst activity starts, **multiple microbursts in the same general area are common and should be expected**. An aircraft entering the centre with a **40 kt headwind** may, on crossing through, face a windshear of **80 kt** (the headwind reverses into an equal tailwind).

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Actions To Counter Wind Shear

RECOVERY PROCEDURE

The **best defence against wind shear is to avoid it altogether** — it could be beyond your or your aircraft's capabilities. However, should you recognise a wind-shear encounter, **prompt action is required**. In all aircraft the recovery could require:

- **Full power** (set maximum take-off thrust);
- A **pitch attitude consistent with the maximum angle of attack** for your aircraft — increase pitch-up to the limit, actuating the stick shaker;
- **Keep the aeroplane's current configuration** (do not change gear/flap);
- **Warn others as soon as possible.**

EXAM HOOKS

When **penetrating wind shear**, the first magnitude to change value is **Indicated Airspeed (IAS)**. During a landing approach with an **increasing tailwind** and no pilot action, the aircraft **flies below the glide path** and has a **decreasing true airspeed**. The amount of control action required to counter wind shear is **substantial**.

Wake Turbulence

Every wing that produces lift also produces a wake. Behind a large aircraft, that wake is a pair of powerful, invisible "horizontal tornadoes" — and the separation tables exist to keep you out of them.

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How Wake Turbulence Forms

CONCEPT

All airfoils produce a wake when producing lift. The **higher-pressure air under the wing flows around the wingtip** and tries to displace the **lower-pressure (lower-energy) air on top**. The greater the pressure differential, the stronger the flow around the wingtip. The air curling up around the wingtip forms a **horizontal tornado** that trails behind the airplane and tends to **sink somewhat below the producing aircraft's flight path** (in level flight).

VORTEX BEHAVIOUR

- The vortex from the **left wing rotates clockwise**; the one from the **right wing rotates counterclockwise** (seen from behind).
- **Larger airplanes produce stronger vortices** — the wings produce more lift to support more weight.
- The vortices settle behind the aircraft to an altitude about **1 000 ft** below the flight path, where they tend to remain.
- Vortices spread laterally from the rear of the aircraft and can persist **up to 5 miles** behind, sinking **500–900 feet**.

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WAKE TURBULENCE IS GREATEST WHEN…

Wake turbulence is created **only when an airplane develops lift**. It is **greatest when the generating airplane is HEAVY, CLEAN and SLOW**. In wake turbulence, the circulation of the vortex is **outward, upward and around each wingtip**.

Aircraft Wake Turbulence Categories

Table 10.1 — Wake Turbulence Categories by MCTOW

Category	MCTOW Criterion	Examples
Super Heavy (S)	Separate designation — currently **only the Airbus A380-800**, MCTOW **5,60,000 kg**.	A380-800
Heavy (H)	**136,000 kg MCTOW or more.**	B777, B767, B747, B787, A300, A310, A330, A340, A350, DC-8, DC-10, MD-11, IL-86, IL-96, AN-225, L-1011
Medium (M)	**More than 7,000 kg and less than 136,000 kg** MCTOW.	B727, B737, B757, A320, A321, Global Express, Challenger, CRJ, Fokker Friendship, Metro 4, BAe-146, Dash 8, ATR-72, Hercules, DC-3, Saab 340
Light (L)	**Less than 7,000 kg** MCTOW.	Bandeirante, Metro 3, Cessna 402 & 421, Islander, Nomad, Piper Navajo, King Air Beech 99

FLIGHT PLAN BOX

The letter **"L"** is written in the wake-turbulence box of a flight-plan form when MCTOW is **less than or equal to 7,000 kg**.

Time-Based Longitudinal Separation Minima

Table 10.2 — ARRIVING Aircraft (Time-Based)

S.No.	Leading Aircraft	Following Aircraft	Separation Minima
1	A380-800	MEDIUM	**3 Minutes**
2	A380-800	LIGHT	**4 Minutes**
3	HEAVY	MEDIUM	**2 Minutes**
4	HEAVY or MEDIUM	LIGHT	**3 Minutes**

Table 10.3 — DEPARTING Aircraft (same runway / parallel < 760 m / specified crossing runways)

S.No.	Leading Aircraft	Following Aircraft	Separation Minima
1	A380-800	Non A380-800 HEAVY	**2 Minutes**
2	A380-800	MEDIUM or LIGHT	**3 Minutes**
3	HEAVY	LIGHT or MEDIUM	**2 Minutes**
4	MEDIUM	LIGHT	**2 Minutes**

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Departing-aircraft minima apply when aircraft use: (a) the same runway; (b) parallel runways separated by less than **760 m (2500 ft)**; (c) crossing runways if the projected flight path of the second aircraft will cross that of the first at the same altitude or less than **1000 ft** below; (d) parallel runways separated by **760 m (2500 ft)** or more, if the second aircraft's projected flight path will cross the first's at the same altitude or less than 1000 ft below.

Table 10.4 — DEPARTING from an INTERMEDIATE part of the same runway / parallel < 760 m

S.No.	Leading Aircraft	Following Aircraft	Separation Minima
1	A380-800	MEDIUM or LIGHT	**4 Minutes**
2	HEAVY	MEDIUM or LIGHT	**3 Minutes**
3	MEDIUM	LIGHT	**3 Minutes**

Displaced Landing Threshold Separation

Table 10.5 — Separation on a Runway With a Displaced Landing Threshold

S.No.	Leading Aircraft	Following Aircraft	Separation Minima
1	A380-800 Arrival	LIGHT or MEDIUM	**3 Minutes**
2	A380-800 Departure	LIGHT or MEDIUM	**3 Minutes**
3	HEAVY Arrival	LIGHT or MEDIUM Departure	**2 Minutes**
4	MEDIUM Arrival	LIGHT Departure	**2 Minutes**
5	HEAVY Departure	LIGHT or MEDIUM Arrival	**2 Minutes**
6	MEDIUM Departure	LIGHT Arrival	**2 Minutes**

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Distance-Based Wake Turbulence Separation Minima

Distance-based minima apply to aircraft provided with an ATS surveillance service in the approach and departure phases, where: (i) an aircraft is operating directly behind another at the same altitude or less than 1000 ft below; (ii) both use the same runway, or parallel runways separated by less than 760 m; or (iii) an aircraft is crossing behind another at the same altitude or less than 1000 ft below.

Table 10.6 — Distance-Based Wake Turbulence Separation Minima

Preceding Aircraft	Succeeding Aircraft	Distance-Based Separation Minima
A380-800 / Non A380-800 HEAVY	A380-800	Not required +
A380-800	Non A380-800 HEAVY	**6 NM** *
	MEDIUM	**7 NM** *
	LIGHT	**8 NM** *
HEAVY	HEAVY	**4 NM** *
	MEDIUM	**5 NM** *
	LIGHT	**6 NM** *

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Preceding Aircraft	Succeeding Aircraft	Distance-Based Separation Minima
MEDIUM	HEAVY	Not required +
	MEDIUM	Not required +
	LIGHT	**5 NM** *
LIGHT	HEAVY	Not required +
	MEDIUM	Not required +
	LIGHT	Not required +

READING THE SYMBOLS

+ — When a wake-turbulence restriction is not required, separation **reverts to the prescribed** ***(laid down by the authority or the rules)*** **radar separation minimum**.
***** — Where the prescribed radar separation minimum is **more than** the applicable wake-turbulence minimum, the **prescribed radar separation minimum has precedence**.

EXAM HOOKS — SEPARATION

- Shortest distance for a **HEAVY preceding a LIGHT** (landing sequence): **6 NM**.
- MEDIUM taking off behind a HEAVY, same runway, wake criteria: **2 minutes**.
- LIGHT taking off behind a MEDIUM, same runway: **2 minutes**.
- Arriving aircraft, timed approaches — MEDIUM behind HEAVY: **2 minutes**; MEDIUM behind MEDIUM: **2 minutes**.

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Opposite-Direction Runway Separation

LOW / MISSED APPROACH CASES

A separation minimum applies between a LIGHT or MEDIUM and a HEAVY aircraft, and between a LIGHT and a MEDIUM aircraft, when the heavier aircraft is making a **low or missed approach** and the lighter aircraft is landing on the same runway in the opposite direction (or on a parallel opposite-direction runway) separated by **less than 760 m**.

Where the lighter aircraft is instead **utilising an opposite-direction runway for take-off**, this minimum is **2 minutes**.

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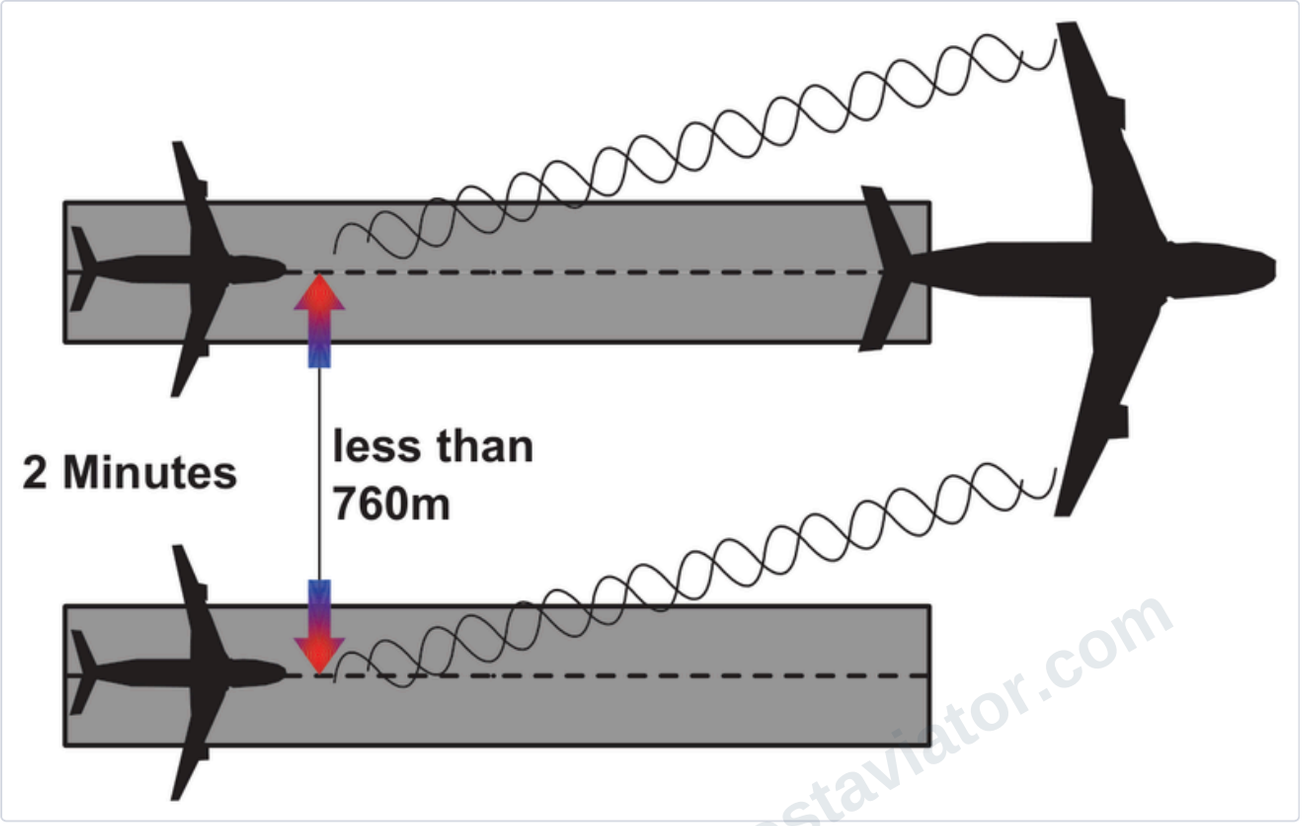

Figure 17.7 *Parallel runways less than 760 m apart: 2 minutes between a heavier aircraft and a lighter one departing from the parallel runway.*

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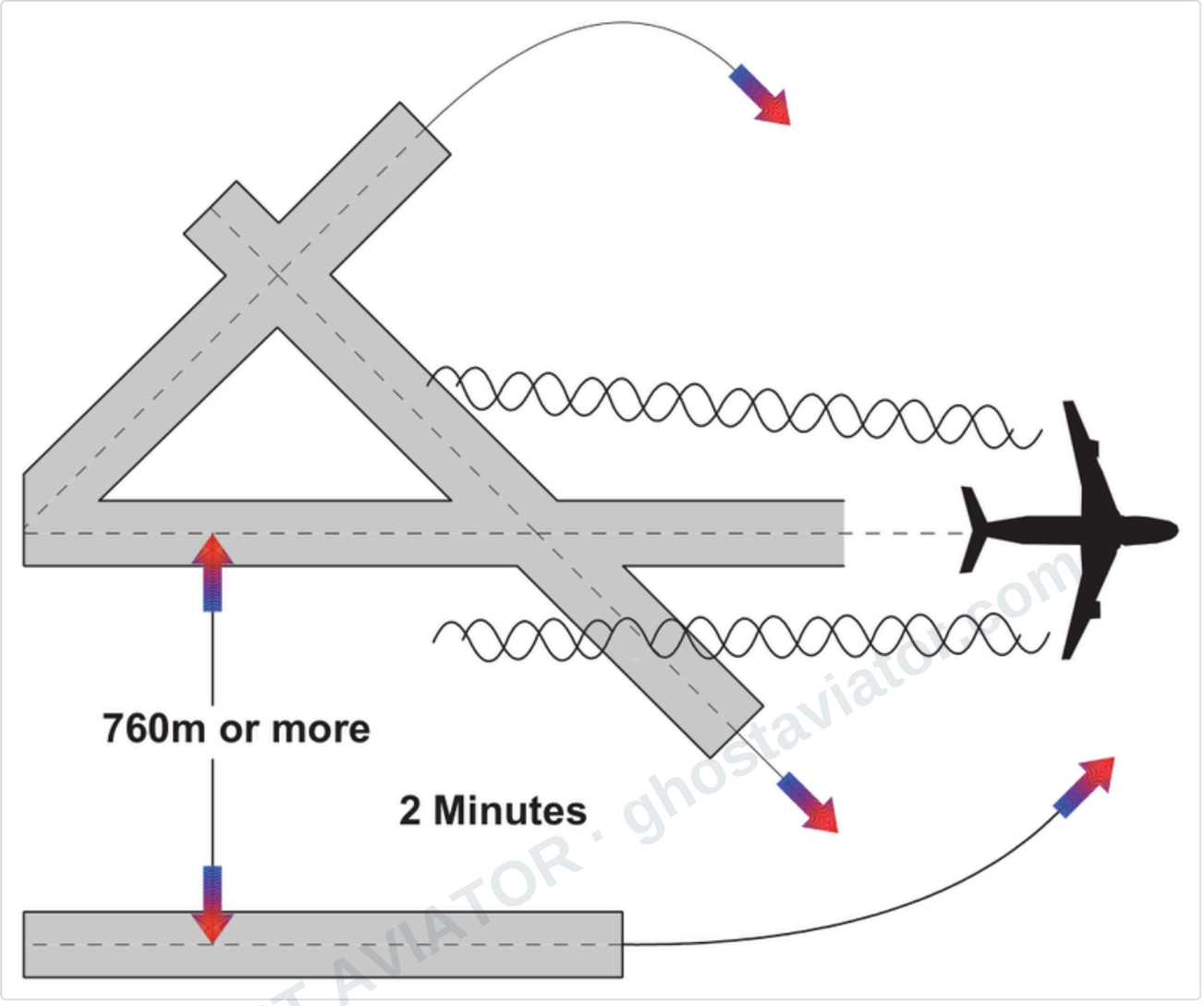

Figure 17.8 *Runways 760 m or more apart, and crossing runways: the flight paths cross, so 2 minutes applies.*

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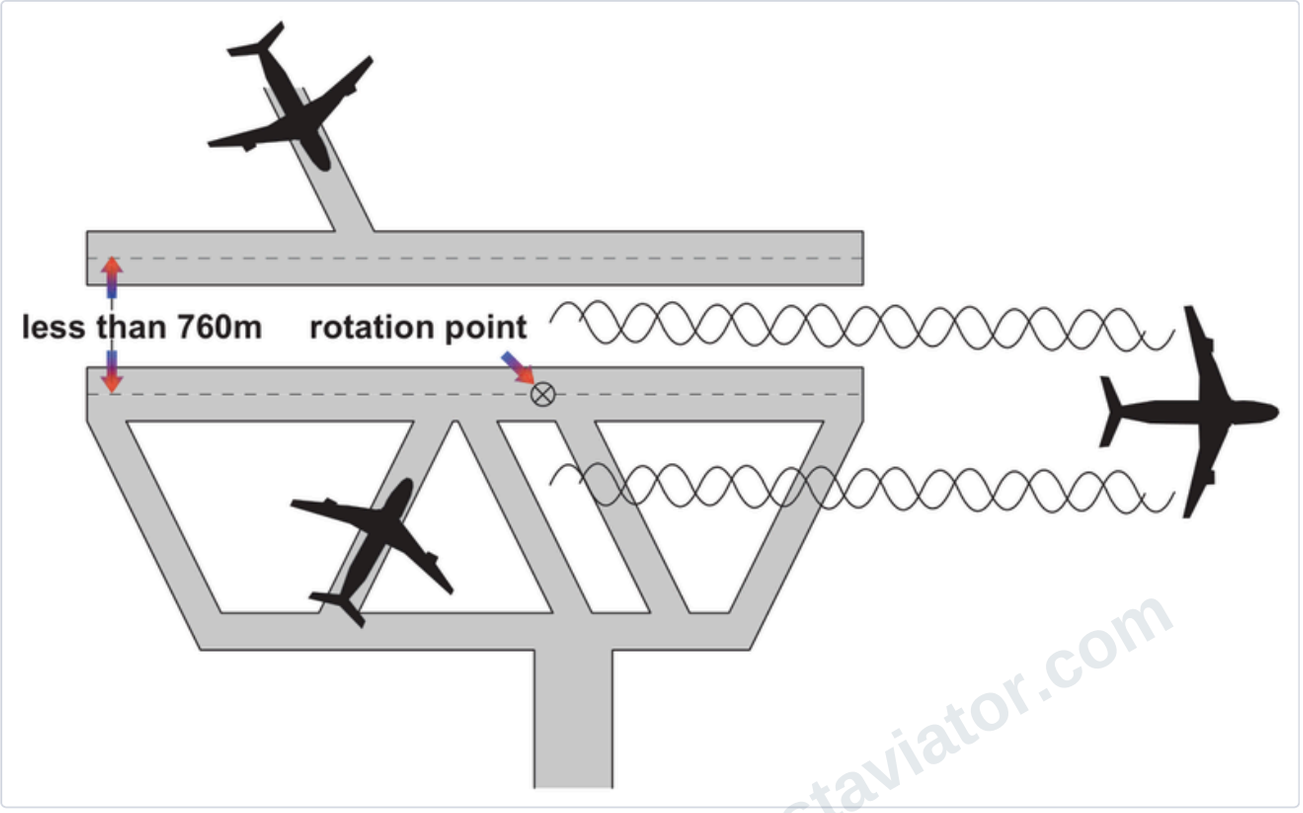

Figure 17.9 *Take-off from an intermediate part of the runway: the vortices of the heavier aircraft are carried beyond its rotation point.*

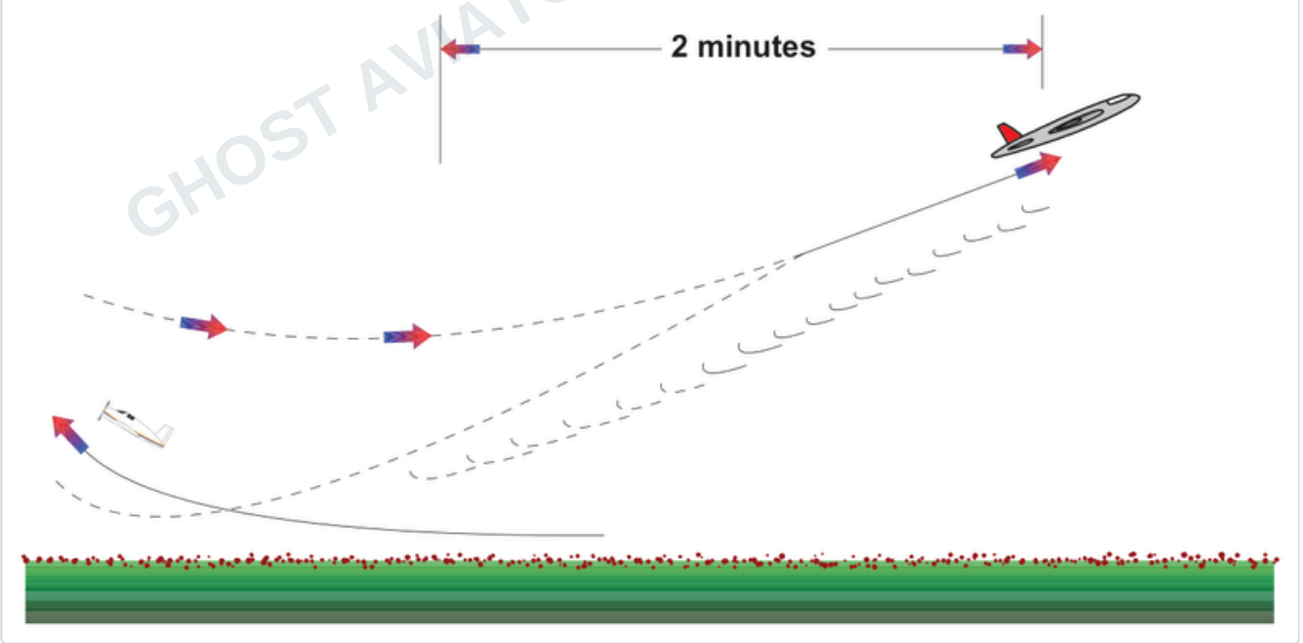

Figure 17.10 *Vertical view: a light aircraft must stay clear of, and preferably above, the flight path of a heavier departing aircraft; 2 minutes separation.*

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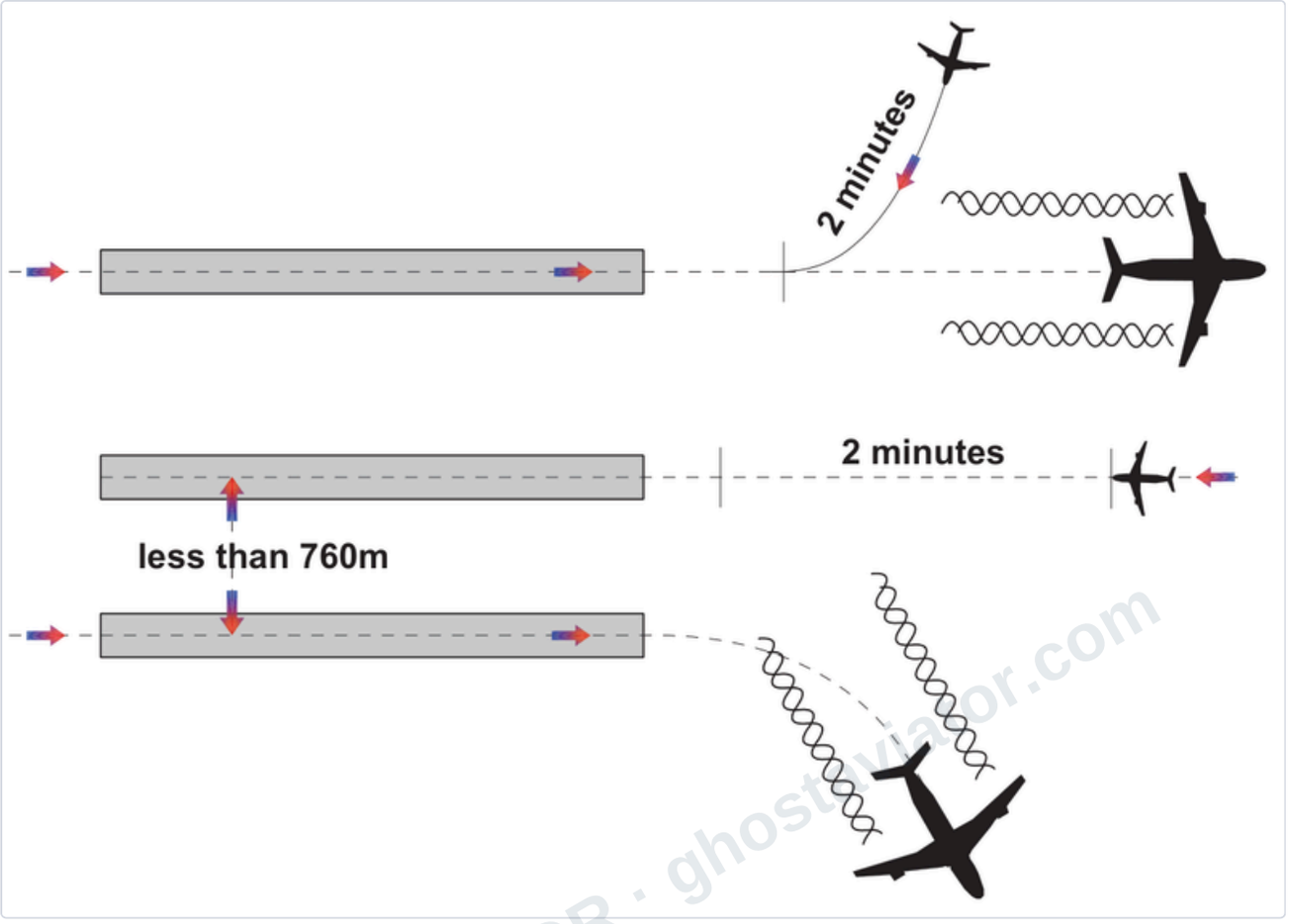

Figure 17.11 *Three 2-minute cases: same-runway departure behind a heavier aircraft, opposite-direction approach, and a parallel runway less than 760 m away.*

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Dealing With Wake Turbulence

HANDLING WAKE TURBULENCE — PRACTICAL RULES

- **Fly upwind** rather than downwind of the flight path of a large aircraft.
- The most dangerous wind while landing/taking off behind a large aircraft is a **light quartering tailwind**.
- When **taking off behind a larger aircraft**, lift off **before the larger aircraft's lift-off point** and climb **above its climb path**.
- When **landing behind a larger aircraft**, touch down **past its touchdown point**.
- Vortices stay on the runway longer in **light or calm winds**.
- In **light crosswinds**, the upwind vortex tends to drift onto and remain on the runway.
- Vortices from a large aircraft landing on a **parallel runway** can be blown onto your runway by a strong crosswind.
- When flying or crossing behind a larger aircraft, maintain your flight path **above** the large aircraft's path rather than below.
- When approaching behind a larger aircraft on an **ILS**, stay **at or above the glide slope** to avoid wake encounters.

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WHEN IS WAKE TURBULENCE RISK HIGHEST?

Risk is highest when a **heavy aircraft has just performed a take-off at a closely situated parallel runway with a light crosswind** — the crosswind drifts the heavy's vortices onto your path.

Security — Acts of Unlawful Interference

REPORTING UNLAWFUL INTERFERENCE

Following an act of unlawful interference on board an aeroplane, the **commander** — or in his absence the **operator** — shall submit, **without delay**, a report of such an act to the **designated local authority** AND the **Authority in the State of the operator**.

NOTIFICATION AFTER LANDING

In addition to informing each State whose citizens are known to be on board, the State of the country in which an aircraft has landed after an act of unlawful interference must **immediately notify**: the **State of Registry of the aircraft, the State of the operator, and ICAO**.

RESPONSIBILITY FOR PASSENGER & CREW SAFETY

The **Contracting State in which the unlawful interference occurs** has the responsibility to take adequate measures for the safety of passengers and crew of an aircraft subjected to an act of unlawful interference, until their journey can be continued.

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Flight Deck Door

FLIGHT CREW COMPARTMENT DOOR

When a commercial transport passenger airplane is equipped with a flight-crew compartment door, that door **must include a locking system to prevent any unauthorized access**. The flight-deck door should be capable of being **locked from within the compartment**.

CRITICAL CODE

In case of a **hi-jack**, the transponder squawk code is **7500**. When flight crew members are at their duty stations, they must **keep the seat belts fastened**.

Emergency & Precautionary Landings, Ditching

Table 12.1 — Types of Non-Routine Landing

Term	Definition
Forced Landing	An **immediate landing**, on or off an airport, necessitated by the **inability to continue further flight**. Typical example — an airplane forced to land due to engine failure.
Precautionary Landing	A **premeditated landing**, on or off the airport, when **further flight is possible but inadvisable**. Examples — deteriorating weather, being lost, fuel shortage, gradually developing engine trouble.
Ditching	A **forced or precautionary landing on water**.

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RESPONSIBILITY & DITCHING CONDUCT

- In the event of a **precautionary landing** (a premeditated landing with time in hand), **ATC** is responsible for alerting the emergency services.
- **If ditching is inevitable:** the use of life jackets is to be **reiterated before the ditching**.
- **In case of ditching, cabin attendants will:** have the passengers **embark directly into the life rafts**; **prevent passenger movements** which may impede the airplane's flotation ability; and **ensure complete evacuation** of the airplane.

LIFE JACKETS

Life jackets should **NOT be inflated inside the aircraft** — an inflated jacket can trap an occupant against the ceiling of a flooding cabin and block escape. They are inflated only on leaving the aeroplane.

Fuel Jettisoning

An aircraft that must land far heavier than its maximum landing weight may dump fuel to get down to a safe landing mass. It is a controlled, coordinated procedure — not a panic action.

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GENERAL & PURPOSE

An aircraft in an **emergency or other urgent situation** may need to **dump fuel so as to reduce to maximum landing weight**, in order to effect a safe landing.

ATC COORDINATION

When an aircraft within controlled airspace requires to dump fuel, the flight crew **shall advise ATC**. The ATC unit should then coordinate with the flight crew:

- (a) The **route to be flown** — if possible clear of cities and towns, preferably over water, and away from areas where thunderstorms have been reported or are expected;
- (b) The **level to be used** — which should be **not less than 1800 m (6000 ft)**;
- (c) The **duration** of the fuel dumping.

Separation From Other Traffic

SEPARATION FROM AN AIRCRAFT DUMPING FUEL

Other known traffic should be separated from the aircraft dumping fuel by:

- (a) At least **19 km (10 NM)** horizontally — but **not behind** the aircraft dumping fuel;
- (b) **Vertical separation** if behind the aircraft dumping fuel within **15 minutes** flying time or a distance of **93 km (50 NM)** by:
  - (i) at least **300 m (1000 ft)** if **above** the aircraft dumping fuel; and
  - (ii) at least **900 m (3000 ft)** if **below** the aircraft dumping fuel.

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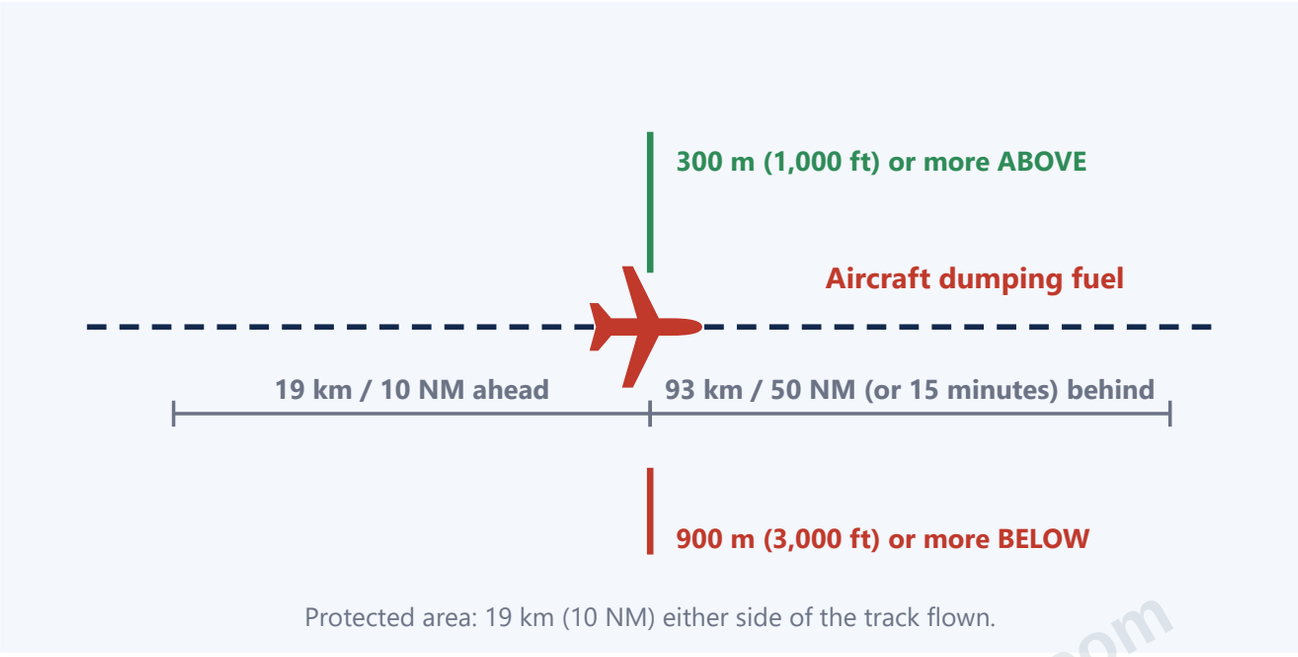

Figure 17.12 *Aircraft dumping fuel*

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COMMUNICATIONS & TEST REQUIREMENTS

If the aircraft will maintain **radio silence** during fuel dumping, the frequency to be monitored and the time when radio silence will terminate should be **agreed**.

Requirements that must be shown to exist during fuel-jettisoning tests:

- The fuel-jettisoning system and its operation are **free from fire hazard**;
- The **fuel discharges clear of any part of the aeroplane**;
- **Fuel fumes do not enter any part of the aeroplane**;
- The jettison operation **does not adversely affect the controllability of the aeroplane**.

EXAM HOOK

Fuel jettison should normally be carried out **above 6,000 ft AGL** — but it may be done **anywhere if unavoidable**.

Transport of Dangerous Goods

CONTEXT

More than half of the cargo carried by all modes of transport in the world is **dangerous cargo** — explosive, corrosive, flammable, toxic and even radioactive. These goods are essential for industrial, commercial, medical and research processes, and a great deal is carried by aircraft.

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GOODS FORBIDDEN FOR AIR TRANSPORT (UNLESS EXEMPTED)

The following shall be **forbidden on aircraft unless exempted** by the States concerned:

- (a) Articles and substances **identified in the Technical Instructions as forbidden for transport in normal circumstances**; and
- (b) **Infected live animals.**

DOCUMENTS & DEFINITIONS

- A list of dangerous goods which may not be transported by air is found in the **Technical Instructions for the Safe Transport of Dangerous Goods by Air**.
- Products/materials are considered dangerous goods if defined as such by the **ICAO document "Technical Instructions for the safe transportation of dangerous goods by air."**
- **ICAO Annex / Appendix 18** deals with the **safety of the air transport of hazardous materials**.
- To carry hazardous materials on board a public transport airplane (per CAR-OPS), they must be accompanied by a **transport document for hazardous materials**, which — if required — shall be **drawn up by the shipper**.
- General information, instructions and recommendations on transporting hazardous materials are specified in the **Operation Manual**.
- Dangerous goods **may be carried in the passenger cabin or on the flight deck — but only goods specified in the Technical Instructions**.

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Contaminated Runways & Hydroplaning

When the runway is wet or contaminated, the friction available to stop the aircraft drops — and so does your safety margin. Knowing the exact definitions and the landing-distance factor is essential.

Braking Action & Brake Coefficient

CONCEPT

Runway surface friction is **directly relevant to the braking action** available to an aircraft decelerating after touchdown or after a decision to reject a take-off. Aircraft braking coefficient depends on the surface friction between the runway and the aircraft tyres — **less friction means less braking coefficient and less braking**.

For most multi-crew aircraft, **anti-skid braking systems** are fitted — these prevent wheel locking and allow more aggressive brake input on wet/slippery runways **without inducing dynamic or viscous aquaplaning**.

REPORTING BRAKING EFFICIENCY

Braking efficiency is information presented as a **combination of the terms: poor, medium, good**.

Runway Surface State — Exact Definitions

Table 15.1 — Runway Surface Conditions

Condition	Definition
Damp	The surface is **not dry**, and surface moisture **does not give a shiny appearance**.

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Condition	Definition
Wet	Covered with water (or loose/slushy snow) **less than or equal to the equivalent of 3 mm of water**; OR surface moisture is sufficient to make it **reflective but does not create large stagnant sheets of water**.
Contaminated	A runway covered with **4 mm** thick water is said to be **contaminated**.

WET RUNWAY LANDING DISTANCE

An operator shall ensure that when weather reports/forecasts indicate the runway may be **wet** at the estimated time of arrival, the **landing distance available is at least 115% of the required landing distance**. Equivalently — if the flight manual gives no specific wet-runway data, the required landing distance on a dry runway must be **increased by 15%**.

Dynamic Hydroplaning (Aquaplaning)

DEFINITION

Aquaplaning (also known as **hydroplaning**) is a condition that exists when landing on a surface with **standing water deeper than the tread depth of the tyres**. When the brakes are applied, the brake may lock up and the tyre rides on the surface of the water, much like a water ski.

WHY HYDROPLANING IS DANGEROUS

When the tyres are hydroplaning, **directional control and braking action are virtually impossible**. An effective **anti-skid system can minimize the effects** of hydroplaning.

Table 15.2 — Types of Hydroplaning

Type	Primary Cause
Dynamic hydroplaning	Depends primarily on the **depth of the standing water** on the runway (and aeroplane speed along the runway lifting the tyre).
Viscous hydroplaning	Caused by a **smooth and dirty runway surface**.

EXAM HOOK

The effect whereby a tyre is **lifted from the runway due to aeroplane speed** along the runway is known as **hydroplaning**.

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PART V

Communication, Safety and Security

Radio language, accident investigation, border formalities and protecting aviation from unlawful acts.

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CHAPTER 18

Aeronautical Communications

18

VFR Communications & Definitions

WHAT THIS CHAPTER COVERS

Radiotelephony (RTF) procedures used for two-way air-ground voice communication between aircraft and Air Traffic Services (ATS) units, including standard phraseology, Q-codes, distress procedures, weather messages, and the underlying radio-wave theory.

Commonly Used Abbreviations Specific to R/T

Certain abbreviations have become so embedded in aviation terminology that they are **spoken using their constituent letters** rather than the phonetic spelling alphabet.

SPOKEN AS LETTERS (NOT PHONETICALLY)

Examples: **ILS**, **QNH**, **RVR**, **VOR**, **DME**, **ATIS**, **METAR**, etc.
i.e. say "I-L-S" — not "India Lima Sierra".

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Q-Code Groups Used in RTF Air-Ground Communications

DEFINITION — Q-CODE

A Q-code is a three-letter group beginning with **"Q"**, used as a shorthand to either **ask a question** or **give an answer/advice**. Same code carries both meanings depending on whether it's transmitted as a question or response.

Q-Code	Question	Answer / Advice
QDL	Do you intend to ask me for a series of bearings?	I intend to ask you for a series of bearings.
QDM	Will you indicate the **MAGNETIC heading** for me to steer towards you (no wind)?	The MAGNETIC heading for you to steer to reach me (no wind) was … degrees (at … hours).
QDR	What is my **MAGNETIC bearing** from you?	Your MAGNETIC bearing from me was … degrees (at … hours).
QFE	What should I set on the altimeter sub-scale so the instrument indicates **height above the reference elevation**?	If you set the sub-scale to … millibars, the altimeter will indicate height above aerodrome elevation (above threshold, runway no. …).

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Q-Code	Question	Answer / Advice
QFU	What is the magnetic direction (or number) of the runway to be used?	The magnetic direction (or number) of the runway in use is … *Note: Runway number = two-figure group; magnetic direction = three-figure group.*
QGH	May I land using … (procedure or facility)?	You may land using … (procedure or facility).
QNE	What indication will my altimeter give on landing at … (place) at … hrs, with sub-scale set to **1013.2 mb (29.92 in)**?	On landing at … your altimeter will indicate … (figures and units).
QNH	What should I set on the sub-scale so the altimeter indicates **elevation** if my aircraft were on the ground at your station?	Set sub-scale to … millibars; altimeter will indicate elevation if on the ground at my station at … hrs. *Note: If setting is given in hundredths of an inch, abbreviation "INS" is used.*
QTE	What is my **TRUE bearing** from you? / from (call sign)? / TRUE bearing of (call sign) from (call sign)?	Your TRUE bearing from me is … degrees at … hours.
QUJ	Will you indicate the **TRUE track** to reach you?	The TRUE track to reach me is … degrees at … hours.

Visual Reference — QDM vs QDR vs QTE

Starting point / condition	Branch	Next step / result
Aircraft	QDM: Mag HDG TO stn	Ground Station
Ground Station	QDR: Mag BRG FROM stn	Aircraft
Ground Station	QTE: TRUE bearing FROM stn	Aircraft
Aircraft	QUJ: TRUE track TO stn	Ground Station

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Categories of Messages (Priority Order)

PRIORITY ORDER — STRICT HIERARCHY

Messages in the Aeronautical Mobile Service follow this **strict priority order**. Higher-priority messages always take precedence over lower-priority traffic.

Priority	Category	Signal / Prefix
a)	Distress messages, distress calls, distress traffic	**MAYDAY MAYDAY MAYDAY**
b)	Urgency messages (incl. medical transports)	**PAN PAN** / **PAN PAN MEDICAL** / SECURITE
c)	Communications relating to direction finding	—
d)	Flight safety messages	—
e)	Meteorological messages	—
f)	Flight regularity messages	—

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General Operating Procedures — Pronunciation Standards

STANDARD ICAO PRONUNCIATION

Numbers are transmitted using **specific ICAO phonetic pronunciations** to avoid confusion over noisy or distorted radio links. Memorise these — they are heavily tested.

Number Pronunciation

Digit	Pronounced as	Digit	Pronounced as
0	**ZE-RO**	5	**FIFE**
1	**WUN**	6	**SIX**
2	**TOO**	7	**SEV-en**

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Digit	Pronounced as	Digit	Pronounced as
3	**TREE**	8	**AIT**
4	**FOW-er**	9	**NIN-er**
Decimal	**DAY-SEE-MAL**	Hundred	**HUN-dred**
Thousand	**TOU-SAND**		

Examples — Aircraft Callsigns, Levels, Headings, Wind

Application	Example	Transmitted as	Pronounced as
Aircraft callsign	AI 235	Air India 235	Air India TOO TREE FIFE
	6E 146	IFLY 146	IFLY WUN FOWer SIX
Flight levels	FL 180	flight level one eight zero	flight level WUN AIT ZE-RO
	FL 200	flight level two zero zero	flight level TOO ZE-RO ZE-RO
	FL 70	flight level seven zero	flight level SEVen ZE-RO
Headings	150	heading one five zero	heading WUN FIFE ZE-RO
	080	heading zero eight zero	heading ZERO AIT ZE-RO
	300	heading three zero zero	heading TREE ZE-RO ZE-RO
Wind direction & speed	020° / 70 kt	wind zero two zero degrees seven zero knots	wind ZE-RO TOO ZE-RO degrees SEVen ZE-RO knots
	100° / 18 kt	wind one zero zero degrees one eight knots	wind WUN ZE-RO ZE-RO degrees WUN AIT knots
	210° / 18 G 30 kt	wind two one zero degrees one eight knots gusting three zero knots	wind TOO WUN ZE-RO degrees WUN AIT knots gusting TREE ZE-RO knots
Runway designator	19	runway one nine	runway WUN NINer
	06	runway zero six	runway ZE-RO SIX
	23L	runway two three left	runway TOO TREE left
Mach number	0.84	Mach decimal eight four	Mach DAY-SEE-MAL AIT FOWer
Altimeter setting	984 hPa	QNH nine eight four	QNH NINer AIT FOWer
	1027 hPa	QNH one zero two seven	QNH WUN ZE-RO TOO SEVen
Frequencies	128.3 MHz	one two eight decimal three	WUN TOO AIT DAY-SEE-MAL TREE
	135.75 MHz	one three five decimal seven five	WUN TREE FIFE DAY-SEE-MAL SEVen FIFE
	5643 kHz	five six four three	FIFE SIX FOWer TREE

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Transmission of Numbers — Hundreds & Thousands Rule

STRICT DGCA / ICAO RULE

All numbers used in transmission of **altitude, height, cloud height, visibility, and runway visual range (RVR)** which contain **whole hundreds and whole thousands** shall be transmitted by:

1. Pronouncing **each digit** in the number of hundreds or thousands,
2. Followed by the word **HUNDRED** or **THOUSAND** as appropriate.

Combinations of thousands and whole hundreds: pronounce each digit of the thousands followed by "THOUSAND", then the hundreds digit followed by "HUNDRED".

Number	Correct Transmission
800	EIGHT HUNDRED
3 500	THREE THOUSAND FIVE HUNDRED
4 500	FOUR THOUSAND FIVE HUNDRED
11 000	ONE ONE THOUSAND
13 500	ONE THREE THOUSAND FIVE HUNDRED

WATCH-OUT — FLIGHT LEVELS USE A DIFFERENT RULE

Flight Levels are **NOT** spoken with "thousand/hundred". Each digit is pronounced separately.
Example: FL 180 → **flight level WUN AIT ZE-RO** (NOT "flight level one eighty").

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Transmission of Time

TIME TRANSMISSION SOP

- When transmitting time, normally **only the minutes of the hour** are required.
- However, the **hour shall be included** if any possibility of confusion exists.
- Time checks shall be given to the **nearest half minute**.
- **Coordinated Universal Time (UTC)** shall be used at all times, unless otherwise specified.
- **2400** hours designates midnight (end of day); **0000** hours = beginning of day.

Time	Transmitted as (minutes only)	Transmitted as (hour + minutes)
0920 (9:20 AM)	TOO ZE-RO	ZE-RO NINer TOO ZE-RO
1643 (4:43 PM)	FOWer TREE	WUN SIX FOWer TREE
175321 (5:53:21 PM)	FIFE TREE AND A HALF	WUN SEVEN FIFE TREE AND A HALF

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Standard ICAO Phraseology (A–W)

WHY STANDARD PHRASEOLOGY?

Standard phraseology eliminates ambiguity, reduces RTF time, and ensures every pilot and controller worldwide understands the same word the same way. **Deviation can be safety-critical.**

ACKNOWLEDGE	"Let me know that you have received and understood this message."
AFFIRM	"Yes."
APPROVED	"Permission for proposed action granted."
BREAK	"I hereby indicate the separation between portions of the message." — *used where there is no clear distinction between the text and other portions of the message.*
BREAK BREAK	"I hereby indicate the separation between messages transmitted to different aircraft in a very busy environment."
CANCEL	"Annul the previously transmitted clearance."
CHANGING TO	When transferring to a pilot-to-controller channel — Aircraft: "CHANGING TO ..." (ATS unit concerned).
CHECK	"Examine a system or procedure." *Not to be used in any other context. No answer is normally expected.*
CLEARED	"Authorized to proceed under the conditions specified."
CLEARED FOR IMMEDIATE TAKE-OFF	"Taxi immediately to runway and commence take-off without stop."
CLIMB	"Climb to a FL, Altitude or Height."
CONFIRM	"I request verification of: *clearance, instruction, action, information*."
CONTACT	"Establish communications with ..."
CORRECT	"True" or "Accurate".
CORRECTION	"An error has been made in this transmission (or message indicated). The correct version is ..."
CORRECTION, I SAY AGAIN	If a correction can best be made by repeating the entire message, the operator shall use the phrase "CORRECTION, I SAY AGAIN" before transmitting the message a second time.
DESCEND	"Descend to a FL, Altitude or Height."
DISREGARD	"Ignore."
HOLD SHORT	"Stop before reaching the specified location." *Only used in limited circumstances where no defined Point exists (e.g. where there is no suitably located holding point), or to reinforce a clearance limit.*
HOW DO YOU READ	"What is the readability of my transmission?"

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I SAY AGAIN	"I repeat for clarity or emphasis."
LANDED	After landing — Aircraft: "LANDED ... (location) ... (time)".
MAINTAIN	"Continue in accordance with the condition(s) specified" or in its literal sense, e.g. "Maintain VFR".
MONITOR	"Listen out on (frequency)."
NEGATIVE	"No" or "Permission not granted" or "That is not correct" or "Not capable".
NEGATIVE, I SAY AGAIN	If, in checking the correctness of a read-back, an operator notices incorrect items, he shall transmit the words "NEGATIVE I SAY AGAIN" at the conclusion of the read-back followed by the correct version of the items concerned.
OPERATIONS NORMAL	When "operations normal" reports are transmitted by aircraft, they should consist of the prescribed *(laid down by the authority or the rules)* call followed by the words "OPERATIONS NORMAL".
OVER	"My transmission is ended, and I expect a response from you." *Not normally used in VHF communications.*
OUT	"This exchange of transmissions is ended and no response is expected." *Not normally used in VHF communications.*
READ BACK	"Repeat all, or the specified part, of this message back to me exactly as received."
RECLEARED	"A change has been made to your last clearance and this new clearance supersedes your previous clearance or part thereof."
REPORT	"Pass me the following information ..."
REQUEST	"I should like to know ..." or "I wish to obtain ..."
ROGER	"I have received all of your last transmission." *Under no circumstances to be used in reply to a question requiring "READ BACK" or a direct answer in the affirmative (AFFIRM) or negative (NEGATIVE).*
SAY AGAIN / SAY AGAIN (item) / SAY AGAIN all after / SAY AGAIN all before / SAY AGAIN all between ... and ...	"Repeat all, or the following part, of your last transmission."
SPEAK SLOWER	"Reduce your rate of speech."
STANDBY	"Wait and I will call you." *The caller would normally re-establish contact if the delay is lengthy. STANDBY is not an approval or denial.*
UNABLE / IMPOSSIBLE	"I cannot comply with your request, instruction or clearance" (normally followed by a reason). / "To indicate that a request cannot be complied with, or that a requested manoeuvre cannot be executed."
VERIFY	"Confirm from originator."
WILCO *(Abbreviation for "will comply")*	"I understand your message and will comply with it."

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WORDS TWICE	**a) As a request:** "Communication is difficult. Please send every word, or group of words, twice." **b) As information:** "Since communication is difficult, every word, or group of words, in this message will be sent twice."

IMPORTANT DELETION — "GO AHEAD"

The phrase **"GO AHEAD"** has been **deleted**. In its place, the use of the **calling aeronautical station's call sign followed by the answering aeronautical station's call sign** shall be considered the invitation to proceed with transmission by the station calling (Annex 10 Vol II 5.2.1.7.3.2.3). The phrase "GO AHEAD" is still used in one place: an aircraft replying to a SELCAL call.

Call Signs for Aeronautical Stations

SUFFIX SYSTEM

Every ATS unit has a defined RTF call-sign suffix that tells the pilot exactly what type of service is being provided. The aerodrome/area name precedes the suffix (e.g. "Delhi Tower", "Mumbai Approach").

Unit / Service Available	Call Sign Suffix
Area Control Centre	**CONTROL**
Upper / Lower Control Area	**UPPER / LOWER CONTROL**
Approach Control	**APPROACH**
Approach Control Radar / Area Control Radar	**APPROACH RADAR / CONTROL RADAR**
Aerodrome Control	**TOWER**
Surface Movement Control	**GROUND**
Radar (in general)	**RADAR**
Precision Approach Radar	**PRECISION**
Direction-finding Station	**HOMER**
Flight Information Service	**INFORMATION**
Clearance Delivery	**DELIVERY**
Apron Control	**APRON**
Company Dispatch	**DISPATCH**
Aeronautical Station	**RADIO**
Flow Control	**FLOW**

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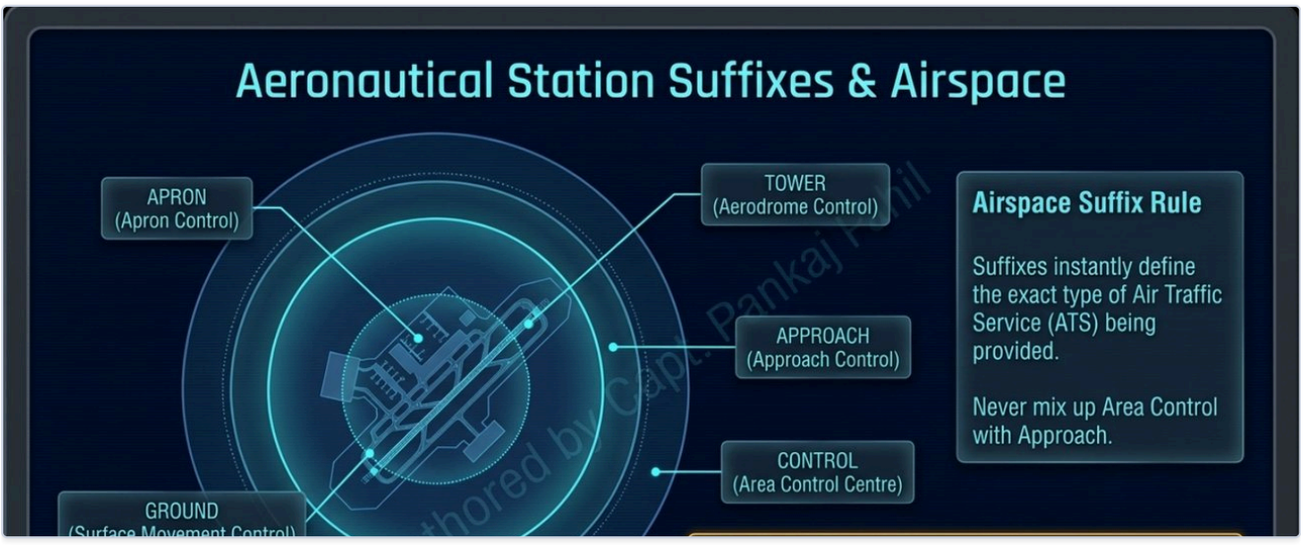

Figure 18.1 *Call sign suffixes of aeronautical stations*

Read-Back & Acknowledgement Requirements

MANDATORY READ-BACK ITEMS

The flight crew **shall read back** to the air traffic controller the **safety-related parts** of clearances and instructions transmitted by voice. Items NOT listed below may be acknowledged by aircraft call sign or by an abbreviated read-back.

QNH IS ALWAYS READ BACK.

The following items shall **always be read back**:

a. **ATC route clearance**;

b. **Clearances and instructions to enter, land on, take off on, hold short of, cross, taxi and back-track on any runway**; and

c. **Runway-in-use, altimeter settings, SSR codes, level instructions, heading and speed instructions** and — whether issued by the controller or contained in ATIS broadcast — **transition levels**.

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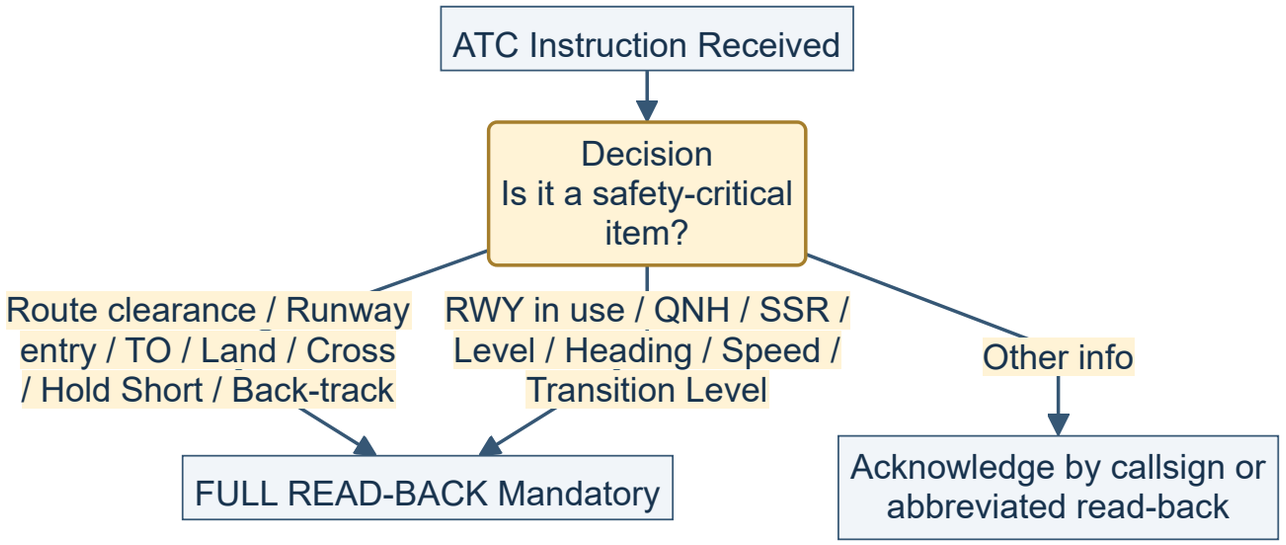

Transfer of Communication

STANDARD PROCEDURE

- An aircraft will **normally be advised** by the appropriate aeronautical station to change from one radio frequency to another in accordance with agreed procedures.
- **In the absence of such advice**, the aircraft shall **notify** the aeronautical station before such a change takes place.
- Aircraft flying in **controlled airspace must obtain permission** from the controlling authority before changing frequency.
- An aircraft may be instructed to **standby** on a frequency when the ATSU will initiate communications, and to **monitor** a frequency on which information is being broadcast.

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Readability Scale & Test Calls

TEST TRANSMISSION TIME LIMIT

All radio transmissions for test purposes shall be of the **minimum duration** necessary for the test and shall **NOT continue for more than 10 seconds**. The recurrence of such transmissions shall be kept to the minimum necessary for the test.

Readability Scale (1–5)

Scale	Readability
1	Unreadable
2	Readable now and then
3	Readable but with difficulty
4	Readable

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Scale	Readability
5	Perfectly readable

Aircraft Call Signs — Full & Abbreviated

Full Call Sign Types

Type	Description	Example
a)	Character corresponding to the registration marking of the aircraft.	**VTEJP** or **CESSNA VTEJP**
b)	Telephony designator of the aircraft operating agency, followed by the **last four characters** of the registration marking.	**AIRINDIA TEJP**
c)	Telephony designator of the aircraft operating agency, followed by the flight identification.	**AIRINDIA 809**

Abbreviated Call Signs

WHEN CAN YOU ABBREVIATE?

After **satisfactory communication has been established**, and provided **no confusion is likely to occur**, aircraft call signs may be abbreviated as follows:

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Type	Description	Example
a)	The **first character** of the registration AND at least the **last two characters** of the call sign.	**VJP** or **CESSNA VJP**
b)	Telephony designator of the operating agency, followed by at least the **last two characters** of the call sign.	**AIRINDIA JP**
c)	Telephony designator of the operating agency, followed by the flight identification.	**NO ABBREVIATED FORM**

Radar Procedural Phraseology

Radar Identification of Aircraft

IDENTIFICATION ≠ RADAR SERVICE

An aircraft must be identified before it can be provided with a radar service. However, **the act of identifying an aircraft is not a service in itself** — pilots should not assume they are receiving a radar service, particularly when flying outside controlled airspace.

When a controller has identified an aircraft, he will inform the pilot, according to circumstances, of the following:

- That the aircraft is **identified**, and
- Of the **position** of the aircraft.

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The pilot will be **warned if identification is lost**, or about to be lost, and appropriate instructions given.

Secondary Surveillance Radar (SSR) Phraseology

MANDATORY ACTION

The pilot **MUST respond to SSR instructions**, and **read back specific settings**.

Radar Service

Where it is not self-evident, pilots will normally be informed by the controller when they are under **radar control**, **radar advisory**, or **information service**.

Radar Vectoring

- Aircraft may be given specific vectors to fly in order to establish separation. Pilots may be informed of the reasons for radar vectoring.
- It may be necessary for a controller to know the heading of an aircraft as separation can often be established by instructing an aircraft to continue on its existing heading.
- A controller may not know the aircraft's heading but does require the aircraft to fly a particular heading.
- When vectoring is complete, pilots will be instructed to **resume their own navigation**, given position information and appropriate instructions as necessary.
- Occasionally an aircraft may be instructed to make a **complete turn** (known as an **ORBIT** or a **360-degree turn**) or given a heading, for delaying purposes or to achieve required spacing behind preceding traffic.

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Traffic Information & Avoiding Action Phraseology

STANDARD TRAFFIC INFORMATION FORMAT

Whenever practicable, information regarding traffic on a possible conflicting path should be given in the following form:

a. **Relative bearing** of the conflicting traffic in terms of the **12-hour clock**; or, if the aircraft under service is established in a turn, the **relative position** of the conflicting traffic in relation to **cardinal points** (i.e. northwest, south etc.);

b. **Distance** from the conflicting traffic;

c. **Direction of flight** of the conflicting traffic; and

d. **Relative speed** of the conflicting traffic or the **type of aircraft** and **level** if known.

RELATIVE MOVEMENT TERMS

Relative movement should be described using one of the following terms as applicable:

closing · converging · parallel · same direction · opposite direction · diverging · overtaking · crossing left to right · crossing right to left

If level is known: **1000 ft above/below**.

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AVOIDING ACTION

Avoiding action is given by the controller when an **imminent risk of collision** will exist if action is not taken immediately. The controller will inform the pilot when the conflict **no longer exists**.

Relevant Weather Information Terms (VFR)

Aerodrome Weather Sources

A pilot may obtain weather through many sources such as: request from **ATC**, **Meteorological Offices**, **ATIS**, **VOLMET**, etc.

Current Weather Messages & Forecasts

Term	Meaning
METAR	Aviation Routine Weather Report
SPECI	Aviation Selected Special Weather Report
TREND	Expected changes in the next **2 hours**
TAF	Aerodrome Forecast
ROFOR	Route Forecast
SIGMET	Information issued by a meteorological watch office concerning the occurrence or expected occurrence of specified **en-route weather phenomena** which may affect the safety of aircraft operations.
AIRMET	Information issued by a meteorological watch office concerning the occurrence or expected occurrence of specified en-route weather phenomena which may affect the safety of **low-level aircraft operations** and which was not already included in the forecast issued for low-level flights in the FIR concerned (or sub-area thereof).

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Cloud Reporting

CLOUD AMOUNT IN OCTAS (EIGHTHS OF SKY)

Term	Octas (eighths)	Coverage
FEW	1 – 2 octas	Few clouds
SCATTERED (SCT)	3 – 4 octas	Less than half
BROKEN (BKN)	5 – 7 octas	More than half, not full
OVERCAST (OVC)	8 octas	Total / 100% cover

Example reading: "*Few at 2000, Scattered at 500, scattered cumulonimbus at 1000, broken at 2500.*"

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CAVOK — CEILING AND VISIBILITY OK

When the following **ALL** conditions are met, the term **"CAVOK"** is used:

- Visibility ≥ **10 km**;
- NO cloud below **1500 m (5000 ft)** or below the highest minimum sector altitude, whichever is greater;
- NO cumulonimbus;
- NO precipitation, thunderstorm, shallow fog, or low drifting snow.

METAR / SPECI / TREND — Format & Decoding

Example 1: VIDP 070200Z 12003KT 2000 BR SKC 10/08 Q1013 NOSIG

Decoded: Delhi (VIDP) weather at 0200 UTC on the 07th — surface wind 120° / 03 kt, visibility 2000 m, mist, sky clear, temp 10 °C, dew point 08 °C, QNH 1013, trend = no significant change.

Example 2: VOBG 070030Z 12003KT 4000 HZ BKN004 18/16 Q1014 TEMPO 3000

Decoded: Bangalore (VOBG) at 0030 UTC on the 07th — surface wind 120° / 03 kt, visibility 4000 m in haze, clouds 5–7 octas at 400 ft, temp 18 °C, dew point 16 °C, QNH 1014. Trend: visibility expected to decrease temporarily to 3000 m.

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Runway Visual Range (RVR) / Visibility

RVR TRANSMISSION

- When transmitting RVR, the abbreviation **"RVR"** is used **without** spelling each letter phonetically. Example: *"RVR runway 27, 800."*
- The runway designator **may be omitted** if there is no possibility of confusion.
- Where **instrumented RVR (IRVR)** observations are available, **more than one reading** may be transmitted (e.g. touchdown, mid-point, stop-end).

CORRECT WAY OF EXPRESSING VISIBILITY

Visibility is expressed in **METRES** (or kilometres if large).
Example: "*Visibility 1200 metres*" — NOT "1.2 nautical miles" and NOT "1200 feet".

Air/Ground Communications

CHANNELS PROVIDED TO INDIAN ATS UNITS

Air-Ground communications between aircraft and Indian ATS units are provided on the channels given below:

- Air/Ground (Voice) over **High Frequency (HF) Radio**.
- Air/Ground (Voice) over **Very High Frequency (VHF) Radio**.

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- Air/Ground (Voice) over **Ultra High Frequency (UHF) Radio**.
- **Remote Controlled Air-Ground (RCAG)** communication — for extended range of VHF/UHF Radio.
- **ADS-CPDLC Service** is available at **Mumbai, Delhi, Kolkata, and Chennai**.

Automatic Terminal Information Service (ATIS)

ATIS — ROUTINE AERODROME INFORMATION

- ATIS is available in **almost all airports in India**.
- This broadcast is related to the **terminal aerodrome** and is **continuous and repetitive**.
- Provides **routine information to arriving and departing aircraft** by means of continuous and repetitive broadcast.
- **Data Link ATIS (D-ATIS)** service is provided through **Pre-FANS Data Link**.

Voice Weather Broadcast (VOLMET)

WHAT IS VOLMET?

VOL (flight) + **MET** (meteorology). Meteorological aerodrome reports for certain aerodromes are **broadcast on specified HF frequencies**. The callsign, frequency, operating hours, aerodromes contained within the group, and contents are published in the **AIP**.

VOLMET is presently broadcast by **Mumbai and Kolkata** stations on HF frequencies.

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Content of a VOLMET Broadcast

- Aerodrome identification (e.g. Mumbai / Kolkata)
- Surface wind
- Visibility
- RVR (if applicable)
- Weather
- Cloud
- Temperature
- QNH
- Trend (if applicable)

D-VOLMET NOTE

Data Link VOLMET (**D-VOLMET**) service is provided through Pre-FANS Data Link. Non-essential words such as "surface wind", "visibility" etc. are **NOT spoken** in D-VOLMET.

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Distress & Urgency Procedures

DISTRESS TRAFFIC FREQUENCY RULE

Distress traffic shall **normally be maintained on the frequency on which such traffic was initiated**, until it is considered that better assistance can be provided by transferring that traffic to another frequency. Any other frequency shall be used to establish contact with any land/mobile or direction-finding station.

International Distress Frequencies

Frequency	Description
121.5 MHz / 243 MHz	International **VHF / UHF** distress frequency
2182 kHz	International distress frequency (MF band)
500 kHz	International **Maritime** distress frequency

Special Purpose SSR Codes

SSR Code	Meaning
7700	**Emergency** / Distress
7600	**Radio Communication Failure**
7500	**Hijack** or Other Act of Violence

Distress, urgency and special transponder codes

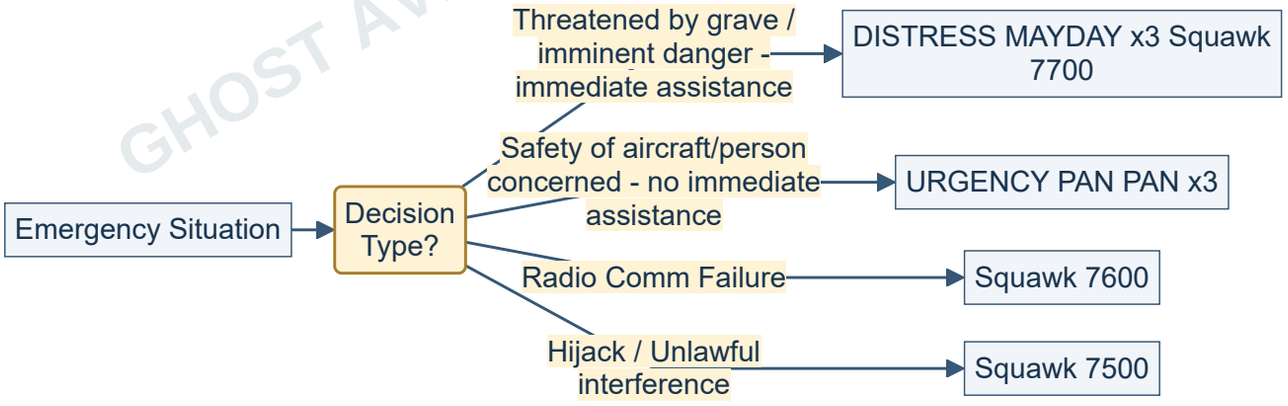

Figure 18.2 *Distress, urgency and special transponder codes*

ELT Transmission Frequencies

EMERGENCY LOCATOR TRANSMITTER (ELT)

ELT transmissions are made on:

- **121.5 MHz**
- **243 MHz**
- **406 MHz**

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SIGNAL WORDS

- **MAYDAY MAYDAY MAYDAY** — Distress signal (spoken 3 times)
- **PAN PAN PAN PAN PAN PAN** — Urgency signal (spoken 3 times)
- **PAN PAN MEDICAL** — Protected medical transport prefix
- **SECURITE SECURITE SECURITE** — Safety signal

IFR Communications

IFR VS VFR PROCEDURES

The procedures specified above for **VFR flights are equally applicable to IFR flights**. IFR flights will maintain **two-way communications** with ground **in ALL classes of airspace**.

An IFR flight operating **outside controlled airspace** shall **maintain listening watch** and **establish two-way communications** with the ATS unit providing **Flight Information Service**.

Wake Turbulence Phraseology

MANDATORY WAKE CATEGORY SUFFIX

Aircraft in the **heavy wake turbulence category** shall include the word **"HEAVY"** immediately after the aircraft call sign **in the initial call to each ATSU**.

For the **Airbus A380**, the word **"SUPER"** is to be included after the call sign on initial contact.

Purpose: confirm the aircraft type and/or wake turbulence category is the same as that stated on the flight progress strip.

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Wake Category	Suffix on initial call	Example
Heavy (e.g. B777, B747)	**HEAVY**	"AIR INDIA 285 HEAVY"
Super (A380)	**SUPER**	"EMIRATES 504 SUPER"

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Level Reporting & Voice Position Reports

Level Reporting

WHOLE NUMBERS & REFERENCE

- All messages relating to an aircraft's climb or descent to a **HEIGHT (QFE)** or **ALTITUDE (QNH)** are given in **whole numbers**.
- The **initial message** in any such RTF exchange will also include the **appropriate QFE or QNH**.
- When transmitting messages containing **flight levels (1013.2 hPa)** each digit shall be transmitted separately.
- Once given an instruction to climb or descend, a **further overriding instruction** may be given to a pilot.

Content of Voice Position Reports

STANDARD POSITION REPORT ELEMENTS

1. Aircraft identification
2. Position
3. Time
4. Flight level or altitude
5. Next position and time over
6. Ensuing significant point

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Meteorological Information at MET Reporting Points

ONCE PER FIR

Pilots shall give the following meteorological information **once in each FIR** along with the position report at **designated MET reporting points** on international and national ATS routes:

- Air temperature
- Wind direction
- Wind speed
- Turbulence
- Aircraft icing
- Humidity (if available)

Model AIREP SPECIAL

SPECIAL AIR-REPORT

An **AIREP SPECIAL** is a special air-report transmitted by an aircraft when an unusual or hazardous meteorological phenomenon is encountered en-route.

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Section	Item	Parameter	Transmit in Telephony
—	—	Message-type designator	(AIREP SPECIAL)
SECTION 1	1	Aircraft identification	(aircraft identification)
	2	Position	POSITION (lat/long) / OVER (sig pt) / ABEAM (sig pt) / (sig pt) (bearing) (distance)
	3	Time	(time)
	4	Level	FLIGHT LEVEL (no.) or (no.) METRES/FEET / CLIMBING TO FL (no.) / DESCENDING TO FL (no.)
	5	Next position and ETO	(position) (time)
	6	Ensuing significant point	(position) NEXT
SECTION 2	7	Estimated time of arrival	(aerodrome) (time)
	8	Endurance	ENDURANCE (hours and minutes)
SECTION 3	9	Moderate turbulence	TURBULENCE MODERATE
		Severe turbulence	TURBULENCE SEVERE
		Moderate icing	ICING MODERATE
		Severe icing	ICING SEVERE
		Severe mountain wave	MOUNTAIN WAVE SEVERE
		Thunderstorms without hail	THUNDERSTORMS
		Thunderstorms with hail	THUNDERSTORMS WITH HAIL
		Heavy dust / sand storm	DUST STORM *or* SAND STORM HEAVY
		Volcanic ash cloud	VOLCANIC ASH CLOUD
		Pre-eruption volcanic activity / volcanic eruption	PRE-ERUPTION VOLCANIC ACTIVITY *or* VOLCANIC ERUPTION
		Runway braking action	GOOD / GOOD TO MEDIUM / MEDIUM / MEDIUM TO POOR / POOR / LESS THAN POOR

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ADDITIONAL — TRANSONIC & SUPERSONIC FLIGHTS

For transonic and supersonic flights, **moderate turbulence** or **hail** or **cumulonimbus clouds** are included in AIREP SPECIAL reports.

Runway Braking Action — Specifications

Category	Specification
Good	Braking deceleration is **normal** for the wheel braking effort applied and directional control is **normal**.
Good to Medium	Braking deceleration or directional control is between Good and Medium.
Medium	Braking deceleration is **noticeably reduced** for the wheel braking effort applied OR directional control is noticeably reduced.

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Category	Specification
Medium to Poor	Braking deceleration or directional control is between Medium and Poor.
Poor	Braking deceleration is **significantly reduced** for the wheel braking effort applied OR directional control is significantly reduced.

Example — Spoken in Radiotelephony:
"Air Rep Special Air India 213, Position zero six seven radial of DPN VOR at seventy five DME, fix zero one three two, Flight Level two nine zero climbing to Flight Level three five zero, thunderstorm with hail."

General Principles of VHF Propagation & Wave Theory

WAVE ENERGY — BASICS

Wave energy is of a **continuous nature** in that it consists of many waves, one following the other at regular intervals, with the wave cycle repeating itself over and over again as the wave motion passes a given point. (A cork floating on water moves up and down as water waves pass — note that the water itself does not move out horizontally with the waves.)

Three Main Wave Parameters

Parameter	Symbol	Definition
Wavelength	λ (lambda)	The length of one single wave (one complete cycle); also the distance travelled by the wave during transmission of one cycle.
Frequency	F	The number of completed waves (or cycles) passing a point in **one second**.
Amplitude	—	The distance from one extremity of the oscillation to the other — "peak to peak".

Wave Equation & Conversions

KEY FORMULA

$F \propto 1/\lambda$ $F = C/\lambda$

Where **$C = 3 \times 10^8$ m/s** (speed of EM waves / light in vacuum).

Higher the wavelength → lower the frequency, and vice-versa.

Unit conversions:

- 1 cycle = 1 Hz
- 1000 Hz = 1 kHz = 10^3 Hz
- 1000 kHz = 1 MHz = 10^6 Hz
- 1000 MHz = 1 GHz = 10^9 Hz
- 1 km = 1000 m; 1 m = 100 cm; 1 cm = 10 mm

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Polarisation

POLARISATION DEFINITION

In an **EM wave**, two components — **electrical** and **magnetic** — travel **perpendicular to each other** in a direction which is **mutually perpendicular to both**. The direction in which the **electrical component** lies is called the **direction of polarisation**.

Polar Diagram

A polar diagram gives the **aerial characteristics** of a transmitter or a receiver. For a transmitter it is drawn up to a distance where signal strength **reduces up to 50% of transmission**.

Modulation

WHAT IS MODULATION?

Modulation is the **process of superimposing audio frequency or information onto a carrier wave**.

FOUR TYPES OF MODULATION

1. **Amplitude Modulation (AM)**
2. **Frequency Modulation (FM)**
3. **Pulse Modulation**
4. **Phase Modulation**

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Properties of Radio Waves

KEY PROPERTIES

i. EM waves travel at the **speed of light = 3×10^8 m/s**.

ii. They get **refracted, reflected, diffracted, and attenuated** during propagation (attenuation = lessening of amplitude in time).

iii. Their **direction and velocity both change** while passing from one medium to another.

iv. While passing from a medium of **lower density to higher density** they **bend TOWARDS the normal**, and vice-versa. They **always bend towards the denser medium**.

Frequency Bands & Allocations

Band	Frequency Range	Wavelength Range	Wavelength Denomination
VLF — Very Low Frequency	3 – 30 kHz	100 km – 10 km	Myriametric
LF — Low Frequency	30 – 300 kHz	10 km – 1 km	Kilometric
MF — Medium Frequency	300 kHz – 3 MHz	1 km – 100 m	Hectometric
HF — High Frequency	3 – 30 MHz	100 m – 10 m	Decametric
VHF — Very High Frequency	30 – 300 MHz	10 m – 1 m	Metric

Band	Frequency Range	Wavelength Range	Wavelength Denomination
UHF — Ultra High Frequency	300 MHz – 3 GHz	1 m – 10 cm	Decimetric
SHF — Super High Frequency	3 – 30 GHz	10 cm – 1 cm	Centimetric (Microwave)
EHF — Extremely High Frequency	30 – 300 GHz	1 cm – 1 mm	Millimetric

Aviation Systems & Their Frequencies

System	Frequency Range	Band
Decca	70 – 130 kHz	LF
Loran C	100 kHz	LF
ADF	190 – 1750 kHz	LF / MF
HF Communications	2 – 25 MHz	HF
ILS Markers	75 MHz	VHF
ILS Localiser	108.1 – 111.95 MHz	VHF
VOR	108.0 – 117.95 MHz	VHF
VHF Communications	118 – 137 MHz	VHF
ILS Glidepath	329.15 – 335.0 MHz	UHF
DME	960 – 1213 MHz	UHF
SSR	1030 and 1090 MHz	UHF
GPS (L1)	1575.42 MHz	UHF
GPS (L2)	1227.6 MHz	UHF
Satcom (Inmarsat) — Aircraft to Sat	1500 – 1600 MHz	UHF
Satcom (Inmarsat) — Sat to Ground	4000 – 6000 MHz	SHF
Radio Altimeter	4200 – 4400 MHz	SHF
Weather Radar	9375 MHz	SHF
MLS	5031 – 5091 MHz	SHF
ATC Surveillance Radars	600 – 1300 MHz	UHF
ATC Ground Manoeuvre Radar	10 – 16 GHz	SHF

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Propagation by Band

HOW EACH BAND TRAVELS

- **VLF, LF** are **absorbed by layers** → More range as **Ground Wave**.
- Part of **MF and HF** are **reflected back** → Travel as **Sky Wave**.
- **VHF, UHF, SHF, EHF** pass through the layers → **Line of Sight (LOS)** only.

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Band	Propagation mode emphasised in this chapter
VLF/LF	Ground-wave propagation.
MF/HF	Ionospheric sky-wave propagation can support long-distance communication.
VHF and higher bands	Predominantly line-of-sight propagation for the applications described.

Figure 18.3 *Frequency bands and typical propagation modes are a comparison, not a sequence.* *Original chapter 18 radio-propagation explanation.*

Satellite Communications (SATCOM / SATVOICE)

SATCOM — DEFINITION

SATCOM refers to the **voice and data service** that allows an aircraft to communicate, **via satellite**, with air traffic control and its airline operations center when **outside the coverage of conventional ground radar and VHF stations**.

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Three Types of Satellite Communication Services

1. **Telecommunications**
2. **Broadcasting**
3. **Data communications**

Satellite communication systems provide **real-time data to aircraft**, allowing for more precise monitoring and improved response times in the event of a maintenance or safety issue. This enhanced capability has led to improved safety and performance, as well as increased efficiency in aircraft maintenance operations.

ICAO DOC 9869 — DEFINITIONS

- **SATCOM** is used only when referring to **BOTH voice AND data** satellite communication.
- **SATVOICE** is for **satellite voice communications only**.
- The SATVOICE system shall be capable of detecting **loss of service, equipment failures, and/or logon failures** and provide indication to the controller / radio operator or flight crew of system status.

SATVOICE STATUS AS LRCS

In addition to **HF**, ICAO recognises **SATVOICE** as a **valid Long-Range Communication System (LRCS)**. *However, it will take some more maturing before SATVOICE replaces HF.*

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Propagation of Radio Waves

TWO MODES OF PROPAGATION

EM waves travel from a transmitter to a receiver in **two ways**:

i. **Ground Waves**

ii. **Sky Waves**

Wave Type	Definition
Ground Wave	A wave received either **directly from transmitter** or **after reflection from ground** is called a ground wave.
Sky Wave	A wave received at the receiver after getting **refracted from atmosphere (ionosphere)**.

Range, Tx Power, LOS & Required Communication Performance

Transmission Power

TX POWER RULE

Higher the transmission power, more is the range. Hence, to **double the range, power must be increased 4 times**.

Range ∝ √(Tx Power)

Line of Sight (LOS) Range Formula

LOS FORMULA — MEMORISE

LOS (range) = 1.25 × (√Ht(Tx) + √Ht(Rx))

Where:

- **Ht** = height in **FEET**
- **Range** = in **Nautical Miles (NM)**

Worked Example: Aircraft at FL80 (8000 ft), Ground VHF at 325 ft:
LOS = 1.25 × (√8000 + √325) = 1.25 × (89.44 + 18.03) = 1.25 × 107.47 ≈ **134 NM**

Required Communication Performance (RCP)

RCP CONCEPT

The RCP concept characterises the **performance required for communication capabilities** that support ATM functions without reference to any specific technology, and is open to new technology. This approach is essential to evolving operational concepts using emerging technologies.

An **ATM function** is an individual operational component of air traffic services. Examples include the application of separation between aircraft, the re-routing of aircraft, and the provision of flight information.

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The number attached to your RCP is the number of seconds it takes for an instruction to travel from the ground to you and your acknowledgement back to the ground.

CPDLC & Automatic Dependent Surveillance (ADS)

Controller Pilot Data Link Communications (CPDLC)

CPDLC — DEFINITION

CPDLC is a **two-way data-link system** by which controllers can transmit **non-urgent strategic messages** to an aircraft as an alternative to voice communications. The message is displayed on a flight-deck visual display.

OPERATIONAL USE

- When communicating with an aircraft operating **within airspace beyond the range of DCPC VHF voice communication**, CPDLC is available, and (unless local ATC procedures state otherwise) controller and flight crew would normally **choose CPDLC** as the means of communication.
- Voice (VHF, HF, or SATVOICE direct or via radio operator) would be used as an **alternative means**. However, the controller and flight crew always determine the communication medium most appropriate at any given time.
- Messages from aircraft to ATC may follow a **standard format** or may be **free-text**.
- Messages from a controller normally follow a **standard format**.
- **Response is required to most messages.**
- Voice and data link shall **co-exist**.
- Implementation of CPDLC is intended as a **supplementary means** of communication to the use of voice communication.

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CPDLC USAGE RESTRICTION

CPDLC shall only be used in the context of **NON-TIME-CRITICAL communications**.

Automatic Dependent Surveillance (ADS)

ADS — DEFINITION

ADS supports **automatic reporting by the Aircraft Flight Management System (FMS)** of:

- Aircraft **position**;
- **Intent** information;
- **Meteorological information** (Spot wind and Temperature).

The FMS reports the required information in accordance with **parameters selected by the ground system**. Messages will be transferred by **VHF and satellite data links**.

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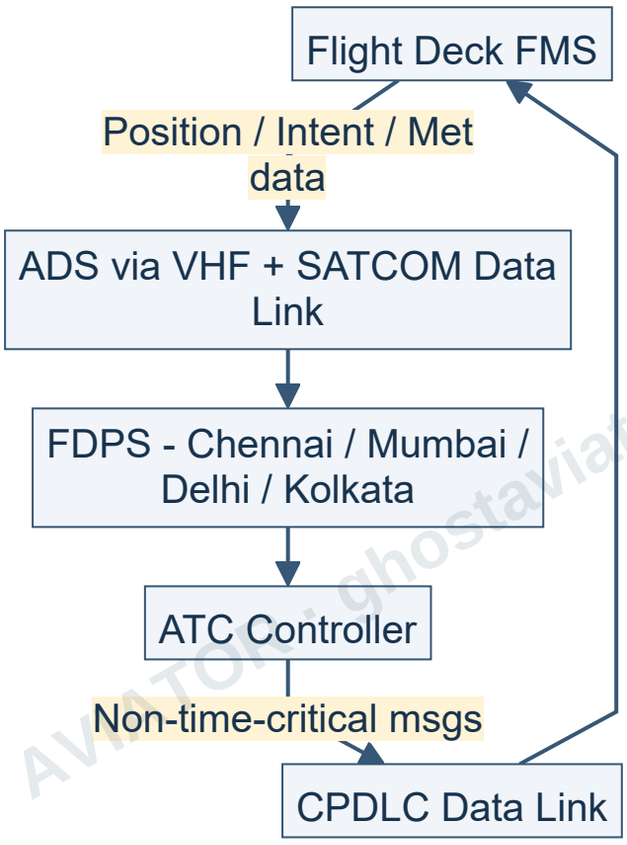
ADS-CPDLC IN INDIA

- ADS/CPDLC is integrated with the **Flight Data Processing System (FDPS)** at **Chennai, Mumbai, Delhi, and Kolkata**.
- ADS-CPDLC Service is available through **FANS-1A Data Link**.
- All aircraft fitted with this facility shall **login 10 minutes prior to entering the concerned FIR**.



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Exam-Prep Quick Cards

HIGH-FREQUENCY EXAM POINTS

- **QDR** = Magnetic bearing FROM the station.
- **Height** → set **QFE**; **Altitude** → set **QNH**; **FL** → set **QNE (1013.2)**.
- **ATIS** = Automatic **Terminal** Information Service (not "Traffic", not "Termination").
- **QDM** = Magnetic heading TO the station (no wind).
- **QTE** = TRUE bearing FROM the station.
- **QUJ** = TRUE track TO the station (D/F use for true heading no-wind).
- Priority order = **Distress** → **Urgency** → **D/F** → **Flight Safety** → **Met** → **Flight Regularity**.
- 3500 ft → "Three Thousand Five Hundred".
- 13500 → "One Three Thousand Five Hundred".
- The decimal point is always transmitted as the word DECIMAL (Annex 10 Vol II 5.2.1.4.1.7).
- 9:20 AM (no confusion) → just "TOO ZE-RO" (minutes only).

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- "CORRECTION QNH 1016" → standard phrase for error in transmission.
- "YES" → **AFFIRM** (NOT "Affirmative", NOT "Roger").
- "Listen out" → **MONITOR**.
- Surface movement = **GROUND**; Approach Radar departures = **APPROACH RADAR**.
- AIR INDIA 285 > 136 t initial call → **"AIR INDIA 285 HEAVY"**.
- Readability 3 = Readable but with difficulty.
- Test call ≤ **10 seconds**.
- Always read back: RWY in use, altimeter settings, SSR codes, level/heading/speed instructions.
- "Overcast" = 100% / 8 octas; "Broken" = 5–7 octas; "Scattered" = 3–4 octas; "Few" = 1–2 octas.
- Visibility expressed in **METRES**.
- CAVOK = ≥10 km vis, no cloud below 5000 ft/MSA, no CB, no precip/TS/fog/drifting snow.
- Comm failure → try alternative frequency.
- Blind transmission → make **twice on designated frequency**.
- Distress: aircraft "threatened by serious / imminent danger" requiring immediate assistance.
- Squawk **7700** = distress.
- MAYDAY x 3 = distress; uses **frequency currently in use**.
- Urgency call: **PAN PAN PAN** on freq in use + station addressed + nature + intentions.
- ATIS = routine continuous repetitive broadcast to arrivals & departures.
- HDG 225 @ 3500 ft → "HEADING TOO TOO FIFE AT TREE THOUSAND FIFE HUNDRED".
- Area Control Centre (no radar) = **...CONTROL**.
- Approach Control Radar Arrivals = **...RADAR**.
- Blind transmission = caller believes called station can still receive though no acknowledgement.
- "Runway vacated" = no longer on active runway.
- IFR + IMC + comm failure → continue per flight plan to designated navaid serving destination.
- Commence descent over destination navaid: **at, or as close to, ETA from current flight plan**.
- Medical transport prefix = **PAN PAN MEDICAL**.
- Urgency > all other comms EXCEPT distress.
- VOLMET includes **METARs**.
- Air India 5345 abbreviated → **No abbreviated form** (telephony designator + flight number cannot be shortened).
- FL80 + 325 ft VHF station → ≈ **134 NM** LOS.
- 295 kHz → **LF** band.
- Frequency = number of complete waveforms passing a spot in **1 second**.
- HF wavelength = **Decametric** (100 m – 10 m).
- Weather radar (9375 MHz) wavelength = **Centimetric**.
- 19 cm → 1.579 GHz, **UHF**.
- Thunderstorms without hail → reported as **"THUNDERSTORMS"**.
- **SATVOICE** — recognised as a valid **LRCS** by ICAO.

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CHAPTER 19

Aircraft Accident and Incident Investigation

19

INVESTIGATION RULES AND RESPONSIBILITIES

Under the 2025 Investigation Rules, reporting is required as soon as reasonably practicable, no later than 24 hours after awareness. Rule 4 specifies the responsible persons, recipients and notification contents. Rule 5 assigns all accidents in India to AAIB; its mass/type threshold applies to serious incidents. Historical rule references below must be distinguished from these current provisions.

Aircraft (Investigation of Accidents and Incidents) Rules 2025, GSR 829(E), Rules 4–5. Gazette pp 28–30.

Objective & Scope of Investigation

CORE PRINCIPLE

The **sole objective** of the investigation of an accident or serious incident is the **prevention of accidents and incidents**. It is not the purpose of this activity to apportion blame or liability. The causes of an aircraft accident or serious incident must be identified in order to prevent repeated occurrences.

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The identification of causal factors is best accomplished through a **properly conducted investigation**. ICAO Annex 13 lays down:

- Which States may **participate** in an investigation.
- The **rights and responsibilities** of each State.

States Entitled to Participate

State of Occurrence	State of Registry State of the Operator State of Design State of Manufacture

Applicability

GENERAL RULE

Unless otherwise stated, the specifications in Annex 13 apply to activities following **accidents and incidents wherever they occurred**.

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SPECIAL CASE – STATE OF THE OPERATOR

The specifications concerning the **State of the Operator** apply only when:

- The aircraft is **leased, chartered or interchanged**, AND
- That State is **not the State of Registry**, AND
- It discharges, in respect of this Annex, in part or in whole, the functions and obligations of the State of Registry.

ICAO Annex 13 – Key Provisions

- P-1 Any State which, on request, provides **information, facilities or experts** to the State conducting the investigation shall be entitled to appoint an **accredited representative** to participate in the investigation.
- P-2 The accident investigation **final report** shall be submitted to appropriate States and to the ICAO, in **one of the working languages of ICAO**.
- P-3 Upon receipt of the modification and a request by the State of Occurrence for participation, the **State of Design** and the **State of Manufacture** shall — in the case of an accident or serious incident — inform the State of Occurrence of the name of its representative to be present at the investigation when the aircraft is of a type for which the rules ask for this.
- P-4 The **Government of the State in which the accident took place** is responsible, under Annex 13 of the Chicago Convention, for the **initiation** of an accident investigation.

Report Language	ICAO Working Lang. Final report submitted in one of
Investigation Initiation	State of Occurrence Government where accident took place

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National Provisions – Definition of "Accident"

ACCIDENT – DEFINITION

An **occurrence associated with the operation of an aircraft** which:

- **(i) Manned aircraft** – takes place between the time any person boards the aircraft with the intention of flight until such time as all such persons have disembarked; OR
- **(ii) Unmanned aircraft** – takes place between the time the aircraft is ready to move with the purpose of flight until such time as it comes to rest at the end of the flight and the primary propulsion system is shut down.

In which any of the following conditions (a), (b) or (c) applies:

Conditions that Constitute an Accident

(a) A person is fatally or seriously injured as a result of

- Being in the aircraft, **or**
- Direct contact with any part of the aircraft, including parts which have become detached from the aircraft, **or**
- Direct exposure to **jet blast**,

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EXCEPT — NOT COUNTED AS ACCIDENT INJURY

Except when the injuries are from:

- **Natural causes**
- **Self-inflicted**
- **Inflicted by other persons**
- When injuries are to **stowaways** hiding outside the areas normally available to passengers and crew

(b) The aircraft sustains damage or structural failure which

- Adversely affects the **structural strength, performance or flight characteristics** of the aircraft, AND

EXCEPT — DAMAGE NOT CONSTITUTING AN ACCIDENT

Damage is **NOT** classified as an accident when limited to:

- Failure of **engine** or damage limited to a **single engine** (including its cowlings or accessories)
- **Propellers**, **wing tips**, **antennas**, **probes**, **vanes**
- **Tires**, **brakes**, **wheels**
- **Fairings**, **panels**, **landing gear doors**
- **Windscreens**, the **aircraft skin** (small dents or puncture holes)
- **Minor damage** to main rotor blades, tail rotor blades, landing gear
- Damage from **hail or bird strike** (including holes in the radome)

(c) The aircraft is missing or is completely inaccessible.

CRITICAL DEFINITIONS

- **"Fatally injured"** — means an injury resulting in **death within 30 days** of the date of the accident.
- **"Aircraft is missing"** — means the wreckage of the aircraft has not been located when the **official search has been terminated**.

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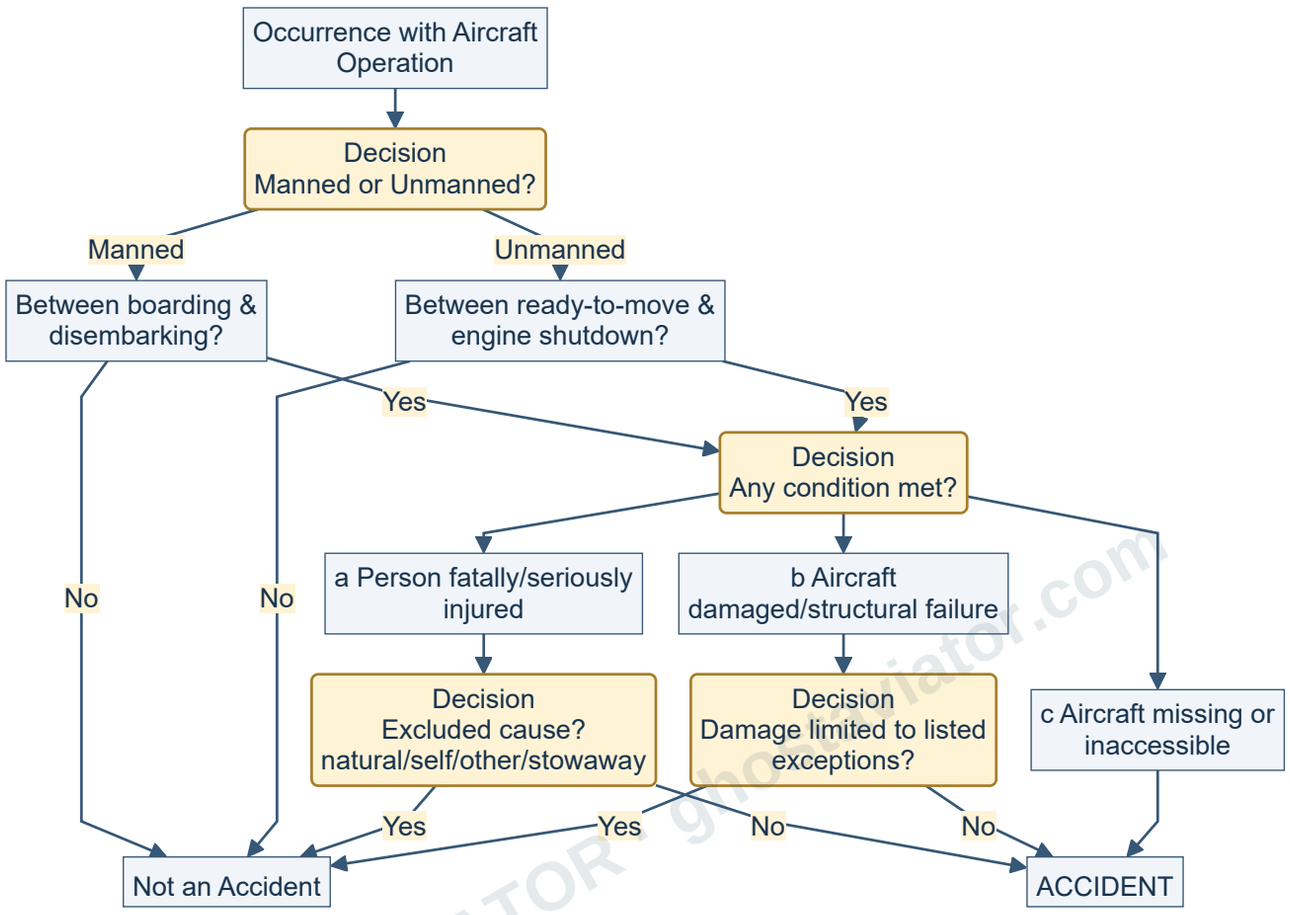
Visual: Accident Classification Flow



Decision flow — Is the occurrence an "Accident"?

Definition of "Serious Incident"

SERIOUS INCIDENT – DEFINITION

An incident involving circumstances indicating that there was a **high probability of an accident** and associated with the operation of an aircraft which:

- **Manned** — takes place between the time any person boards the aircraft with the intention of flight until all such persons have disembarked;
- **Unmanned** — between the time the aircraft is ready to move with the purpose of flight until it comes to rest at the end of the flight and the primary propulsion system is shut down.

NOTE ON THE LIST BELOW

The incidents listed below are **typical examples** of incidents likely to be serious incidents. The list is not exhaustive and only serves as guidance to the definition of "serious incident".

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Typical Examples of Serious Incidents

Item	Description
a)	Near collisions requiring an avoidance manoeuvre to avoid a collision or an unsafe situation, or when an avoidance action would have been appropriate.
b)	Collisions not classified as accidents.
c)	Controlled flight into terrain (**CFIT**) only marginally avoided.
d)	**Aborted take-offs** on a closed or engaged runway, on a taxiway or unassigned runway.
e)	**Take-offs** from a closed or engaged runway, from a taxiway or unassigned runway.
f)	**Landings or attempted landings** on a closed or engaged runway, on a taxiway or unassigned runway.
g)	Gross failures to achieve predicted performance during take-off or initial climb.
h)	Fires and/or smoke in the cockpit, in the passenger compartment, in cargo compartments or engine fires — *even though such fires were extinguished by extinguishing agents*.
i)	Events requiring the **emergency use of oxygen** by the flight crew.
j)	Aircraft structural failures or engine disintegrations, including **uncontained turbine engine failures**, not classified as an accident.
k)	Multiple malfunctions of one or more aircraft systems seriously affecting the operation of the aircraft.
l)	**Flight crew incapacitation** in flight.
m)	Fuel quantity level or distribution situations requiring the declaration of an emergency by the pilot — such as **insufficient fuel, fuel exhaustion, fuel starvation**, or inability to use all usable fuel on board.
o)	**Runway incursions classified with severity A.** (*The Manual on the Prevention of Runway Incursions — ICAO Doc 9870 — contains information on the severity classifications.*)
p)	Take-off or landing incidents — **under-shooting, overrunning, or running off the side of runways**.
q)	System failures, weather phenomena, operations outside the **approved flight envelope**, or other occurrences which caused or could have caused difficulties controlling the aircraft.
r)	Failures of **more than one system** in a redundancy system mandatory for flight guidance and navigation.
s)	The **unintentional** or, as an emergency measure, the **intentional release** of a slung load or any other load carried external to the aircraft.

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Definition of "Serious Injury"

SERIOUS INJURY – DEFINITION

An **injury sustained by a person in an accident** and which meets any one or more of the following six criteria:

- A Requires **hospitalization for more than 48 hours**, commencing **within 7 days from the date the injury was received**; OR
- B Results in a **fracture of any bone** — *except simple fractures of fingers, toes or nose*; OR
- C Involves **lacerations** which cause **severe haemorrhage**, or **nerve, muscle or tendon damage**; OR
- D Involves **injury to any internal organ**; OR

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- E Involves **second or third degree burns**, or any burns affecting **more than 5 percent of the body surface**; OR
- F Involves **verified exposure to infectious substances or injurious radiation**.

Hospitalization Limit	> 48 Hours Within 7 days of injury
Burns Threshold	> 5 % Body Or 2nd / 3rd degree burns
Fatal Injury Window	30 Days Death within 30 days = fatal

Definition of "Incident"

INCIDENT – DEFINITION

In relation to an aircraft, means an occurrence which takes place **either on the ground or in flight**, in which any of the following applies:

- A The aircraft **suffers damage**, or a person associated either with the **maintenance or operation of aircraft** (or both) suffers injury, in circumstances other than those specified in the definition of "accident".
- B The aircraft makes a **forced landing**.
- C The aircraft **lands at an aerodrome in an un-airworthy condition**.
- D The aircraft is **compelled to land at the aerodrome of departure** without completing the scheduled flight.
- E The aircraft lands owing to conditions which make **continuance of the flight inadvisable**.
- F The **position of the aircraft becomes "unknown for any period"**.
- G The **safety of the aircraft, its occupants, or any other person or property is jeopardised** in any manner.

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Definition of "Significant Incident"

SIGNIFICANT INCIDENT – DEFINITION

An incident involving circumstances indicating that an **accident or a serious incident could have occurred**, if the **risk had not been managed within safety margins**.

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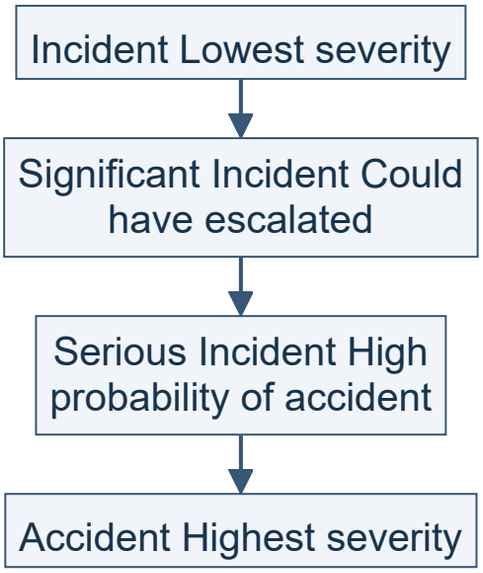
Hierarchy of Occurrences (Severity Scale)



Occurrence severity ladder — increasing left to right

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Air Traffic Incidents

AIR TRAFFIC INCIDENT – DEFINITION

"Air traffic incidents" is used to mean a **serious occurrence related to the provisions of air traffic services**, such as:

- **Aircraft proximity (AIRPROX);**
- Serious difficulty resulting in a **hazard to aircraft** caused, for example, by:
  - i) **Faulty procedures**
  - ii) **Non-compliance with procedures**
  - iii) **Failure of ground facilities**

Reporting Procedures (including In-Flight Procedures)

The following are the procedures to be followed by a **pilot who is or has been involved in an incident**:

DURING FLIGHT

Use the appropriate **air/ground frequency** for reporting an incident of **major significance**, particularly if it involves other aircraft, so as to **permit the facts to be ascertained immediately**.

AFTER LANDING

As promptly as possible after landing, and in any case **within 24 hours**, submit a completed **Air Traffic Incident Report Form** by the quickest means available to:

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- **Director General of Civil Aviation**
  (Attn: Director Air Safety, HQ)
- **With a copy to**: Executive Director (ATM), Airports Authority of India, Rajiv Gandhi Bhawan, Safdarjung Airport, New Delhi – 110003.

This submission is required:

- **i)** For **confirming a report** of an incident made **initially by radio**, or for making the initial report on such an incident if it had not been possible to report it by radio;
- **ii)** For reporting an incident which did **not require immediate notification** at the time of occurrence.

Information in an Initial Radio Report

INITIAL REPORT BY RADIO – CONTENTS

An initial report made by radio should contain the following information:

1. **Aircraft identification**
2. **Type of incident** — e.g. aircraft proximity
3. **The incident** — give information as laid down in the Air Traffic Incident Report Form
4. **Miscellaneous**

Reporting Timeline

T = 0 (During flight) Use air/ground frequency to report incident of major significance — facts ascertained immediately.
As soon as practicable after landing Begin compiling the Air Traffic Incident Report Form.
Within 24 hours (maximum) Submit completed Air Traffic Incident Report Form to DGCA (Attn: Dir. Air Safety, HQ) with copy to ED (ATM), AAI.

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Investigation Authority

INVESTIGATION – AUTHORITY

In case of an accident or incident to an aircraft **in the territory of India**, **notwithstanding** ***(in spite of)*** **its registration**, the **DG, Aircraft Accident Investigation Bureau (DG, AAIB)** shall institute an investigation into the circumstances of the accident or incident, and shall be responsible for conducting the investigation.

WHO IS "DG, AAIB"?

"DG, AAIB" means the **Director General** who shall be the **head of the Aircraft Accident Investigation Bureau**.

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Notification of Accident / Incident

MANDATORY NOTIFICATION – BY PIC

Where an accident or incident occurs to an aircraft, the following persons (in order of responsibility) shall give notice:

- The **Pilot-in-Command (PIC)** of the aircraft; or
- If the PIC is **killed or incapacitated**, then:
  - The **Owner**,
  - The **Operator**,
  - The **Hirer**, or other person on whose behalf he was in command, or
  - Any relevant person, as the case may be.

Notice shall be given **as soon as is reasonably practicable**, but **in any case NOT LATER THAN 24 HOURS** after he becomes aware of the accident or incident.

Who must be Notified by the PIC

- A Send notice to the **Aircraft Accident Investigation Bureau (AAIB)** and the **Directorate General of Civil Aviation (DGCA)** — by the **quickest means of communication available**; AND
- B In the case of an accident or incident occurring **in India**, give information to:
  - The **District Magistrate (DM)**, AND
  - The **Officer-in-Charge (O-i-C) of the nearest Police Station** of the accident or incident and of the place where it occurred.

Additional Notification to AAIB

NOTIFICATION SHALL ALSO BE SUBMITTED TO AAIB BY

- **(a)** Aerodrome operator
- **(b)** Air Traffic Services in-charge concerned
- **(c)** DG, CA — wherever applicable

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Visual: Notification Flow

Who / event	Required action
Pilot-in-command; substitute person if killed or incapacitated	Notify AAIB and DGCA as soon as reasonably practicable, no later than 24 hours after awareness; use the quickest available means.
Unmanned aircraft	Remote pilot-in-command notifies under the same provision.
Accident or serious incident in India	Also inform the District Magistrate and nearest police-station officer in charge.
Aerodrome operator; ATS in-charge; DG, DGCA where applicable	Also submit the prescribed notification to AAIB under Rule 4(4).

Figure 19.1 *Occurrence notification: the deadline is an outer limit, not permission to wait.* *2025 Investigation Rules, Rule 4(1)–(4).*

Mandatory 24-hour notification chain following an accident/incident

Protection of Evidence, Custody, Removal & Preservation of Damaged Aircraft

Access & Control by Investigator-in-Charge

CLAUSE (1) — INVESTIGATOR'S RIGHTS

In the case of an accident or a serious incident which is required to be notified, the **Investigator-in-Charge** shall have:

- **Unhampered access** to the wreckage and all relevant material and information, including **flight recorders** and **Air Traffic Services records**; AND
- **Unrestricted control** over it, to ensure that a **detailed examination** can be made **without delay** by authorised personnel participating in the investigation.

Removal Only Under Authority of DG, AAIB

CLAUSE (2) — GENERAL PROHIBITION

The aircraft and the contents thereof **shall NOT**, except by a person **under the authority of the DG, AAIB**, be **removed or otherwise interfered with**.

Permitted Exceptions (Provisos)

Provided that *(an exception or condition added to a rule)* the aircraft / its parts / its contents may be moved in the following circumstances:

- A **Search & Rescue Operations** — by persons authorised to conduct search and rescue operations, for the purpose of:
  - Extricating **persons or animals dead or alive**;

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  - Preventing destruction of the aircraft and its contents by **fire or other cause**;
  - Preventing damage or obstruction to the **public**, to **air navigation**, or to **other transport**.
- B **Aircraft Wrecked on Water** — may be removed (to the extent necessary) for bringing it to a **place of safety**, by persons authorised to conduct search and rescue operations.
- C **Goods** — may be removed under the **supervision and with the concurrence** of:
  - An officer of the AAIB, **OR**
  - A person authorised by the DG, AAIB.
- D **Personal Luggage** of passengers and crew — after **photography, weighing, etc.** — may be removed under the supervision of:
  - A **Police Officer**, OR
  - A **Magistrate**, OR
  - An **Officer of the AAIB**, OR
  - A person **authorised by DG, AAIB**.
- E **Mails** — after photography, weighing, etc. — may be removed under the supervision of:
  - A **Police Officer**, OR
  - A **Magistrate**, OR
  - An **Officer of the Department of Posts & Telegraphs**, OR
  - An **Officer of the AAIB**, OR
  - A person **authorised by DG, AAIB**.
- F **CVR and Airborne Image Recording Data — Special Protection.**

> **USE OF CVR / AIRBORNE IMAGE DATA**
>
> **CVR and airborne image recording data shall NOT be used for:**
> - Disciplinary proceedings
> - Civil proceedings
> - Administrative proceedings
> - Criminal proceedings
>
> — against the **employees, operational personnel or organizations**.

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Visual: Who Can Move What?

Item Being Removed	Authorised Supervisor / Authority
Aircraft / parts / contents (general)	Only under authority of **DG, AAIB**
Aircraft for SAR (life, fire, public safety)	Search & Rescue authorised personnel
Aircraft wrecked on water	SAR authorised personnel — for safety
Goods	Officer of AAIB OR person authorised by DG, AAIB
Personal luggage	Police Officer / Magistrate / AAIB Officer / person authorised by DG, AAIB
Mails	Police Officer / Magistrate / Officer of Posts & Telegraphs / AAIB Officer / person authorised by DG, AAIB

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Item Being Removed	Authorised Supervisor / Authority
CVR & Airborne Image Data	**Not usable** against personnel in any disciplinary / civil / admin / criminal proceeding

Quick Reference – Critical Values & Limits

Item	Value / Rule	Source
Final report submission language	One of the **ICAO working languages**	ICAO Annex 13
Fatal injury — definition	Death within **30 days** of accident	National Provisions
Serious injury — hospitalisation	More than **48 hours**, beginning within **7 days**	National Provisions
Serious injury — burns	2nd / 3rd degree, OR more than **5%** body surface	National Provisions
Air Traffic Incident Report — deadline after landing	Within **24 hours**	Reporting Procedures
Notification by PIC after accident/incident	Not later than **24 hours** after becoming aware	National Provisions
Authority to initiate investigation in India	**DG, AAIB**	National Provisions
Authority responsible (under Annex 13) for initiation	Government of **State of Occurrence**	ICAO Annex 13
CVR & Airborne Image Data	NOT usable in disciplinary, civil, administrative, criminal proceedings	National Provisions

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CHAPTER 20

Facilitation

20

WHAT "FACILITATION" MEANS IN AVIATION

Facilitation is the body of regulations that ensures the **smooth, efficient, and delay-free** movement of aircraft, crew, passengers, baggage, cargo, and mail across international borders. It is governed globally by **ICAO Annex 9** ("Facilitation") and nationally by the **Aeronautical Information Publication (AIP) of India**. The focus is on standardising customs, immigration, public health, and agricultural-quarantine procedures so that border formalities do not strangle the success of international civil aviation.

Key Definitions (Annex 9 Vocabulary)

Examination questions on Facilitation almost always begin by testing your **precise grasp of definitions**. Memorise these word-for-word.

Cargo	Any property carried on an aircraft **other than** mail, stores, and accompanied or mishandled baggage.
Clearance of Goods	The accomplishment of the customs formalities necessary to allow goods to **enter home use, be exported, or be placed under another customs procedure**.
Commencement of Journey	The point at which the person *began* his journey, **without taking into account any airport at which he stopped in direct transit**, either on a through-flight or a connecting flight, if he did not leave the direct transit area of the airport in question.
Commissary Supplies	Items, either **disposable or intended for multiple use**, used by the aircraft operator for provision of services during flights — particularly for **catering** and **passenger comfort**.
Deportee	A person who had been *legally* admitted to a State by its authorities, **OR** who had entered a State *illegally*, and who at some later time is **formally ordered by the competent authorities to leave** that State.
Direct Transit Area	A **special area** established in an international airport, **approved by the public authorities** and under their **direct supervision or control**, where passengers can stay during transit or transfer **without applying for entry** to the State.
Disembarkation	The **leaving of an aircraft after landing**, except by crew or passengers continuing on the next stage of the same through-flight.
Embarkation	The **boarding of an aircraft** for the purpose of commencing a flight, **except** by such crew or passengers as have embarked on a previous stage of the same through-flight.

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International Airport	Any airport designated by the Contracting State in whose territory it is situated as an **airport of entry and departure for international air traffic**, where the formalities incident to **customs, immigration, public health, animal and plant quarantine** and similar procedures are carried out.
Lading	The **placing of cargo, mail, baggage, or stores on board** an aircraft to be carried on a flight.
Unlading	The **removal of cargo, mail, baggage, or stores from an aircraft** after a landing.

Overview & Purpose of the Annex

WHAT THE SARPS COVER

The **Standards and Recommended Practices (SARPs)** of Annex 9 pertain **specifically** to facilitation of **landside formalities** for clearance of aircraft and commercial traffic through the requirements of:

- Customs authorities
- Immigration authorities
- Public health authorities
- Agriculture authorities

The Annex is a **wide-ranging document** which reflects the flexibility of ICAO in keeping pace with international civil aviation. ICAO is recognised as the **first international body** to make a real start on facilitation by developing Standards which bind its Contracting States.

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ORIGINAL THRUST OF ANNEX 9

Initially, the main thrust of the Annex consisted of efforts to:

1. **Reduce paperwork**
2. **Standardize internationally** the documents accompanying traffic between States
3. **Simplify the procedures** required to clear aircraft, passengers, and cargo

Why? Because delays from cumbersome formalities must be reduced — not just because they are unpleasant, but because they are **costly to every "customer group"** in the community and **interfere with the success of everyone**.

Entry & Departure of Aircraft

General Description

STANDARD OBLIGATION ON CONTRACTING STATES

Contracting States shall adopt **appropriate measures** for the clearance of aircraft arriving from or departing to another Contracting State, and shall implement them in such a manner as to **prevent unnecessary delays**.

While developing such procedures, Contracting States shall take into account the application of:

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- **Aviation security** measures
- **Narcotics control** measures, where appropriate

Purpose & Use of Aircraft Documents

NO EXTRA DOCUMENTS RULE

Contracting States shall **NOT** require any documents **other than those provided for in this Chapter**, for the entry and departure of aircraft.

NO VISA, NO FEE FOR AIRCRAFT DOCUMENTS

Contracting States shall **NOT** require a visa, nor shall any visa or other fee be collected, in connection with the use of any documentation required for the entry or departure of aircraft.

Acceptable Forms of Aircraft Documents

Subject to the technological capabilities of the Contracting State, documents for the entry and departure of aircraft shall be accepted when presented:

Option	Form of Presentation
(a)	In **electronic form**, transmitted to an information system of the public authorities
(b)	In **paper form**, produced or transmitted electronically
(c)	In **paper form**, completed manually following the formats depicted in **Annex 9**

E-DOCUMENT ACCEPTANCE RULE

When a particular document is transmitted by (or on behalf of) the aircraft operator and **received electronically** by the public authorities, the Contracting State **shall NOT** require the presentation of the **same document in paper form**.

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Information & Manifest Requirements

Annex 9 — Aircraft Document Requirements at a Glance

Document	Standard Requirement	Form
General Declaration	Information limited to elements indicated in Annex 9	Electronic OR paper
Passenger Manifest	Contracting States shall **NOT normally require** presentation	—
Cargo Manifest	When required in paper form, must accept the format shown in Annex 9	Annex 9 format
Declaration of Stores on Board	Contracting States shall **NOT require** a written declaration	—
List of Accompanied / Mishandled Baggage	Contracting States shall **NOT require** presentation	—
Mail Declaration	Only the form(s) prescribed *(laid down by the authority or the rules)* by the **Acts in force of the Universal Postal Union (UPU)** may be required	UPU forms only

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MAXIMUM COPIES LIMIT — STRICT RULE

Contracting States shall **NOT** require the aircraft operator to deliver to the public authorities **more than three (3) copies** of any of the above-mentioned documents at the time of entry or departure of the aircraft.

"EMPTY" AIRCRAFT EXEMPTION

If the aircraft is **NOT** embarking/disembarking passengers **OR** lading/unlading cargo, stores, or mail, the relevant document(s) shall **not be required**, provided an **appropriate notation is included in the General Declaration**.

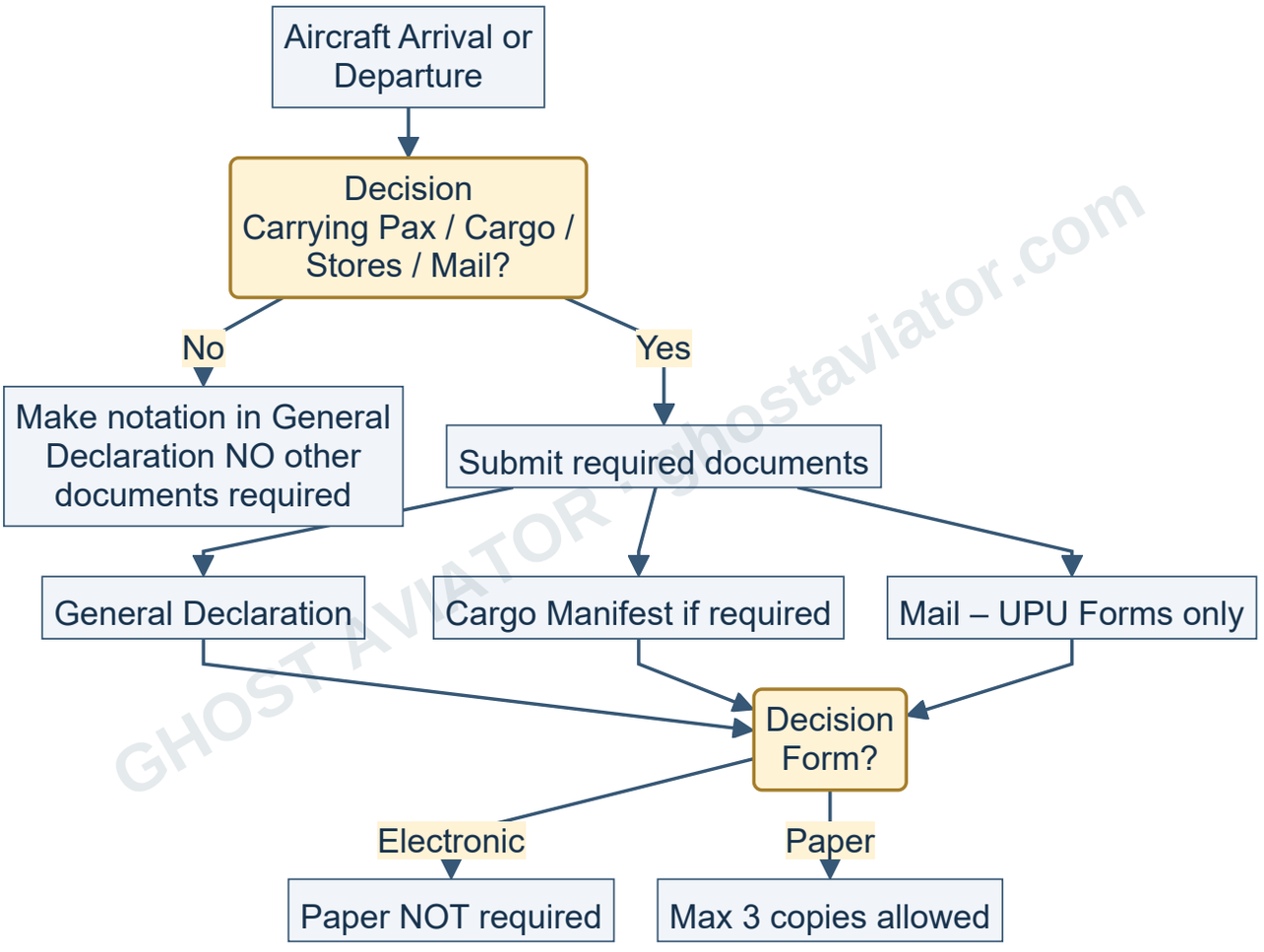

Figure 3.1 — Aircraft Document Submission Flow (Annex 9)

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General Declaration

WHAT IS THE GENERAL DECLARATION?

On those occasions when a General Declaration is required, the information required shall be **limited to the format given in Annex 9**. The information is acceptable in **electronic OR paper form**.

The General Declaration is the **"identity card"** of the flight — it tells customs, immigration, and health authorities exactly *who flew the aircraft, where it came from, where it's going, how many crew & passengers it carries, and the public health status on board*.

Contents Typically Required on a General Declaration

- Operator name
- Aircraft marks of nationality and registration
- Flight number and date
- Departure and arrival places
- Flight routing (origin, every en-route stop, destination)
- Total number of crew at each stage
- Number of passengers embarking, disembarking, and through on same flight
- Declaration of Health (illnesses, conditions on board, disinsecting details)
- Signature of authorised agent or pilot-in-command

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Entry/Departure of Persons — Crew Member Certificate (CMC)

Crew Inspection

EXPEDITING CREW INSPECTIONS

Contracting States shall establish measures — with the cooperation of **aircraft operators** and **airport operators** — to **expedite the inspection of crew members and their baggage**, as required at departure and upon arrival.

Crew Member Certificate (CMC) Issuance

WHAT IS A CMC?

Contracting States shall facilitate and expedite the process under which aircraft operators based in their territories can apply for **Crew Member Certificates (CMCs)** for their crew members.

Note: The CMC has developed as a **card for use for identification purposes by crew members**, leaving the **crew licences** to serve their primary purpose of **attesting to the professional qualifications of the flight crew members**.

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TWO DISTINCT DOCUMENTS — DON'T CONFUSE THEM

Document	Primary Purpose
Crew Member Certificate (CMC)	**Identification** of the crew member at borders/airports
Crew Licence (FCL)	**Attesting the professional qualifications** of the flight crew member

CMC Validity Review & Correction

CREW'S RIGHT TO REVIEW CMC DATA

Contracting States **should** put in place procedures which will enable any crew member issued with a Crew Member Certificate to **examine and review the validity** ***(the period during which a document is accepted)*** **of the data held**, and to **provide for correction if necessary, at no cost to the crew member**.

Crew Identity Card Format

STANDARDISED LAYOUT REQUIREMENT

To the extent that aircraft operators issue crew identity cards, Contracting States **should require** the production of such identity documents in the format equivalent to the **visual zone of the machine readable crew member certificate**, and having the capability to support:

- Machine assisted identity confirmation
- Document security verification

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Electronic Database of Crew Records

DATABASE REQUIREMENT

Contracting States shall ensure that a record of each crew member's certificates and other official identity documents **issued, suspended, or withdrawn** is stored in an **electronic database**, secure from **interference and unauthorised access**.

Anti-Fraud Controls on CMC Issuance

ADEQUATE CONTROLS — MANDATORY

Adequate controls **shall** be placed on the issuance of CMCs and other official crew identity documents to **prevent fraud**. Examples:

- **Background check** of the applicant
- **Certification of employment status** of the applicant prior to issuance
- **Controls on blank card stock**
- **Accountability requirements** for issuing personnel

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Visa Waiver for Crew Holding CMCs

MANDATORY VISA WAIVER — DUTY STATUS (SAME OPERATOR)

Contracting States **shall waive the visa requirement** for arriving crew members presenting CMCs, when:

- Arriving in a **duty status**
- On an **international flight**
- Seeking **temporary entry** for the period allowed by the receiving State
- In order to **join their next assigned flight in a duty status**

RECOMMENDED VISA WAIVER — POSITIONING CREW

Contracting States **should waive the visa requirement** for arriving crew members presenting CMCs, when:

- Arriving on **another aircraft operator** or **another mode of transport**
- Seeking temporary entry to join their assigned flight in a duty status

(This covers crew "positioning" — e.g., flying as a passenger to start their next duty.)

Temporary Entry of Technical Personnel

ENGINEERS & MAINTENANCE PERSONNEL

Contracting States **shall** establish measures to provide for the **temporary entry without delay** into their territories of **technical personnel of foreign aircraft operators** operating to or through such territories who are **urgently required for the purpose of converting to an airworthy condition any aircraft** which is, for technical reasons, unable to continue its journey.

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Biometric & Machine Readable Travel Documents

BIOMETRIC DATA IN EMRTDS

Contracting States **should incorporate biometric data** in their machine readable travel documents in a **contactless integrated circuit chip**, as specified in **ICAO Doc 9303 — Machine Readable Travel Documents (MRTD)**.

Automated Border Control (ABC) Systems

ABC SYSTEM IMPLEMENTATION

Each Contracting State **should consider** the introduction of **Automated Border Control (ABC) systems** in order to facilitate and expedite the clearance of persons entering or departing by air.

DATA SOURCES FOR ABC SYSTEMS

Contracting States utilising ABC systems should use the information available from:

- **ICAO Public Key Directory (PKD)** — to validate eMRTDs
- **Biometric matching** — to establish that the passenger is the rightful holder of the document
- **INTERPOL's Stolen and Lost Travel Documents (SLTD)** database
- **Other border control records** — to determine eligibility for border crossing

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Starting point / condition	Branch	Next step / result
Passenger at ABC Gate	→	Present eMRTD
Present eMRTD	→	Read Contactless Chip per Doc 9303
Read Contactless Chip per Doc 9303	→	Validate with ICAO PKD
Validate with ICAO PKD	Valid	Biometric Capture Face / Fingerprint
Validate with ICAO PKD	Invalid	Refer to Manual Counter
Biometric Capture Face / Fingerprint	→	Match with Chip Data
Match with Chip Data	Match	Check INTERPOL SLTD and other records
Match with Chip Data	No Match	Refer to Manual Counter
Check INTERPOL SLTD and other records	Clear	Border Crossing Approved
Check INTERPOL SLTD and other records	Flag	Refer to Manual Counter

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Figure 5.1 — ABC System Logic Flow

National Provisions (India) — AIP India

APPLICABILITY OF INDIAN REGULATIONS

International flights into, from, or over **Indian territory** shall be subjected to the **current Indian regulations** relating to:

- Civil aviation
- Immigration
- Customs
- Passport
- Health, etc.

These regulations **correspond in all essentials** to the SARPs contained in **Annex 9** to the Convention on International Civil Aviation.

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MANDATORY COMPLIANCE — ALL AIRCRAFT

Additionally, **every aircraft entering or leaving India** must comply with regulations relating to:

- **Immigration**
- **Customs**
- **Quarantine**
- **Health**

...as laid down by the Government from time to time.

Landing & Departure Aerodromes

FIRST LANDING / FINAL DEPARTURE RULE

Aircraft flying into or departing from Indian territory shall make their **first landing at** — or **final departure from** — an **International Aerodrome** (see AIP India, AD 1.3 and AD 2).

Exception: Aircraft **may be permitted** to land or depart from any **notified customs aerodrome**.

Cabotage Restriction

NO CABOTAGE ON INTERNATIONAL FLIGHTS

International flights are **NOT permitted** to:

- **Pick up** passengers/load at any place in India **AND**
- **Disembark/discharge** at any other place in India

(This is the classic prohibition on cabotage (carrying passengers or cargo between two places inside one country) — domestic carriage by a foreign-flagged carrier.)

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Aircraft Documents Required in India (Arrival/Departure)

India — Document Copies Required by Each Authority

Required By	General Declaration	Passenger Manifest	Cargo Manifest
Customs	**1**	**1**	**1**
Immigration	**1**	**1**	**1**
Health	**1**	**1**	**1**
TOTAL COPIES	**3**	**3**	**3**

Why "3 Copies" Aligns With Annex 9

Annex 9 caps the maximum at **3 copies of any document**. India requires **1 each to Customs, Immigration, and Health** — which gives exactly **3 copies total per document**. This is the maximum permitted under ICAO Standards.

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Outbound Clearance

NO DEPARTURE WITHOUT CLEARANCE

No flight shall leave India without obtaining clearance of **Immigration AND Customs** authorities on the General Declaration.

Customs Duty on Aircraft

6-MONTH DUTY-FREE AIRCRAFT RULE

No customs duty is levied on an aircraft **not registered in India** which is brought into India for the purpose of a flight to or across India, **provided**:

- The aircraft is **not intended to be registered in India**; AND
- It is intended to be **removed from India within 6 months** from the date of entry; AND
- The **person-in-charge of the aircraft makes a written declaration** to that effect to the **Customs Collector on arrival**.

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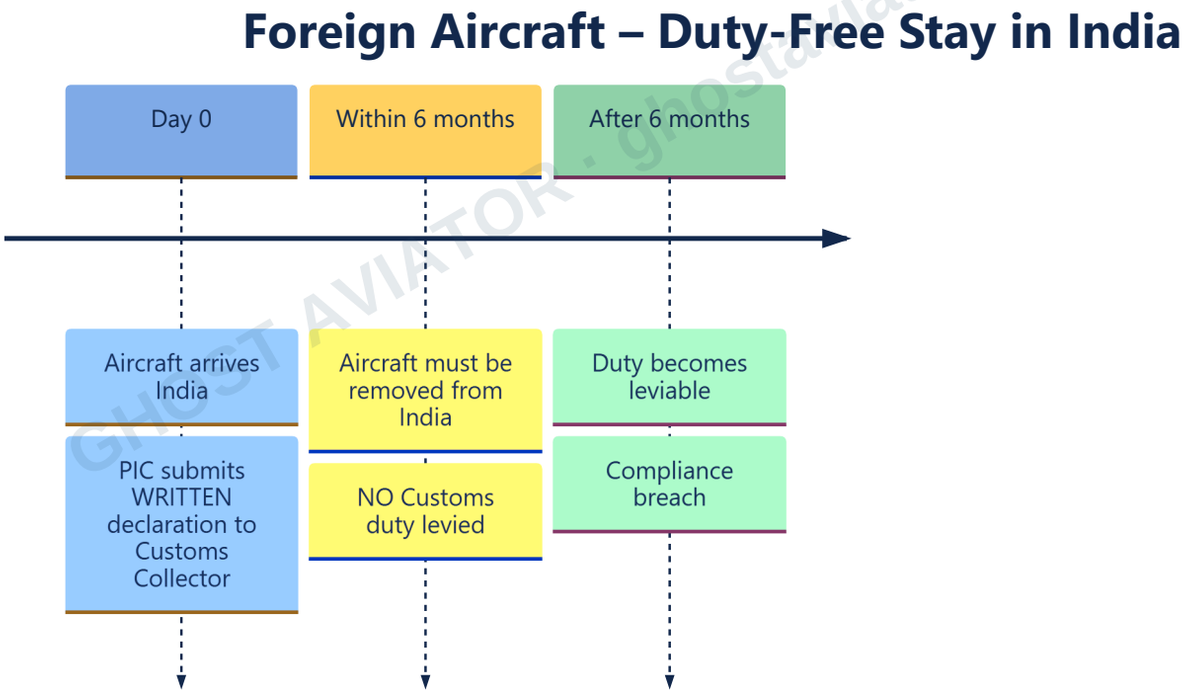

Figure 20.1 *Day 0*

Figure 6.1 — Validity Timeline for Duty-Free Foreign Aircraft in India

Standard Forms (Visual Reference)

The three standard documents prescribed by Annex 9 (and reproduced in AIP India) are illustrated below. Familiarise yourself with the layout — exam questions often ask "*what data appears on a General Declaration?*"

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GENERAL DECLARATION (Outward / Inward)

Operator: _____

Marks of Nationality & Registration: _____

Flight No.: _____

Date: _____

Departure from: _____

Arrival at: _____ (Place)

FLIGHT ROUTING — Place (origin, every en-route stop, destination)	Total No. of Crew	No. of Passengers on this Stage
		Departure Place: __ Embarking: __ Through: __ *Arrival Place: __ Disembarking: __ Through: __*

Declaration of Health: Persons on board with illnesses other than airsickness or accidents (rash, fever, chills, diarrhoea), other conditions which may spread disease, details of disinsecting/sanitary treatment (place, date, time, method).

Signed: _____ (Crew member concerned)

"I declare that all statements and particulars contained in this General Declaration… are complete, exact, and true to the best of my knowledge…"

SIGNATURE: _____ (Authorised Agent or Pilot-in-Command)

PASSENGER MANIFEST

Operator: _____

Marks of Nationality & Registration: _____

Flight No.: _____

Date: _____

Point of Embarkation: _____

Point of Disembarkation: _____

Surname & Initials	For use by Operator only	For Official use only

CARGO MANIFEST

Operator: _____

Marks of Nationality & Registration: _____

Flight No.: _____

Date: _____

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Point of Lading: _____

Point of Unloading: _____

Air Waybill Number	Number of Packages	Nature of Goods	For use by Operator only	For Official use only

CARGO MANIFEST — INFORMATION LIMITS

A Contracting State which continues to require the Cargo Manifest, apart from the heading information, **shall NOT require more than**:

- The **airway bill number**
- The **number of packages** related to each airway bill number
- The **nature of the goods**

Customs Requirements (India)

Incoming Passengers — The Two-Channel System

TWO-CHANNEL CLEARANCE

All goods imported into India by air are subject to clearance by Customs authorities, **except goods within the limits of duty-free allowance**. For the purpose of Customs Clearance of arriving passengers, a **two-channel system** has been adopted:

- **GREEN CHANNEL** — for passengers **NOT having any dutiable goods**
- **RED CHANNEL** — for passengers **having dutiable goods**

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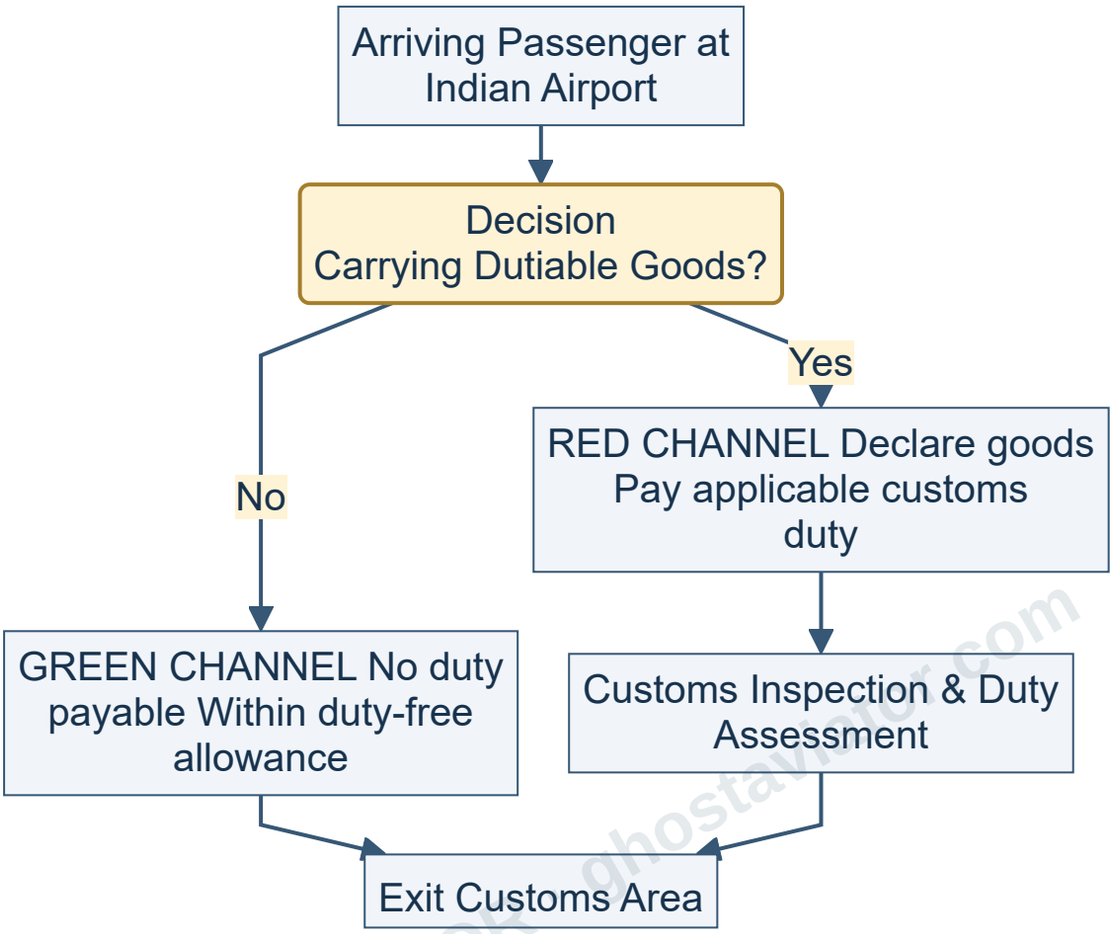

Figure 8.1 — Two-Channel Customs Clearance System

Regulations for Airline Crew

CREW DECLARATION OBLIGATIONS

Crew members of aircraft are subject to submit a **correct declaration** before Customs authorities with respect to:

- **Currency**
- **Gold ornaments**
- **Electronic goods, etc.**

...in their possession **on arrival as well as departure**.

Outgoing Passengers

EXPORT CERTIFICATE FOR HIGH-VALUE ITEMS

All passengers leaving India by air are subject to clearance by Custom Authorities. **Only bonafide baggage** is allowed to be cleared by passengers. There is a procedure prescribed whereby passengers leaving India can take the **Export Certificate** for various **high-value items as well as jewellery** from the Customs authorities.

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Regulations for Transit Passengers

TRANSIT PASSENGER RESTRICTIONS

Transit passengers are **kept in the Customs area** and Customs authorities **keep watch** on such passengers so that:

- They should **NOT hand over anything to any other person**
- They are **NOT allowed to go outside the airport building**

TRANSIT VISA (AIP INDIA GEN 1.3)

A transit visa is **not required** by a foreigner passing through India in **direct transit by air**, provided the foreigner **does not leave the specified precincts of the airport**.

Customs Requirements — Cargo & Other Articles

PRE-DEPARTURE CARGO EXAMINATION — STRICT TIMING

Goods for export by air (**including unaccompanied baggage**) are required to be presented for examination to the Customs authorities **sufficiently in advance** for such goods to be examined **before the departure** of the aircraft on which they are consigned.

General rule: Such presentation should be made **NOT less than 4 working hours** before the scheduled time of departure.

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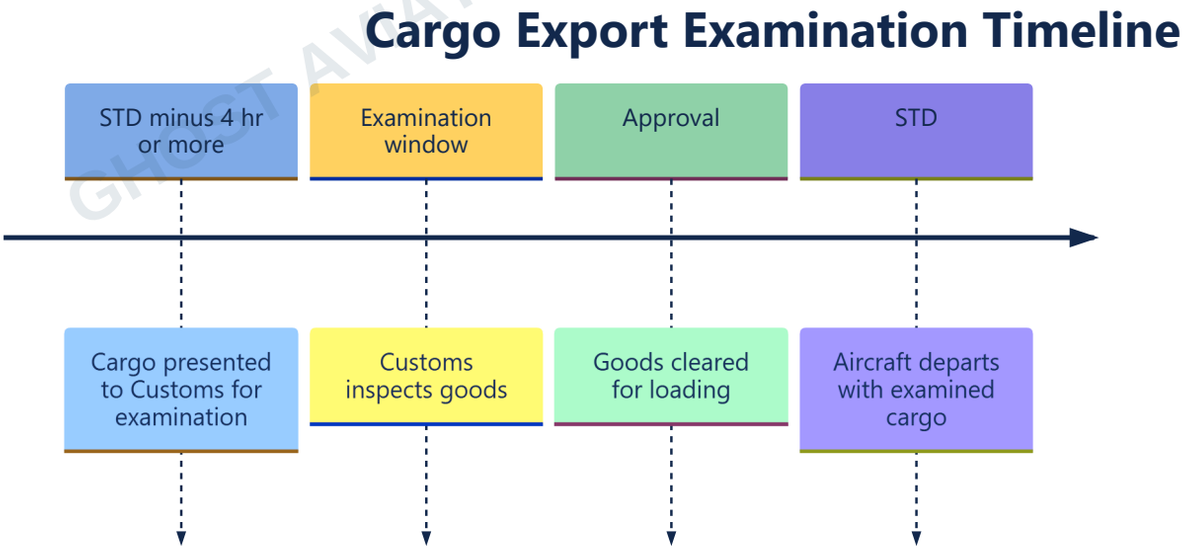

Figure 20.2 *STD minus 4 hr or more*

Figure 8.2 — 4-Hour Cargo Examination Rule

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Top Numerical provisions — reference These

Critical Regulatory Values You Must Know Cold

Value	Subject
Annex 9	ICAO Annex on Facilitation
Doc 9303	Machine Readable Travel Documents (MRTD)
3 copies	Maximum copies of any aircraft document required (Annex 9)
2 hours	Advance flight-plan notice for non-scheduled ops crossing territory
2 days	Visa-free transit stay for stranded transit passenger
4 working hours	Cargo presentation before STD for Customs exam (India)
6 months	Duty-free stay of foreign aircraft in India
Green / Red Channels	Two-channel customs system in India
PKD / SLTD	ICAO Public Key Directory / INTERPOL Stolen & Lost Travel Docs

FINAL INSTRUCTOR'S NOTE

Facilitation questions test **specific numbers** (2 hrs, 4 hrs, 3 copies, 6 months, 2 days), **document hierarchy** (CMC vs Licence; General Declaration vs Manifest), and **document acceptance modes** (paper, electronic, handwritten block lettering). Always read the question stem carefully — Facilitation MCQs are notorious for swapping "shall" with "should" and "passenger" with "passenger AND crew".

Fly safe, study smart. — *Capt. Pankaj Pahil*

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CHAPTER 21

Security and Unlawful Interference

21

SECURITY AUTHORITY AND REPORTING

The 2023 Security Rules superseded the 2011 Rules. The Director General, BCAS is the appropriate authority for Annex 17. Every entity must report as specified by the Director General. Under Rule 52(1), the Inquiry Officer must be at least Assistant Director rank.

Aircraft (Security) Rules 2023, Rules 2(1)(s),2(1)(t),3,51 and 52(1).

Historical Background

The story of aviation security begins with a problem the world could no longer ignore. During the **late 1960s**, civil aviation experienced a dramatic increase in crimes of violence which adversely affected the safety of flight operations across the globe. Aircraft were being hijacked at an alarming rate, passengers were being held hostage, and the world's airline industry was forced to confront a threat for which existing international law had no comprehensive answer.

This crisis forced the International Civil Aviation Organization (ICAO) to take extraordinary action. An **Extraordinary Session of the ICAO Assembly** was convened in June 1970. One of the resolutions of that Assembly called for specifications — either in existing Annexes or in a brand-new Annex to the **Chicago Convention** — to specifically deal with the problem of unlawful interference, in particular with the **unlawful seizure of aircraft (hijacking)**. The result was the birth of **Annex 17**.

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WHY ANNEX 17 WAS CREATED

Annex 17 seeks to **co-ordinate the activities** of those involved in security programmes. The international community recognised that *airline operators themselves have a primary responsibility* for protecting their passengers, assets and revenues. Therefore, States must ensure that the carriers develop and implement effective **complementary security programmes** compatible with those of the airports out of which they operate.

The Pre-1985 Era — Hijacking-Centric Focus

Prior to 1985, the significant threat to civil aviation was seen overwhelmingly as **the hijacking**. Because of this perception, the Standards and Recommended Practices (SARPs) of the day tended to focus on:

- **Hijacking** as the primary threat
- Rather than on **sabotage**, **in-flight attack**, or **facility attack**

By modifying existing technology and applying agreed-upon specifications and procedures, the worldwide aviation community established a **reasonably effective screening system** for passengers and their carry-on luggage. The annex was further amended over the years to deal with:

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PROGRESSIVE AMENDMENTS TO ANNEX 17 — PRE-1985 ADDITIONS

- **Reconciliation of baggage with passengers** (no bag flies unless its owner flies)
- Controls over **items left behind on the aircraft** by disembarking passengers
- Security controls for **commercial courier services**
- Controls over **cargo and mail** under certain situations

The Modern Amendment — Post-9/11 & Beyond

The later amendment of Annex 17 introduced a significantly broader set of provisions. Today the Annex includes various definitions and new provisions in relation to:

MODERN SCOPE OF ANNEX 17

- Applicability of the Annex to **domestic operations** (not just international)
- **International cooperation** relating to threat information
- **National quality control**
- **Access control**
- Measures related to **passengers and their cabin and hold baggage**
- **In-flight security personnel** and protection of the cockpit
- **Code-sharing / collaborative arrangements**
- **Human factors**
- **Management of response** to acts of unlawful interference

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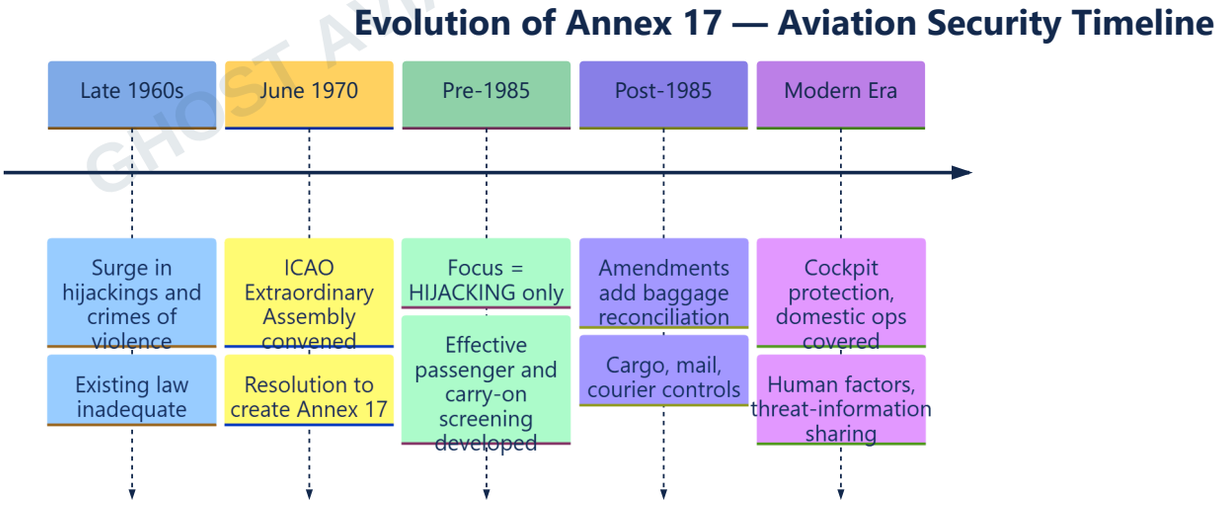

Figure 21.1 *Historical evolution of ICAO Annex 17*

Key Definitions

Before working through the operational rules, students must lock down the precise legal definition of "Security" as it appears in Annex 17, because it is examined directly in MCQs.

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DEFINITION — "SECURITY" (ICAO ANNEX 17)

Security is a **combination of measures** *AND* **human and material resources** intended to safeguard international civil aviation against acts of unlawful interference.

Memory trick: Three parts must appear in the definition:

1. Combination of **measures**
2. Combination of **human resources**
3. Combination of **material resources**

All three together safeguard *international civil aviation* against *acts of unlawful interference*. If an option leaves out "human and material resources", it is incorrect.

ACTS OF UNLAWFUL INTERFERENCE — WHAT THEY INCLUDE

Acts of unlawful interference are acts (or attempted acts) which jeopardise the safety of civil aviation, such as:

- Unlawful seizure of an aircraft in flight (**hijacking**)
- Unlawful seizure of an aircraft on the ground
- Hostage-taking on board an aircraft or on aerodromes
- Forcible intrusion on board an aircraft, at an airport or on the premises of an aeronautical facility
- Introduction on board an aircraft or at an airport of a weapon, hazardous device or material intended for criminal purposes
- Communication of false information so as to jeopardise the safety of an aircraft in flight or on the ground

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Scope of Annex 17 Today

The ICAO Annex 17 comprises rules in order to establish security measures for passengers covering the following items:

Security Element	What it Covers
Cabin Baggage	Items the passenger carries into the aircraft cabin — screened at the security check point.
Checked Baggage	Hold baggage tendered for carriage in the aircraft hold — must be reconciled with the passenger.
Cargo & Other Goods	Includes commercial cargo, mail, courier shipments — controlled by regulated agents.
Access Control	Restricting entry to security restricted areas; airside access; staff and crew vetting.
Airport Design	Designing terminals to channel passenger flows safely and to permit effective screening.

Aims & Objectives of Each Contracting State

This section is the regulatory heart of Annex 17. Every Contracting State (i.e. every country that has signed the Chicago Convention, including India) must comply with these obligations.

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PRIMARY OBLIGATION OF EVERY CONTRACTING STATE

Each Contracting State shall **establish an organization** and **develop and implement regulations, practices and procedures** to safeguard civil aviation against acts of unlawful interference, taking into account the:

- **Safety** of flights
- **Regularity** of flights
- **Efficiency** of flights

TWO MANDATORY CAPABILITIES — SUB-CLAUSES (A) AND (B)

Each Contracting State shall ensure that such an organization and such regulations, practices and procedures:

a. **Protect the safety** of *passengers, crew, ground personnel and the general public* in all matters related to safeguarding against acts of unlawful interference with civil aviation; **AND**

b. Are **capable of responding rapidly** to meet any increased security threat.

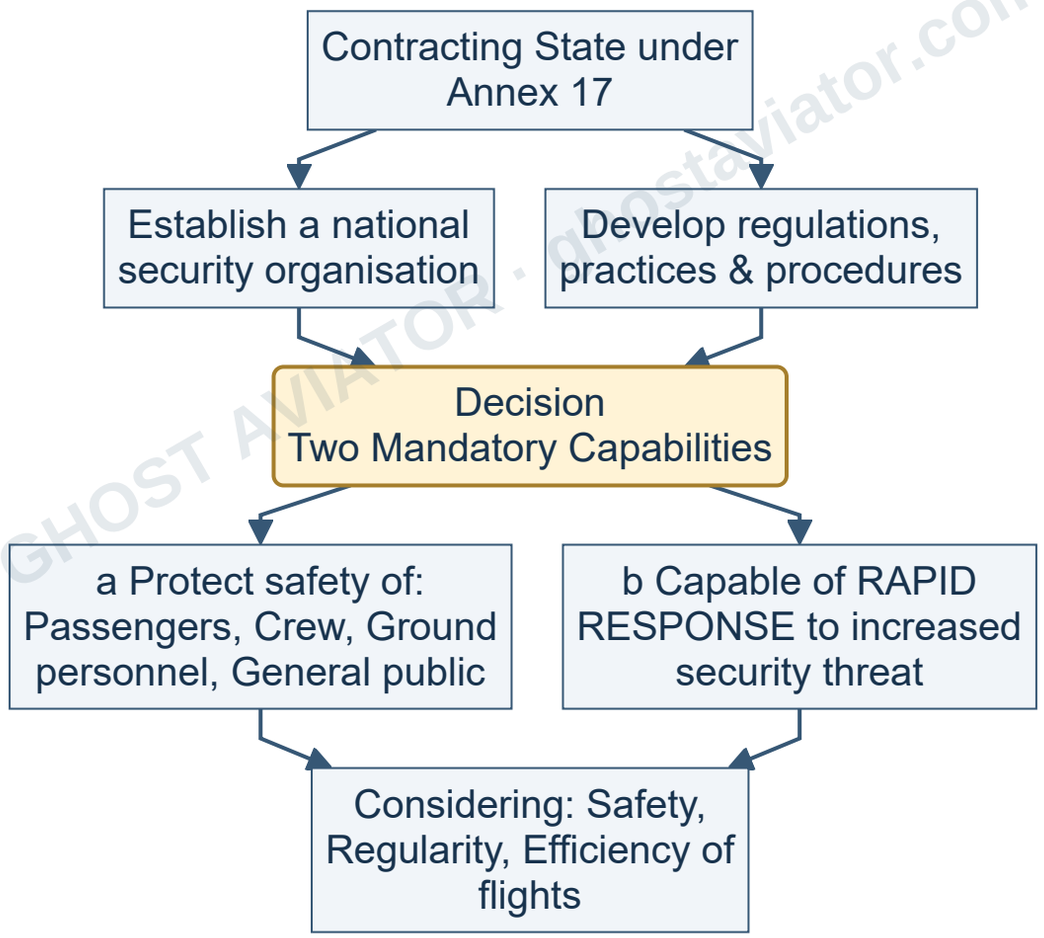

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Figure 21.2 *Annex 17 obligations on every Contracting State*

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Operational Provisions on Unlawful Interference

When an act of unlawful interference actually occurs — most importantly an unlawful seizure / hijacking — Annex 17 lays down a sequence of mandatory actions. Students must memorise each clause word-for-word because exam options test fine distinctions.

Retention of the Hijacked Aircraft

MANDATORY RETENTION RULE

The Contracting State **will make provisions to ensure that an aircraft affected by an unlawful seizure act**, which has landed in their territory, **would be retained** — *unless its departure is justified to protect lives*.

Why this matters: The default action is RETAIN the aircraft. The *only* exception is when allowing it to depart will protect human lives (e.g. hijackers credibly threaten to kill hostages unless permitted to leave).

Safety of Passengers & Crew After Landing

DUTY OF CARE — UNTIL THE JOURNEY CONTINUES

A State shall take adequate measures for the **safety of passengers and crew** of an aircraft which is subjected to an act of unlawful interference — **until their journey can be continued**.

The phrase "*until their journey can be continued*" is examined verbatim. It is **not** "during a period of investigation" and it is **not** "until returned to country of origin".

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Notification Obligations After Landing

When an aircraft subject to unlawful interference has landed, the State of landing must, by the **most expeditious means**, notify a specific list of parties. *This list is examined directly — memorise it.*

MANDATORY NOTIFICATION LIST — FIVE PARTIES

The State of landing shall notify **by the most expeditious means**:

1. The **State of Registry** of the aircraft
2. The **State of the Operator**

...and shall similarly transmit all other relevant information to:

3. The **two aforementioned States** (Registry + Operator) — i.e. complete info, not just notification
4. Each **State whose citizens suffered fatalities or injuries**
5. Each **State whose citizens were detained as hostages**
6. Each **State whose citizens are known to be on board** the aircraft
7. The **ICAO**

Watch-out: A wrong option will *omit* "citizens detained as hostages" or omit "ICAO". Cross-check every option against this complete list.

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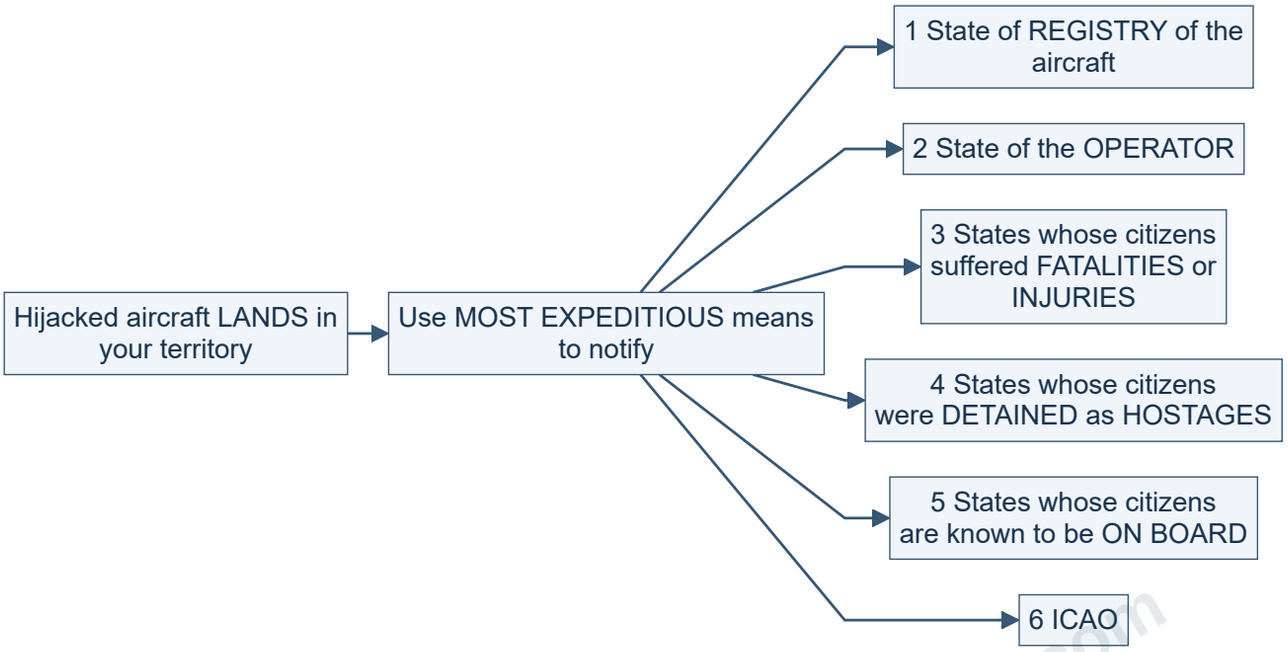

Figure 21.3 *Notification flowchart following landing of a hijacked aircraft*

Assistance to a Hijacked Aircraft

ASSISTANCE REQUIRED FROM A STATE — EXACT LIST

A State shall provide assistance to an aircraft subjected to an act of unlawful seizure. This assistance **includes**:

- **Provision of navigation aids**
- **Air traffic services**
- **Permission to land**

Watch-out: "Only permission to land" is *wrong*. "Catering for passengers" is *not* part of this prescribed *(laid down by the authority or the rules)* list, although a State may provide it humanely — exam answer must match the regulation, not common sense.

Potentially Disruptive Passengers

SUPPLEMENTARY SAFEGUARDS FOR DISRUPTIVE PASSENGERS

For the transport of **potentially disruptive passengers**, some supplementary safeguards are to be observed, such as **boarding prior to all passengers**.

The disruptive passenger (e.g. a deportee, person in custody, or a known disorderly individual) is boarded *first*, before all other passengers — not at the pilot-in-command's discretion, not at the State's discretion, but as a prescribed safeguard.

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Mixing & Contact Post-Screening

RE-SCREENING RULE — CROSS-CONTAMINATION

When **mixing or contact does take place** between passengers who have been subjected to security control, and other persons *not* subjected to such control, **after** the security screening points at airports serving international civil aviation have been passed — then:

The passengers concerned AND their cabin baggage shall be RE-SCREENED before boarding an aircraft.

Why? Once a screened passenger contacts an unscreened person, the integrity of the sterile area is compromised — a prohibited item could have been passed across. The only way to restore security integrity is to re-screen passenger *and* cabin baggage.

What the pilot does (Annex 2, 3.7 and Attachment B)

- An aircraft subjected to unlawful interference shall **endeavour to notify the ATS unit**: the fact, any significant circumstances and any deviation from the flight plan, so ATS can give it priority and minimise conflict (Annex 2, 3.7).
- Unless conditions on board dictate otherwise, the PIC should **continue on the assigned track and level** at least until able to notify ATS or within radar coverage.
- If forced to leave the assigned track or level without radio contact, whenever possible: broadcast warnings on the VHF emergency frequency and other appropriate frequencies, use the **transponder (SSR code 7500)** and data link, and follow any published regional contingency procedure.
- If no regional procedure exists, fly a level that differs from the normal IFR cruising levels by **500 ft** where vertical separation minimum is 1,000 ft, or by **1,000 ft** where it is 2,000 ft.
- SSR codes **7500 / 7600 / 7700** are reserved internationally for unlawful interference, radio failure and emergency respectively (Annex 10 Vol IV 8.5.2.1).

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Carriage of Weapons by Foreign Agents

Some States allow armed personnel from foreign States (e.g. sky marshals, diplomatic protection officers, presidential bodyguards) to carry weapons on board aircraft within their territory. Annex 17 imposes **two strict conditions** before such carriage is permitted.

CONDITIONS FOR CARRIAGE OF WEAPONS BY FOREIGN PERSONNEL

When a member state allows **police officers, security staff, bodyguards or other agents of foreign states** to carry weapons in their territory for the protection of aircraft in flight, permission for the carriage of weapons should be conditional upon:

1. **Prior notification** by the State of *embarkation* to the foreign State in which the weapons will be carried on the airport of arrival; **AND**
2. **Notification of the Pilot-in-Command** of a decision to permit a weapon to be carried on board his aircraft.

Both conditions are mandatory. Notification of only the pilot, without prior State-to-State notification, is *insufficient*. A mere agreement between embarkation and destination States, without informing the PIC, is also *insufficient*.

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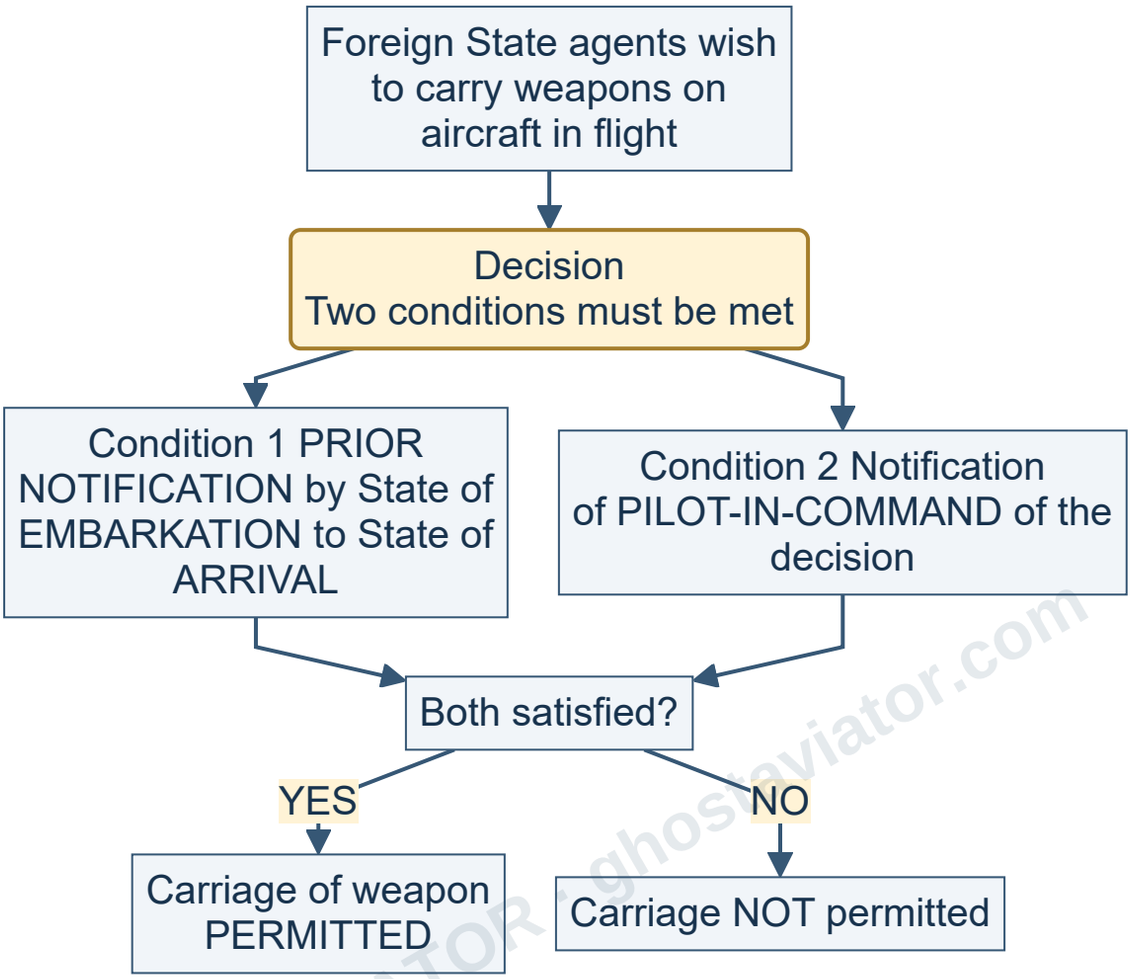

Figure 21.4 *Decision flow for foreign-agent weapons carriage*

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National Provisions — The Indian Framework

India implements its Annex 17 obligations through a dedicated national framework. Every student pilot must know the chain of command and the regulatory instruments involved, because RTR examiners ask **who reports to whom** and **which rules govern**.

Bureau of Civil Aviation Security (BCAS)

BCAS — CONSTITUTION & MANDATE

The **Central Government** has constituted a body known as the **Bureau of Civil Aviation Security (BCAS)**, which shall be **headed by an officer designated as the Director General of Bureau of Civil Aviation Security**, to be appointed in this behalf by the Central Government by **notification in the Official Gazette**.

The BCAS shall be responsible for carrying out the **regulatory and oversight functions** in respect of matters relating to civil aviation security.

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The Governing Rules — Aircraft (Security) Rules, 2011

APPROPRIATE AUTHORITY FOR AVIATION SECURITY

The Aircraft (Security) Rules,2023 superseded the 2011 Rules. Rule 2(1)(s) defines the Director General of BCAS as the appropriate authority for Annex 17. Rule 3 assigns regulatory and oversight responsibilities.

Aircraft (Security) Rules 2023, Rules 2(1)(s),2(1)(t),3,51 and 52(1).

Functions of the Commissioner of Security (CA)

DUTIES OF THE COMMISSIONER — FIVE-VERB RULE

Rule 3(2)(a) empowers the Director General to develop, implement, maintain and review the National Civil Aviation Security Programme, consistent with Annex 17, taking account of the safety, regularity and efficiency of flights.

- **Safety** of flights
- **Regularity** of flights
- **Efficiency** of flights

Aircraft and aerodrome operators must comply with the applicable directions issued by the Director General, BCAS.

Airport-Level Responsibility — AAI vs Private/JVC Airports

India has a two-track system at airport level. AAI-operated airports are looked after by the AAI's own Directorate of Security, while major private and joint-venture airports have their own security set-ups.

Category	Nodal/Responsible Body	Airports / Notes
AAI Airports	**Directorate of Security (AAI)** acts as the nodal agency	Covers AAI airports only. At individual airports, the **Airport Director** is responsible for security functions.
JVC / Private Airports	**DIAL** — Delhi International Airport Ltd.	Delhi (IGI) — Operates own security set-up
	MIAL — Mumbai International Airport Ltd.	Mumbai (CSMI)
	CIAL — Cochin International Airport Ltd.	Cochin
	HIAL — Hyderabad International Airport Ltd.	Hyderabad (RGIA)
	BIAL — Bangalore International Airport Ltd.	Bangalore (KIA)

These JVC/private airports have their own security set up to monitor and carry out security functions.

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Authority	Role
Central Government	Constitutes the statutory framework and appoints the Director General, BCAS.
DG, BCAS / BCAS	National civil-aviation security regulation and oversight.
Aerodrome and aircraft operators; other entities	Implement applicable security requirements and comply with directions.

Figure 21.5 *Security oversight and implementation responsibilities.* *BVA 2024 §5; Security Rules 2023 §§2(1)(s),3.*

Reporting & Investigation of Security Incidents

Who Must Report

REPORTING SECURITY OCCURRENCES

Rule 51 requires every entity to report security occurrences or incidents as specified by the Director General. Follow the applicable BCAS directions and approved security programme for the reporting channel and timing. Rule 51 itself does not set a five-category limit or state a universal deadline.

Aircraft (Security) Rules 2023, Rules 2(1)(s),2(1)(t),3,51 and 52(1).

Investigation Procedure

SECURITY INVESTIGATION

Under Rule 52(1), the Director General may order a security investigation in writing and appoint an officer not below the rank of Assistant Director as Inquiry Officer.

Aircraft (Security) Rules 2023, Rules 2(1)(s),2(1)(t),3,51 and 52(1).

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Stage	Statutory provision
Report	Every entity reports as specified by the Director General — Rule 51.
Order an investigation	Director General may direct investigation by an order in writing — Rule 52(1).
Appoint Inquiry Officer	Officer not below Assistant Director — Rule 52(1).
Conduct inquiry and report	Give concerned parties an opportunity to be heard; submit a written report to the Director General — Rule 52(2).

Figure 21.6 *Reporting and inquiry under the 2023 Security Rules.* *Aircraft (Security) Rules 2023, Rules 51–52.*

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PART VI

Human Performance and Limitations

The pilot as part of the system: how people make errors, how crews work together, how the mind handles information, and what altitude does to the body.

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CHAPTER 22

Human Factors: The Basics

22

The Knowledge Requirement

WHAT THIS SECTION COVERS
Why human factors knowledge is mandatory for flight crew, who else benefits, and the scale of human error in aviation accidents.

This part of the book is designed to meet the **ICAO requirements** of human factor knowledge for the flight crew. The relevant portions of the information are also useful to other key aviation personnel:

- Cabin Crew
- Air Traffic Controllers (ATCs)
- Maintenance personnel

Why the Human Element Matters

The human element is the **most flexible, adaptable and valuable** part of the aviation system — but it is also the **most vulnerable** to influences which can adversely affect its performance.

KEY STATISTIC — ACCIDENT CAUSATION
Throughout the years, **three out of four** accidents have resulted from less than optimum human performance. This has commonly been classified as *human error*.

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Human errors are now considered as being **inherent to the cognitive function** of humans and are **generally inescapable**. A sound understanding of Human Performance and Limitations will help enhance performance and safety.

EXAM TIP
Remember the framing: human error is **not** a moral failing or laziness — it is an inherent cognitive feature. The objective of HPL training is to *manage* error, not eliminate it.

Quick Revision — Section 1

- Part 2 of Air Regulations meets **ICAO** human factor knowledge requirements.
- Relevant to: Flight Crew, Cabin Crew, ATCs, Maintenance.
- **3 out of 4** accidents result from less-than-optimum human performance.
- Human error is **inherent** and **generally inescapable**.

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Disciplines Frequently Involved in Human Factors Activities

WHAT THIS SECTION COVERS
The multidisciplinary nature of Human Factors and the academic fields contributing to it.

Human Factors are **multidisciplinary in nature**. Information is drawn from many fields:

- **Psychology** — to understand how people process information and make decisions.
- **Psychology & Physiology** — sensory processes for detecting and transmitting information.
- **Anthropometry & Bio-mechanics** — body measures & movement for optimizing flight deck and cabin design.
- **Biology & Chronobiology** — body rhythms and sleep, effects in night flying and time-zone changes.
- **Statistics** — proper analysis or presentation of data from surveys/studies.

PRACTICAL ORIENTATION
Human Factors is **practical in nature**; it is **problem-oriented** rather than discipline-centered. It exists to solve real-world aviation problems.

Scope of Disciplines

Human Factors disciplines are **inclusive of, but not limited to**:

- Psychology
- Engineering
- Physiology
- Sociology
- Anthropometry

Other disciplines with active representatives in Human Factors activities include:

- Education
- Physics
- Biochemistry
- Mathematics
- Biology
- Industrial design
- Operations research

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Disciplines Table (Authoritative)

Discipline	Definition	Specific Area of Interest	Typical Area of Application
Psychology	The science of mind and behavior.	Sensory characteristics, perceptual laws, learning principles, information processing, motivation, emotion, research methods, psychomotor skills, human errors.	Display requirements and design, control systems design, allocation of function, training systems requirements and methods, selection methods, effects of emotional and environmental stress on performance, simulation requirements.
Engineering	Applying the properties of matter and the sources of energy in nature to the uses of man.	Hydraulics, mechanical, structural, electrical, electronic, and aerodynamics design, systems analysis, simulation, optics.	Design of displays, design of controls, design of control systems, design of complex systems, design of optical systems, simulator design.
Human Physiology	Deals with the processes, activities, and phenomena characteristic of living matter, particularly appropriate to healthy or normal functioning.	Cell structure and chemistry, organ structure and chemistry, interaction of the various body constituents to promote health and function, functions and requirements of body systems.	Environmental systems, diet and nutrition, effects of environmental factors (heat, cold, hypoxia *(too little oxygen reaching body tissues)*), establishment of environmental requirements.
Medicine	The science and art of preventing, alleviating, or curing disease and injuries.	Effects of various forces, radiation, chemical and disease agents; appropriate preventive methods of protecting health and well-being.	Toxicology of smoke, chemicals, impact protection, maintenance of health.
Sociology	The study of the development, structure, and function of human groups.	Small and large groups or "teams"; crew composition; behavior of passengers in emergency situations.	Crew selection, passenger safety.
Anthropometry	Study of human body sizes and muscle strength.	Anatomy, biodynamics, kinesiology.	Ground support equipment, access door size for maintenance, work station layout (reach, range of adjustment of seats, etc.)

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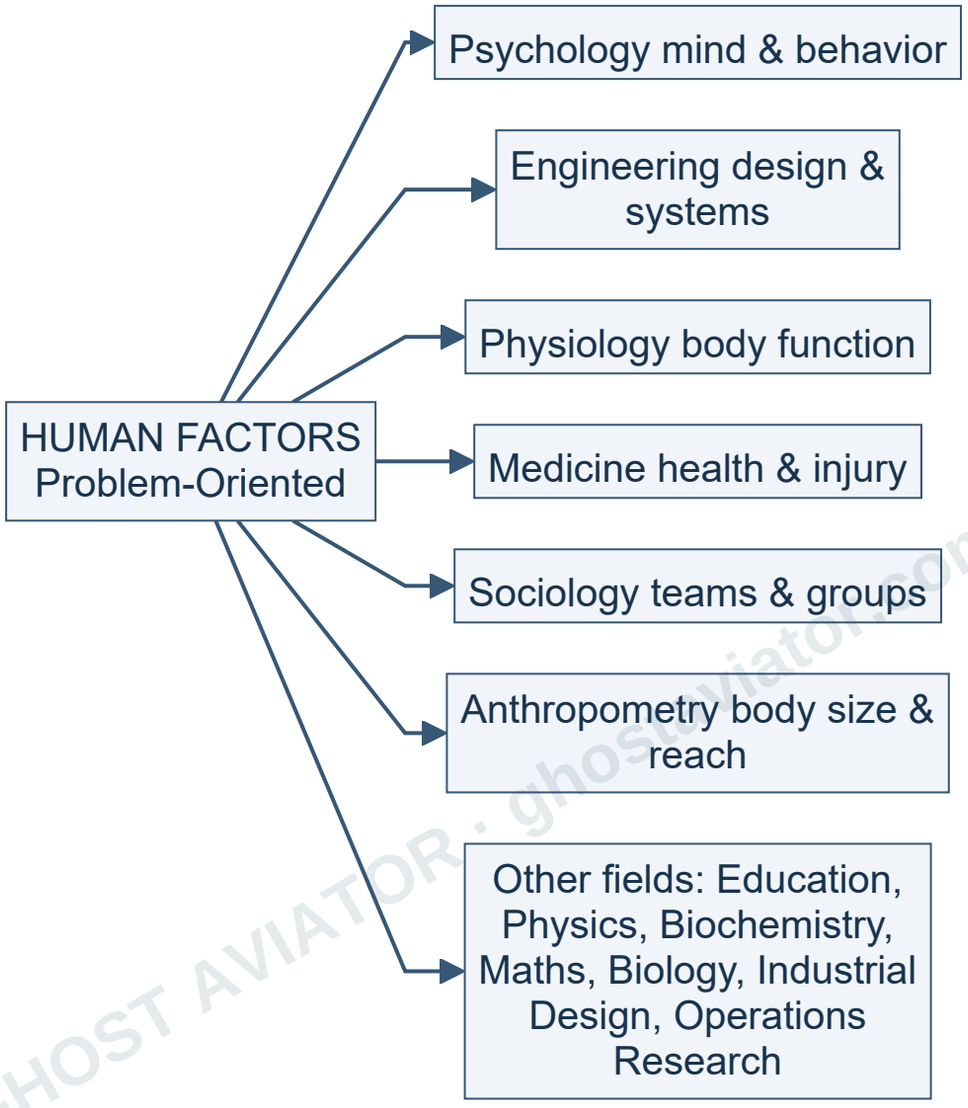
Visualising the Multidisciplinary Web



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Quick Revision — Section 2

- HF is **multidisciplinary** & **problem-oriented**.
- Core 5: **Psychology, Engineering, Physiology, Sociology, Anthropometry**.
- Plus: Medicine, Education, Physics, Biochemistry, Maths, Biology, Industrial Design, Operations Research.

Qualities of a Professional Pilot

> **WHAT THIS SECTION COVERS**
> The DGCA-recognised attributes that define a professional pilot — examinable as a list and as conceptual groupings.

The following are the **qualities of a professional pilot**:

1. Situation awareness

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2. Flexibility
3. Task management
4. Effective communications
5. Threat and error management
6. High sense of responsibility
7. Aircraft handling skills
8. Sound knowledge of flying theory
9. Motivation
10. Physical fitness
11. Reliability
12. Balanced personality
13. Teamwork
14. Stress management
15. Quick reflexes
16. Risk assessment capabilities
17. Decision making
18. Crew resource management

Grouping for Easier Recall

Cluster	Qualities
Cognitive / Mental	Situation awareness, Decision making, Risk assessment, Sound knowledge of flying theory, Threat & error management
Behavioural / Personal	Motivation, Reliability, High sense of responsibility, Balanced personality, Flexibility
Operational / Skill	Aircraft handling skills, Quick reflexes, Task management, Stress management
Social / Team	Effective communications, Teamwork, Crew Resource Management (CRM)
Physical	Physical fitness

> **STANDARD PRACTICE**
> A professional pilot continually develops these qualities through structured training, recurrent checks, simulator exercises and personal reflection. CRM training in particular is mandated for two-crew operations.

Quick Revision — Section 3

- 18 qualities listed in the syllabus — learn the full list verbatim.
- Five mental groupings: **Cognitive, Behavioural, Operational, Social, Physical**.
- Critical examinable items: SA, CRM, Decision Making, Threat & Error Management.

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Causes of Aircraft Accidents

WHAT THIS SECTION COVERS
The base rate of accidents in commercial aviation, the dominant role of human factors, the specific deficiencies cited, and the most common form of accident.

Accident Rate

ACCIDENT RATE — COMMERCIAL AVIATION
The rate of accidents in commercial aviation (*excluding sabotage and acts of terrorism*) is approximately **1 accident per million airport movements**.

Human Performance as Root Cause

Most aircraft accidents are linked to deficiencies in human performance. These deficiencies may involve a variety of factors. The factors include:

- Poor lookout
- Situation awareness (SA)
- Decision-making
- Task organization
- Communication
- Failure to recognize threats to safety
- The commission of errors

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Most Common Form of Air Accident

CRITICAL EXAM POINT
Controlled Flight Into Terrain (CFIT) is the **most common form of air accident**.

CFIT occurs when an airworthy aircraft, under the control of a qualified pilot, is flown unintentionally into terrain, water, or an obstacle — usually with no prior awareness by the crew. Because the aircraft is functioning normally, CFIT is overwhelmingly a **human factors** accident category (loss of situation awareness, poor decision-making, communication breakdown, navigation errors).

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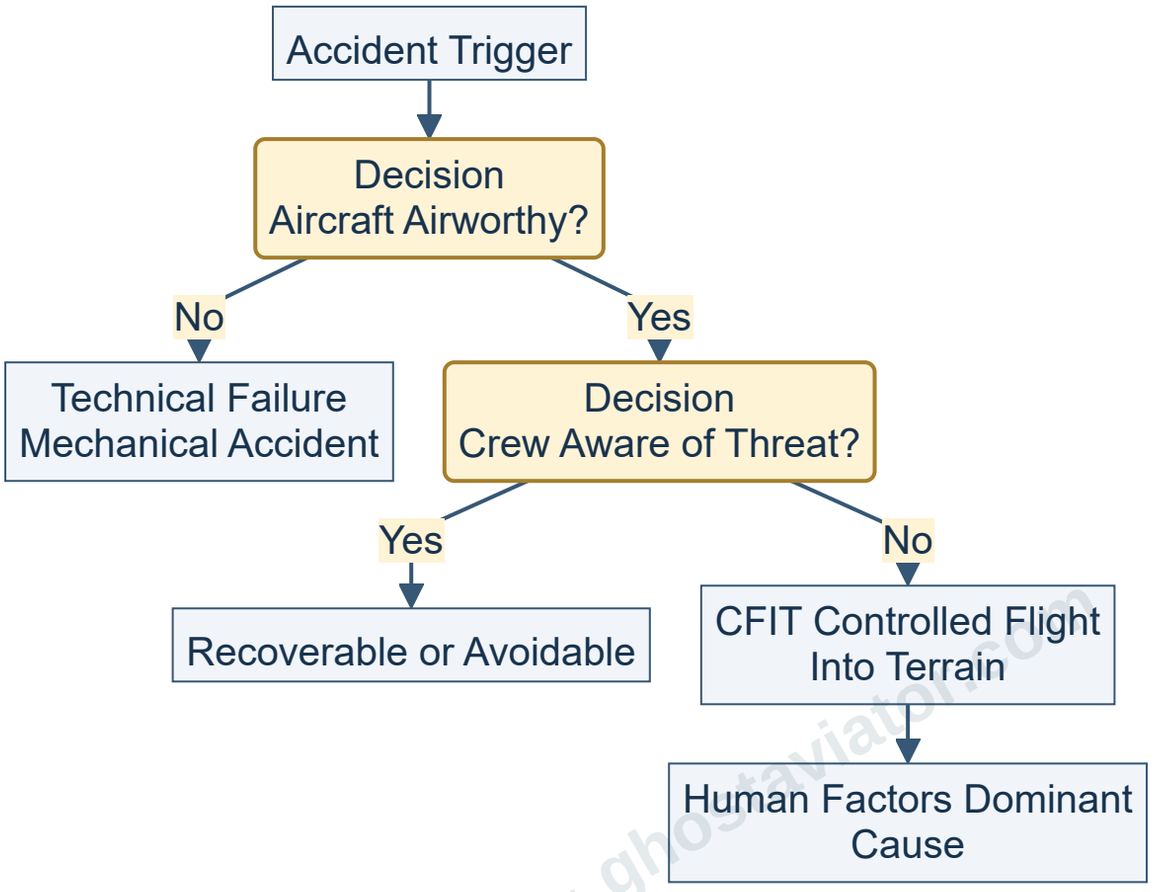

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COMMON CONFUSION

Don't confuse **CFIT** (Controlled Flight Into Terrain — aircraft serviceable) with **LOC-I** (Loss Of Control In-flight — aircraft departs controlled flight). Both are human-factors heavy, but CFIT is explicitly named in the syllabus as "the most common form" — that's the testable phrase.

Quick Revision — Section 4

- Accident rate: **~1 per million airport movements** (excluding sabotage & terrorism).
- Most accidents linked to **deficiencies in human performance**.
- Specific factors: poor lookout, SA, decision-making, task organization, communication, failure to recognise threats, commission of errors.
- **CFIT is the most common form of air accident.**

Human Factors — Physiological Demands & Adaptation

WHAT THIS SECTION COVERS

The environmental stresses imposed by flight, the body's compensatory mechanisms, and the absolute limit of oxygen deprivation.

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Environmental Changes of Greatest Physiological Significance

In aviation, the demands upon the **compensatory mechanisms** of the body are numerous and of considerable magnitude. The environmental changes of greatest physiological significance involved in flight are:

a) marked changes in barometric pressure,
b) considerable variation in temperature, and
c) movement at high speed in three dimensions.

The **limiting factors** in adjustment of the human body to flight must be appreciated.

The Body's Remarkable Adaptive Capacity

Human beings have the remarkable ability to adapt to their environment. The human body:

- Makes adjustments for changes in **external temperature**.
- Acclimates to **barometric pressure variations** from one habitat to another.
- Compensates for **motion in space** and **postural changes** in relation to gravity.
- Performs all these adjustments **while meeting changing energy requirements** for varying amounts of physical and mental activity.

Compensating for Reduced Oxygen Supply

The human body can adjust to **acute and chronic reductions** in its oxygen supply by:

1. Increasing respiratory rate
2. Chemical changes in the blood
3. Increasing the production of red blood cells

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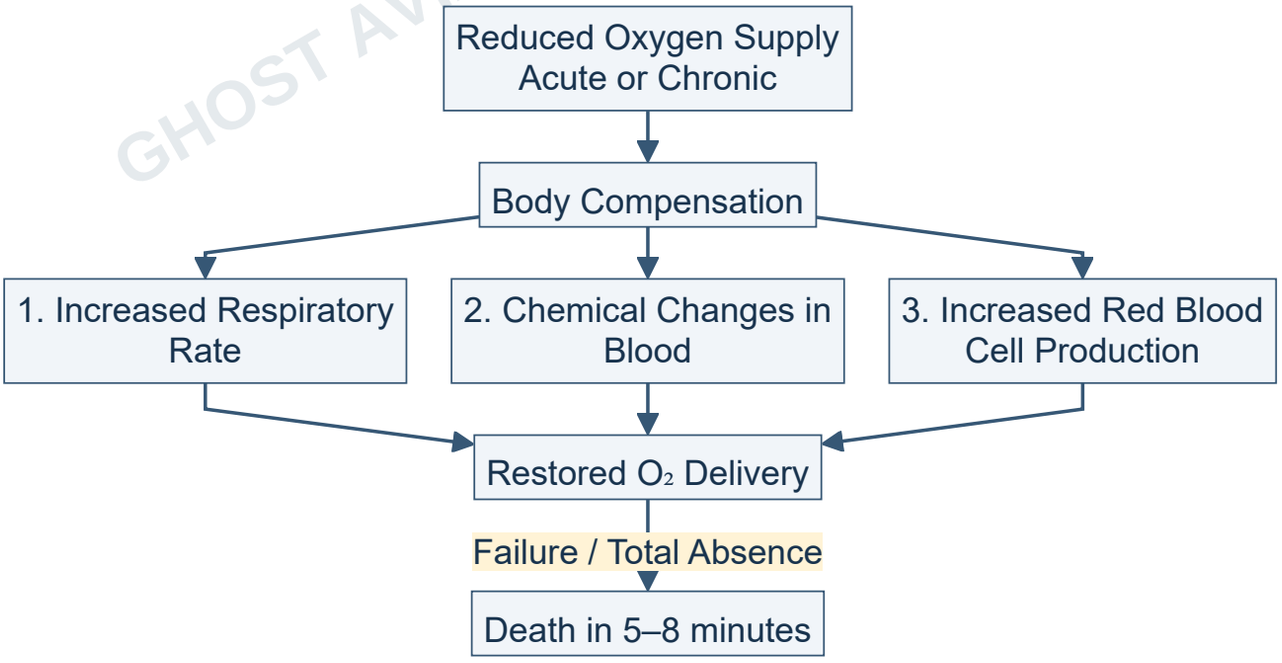

ABSOLUTE PHYSIOLOGICAL LIMIT

As efficient as the body is, a **complete absence of oxygen** will cause death in approximately **5 to 8 minutes**.

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Why HPL Training is Mandatory

THE 73% STATISTIC

Human factors are cited as a **major cause in over 73% of accidents**. This figure **has not changed since the 1950's**.

If the accident rate is to be decreased, Human Factors issues in aviation must be:

- Better understood, and
- Human Factors knowledge more broadly and proactively applied.

TRAINING OBJECTIVES

It is necessary that a better understanding of human capabilities and limitations — **both physical and psychological** — are known to aviators. It is also necessary for the flight crew personnel to train to ensure:

- Safe and efficient operation
- Reducing error
- Avoiding stress
- Increasing efficiency

Worked Example — Oxygen-Free Survival Window

Scenario: Rapid cabin depressurisation at FL350 with both pilots experiencing simultaneous oxygen mask failure (a worst-case *complete absence of oxygen* at altitude).

Available time before death: 5 to 8 minutes.

However — Time of Useful Consciousness (TUC) at FL350 is only ~30–60 seconds, far shorter than the death window. **Useful actions (donning mask, initiating emergency descent) must be completed within seconds, not minutes.**

Lesson: The "5–8 minutes to death" figure does not mean you have 5–8 minutes to act. Useful function ends in well under a minute at cruising altitudes — hence the absolute priority of the "Oxygen Mask — ON, 100%" memory item.

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Quick Revision — Section 5

- Three flight environment stressors: **barometric pressure, temperature, high-speed 3-D motion**.
- Body adjusts to: temperature, pressure, motion, posture, energy demand.
- Oxygen compensation: ↑ **respiratory rate, blood chemistry change,** ↑ **RBC production**.
- Complete oxygen absence → death in **5–8 minutes**.
- Human factors cause **>73%** of accidents — unchanged since the **1950's**.

Accident Distribution by Phase of Flight

WHAT THIS SECTION COVERS

The recreated DGCA chart of accident percentages versus phase of flight, with the critical insight that *accident risk is wildly disproportionate to flight time*.

The Headline Insight — Risk vs. Time

Phase Group	% of Flight Time	% of Accidents	Risk Indicator
Take-off & Initial Climb	**2%**	23.4%	Disproportionately high
Cruise (Climb + Cruise + Descent)	**83%**	~21.6%	Low (relative to time)
Approach & Landing	**15%**	~33.8%	Disproportionately high

CRITICAL PATTERN

Only **17%** of flight time (take-off + approach + landing) accounts for the majority of accidents. Conversely, the cruise — comprising **83%** of flight time — sees a far smaller share. The take-off and landing phases are the highest-risk operational windows.

Phase-by-Phase Accident Percentages (DGCA chart, recreated)

Numerical Breakdown

Phase	% of Accidents	Rank (highest = 1)
Landing	**24.1%**	1
Take-off & Initial Climb	**23.4%**	2
Cruise	**15.7%**	3
Maneuvering	**13%**	4
Approach	**9.7%**	5
Preflight / Taxi	**9.5%**	6
Other	**4.7%**	7
Climb	**3.3%**	8
Descent	**2.6%**	9

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EXAM TIP — "MOST ACCIDENTS TAKE PLACE DURING…"

The DGCA-correct answer is **Approach and Landing**. Even though "Take-off and Initial Climb" is a single very high figure (23.4%), the combined Approach + Landing phases (9.7% + 24.1% = **33.8%**) represent the largest share of all accidents.

SOP IMPLICATION

The "sterile cockpit" rule below **10,000 ft** exists precisely because approach/landing and take-off/climb dominate the accident statistics. Crew workload management and discipline in these phases is the single biggest accident-prevention lever.

Quick Revision — Section 6

- Flight time split: T/O+Climb **2%** | Cruise group **83%** | App+Landing **15%**.
- Highest accident phase: **Landing (24.1%)**, followed by Take-off/Initial Climb (23.4%).

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- Combined Approach + Landing = **33.8%** → "Most accidents take place during Approach and Landing."
- Cruise sees only 15.7% of accidents despite occupying most flight time.

> **HOW TO USE THIS SECTION**
>
> Attempt each question *before* reading the answer. If you got it wrong, read the explanation and the distractor analysis — that is where the deepest learning happens. Use the anchor links to revisit the relevant study section for any concept you are unsure of.

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CHAPTER 23

Crew Resource Management, Threat and Error Management, LOFT

Crew Resource Management (CRM) — Introduction & Applicability

WHAT THIS SECTION COVERS

The definition and purpose of CRM, the cognitive vs interpersonal skill distinction, why CRM applies even to single-pilot operations, the link to the Tenerife disaster (**1977**), and the ICAO Standard compliance requirement.

Definition of CRM

DEFINITION — CRM

CRM is the **effective use of all available resources** for flight crew personnel to assure a safe and efficient operation, **reducing error, avoiding stress, and increasing efficiency**.

CRM is concerned *not* with technical knowledge and stick-and-rudder skills, but with the **cognitive and interpersonal skills** needed to manage a flight within an organized aviation system.

Skill Type	Definition
Cognitive skills	Mental processes used for gaining and maintaining *situational awareness*, for *solving problems* and for *taking decisions*.
Interpersonal skills	Communications and a range of behavioural activities associated with *teamwork*.

EXAM TIP — OVERLAP RULE

These skill areas **often overlap with each other AND with the required technical skills**. CRM is **not confined to multi-crew aircraft** — it also relates to **single-pilot operations**, which invariably interface with other aircraft and ground support agencies.

Applicability of CRM — Human Factors Framework

CRM is an **application of Human Factors (HF)** to the flight crew domain. The term "Human Factors" includes CRM along with other aspects of human performance.

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WORKING DEFINITION

CRM is the utilization of all resources available to the crew to manage human error. The objective of CRM training is to **contribute to incident and accident prevention**.

CRM is one practical application of Human Factors training — concerned with supporting **crew responses to threats and errors** that manifest in the operating environment.

SOP — TRAINING WEIGHT

The requirement to develop human-performance knowledge and skills among flight crew members and other operational personnel has the **same weight** as that related to systems, normal, abnormal, and emergency procedures. Most operators comply mainly through **CRM Training** and **LOFT**.

Historical Origin — The Tenerife Catalyst

It took one of history's worst crashes — the **Tenerife disaster of 1977** — to raise CRM from a mostly unpracticed principle to a full-fledged program. It became apparent that pilots needed to learn more about how best to manage all the resources available to them in the cockpit, including:

- Other crew members
- Procedures
- The machine interface
- Themselves

Regulatory Status

ICAO COMPLIANCE WARNING

CRM is now considered **essential training** for most aviation professionals who make an operational contribution — including **air traffic controllers and engineers**. **Non-compliance with the requirement to provide human-performance training would mean non-compliance with an ICAO international Standard.**

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Starting point / condition	Branch	Next step / result
Aviation System	→	Human Factors HF
Human Factors HF	→	CRM Training
Human Factors HF	→	Other HF Aspects
CRM Training	→	Error Avoidance
CRM Training	→	Threat Management
CRM Training	→	Error Management

Starting point / condition	Branch	Next step / result
CRM Training	→	Undesired A/C State Mgmt
Error Avoidance	→	Accident/Incident Prevention
Threat Management	→	Accident/Incident Prevention
Error Management	→	Accident/Incident Prevention
Undesired A/C State Mgmt	→	Accident/Incident Prevention

QUICK REVISION SUMMARY — SECTION 1

- **CRM = Effective use of ALL resources** → reduce error, avoid stress, increase efficiency.
- Focus on **cognitive + interpersonal skills**, not stick-and-rudder.
- Applies to **single-pilot operations too**.
- Driver event: **Tenerife 1977**.
- Mandatory for compliance with an **ICAO Standard**.
- Aim: **contribute to incident and accident prevention**.

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A Conceptual Model of Human Factors — The ICAO SHEL Model

WHAT THIS SECTION COVERS

The SHEL model's origin (**Edwards, 1972**), its ICAO basis (**Doc 9859**, **Circular 216-AN31**), the four components (S–H–E–L), the four interfaces, why Liveware is the hub, and why mismatches matter.

Purpose of the SHEL Model

The SHEL model is a conceptual model of Human Factors that:

- Clarifies the **scope** of aviation human factors.
- Assists in understanding the relationships between the **flying subsystem** (system resources/environment) and the **human subsystem**.
- Provides a conceptual framework illustrating the various constituents and their **interfaces — the points of interaction**.

ICAO Reference — Source Authority

Reference	Detail
ICAO Source Document	**ICAO Doc 9859** — Safety Management Manual (SMM)
Earlier Framework Reference	**ICAO Circular 216-AN31**

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Reference	Detail
Original Developer	**Edwards — 1972**
Source Disciplines	Psychology · Engineering · Physiology · Anthropometry

The Four SHEL Components

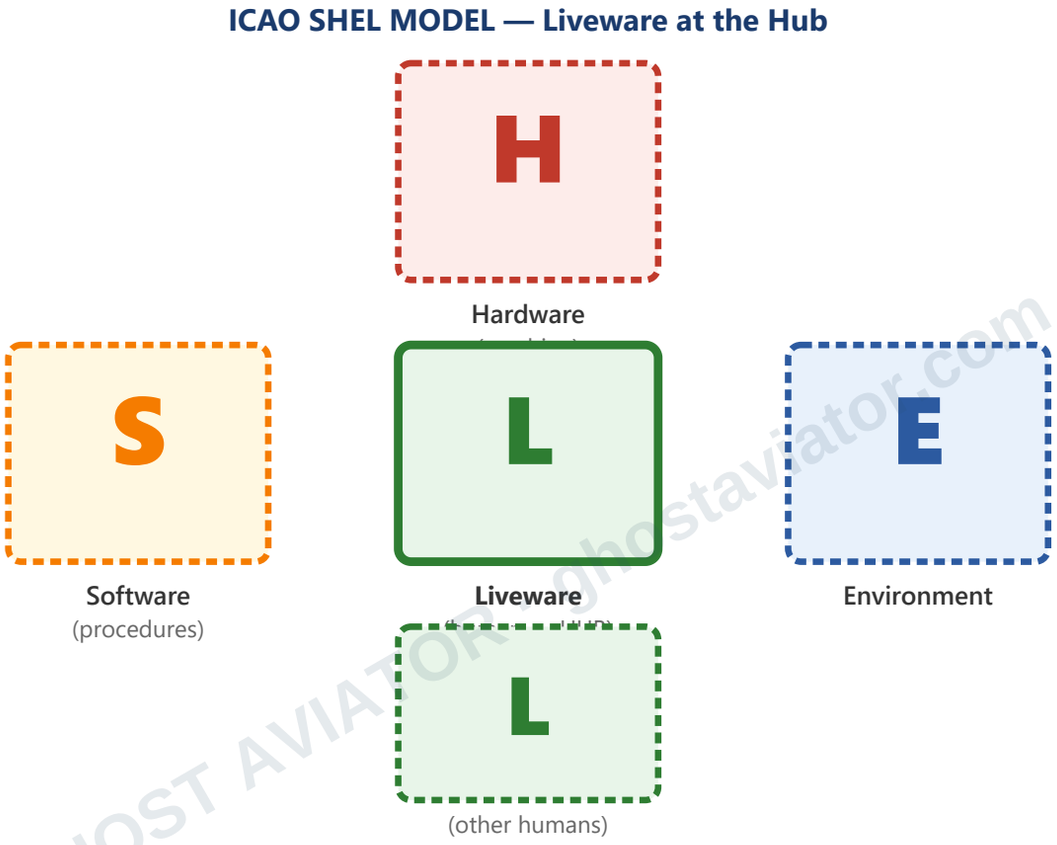

Figure 23.1 *H*

Letter	Stands For	Description
S	Software	Procedures, manuals, checklist layouts, documents, symbology, computer programmes, maps and charts.
H	Hardware	Design of flight decks, physical structure of the aircraft, presentation of instruments, positioning and operating sense of controls, machinery, equipment, etc.
E	Environment	Both **internal and external** to the workplace.
L	Liveware (1st)	Man — the most **critical, valuable and flexible** component of the system.
L	Liveware (2nd)	Other humans in contact with the pilot.

The Four Interfaces

Interactions between human beings and other SHEL elements are at the heart of Human Factors. The interfaces are:

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Interface	Pairing	Examples (operational)
L–H	Humans & Machines	Cockpit layout, controls, displays, instruments, congested cockpit, eye reference position
L–S	Humans & Materials	Checklists, manuals, SOPs, NOTAMs, automation, computer software, warning logic/symbology
L–L	Humans & Colleagues	Crew co-ordination, briefings, communication (incl. noise interference), worker-management
L–E	Humans & Operating Environment	Weather, time of day, circadian rhythm/jet lag, illumination, vibration, aerodrome facilities

Liveware as the Hub — Why the Match Matters

CORE SHEL PRINCIPLE

The Liveware (human element) is the HUB of the SHEL model. The remaining components must be **adapted and matched** to this central component.

MISMATCH = SOURCE OF HUMAN ERROR

In this model the **match or mismatch** of the blocks (the interface) is **just as important as the characteristics of the blocks themselves**. A mismatch can be a source of human error **leading to an accident or an incident**.

Humans suffer considerable performance variation and many limitations — most of which are now predictable. The Liveware must be carefully matched to other components to **avoid stress in the system and eventual breakdown**.

QUICK REVISION SUMMARY — SECTION 2

- **SHEL** = Software, Hardware, Environment, Liveware (+ second Liveware).
- Developed by **Edwards, 1972**; described in **ICAO Doc 9859** and Circular **216-AN31**.
- Four interfaces: **L–H, L–S, L–L, L–E**.
- **Liveware is the HUB** → all other components must adapt to humans.
- **Mismatch at any interface = source of human error**.

Factors Relating to the Individual (Liveware)

WHAT THIS SECTION COVERS

The four factor categories that shape pilot performance — **Physical, Physiological, Psychological, Psychosocial** — and every sub-element listed in the DGCA source. Note that the **physiological and psychosocial factors are more pronounced** in effect.

Many factors influence aviation personnel: **Physical, Physiological, Psychological, Psychosocial**. However, the effect of **physiological and psychosocial** factors is more pronounced.

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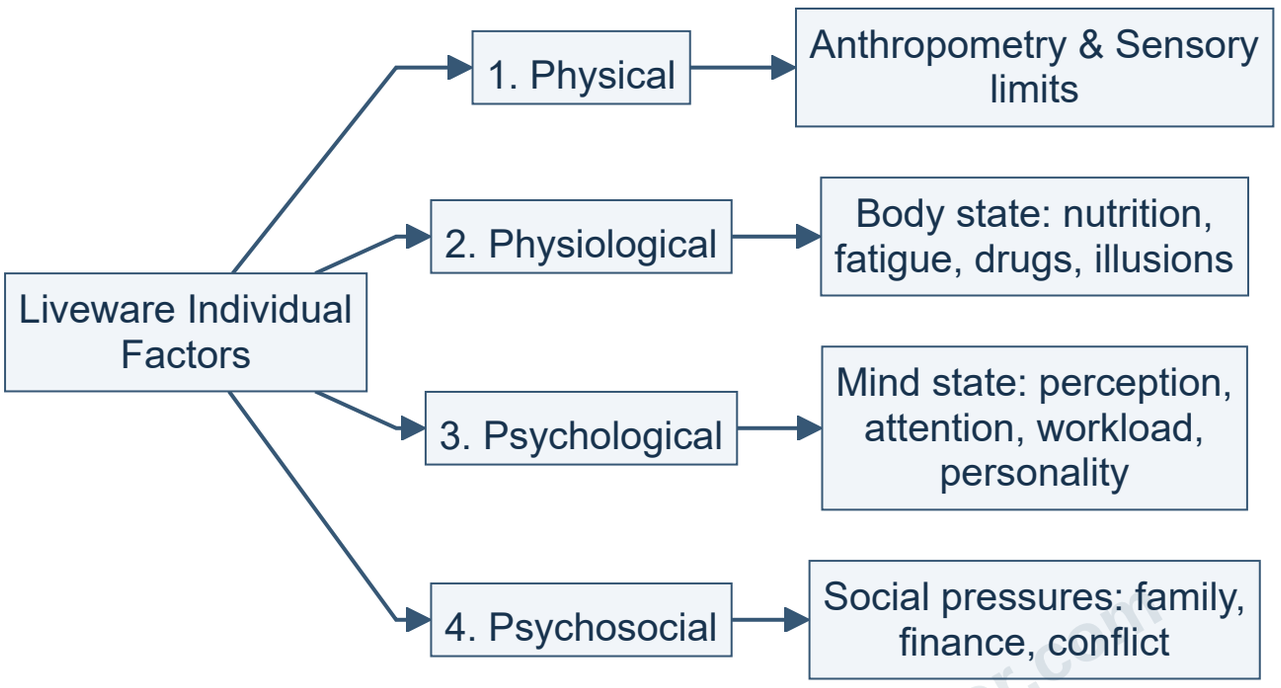

Physical Factors

Anthropometric attributes: Height, weight, age, sex, build, sitting height, functional reach, leg length, shoulder width, strength, co-ordination.

Sensory Limitations — Vision

- Visual threshold
- Visual acuity (seeing details)
- Focus time, light adaptation
- Peripheral vision, speed, depth perception
- Empty field myopia
- Glasses, contacts
- Auditory threshold, understanding
- Vestibular (ear senses), smell, touch, kinesthetic (body feelings)
- G-tolerances

Physiological Factors

Sub-Factor	Elements
Nutritional	Food intake **24 hours**, hours since last meal, dehydration, on a diet / weight loss.
Health	Disease, fitness, pain, dental conditions, blood donation, obesity, pregnancy, stress coping (emotional/behavioural signs), smoker.
Lifestyle	Friendships, relations with others, change in activities, life habits.
Fatigue	**Acute** (short term), **chronic** (long term), skill (due to task), activity level (mental/physical).
Duty	Duration of flight, duty hours, leave periods, activities.

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Sub-Factor	Elements
Sleep	Crew rest, nap duration, sleep deficit, disruption, **circadian dysrhythmia (jet lag)**.
Drugs	Medication over the counter, medication prescription, illicit drugs, cigarettes, coffee, others.
Alcohol	Impairment, hangover, addiction.
Incapacitation	Carbon monoxide poisoning, hypoxia *(too little oxygen reaching body tissues)*/anoxia, hyperventilation, loss of consciousness, motion sickness, food poisoning, nauseating fumes, toxic fumes.
Decompression / Diving	Decompression, trapped gas effects, underwater diving.

Physiological Illusions

Type	Examples
Vestibular	Somatogyral (vertigo), somatogravic, the leans, Coriolis illusion, elevator illusion, giant hand.
Visual	Black hole, autokinesis, horizontal misplacement, circularvection, linearvection, landing illusions, chain-link fence illusion, flicker vertigo, geometric perspective illusion.

SELF-MEDICATION PROHIBITION

Over-the-counter (OTC) medicines from a chemist **are NOT permitted** for a pilot before flight. Even OTC items from an aircraft first-aid box **must NOT** be taken to fly. **All medication must be prescribed *(laid down by the authority or the rules)* by an aviation medicine specialist, whose advice on fitness to fly is binding.**

Psychological Factors

Sub-Factor	Elements (source)
Perceptions	*Types:* Non-perception, misperception, delayed perception.
Reaction Time	To detect → to make an appropriate decision → to take the appropriate action.
Disorientation	Situational awareness, spatial, visual, temporal, geographic (lost).
Attention	Attention span, inattention (general/selective), distraction (internal/external), channelized attention, fascination, fixation, vigilance, boredom, monotony, habit pattern interference/substitution, time distortion.
Information Processing	Mental capacity, decision making (delayed/poor), judgment (delayed/poor), memory capacity, forgetting, co-ordination — timing.
Workload	Task saturation, **underload**, prioritization, task components.
Experience / Recency	In position, in aircraft type, total time, on instruments, on route, aerodrome, night time, emergency procedures.
Knowledge	Competence, skills/techniques, airmanship, procedures.
Training	Initial, on-the-job, ground, flight, transition, learning transfer, recurrent, problem areas, emergency procedures.
Planning	Pre-flight, in-flight.

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Sub-Factor	Elements (source)
Attitudes / Moods	Mood, motivation, habituation, attitude, boredom, complacency.
Expectations	Mind set / expectancy, false hypothesis, "get-home-itis", risk-taking.
Confidence	In aircraft, equipment, self; overconfidence, showing off.
Mental / Emotional State	Emotional state, anxiety, apprehension, panic, arousal level / reactions, self-induced mental pressure / stress.
Personality	Withdrawn, grouchy, inflexible, hostile, sarcastic, negative, aggressive, assertive, impulsive, excitable, careless, immature, risk-taker, insecure, follower, disorganized, late, messy, anti-authoritative, resigned, invulnerable, "macho".

HIGH AROUSAL OVERLOAD — ATTENTION NARROWING

Under **high arousal / cognitive overload**, attention does **NOT** sharpen — there is a **real danger of attention becoming narrowed** (channelized attention, fixation). Optimum arousal applies to *moderate* difficulty tasks; for highly difficult cognitive tasks the optimum arousal is *lower*.

Psychosocial Factors

- Mental pressure
- Interpersonal conflict
- Personal loss
- Financial problems
- Significant lifestyle changes
- Family pressure

QUICK REVISION SUMMARY — SECTION 3

- Four Liveware factor families: **Physical, Physiological, Psychological, Psychosocial**.
- **Physiological + Psychosocial** effects are most pronounced.
- Personality red flags: **5 hazardous attitudes** ⇒ anti-authoritative, impulsive, invulnerable, macho, resigned.
- **OTC self-medication = prohibited**; only aviation-medicine-specialist prescription is acceptable.
- High arousal ⇒ **attention narrows**, not sharpens.

Factors Related to Individuals and Their Work — The Four Interfaces

Liveware–Liveware (Human–Human) Interface

Sub-Element	Items
Oral Communication	**Noise interference**, misinterpretation, phraseology (operational), content, rate of speech, language barrier, readback/hearback.
Visual Signals	Ground/hand signals, body language.

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Sub-Element	Items
Crew Interactions	Supervision, briefings, co-ordination, compatibility/pairing, resource management, task assignment, age, personality, experience.
Controllers	Supervision, briefing, co-ordination.
Passengers	Behavior, briefing, knowledge of aircraft, procedures.

Worker–Management (still within L–L)

Sub-Element	Items
Personnel	Recruitment/selection, staffing requirements, training, policies, remuneration/incentives, crew pairing, scheduling, seniority, resource allocation, operational support/control, instructions/directions/orders, managerial operating pressure.
Supervision	Operational supervision, quality control, standards.
Labour Relations	Employee/employee-management, industrial action, unions/professional group.
Pressures	Mental pressure — operational, morale, peer pressure.
Regulatory Agency	Standards, regulations, implementation, audit, inspection, monitoring, surveillance.

EXAM TIP — NOISE IN COCKPIT

High noise level in a cockpit is treated as a Liveware–Liveware mismatch because it primarily disrupts oral communication between humans. (Noise is listed as the very first item under "Oral communication" in L–L.)

Liveware–Hardware (Human–Machine) Interface

LANDMARK EQUIPMENT — GPWS

In the **1980s**, the **Ground Proximity Warning System (GPWS)** marked a **substantial decrease in hull loss rates**. This is a classic L–H interface improvement.

Sub-Element	Items
Switches, Controls, Displays	Instrument/controls design, location, movement, colors, markings, illumination, confusion, standardization.
Workspace	Workspace layout, standardization, communication equipment, eye reference position, seat design, restrictions to movement (**congested cockpit**), illumination level, motor workload, information displays, visibility restrictions, alerting and warnings, personal equipment interference (comfort), data link, operation of instruments ("finger trouble").

EXAM TIP — "CONGESTED COCKPIT RESTRICTING MOVEMENT"

This is a **Liveware–Hardware** mismatch (workspace/physical layout). Do **not** confuse it with Environment (which would be heat/cold/glare etc.).

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Liveware–Software (Human–System) Interface

Sub-Element	Items
Written Information	Manuals, **checklists**, publications, regulations, maps and charts, NOTAMs, standard operating procedures, signage, directives.
Computers	Computer software, user friendliness.
Automation	Operator workload, monitoring task, task saturation, situational awareness, skill maintenance, utilization.
Regulatory Requirements	Qualification in position, qualification in management, certification, medical certificate, licence/rating, non-compliance, infraction history.

EXAM TIP — "ERROR IN CHECKLIST"

A checklist is part of **Software**. Therefore an error in a checklist is a **Liveware–Software** mismatch. Inefficient warning logic / symbology is also classified under L–S because warning systems are software-driven.

Liveware–Environment (Human–Environment) Interface

Interface	What the pilot interacts with	Example of a mismatch
L–H: liveware–hardware	Equipment and controls	A poorly located or confusingly coded control.
L–S: liveware–software	Procedures and information	A cluttered checklist that makes vital information hard to find.
L–E: liveware–environment	Surroundings and working conditions	Noise, heat, vibration or disrupted sleep patterns.
L–L: liveware–liveware	Other people	A misunderstanding or leadership problem within the crew.

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Figure 23.2 *SHEL interfaces — ICAO Doc 9683, Part 1, Chapter 1, paragraph 1.4.2. Noise is an environment issue even when it disrupts communication.*

Category	Sub-Element	Items
Internal	—	Heat, cold, humidity, ambient pressure, illumination, glare, acceleration, noise interference, vibrations, air quality, pollution, fumes, ozone, radiations.
External	Weather	Weather briefing, FSS facilities, weather actual/forecasts, visibility, ceiling, turbulence (wind, mechanic), whiteout.
	Other Factors	Time of day, lighting/glare, other air traffic, wind blast, terrain/water features, obstacles.
	Infrastructure	*Dispatch facilities:* type, use, quality. *At the gate:* APU, towing, refueling, support. *Aerodrome:* runway/taxiway characteristics, markings, lighting, obstructions, approach aids, emergency equipment, radar, ATC, FSS, weather, airfield facilities. *Maintenance:* support equipment, parts availability, operational standards, procedures and practices, QA practices, servicing and inspection, training, documentation.

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CRITICAL INTERFACE MISMATCH EXAMPLES (MEMORIZE FOR EXAM)

- **Error in checklist → L–S**
- **Congested cockpit / restricted movement → L–H**
- **High cockpit noise → L–L** (disrupts communication)
- **Inefficient warning system → L–S** (warning logic/symbology is software)
- **Disturbance of biological rhythm (jet lag) → L–E** (caused by the time-zone environment)

QUICK REVISION SUMMARY — SECTION 4

- L–L = humans + humans (comm, briefings, worker-management, noise interference).
- L–H = humans + machines (cockpit, controls, displays; GPWS milestone **1980s**).
- L–S = humans + procedures/software (checklists, NOTAMs, SOPs, automation, warning logic).
- L–E = humans + environment (weather, jet lag, glare, vibration, aerodrome).

Effectiveness of the CRM System

WHAT THIS SECTION COVERS

The PIC's authority recalibration, the three foundations of CRM success, the four determinants of an effective CRM programme, and the four formal aims of CRM training.

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Authority Recalibration — Captain, F/O, Cabin Crew

Both tradition and regulation give full authority to the **Pilot-in-Command**, who must understand and accept that **questions and observations are NOT a threat to his final authority**.

THE "OLD DAYS" FAILURE MODE

The role of the First Officer in the old days was to obey the captain's "*gear up and shut up*" commands without question. The role of the cabin crew was to bring coffee and meals to the flight deck. **No one dared contravene *(breaking a rule)* the captain's dictates, even to provide essential safety information.** Predictable result: **lots of unwanted accidents and many lost lives.**

FUNDAMENTAL CRM CONCEPT

No single human being has the full picture, and every single person has something useful to contribute to developing it. Working with other people to assemble a complete picture is absolutely necessary to take advantage of all available information needed for safe operation.

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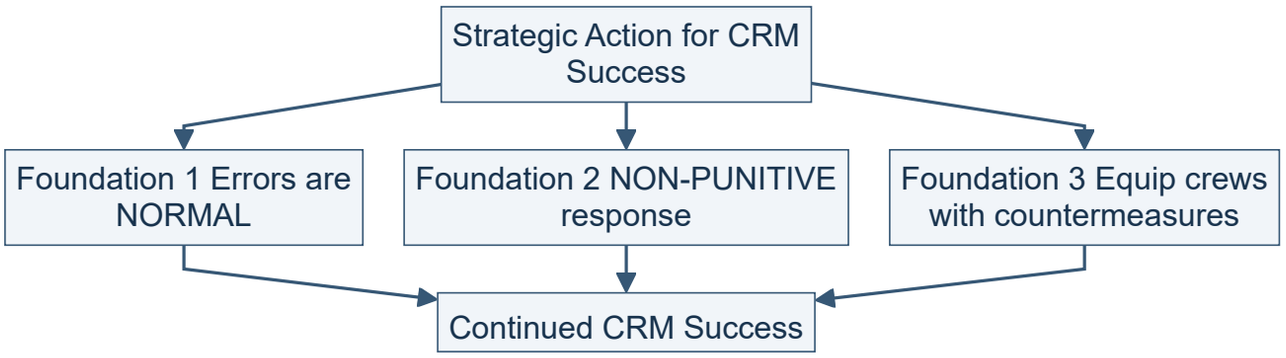
Three Foundations of Continued CRM Success



1. **Operational error passes through the complete human and technology operational process.** CRM aims to present errors as **normal occurrences** and to develop strategies for managing them.
2. A **non-punitive response** to operational error sets the best foundation to identify conditions that breed errors within an organization.
3. Airlines that **tolerate operational error and implement non-punitive policies** are likely to better equip flight crews with appropriate countermeasures to deal with operational errors.

Four Determinants of an Effective CRM Programme

Determinant	Source Description
a) Operational Relevance	Deliberate **avoidance of classroom games**, non-operational activities, and personality assessment.
b) Use of Own Experience	Utilization of own incidents and accidents that reflect the typical safety issues of the airline — the airline is determined to **learn from its own errors**.
c) Threat Assessment	Crews allowed to assess threats and their management. **Open discussion** of threats within the airline and how these are detected, addressed and mitigated.
d) Error Management	Examination of **both effective and ineffective** error countermeasures — thus maximizing learning.

Four Aims of CRM Training

FORMAL AIMS

CRM training has clear aims in terms of specific skills to develop for the purpose of:

1. **Error avoidance**
2. **Threat management**
3. **Error management**
4. **Undesired aircraft state management**

EXAM TRAP — WRONG COMBINATIONS

Beware options like "system management" or "environment management" — these are NOT among the four aims. The correct list always ends with **"undesired aircraft state management"**.

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QUICK REVISION SUMMARY — SECTION 5

- PIC's authority is final, but **questions/observations are not a threat** to it.
- 3 Foundations: **Errors normal · Non-punitive · Equip crews**.
- 4 Determinants: **Operational relevance · Own experience · Threat assessment · Error management**.
- 4 Aims: **Error avoidance · Threat mgmt · Error mgmt · Undesired aircraft state mgmt**.

CRM Skills — The Fifteen Competencies

WHAT THIS SECTION COVERS

The 15 specific CRM skill competencies that govern the scope of CRM training, including the important concept of the **Trans-Cockpit Authority Gradient (TCAG)**.

Skill	Competency Description (source-aligned)
1. Leadership / Command	Uses appropriate authority to ensure focus on task and crew member concerns. Supports others in completing tasks.
2. Decision Making	Detects deviation from desired state, assesses the problem, generates alternatives, identifies risks, selects best course of action; subsequently reviews chosen course for learning.
3. Communication	Clear and effective language; plans stated, ambiguities resolved; demonstrated in interactive briefings and debriefings.
4. Situation Awareness	Comprehends present system & environmental conditions; anticipates future changes; projects changes as the flight progresses.
5. Team Building	Establishes task priorities; utilizes crew resources; improves interpersonal relations.
6. Workload Management	Prioritizes and delegates; keeps everyone "in the loop"; continuously monitors flight progress.
7. Vigilance	Consciously avoids complacency; watches over system/environment changes; informs other crew members of potential threats and errors.
8. Automation Management	Uses automation to assist in managing flight, especially in high workload. Tracks mode changes; "keeps ahead of the curve".
9. Human Performance	Aware of personal and human limitations; recognizes stress loads; assertive when approaching limitations.
10. Trans-Cockpit Authority Gradient (TCAG)	**The authority relationship between captain and first officer.** Steep with domineering captain + unassertive F/O; shallow with two captains; **desired** = balanced gradient where captain's authority is recognized but good relationships and synergy retained.
11. Briefings	Sets open and interactive communication; checks understanding by soliciting questions/comments; focuses on operational issues.
12. Setting Bottom Lines	Aware of crew actions and potential breaches of minima; verbalizes concerns when risks increase vulnerability to error.
13. Contingency Management	Constant awareness of changes; assesses threats and plans for contingent actions.
14. Evaluation of Plans	Examines course of action; solicits input to analyse how threats/errors were managed.

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Skill	Competency Description (source-aligned)
15. Assertiveness	Queries others in ambiguous situations; constructively asserts views; contributes to overall team effectiveness.

Spotlight — Trans-Cockpit Authority Gradient (TCAG)

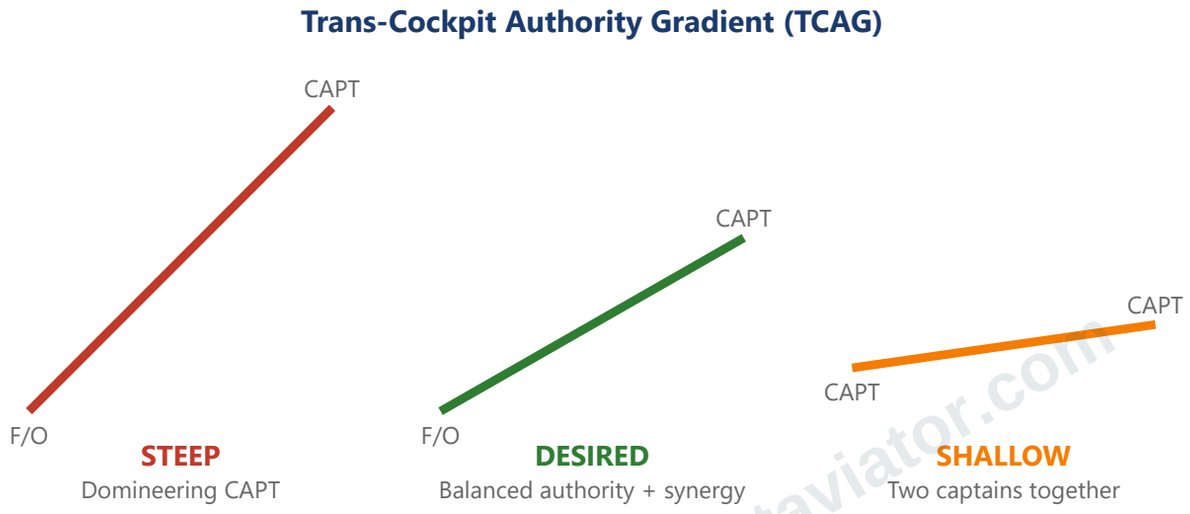

Figure 23.3 *STEEP*

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DESIRED TCAG

A gradient where the captain's authority is recognized, but **good relationships and synergistic working are retained**. This requires a balance of **task-oriented and relationship-oriented behaviors**, the amount of which depends on the crew-members and the situation.

QUICK REVISION SUMMARY — SECTION 6

15 CRM skills to remember: **Leadership · Decision Making · Communication · Situation Awareness · Team Building · Workload Management · Vigilance · Automation Management · Human Performance · TCAG · Briefings · Setting Bottom Lines · Contingency Management · Evaluation of Plans · Assertiveness.**

Threat and Error Management (TEM) — Threats

WHAT THIS SECTION COVERS

The TEM concept (CRM = manage resources; TEM = manage threats), the foundational acceptance that threats and errors will occur, the formal definition of a **threat**, and the **six types of threats** per the DGCA source.

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TEM Concept

DEFINITION — TEM

CRM is about managing resources; TEM is about managing threats. Safe flight operations require the recognition of threats and the appropriate use of error-management countermeasures to **avoid, trap and mitigate** the effects of human error.

The fundamental purpose of CRM training is to improve flight safety through the effective use of error-management strategies in individual as well as systemic areas of influence. Hence, it is only reasonable to refocus CRM as **Threat and Error Management (TEM) training**.

PRACTICAL SUMMATION OF TEM

TEM is the practice of **thinking ahead** in order to predict and avoid errors and operational threats, and to manage any that occur. A foundation of TEM is the **acceptance that threats will occur, and errors will be made.** Hence TEM is **NOT an attempt to eliminate** threats and errors but is concerned with the **management of them**.

Definition of a Threat

DEFINITION — THREAT

Threats are **external events, outside the influence of the crew, that require crew management**. They increase the complexity of the operational environment and have the potential to foster flight crew errors.

Examples of real-life threats include bad weather, time pressures to meet departure/arrival slots, delays, and (more recently) security events.

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The Six Types of Threats

#	Type	Examples (source)
1	**Environmental**	Adverse weather, terrain, airport conditions, heavy traffic/TCAS events, unfamiliar airports.
2	**ATC**	Command events/errors, language difficulties, similar call signs.
3	**Aircraft**	Aircraft malfunctions, automation events.
4	**Crew Support**	Dispatch events/errors, ground events/errors, maintenance events/errors.
5	**Operational**	Time pressure, irregular operations, flight diversions, missed approaches.
6	**Cabin Events**	Flight attendant errors, passenger events.

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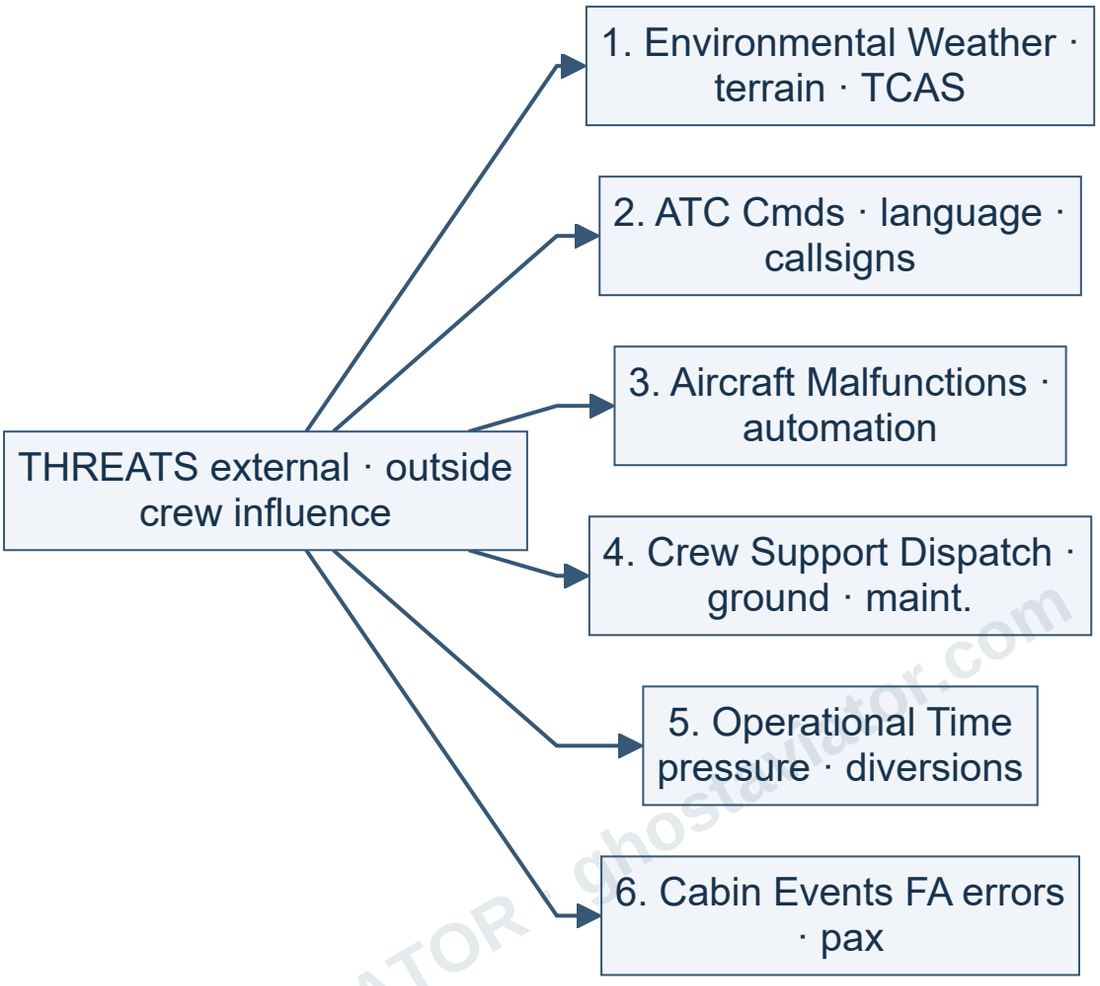

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QUICK REVISION SUMMARY — SECTION 7

- **CRM manages resources · TEM manages threats.**
- TEM accepts that threats/errors **will occur** — it does NOT seek to eliminate them.
- Threat = **external, outside crew influence, requires crew management**.
- 6 threat types: Environmental, ATC, Aircraft, Crew Support, Operational, Cabin Events.
- TEM countermeasures: **avoid · trap · mitigate**.

Error Management — The Five Categories of Errors

WHAT THIS SECTION COVERS

The definition of operational error, the five formal error categories under TEM, and what happens when errors go unmanaged — the **Undesired Aircraft State**.

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Definition

DEFINITION — OPERATIONAL ERROR

Within the TEM concept, flight crew operational error is defined as **an action or inaction by the crew that leads to deviations from organizational or flight crew intentions or expectations**. Operational errors **may or may not lead to adverse outcomes**.

The Five Error Categories

#	Category	Definition
a	**Intentional Non-compliance Error**	**Willful** deviation from regulations and/or operator procedures.
b	**Procedural Error**	Deviation in the **execution** of regulations and/or operator procedures. The intention is correct, but the execution is flawed. **Includes errors where the crew forgot to do something.**
c	**Communication Error**	Miscommunication, misinterpretation or failure to communicate pertinent information within the flight crew, or between the flight crew and an external agent (e.g., ATC or ground operations).
d	**Proficiency Error**	Lack of knowledge or psycho-motor ("stick and rudder") skills.
e	**Operational Decision Error**	A decision-making error **not standardized** by regulations or operator procedures that, as such, unnecessarily compromises safety.

The Three Conditions That Define an Operational Decision Error

DECISION ERROR — AT LEAST ONE CONDITION MUST EXIST

1. The crew had **more conservative options** within operational reason and decided NOT to take them.
2. The decision was **NOT verbalized** and therefore **not shared** between crew members.
3. The crew had time but **did not use it effectively to evaluate** the decision.

Source example: A crew's decision to fly through known windshear on an approach instead of going around.

The Undesired Aircraft State (UAS)

CONSEQUENCE OF UNMANAGED ERRORS

If the crew is unable to **avoid, trap or mitigate** the error (i.e., unmanaged errors), the consequential outcome may lead to an **Undesired Aircraft State**. Typical situations:

- Incorrect aircraft configurations
- Unstable approaches
- Vertical, lateral or speed deviations

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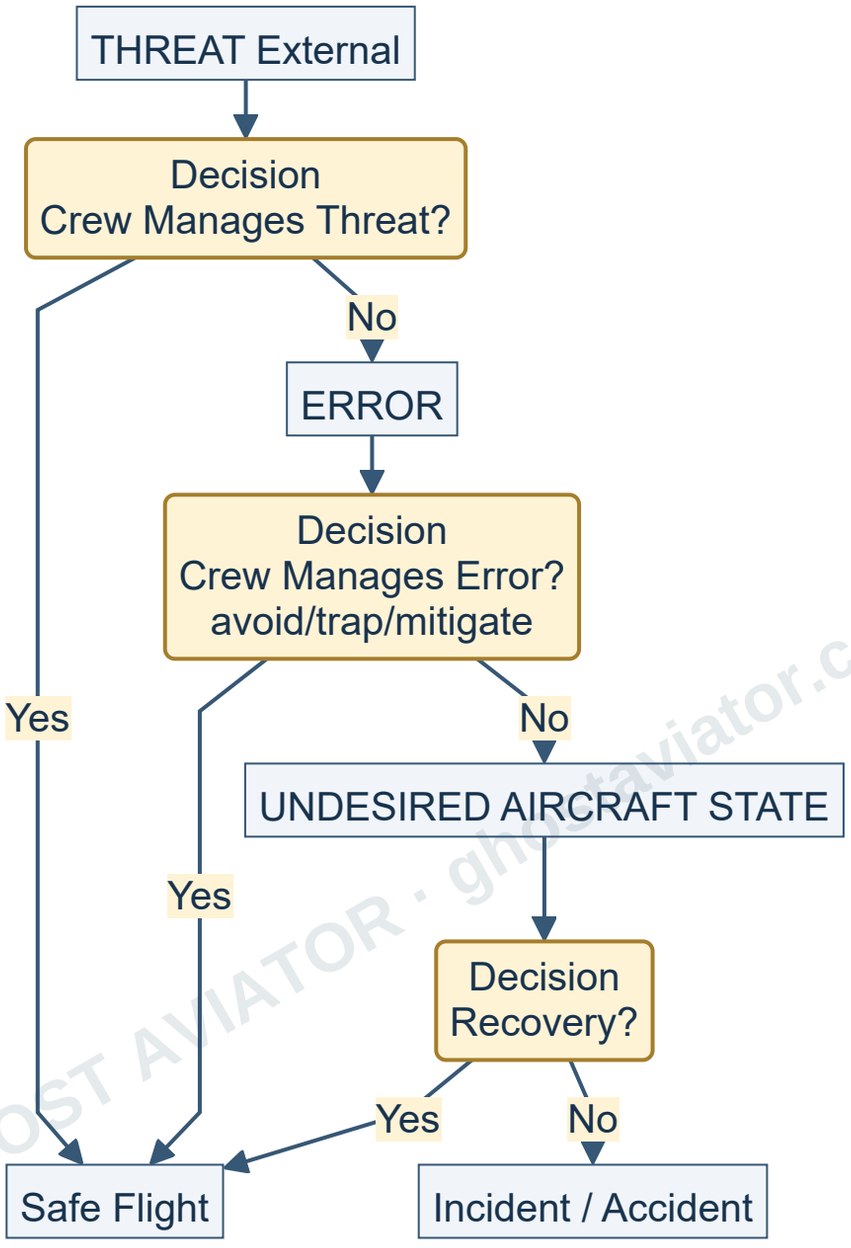

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QUICK REVISION SUMMARY — SECTION 8

- Operational error = **action or inaction** → **deviation**.
- 5 error categories: **Intentional non-compliance · Procedural · Communication · Proficiency · Operational decision**.
- "Forgot to do something" → **Procedural error**.
- Unmanaged errors → **Undesired Aircraft State** (configuration / unstable approach / deviations).

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Line-Oriented Flight Training (LOFT)

WHAT THIS SECTION COVERS

The definition of LOFT, the syllabus of scenarios, and the critical distinction that LOFT is **NOT a check-ride** but a validation of training programmes.

Definition

DEFINITION — LOFT

Line-Oriented Flight Training refers to **non-jeopardy, facilitated aircrew training** which involves a **full mission simulation** of situations representative of line operations. LOFT places special emphasis on situations involving **communications, management, and leadership**.

In short — LOFT means realistic, real-time, full-mission training.

Scenario Sources

LOFT scenarios may be developed from many sources, but **accident and incident reports provide a realistic and appropriate starting point**.

LOFT is NOT a Check-Ride

CRITICAL LIMITATION — NON-JEOPARDY

LOFT **should NOT be used as a method of checking the performance of individuals**. Instead, it is a **validation of training programmes and operational procedures**.

POST-LOFT SOP

An individual or crew needing additional training after a LOFT session shall be afforded that opportunity **immediately, with no stigma or recrimination**.

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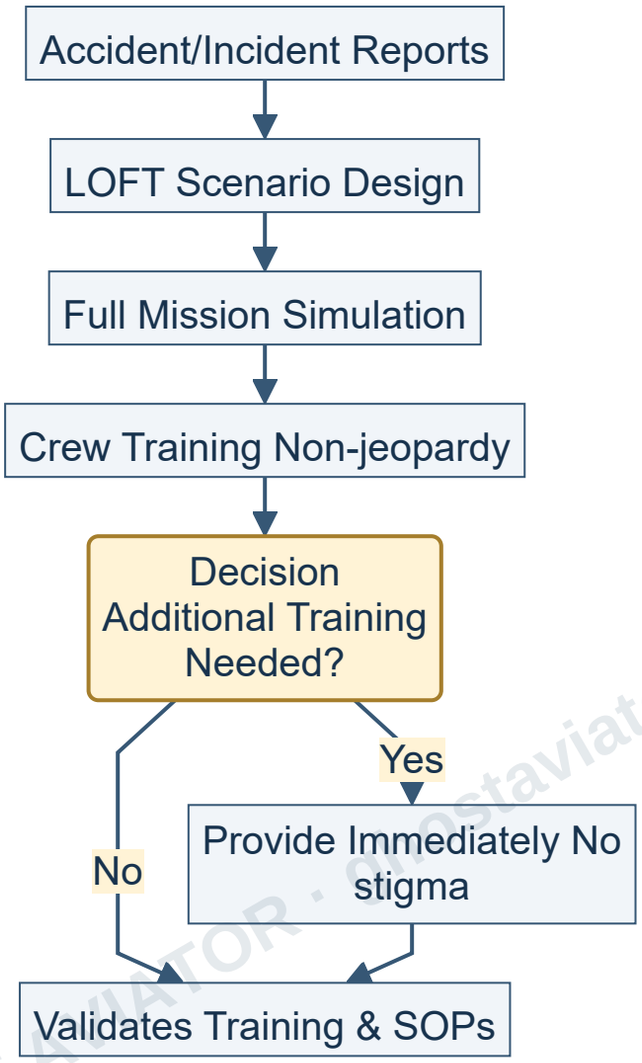

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QUICK REVISION SUMMARY — SECTION 9

- LOFT = **non-jeopardy**, full-mission, **realistic real-time** training.
- Emphasis: **communications, management, leadership**.
- Sources: **accident/incident reports**.
- **NOT a check on individuals** — it validates training and procedures.
- Additional training after LOFT = **immediate, no stigma**.

CHAPTER 24

Aviation Psychology

24

Introduction to Aviation Psychology

WHAT THIS SECTION COVERS
Definition of aviation psychology, its aims, and how its tools support flight safety and efficiency.

Aviation Psychology is a specialty in applied psychology that focuses on understanding human behaviour, emotions and mental states as they relate to the operation and control of aviation systems and their influence on the safety and efficiency of flight.

The *aim* of aviation psychology is to understand and to predict the behaviour of individuals in an aviation environment. Its study is aimed at improving **safety, efficiency and comfort**.

Practitioners of aviation psychology bring the tools and techniques of psychology to bear in order to **describe, predict, understand and influence** the aviation community to achieve those aims.

EXAM TIP — THE FOUR ACTION VERBS
Remember the four practitioner verbs: **D – P – U – I** → *Describe, Predict, Understand, Influence*.

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Human Error & Human Reliability

WHAT THIS SECTION COVERS
The inevitability of human error and the four areas where pilot errors usually originate.

It has long been accepted that **human errors are inevitable**. The majority of pilot-related errors are considered to be failures of:

- Interpersonal skills
- Communications
- Decision-making
- Leadership

This does **not** mean that the frequency of error cannot be reduced, or that the effects cannot be avoided or mitigated.

GOLDEN MAXIM — "SUPERIOR PILOT"
"A superior pilot uses his superior judgment to avoid situations that would require his superior skills."

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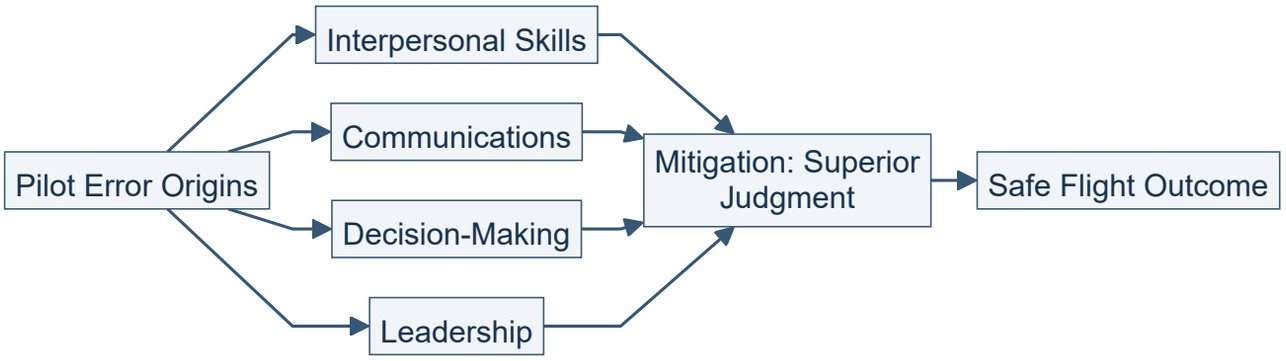

Workload

WHAT THIS SECTION COVERS
Cognitive (mental) workload definition, the four task factors that drive it, indirect factors and observable symptoms.

Workload = the amount of mental effort needed (and expended) to process information. Here we discuss **cognitive (mental)** workload, as opposed to physical workload. Workload is linked to almost all other areas within cognition and performance, particularly *attention, vigilance, fatigue, decision-making and multi-tasking*.

EFFECTS OF HIGH WORKLOAD
High workload is associated with **increased errors, fatigue, task degradation and poor performance**.

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Four General Task Factors Affecting Workload

1. 'Difficulty' of the task
2. Number of tasks running in parallel (concurrently)
3. Number of tasks in series (switching from task to task)
4. The time available for the task (speed of task)

Indirect factors also play a role: *duration of task, fatigue and level of arousal*.

Symptoms of Increasing Workload

- Attention and task focusing
- Task shedding and re-prioritization
- Implications for Situation Awareness
- Increased use of decision short-cuts and less scrutiny or review
- Increased fatigue and chance of error

The area that deals with all such activity is called the **working memory** ('Information Processing').

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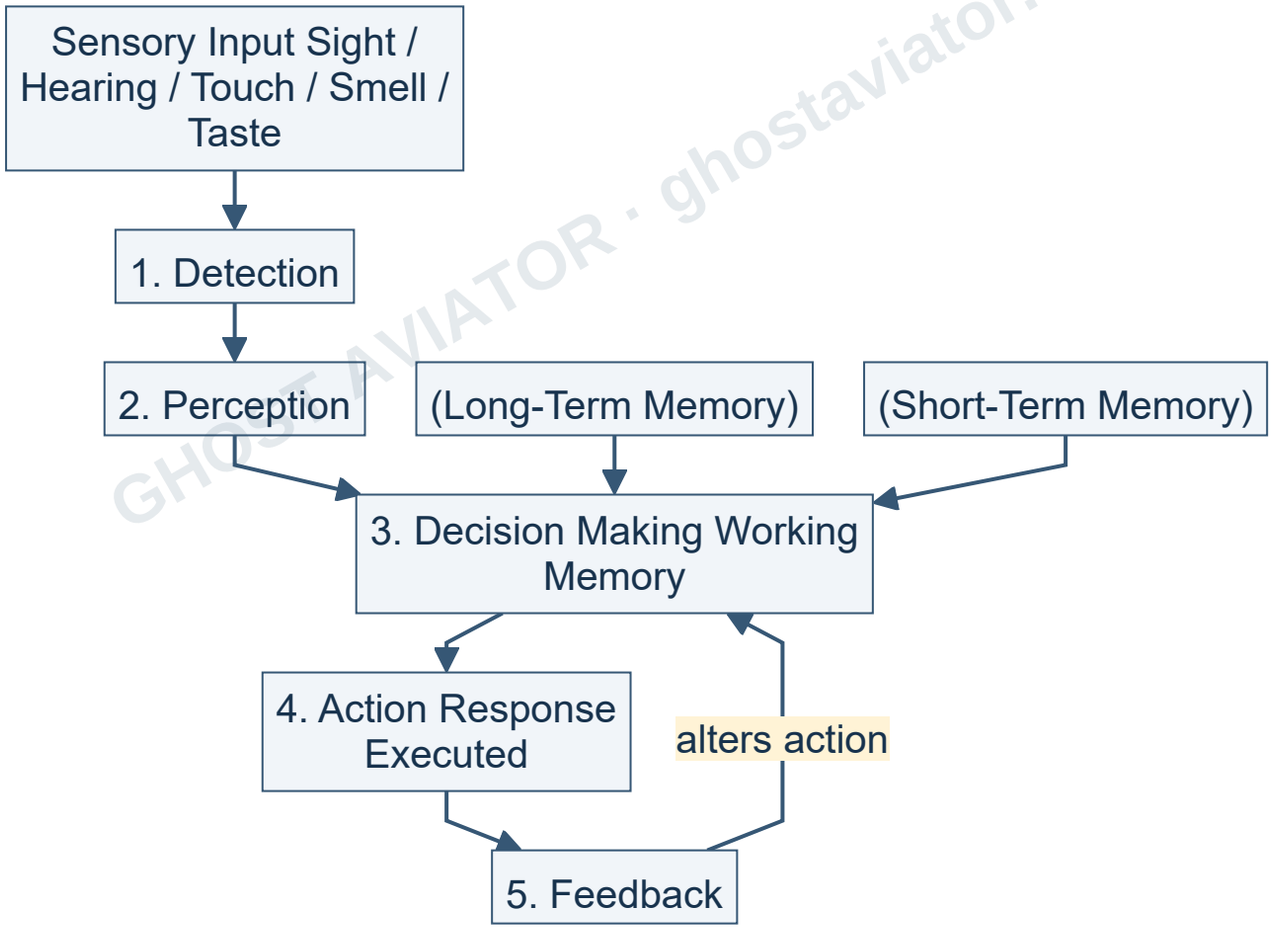
Information Processing — A Functional Model

WHAT THIS SECTION COVERS
How sensory inputs are converted into reasoned actions through five mental stages, and the central role of working memory.

We receive information from the world around us through our senses: **sight, hearing, touch, smell and taste**. The "Gestalt laws" formulate basic principles governing how objects are mentally organised and perceived.

The Five Stages of Reasoning

1. **Detection**
2. **Perception**
3. **Decision Making**
4. **Action** (responses are selected and executed)
5. **Feedback**



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The Brain — Central Decision Maker & Response Selection

Once information has been perceived a decision must be made as to the response. Information is continuously entered into and withdrawn from both long-term and short-term memories to assist the decision process. The decision involves **input, processing, actions and feedback**. The information comes from many sources and

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requires *conscious processing* in working memory. A problem at any stage of information processing could affect the outcome.

To carry out multi-tasks we must learn skills through **Motor Programmes**.

Stimuli

The senses provide stimuli to our brain which has the ability to retain them for a short time, from the time they arrive. We may not have the processing capacity to deal with them all.

> **EXAM TIP — FIRST STAGE**
> First stage in the information process = **Sensory stimulation** (not "Attention", not "Perception"). Detection is what follows.

Types of Memory & Sensory Memories

Mnemonics — the practice of improving or helping the memory, or the systems used to achieve this.

> **WHAT THIS SECTION COVERS**
> Three memory types, key features of sensory memory, iconic memory for sight, and sensory adaptation.

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The Three Types of Memory

- **Sensory memory**
- **Short-term (or working) memory**
- **Long-term memory**

Receptors & Sensory Memories — Key Features

- There is a **separate memory store for each sensory system**.
- The input **decays rapidly**.

The sensory stores for **sight and sound** are the most important, and knowledge of these is necessary.

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Information Processing Model — Diagrammatic View

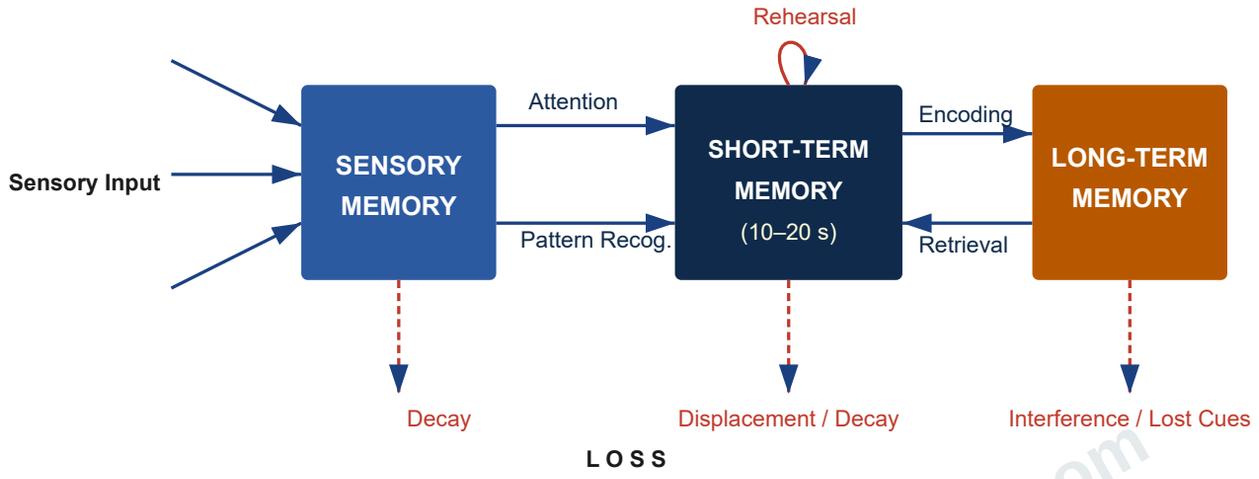

Figure 24.1 *Sensory Input*

Sensory Memory for Sight — Iconic Memory

The **Iconic Memory** is the visual sensory store. It only lasts for between **0.5 and 1 second**.

70–80 % of the information processed by humans is received through the **visual channel**.

Sensory Adaptation (Habituation)

All senses adapt, either partially or completely, to their stimuli after a period of time.

> **EXAM TIP — SENSORY MEMORY VALUES**
>
> **Iconic memory = 0.5–1 second.** Visual channel carries **70–80 %** of all received information.

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Anticipation & Perception

> **WHAT THIS SECTION COVERS**
>
> How sensory input becomes meaning, the role of expectation, mental models and the concept of funneled perception.

Perception

Perception involves the **conversion of the sensory information received into a meaningful structure**.

Sensory information that we *expect* to receive is more easily perceived and integrated when it actually occurs, compared with totally unexpected information.

> "We can 'perceive' only that which we can 'conceive', but we perceive only a fraction of the information reaching our senses at any moment."

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The process of perception is greatly assisted by our ability to form **mental and three-dimensional visual models**.

Funneled Perception

Perception of a situation can differ depending upon the starting point of an observer.

> **HAZARD — CONFIRMATION BIAS / FUNNELING**
> In an abnormal situation, if a pilot has an apparently correct explanation for the problem, the chance that he/she now **ignores or devalues other relevant information** not fitting into the mental picture is *increasing*. Stay open-minded; verify, don't rationalize.

Attention

> **WHAT THIS SECTION COVERS**
> Definition, the two types of attention, why an attention mechanism is required, the cocktail party effect and the dangers of attention mismanagement.

Attention is the process of directing and focusing psychological resources to enhance perception, performance and mental experience. To pay attention to something is to concentrate on it — a stimulus, threat, decision or calculation. Critically, **attention is limited**; if some attention is being used for one thing, it cannot be used for something else.

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Types of Attention

1. **Selective Attention** — inputs are sampled continually to decide their relevance to the present task at hand.
2. **Divided Attention** — when our central decision-making channel can *time-share* between a number of tasks, executing several mental activities at almost the same time (e.g. switching attention from outside the aircraft to the airspeed indicator and monitoring the progress of a motor programme such as flying or taxiing on a relatively subconscious level, while making a radio call at the same time).

The human brain has **different reservoirs of resources** for information-gathering, information-processing, or the action phase.

Lack of Attention

The major danger for pilots is the **poor management of attention**.

Attention Mechanism — Why It Is Required

Because of two potentially limiting stages in processing information:

a. There is a limit to the number of items held or maintained in short-term memory (working memory).

b. Our channel capacity is limited.

Choice of Item

Attention is the deliberate devotion of cognitive resources to a specific item. Although attention can move very quickly from item to item, it can only deal with **one at a time** — so the pilot must consciously prioritize.

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The Cocktail Party Effect

With many conversations occurring all around, one conversation can suddenly break through to attention because a word or phrase was particularly meaningful — the classic example is a person hearing their **own name** mentioned in another conversation that they were not paying attention to.

> **EXAM TIP — COCKTAIL PARTY EFFECT**
> Defined as the ability to *pick up relevant information unintentionally*. It demonstrates that the brain monitors even unattended channels for personally significant cues.

Vigilance & Complacency

> **WHAT THIS SECTION COVERS**
> Definition of vigilance, the three states (hypo, optimum, hyper), causes and signs, plus the trap of complacency in automated cockpits.

Vigilance = the capability to be sensitive to potential changes in one's environment, i.e. the capability to reach a level of alertness above a threshold for a certain period of time — rather than the state of alertness itself.

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Hypo (Low) Vigilance

Mainly due to the **monotony of the task, tiredness, the need for sleep and a lack of stimulation**. It may occur at any moment of the flight. Decrease in sensory perception and sensation of muscular heaviness are indications of low vigilance.

A state of *high fatigue* is liable to cause phases of drowsiness and hypo-vigilance, which result in a reduction of performance and unconscious phases of **micro-sleep**.

> **SOP — MANAGING HYPO-VIGILANCE**
> **Healthy food and organising periods of rest** during the flight can mitigate the effects of low vigilance.

Hyper Vigilance

In its most acute forms, an extremely **agitated state of panic or near panic**. Characterised by indiscriminate attention to all sorts of minor and major threat cues, as the person frantically searches for means of escaping the anticipated danger.

> **HAZARD — HYPER VIGILANCE**
> Hyper vigilance is a state of **panic** that often results in *regrettable decision-making*.

Optimum Vigilance

Lies somewhere between hypo and hyper vigilance.

Complacency

A state of **self-satisfaction** with one's own performance coupled with an **unawareness of danger, trouble or controversy**.

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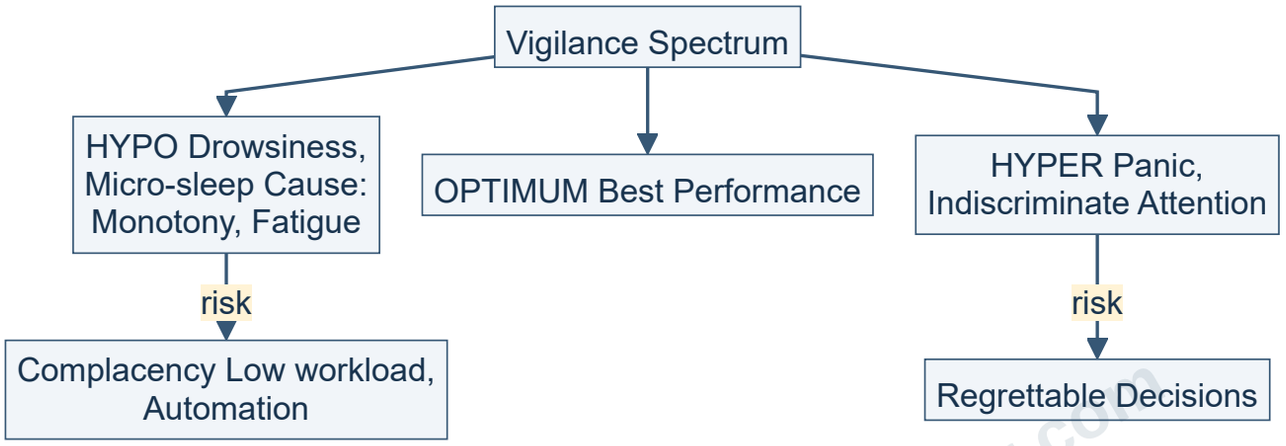
HAZARD — COMPLACENCY

Complacency is **extremely dangerous in aviation**. You get so used to things being done the same way that you become less vigilant to hazards in your surroundings. **Automation may make pilots complacent due to low workload.**



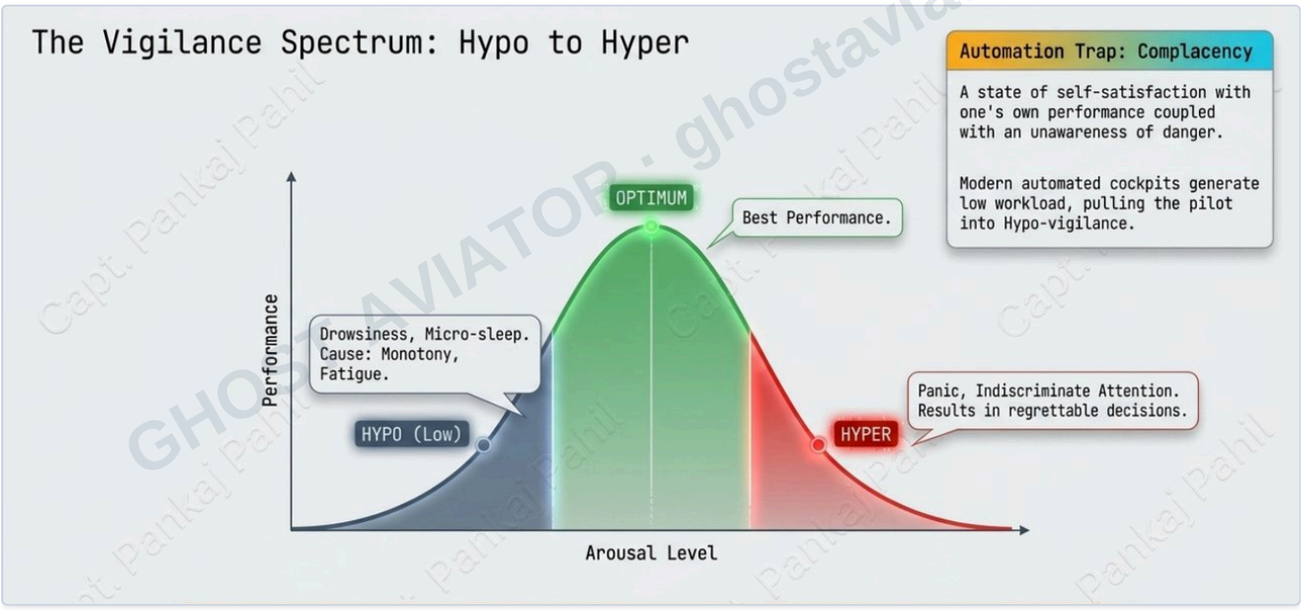


Figure 24.2 *Vigilance from too low to too high*

Motor Programmes (Skills) — Anderson Model

WHAT THIS SECTION COVERS

What motor programmes are, the three phases of skill development per the Anderson model, and how skill enables resource management.

Motor Programmes, or "Skills", are behavioural sub-routines which are **learnt by practice and/or repetition** and are **held within the Long-Term Memory**. They can be carried out without conscious thought.

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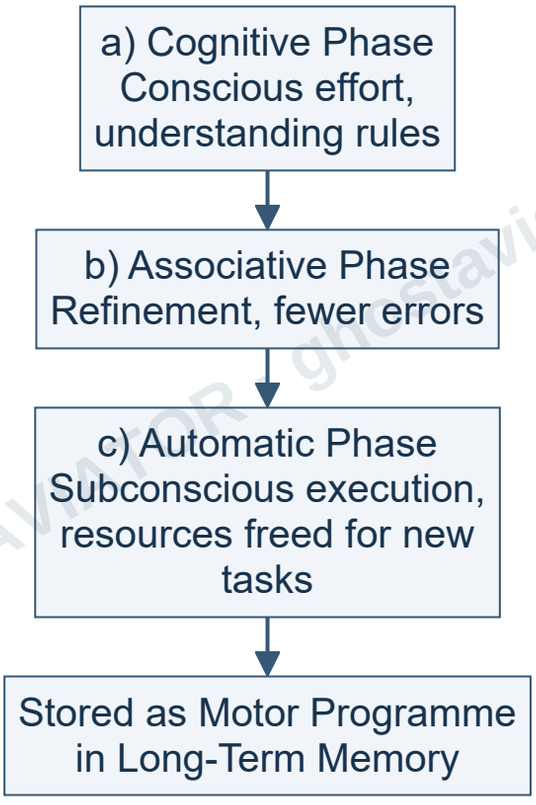
In the *initial phase* of flight training the pilot is competent enough to fly the aircraft at this stage but does *neither* have a great deal of confidence in his/her abilities *nor* in the whole system.

WHAT MAKES A PILOT "SKILLED"

A pilot is skilled when he trains or practices regularly, knows how to manage himself/herself and knows how to **keep resources in reserve for coping with the unexpected**.

Anderson Model — Three Phases

a. **The Cognitive Phase**
b. **The Associative Phase**
c. **The Automatic (Autonomous) Phase**



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EXAM TIP — "FLYING A COORDINATED TURN IS …"

Skill-based behaviour — not attention-based, not experience-based. It draws from procedural memory in the LTM.

Reflexes, Reaction Time, Surprise & Startle

WHAT THIS SECTION COVERS

Reflex types, what reaction time depends on, and the operational impact of the startle reflex on pilots.

Reflexes occur with little or no involvement of the central nervous system.

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Reaction Time

There is a delay between detection, stimulus and muscle contraction called **reaction time**. Reaction time depends on the type of reflex action being used.

Three Types of Reflex Actions

Type	Description	Example
Unconditioned	Instinctive natural reflexes	Blinking
Conditioned	Reflexes that may be *learned*	Foot-on-rudder on swing
Trained	Reflexes that may be *increased by repeated practice*	Stall recovery inputs

Surprise and Startle

The **startle reflex** is a reflex-like event that blinks the eyes and causes a whole-body 'jerk' to occur (similar to that sometimes caused in sleep). The reflex has a relatively basic neural pathway from the sense organ. Many things can cause (or contribute to) a startle reflex:

- Sudden noises
- Unexpected tactile sensations
- Abrupt shocking perceptions
- The sensation of falling
- An abrupt visual stimulus

There is little evidence that a startle reflex alone creates much of a sustained or lasting impact on cognitive functions, although there are some minor and short-lived physiological changes such as *raised heart rate*.

> **KEY VALUE — STARTLE RECOVERY**
>
> A skilled motor task will be momentarily disrupted by a startle reflex but return to normal within **5 to 10 seconds**.

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Effects on Pilots

The main effects of the pure startle reflex on pilots are:

- **Interruption** of the on-going process.
- **Distraction of attention** toward the stimulus.

These happen almost immediately and can be quickly dealt with if the cause is found to be *non-threatening*.

> **HAZARD — FEAR POTENTIATES STARTLE**
>
> A perception of fear can cause a startle reflex to be **potentiated** (more pronounced) should it occur, and attention to become more focused. In a state of fear, very little is required to trigger a full **'fight or flight'** response.

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Quantitative Overload & Human Reliability

WHAT THIS SECTION COVERS
When workload is acceptable, what overload looks like, error rates with practice, and the cumulative nature of errors.

Quantitative Overload — a very high workload can be interpreted as **'Stress'**.

SOP — ACCEPTABLE WORKLOAD
Workload may be said to be acceptable if it requires about **60 %** of the crew resources, depends on the pilot's expertise and corresponds to the amount of resources available.

Most Common Symptoms of Overload

- A sharp degradation of performance
- Funneling of attention or focus
- Regression
- Mental "blocking"
- Mood swings
- Restlessness and trembling
- Panic

Human Reliability

The rate of human error during simple and repetitive tasks might be expected to be **1 in 100**, but after practice, a rate of **1 in 1,000** could be achieved.

DEFINITION — HUMAN RELIABILITY
The individual functioning in the manner in which he or she is supposed to function.

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Error Generation

Errors tend to be cumulative. One slip rarely stays isolated — it sets up the next.

Worked Example — Human Error Rate A pilot performs 200 routine pre-flight checks per week.
• Before practice (1 in 100): expected errors ≈ 200 ÷ 100 = **2 errors / week**.
• After practice (1 in 1,000): expected errors ≈ 200 ÷ 1,000 = **0.2 errors / week** (i.e., about 1 every 5 weeks).
Practice yields a 10-fold reduction in error rate, but never reaches zero — hence the value of cross-checking and SOPs.

Short-Term & Long-Term Memory

WHAT THIS SECTION COVERS
Duration and capacity of short-term memory, methods to expand it, the three subtypes of long-term memory and its capacity characteristics.

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Short-Term Memory (Working Memory) — 10 to 20 Seconds

Short-term memory enables information to be retained for a short period of time. That information will be lost in **10 to 20 seconds** unless it is actively rehearsed and deliberately placed in our long-term memory.

Auditory information is considered easier to retain than visual information as it is easier to rehearse sounds than data in a visual form.

Limitations of Short-Term Memory

The capacity of our short-term memory is limited. The maximum number of unrelated items which can be maintained in the short-term memory is about **7 ± 2**. Once this limit is exceeded one or more of the items are likely to be lost or transposed.

Methods of Increasing Short-Term Memory

Method	Description
Chunking	Breaking items to be remembered into small pieces and remembering them one at a time.
Association	Used to remember spoken lists of items. A wild and bizarre association is imagined and attached to each item on the list.

Worked Example — Chunking an ATC Clearance Raw clearance string: `"VTBL VTBS RWY01L SQK4732 ALT FL310 HDG270"` — 7+ unrelated tokens, near STM limit.
Chunked: `"VT-BL → VT-BS"` (route), `"RWY 01L"` (departure), `"4732"` (squawk), `"FL310"` (level), `"HDG 270"` (heading) = **5 chunks** — comfortably within 7 ± 2.

Long-Term Memory — Unlimited Time Period

It is believed that information is stored in the Long-Term Memory for an **unlimited time period**, although frequently there may be *retrieval problems*.

Three Subtypes of Long-Term Memory

Type	Definition	Example
Semantic Memory	Stores general knowledge of the world	Knowing that "QNH is altimeter setting"
Episodic Memory	Memory of events or 'episodes' in our life	Recalling your first solo flight
Procedural Memory	Skills are included within the make-up of the LTM	Executing a coordinated turn

EXAM TIP — MEMORY QUICK FACTS

- STM duration = **10–20 seconds**, capacity = **7 ± 2** items.
- LTM capacity = **unlimited**, but retrieval problems may exist.
- Motor programmes live in **LTM (Procedural)**, not STM.

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Learning, Retention & Motivation

WHAT THIS SECTION COVERS
Types of learning, methods to retain information, and the role and sources of motivation.

The Learning Process

Learning is an **internal process** which allows the mental acquisition and retention of data. Types include:

Insight

The data is intellectually and cognitively understood and is retained. *Observational Learning / Imitation* — data from an outside source is replicated. *Experience* — learning from our mistakes.

Skill Learning

Involves **motivation, attention, observation, much practice and corrective feedback**.

Retention of Information

Retention can be increased by:

a. **Memory Training**

Motivation

Motivation is the combination of a person's desire and energy directed at achieving a goal. It is the **cause of action**.

Type	Examples
Intrinsic	Satisfaction and feelings of achievement
Extrinsic	Rewards, punishment, and goal obtainment

Not all people are motivated by the same thing and over time their motivations might change.

SOP — MOTIVATION AND PERFORMANCE

The learning process is vastly improved with motivation, and **high performance is rarely achieved without it**.

Experience, Response, Cognition & Hallucination

WHAT THIS SECTION COVERS
How experience shapes us, response characteristics with age, error of commission, response times, cognition in flight, and hallucinations.

Experience

We all have the ability to learn from our experiences and mistakes, and from those of others.

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Response

Any response will cause a detectable change which, in turn, will be noted by the senses. The feedback may alter the action being taken.

- **Auditory stimuli (noises)** are more likely to attract attention than visual stimuli, but they are also more likely to be responded to in error.
- An increase in age between **20 and 60 years** tends to lead to *slower but more accurate* responses.

Response Error (Error of Commission)

If an unexpected stimulus occurs, we will be more likely, under pressure, to make an error of commission.

Response Times

Response to reaction time is the interval between the onset of a given signal and the production of a response to that signal.

Cognition in Aviation

Cognition = the mental faculty or process of **acquiring knowledge by the use of reasoning, intuition or perception**. In aviation, flight puts the pilot into an environment which can **distort sense organs**, and the changed perspective which is experienced in flight can result in information being presented which is *outside the individual's expectations*.

Hallucination

A hallucination is actually a **false perception characterised by a distortion of real sensory stimuli**. (Not a fabrication of stimuli — the stimuli are real, but their perception is distorted.)

Workload and Limitations

Too high *or* too low a workload can result in degraded performance. Several types of situations may cause mental overload.

> **EXAM TIP — AGE AND RESPONSE**
>
> Between **20 and 60 years** → responses become *slower but more accurate*. Experience compensates for slower reflexes.

Judgment & Decision Making

> **WHAT THIS SECTION COVERS**
> Why decisions matter, how to seek advice, risk assessment fundamentals, and the dynamic nature of "good" decisions.

Making good decisions is one of the important aspects of piloting an aircraft. **Good decisions** can help lead to safe, successful flights, whereas **bad decisions and even indecision** have led to many aircraft accidents. Talk to your instructor and other pilots about different flying situations to obtain advice about what a good decision would be in a given circumstance.

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When flying with another pilot, it is possible to increase the chance of a good decision being made by **discussing the potential problems beforehand**. If as an inexperienced pilot you are flying with someone of more experience and you see him doing something you consider to be dangerous, you should **immediately question their course of action**.

BEST PRACTICE — ALWAYS MORE TO LEARN

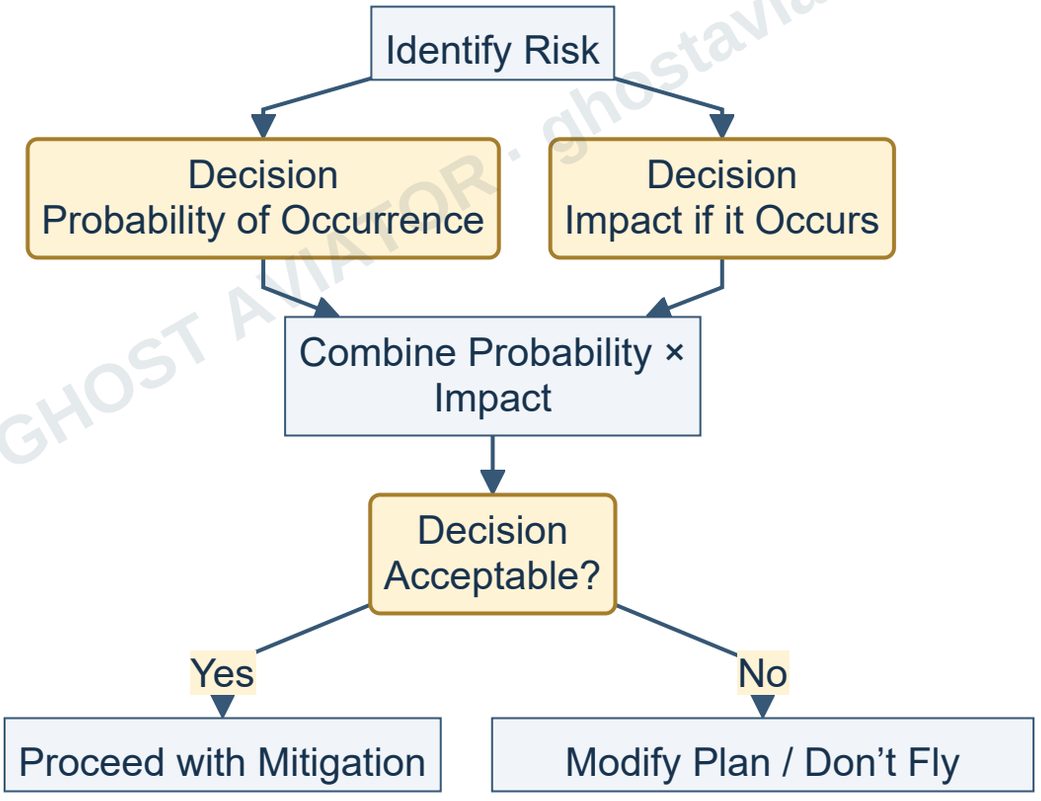
A good pilot is one who always realizes there is more to learn. When making a decision a pilot will be influenced by his **previous experience**, the **probability of an occurrence**, and whether the information he is receiving **matches that expected**.

Risk Assessment

Risk assessment is based on:

- The **probability of the risk occurring**.
- The **impact of that risk** if it did occur.

You will have to weigh up the risk and the possible consequences of the risk in order to determine your course of action.



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REMEMBER

Making good decisions is one of the most important aspects of piloting an aircraft. Good decisions can help lead to safe, successful flights, whereas bad decisions and even indecision have led to many aircraft accidents. **Always strive to make good decisions.**

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Personality, Attitude & Situational Awareness

WHAT THIS SECTION COVERS
Desirable pilot traits, the meaning of being Pilot-in-Command, factors that destroy situational awareness, and how to maintain it.

Personality, Attitude and Behaviour

Someone who is **too introverted or anxious may not make a good pilot**. Examples of desirable characteristics in a pilot are:

- The ability to cope with stress and adventure.
- The ability to assess and control risks.

As captain of an aircraft, you will need to show **good leadership skills**.

SOP — BEING PILOT-IN-COMMAND
While acting as pilot-in-command you should ensure that you **stay in command** — you should be **properly organized and prepared**.

Situational Awareness

A good leader displays good situational awareness and accurately assesses his own performance. The following factors can interfere with accurate situational awareness:

1. Stress
2. Boredom
3. Fatigue
4. Emotional disturbance
5. Poor communication
6. Interruptions

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MAINTAINING GOOD SA — SOP
In order to maintain good situational awareness, **gather as much information as possible**. Do not rush into making a decision. Make sure you consider all the options. **Do not make the mistake of seeing something that is not really there just because you want it to be there.**

Mechanics of Decision Making

WHAT THIS SECTION COVERS
The seven-step decision cycle and the dynamic nature of decisions in flight.

Pilots operate in a **dynamic and constantly changing environment**. A good decision reached a minute ago will *not necessarily* be the same good decision in two minutes' time.

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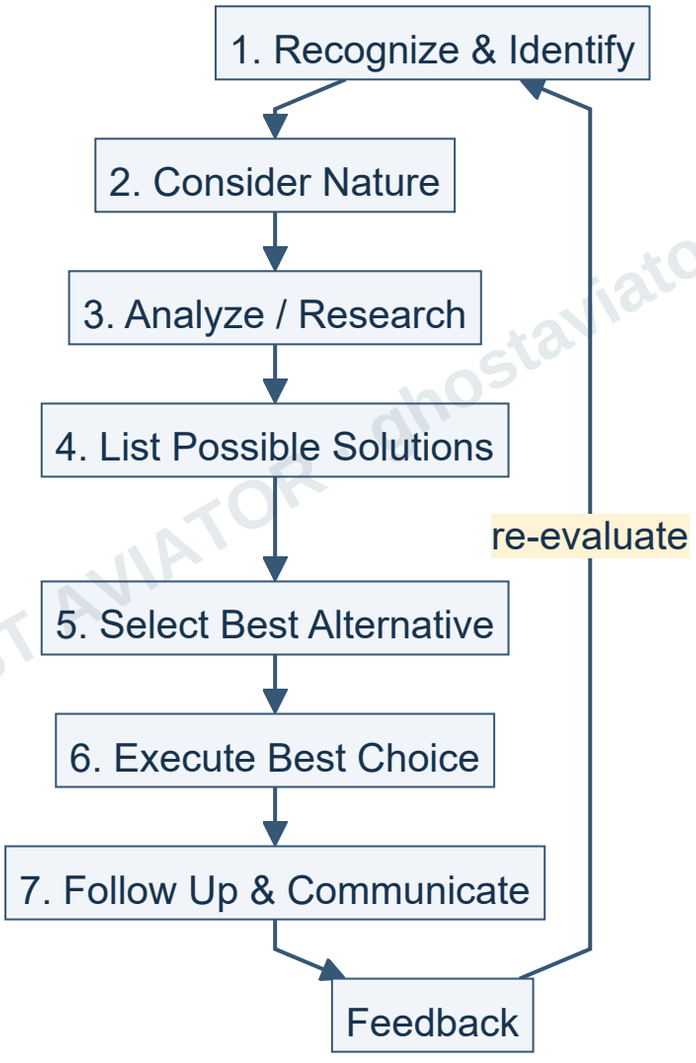
The Seven Steps

1. **Recognize and identify the problem** — have a clear definition of what needs addressing.
2. **Consider the nature of the problem** — type of issue, why it needs a decision, hoped-for results.
3. **Analyze or research the problem** — gather all information so informed choices can be made.
4. **Develop a list of possible solutions** — list decisions and their consequences.
5. **Select the best alternative** — choose the best solution for the situation.
6. **Execute the best choice** — taking action is often the hardest part.
7. **Follow up and communication** — keep the process going with constant communication; **Feedback** determines the overall success of and reaction to the decision.



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EXAM TIP — DYNAMIC DECISIONS

A good decision a minute ago may not be a good decision two minutes from now. **Re-evaluate continuously.**

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Communication & Team Work

WHAT THIS SECTION COVERS
Why communication matters, attitudes that destroy teamwork, and the danger of "expectation".

When piloting an aircraft — whether with a passenger, another pilot or an instructor — **good verbal communication** will help the flight go more smoothly and contribute to flight safety.

Concise and unambiguous communication is essential to the safe conduct of air traffic.

Barriers to Communication & Team Work

Several things can be done by team leaders to facilitate good teamwork. *In the end, it is the team leader who takes the decisions on behalf of the team.*

HAZARDOUS ATTITUDES — BARRIERS

Some of the barriers to communication and teamwork are:

- Aggressiveness
- Arrogance
- Anti-authoritarian behaviour
- Impulsiveness
- Feelings of invulnerability
- Resignation

Expectation

HAZARD — EXPECTATION BIAS

Do **not assume** you have heard what you are expecting to hear. **Actively listen.** Clear, concise and unambiguous communication is essential to the safe conduct of aviation.

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Automation

WHAT THIS SECTION COVERS
Definition and evolution of automation, the mismatch between automation and the human element, and qualities of human-centred automation.

Automation refers to a system or method in which many of the processes of production are automatically performed or controlled by self-operating machines, electronic devices, etc.

Through microprocessor technology, navigational tasks and aircraft system management have been automated, making the flight crew more *peripheral* to the actual operation of the aircraft. Pilots who at one time had direct authority over all aspects of aircraft control and management have now become responsible for the **management of complex hardware and software interfaces**.

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These technological advances have given rise to **new forms of error**. Automation is almost always introduced with the expectation of reducing human error and workload, but what frequently happens is that **the potential for error is simply relocated**. More often than not, automation does not replace people in systems; rather, it places the person in a different, and in many cases, *more demanding* role.

Definition of Automation

> "The technique of controlling an apparatus, a process or a system by means of electronic and/or mechanical devices that replaces the human organism in the sensing, decision-making and deliberate output."

Evolution of Transport Aircraft Automation

Generation	Pilot / control interfaces
1	Pilot controls the aircraft
2	Pilot → autopilot → aircraft
3	Pilot → controller → autopilot → aircraft; navigation aids
4	Pilot → CDU → FMS → controller → autopilot → control systems; PMS, navigation aids, INS and CADC

Increasing peripheralization (greater separation) of the pilot.

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Mismatch Between Automation and the Human Element

Nevertheless, a few accidents point to a mismatch between automation and the human element. Studies identified **nine categories** to focus on:

1. Situational awareness
2. Automation complacency
3. Automation intimidation
4. Captain's command authority
5. Crew interface design
6. Pilot selection
7. Training and procedures
8. The role of pilots in automated aircraft
9. (Studies concluded scope for improvement in fields like: human capabilities and limitations, ergonomics, cognitive suitability and instrument standardization.)

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HAZARD — AUTOMATION SIDE-EFFECTS

Although cockpit automation may provide pilots with more time to think, it may encourage pilots to reinvest only some of this mental free time in flight-related thoughts. While manual workload has reduced, **mental workload has not been reduced by the same amount — in fact it may have been increased**. Automation may not always reduce workload in phases of flight in which it is usually high (e.g. arrivals/landings at busy terminals). High cockpit automation may reduce pilot attention with the consequence of **'being out of the loop'**.

HAZARD — RELIABILITY ≠ SAFETY

All forms of automated assistance for the human operator must be highly reliable, but this may also induce **complacency**. Human expertise may gradually be lost and if the machine fails, the human operator may accept an inappropriate solution or become unable to formulate a satisfactory alternative.

Qualities of Human-Centred Automation

Human-centred automation should possess these qualities in proper measure:

- Accountable
- Subordinate
- Predictable
- Adaptable
- Comprehensible
- Flexible
- Dependable
- Informative
- Error-resistant
- Error-tolerant

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BEST PRACTICE — COMPLEMENTARY, NOT ALTERNATIVE

Pilots, computers and machines are **not alternatives, but complementary factors** in ensuring flight safety. Achieving the correct balance benefits safety. Automation has undeniably led to an improvement in flight safety.

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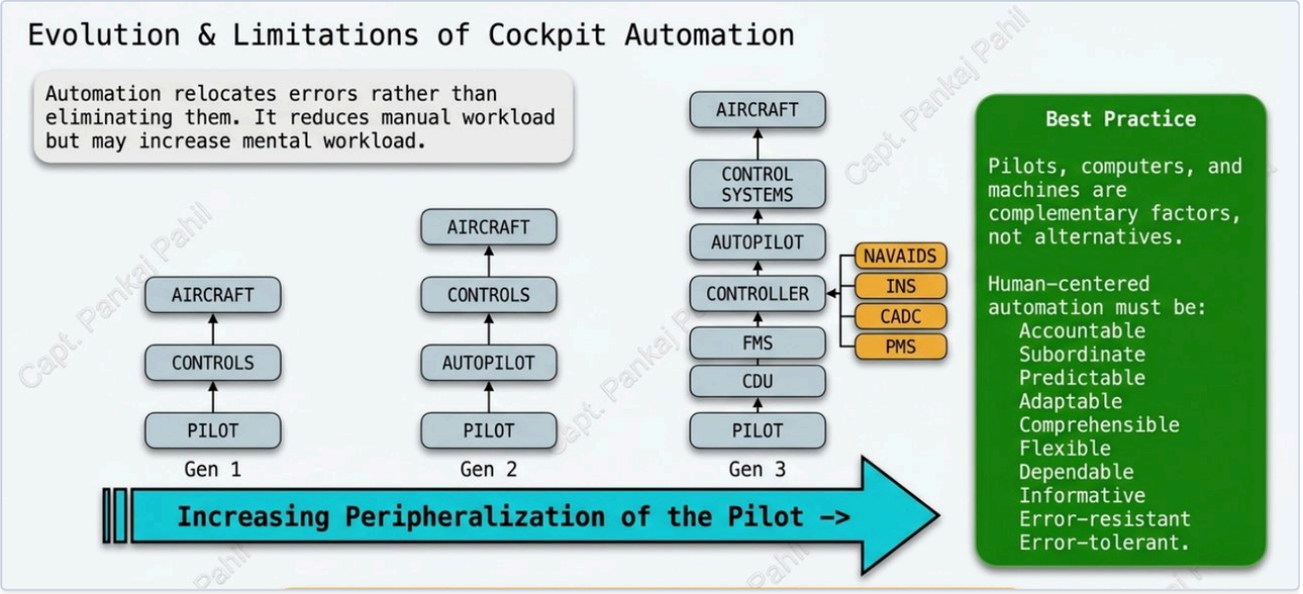

Figure 24.3 *Three generations of cockpit automation*

Human Characteristics & Cockpit Design

> **WHAT THIS SECTION COVERS**
> Personality types, cultural differences, anthropometry, eye datum, cockpit seats and the design considerations driving them.

Personality

Personality involves two major factors:

- **Neuroticism – Stable** continuum
- **Introversion – Extroversion** continuum

Extroverts are said to be impulsive and sociable; **introverts** are more withdrawn and cautious.

A low-neuroticism personality is one of an emotionally stable person, whereas a high-neuroticism person will worry and get upset easily.

> **TRAIT VS STATE ANXIETY**
> **High neuroticism = high trait anxiety** (a personal trait of that person). As opposed to **state anxiety**, which is a *transient state of anxiety present in anyone at any time.*

Cultural Differences

The way people behave, think and interact with each other generally — as well as what motivates them — will be partly a factor of their national and cultural background. One culture may value subordinates speaking up, whereas another culture may value subordinates who obey their superiors unquestioningly. The latter might perceive a subordinate who points out a concern as being impolite, aggressive or disruptive.

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Competencies

The main area where personality and background culture are important is **interpersonal interaction**. In terms of competencies, this means **communication, leadership and teamwork** — crew interactions or interactions with outside agencies such as air traffic control.

Design Considerations

The following human characteristics must be taken into account in the design of aviation systems:

- Physical dimensions
- Capability for data sensing
- Capability for data processing
- Capability for motor activity
- Capability for learning
- Physical and psychological needs
- Sensitivities to physical environment
- Sensitivities to social environment
- Coordinated action
- Differences among individuals

Hardware, Design of Flight Decks, Size & Anthropometry

> **MURPHY'S LAW**
>
> *"If equipment is designed in such a way that it can be operated wrongly, then sooner or later, it will be."*

Therefore a great deal of attention is required to design the equipment. **The most important requirement in the design of both displays and controls is STANDARDIZATION.**

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Anthropometry

This is the study of **human measurement**. The information is grouped into:

- Static measurements
- Dynamic measurements
- Contour surface measurements

It is not practical to design a cockpit for both the very short and the very tall individual. Those in the **central 90 % of size distribution** will be catered for. Normally the design of aircraft uses measurements taken from the entire population **disregarding both the 5 % lowest and the 5 % highest**.

Eye Datum

Cockpit space must be designed around a defined position of the pilot's eye. This may be called the **Eye Datum**, the **Design Eye Position** or the **Reference Eye Point**.

Position	Effect
Too high	Poor view of instruments, obstructed high view, good downward view

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Just right	Optimum view of outside and instruments
Too low	Good view of instruments but poor forward and downward view

Once the design eye position has been set, the size of the cockpit can be established.

Design of Cockpit Seats

It is of the utmost importance that the seating is comfortable and adjustable to the individual pilot's size and shape. Pilots should adjust their seats to establish a comfortable position giving full control movement, with **optimum instrument scan and outside visibility**. This position should be used for **all phases of the flight**.

> **SOP — RESTRAINT**
> Restraint should be provided by a **5-point harness with a negative 'G' strap**.

Hardware — Displays & Controls

> **WHAT THIS SECTION COVERS**
> Digital vs analogue display selection, the standard "T" layout, three-pointer altimeter pitfalls, and the eight basic considerations for control design.

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Presentation Requirements

Information Type	Best Display
Purely quantitative information	**Digital**
Qualitative information	**Analogue** (more easily assessed)

Conventional Analogue Standard 'T' Display

A standard 'T' layout has the **artificial horizon / attitude indicator** at the centre with the **altimeter, airspeed indicator and direction indicator** grouped around it. **Digital Display** and the **Compass**. The conventional analogue type of compass card gives a better picture of the aircraft orientation than would a digital readout.

Combination or Analogue and Digital Displays

In some instances, both digital information and analogue information can be combined in a single instrument.

> **HAZARD — THREE POINTER ALTIMETER**
> The three-pointer altimeter can easily be **misread** and produces significantly **more reading errors** than the single-pointer altimeter.

Hardware, Controls — Basic Considerations

Principle	Description
Standardization	Control of location and sense of use from one aircraft to another should be standardized.

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Frequency of Use	Controls used frequently or for protracted periods should be located so that they do not require the pilot to adopt an awkward or fatiguing position.
Sequence of Use	Controls that should normally be used in a given order should be laid out so that the sequence of use is represented in that layout.
Importance	Important controls must be located in easily reached and unobstructed positions.
Visual/Tactile Dissimilarity	Switches and knobs that control different systems or functions should look and feel different from each other.
Symbolism	Controls, if possible, should be designed to contain some reference to their function.
Simultaneous Use	Controls which may require simultaneous use should be located to enable this to take place.
Warnings	All warnings should be 'attention getting' without being startling. The most conspicuous visual warnings rely on **head and gaze orientation**.

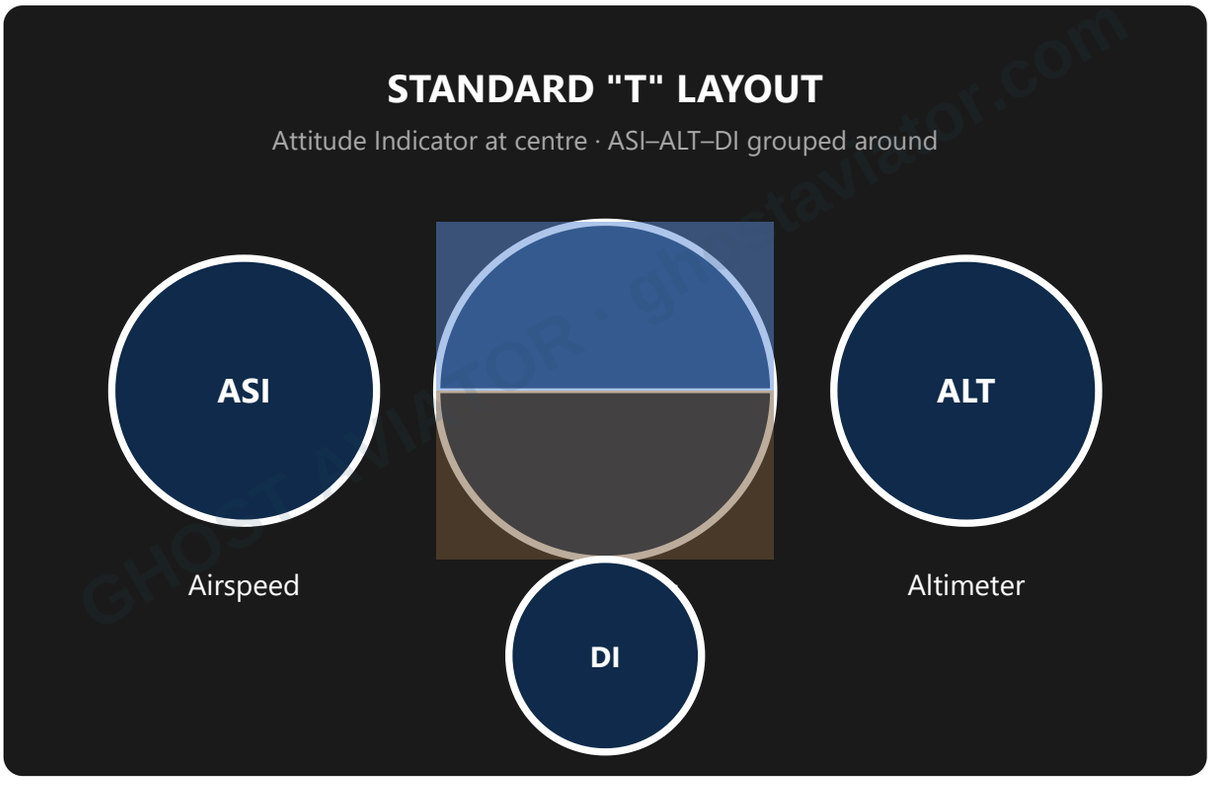

Figure 24.4 *ATTITUDE*

Stress Tolerance, Fatigue & Arousal — Well-Being of Operational Personnel

WHAT THIS SECTION COVERS
Stress, arousal, fatigue, environmental tolerances, life stress scoring, stress management, and the arousal-performance relationship.

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Three of the many factors which may influence the well-being of operational personnel are:

1. Stress and fatigue
2. Body rhythm disturbance
3. Sleep deprivation or disturbance

Other factors affecting physiological or psychological well-being include: *temperature, noise, humidity, light, vibration, workstation design and seat comfort*.

Stress, Arousal and Performance

The different stress levels generated within individuals by a particular stressor will differ. A **moderate level of stress may improve performance**. Stress promotes an **increase in physical strength** rather than promoting mental performance.

- The response of a person to the event or circumstances to which he is exposed is known as **arousal**.
- Arousal is a measure of a person's **readiness to respond effectively** to a given stress factor.
- Personality, character, training and experience will influence levels of arousal in a given individual.

The Effects of Stress

In an active, outward-going, highly trained person, *too little* stimulation or stress arousal will lead to the onset of **boredom and even drowsiness**. An introspective, under-confident person, if highly aroused, might be unable to function at all — even in circumstances that he is competent to deal with.

Stress Factors (Stressors)

Stress factors are **cumulative**. Stress occurs under various conditions:

- Pilot must revise his plan of action and does not immediately have a solution
- Inexperienced pilots when situational demands exceed their individual capabilities
- Pilot is convinced he will not be able to find a solution to the problem
- Physiological factors — hunger, thirst, pain, lack of sleep, fatigue

> **SOP — KEY TO REDUCED WORKLOAD**
>
> **Thorough flight planning is the key to reducing cockpit workload.**

Environmental Tolerances

Temperature, pressure, humidity, noise, time of day, light and darkness can all be reflected in performance and also in well-being. Heights, enclosed spaces, and a boring or stressful working environment can also influence performance.

Managing Stress

Recognising stress, accepting it and developing a coping strategy is essential for stress management. Training and experience help to ward off stress and high levels of arousal. Successful completion of a stressful task will reduce the amount of stress experienced when a similar situation arises in the future.

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SOP — STRESS SELF-MANAGEMENT

To avoid stress, know your limits and fly well inside them. If you are aware that you are under stress, consider whether it would not be wiser for you to stay on the ground rather than to fly when you are not up to it. Make sure that **cockpit housekeeping** is of a high order and that all documents, charts and associated equipment are appropriately stored and accessible. **High-quality headsets** will reduce cockpit noise levels. Temperature can often be regulated using the cabin heating or ventilation system.

Life Stresses — Psychological Scoring

Pilots suffering from life stress should be aware that this can affect their concentration and performance when at the controls of an aircraft. The descending order in which the factors affect a person:

Death of spouse/child → Divorce → Marital separation → Death of a close family member → Injury / illness → Marriage → Loss of job → Retirement → Pregnancy → Sexual problems → Birth → Change of financial status → Siblings leaving home → Change of eating habits → Change of residence → Loan/debt/mortgage → Vacations → Minor violations of law.

Life Event	Score
Death of a spouse or partner	**100**
Divorce	**73**
Marital separation	**65**
Death of a close family member	**63**
Personal injury or illness	**53**
Loss of job	**47**
Retirement	**45**
Pregnancy	**40**
Sexual problems	**40**
Son or daughter leaving home	**29**
Change of residence	**20**
Bank loan or credit card debt	**17**
Vacation	**13**
Minor law violation	**11**

Cumulative Score	Interpretation
< 60	Free of life stress
60 – 80	Normal life stress
80 – 100	High life stress
> 100	Under serious life stress

Note: such schemes need to be treated with caution because of wide individual variability.

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Indications of Stress

Stress causes: **mental blocks, confusion, channelized attention, resignation, frustration, rage, deterioration in motor coordination, high pitch voice and fast speaking**.

EASILY OBSERVABLE INDICATIONS OF STRESS

Perspiration, flushed skin, dilated pupils and fast breathing.

Tiredness and Fatigue

Tiredness and fatigue, though related concepts, differ in their long-term physical effect on the body. To deal with normal tiredness it is sufficient to ensure that periods of activity and periods of restful sleep comply with the normal pattern for a person's age and physical condition.

Ordinary tiredness results from normal physical and/or mental exertion over a normal waking period. If a person is tired, *a good night's sleep* is the only requirement for that person to be fit the following morning.

DEFINITION — FATIGUE

Fatigue is a **very deep tiredness due to the cumulative effects of a stressful lifestyle and/or living and working environment**.

- The effects of fatigue can be *insidious*.
- They can creep up slowly on the sufferer.
- The sufferer may be unwilling to recognize or admit that he is suffering from fatigue.
- Tiredness and fatigue can kill, but fatigue is harder to recognize.

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Causes of Fatigue

- Long-term ill health, either physical or mental
- Regular sleep deprivation, for whatever reason
- High levels of stress and/or anxiety over a prolonged period
- A disturbed body cycle (e.g. jet lag, changing work shift patterns)
- Long-term difficulties in personal or work-based relationships
- A stressful living and/or working environment (e.g. high levels of noise, or unreasonable physical or mental demands)

Symptoms of Fatigue

- Memory lapses
- Frequent mistakes in speech and actions
- Rapid changes of mood
- Diminished awareness
- Tiredness
- Lack of coordination

Minimum Self-Help Program

- Accept the fact that you are fatigued
- Eat a balanced diet and healthy food
- Try to take the stress out of your life

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- Make sure that you get regular, undisturbed sleep
- Avoid alcohol and caffeine
- Always consult a doctor

Arousal

Arousal is a major aspect of many learning theories and is closely related to other concepts such as *anxiety, attention, agitation, stress and motivation*.

Arousal Level	Effect
Low Arousal	In cruise, attention can wander; information missed or misinterpreted
Optimum Arousal	Central Decision Maker at its most efficient. Lower for difficult/cognitive tasks; higher for tasks requiring endurance and persistence
High Arousal (Overload)	Real danger of attention becoming **narrowed**

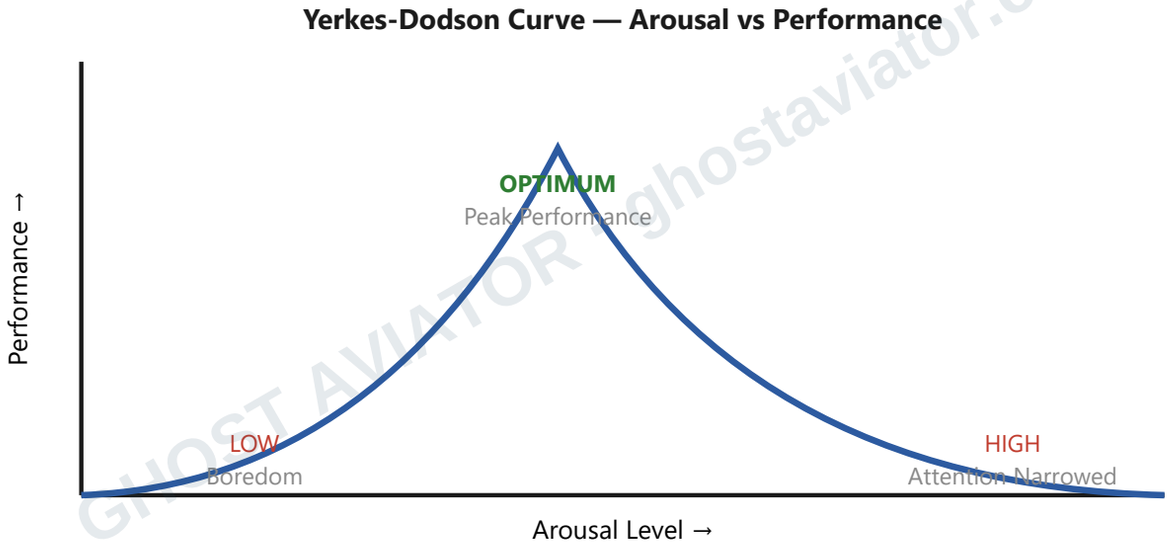

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Figure 24.5 *Yerkes-Dodson Curve — Arousal vs Performance*

> **EXAM TIP — STRESS PROMOTES...**
> Stress promotes an **increase in physical strength** rather than mental performance. A moderate level of stress may **improve** performance.

Body Rhythm Disturbance — The Circadian Cycle

> **WHAT THIS SECTION COVERS**
> The biological clock, the 24-hour vs 25-hour cycle, melatonin's role, jet-lag rules and re-synchronization rates.

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Stimuli and Attention

Our bodies are continuously receiving stimuli through our **five senses**. This information is stored briefly in our sensory memory and, if we perceive it to be important, it is transferred to our short-term memory or Central Decision Maker. Some stimuli are better than others at getting our attention. We can split our attention between several different things by concentrating on them in rapid succession.

The Circadian Circle

The Circadian Circle represents our level of alertness throughout the day. Circadian rhythms are internally generated by a **self-sustaining (autonomous) biological clock** located in the **hypothalamus**, which functions as the main control centre for the autonomic nervous system by regulating sleep cycles, body temperature, appetite, etc., and acts as an endocrine gland by producing hormones. It takes into account biological elements such as **body temperature, heart rate and blood pressure** — which affect our level of alertness during the day.

> **HAZARD — CIRCADIAN LOWS**
>
> **Human performance degradation at circadian lows is one of the major challenges for the aviation industry.**

24-Hour Circadian Reference

Time	Body Event
00:00	Midnight
02:00	Deepest sleep
04:30	**Lowest body temperature**
06:45	Sharpest rise in blood pressure
07:30	Melatonin secretion stops
08:30	Bowel movement likely
09:00	Highest testosterone secretion
10:00	High alertness
12:00	Noon
14:30	Best coordination
15:30	Fastest reaction time
17:00	Greatest cardiovascular efficiency and muscle strength
18:30	Highest blood pressure
19:00	Highest body temperature
21:00	Melatonin secretion starts
22:30	Bowel movements suppressed

As we sleep, our heart rate is lowered and hence our level of alertness is reduced. Blood pressure is also often lowered after mealtimes. **Human performance declines at night when the body and mind desire rest.**

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Melatonin — "The Light of Night"

Melatonin is secreted from the **pineal gland** principally at night. The hormone is involved in **sleep regulation**, as well as in a number of other cyclical bodily activities and circadian rhythm in humans.

- Secretion increases soon after the onset of darkness.
- **Peaks in the middle of the night, between 2 and 4 a.m.**
- Gradually falls during the second half of the night.

This circadian rhythm of secretion plays an important role in its hormonal activity. Melatonin is exclusively involved in signalling the **'time of day' and 'time of year'** (hence considered to help both clock and calendar functions) to all tissues, and is thus considered to be the body's chronological pacemaker or **'Zeitgeber'**.

Trans-Meridian Flight — Pilot Considerations

Factor	Value / Rule
Free-running circadian cycle (no time cues)	**~ 25 hours**
Cycle with normal time cues	**~ 24 hours**
Stop-over rule	If stop-over **> 24 hours** → move to new time as soon as possible
Crossing > 3–4 time zones with layover > 24 hours	Keep in swing with rhythm of the **departure** country for as long as possible; maintain regular living patterns
Re-synchronization rate	**1 – 1.5 hours per day**
Eastbound adaptation	**~ 50 % slower** than westbound; **1.5 days per time-zone** east
Westbound adaptation	**~ 1 day per time-zone**
Difficulty	Readjustment after a time shift is normally **more difficult with flights towards the East**
Sleep duration	Governed primarily by the point within your circadian rhythm at which you try to sleep
Sensorimotor vs intellectual performance	Sensorimotor performance is better in the **evening**; intellectual performance is better in the **morning**

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Worked Example — Hyderabad → Berlin Layover Hyderabad (IST, UTC+5:30) to Berlin (CET, UTC+1) — a westbound trip of about 4.5 time zones with a **22-hour layover**.
• Layover < 24 hours → *do NOT shift to local time*. Stay on departure (IST) rhythm.
• Relevant time measure for your circadian rhythm: **IST (Indian Standard Time)**.
Reason: short layovers do not give the biological clock time to re-synchronize (only 1–1.5 hours/day shift is possible); attempting to shift will degrade alertness on the return leg.

> **EXAM TIP — EASTBOUND = WORSE**
>
> Eastbound travel adaptation is **50% slower** than westbound. East → 1.5 days per time-zone; West → 1 day per time-zone. "*Lose hours, lose sleep.*"

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Sleep — NREM & REM

WHAT THIS SECTION COVERS
The two components of sleep, the four NREM stages, REM sleep timing and function, sleep inertia and the implications of sleep loss for pilots.

Sleep is basically divided into **two components**:

- **Rapid Eye Movement (REM) sleep**
- **Non-Rapid Eye Movement (NREM) sleep**

Component	Purpose
NREM sleep	**Body restoration** — repair tissues, build bone and muscle, strengthen immune system
REM sleep	**Brain restoration** — strengthening, refreshing and organizing memory

NREM sleep is further divided into **four stages** from lightest to deepest. **Both types of sleep are required to recoup physical and mental energy.**

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NREM Sleep Stages

Stage	Description	Duration
NREM Stage 1	Transition phase between wakefulness and sleep. Brain activity, eye movement and muscle activity become slower. A person is easily awakened. Waking up in this stage causes a person to feel that he/she has *not slept*.	**10 minutes** each time
NREM Stage 2	Light sleep — the first stage of true sleep. Occupies **50 %** of the sleep patterns. Brain activity, eye movement become even slower; cardiac activity decreases.	**10–25 minutes** each time
NREM Stage 3	Beginning of deep sleep — slow-wave delta sleep. Brain activity and eye movement approaching zero. If awoken, the person may feel groggy and disoriented for a few minutes.	—
NREM Stage 4	**Deep sleep** — slow-wave delta sleep. **No eye movement or muscle activity.** If awoken, the person may feel groggy or disoriented for a few minutes.	—

REM Sleep (Paradoxical Sleep)

- Usually occurs **70–90 minutes** into a sleep.
- Happens in cycles lasting for approximately **10 minutes initially to 1 hour**.
- The eye moves rapidly from side to side and the brain wave form is similar to that when a person is awake.
- **Dreams occur.**

As we grow older, the time spent in REM sleep declines from **50 %** of our sleep for infants, to **20 %** of our sleep for adults.

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Sleep Inertia

DEFINITION — SLEEP INERTIA

Refers to the transitional state between sleep and wake, marked by **impaired performance, reduced vigilance, and a desire to return to sleep**.

Key Points to Remember

SLEEP — CRITICAL FACTS

- The effects of sleep deprivation on performance **increase with altitude and with higher workload**.
- During paradoxical sleep rapid eye movements can be observed.
- The duration of a period of sleep is governed primarily by the point within your circadian rhythm at which you try to sleep.
- **Deep sleep allows for physical recovery and the reconstitution of neuron energy reserves.**
- Lack of sleep increases fatigue, concentration and attention. It has risks of **sensory illusions and mood disorders**. Disruption of sleeping patterns can lead to symptoms of drowsiness, irritability and lack of concentration — making an individual more prone to make errors.
- A person experiencing sleep loss is **unlikely to be aware** of personal performance degradation.
- Performance loss may be present up to **20 minutes** after awaking from a short sleep (nap).
- The sleep cycles repeat during the course of a night's sleep. Each succeeding cycle contains a greater amount of REM sleep. **Frequent interruption of the REM sleep can harm a human being in the long run.** REM sleep is more important for the regeneration of mental and physical functions than all the other sleep stages are.

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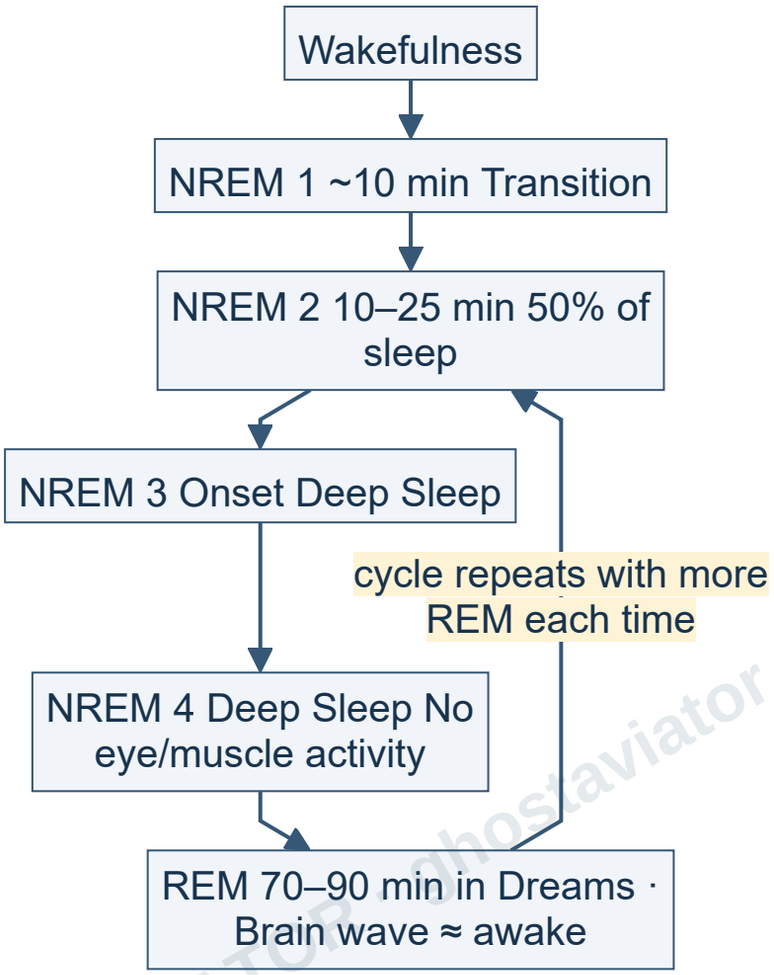

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Mental Health Problems & Disorders

WHAT THIS SECTION COVERS

The risk of mental health problems among aviation personnel, the Germanwings 9525 case study, and where preventive efforts should focus.

Mental health problems and disorders among pilots, ATCOs, maintenance and other personnel in aviation may impair performance and therefore be a **threat to flight safety**.

Case Study — Germanwings Flight 9525

GERMANWINGS 9525 — 24 MARCH 2015

- **Route:** Barcelona–El Prat Airport (Spain) to Düsseldorf Airport (Germany).
- **Aircraft:** Airbus A320.
- **Date:** 24 March 2015.
- **Crash location: 100 km** north-west of Nice in the French Alps.
- **Casualties:** All **144 passengers** and all **6 crew members** were killed.

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- **Cause:** The crash was deliberately caused by the co-pilot, Andreas Lubitz, who had previously been treated for suicidal tendencies and declared unfit to work by his doctor. Lubitz kept this information from his employer and instead reported for duty. Shortly after reaching cruise altitude and while the captain was out of the cockpit, Lubitz locked the cockpit door and initiated a controlled descent that continued until the aircraft hit a mountainside.

Why Pilots May Not Report Mental Health

There may be many reasons why a pilot may be reluctant to discuss mental health problems with the examining physician during the annual medical assessment, including **fear of losing his or her licence** with both personal and financial costs as a result. This may prevent the pilot from receiving adequate and timely help, and this could potentially make the problems worse and prolong the time for recovery.

BEST PRACTICE — REDUCING STIGMA

More could probably be done to increase knowledge about mental health problems and reduce stigma so that more pilots and other professionals working in aviation report mental health problems, get treatment, and return to work. This would probably increase individual well-being, but also **promote safety**.

Where to Focus Preventive Efforts

Serious mental health disorders (e.g. *psychosis*) are relatively rare and their onset is difficult to predict. **Preventive efforts should be aimed at more common mental health problems such as:**

- **Depression**
- **Anxiety**
- **Substance misuse**

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M-1. Master Numerical Reference

Topic	Value	Section
Iconic memory duration	**0.5 – 1 second**	5
Visual channel share of information processing	**70 – 80 %**	5
Short-term memory duration	**10 – 20 seconds**	12
Short-term memory capacity	**7 ± 2** items	12
Long-term memory capacity	**Unlimited** (retrieval may fail)	12
Acceptable workload (crew resources)	**~ 60 %**	11
Human error rate – simple/repetitive	**1 in 100**	11
Human error rate – after practice	**1 in 1,000**	11
Startle reflex – motor task recovery	**5 – 10 seconds**	10
Age-related response slowing	**20 – 60 years**	14
Anthropometric design population	Central **90 %** (disregard 5% lowest & 5% highest)	20
Free-running circadian cycle	**~ 25 hours**	23

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Cued circadian cycle	**~ 24 hours**	23
Circadian re-sync rate	**1 – 1.5 hours/day**	23
Eastbound adaptation	**~ 1.5 days/time-zone** (50% slower than west)	23
Westbound adaptation	**~ 1 day/time-zone**	23
Stop-over threshold for adjusting to local time	**> 24 hours**	23
Lowest body temperature time	**04:30**	23
Highest body temperature time	**19:00**	23
Melatonin peak	**02:00 – 04:00**	23
Melatonin secretion starts	**21:00**	23
Melatonin secretion stops	**07:30**	23
NREM Stage 1 duration	**10 minutes**	24
NREM Stage 2 duration / share	**10–25 min** / **50 %** of sleep	24
REM onset after sleep start	**70 – 90 minutes**	24
REM cycle duration	**10 min → up to 1 hour**	24
REM share – infant vs adult	**50 %** vs **20 %**	24
Post-nap performance loss	**Up to 20 minutes**	24
Life stress scoring – free / normal / high / serious	**<60 / 60–80 / 80–100 / >100**	22
Germanwings 9525 — date	**24 March 2015**	25
Germanwings 9525 — crash distance	**100 km** NW of Nice	25
Germanwings 9525 — casualties	**144 pax + 6 crew**	25
Cockpit restraint	**5-point harness with negative-G strap**	20

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M-3. Quick Definitions Recall Sheet

Term	One-Line Definition
Aviation Psychology	Applied psychology focused on human behaviour in aviation systems.
Workload	Mental effort needed to process information.
Perception	Conversion of sensory information into a meaningful structure.
Selective Attention	Continual sampling of inputs to judge relevance.
Divided Attention	Time-sharing of central decision channel between tasks.
Vigilance	Capability of remaining alert above a threshold for a period.
Complacency	Self-satisfaction with one's performance + unawareness of danger.
Motor Programme	Behavioural sub-routine learnt by practice and held in LTM.
Reaction Time	Delay between detection, stimulus and muscle contraction.
Startle Reflex	Reflex-like blink + body jerk to abrupt stimulus.

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Human Reliability	Individual functioning in the manner he/she is supposed to.
Hallucination	False perception characterised by distortion of real sensory stimuli.
Cognition	Mental process of acquiring knowledge by reasoning, intuition or perception.
Risk Assessment	Probability of risk × Impact if it occurs.
Situational Awareness	Accurate appraisal of self, environment and own performance.
Automation	Controlling apparatus/process by electronic/mechanical devices replacing the human organism.
Anthropometry	Study of human measurement (static, dynamic, contour-surface).
Eye Datum	Defined eye position around which the cockpit is designed.
Trait Anxiety	Personality trait of high neuroticism (persistent worry).
State Anxiety	Transient anxiety present in anyone at any time.
Arousal	Person's readiness to respond effectively to a stress factor.
Fatigue	Deep tiredness from cumulative stressful lifestyle/environment.
Zeitgeber	Body's chronological pacemaker — melatonin signals time of day/year.
Sleep Inertia	Transitional state between sleep and wake with impaired performance.

M-4. Risk Formula

Risk Assessment Formula `Risk = Probability of Occurrence × Impact if it Occurs`

Variables:
• *Probability* — likelihood the hazard manifests (low / medium / high).
• *Impact* — severity of consequence (minor / major / catastrophic).

Worked example — Marginal weather decision:
Probability of icing on route = High (forecast supports it).
Impact if encountered without anti-ice = Catastrophic.
→ Risk = High × Catastrophic = **Unacceptable**. Decision: **Don't fly**, or re-route below freezing level / with active anti-ice.

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M-5. Final Quick-Revision Summary Box

EXAM-NIGHT ONE-PAGE RECAP

Memory: Iconic 0.5–1 s · STM 10–20 s, 7±2 · LTM unlimited.
Workload: Acceptable ≈ 60 % · DPST drivers · symptoms = funneling, regression, mental blocking, panic.
Error rate: 1/100 raw → 1/1,000 after practice. Errors are cumulative.
Startle: Motor task back to normal in 5–10 s. Fear potentiates.
Anderson Skill: Cognitive → Associative → Automatic.
Anthropometry: central 90 %, discard top & bottom 5 %.
Restraint: 5-point harness + negative-G strap.
Display rule: Digital for quantitative · Analogue for qualitative. Standardization is the #1 design rule.
Three-pointer altimeter: more misreads than single-pointer.
Stress observables: perspiration, flushed skin, dilated pupils, fast breathing.

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Life stress: <60 free · 60–80 normal · 80–100 high · >100 serious. Spouse death = 100.
Circadian: Free-running 25 h · cued 24 h · Lowest body temp 04:30 · Highest 19:00 · Melatonin peak 02–04 · Zeitgeber = Melatonin (pineal gland).
Jet-lag: 1–1.5 h/day re-sync · East 1.5 d/zone (50% slower) · West 1 d/zone · >24 h stopover → shift; <24 h → stay on departure rhythm.
Sleep: NREM 1 = 10 min · NREM 2 = 10–25 min, 50% · REM at 70–90 min · REM 50% infant → 20% adult · Post-nap loss up to 20 min.
Mental health priority: Depression, Anxiety, Substance misuse. Germanwings 9525 (24 Mar 2015, A320, 100 km NW Nice, 144 pax + 6 crew).
Communication barriers: Aggressiveness, Arrogance, Anti-authoritarian, Impulsiveness, Invulnerability, Resignation.
Murphy's Law: if it can be operated wrongly, sooner or later it will be.
Golden Maxim: A superior pilot uses his superior judgment to avoid situations that would require his superior skills.

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CHAPTER 25

Aviation Physiology

25

WHY THIS CHAPTER MATTERS TO A PILOT

Aviation Physiology deals with the **physical and mental effects of flight on aircrew personnel and passengers**. In aviation, the demands upon the compensatory mechanisms of the body are *numerous and of considerable magnitude*. The environmental changes of greatest physiological significance involved in flight are:

- Marked changes in **barometric pressure**
- Considerable variation in **temperature**
- **Movement at high speed in three dimensions**

This part of the syllabus familiarises you with the physiological problems of flight and assists in human compensation for the numerous environmental changes encountered in flight. **Remember: every human is physiologically different and can react differently in any given situation.**

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1 The Atmosphere

Human beings live their lives in the **lower reaches of the atmosphere** where temperatures, pressures and oxygen supply are able to support life. The moment we climb, every one of those three variables shifts against us. To fly safely, you must first understand the layer you're flying in and what it is made of.

1.1 The Layers — Troposphere & Stratosphere

Troposphere – Average Top	13 km · 8.1 mi · 43,000 ft
Stratosphere – Reaches Above	100,000 ft
Where Almost All Weather Lives	Troposphere
Where Air is Densest	Lowest Layer (Surface)

The Troposphere height **varies**. On average it stretches from the Earth's surface to about **13 km (8.1 mi; 43,000 ft)**. The Stratosphere reaches up to over **100,000 ft**. The troposphere contains **almost all the weather**, the air is densest in this lowest layer, and in fact the troposphere contains **three-quarters of the mass of the entire atmosphere**.

1.2 Constituent Gases of the Atmosphere

The earth's atmosphere near the surface is composed primarily of **Nitrogen** and **Oxygen**. Together, the two comprise **about 99%** of the gas in the atmosphere. The remaining 1% is made up of Argon plus traces of other gases.

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Composition of the lower atmosphere — exact percentages

Gas	Percentage	Gas	Percentage
Nitrogen (N_2)	**78.084 %**	**Oxygen (O_2)**	**20.95 %**
Argon (Ar)	**0.934 %**	Carbon Dioxide (CO_2)	**0.036 %**
Neon (Ne)	**0.0018 %**	Helium (He)	**0.0005 %**
Methane (CH_4)	**0.00017 %**	Hydrogen (H_2)	**0.00005 %**
Nitrous Oxide (N_2O)	**0.00003 %**	Ozone (O_3)	**0.000004 %**

In addition, **water vapour** is variable but typically makes up about **1 – 4 %** of the atmosphere.

CRITICAL PRINCIPLE TO REMEMBER

The relative proportions of these gases remain CONSTANT in the Troposphere and Stratosphere. What changes with altitude is not the *percentage* of oxygen — it is the *total pressure*, and therefore the **partial pressure** of each gas. Oxygen continues to make up 21 % of the air by volume **at every altitude**; only the partial pressure of oxygen, pO_2, falls.

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2 The Gas Laws

The body responds to barometric pressure changes in temperature, pressure, and volume. These changes are **rapid and continuous** in the aviation environment. It is therefore essential to know the implication of these changes on our body and take preventive measures to counter them. The gas laws explain to us the **science behind what goes on within our body** when exposed to changes in pressure and temperature.

2.1 Boyle's Law

DEFINITION

At a **constant temperature**, a given volume of gas is **inversely proportional to the pressure** surrounding the gas. The volume of gas **expands as the pressure surrounding the gas is reduced**.

- **As altitude increases → gas expands.**
- **As altitude decreases → gas compresses.**
- The amount of volume expansion is **limited by the pliability** of the structure or membrane which encloses the gas.

EXAMPLES IN AVIATION & MEDICINE

PASG / Air Splints · Respiratory Rate & Depth changes · Flow rates of **IV sets** · **ETT** or **Tracheal cuff** pressures · **Trapped gas effects** within the body (middle ear, sinuses, GI tract — every pilot has felt these).

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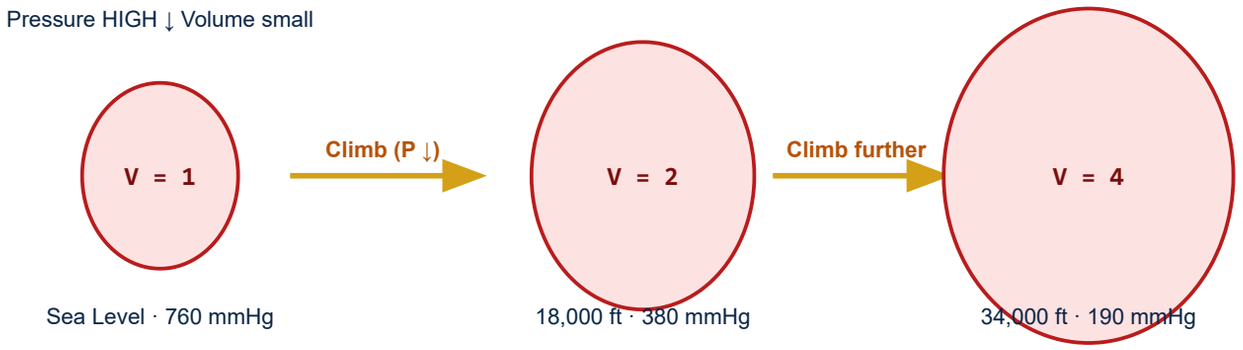

Figure 25.1 *Boyle's law: for a fixed mass of gas at constant temperature, pressure multiplied by volume is constant. Falling ambient pressure can expand gas trapped in body cavities.*

2.2 Henry's Law

DEFINITION

The amount of **gas in solution** is proportional to the **partial pressure of that gas over the solution**. As the pressure of the gas above a solution *increases*, the amount of that gas dissolved in the solution *increases*; the reverse is also true — as the pressure of the gas above a solution *decreases*, the amount of gas dissolved *decreases* and forms a **"bubble" of gas within the solution**.

In normal physiologic function, this law can be seen in the **transfer of gas between the alveoli** ***(tiny air sacs in the lungs)*** **and the blood**. This is **significant physiologically** for the occurrence of evolved gas disorders, e.g. **decompression sickness**. It explains the hypoxia *(too little oxygen reaching body tissues)* experienced with increasing altitude — as the pressure of gases is reduced with ascent, the amount of gases dissolved in solution decreases, and this leads to hypoxia and may lead to **nitrogen bubble formation**.

EVERYDAY ANALOGY — A FIZZY DRINK

Opening a fizzy drink reduces the pressure above it and dissolved gas can form bubbles. In the body, a fall in ambient pressure can similarly allow dissolved nitrogen to form bubbles. This explains a risk of decompression sickness; it does not mean that every cabin depressurisation produces bubbles or illness.

CLINICAL LINK — DCS (DECOMPRESSION SICKNESS)

A pilot who has been SCUBA diving cannot climb shortly after surfacing because Henry's Law is still working against them — dissolved nitrogen in their blood will come out of solution as bubbles when ambient pressure falls. *(Strict wait-times before flying after diving are covered in a later section.)*

2.3 Charles' Law

CHARLES' LAW — VOLUME AND TEMPERATURE

For a **fixed mass of gas at constant pressure**, volume is proportional to absolute temperature: **V / T = constant**. Use kelvin for T. Heating the gas increases its volume if it can expand while the pressure remains constant.

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A flexible gas container can expand as its contents warm. A rigid, sealed cylinder cannot expand appreciably: its pressure rises as temperature rises. This fixed-volume pressure–temperature relationship is a different special case of the gas equation, **P / T = constant**.

KEEP THE CONDITIONS CLEAR

Charles' law keeps **pressure** constant; the pressure–temperature law keeps **volume** constant. Rapid filling or decompression involves changing conditions and heat transfer, so Charles' law alone does not describe the whole process. Never heat a pressurised container.

2.4 Graham's Law — Law of Gaseous Diffusion

GRAHAM'S LAW — RELATIVE RATES OF EFFUSION

Effusion is the escape of gas through a very small opening. At the same temperature and pressure, a gas with lower molar mass effuses faster: **rate ∝ 1 / √(molar mass)**.

Diffusion is the net spread of molecules from a region of higher concentration towards a region of lower concentration. Its rate also depends on the temperature, available area and distance travelled. In the lungs, oxygen passes from the alveoli into the blood and carbon dioxide passes in the opposite direction. Graham's rate relationship must not be used as a complete explanation of exchange through the lung membrane.

2.5 Dalton's Law — Law of Partial Pressures

DEFINITION

Partial pressures describe the **distribution of certain gases in a mixture** and follows Dalton's law which states that the **total pressure of a mixture of gases is equal to the sum of the partial pressures of the gases which compose the mixture**. As altitude increases — gases exert **less pressure**. This explains the **hypoxia that occurs with flight to higher altitudes**.

WORKED EXAMPLE — OXYGEN AT DIFFERENT ALTITUDES (DALTON'S LAW IN ACTION)

At Sea Level: O_2 = 21 % & **pO_2 = 21 % × 760 mm = ≈ 160 mmHg**

At 8,000 ft: O_2 = 21 % & **pO_2 = 21 % × 565 mm = ≈ 119 mmHg**

The barometric pressure at **36,000 ft** is **one-fourth** of that at sea level. Hence the quantity of oxygen available is **proportionately low**. *Whatever the air pressure, oxygen continues to make up 21 % of the air by volume.* In other words, the **proportion of oxygen in the air always stays the same whatever the altitude**. **The partial pressure of Oxygen decreases with altitude as does the total pressure of air.**

EXAM-CRITICAL TAKEAWAY

The percentage of O_2 in air does **NOT** change with altitude — it is always **21 %**. What kills the pilot is the falling **partial pressure** of that oxygen, because diffusion across the alveolar membrane depends on *pressure gradient*, not percentage.

Quick-Reference — The Five Gas Laws of Aviation Physiology

Law	One-line Statement	What it Explains in Flight

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Law	One-line Statement	What it Explains in Flight
Boyle's	P × V = constant (at fixed T) — pressure ↑ → volume ↓	Trapped-gas pain in ears, sinuses, teeth, GI tract; tracheal-cuff & IV behaviour
Henry's	At fixed temperature, dissolved gas at equilibrium is proportional to its partial pressure above the liquid	Gas coming out of solution as pressure falls; decompression-sickness risk.
Charles'	V / T = constant for a fixed mass at constant pressure; T in kelvin	Expansion of a gas as it warms when pressure is kept constant.
Graham's	Effusion rate ∝ 1 / √(molar mass), at the same temperature and pressure	Lighter gas molecules effuse faster; lung exchange also depends on other factors.
Dalton's	Total P = Σ partial pressures	Why high-altitude hypoxia happens despite air being "21% oxygen"

3 Variation of Pressure & Temperature with Altitude

3.1 The International Standard Atmosphere (ISA)

WHAT ISA ACTUALLY IS

The International Standard Atmosphere is used to show **standardised values for temperature, pressure, density and lapse rate**. Pressure decreases with altitude *throughout* the atmosphere. Temperature decreases with altitude *in the Troposphere*.

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ISA Sea-Level Values (reference these cold)

Sea-Level Pressure	1013.25 hPa
Equivalent (mb)	1013.25 mb
Equivalent (psi)	14.7 lb/in²
Equivalent (Hg)	29.92 in Hg
Equivalent (mmHg)	760 mm Hg
ISA Lapse Rate	1.98 °C / 1000 ft

STRICT ISA LIMITS — EXAMINATION VALUES

Temperature reduces at 1.98 °C per 1000 ft up to 36,090 ft. Thereafter it remains **constant** at **–56.5 °C**. *(The 1.98 °C figure is the precise ISA lapse rate; many texts round to 2 °C / 1000 ft for mental maths — never round in your exam paper.)*

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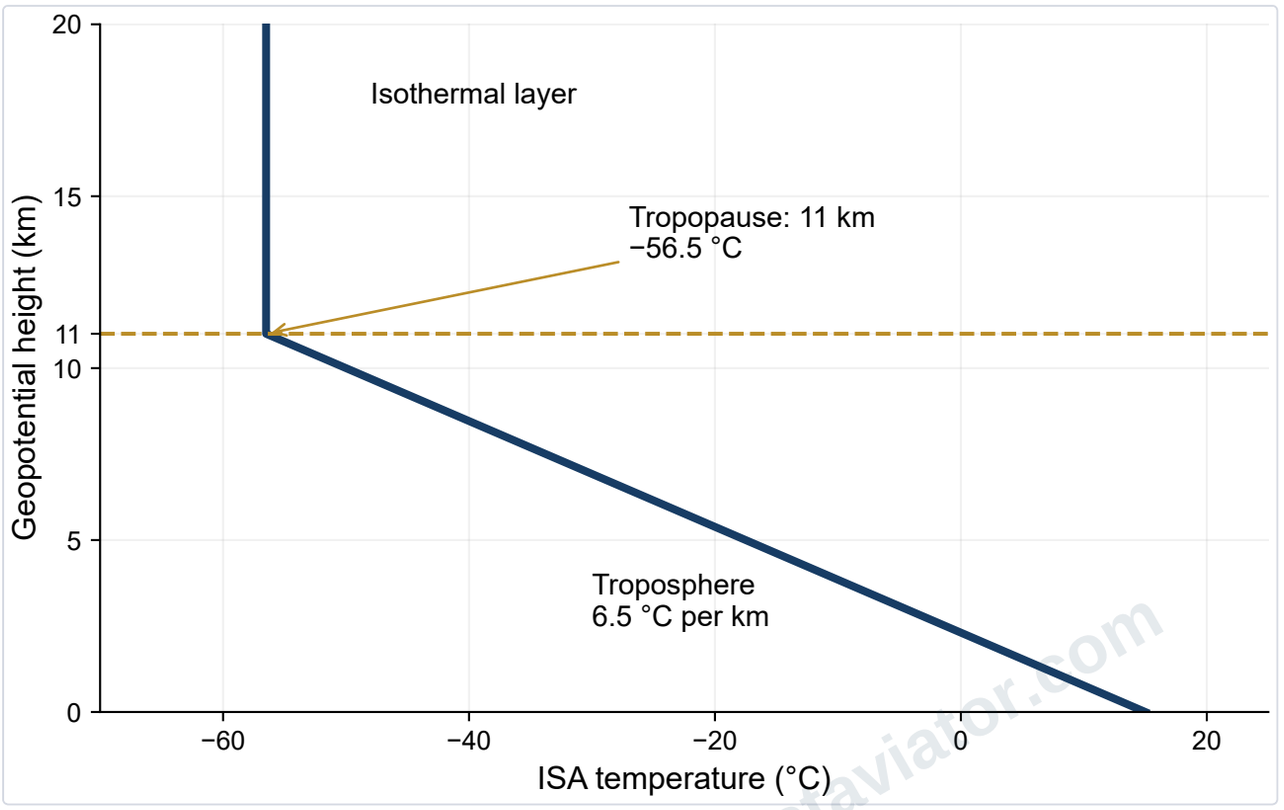

Figure 25.2 *International Standard Atmosphere temperature profile to20km. Actual atmospheric conditions differ from this reference model.*

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4 Partial Pressure of Oxygen — Effects of Increasing Altitude

As you climb, the **total barometric pressure** drops, and with it the **partial pressure of oxygen** (pO_2). Because the diffusion of oxygen across the alveolar membrane depends on the pressure gradient, what matters to your blood is not the percentage of O_2 — it is the pO_2. Here are the standard values the DGCA expects you to know:

Standard barometric pressure & oxygen availability at altitude

Altitude	Standard Barometric Pressure	O_2 Available (% of Sea Level)	Pilot Significance
Sea Level (0000 ft)	**101 kPa** (**760 mmHg**)	**100 %**	Normal physiological baseline.
8,000 ft	**77 kPa** (**574 mmHg**)	**76 %**	**Cabin pressure is maintained between 6,000 – 8,000 ft.**
12,000 ft	**66 kPa** (**496 mmHg**)	**65 %**	Lower edge of the Physiological-Deficient Zone.
18,000 ft	**53 kPa** (**395 mmHg**)	**52 %** (≈ half)	**Half the O_2 of sea level.**
24,000 ft	**41 kPa** (**311 mmHg**)	**41 %**	Without supplemental O_2 — incapacitation imminent.
36,000 ft	**25 kPa** (**187 mmHg**)	**25 %** (one-fourth)	**Only ¼ of sea-level oxygen.**

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THREE DEDUCTIONS YOU MUST BE ABLE TO STATE IN THE EXAM

1. The **availability of oxygen and barometric pressure DECREASE with altitude**.
2. The oxygen available is **one-fourth** at **36,000 ft** and **half** at **18,000 ft** of the oxygen available at sea level.
3. **Atmospheric pressure drops faster at lower altitudes** in comparison to the same altitude changes at higher altitudes (the pressure–altitude curve is exponential, not linear).

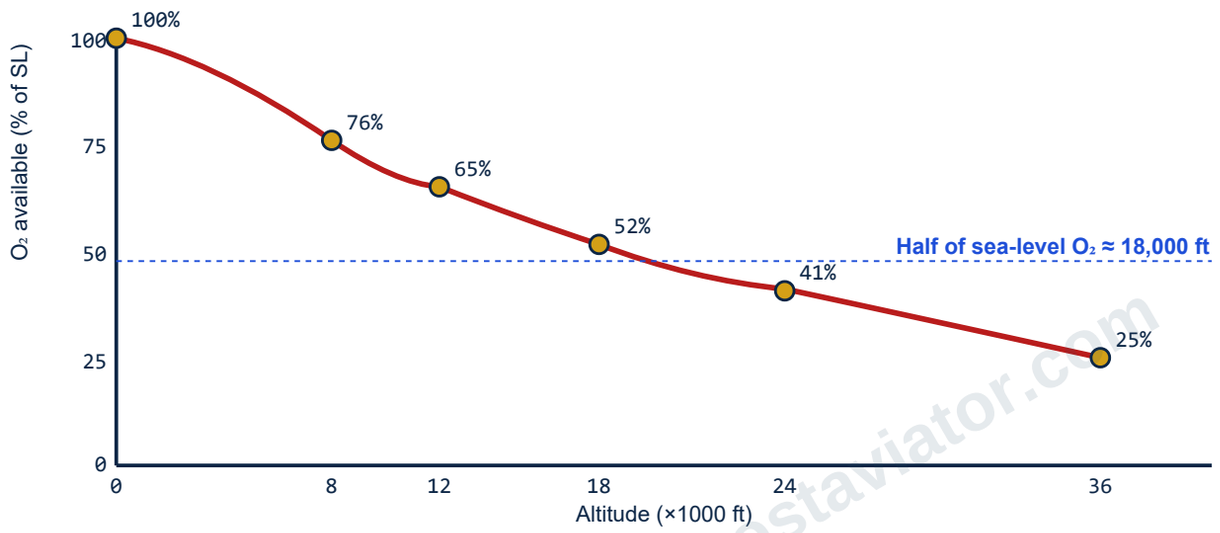

Figure 25.3 *Oxygen stays about 21% of the air; its partial pressure falls as atmospheric pressure falls.*

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5 The Physiological Zones of the Atmosphere

The atmosphere is divided — from the pilot's body's point of view — into four functional zones, each demanding a different protective strategy. **You must know the boundaries and the SOP for each.**

5.1 The Physiological Zone

SOP — WITHIN THIS ZONE

Normal healthy human beings used to living near sea level will need **supplementary oxygen to function normally at altitude exceeding 10,000 to 12,000 ft**. **Pilots will normally begin breathing supplementary oxygen from 10,000 ft above sea level.**

5.2 The Physiological-Deficient Zone

LIMITS & SYMPTOMS — STRICT

Exists from **12,000 ft to 50,000 ft**. **The body is not used to this environment.** The adverse effects include:

- Middle ear and sinus blockage
- Shortness of breath
- Dizziness
- Headache

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COMPENSATORY THRESHOLD

At **6,000 – 7,000 ft** altitude the human organism starts with remarkable measures to compensate for the drop in pO_2 when climbing — this is the **threshold for compensatory reactions**. (Hyperventilation *(breathing more than the body needs)*, raised heart rate, raised cardiac output, etc. — all later sections.)

CRITICAL-THRESHOLD LIMITS — NON-PRESSURISED FLIGHT

- A pilot climbing in a **non-pressurised aircraft and WITHOUT using supplemental oxygen** will pass the **"critical threshold" at approximately 22,000 ft**.
- **Breathing 100 % oxygen** will lift the pilot's physiological safe altitude to approximately **38,000 ft**.

5.3 The Partial Space-Equivalent Zone

DEFINITION & EXTENT

This zone extends from **50,000 ft to 120 nm** (nautical miles). Even 100 % oxygen ceases to be enough at these heights — total pressure becomes the limiting factor.

5.4 The Total Space-Equivalent Zone

DEFINITION

The Total Space-Equivalent Zone extends **outwards from 120 nm**.

MANDATORY LIMITS IN THE UPPER TWO ZONES

- **100 % oxygen does NOT protect from hypoxia** — pressure-breathing required.
- **Sealed cabins and pressure suits are a MUST.**
- Blood and body fluids **BOIL above 63,000 ft** (Armstrong's Line — vapour pressure of body fluids equals ambient pressure).
- **Gravitational changes affect the body** (micro-G adaptation, fluid shift, motion-sickness, etc.).

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6.1 Numbers you Must know by heart

Master cheat-sheet — numbers

Parameter	Exact Value	Where it comes from
Top of troposphere (average)	**13 km / 8.1 mi / 43,000 ft**	§1 Atmospheric layers
Top of stratosphere (over)	**100,000 ft**	§1
Troposphere holds	**75 %** of atmospheric mass	§1
Nitrogen / Oxygen / Other	**78.084 / 20.95 / ~1 %**	§1.2
Water vapour content	**1 – 4 %** (variable)	§1.2
ISA sea-level pressure	**1013.25 hPa = 760 mmHg = 29.92 inHg = 14.7 psi**	§3.1
ISA lapse rate	**1.98 °C / 1000 ft**	§3.1

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Parameter	Exact Value	Where it comes from
Tropopause altitude (ISA)	**36,090 ft**	§3.1
Isothermal temperature	**−56.5 °C**	§3.1
Cabin altitude maintained	**6,000 – 8,000 ft**	§4
O_2 available at 18,000 ft	**52 %** (half of SL)	§4
O_2 available at 36,000 ft	**25 %** (one-fourth of SL)	§4
Compensatory reactions begin	**6,000 – 7,000 ft**	§5.2
Pilot starts supplementary O_2	**10,000 ft**	§5.1
Physiological-Deficient Zone	**12,000 – 50,000 ft**	§5.2
Critical threshold (no O_2, unpressurised)	**~22,000 ft**	§5.2
Safe altitude on 100% O_2	**~38,000 ft**	§5.2
100% O_2 Under Pressure (PBA) from	**Above 40,000 ft**	§5.1 fig.
Partial Space-Equivalent Zone	**50,000 ft – 120 nm**	§5.3
Armstrong's Line (body fluids boil)	**63,000 ft**	§5.4
Total Space-Equivalent Zone	**Outwards from 120 nm**	§5.4

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7 The Brain & the Nervous System

The brain is the master controller of your body — every input you receive in flight (visual, vestibular, proprioceptive, auditory) is processed here, and every output (control inputs, R/T calls, decisions) originates here. The nervous system is the wiring that connects the brain to every organ and muscle.

7.1 The Brain

> **DEFINITION**
>
> **The brain controls all bodily functions.** It performs an incredible number of tasks, every one of which is potentially flight-critical.

What the brain does — exam list

- It controls **body temperature, blood pressure, heart rate and breathing**.
- It accepts information about the world around you from your various **senses** — seeing, hearing, smelling, tasting and touching.
- It handles your **physical movement** when walking, talking, standing or sitting.
- It lets you **think, dream, reason and experience emotions**.
- The brain controls **decision-making and speech**. Through the brain we experience **consciousness, vision, hearing, taste, smell, thought and memory**.

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WHY THIS MATTERS IN THE COCKPIT

Every one of these functions is **oxygen-dependent**. The brain consumes about **20 % of the body's oxygen** despite being only ~2 % of body mass. Even mild hypoxia degrades judgement, decision-making and speech long before the pilot is consciously aware of it. *(Detailed in §10 Hypoxia.)*

7.2 The nervous system — structure and automatic control

The main structural division is between the CNS and PNS. Autonomic control describes a function of connected nervous-system pathways.

Term	Meaning
Central nervous system	Brain and spinal cord.
Peripheral nervous system	Nerves outside the brain and spinal cord; carries information between the CNS and body.
Autonomic control	Connected central and peripheral pathways regulate internal organs without a conscious command for each action.

Figure 25.4 *Structural divisions and autonomic function: these are not three equal anatomical branches.*

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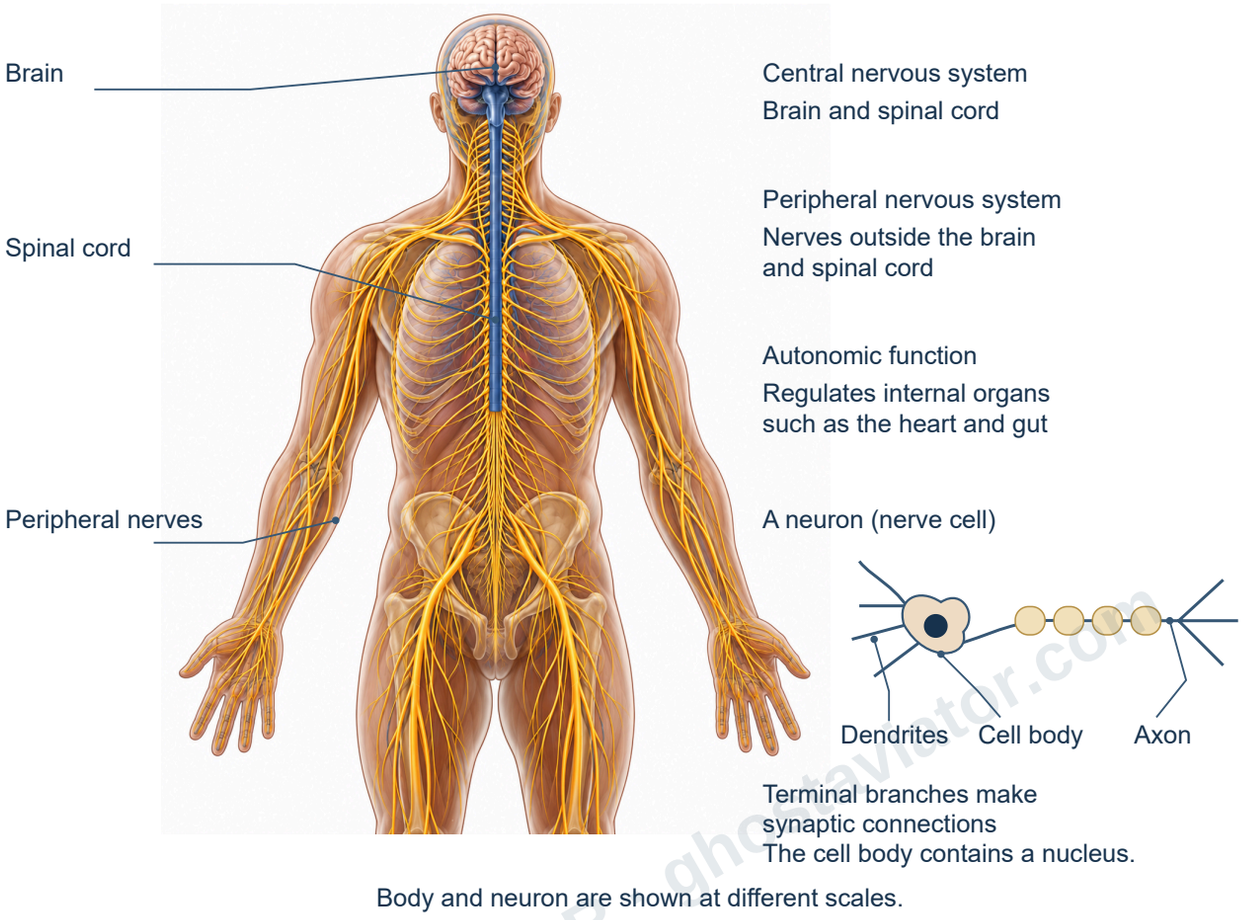

Figure 25.5 *Central and peripheral nerve pathways, with an enlarged neuron inset. Autonomic functions regulate internal organs.*

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The Central Nervous System (CNS)

> **DEFINITION**
>
> The **central nervous system (CNS)** consists of the **brain and spinal cord**. The CNS and the peripheral nerves together form the nervous system, which receives information and coordinates the body's responses.

- The scientific study of the brain and nervous system is called **neuroscience** or **neurobiology**.
- The **brain** functions to **receive nerve impulses from the spinal cord and cranial nerves**.
- The **spinal cord** contains the nerves that **carry messages between the brain and the body**.

The Peripheral Nervous System (PNS)

> **FUNCTION**
>
> The **peripheral nervous system (PNS)** contains the nerves outside the brain and spinal cord. Sensory nerves carry information towards the CNS; motor nerves carry commands towards muscles and organs. Reflexes involve connected sensory, processing and motor pathways.

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The Autonomic Nervous System (ANS)

DEFINITION

The organs (the **"viscera"**) of our body — such as the **heart, stomach and intestines** — are regulated by a part of the nervous system called the **Autonomic Nervous System (ANS)**. The ANS is **part of the peripheral nervous system**, and it controls many organs and muscles within the body.

AUTOMATIC CONTROL

The autonomic nervous system regulates functions such as heart rate, digestion and changes in blood-vessel size without a conscious command for each action. It has connections with the CNS: **automatic does not mean independent of the brain and spinal cord**.

Examples of autonomic functions

Sensory	Eye
Cardiac	Heart
Respiratory	Breathing
Thermoregulation	Temperature
Vascular	Blood Pressure
Digestive	Stomach & Intestines
Excretory	Urinary Output & Bladder
Cooling	Sweating & Glands
Stress Response	Fight-or-Flight Reaction

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FLIGHT-RELEVANCE — THE "FIGHT OR FLIGHT" REFLEX

The ANS controls the **fight-or-flight response or reaction to stress**. In a sudden emergency — engine failure, severe turbulence, near-miss — the ANS dumps adrenaline into your bloodstream automatically. Heart rate, breathing rate, BP and sweating all spike **without your conscious permission**. *This is why a well-trained pilot's drilled checklist beats raw reaction: the conscious brain must override an autonomic surge.*

Three divisions of the Nervous System — quick comparison

Division	What it Contains	What it Controls	Conscious / Involuntary
Central NS	Brain + Spinal Cord	Receives, processes & commands all activity	Both
Peripheral NS	All peripheral nerves	Skin, eye, blood vessels, heart, stomach & muscle action	Both — voluntary & reflex
Autonomic NS	Sub-system of PNS	Eye · Heart · Breathing · Temp · BP · GI · Bladder · Sweat · Stress	**Involuntary only**

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8 The Respiratory System

OVERALL PURPOSE — ONE SENTENCE

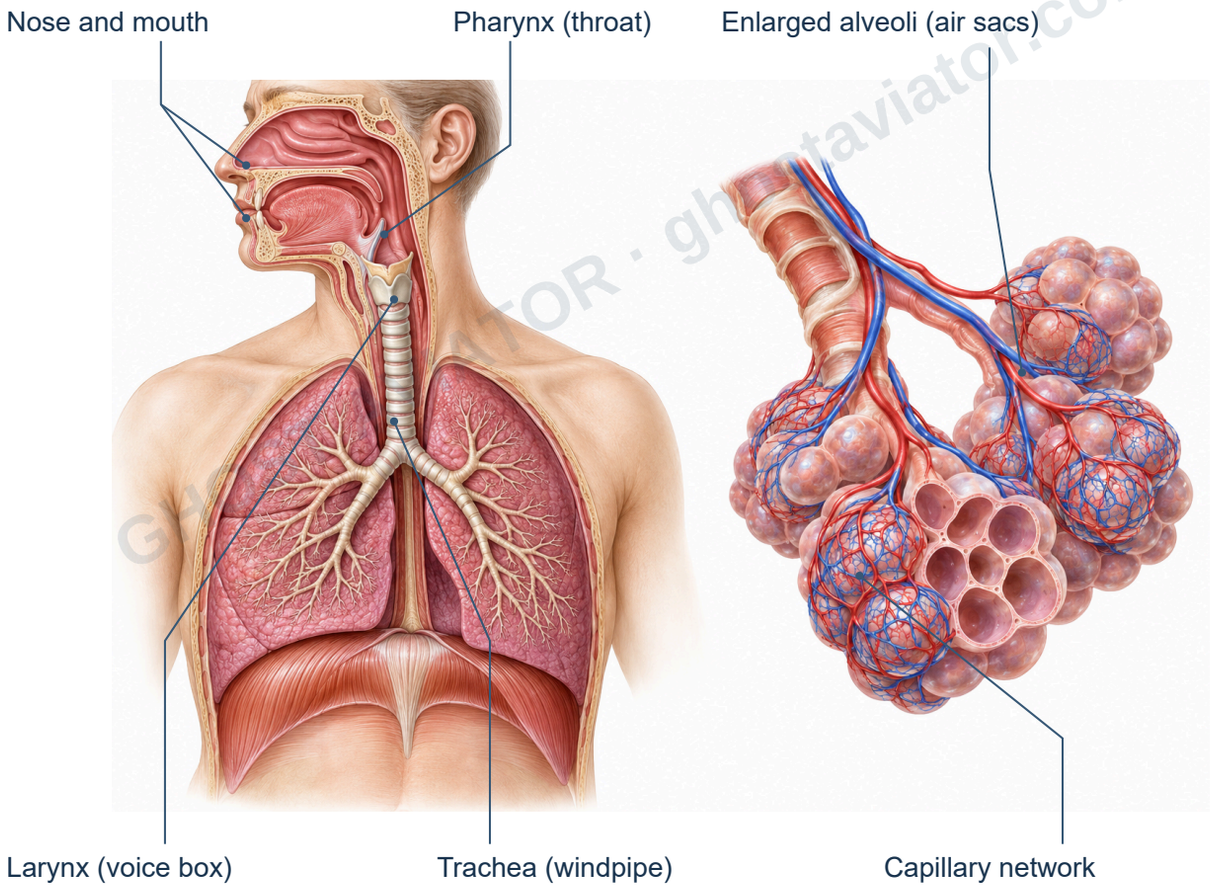
Lungs are complex organs, but what they do is to get rid of carbon dioxide and exchange it for oxygen.

The respiratory process consists mainly of:

- the **diffusion of oxygen through the respiratory membranes into the blood**,
- **transportation to the cells**,
- **diffusion into the cells**, and
- **elimination of carbon dioxide from the body**.

8.1 The Pathway of Air — Nose to Alveoli

As you breathe air in through your nose or mouth, it travels along this path:



Air route: nose / mouth → pharynx → larynx → trachea
→ bronchi → bronchioles → alveoli

At the alveoli: oxygen enters the blood; carbon dioxide leaves the blood.

Figure 25.6 *The route of inhaled air and an enlarged view of alveoli and surrounding capillaries.*

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ANATOMICAL LANDMARKS

Air passes through the nose or mouth, pharynx (throat), larynx (voice box) and trachea (windpipe), then through the branching bronchi and bronchioles to the alveoli (air sacs). The diaphragm is the main breathing muscle below the lungs. The enlarged inset shows alveoli surrounded by capillaries (very small blood vessels).

8.2 What Happens Inside the Alveolus — Gas Exchange

GAS EXCHANGE: FOLLOW THE PARTIAL-PRESSURE DIFFERENCE

Partial pressure (the pressure contributed by one gas) helps explain gas exchange.

1. Oxygen moves from the alveoli into the blood, where its partial pressure is lower.
2. Most oxygen is carried by haemoglobin in red blood cells.
3. Carbon dioxide moves from the blood into the alveoli, where its partial pressure is lower.
4. Most carbon dioxide travels in the blood as bicarbonate. At the lungs, it is converted back to carbon dioxide and breathed out.
5. Blood leaving the lungs carries more oxygen towards the heart.

THE PURPOSE, IN ONE LINE

The purpose of breathing is to keep the **oxygen concentration HIGH** and the **carbon dioxide concentration LOW** in the alveoli, so this gas exchange can occur.

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8.3 External vs Internal (Tissue) Respiration

External Respiration	External Respiration takes place through the **lungs** and refers to: • The **absorption of Oxygen from the air into the blood**. • The **excretion of Carbon Dioxide from the blood to the air**.
Internal / Tissue Respiration	Internal or Tissue Respiration refers to the **transfer of Oxygen from the blood to the tissues** of the body. At the same time as this occurs, the tissues **give up Carbon Dioxide to the blood**.

QUICK TEST **External = at the LUNGS** (air ↔ blood). **Internal = at the TISSUES** (blood ↔ cell). Don't mix these up — examiners love this distinction.

8.4 Breathing

DEFINITION

Normal breathing is a **purely automatic process under the unconscious control of the nervous system**. The normal rate of respiration in adults is **14 to 18 breaths per minute**.

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WHAT REGULATES YOUR BREATHING RATE — COUNTER-INTUITIVE BUT VITAL

The level of carbon dioxide in the blood effectively regulates the rate and depth of breathing.

It is **NOT oxygen** that triggers your urge to breathe — it is **rising CO_2**. This is the core mechanism behind **Hyperventilation** (over-breathing → CO_2 washes out → urge to breathe drops → tingling, dizziness) which is dealt with separately later in this chapter.

Normal Adult Respiration Rate	14 – 18 / min
Control Centre	Medulla (Autonomic)
Primary Stimulus	Blood CO_2 Level
Conscious Override	Possible (limited)

9 The Circulatory System

WHY IT EXISTS

Blood supplies our organs with life-giving oxygen and carries away waste products. The circulatory system consists of the **heart and the blood vessels**, and maintains the **flow of blood throughout the body**. Without it, gas exchange at the lungs is pointless — there must be a delivery network.

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9.1 The Heart

WHAT THE HEART DOES — CONCISE

The arteries **carry blood from the heart at HIGH pressure**, and the veins **return blood to the heart at LOW pressure**. The heart is a **pumping system** which:

- Intakes **de-oxygenated blood through the veins**.
- Delivers it to the **lungs for oxygenation**.
- Then pumps it into the various **arteries to be transmitted** where it is needed throughout the body for energy.

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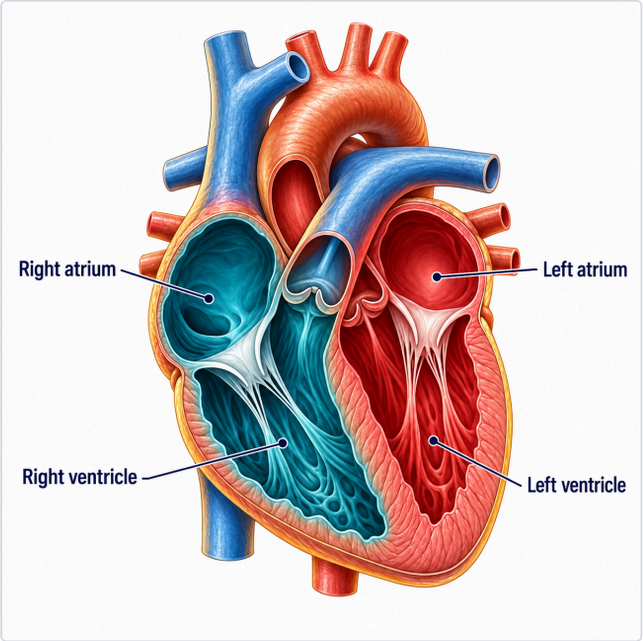

Follow the blood through the heart

1. Blood returns from the body through the **venae cavae** (the large returning veins).
2. The **right atrium** receives it.
3. It passes through a valve to the **right ventricle**.
4. The right ventricle pumps it through the **pulmonary artery** to the lungs, where it gains oxygen.
5. The **pulmonary veins** bring it to the left atrium.
6. It passes through a valve to the **left ventricle**.
7. The left ventricle pumps it through the **aorta** and arteries to the body, where tissues use oxygen.

Figure 25.7 *The four chambers and the circulation route. Anatomical right is on the viewer's left. Blue indicates blood with less oxygen; red indicates blood with more oxygen. Blood itself is not blue. Schematic illustration, not to scale.*

Physiology source: NIH/NHLBI, How Blood Flows through the Heart, "Adding oxygen to blood".

MEMORISE — CARDIAC OUTPUT VALUES

At rest the **cardiac output** (the quantity of blood the heart pumps in one minute) of an adult with:

- Heart Rate = **72 beats per minute**
- Stroke Volume = **70 ml**

… is **about 5 litres per minute**. (Stroke volume × Heart rate = Cardiac output.)

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9.2 How the Heart Works — Systole & Diastole

TWO PHASES OF THE CARDIAC CYCLE

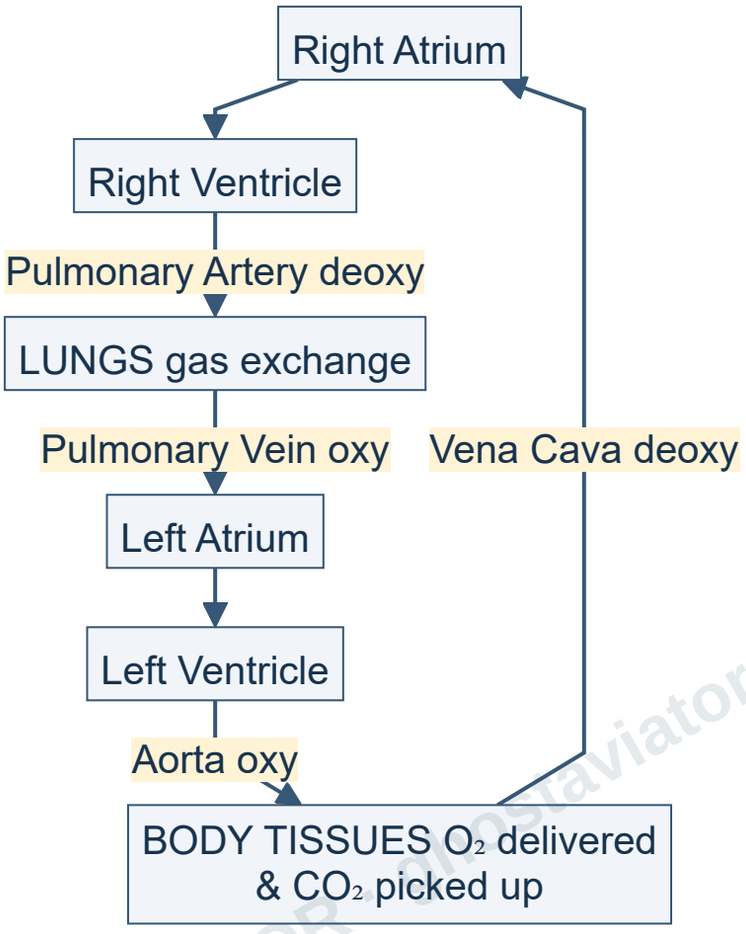
When the heart muscle contracts or beats (called **SYSTOLE**), it **pumps blood out of the heart**. Then the heart muscle relaxes (called **DIASTOLE**) before the next heartbeat. This allows blood to **fill up the heart again**.

The two-stage contraction of systole

1. **Stage 1:** The **right and left atria contract at the same time**, pumping blood to the right and left ventricles.
2. **Stage 2:** The **ventricles contract together** to propel blood out of the heart.

The two-sided traffic plan

Right Side of the Heart	Collects **oxygen-poor blood from the body** and pumps it to the **lungs**, where it picks up oxygen and releases carbon dioxide.
Left Side of the Heart	Collects **oxygen-rich blood from the lungs** and pumps it to the **body**, so that the cells throughout your body have the oxygen they need to function properly.



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BLOOD'S JOURNEY FROM LEFT VENTRICLE TO CAPILLARIES

Blood containing oxygen is pumped around the body from the **left ventricle** ***(a lower pumping chamber of the heart)***. The oxygenated blood passes through the **aorta into the arteries** before arriving at the smallest vessels of the system, the **capillaries**. The **Oxygen–Carbon Dioxide exchange takes place through the walls of the capillaries**.

9.3 Pulse Rate

DEFINITION

The normal rate of the **pulse is the rate of the heartbeat**. A healthy person at rest has a pulse rate of between **60 and 80 beats per minute**. The rate is increased by:

- Exercise
- Emotional inputs
- Disease

STRESS & FEAR — AUTONOMIC ADRENALINE SURGE

When the body experiences **stress or fear**, **adrenaline is released into the bloodstream** causing an **immediate increase in the pulse rate**. (This is the ANS "fight-or-flight" response from §7.4.)

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Resting Pulse (Healthy)	60 – 80 bpm
Cardiac Output Reference	72 bpm × 70 ml
Output Per Minute	≈ 5 L/min
Adrenaline Effect	Immediate ↑ pulse

9.4 Composition & Function of the Blood

TWO MAIN COMPONENTS OF BLOOD

Blood has **two main components — PLASMA and FORMED ELEMENTS**.

- **Plasma:** Nearly everything that blood carries — including **nutrients, hormones and waste** — is dissolved in plasma, which is **mostly water**.
- **Formed elements:** Cells and parts of cells that also **float in plasma**.

Formed elements — the three cell types

Cells and cell-fragments in the blood

Element	What it is / does	Notes
White Blood Cells (WBCs)	Part of the **immune system**	White corpuscles **produce antibodies** to fight bacteria.
Platelets	Help form **clots**	**Smallest of the blood cells**; assist in the blood-clotting process.
Red Blood Cells (RBCs)	Carry **oxygen & carbon dioxide**	Numerous — make up **more than 90 % of the formed elements in the blood**. Virtually everything about them helps them carry oxygen more efficiently.

Inside a Red Blood Cell — Haemoglobin (Hb)

STRUCTURE OF HAEMOGLOBIN — DGCA-FAVOURITE QUESTION

A red blood cell's lack of nucleus also gives it **more room for haemoglobin (Hb)**, a complex molecule that carries oxygen. It is made of:

- A protein component called **GLOBIN**.
- **FOUR pigments called HEMES**.
- The hemes use **IRON** to bond to oxygen.

Inside each RBC are **≈ 280 million haemoglobin molecules**.

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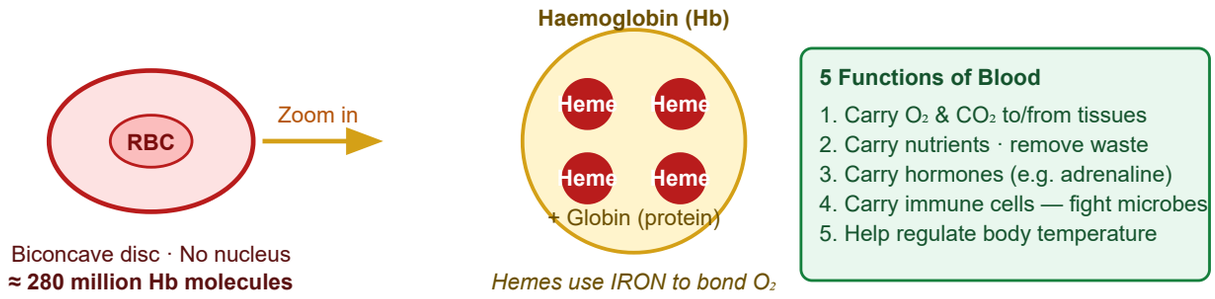

Figure 25.8 *Each RBC packs ~280 million haemoglobin molecules; each Hb has 4 iron-bearing hemes that grip oxygen, plus the globin protein backbone.*

If you lose a lot of blood...

WHY BLOOD LOSS IS DANGEROUS

If you lose a lot of blood, you lose a lot of your **oxygen delivery system**. The immune cells, nutrients and proteins that blood carries are important too, but doctors are generally **most concerned with whether your cells are getting enough oxygen**.

EMERGENCY MANAGEMENT

In an emergency situation, doctors will often give patients **volume expanders**, like **saline**, to make up for lost blood volume. This:

- Helps restore **normal blood pressure**.
- Lets the remaining red blood cells **continue to carry oxygen**.
- Sometimes is enough to keep the body going until it can produce new blood cells and other blood elements.

If not, doctors give **blood transfusions** to replace some of the lost blood. Blood transfusions are also fairly common during some **surgical procedures**.

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The principal functions of the blood — factors discussed in this chapter

The five principal functions of blood

#	Function	Mechanism
1	Carry **oxygen to, and carbon dioxide from**, the various tissues and organs of the body.	Haemoglobin in RBCs
2	Carry **nutrients to tissues** and remove **waste products** from these tissues.	Dissolved in plasma
3	Carry **chemical messengers, such as hormones including ADRENALINE**, to regulate the actions and secretions of various organs.	Plasma transport
4	Transport **cells which can attack and destroy invading micro-organisms**, enabling the body to resist disease.	WBCs / antibodies
5	**Assist in temperature control** of the body.	Vasodilation/constriction

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9.5 Failures or Malfunctions of the Circulatory System

TWO PRINCIPAL FAILURE MODES

The circulatory system can malfunction in **two principal ways**:

1. The main component of the system, the **heart and the blood vessels**, may develop a fault.
2. The **blood may become unable to carry enough Oxygen** for the need of the organs and tissues of the body.

Angina & Heart Attack

DEFINITIONS

A **lack of oxygen supply to the heart** may give rise to symptoms of **Angina**. A **heart attack** is when **low blood flow causes the heart to starve for oxygen**. Heart muscle **dies or becomes permanently damaged**. Your doctor calls this a **myocardial infarction**.

Causes of angina and heart attack

THE BLOOD-CLOT MECHANISM

- Most heart attacks are caused by a **blood clot that blocks one of the coronary arteries**.
- The **coronary arteries bring blood and oxygen to the heart**. If the blood flow is blocked, the heart starves for oxygen and heart cells die.
- A clot most often forms in a coronary artery that has become **narrow because of the build-up of a substance called PLAQUE** along the artery walls.
- Sometimes, **the plaque cracks and triggers a blood clot to form**.
- Occasionally, **sudden overwhelming stress can trigger a heart attack**.

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Risk factors for angina and heart attack — the factors discussed in this chapter

Risk factors — pilots should be screened for these at every medical

Risk Factor	Why it matters
Bad genes (hereditary factors)	Family history is non-modifiable but flagged in flight-medical examinations.
Being male	Higher statistical risk than females (until menopause for women).
Diabetes	Damages blood-vessel linings and accelerates plaque build-up.
Getting older	Arterial elasticity falls; plaque accumulation rises.
High blood pressure	Damages arterial walls — central to plaque mechanism. *(See §14.)*
Smoking	Reduces O_2-carrying capacity 5–8 %, raises BP, accelerates atherosclerosis. *(§12 — next part.)*
Too much fat in your diet	Raises LDL cholesterol — primary feedstock for plaque.

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Risk Factor	Why it matters
Unhealthy cholesterol levels — especially **HIGH LDL ("bad") cholesterol** and **LOW HDL ("good") cholesterol**	Direct correlation with coronary artery narrowing.
Lack of exercise	Increases all of the above; lowers cardiac reserve.
Stress	Sustained adrenaline release damages the heart and vessels.
Obesity	Independent risk factor + driver of diabetes, BP, lipids.
Alcohol	Raises BP, damages heart muscle, raises triglycerides. *(§13 — next part.)*

WHY PILOTS CARE — DGCA MEDICAL IMPLICATIONS

Every one of these risk factors is screened at Class-I & Class-II DGCA medicals via ECG, lipid panel, fasting glucose, BMI, blood pressure measurement and history. *Even an unsuspected silent angina is a fitness-defeating condition; routine cardiac screening exists precisely because the in-flight failure mode is sudden incapacitation.*

Insufficiency of Oxygen — the second failure mode

WHY THIS IS ITS OWN CATEGORY

The second principal way the circulatory system can malfunction is **"Insufficiency of Oxygen"** — the blood becoming **unable to carry enough oxygen for the needs of the organs and tissues**. This is the gateway concept that leads directly into the next major topic — **HYPOXIA** — covered in detail in §10. Hold the term in mind.

Numbers you must know cold

Master cheat-sheet — numbers

Parameter	Exact Value	Where
Normal adult respiration rate	**14 – 18 breaths/min**	§8.4
Primary regulator of breathing rate	**Blood CO_2 level**	§8.4
Resting pulse (healthy)	**60 – 80 bpm**	§9.3
Reference heart rate (for CO calc)	**72 bpm**	§9.1
Stroke volume (reference)	**70 ml**	§9.1
Cardiac output at rest	**≈ 5 L/min**	§9.1
RBCs as % of formed elements	**> 90 %**	§9.4
Haemoglobin molecules per RBC	**≈ 280 million**	§9.4
Hemes per Hb molecule	**4**	§9.4
Metal that bonds O_2 to heme	**Iron**	§9.4
Protein component of Hb	**Globin**	§9.4
Heart attack medical term	**Myocardial Infarction**	§9.5

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Parameter	Exact Value	Where
"Bad" cholesterol	**LDL (high = bad)**	§9.5
"Good" cholesterol	**HDL (high = good)**	§9.5

WHY THIS SECTION IS THE SINGLE MOST IMPORTANT BLOCK IN THIS CHAPTER

The DGCA HPL paper draws the **largest single share of questions** from the three topics covered in this section. More importantly — **these three conditions kill pilots every year**. Each one is *insidious*: by the time the symptoms register, judgement is already impaired. You must be able to **identify · diagnose · respond** with reflex speed.

Read every line. Memorise every list. Practise the immediate-action steps until they're checklist-instinctive.

10 Hypoxia

TECHNICAL DEFINITION — "CEREBRAL HYPOXIA"

The term **cerebral hypoxia** technically refers to **lack of oxygen supply to the cerebral hemispheres** (the outer portion of the brain). However, it is more typically used to refer to a **lack of oxygen supply to the entire brain**.

10.1 Why Hypoxia Matters to a Pilot

TWO WAYS THE BODY CAN BE STARVED OF OXYGEN

1. The body's organs and tissues may be **deprived of the oxygen they need because of illness or disease**.
2. **But the pilot must know that oxygen deprivation or hypoxia can be caused by breathing air at low pressures at high altitude.**

This second category — **altitude-induced hypoxia** — is the aviator's main concern, and the form most commonly tested in DGCA exams.

EFFECT ON THE PILOT — DIRECT QUOTE, MEMORISE

Hypoxia will cause a pilot's intellectual and sensory judgment to become impaired. In **mild cases**, hypoxia causes only:

- **Inattentiveness**,
- **Poor judgment**, and
- **Uncoordinated movement**.

10.2 Stages & Clinical Course — from "tipsy" to lethal

SEVERE CASES — THE CASCADE TO DEATH

Severe cases result in a state of complete unawareness and unresponsiveness (coma) — brain-stem reflexes, including **response to light and the breathing reflex**, stop. Only **blood pressure and heart function are maintained**. If this persists, **brain death is inevitable**.

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If the lack of oxygen to the brain is limited to a **very brief period of time**, coma **may be reversible** with varying levels of return to function, depending on the extent of injury. Sometimes **seizures** may occur, which may be **continuous with no stop between them** — a condition called **STATUS EPILEPTICUS**.

Serious oxygen deprivation can kill a pilot within minutes.

ONSET CHARACTERISTIC — NOTE THE WORD "INSIDIOUS"

The symptoms are **slow but progressive, insidious in onset**, and are most marked at altitudes **starting above 10,000 ft (3,500 m)**. **Night vision, however, can be impaired starting at altitudes 5,000 ft**.

"Insidious" means it *sneaks up on you*. You will not realise you are hypoxic — that is the entire problem. Your buddy in the right seat may notice it before you do.

WHY "EUPHORIA" IS THE MOST DANGEROUS EARLY SYMPTOM

The onset of hypoxia **may be accompanied by a feeling of well-being, known as EUPHORIA**. The pilot feels good — even *great* — and is therefore the **last person in the cockpit** who will suspect anything is wrong. This is why CRM and crew cross-checks are critical above 10,000 ft.

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10.3 The Four Types of Hypoxia

CLASSIFICATION — OPENING LINE OF EVERY DGCA HYPOXIA QUESTION

HYPOXIA IS CLASSIFIED INTO FOUR DIFFERENT TYPES:

TYPE (a)	**Hypoxic Hypoxia** **Cause:** low oxygen *levels* in the bloodstream. **The pilot's type:** in pilots, this most often occurs with **exposure to altitude** — **hypobaric hypoxia**. At low altitudes the partial pressure of oxygen is adequate to maintain brain function at peak efficiency. **Atmospheric pressure and the partial pressure of oxygen both decline at higher altitudes.**
TYPE (b)	**Anemic Hypoxia** **Cause:** blood cannot carry enough oxygen even though atmospheric oxygen may be plentiful. Oxygen in blood is carried by **haemoglobin**, which is found in red blood cells. When the **RBC count decreases**, or the **haemoglobin does not function properly**, less oxygen can be carried by the blood. **Occurs in:** heavy bleeding · some cancers · sickle-cell anemia · **carbon monoxide poisoning**, to name a few. Symptoms: **breathlessness, fatigue, chest pain** — and they **worsen at higher altitudes**, as the effects of hypoxia and anemia are **additive**.

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TYPE (c)	**Ischemic / Stagnant Hypoxia** **"Ischemia"** = inadequate supply of blood. **Ischemic hypoxia** occurs when there is **inadequate blood flow to body tissues**. **Occurs in:** • **Constriction of blood vessels** (e.g. fingers and toes exposed to cold). • **Low blood pressure / low cardiac output** such as fainting. • **Exposure to high sustained accelerations** like excessive G-forces (stagnant hypoxia). **Oxygen therapy is not very helpful** in this form. The **best remedy is to correct the underlying cause**.
TYPE (d)	**Histotoxic Hypoxia** **Cause:** the **inability of the cells of the body to use the oxygen available**. The oxygen is there in the blood, in the tissues — but the cells can't metabolise it. **Rare in pilots,** but can occur with: • **Cyanide poisoning**, • **Chemical poisoning**, • **Intoxication with certain drugs**. **Can also be caused by HIGH BLOOD ALCOHOL LEVELS.** (Direct link to §14.)

10.4 Causes · Symptoms · Immediate Actions — Master Table

The DGCA syllabus consolidates Hypoxia management into **one master tabulation** of all causes, all symptoms, and the immediate actions if hypoxia is suspected. It is set out below.

Hypoxia — Causes, Symptoms & Immediate Actions

Hypoxia Causes	Symptoms	Immediate Actions if Hypoxia is suspected

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Hypoxia Causes	Symptoms	Immediate Actions if Hypoxia is suspected
• Drug overdose / alcohol • Asphyxiation caused by smoke inhalation • Very low blood pressure • Strangulation • Cardiac arrest (when the heart stops pumping) • Carbon monoxide poisoning • High altitudes • Choking • Compression of the trachea • Diseases that paralyze the respiratory muscles • **A tobacco smoker is likely to experience the effects of hypoxia at a lower altitude than a non-smoker**	• The symptoms are **slow but progressive, insidious in onset**, and are most marked at altitudes **starting above 10,000 ft (3,500 m)**. **Night vision can be impaired starting at altitudes 5,000 ft**. • Its onset may be accompanied by a feeling of well-being, known as **EUPHORIA**. • Even minor hypoxia **impairs night vision and slows reaction time**. • More serious hypoxia **interferes with reasoning**, gives rise to **unusual fatigue** and, finally, results in **loss of consciousness / death**. • **Impaired Judgment** • **Headache** • **Tingling in hands & feet** • **Hyperventilation** (yes, hypoxia can *trigger* hyperventilation) • **Muscular Impairment** • **Memory Impairment** • **Sensory Loss** • **Tunnel Vision** • **Cyanosis** (a bluing of the body extremities) • **Formication** (a feeling of ants under the skin)	• **Provide Oxygen**. • **Descend below 10,000 feet** or **Minimum Safe Altitude if it is higher than 10,000 ft**.

MEMORISE — THE IMMEDIATE-ACTION DRILL (A 2-STEP REFLEX)

1. **PROVIDE OXYGEN** — don the mask, 100 % flow, regulator to "EMERGENCY" if available.
2. **DESCEND BELOW 10,000 ft** (or below MSA — Minimum Safe Altitude — if MSA is higher than 10,000 ft).

This is the standard DGCA-examined answer. Do not over-elaborate in the exam — write these two steps in this order.

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10.5 Altitude Thresholds & Night Vision

Night vision impairment from	5,000 ft
Day symptoms marked from	10,000 ft (3,500 m)
Action threshold — descend below	10,000 ft
Smoker effective altitude (recap)	≈ 7,000 ft

DGCA REASONING THE EXAMINER IS TESTING

The reason the action threshold is **10,000 ft** and not, say, 14,000 ft is because **symptoms are most marked at altitudes starting above 10,000 ft (3,500 m)**. The reason supplemental O_2 is required **at night above 5,000 ft** in many jurisdictions is because **night vision is impaired starting at 5,000 ft** — the rod cells in the retina are extraordinarily sensitive to even mild hypoxia. *(Vision pathophysiology is dealt with in a later part of this chapter.)*

11 Carbon Monoxide (CO) Poisoning

11.1 The Colourless Killer

WHAT CO IS

Carbon monoxide is a **colourless, odourless, tasteless gas** that is a **product of incomplete combustion**.

WHY CO IS SO DANGEROUS — THE 200× RULE

Haemoglobin, the oxygen-carrying chemical in the blood, picks up carbon monoxide **over 200 times more readily** than it picks up oxygen.

Thus, even **minute quantities in the cockpit** (often from improperly vented exhaust fumes) **may result in pilot incapacitation**. Exhaust gases from piston engines can consist of **as much as 9 % carbon monoxide**. So, gases from **leaking exhausts** can cause carbon monoxide poisoning in pilots.

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Carbon monoxide effect	Why it matters
Binding to haemoglobin	CO binds much more strongly than oxygen and reduces the haemoglobin available to carry oxygen. Affinity is not the same quantity as a reaction speed.
Oxygen delivery to tissues	A hazardous exposure can impair oxygen delivery even when ambient air contains oxygen.
Suspected contamination	Use the aircraft's approved emergency procedure and seek medical help.

Figure 25.9 *Carbon monoxide impairs oxygen transport; the amount and duration of exposure matter.* *Human-performance reference, carbon-monoxide section; approved aircraft procedures govern the response.*

SOURCES OF COCKPIT CO — THE USUAL SUSPECTS

- **Improperly vented exhaust fumes** entering the cabin via the heating system.
- **Leaking exhaust manifolds / muffler shrouds** (especially common in piston singles where cabin heat is drawn over the exhaust).
- **Engine combustion gases**, which can contain up to 9 % CO.
- **Tobacco smoke** in confined cabins.

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11.2 Symptoms · Actions · Prevention

Carbon Monoxide Poisoning – Symptoms	• **Initially, there is an INABILITY TO CONCENTRATE** • Headache • Dizziness • Nausea • Impaired vision • Lethargy or weakness • Impaired judgment • Personality change • Impaired memory • **Flushed cheeks and cherry-red lips** *(classic CO sign)* • Convulsions
Actions if CO Poisoning is Suspected	1. **Turn off cabin heating** 2. **Open cabin ventilators** 3. **Consider using oxygen if available** 4. **Land as soon as possible** 5. **Take medical aid** 6. **Do not fly till cleared by doctor**

PREVENTION RULE

AT ALL TIMES WHEN THE CABIN HEATING IS USED, FRESH AIR MUST BE CIRCULATED TO REDUCE PRESENCE OF CO.

WHY THE "CHERRY-RED" COLOUR?

Carboxyhaemoglobin (Hb-CO) is a *bright cherry red* — much brighter than oxygenated Hb. The pilot's cheeks and lips therefore flush bright pink/red. **Don't confuse this with the bluish CYANOSIS seen in hypoxic hypoxia.** They look opposite:

- **Hypoxic Hypoxia** → blue extremities (cyanosis)
- **CO Poisoning** → bright cherry-red lips/cheeks

CO affinity for Hb (vs O_2)	> 200 ×
CO in piston-engine exhaust	up to 9 %
First symptom	Inability to concentrate
Classic skin sign	Cherry-red lips
First action	Turn off cabin heat
Hypoxia type involved	Anemic (Type b)

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12 Hyperventilation

DEFINITION — "OVER-BREATHING"

"Hyperventilation" is another word for "over-breathing" and may be defined as **lung ventilation in excess of the body's needs**. **Good training is the best way to avoid Hyperventilation in pilots.**

MITIGATION FOR PASSENGERS

The chances of Hyperventilation affecting your passengers can be reduced by **giving them a thorough pre-flight briefing on every aspect of the flying sortie**. (Briefed passengers = calm passengers; calm passengers don't hyperventilate.)

12.1 Causes at Low Altitude

At low altitude — where hypoxia is *not* a factor — the most common causes of hyperventilation are psychological or environmental rather than physiological.

The most common causes of Hyperventilation at low altitude

Category	Cause
Cognitive load	**Intense concentration on a difficult task**
Emotional	**Fear**
Emotional	**Anxiety**
Physical	**Motion sickness**
Physical	**Shock**
Environmental	**Vibration**
Environmental	**Heat**
Aerodynamic / Manoeuvring	**High G-forces**

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12.2 Symptoms · Treatment · the Paper-Bag Trick

Symptoms of Hyperventilation	• Dizziness • Tingling • Visual disturbances • Hot or cold sensation • Anxiety • Loss of muscular co-ordination • Increased heart rate • Spasms • Loss of consciousness • **Cramping and spasms of the hands and feet** • Cold clammy skin • Paleness
Treatment of Hyperventilation	Follow the approved in-flight procedure, address possible hypoxia, and obtain medical help. Rapid breathing alone does not establish the cause.

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RAPID BREATHING: RECOGNISE THE UNCERTAINTY

Hyperventilation can lower blood carbon dioxide and cause symptoms such as dizziness or tingling. Rapid breathing can also accompany serious illness or inadequate oxygen delivery; symptoms alone do not reliably identify the cause.

Do not use paper-bag rebreathing. Follow the approved in-flight procedure, consider possible hypoxia, and obtain medical help. ANZCOR Guideline9.2.8, recommendation5, advises against rebreathing because it can cause dangerously low oxygen levels.

12.3 Hypoxia vs Hyperventilation — Telling Them Apart

SIMILAR SYMPTOMS: DO NOT DIAGNOSE BY ALTITUDE ALONE

Hypoxia (too little oxygen reaching the tissues) and hyperventilation (breathing more than the body needs) can cause similar early symptoms. If the cause is uncertain, treat possible hypoxia first: use the oxygen system as prescribed *(laid down by the authority or the rules)* and check the supply and equipment. If symptoms continue, consciously control the rate and depth of breathing. Follow the aircraft's approved procedures and obtain help as needed.

Hypoxia vs Hyperventilation — side-by-side

Parameter	Hypoxia	Hyperventilation
Typical altitude	**> 10,000 ft**	**Any altitude** (often below 10,000 ft)
Onset	**Insidious, slow**	Rapid — linked to a trigger (stress, fear)
Skin sign	**Cyanosis** (bluish)	**Paleness · cold clammy skin**
Heart rate	Often increased	**Increased**
Feeling of well-being?	**Yes — euphoria**	No — anxiety, fear
Blood CO_2 level	Variable / often normal	**LOW**
First-line treatment	Use approved oxygen and emergency procedures. Any descent must respect terrain and obstacle clearance.	Follow the approved in-flight procedure, address possible hypoxia, and obtain medical help. Rapid breathing alone does not establish the cause.
How to confirm in flight	Recovery may be rapid with oxygen; do not use a fixed number of breaths as a diagnostic test.	If symptoms persist after addressing possible hypoxia, control the breathing rate and depth.

PLAIN MEANINGS

Hypoxia means too little oxygen reaching the tissues. Hyperventilation means breathing more than the body needs. Similar early symptoms make it unsafe to diagnose either condition from altitude alone.

Master cheat-sheet — numbers

Every regulatory / clinical number from this part

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Parameter	Exact Value	Where
Day-time hypoxia symptoms marked above	**10,000 ft (3,500 m)**	§10.5
Night vision impairment begins at	**5,000 ft**	§10.5
Smoker effectively at altitude of (vs non-smoker 10,000 ft)	**7,000 ft**	§10.4
Action on hypoxia — descend below	**10,000 ft** (or MSA if higher)	§10.4
Number of hypoxia types	**FOUR**	§10.3
CO affinity for Hb vs O_2	**> 200 ×**	§11.1
Max CO content of piston exhaust gases	**9 %**	§11.1
Hyperventilation treatment — slow to	**10 – 12 breaths/min**	§12.2
Hypoxia confirmed if O_2 relieves symptoms in	**3 – 4 breaths**	§12.2
Altitude alone does not establish the cause.	**< 10,000 ft**	§12.2

13 Smoking

13.1 The 3,000-ft Penalty — 10,000 ft becomes 7,000 ft

THE SINGLE MOST-ASKED DGCA FACT ABOUT SMOKING

A pilot who is also a smoker **may experience the symptoms of oxygen deprivation, or hypoxia, at a lower altitude (7,000 ft) than a non-smoker (10,000 ft)**.

That is a **3,000-ft penalty**. The smoker walks into the cockpit already partway up the hypoxia ladder. Combine that with an unpressurised climb to FL080 and the smoker can be hypoxic *at cruise altitude* when the non-smoker beside him is still asleep.

7,000ft **SMOKER'S EFFECTIVE HYPOXIA THRESHOLD**

5–8% **O_2-CARRYING CAPACITY REDUCTION — 1 PACK/DAY**

↑ **INCREASED SUSCEPTIBILITY TO CO POISONING**

13.2 The 5–8 % O_2-Capacity Hit

WHY SMOKING HANDICAPS YOU — DIRECT QUOTE

A person who smokes **one packet of cigarettes per day** will **reduce his capacity to carry oxygen by 5–8 %**. A smoker also has **increased susceptibility to Carbon Monoxide poisoning**.

THE MECHANISM — ANEMIC HYPOXIA, SELF-INFLICTED

Cigarette smoke contains **carbon monoxide**. Recall from §11.1 that CO binds to haemoglobin **over 200 times more readily** than oxygen. Every cigarette saturates a portion of your haemoglobin with CO — and that Hb is then **unavailable to carry oxygen** until the body slowly clears it (hours). A heavy smoker effectively walks around with **permanent low-grade anemic hypoxia (Type b)**. This is why the smoker's hypoxia threshold drops 3,000 ft.

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13.3 The Full Medical-Impact List

HEALTH EFFECTS ON A SMOKING PILOT

The smoker pilot is exposed to **all six** of the following — every one with direct flight-safety implications:

- **Lung cancer**
- **Breathing problems**
- **Circulatory problems**
- **Reduced tolerance to G forces**
- **Increased risk of heart attack**
- **Degradation of night vision**

Smoking — what it breaks and why it matters in the cockpit

Health Effect	Flight-Safety Implication
Lung cancer	Eventual respiratory failure — medical disqualification.
Breathing problems (COPD, chronic bronchitis)	Reduced ventilation → reduced O_2 uptake → lowered hypoxia threshold (the 7,000-ft figure above).
Circulatory problems	Plaque, atherosclerosis, peripheral vascular disease → narrower arteries → ischemic hypoxia risk (§10.3c).
Reduced tolerance to G forces	Critical in aerobatics, fighter pilots, upset recovery. Less G-tolerance → earlier G-LOC.
Increased risk of heart attack	Direct link to angina/MI risk-factor list in §9.5 (smoking was one of the 12 listed).
Degradation of night vision	Recall night vision is impaired at 5,000 ft for non-smokers (§10.5). Smokers see degradation at *even lower altitudes* — devastating for night IFR and night VFR ops.

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PRACTICAL CRM NOTE

DGCA Class-1 / Class-2 medicals do *not* automatically disqualify a smoker, but the lifestyle is flagged. **If you smoke and you are reading this, you are accepting a permanent flight-performance handicap.** Even one cigarette before takeoff materially lowers your hypoxia threshold for that sortie. Best practice for examined fitness: **do not smoke at all, especially in the 24 hours before flight**.

14 Alcohol

THE HEADLINE — ALCOHOL IS A DEPRESSANT

Alcohol acts primarily as a depressant. Do not fly while under the influence of alcohol. **Even small amounts of alcohol in the system can adversely affect judgment and decision-making abilities.**

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14.1 The 24-Hour Bottle-to-Throttle Rule

LAW AND ADVICE ARE DIFFERENT NUMBERS

The law in India is **12 hours**: Rule 24 of the Aircraft Rules, 1937 forbids crew from taking alcohol within twelve hours before a flight (see the National Law chapter). The **24 hours** taught in this section is the human-performance recommendation. Answer a law question with 12 hours and a human-performance question with 24 hours.

14.2 Alcohol Metabolism — The Fixed Rate

METABOLISM RATE

Remember that your body metabolises alcohol at a **fixed rate**, and **no amount of coffee or medication will alter this rate**. Alcohol leaves the body at an average rate of **0.015 g/100 mL/hour**, which is the same as reducing your **Blood Alcohol Content (BAC) level by 0.015 per hour**.

For men, this is usually a rate of about ONE STANDARD DRINK PER HOUR.

24hr **BOTTLE-TO-THROTTLE SEPARATION**

0.015 **G/100 ML/HR METABOLISM RATE (≈ 1 DRINK/HR IN MEN)**

40mg% **BAC AT WHICH PILOT ERRORS SPIKE SIGNIFICANTLY**

14.3 Altitude Amplifies the Effect — "1 = 2" Rule

THE "ONE DRINK AT ALTITUDE EQUALS TWO AT SEA LEVEL" RULE

Relatively small amounts of alcohol **significantly decrease a pilot's tolerance to hypoxia (oxygen lack)**. **At 6,000 ft (1,800 m), the effect of ONE drink is that of TWO drinks at sea level.** Even at sea level alcohol impairs judgment and reaction time. **Therefore, alcohol and flying do not mix.**

WHY? — HISTOTOXIC + HYPOXIC STACKING

Recall **Histotoxic Hypoxia (§10.3d)**: high blood alcohol levels prevent cells from *using* the oxygen they receive. Layer that on top of **Hypoxic Hypoxia (§10.3a)** from altitude, and you have two oxygen-starvation mechanisms operating *at the same time*. They are **additive** — exactly the same principle that makes a smoker's hypoxia worse at altitude.

14.4 Recommended Maximum Alcohol Intake — Units & Rules

DEFINITION OF "ONE UNIT"

ONE UNIT of alcohol = half a pint of beer = a standard glass of wine = one measure of spirits.

MEN	5 units / day · 21 units / week Maximum recommended consumption.
WOMEN	3 units / day · 14 units / week Maximum recommended consumption.

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1 unit equals	½ pint beer
1 unit equals	1 glass wine
1 unit equals	1 measure spirits
Men – daily/weekly	5 / 21
Women – daily/weekly	3 / 14

14.5 Bac 40 mg/100 ml — The Pilot Error-Spike Threshold

THE BAC LEVEL AT WHICH ERRORS BALLOON

Blood alcohol concentrations of 40 mgs per 100 ml results in significant increases in errors committed by pilots due to the following effects:

- **Impaired judgment and impaired ability to reason**
- **Degraded muscular coordination** and **degraded vision**
- **Lack of inhibitions and self-control**
- **Increased susceptibility to Hypoxia**
- **Damages to the liver, heart, brain and blood cells**
- **Affects short and long-term memory**
- **Slows reaction time**
- **Sufferer may feel that performance is improved** *(the alcohol-euphoria trap)*
- **Balance and sensory illusions**
- **Irregular sleep patterns**

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THE EUPHORIA TRAP — ANALOGOUS TO HYPOXIA

"**Sufferer may feel that performance is improved**" is the alcohol equivalent of *hypoxia's euphoria* (§10.2). The pilot is the *least* qualified person in the cockpit to judge whether he's safe to fly. This is precisely why **fixed bottle-to-throttle rules exist** — they take the judgement out of the impaired pilot's hands.

14.6 Hangover & the "Masked Hangover" Warning

HANGOVER RULE

Do not fly with a hangover, or a "masked hangover" (symptoms suppressed by aspirin, caffeine or other medication). High altitude, where oxygen is deficient, aggravates these effects.

WHY A "MASKED" HANGOVER IS WORSE THAN AN OBVIOUS ONE

A normal hangover at least gives you a warning sign — you feel awful, you don't fly. A **masked hangover**, where caffeine, paracetamol, ibuprofen or other OTC medication has suppressed the headache and nausea, leaves you **still impaired** in judgement, coordination, vision and reaction time — but **without the warning**. Several aviation accidents have been traced to a pilot who took an aspirin, "felt fine," and flew with substantial residual BAC.

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15 Blood Pressure

15.1 What Determines Blood Pressure

THE FOUR DETERMINANTS

Blood pressure depends on:

- The **cardiac output**,
- The **resistance of the capillaries** (peripheral resistance),
- The **elasticity of the arterial walls**, and
- The **blood volume and viscosity**.

15.2 Definitions — pressure on the arteries

THREE KEY FACTS

- **Blood pressure is the pressure exerted by blood on the walls of the main arteries**.
- The blood-pressure which is measured during flight medical checks is the **pressure in the artery of the upper arm** (representing the pressure at heart level).
- The **permanent pressure against the arterial wall is called DIASTOLIC pressure**.
- The **increased pressure occurring with each beat of the heart is called the SYSTOLIC pressure**.

SYSTOLIC PRESSURE	The **increased pressure occurring with each beat of the heart**. This is the higher number in the BP reading — generated during ventricular contraction (*systole*, §9.2). It is the pressure peak as blood is ejected from the left ventricle into the aorta.
DIASTOLIC PRESSURE	The **permanent pressure against the arterial wall**. This is the lower number — the resting pressure between heartbeats, when the heart muscle relaxes (*diastole*) and refills with blood. It represents the baseline strain on the arteries.

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15.3 120/80 — the Normal Benchmark

THE NUMBER EVERY PILOT MUST KNOW

120/80 is a normal blood pressure for a healthy young adult.

The notation reads "120 over 80" — meaning **systolic 120 mmHg, diastolic 80 mmHg**. This is the figure that DGCA aero-medical examiners measure against at every Class-1 and Class-2 medical.

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Figure 25.10 *Equipment used for manual upper-arm blood-pressure measurement. This view identifies the cuff, chestpiece and gauge; it does not prescribe positioning or diagnostic thresholds.*

15.4 High Blood Pressure (Hypertension) — Causes & Pilot Implications

HIGH BLOOD PRESSURE AND MEDICAL FITNESS

High blood pressure needs assessment and appropriate management. Fitness is decided through the applicable DGCA medical assessment process; a diagnosis alone is not a complete statement of the licensing decision.

DGCA CAR 7-C-I, §§2.2 and 3.1.2–3.1.4.

Causes of Blood Pressure (Hypertension) — factors discussed in this chapter

Seven causes of hypertension a pilot must remember

#	Cause	Pilot-Specific Note
1	**Stress**	Sustained ANS / adrenaline activation. Pilots are exposed to chronic occupational stress — duty rosters, weather, currency requirements.
2	**Smoking**	Direct link to §13. Nicotine constricts vessels; CO damages endothelium.
3	**Poor diet** (excess fat or salt)	Salt → fluid retention → ↑ blood volume. Fat → plaque → ↑ peripheral resistance.
4	**Obesity**	Greater body mass requires more cardiac output; metabolic syndrome adds insulin resistance.

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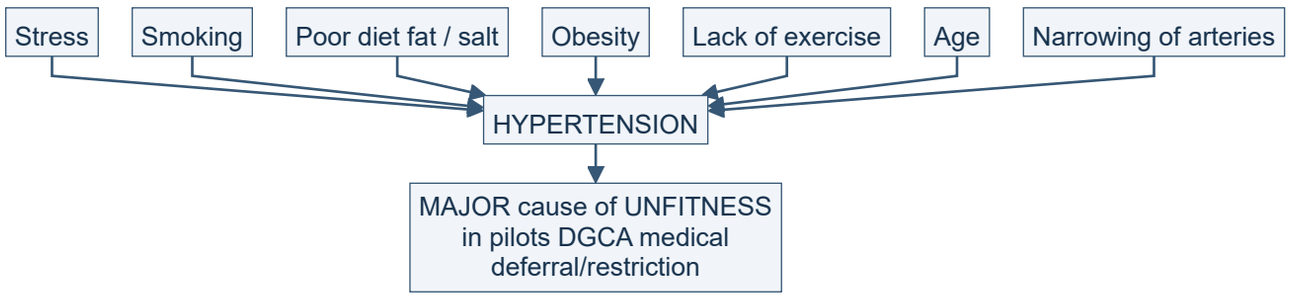
#	Cause	Pilot-Specific Note
5	**Lack of exercise**	Deconditioned heart, stiffer vessels, higher resting BP.
6	**Age**	Arterial elasticity falls with age (recall §9.5 risk-factor list). Non-modifiable.
7	**Narrowing of the arteries**	Atherosclerosis — plaque-narrowed vessels create higher peripheral resistance. Same plaque that causes angina/MI.



WHAT A PILOT CAN DO — THE MODIFIABLES

Of the seven causes, **five are modifiable** (stress management, no smoking, low-salt/low-fat diet, weight control, regular exercise). Only **age** and (to a large extent) **narrowing of arteries from existing plaque** are non-modifiable. **The pilot who fails his Class-1 medical because of hypertension has lost his career to a condition that was, in most cases, preventable.**

CROSS-LINKS — SEE ALSO

- **§9.1 / 9.2:** cardiac output = HR × stroke volume — directly drives systolic pressure.
- **§9.5:** hypertension is one of the 12 risk factors for angina / heart attack.
- **§13.3:** smoking causes circulatory problems and accelerates plaque.
- **§14.5:** alcohol contributes to liver/heart/brain damage.

The DGCA HPL paper frequently *combines* these topics in one MCQ stem — e.g. "A 48-year-old smoker with BP 145/95 …" — testing whether you can connect the dots.

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Master cheat-sheet — numbers

Every numeric / regulatory fact from this section

Parameter	Exact Value	Where
Smoker's effective hypoxia altitude (vs non-smoker 10,000 ft)	**7,000 ft**	§13.1
O_2-capacity reduction — 1 packet of cigarettes/day	**5 – 8 %**	§13.2
Number of medical effects of smoking listed by DGCA	**6**	§13.3
Bottle-to-throttle rule	**24 hours**	§14.1
Alcohol metabolism rate	**0.015 g/100 ml/hr**	§14.2
Approximate male metabolism rate	**≈ 1 standard drink / hr**	§14.2
"1 = 2" altitude	**6,000 ft (1,800 m)**	§14.3

Parameter	Exact Value	Where
1 unit of alcohol equals	**½ pint beer / 1 glass wine / 1 spirit measure**	§14.4
Men — daily / weekly maximum	**5 / 21 units**	§14.4
Women — daily / weekly maximum	**3 / 14 units**	§14.4
BAC at which pilot errors spike	**40 mg/100 ml**	§14.5
Normal BP for healthy young adult	**120 / 80 mmHg**	§15.3
Site of BP measurement at medical	**Upper-arm artery**	§15.2
Number of listed causes of hypertension	**7**	§15.4

16 Baroreceptor Reflex

WHAT IT IS

The baroreceptor reflex is the body's **automatic blood-pressure regulator**. Any change in your body's **demand for blood** can **trigger your baroreceptor reflex**. Specialised pressure-sensors ("baroreceptors") in the carotid arteries and aortic arch detect changes in BP and signal the brain to adjust heart rate and vessel diameter accordingly.

17 Donating Blood — the 48-hr rule

EXAM POINT

In a **completely healthy individual**, the fluid reduction caused by donating **one unit of blood** is replaced within **several hours**. In some people, however, the loss of blood causes **disturbances to the circulation that may last for several days**.

While the effects at **ground level are minimal**, **flying during this period may entail a risk**.

THE DGCA RULE — EXACT PHRASING

Generally, active pilots should NOT donate blood, but if blood has been donated, they should wait at LEAST 48 hours before flying.

Recommended for active pilots	DO NOT donate
If donation has occurred	Wait ≥ 48 hr
Reason	Circulatory disturbance
Ground-level effect	Minimal

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WHY BLOOD DONATION INTERACTS DANGEROUSLY WITH FLIGHT

A single unit of blood (~450 ml) represents about **9 % of total blood volume** in an average adult. While the plasma volume is restored in hours, the **red blood cell mass** takes weeks to fully recover. Reduced RBC mass = reduced haemoglobin = reduced oxygen-carrying capacity = **anemic hypoxia (§10.3b)**. Add altitude (hypoxic hypoxia, §10.3a) and the two stack — exactly the same principle as smoker-at-altitude and alcohol-at-altitude earlier in this chapter.

18 Cabin Pressurization & Time of Useful Consciousness (TUC)

18.1 What TUC / EPT is

DEFINITION — MEMORISE VERBATIM

Time of useful consciousness (TUC), also **effective performance time (EPT)**, is defined as the **amount of time an individual is able to function effectively (e.g., perform flying duties) in an environment of inadequate oxygen supply**.

It is the **period of time from the interruption of the oxygen supply** or exposure to an oxygen-poor environment **to the time when useful function is lost**, and the individual is **no longer capable of taking proper corrective and protective action**.

CRITICAL CONCEPT — TUC IS NOT TIME TO UNCONSCIOUSNESS

TUC is the time during which a pilot can still **do useful work** — don a mask, push a checklist, initiate descent. After TUC ends, the pilot may still be conscious for several more seconds, but is no longer **functional**. *This distinction is precisely why the DGCA emphasises "corrective and protective action" — the mask must be on the face before TUC expires.*

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18.2 Why TUC Drops Sharply at Altitude

EXAM POINT

At the higher altitudes, the **TUC becomes very short; considering this danger, the emphasis is on PREVENTION rather than CURE**.

Cabin pressurization eliminates many problems associated with high altitude flying, but it introduces other potential problems, the most important being the **risk of rapid decompression**.

WHAT DETERMINES TUC AFTER A RAPID DECOMPRESSION

The time of useful consciousness (TUC) following a rapid decompression depends on:

1. **Aircraft altitude**,
2. **The rate at which pressure falls**, and
3. **The level of physical activity of the individual at the time of the event**.

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INTERPRETING USEFUL-CONSCIOUSNESS TIMES

These times are averages under particular conditions. Altitude, activity and the pattern of decompression affect the result. A single fixed percentage must not replace the condition-specific table below.

18.3 Time of useful consciousness: conditions matter

Time of useful consciousness varies with altitude, activity and the type of decompression. The source table below keeps those conditions separate; its averages are not guaranteed response times.

ALTITUDE (ft)	Progressive decompression		Rapid Decompression
	Sitting	Moderate Activity	
18,000	About 40 min.	About 30 min.	**20 to 25 min.**
20,000	10 min	5 min.	**3 min.**
25,000	5 min.	3 min.	**2 min.**
30,000	1.5 min.	45 sec.	**30 sec.**
35,000	45 sec.	30 sec.	**20 sec.**
40,000	25 sec.	18 sec.	**12 sec.**
43,000	18 sec.	12 sec.	**12 sec.**

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Figure 25.11 *Average time of useful consciousness depends on altitude, workload and the pattern of decompression. These illustrative values are not guaranteed response times; effective performance may be lost earlier.* *Human-performance reference, printed p 45, condition-specificTUC table.*

18.4 Rapid decompression: read the conditions

USEFUL TIME VARIES

Rapid decompression may shorten useful-consciousness time. The source table gives different values for sitting, activity and rapid decompression; it does not establish one universal 50% rule.

Use the aircraft's approved oxygen and emergency procedures. Do not wait for symptoms.

Source: human-performance reference, printed p 45.

OPERATIONAL PILOT RESPONSE — THE DRILL

After a rapid decompression in a transport jet you have **seconds, not minutes**:

1. **Don oxygen masks** immediately — quick-don, regulator to EMERGENCY/100 %.
2. **Establish crew communication**.
3. **Initiate emergency descent** — target **10,000 ft** or MSA, whichever higher.
4. **Squawk 7700**, declare MAYDAY.

This drill is muscle-memory in transport flight training precisely because there is no time to think it through at FL350+.

19 Decompression — Theory

DEFINITION

"Decompression" means the lowering of pressure.

19.1 The 8,000 ft "Maximum Comfort" Rule

EXAM POINT

The maximum altitude without oxygen at which flying efficiency is not impaired is 8,000 feet.

FLIGHT RISK BELOW 10,000 FT

When flying at altitude below 10,000 feet, the risk of suffering from conditions related to decompression is low.

19.2 Atmospheric Pressure is Halved at 18,000 ft

NITROGEN SATURATION & PRESSURE FACTS

- At ground level, the body tissues are **saturated with nitrogen**, the inert gas which makes up **80 % of our atmosphere**.
- As the aircraft climbs, atmospheric pressure is reduced.
- **By 18,000 ft ASL (5,486 m) atmospheric pressure is HALVED.**
- **Decompression sickness symptoms may develop at 18,000 ft and above.**
- Pilots flying aircraft with **unpressurised cabins at altitudes greater than 25,000 feet ASL (7,620 m)** may be subject to "the bends".

19.3 The Bends — Physiology of Nitrogen Bubbles

THE BOTTLE-OF-SODA MECHANISM — DGCA'S OWN ANALOGY

"The bends" condition is caused by **bubbles of nitrogen forming in the tissues because the ambient (atmospheric) pressure is less than the pressure at ground level**. (An example of this phenomenon is the **bubbles formed when a bottle of soda pop is opened**, and the pressure is reduced.)

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The bubbles may:

- **Track into joint spaces** causing a **dull, sickening pain**.
- **More dangerously** they may be **released into the lungs or the brain**, giving rise to **chest pain and/or collapse**.

Possible presentation	Plain meaning
Bends	Joint pain.
Creeps	Skin itching or crawling sensations.
Chokes	Breathing difficulty or chest symptoms.
Neurological symptoms	Brain or nerve involvement can cause serious impairment.

Figure 25.12 *Decompression sickness can involve several systems; symptoms may develop after exposure. Suspected decompression illness needs urgent medical assessment.* *Human-performance reference, printed p 48, decompression-sickness section.*

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19.4 Cabin Pressurization & Rapid vs Slow Decompression

HOW AIRLINE CABINS SOLVE THE PROBLEM

Airlines and high-performance aircraft have cabin pressurization systems to maintain an artificially "low altitude" within the cabin or cockpit. **The pressurization of a commercial airliner flying at 30,000 ft maintains an internal cabin pressure equivalent to about 6,000 ft, with a maximum pressure of 8,000 ft.**

FAST (Rapid) Decompression	A fast decompression is recognizable by: • **Mist in the cabin** • **Blast towards the exterior of the aircraft** • **Expansion of body gases** • **Blast of air released violently from the lungs**
SLOW Decompression	A slow decompression may be caused by: • A **slight air-tightness defect**, or • **Bad functioning of the pressurization**. Insidious onset — no warning blast, no mist, but cabin altitude creeps up over minutes. Detected via the cabin-altitude warning horn (typically at ~10,000 ft cabin altitude in transport jets).

19.5 Factors that Increase the Tendency to Develop the Bends

FIVE PREDISPOSING FACTORS

The tendency to develop the bends increases with:

1. **High rates of climb**,

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2. **Age**,
3. **Obesity**,
4. **Physical activity**,
5. **Low temperatures**.

Max altitude without O_2 — no impairment	8,000 ft
Low DCS risk below	10,000 ft
Atmospheric pressure halved at	18,000 ft
Unpressurised flight — bends threshold	> 25,000 ft
Typical airliner cabin altitude	≈ 6,000 ft
Maximum cabin altitude	8,000 ft
N_2 % in atmosphere	≈ 80 %
DCS risk factors count	5

20 Decompression Sickness (DCS)

INTRODUCTION

Without a cabin pressurization system, pilots and passengers in high flying aircraft would be exposed to high altitude, like hypoxia, low temperatures and Decompression Sickness/Illness.

20.1 Henry's Law Revisited — the underlying physics

HENRY'S LAW STATEMENT FROM DGCA

Henry's Law explains the occurrence of decompression sickness. The principle that **at a constant temperature the concentration of a gas dissolved in a fluid with which it does not combine chemically is almost directly proportional to the partial pressure of the gas at the surface of the fluid**.

A rapid reduction in ambient pressure may cause the nitrogen in our blood to come out of solution as small bubbles leading to decompression sickness.

SYMPTOM LIST & THE "DELAYED ONSET" WARNING

Symptoms of decompression sickness are:

- **Bends**
- **Chokes**
- **Skin manifestations**
- **Neurological symptoms**
- **Circulatory shock**

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Symptoms may appear several hours after the exposure.

20.2 Effects of Nitrogen Bubbles — Four Syndromes

The DGCA syllabus names **four specific syndromes** based on the tissue affected. Memorise the names and the body region.

Joints	The BENDS **Joints:** Bubbles in the joints cause **rheumatic-like pain**, called **the Bends**. Classic shoulder & knee deep, dull ache — diver's bends, identical mechanism.
Skin	The CREEPS **Skin:** Nitrogen bubbles released under the skin cause **the Creeps**, a **sensation of movement under the skin**. Linked to formication (§10.4) — the "ants crawling under the skin" feeling.
Respiratory System	The CHOKES **Respiratory System: Shortness of breath and a feeling of burning, gnawing and piercing pain**. Known as **the Chokes**. Bubbles in pulmonary capillaries → ventilation/perfusion mismatch. Burning chest pain.
The Brain	The STAGGERS **The Brain: Loss of mental functions and control of movement.** Known as **the Staggers**. Neurological DCS — confusion, ataxia, weakness, paralysis. Most dangerous form.

DCS Syndromes — quick-reference one-liner

Tissue	Syndrome Name	Symptom Type
Joints	**The Bends**	Rheumatic-like joint pain
Skin	**The Creeps**	Sensation of movement under skin
Respiratory System	**The Chokes**	Shortness of breath; burning, gnawing, piercing chest pain
The Brain	**The Staggers**	Loss of mental function & movement control

20.3 Treatment of Decompression Sickness

FOUR-STEP TREATMENT

1. **Keep patient warm and on 100 % Oxygen.**
2. **An immediate descent must be initiated.**
3. **Land as soon as possible.**
4. **Seek medical assistance immediately on landing.**

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20.4 The 12-Hour No-Fly Rule After Rapid Decompression

EXAM POINT

Do not fly for at least 12 hours after experiencing rapid decompression even though you may be feeling fit.

WHY 12 HOURS?

DCS symptoms can be **delayed for several hours**. Feeling "fit" immediately after a decompression event does **not** mean nitrogen bubbles are not still forming or growing in tissues. The 12-hour pause allows the body to off-gas the residual nitrogen at normal pressure, much like a diver waiting before flying or driving up a mountain.

21 Flying After Diving — the 24-hour rule

MECHANISM

Decompression sickness can occur when flying at low altitude in individuals who have been diving, using compressed air breathing apparatus shortly before flight at a depth of 30 feet or more.

THE HARD RULE — MEMORISE VERBATIM

As a general rule, individuals should NOT fly within 24 hours following diving and certainly not the same day.

30ft DIVING DEPTH THAT CREATES DCS RISK ON FLYING

24hr MINIMUM WAIT BETWEEN DIVING & FLIGHT

15min – 12hr SYMPTOM-ONSET WINDOW AFTER SURFACING

WHY DIVING + FLIGHT IS SO DANGEROUS

Occasionally a "medical emergency" arises as a result of compressed-air diving, when a diver is in danger of developing **air-embolism (bends) at the surface altitude**, as a result of being unable to decompress before surfacing. In some of these cases **air-evacuation is the only feasible method of getting the patient to a decompression chamber in time to treat this condition**. Flight, however, should be at **the lowest possible altitude to avoid aggravating the condition**.

SYMPTOM ONSET TIMING — DGCA QUOTE

Symptoms and signs **usually appear within 15 minutes to 12 hours after surfacing**; but in severe cases, symptoms may **appear before surfacing or immediately afterwards**. **Delayed occurrence of symptoms is rare** (the syllabus text terminates mid-sentence at the page break — continued in §21+).

CROSS-LINK RECAP — THE THREE "WAIT BEFORE FLYING" RULES

You now have three numbers in your head that the DGCA examiner loves to swap around:

- **Alcohol — bottle to throttle: 24 hr** (§14.1)
- **Blood donation: 48 hr** (§17)

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- **Flying after diving: 24 hr** (§21)
- **Flying after rapid decompression: 12 hr** (§20.4)

Do not mix them up. Diving and alcohol share the 24-hour figure; donation gets the longest pause; rapid decompression "only" 12 — but with mandatory medical clearance afterwards.

Master cheat-sheet — numbers

Every numeric / regulatory fact from this section

Parameter	Exact Value	Where
Number of DGCA-listed baroreceptor triggers	**3**	§16.1
Active-pilot blood donation rule	**Don't donate; if done, wait 48 hr**	§17
Max altitude without O_2, no efficiency loss	**8,000 ft**	§19.1
Low DCS risk altitude — below	**10,000 ft**	§19.1
Atmospheric pressure halved at	**18,000 ft (5,486 m)**	§19.2
DCS symptoms may develop at and above	**18,000 ft**	§19.2
Unpressurised flight bends threshold	**> 25,000 ft (7,620 m)**	§19.2
N_2 % of atmosphere	**80 %**	§19.2
Typical airliner cabin altitude (30,000 ft cruise)	**≈ 6,000 ft**	§19.4
Maximum cabin altitude	**8,000 ft**	§19.4
TUC at FL180 (normal)	**20–30 min**	§18.3
TUC at FL220 (normal)	**10 min**	§18.3
TUC at FL250 (normal)	**3–5 min**	§18.3
TUC at FL280 (normal)	**2.5–3 min**	§18.3
TUC at FL300 (normal)	**1–2 min**	§18.3
TUC at FL350 (normal)	**30 s – 1 min**	§18.3
TUC at FL400 (normal)	**15–20 s**	§18.3
TUC at FL430 (normal)	**9–12 s**	§18.3
TUC at FL500 (normal)	**8–10 s**	§18.3
TUC at FL350 rapid decompression	**15–30 s**	§18.3
TUC reduction by rapid decompression	**½ (halved)**	§18.4
TUC range at FL350 (jet typical)	**33–54 s**	§18.2
Number of factors that determine TUC after RD	**3**	§18.2
Number of factors that ↑ tendency to bends	**5**	§19.5
Number of named DCS symptoms (DGCA list)	**5**	§20.1
Number of named bubble syndromes	**4 (B-C-C-S)**	§20.2
DCS treatment steps	**4**	§20.3

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Parameter	Exact Value	Where
No-fly after rapid decompression	**≥ 12 hr**	§20.4
Diving depth that triggers fly-risk	**≥ 30 ft**	§21
No-fly after diving	**24 hr (not same day)**	§21
DCS symptom onset window post-dive	**15 min – 12 hr**	§21

22 Delayed DCS After Diving

DELAYED SYMPTOMS

The rule on flying after diving ends with "*Delayed occurrence of symptoms is rare…*" — the rest of the sentence reads: **"…but it does occur, especially if air travel follows diving."**

In plain English: **even if a diver feels perfectly fine 24+ hours after surfacing, a subsequent flight can re-precipitate DCS symptoms**. This is because residual nitrogen still dissolved in slow-perfused tissues (fat, bone marrow) can come out of solution when the pilot enters the lower cabin pressure of a transport flight. *The 24-hour rule is a minimum, not a guarantee.*

23 Humidity

23.1 Why Humidity Matters Up High

EXAM POINT

Humidity may become an issue with high-altitude jet transport aircraft because of the low relative humidity at their operational altitudes. The discomfort arising from low relative humidity may not imply physical indisposition.

HUMIDITY RANGES

- **40 – 60 % relative humidity is NORMAL.**
- **< 20 %** may create **minor discomfort**, such as **skin, eye, nose, throat dryness**.

40–60% **NORMAL RELATIVE HUMIDITY**

< 20% **MINOR DISCOMFORT — SKIN, EYE, NOSE, THROAT DRYNESS**

3–8% → 22% **MODERN HUMIDIFIERS RAISE CABIN RH FROM 3–8% TO ~22%**

PREVENTION & MANAGEMENT

- **Overall dehydration can be prevented with adequate fluid intake.**
- **Diuretics like coffee or tea should be avoided.** They drive fluid out of the body via increased urination — the opposite of what is needed at altitude.

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23.2 Aircraft Humidifier Systems

EXAM POINT

The installation of humidifiers on aircraft **raises cabin/cockpit humidity**, but there are potential problems such as:

- **Weight penalty**,
- **Condensation**, and
- **Mineral contamination** that the designer must consider.

Aircraft such as the **Airbus A350, Boeing 787, and the future Boeing 777X** are compatible with onboard humidifiers, either in **crew areas or the entire cabin**. These technologies **increase humidity from 3–8 % to 22 %** and **reduce contaminants** in the air we breathe on a flight.

24 Temperature

EXAM POINT

Temperature affects performance as follows:

24.1 The Three Reference Temperatures — reference verbatim

Temperature effects on pilot performance

Temperature	Effect on Performance
20 °C	**Comfortable temperature for most people in normal clothing.**
30 °C	**Increased heart rate, blood pressure, and sweating.**
> 15 °C	**Discomfort, loss of feeling in hands, poor control of fine muscle movement.** *(Note: the usual phrasing uses "> 15 °C" — referring to deviations beyond 15 °C from a comfortable baseline, i.e. cold-stress.)*

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COCKPIT-TEMPERATURE CONSIDERATIONS

- **At 20 °C** the pilot is comfortable — fine motor skills, judgement and reaction time are optimal.
- **At 30 °C** the body's ANS responds to heat stress by raising HR, BP and triggering sweating to cool the skin. Cognitive performance starts to degrade.
- **In cold conditions** — far below the comfort baseline — the pilot loses sensation in the fingers, struggles with fine-motor tasks like switch selection or fast checklist scanning, and may shiver involuntarily.
- For both extremes, **cabin temperature management is a flight-safety task**, not merely a comfort one. Climate control should be set to keep cockpit temperature near 20 °C for sustained operations.

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Condition	Possible effect on performance
Cold	Discomfort, reduced sensation and impaired fine movement.
Comfortable cabin conditions	Support effective work; clothing, workload and individual factors also matter.
Heat	Discomfort and physiological strain; sweating can increase fluid loss.

Figure 25.13 *Thermal comfort supports performance. This comparison does not prescribe a universal performance curve or a cockpit temperature limit.*

25 Vibration

25.1 Natural Resonance — Why Frequency Matters

OPENING

Different parts of the body show a natural resonance at different periods of vibration. For example:

- The **natural resonance of the eyeball is 30 – 40 Hz**, and
- The **skull is 1 – 4 Hz**.

WHAT "NATURAL RESONANCE" MEANS IN PLAIN ENGLISH

Every physical structure has a frequency at which it *wants* to vibrate — push it at that frequency and small inputs build up into huge oscillations (think of pushing a child on a swing in time with the swing's natural period). When external aircraft vibration matches a body part's resonant frequency, **that part suffers maximum disturbance**. The eyeball resonating at 30–40 Hz is precisely why **vision becomes blurred at certain helicopter or propeller-vibration frequencies**.

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25.2 Effects of Vibration — Full DGCA Frequency Table

MEMORISE THIS TABLE — DIRECT EXAM TARGET

Effects of vibration include:

Vibration frequency vs body effect

Frequency Range	Effect on the Body
1 – 4 Hz	**Interference with breathing; neck pain.** *(Skull resonance range.)*
4 – 10 Hz	**Chest and abdominal pain.**
8 – 12 Hz	**Backache.**
10 – 12 Hz	**Headache, eyestrain, throat pain, speech difficulty, muscle tension.**
30 – 40 Hz	**Interference with vision.** *(Eyeball resonance range.)*

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Frequency	Effects/Symptoms
1 to 4 Hz.	Interferes with breathing.
4 to 10 Hz.	Chest and abdominal pains.
8 to 12 Hz.	Backache.
10 to 20 Hz.	Headache, eyestrain, pains in the throat, speech difficulties and muscular tensions plus degradation of visual acuity.

Figure 25.14 *Examples of vibration frequency ranges and associated effects. Actual response depends on exposure, posture, direction and individual factors.* *Human-performance reference, printed p 125, frequency/effects table.*

OPERATIONAL VIBRATION SOURCES IN AVIATION

- **Helicopter rotors:** typically 4–7 Hz (main rotor) → chest/abdominal pain & backache range.
- **Piston / turboprop engines:** 10–30 Hz on the airframe → headache, eyestrain, vision interference.
- **Jet engine buffet:** higher-frequency, but resonant peaks in the seat may still strike critical body-part frequencies.
- **Turbulence:** low-frequency buffeting (1–10 Hz) → fits the skull / neck / chest range.

This is why aircraft seats incorporate **damping and isolation systems** — to keep the airframe's vibration spectrum from delivering energy at the worst body-resonance frequencies.

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26 Glare & UV Radiation

EXAM POINT

UV radiation from sunlight can cause visual fatigue, as well as affect visual health.

WHY GLARE AND UV DESERVE THEIR OWN SECTION

At cruise altitudes (FL300+), the protective filtering of the atmosphere is reduced. Cockpit windows admit **significantly more UV-A and UV-B** than ground-level glass would. Over a long career, this contributes to:

- **Visual fatigue** — the immediate effect of glare on the working pilot. Squinting, eyestrain, reduced contrast sensitivity, headache.
- **Cumulative visual health damage** — cataract formation, macular degeneration risk, conjunctival/skin damage.

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PRACTICAL PILOT PRECAUTIONS

- Use **aviation-grade polarised or non-polarised UV-blocking sunglasses** (note: certain LCD instruments may distort under polarised lenses — check before adoption).
- Use **cockpit sun-visors** appropriately — don't leave them stowed when the sun is over the nose.
- For long-haul crews, schedule **vision rest breaks** on autopilot legs.
- Periodic **aero-medical eye examinations** — already mandated by DGCA — pick up early cataract / retinal changes.

27 Incapacitation in Flight

27.1 Why Pilot Incapacitation is a Flight-Safety Issue

INTRODUCTION

The risk of seizure in flight is obvious. Incapacitation is in most cases:

- **Sudden**,
- **Unpredictable**,
- **Unavoidable**,
- **Prolonged**,
- **Complete**, and
- **Potentially more frequent in the stressful flying environment**,

... and constitutes a **direct threat to the health and safety**.

PREVENTION MEASURE

Periodical medical examinations minimize the risk of incapacitation in flight. The frequency of medical checks increases with advancing age.

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Assessment	Validity under the Indian CAR
Class 1	One year up to 60 years; six months thereafter. For single-crew operations, six months above 40 years.
Class 2	Two years up to 50 years; one year thereafter.
Class 3	Four years up to 40 years; two years above 40 up to 50 years; one year above 50 years.

27.2 Obvious Incapacitation

DEFINITION

Obvious incapacitation normally refers to a state in which **all of a crew member's physical or mental functions are lost**, including:

- **Loss of consciousness**, OR
- **Being unable to move while retaining consciousness**,

... **rendering them completely unable to carry out their duties**.

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ONSET PATTERNS & PRESENTATION

- Obvious incapacitation can sometimes involve **convulsions**,
- OR the victim may **lapse into unconsciousness several minutes after the incapacitation occurs**.

27.3 Subtle / Insidious Incapacitation

DEFINITION

SUBTLE (develops slowly and gradually), incapacitation refers to a state of a **partial or temporary loss of physical or mental function** which manifests itself in the form of:

- **Partial paralysis**,
- **A dulling of perception, judgment, or responses** or lack thereof,
- **A state of absence of mind, distraction of attention**,
- **Trouble with speech**,
- **Inadequate responses**,
- **Meaningless utterances**, etc.

THE HIDDEN-DANGER WARNING

Particular attention must be paid to the fact that a crew member may become incapacitated EVEN THOUGH THEIR APPEARANCE IS NO DIFFERENT FROM NORMAL.

POSSIBLE CAUSES OF SUBTLE INCAPACITATION — DGCA LIST

- **Temporary hypoglycemia** (low blood sugar — pilot skipped meals)
- **Reduction in blood pressure**
- **Cerebropathy or psychopathy**
- **Excessive muscular fatigue**
- **Excessive drinking** (see §14)
- **Insufficient sleep**
- **Emotional instability**
- **Toothache**
- **Stomachache**
- **Headache**

27.4 Obvious vs Subtle — Why Subtle is More Dangerous

THE COUNTERINTUITIVE TRUTH

Since other crew members are often unable to detect subtle incapacitation quickly, from a flight safety point of view, it is possible that SUBTLE INCAPACITATION MAY LEAD TO A SITUATION OF COMPARATIVELY GREATER DANGER THAN OBVIOUS INCAPACITATION.

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WHY THIS MATTERS IN TWO-CREW OPERATIONS

With **obvious incapacitation**, the other crew member sees the captain slumped or convulsing, declares emergency, takes control, gets the aircraft on the ground. The hand-off is instant.

With **subtle incapacitation**, the captain is still moving switches, talking on the radio, even responding to questions — but his judgement is impaired. He may set the wrong altitude, mis-tune a NAV, mishear an ATC clearance, or fail to initiate a missed approach. The other pilot may not detect this until *after* the wrong outcome unfolds. This is the rationale behind **two-crew CRM**: a healthy independent cross-check that catches subtle drift before it becomes an accident.

OBVIOUS Incapacitation	• **All functions lost** — LoC or paralysed-conscious • May include convulsions / delayed LoC • **Detected instantly** by the other crew • Causes: cardiac arrest, MI, intracranial bleed, stroke, epilepsy • **Handover happens fast**
SUBTLE Incapacitation	• **Partial or temporary** loss of function • Appearance may be normal • **Often undetected for some time** by the other crew • Causes: hypoglycemia, ↓ BP, fatigue, sleep loss, drinking, emotional/dental/stomach/headache • **Greater overall flight-safety danger**

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28 Fits & Faints

28.1 Epilepsy — Grand Mal & Petit Mal

DEFINITION

A **fit or seizure is usually referred to as "epilepsy"**. A fit or a seizure is **not a specific disease** but **a set of signs or symptoms in response to a disturbance of the electrical activity in the brain**.

Grand Mal Epilepsy	• Manifests as a **generalized seizure** • **Associated with a transient loss of consciousness** • May be associated with a **prodromal phase** *(warning aura before the seizure)* • Normally accompanied by **convulsions and uncontrolled physical movement**
Petit Mal Epilepsy	• Also a **generalized seizure** • **Not associated with a loss of consciousness** • **Petit Mals are a MINOR attack** • Often presents as brief "absence spells" — staring, blanking out for a few seconds

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THE ABSOLUTE BAR

A seizure may or may not be associated with a loss of consciousness... but **ANY FIT, MAJOR OR MINOR, IS ASSOCIATED WITH AN UNPREDICTABLE LOSS OF CONSCIOUSNESS AND IS THEREFORE AN ABSOLUTE BAR TO THE HOLDING OF A FLYING LICENCE.**

PILOT-MEDICAL IMPLICATION

This is one of the very few *absolute* medical disqualifications in aviation. Even controlled epilepsy on medication — even a single documented seizure event in adult life — is grounds for medical refusal. The reason is the word "**unpredictable**" — DGCA cannot certify the probability of an in-flight seizure as acceptably low.

28.2 Faint & Vasovagal Syncope

DEFINITION

Faint is a common cause of a loss of consciousness in adults. The most common causes of faints are:

Common causes of faint

#	Cause
1	**Standing up quickly after prolonged sitting especially when hot or dehydrated**
2	**A sudden shock**
3	**Loss of blood after an accident**
4	**Lack of food or fluid**
5	**Other physiological stress**

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SYNCOPE & VASOVAGAL SYNCOPE

Syncope is a temporary but sudden loss of consciousness when blood flow to the brain is compromised. In young individuals, **fear, anxiety, sight of blood, etc., can result in a temporary loss of consciousness**. This is referred to as **VASOVAGAL SYNCOPE**.

Frequently, syncope is associated with symptoms like:

- **Light-headedness**,
- **Muscle weakness**, and
- **Dizziness** before the actual fainting occurs.

FLIGHT-LICENCE IMPLICATION

A faint has NO SIGNIFICANCE as far as future flying is concerned, so long as the cause is clearly understood.

EPILEPSY VS FAINT — THE CRITICAL COMPARISON

This is the most frequently tested contrast in DGCA HPL papers. Memorise it:

Epilepsy vs Faint (Syncope) — side-by-side

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Parameter	Epilepsy (Grand/Petit Mal)	Faint (Syncope)
Underlying cause	**Disturbance of brain electrical activity**	**Compromised blood flow to the brain**
Loss of consciousness	Often (Grand Mal) / Absent (Petit Mal)	**Yes, temporary & sudden**
Convulsions / uncontrolled movement	**Yes (Grand Mal)**	No
Predictability	**Unpredictable**	Cause is usually identifiable (heat, hunger, fear, blood loss)
Typical warning signs	Prodromal aura (Grand Mal)	**Light-headedness, muscle weakness, dizziness**
Effect on flying licence	**ABSOLUTE BAR — no licence**	**No significance — fit to fly if cause is clear**

Master cheat-sheet — numbers

Every numeric / regulatory fact from this section

Parameter	Exact Value	Where
Normal cabin RH	**40 – 60 %**	§23.1
Minor discomfort RH threshold	**< 20 %**	§23.1
Native jet-cabin RH (no humidifier)	**3 – 8 %**	§23.2
Cabin RH with modern humidifier (A350/787/777X)	**22 %**	§23.2
Comfortable cockpit temperature	**20 °C**	§24.1
Heat-stress temperature (↑ HR, ↑ BP, sweating)	**30 °C**	§24.1
Skull natural resonance frequency	**1 – 4 Hz**	§25.1
Eyeball natural resonance frequency	**30 – 40 Hz**	§25.1
Breathing interference / neck pain frequency	**1 – 4 Hz**	§25.2
Chest & abdominal pain frequency	**4 – 10 Hz**	§25.2
Backache frequency	**8 – 12 Hz**	§25.2
Headache · eyestrain · throat · speech difficulty · muscle tension	**10 – 12 Hz**	§25.2
Vision interference frequency	**30 – 40 Hz**	§25.2
Number of "obvious-incapacitation" defining attributes	**6** (sudden, unpredictable, unavoidable, prolonged, complete, frequent in stress)	§27.1
Number of DGCA-listed causes of obvious incapacitation	**5**	§27.2
Number of DGCA-listed causes of subtle incapacitation	**10**	§27.3
Number of DGCA-listed causes of fainting	**5**	§28.2

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Parameter	Exact Value	Where
Effect of epilepsy on flying licence	**ABSOLUTE BAR**	§28.1
Effect of a faint (cause understood) on flying licence	**No significance**	§28.2

29 The Eye

OPENING — THE EYE IS THE MOST IMPORTANT SENSOR

The eye delivers to the brain information about the outside world at a much faster rate than any other sensory organ. The eye is the organ of sight. The eye is the most sensitive of our sensory organs.

Its basic structure is similar to a **camera with an aperture (Iris), a lens (lens) and a light sensitive screen (Retina)**.

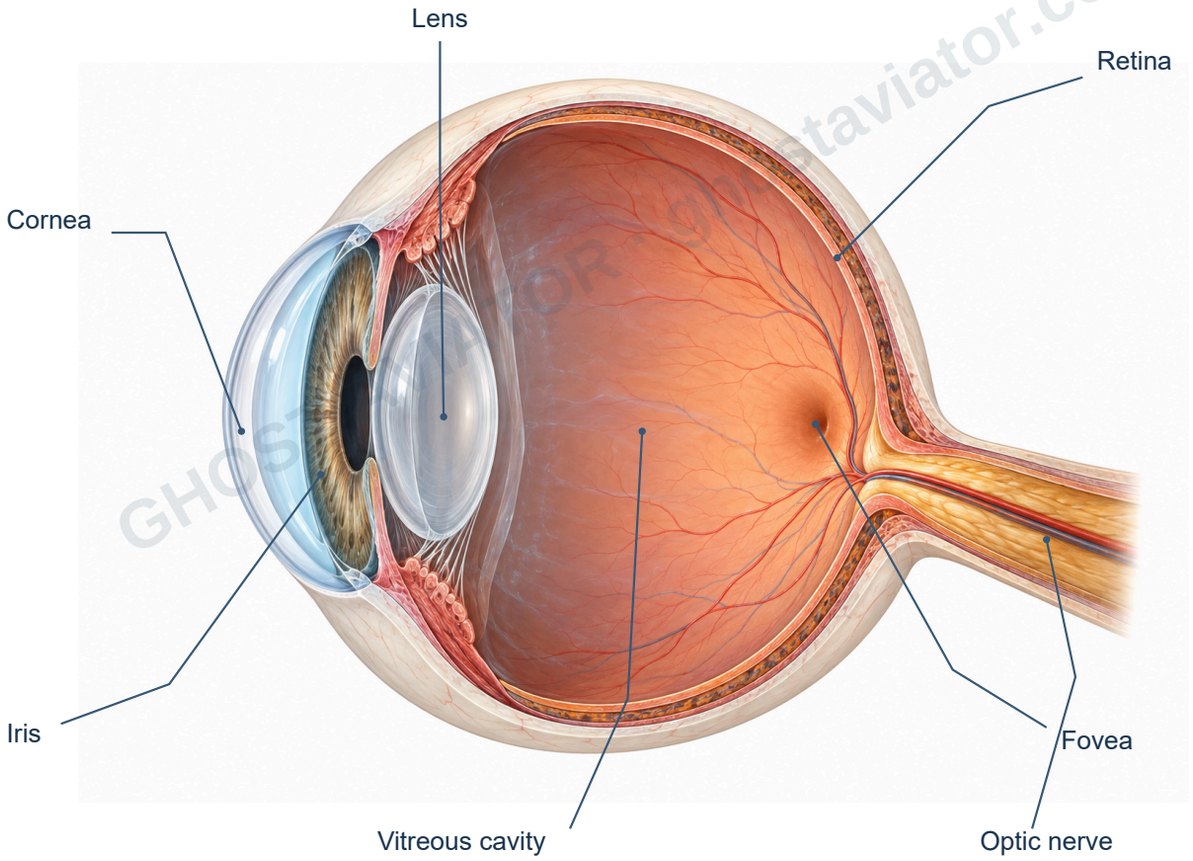

Figure 25.15 *The eye in section. The retina lines the back of the eye; the fovea is a small central region used for sharp vision.*

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29.1 The Cornea, Iris & Pupil, Lens

THE CORNEA

Light enters the eye through the **Cornea**, a **clear window at the front of the eyeball**. The Cornea is capable of contributing **70 % to 80 % towards the total focusing ability of the eye**.

THE IRIS AND THE PUPIL

The **amount of light allowed to enter the eye is controlled by the Iris**. The **Pupil**, the **hole in the center of the Iris**, **adjusts to the flow of light**.

THE LENS

The **shape of the lens is changed by muscles**. This controls **final focusing onto the fovea**.

29.2 The Retina — Rods, Cones, and the Fovea

THE RETINA

The **Retina is a light sensitive screen lining the inside of the eyeball**. On this screen are **light-sensitive cells**. When light falls on them, it generates a **small electrical charge which is passed to the brain by nerve fibres (neurons) which combine to form the Optic Nerve**.

CRITICAL NUMBERS — BURN INTO MEMORY

The retina has **rods in its peripheral zone** and **cones in its central zone**. The retina contains the **receptors for vision**:

- About **100 million rods**
- About **6 million cones**

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100M RODS (PERIPHERAL · NIGHT VISION · BLACK & WHITE)

6M CONES (CENTRAL · DAY VISION · COLOUR & DETAIL)

~17:1 ROD-TO-CONE RATIO

The Rods	The Rods can **only detect black and white** but are **much more sensitive at lower light levels**. **Rods are responsible for our PERIPHERAL VISION.** • Located in the peripheral retina • ~ 100 million in number • Sensitive in dim light → night vision • No colour information — monochrome • Slow dark-adaptation (~ 30 min)
The Cones & The Fovea	The **central part of the retina is called the FOVEA**. The fovea centralis is a **small, central pit composed of closely packed cones** in the eye. This is the area with the **best day vision and NO night vision at all**. Any object that **needs to be examined in detail is automatically brought to focus on the fovea**. This is called **"Central Vision"**. The rest of the retina fulfils the function of **attracting our attention to movement and change**.

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Retinal region / receptor	Function
Fovea	Central retinal region used for sharp detail; its centre is densely packed with cones.
Cones	Colour and detailed vision, especially in good light.
Rods	Sensitive to dim light; more important away from the foveal centre.
Optic disc — blind spot	Where nerve fibres leave the eye; it has no photoreceptors.

Figure 25.16 *The retina's regions and receptors. Use the preceding eye section for their anatomical positions.*

29.3 Eye Movement & Binocular Vision

EYE MOVEMENT

To **track an object successfully, or to focus on an object, the eyes need to move in harmony with one another**. This means that the brain must **co-ordinate control of the muscles of the two eyes**. **In a fatigued person, double vision can occur.**

BINOCULAR VISION

Binocular Vision means seeing with two eyes. With binocular vision **each eye sees an object from two slightly different angles**. The brain **merges the two images into one** and is thus able to **perceive that the image has depth**.

A further advantage of binocular vision is that the **blind spot of one eye is covered by the other eye**. **Depth perception when objects are close is achieved through binocular vision.**

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29.4 Visual Acuity

DEFINITION

Visual acuity is a measure of the capacity of the eye to determine SMALL DETAIL, UNDISTORTED, at a given distance.

WHY ACUITY MATTERS IN FLIGHT

The **sharpest visual acuity occurs when the retinal image is sharply focused on the fovea**, so that the pilot needs to look **exactly in the direction of the on-coming aircraft to detect it**. It is thus **essential for pilots to have normal visual acuity**, either with the **naked eye, or by wearing spectacles**, in order that they may detect **objects clearly at safe distance**.

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29.5 Accommodation & Reading Glasses

ACCOMMODATION

As well as being able to see objects clearly at a distance, **pilots also need good near vision in order to read instruments and maps**. Being able to **focus on close objects is a function of the eye's ability to accommodate**.

READING GLASSES

Pilots and drivers who have reached middle-age normally wear bi-focal spectacles to allow them to **see clearly at a distance and to read their instruments and maps, while wearing the same spectacles**.

Cornea — % of total focusing	70 – 80 %
Rods total	~ 100 million
Cones total	~ 6 million
Best day vision area	Fovea (all cones)
Fovea night vision	NONE
Pilot middle-age glasses	Bi-focal

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30 Limitations of Acuity — Sharpness of Central Vision

EXAM POINT

The sharpness of central vision **drops as light falls on retina at increasing angles from the fovea**. The following factors affect the sharpness of central vision:

Twelve factors that affect the sharpness of central vision

#	Factor	Pilot Note
1	**Angular distance from the fovea**	Acuity is maximum on the fovea, drops sharply off-axis.
2	**Physical imperfections within the visual system**	Refractive errors, cataract, retinal disease.
3	**Age**	Presbyopia, lens stiffening, declining contrast sensitivity.
4	**Hypoxia**	Retina has very high O_2 demand — first system to suffer (§10).
5	**Smoking**	CO + nicotine reduce retinal perfusion (§13).
6	**Alcohol**	Histotoxic mechanism degrades cells including retinal photoreceptors (§14).
7	**Amount of light available**	Photopic / mesopic / scotopic — acuity scales with luminance.
8	**Size and contours of an object**	Large object with clear contour = easier; small or amorphous = harder.
9	**Distance of the object from the viewer**	Resolvable detail falls off with distance.

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#	Factor	Pilot Note
10	**Contrast of an object with its surrounding**	Low contrast = hard to see (e.g. white aircraft in cloud).
11	**Relative motion of a moving object**	A constant-bearing target on collision course is *hardest* to detect.
12	**Drugs or medication**	Many OTC and prescription meds cause blurred vision or accommodation problems.

31 Limitations of the Visual System

31.1 Night Vision & Dark Adaptation — the critical 7 vs 30 minutes

DEFINITION

Adaptation is the adjustment of the eyes to high or low levels of illumination.

The time required for:

- **Complete adaptation for HIGH levels of illumination is 10 seconds**, and
- **For full DARK adaptation, 30 minutes**.

WHY PILOTS CARE

When **passing from bright ambient surroundings into the dark**, **visual capacity is severely reduced** until the eyes have adapted to the dark. It is **especially important for pilots to allow sufficient time for dark adaptation to take place before flying at night**.

CONES VS RODS — THE TWO-STAGE DARK ADAPTATION

Dark adaptation takes time:

- **About 7 minutes for the CONES**,
- **And 30 minutes for the RODS**.

10s BRIGHT-LIGHT ADAPTATION (DARK → BRIGHT)

7min CONES — PARTIAL DARK ADAPTATION

30min RODS — FULL DARK ADAPTATION

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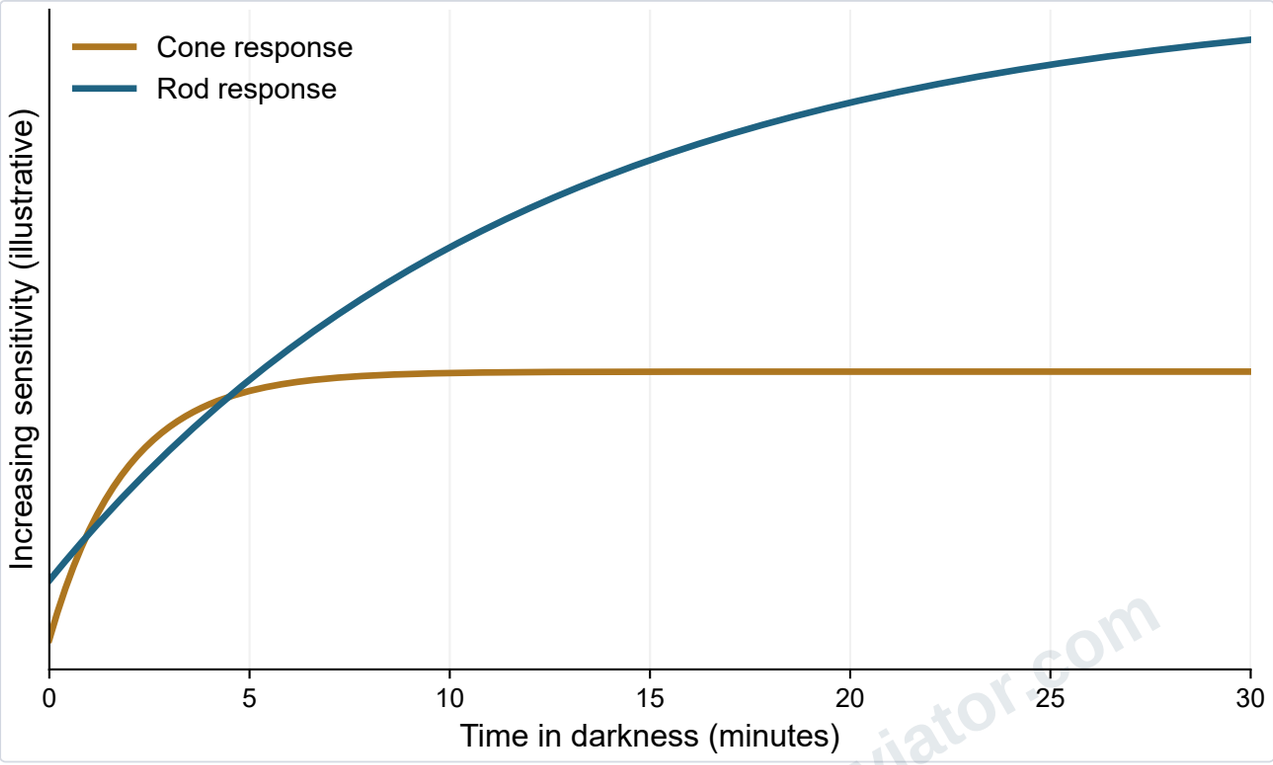

Figure 25.17 *Dark adaptation involves faster cone adjustment and slower rod adjustment. Shapes are explanatory, not measured sensitivity data. Bright light can reduce dark adaptation.*

31.2 Vision Under Dim & Bright Illumination

EXAM POINT

Under conditions of **dim illumination**, **small print and colors on aeronautical charts and aircraft instruments become unreadable unless adequate cockpit lighting is available**. Moreover, another aircraft must be much closer to be seen **unless its navigation lights are on**.

In darkness, vision becomes more sensitive to light, a process called **dark adaptation**. Although exposure to **total darkness for at least 30 minutes is required for complete dark adaptation**, a pilot can achieve a **moderate degree of dark adaptation within 20 minutes under dim red cockpit lighting**.

WHY RED LIGHT — AND WHEN NOT TO USE IT

Since red light severely distorts colors, especially on aeronautical charts, and can cause serious difficulty in focusing the eyes on objects inside the aircraft, its use is advisable ONLY where optimum outside night vision capability is necessary.

Even so, white cockpit lighting must be available when needed for map and instrument reading, especially under IFR conditions.

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DARK-ADAPTATION KILLERS

Dark adaptation is impaired by:

- **Exposure to cabin pressure altitudes above 5,000 feet**,
- **Carbon monoxide inhaled in smoking and from exhaust fumes**,
- **Deficiency of Vitamin A in the diet**, and
- **By prolonged exposure to bright sunlight**.

THE "CLOSE ONE EYE" PRESERVATION TRICK

Since any degree of dark adaptation is lost within a few seconds of viewing a bright light, a pilot should close one eye when using a light to preserve some degree of night vision.

31.3 Off-Centre Viewing — "Look 15–20° to the Side"

THE NIGHT-VISION TECHNIQUE

Look to the side (15 – 20 deg) of the object.

WHY OFF-CENTRE VIEWING WORKS AT NIGHT

The fovea has **no rods — and therefore no night vision**. When you stare *directly* at a dim object at night, its image falls on the fovea, where there are zero rods → you cannot see it. By looking **15–20° to the side**, you place the image on the **rod-rich peripheral retina** where night vision is sharpest.

Counterintuitive but vital: at night, the way to see something is NOT to look at it. This is taught in night-VFR training as "**off-centre / scan viewing**".

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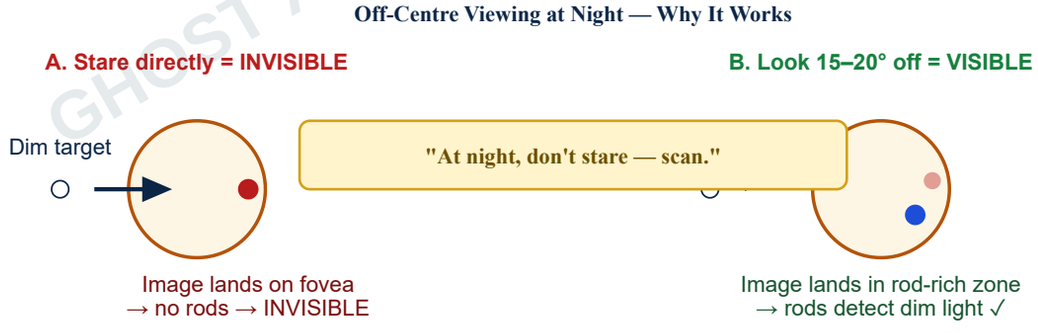

Figure 25.18 *Direct stare puts the target on the fovea (no rods, no night vision). Looking 15–20° to the side places it on the rod-rich peripheral retina — and the dim object becomes visible.*

31.4 The Blind Spot

EXAM POINT

The **blind spot is point on the retina where the optic nerve enters the eyeball**. Here the retina has **no covering of light-detecting cells**.

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WHY THIS MATTERS FOR SEE-AND-AVOID

If the eye remains **looking straight ahead**, it is possible for a **closing aircraft to remain in the blind spot until a very short time before impact**. To lessen the danger of collision, pilots are taught to **carry out a SYSTEMATIC LOOK OUT at all times**.

With both eyes open, the blind spot of one eye is covered by the other eye. But **be aware of obstructions to your visual field** such as passengers or canopy structures.

31.5 Empty Visual Field (Empty Field Myopia)

EXAM POINT

In the **absence of anything to focus on** (that is when your visual field is empty), the **natural focus point of the eye is, on average, at a distance of between 1 and 2 meters in front of the eye**.

Pilots should **minimize the risks associated with empty visual field by periodically and deliberately focusing on objects, both close and at a distance**.

THE DANGER — "EMPTY FIELD MYOPIA"

At high altitude with a clear blue sky and no clouds, your eye relaxes to its **resting focal length of 1-2 m** — well inside the cockpit. A distant aircraft becomes **blurry and easy to miss**. Mitigation: deliberately re-focus the eyes by looking at the wingtip, then at the horizon, then at a distant ground feature. This pumps the ciliary muscles and breaks empty-field myopia.

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31.6 Damage to the Visual System — UV at Altitude

EXAM POINT

Very high light occurs at altitude. At altitude, light contains **more of the high energy blue and ultraviolet wavelengths than is experienced at sea level**. Over a long period, such light can cause **cumulative damage to the retina and lens of the eye**. However, **most harmful wavelengths are filtered out by the cockpit windows**.

31.7 Vibrations & Blurred Vision

EXAM POINT

Vibrations can cause blurred vision. This is due to TUNED RESONANCE oscillations of the eyeballs.

(Recall §25 — the eyeball's natural resonance is **30–40 Hz**. Aircraft vibration that hits that band → vision-blur. This is one of the most direct cross-links between vibration and vision.)

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32 Protection of the Visual System — Sunglasses

FLASH-BLINDNESS PROTECTION IN THUNDERSTORMS

When flying through a thunderstorm with lightning you can protect yourself from FLASH BLINDNESS by:

- **Turning up the intensity of cockpit lights**,
- **Looking inside the cockpit**,
- **Wearing sunglasses**, and
- **Using face blinds or face curtains when installed**.

WHAT GOOD SUNGLASSES MUST DO

The requirement of good sunglasses is to:

- **Absorb at least 85 % of visible light to eliminate glare without decreasing visual acuity**,
- **Absorb UV and IR radiation**,
- **Absorb all colors equally**.

Make sure you avoid using cheap sunglasses. Light sensitive lenses (Photo chromatic) are also generally forbidden for use in flight.

SUNGLASSES CHARACTERISTICS (THE "SIX" LIST)

Sunglasses should have the following characteristics:

1. **Be impact resistant**.
2. **Have thin frames** (minimum visual obstruction).
3. **Be coated with poly carbonate for strength**.
4. **Be of good optical quality**.
5. **Have a luminescence transmittance of 10 – 15 %**.
6. **Possess appropriate filtration characteristics**.

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Sunglass light absorption	≥ 85 %
Sunglass luminescence transmittance	10 – 15 %
Photochromatic lenses	Forbidden in flight
Frame coating for strength	Polycarbonate

33 Visual Defects

OPENING

The most common visual defects are caused by the distorted shape of the eyeball.

33.1 The Four Refractive Defects — Myopia, Hypermetropia, Presbyopia, Astigmatism

MYOPIA (Short-sightedness)	Myopia is more commonly known as **short-sightedness**. In a myopia eye, the **eyeball is LONGER than normal** causing the **image to fall in FRONT of the retina**. **Correction:** A **CONCAVE lens** will correct Myopia by **bending the light from distant objects OUTWARDS before it hits the cornea**. **Normal pilot distance vision:** "may be very approximately assessed as the ability to **read a car number plate at 40 meters**".
HYPERMETROPIA (Long-sightedness)	Hypermetropia is also known as **long-sightedness**, because **only objects at a distance can be seen clearly**. **Correction:** A **CONVEX lens** will overcome Hypermetropia by **bending the light rays from near objects INWARDS before they meet the cornea**. *(Eyeball is shorter than normal — image would form behind the retina. Convex lens converges the rays earlier.)*
PRESBYOPIA	Presbyopia is the **inability of the lens to change its shape to accommodate adequately**, to focus as an image from a near object onto the retina. This condition normally arises in people **between the ages of 40 and 50**. It is a **form of long-sightedness** and is **corrected using a CONVEX lens**. *(This is why pilots at middle age often need bi-focals — see §29.5.)*
ASTIGMATISM	Astigmatism is caused by a **misshapen or oblong cornea**. For a person with astigmatism **objects will appear irregularly shaped**. *(Corrected with a cylindrical lens that compensates for the corneal irregularity.)*

Refractive Defects — Memory Table

Defect	Common Name	Mechanism	Correction
Myopia	Short-sightedness	Eyeball **longer** → image in front of retina	**Concave lens**
Hypermetropia	Long-sightedness	Eyeball shorter → image behind retina (only distance clear)	**Convex lens**
Presbyopia	Age-related (40–50 yrs)	Lens can't change shape — accommodation fails	**Convex lens** (often bi-focal)
Astigmatism	—	Misshapen / oblong cornea — objects irregular	Cylindrical lens

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33.2 Wearing of Corrective Spectacles by Pilots

RULE

Pilots who wear corrective spectacles or contact lenses, for whatever reason, must carry a SPARE PAIR at all times when they are exercising the privileges *(what a licence allows its holder to do)* of their license.

33.3 Glaucoma

GLAUCOMA (THE MOST AVIATION-MEDICALLY IMPORTANT VISUAL DEFECT)

Glaucoma is characterized by:

- **Progressive narrowing of the visual field**,

- **Insidious onset and concealed progression**,
- **An increase in intra-ocular pressure**.

Glaucoma can lead to total blindness and undetected reduction of the visual field. It reduces visual acuity in its final stage.

WHY GLAUCOMA IS SCREENED AT EVERY PILOT MEDICAL

The word "**insidious**" + "**concealed**" + "**undetected reduction of visual field**" is what makes glaucoma so dangerous to aviation. A pilot may be losing peripheral vision over months or years and not realise it — until a critical traffic conflict or runway-incursion-detection failure exposes it. **Intra-ocular pressure (IOP) measurement (tonometry) is part of every routine Class-1 / Class-2 DGCA medical** precisely to catch glaucoma early.

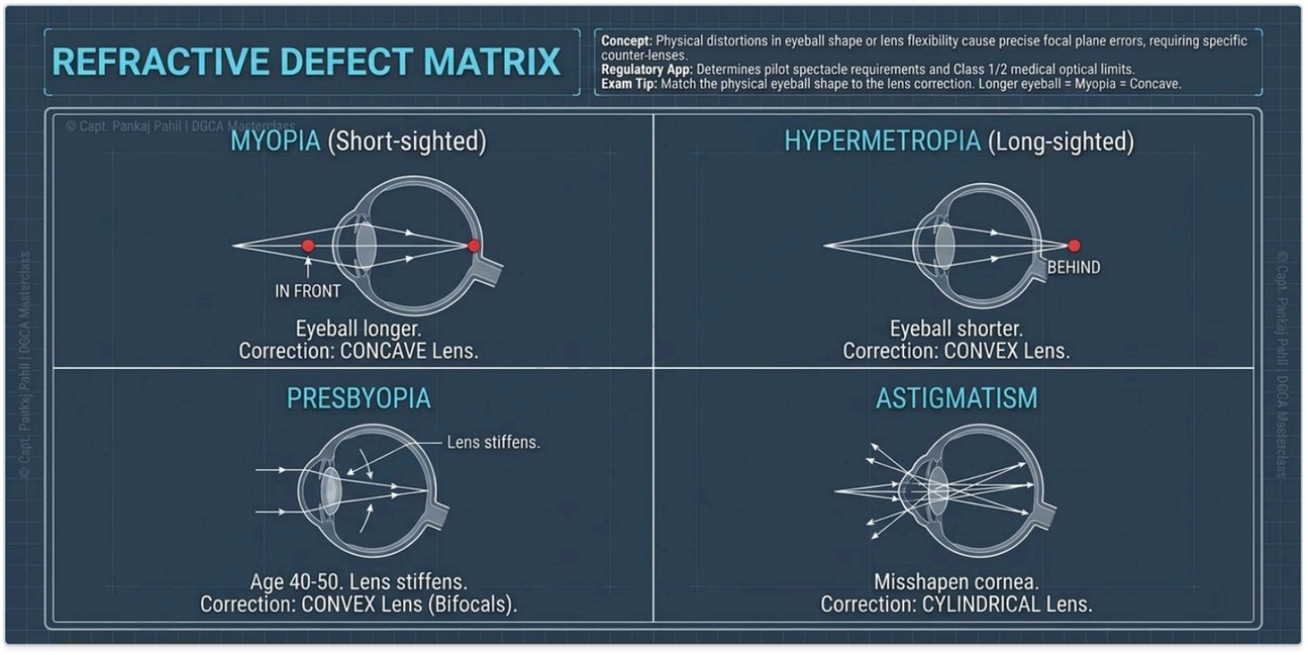

Figure 25.19 *The four refractive defects and the lens that corrects each*

Master cheat-sheet — numbers

Every numeric / regulatory fact from this section

Parameter	Exact Value	Where
Cornea — % of total focusing ability	**70 – 80 %**	§29.1
Number of rods in retina	**~ 100 million**	§29.2
Number of cones in retina	**~ 6 million**	§29.2
Rod location	**Peripheral retina**	§29.2
Cone location	**Central retina (Fovea)**	§29.2
Best day vision area	**Fovea (all cones)**	§29.2

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Parameter	Exact Value	Where
Fovea night-vision capability	**NONE**	§29.2
Bi-focal age (DGCA implied)	**Middle-age**	§29.5
Number of factors that affect central-vision sharpness	**12**	§30
Adaptation: dark → bright	**10 seconds**	§31.1
Adaptation: bright → dark (FULL)	**30 minutes**	§31.1
Cone dark-adaptation	**7 minutes**	§31.1
Rod dark-adaptation (full)	**30 minutes**	§31.1
Moderate dark adaptation under dim red light	**20 minutes**	§31.2
Dark adaptation impaired above cabin altitude	**5,000 ft**	§31.2
Off-centre night-viewing angle	**15 – 20 degrees**	§31.3
Eye natural resting focal length (empty field)	**1 – 2 m in front of eye**	§31.5
Eyeball natural resonance (vision blur)	**30 – 40 Hz**	§31.7
Sunglasses — visible light absorption	**≥ 85 %**	§32
Sunglasses — luminescence transmittance	**10 – 15 %**	§32
Photochromatic lenses in flight	**Forbidden**	§32
Sunglasses characteristics — count	**6**	§32
Normal distance vision benchmark	**Read car plate at 40 m**	§33.1 (Myopia)
Presbyopia onset age	**40 – 50 years**	§33.1
Spare-spectacles rule	**Mandatory carry**	§33.2
Number of visual defects covered	**5 (Myopia, Hypermetropia, Presbyopia, Astigmatism, Glaucoma)**	§33

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34 Colour Vision

EXAM POINT

Good color vision is essential for pilots because of the use of color associated with the items listed below.

Why pilots need good colour vision — DGCA-listed items

#	Item	Operational use of colour
1	**Navigation light of aircraft**	Red (left wingtip) · Green (right) · White (tail) — defines aircraft direction at night.

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#	Item	Operational use of colour
2	**Runways and airfields**	White runway edge lights · red runway end · green threshold · PAPI (white/red) glide-path.
3	**Ground obstructions**	Red obstruction lights on towers · obstacle markings.
4	**Cockpit displays and instruments**	Red = warning · amber = caution · green = normal · cyan = sky · brown = ground (PFD).
5	**Maps and charts**	VFR/IFR charts use colour-coded airspace, terrain, MSA values.
6	**Emergency flares**	Distress flares — red = distress · white = signalling.
7	**Light signals**	ATC light-signal codes (steady green = clear to land · red flashing = airport unsafe, etc.)

35 Colour Blindness

DEFINITION

Color blindness or, more accurately, color-defective vision, is caused by a defect in the structure of the color-sensitive cones in the retina.

CONES AND COLOUR VISION

Cones are the photoreceptors (light-sensitive cells) used for colour and fine detail. They are concentrated in the fovea, but cones also occur elsewhere in the retina. A total retinal cone count must not be described as a count inside the fovea. Inherited differences in cone pigments can make particular colours difficult to distinguish.

DGCA-MEDICAL IMPLICATION

Colour-defective vision (red-green most commonly) is screened at every DGCA Class-1 / Class-2 medical using the **Ishihara colour plates**. Significant deficiency can disqualify a candidate, or restrict the licence to "day-VFR only" because of the night colour-signal interpretation problem (nav lights, PAPI, ATC light signals all rely on red/green discrimination).

36 Vision & Speed — Reaction Times

CLOSING SPEED REDUCES THE TIME AVAILABLE

Higher closing speed gives the crew less time to detect a conflict and take effective action. Detection, recognition, decision-making and aircraft response all take time; their duration varies with the situation.

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WORKED EXAMPLE — AN ILLUSTRATIVE SIX-SECOND DELAY

Two aircraft approaching head-on at 500 kt each have a closing speed of **1,000 kt**.

Distance closed each second = 1,000 ÷ 3,600 = 0.278 NM.
Distance closed in six seconds = 1,000 × 6 ÷ 3,600 = 1.67 NM.

Six seconds is an example, not a universal pilot reaction time. A systematic lookout and timely use of available traffic information help the crew recognise a developing conflict earlier.

37 The Ear — Two Functions, Three Sections

THE TWO FUNCTIONS OF THE EAR

The ear performs two quite separate functions:

- **To receive vibrations or sound waves in the air which it transmits to the brain.**
- **To act as a balance organ and acceleration director.**

WHY THIS DISTINCTION IS CRITICAL

Most students think of the ear as the "hearing organ" only. The second function — **balance** — is arguably more important for pilots, because **spatial disorientation kills pilots, hearing loss usually doesn't**. The vestibular apparatus inside the inner ear (covered in §42–43) is what tells you which way is up — when it lies to you, you crash.

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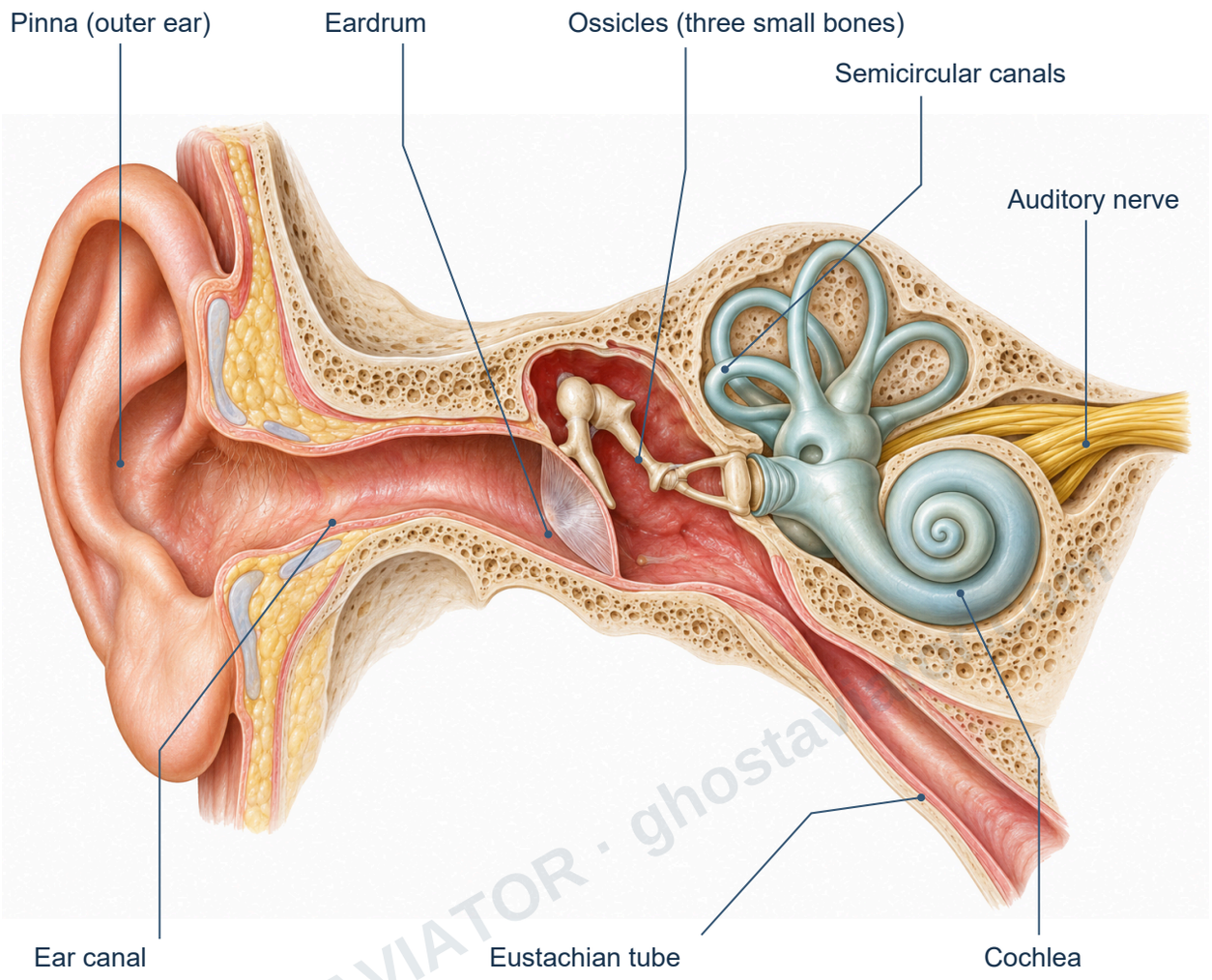

Figure 25.20 *A cutaway teaching view of the ear: the sound-conducting structures and inner-ear structures are connected.*

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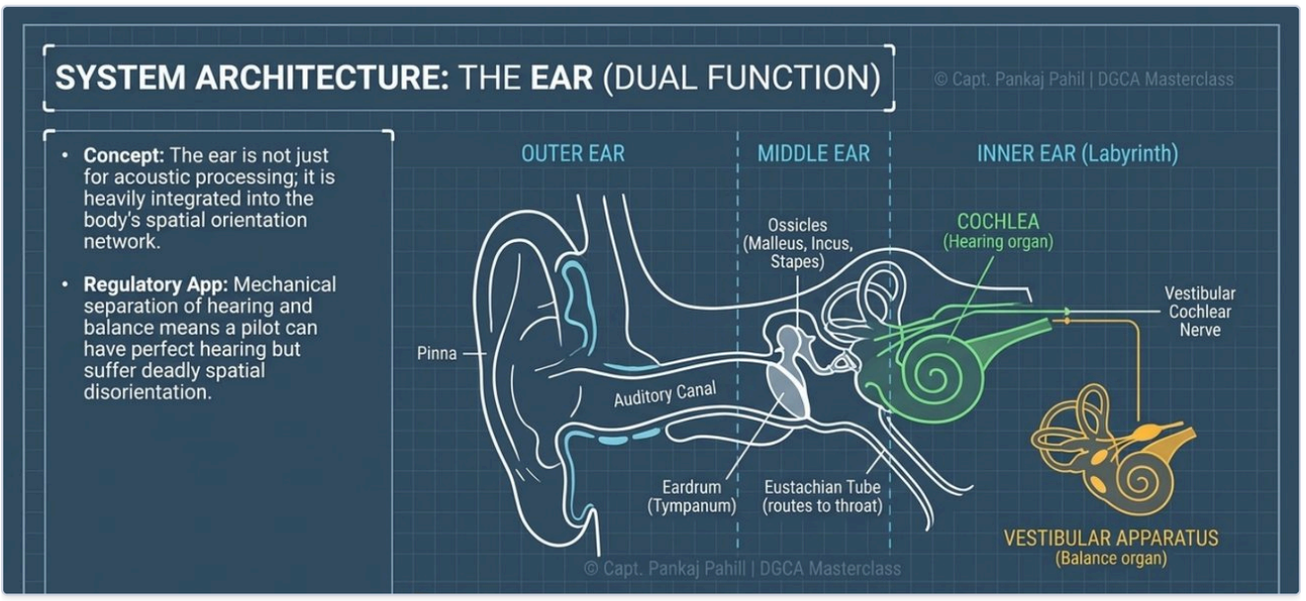

Figure 25.21 *The ear: outer, middle and inner*

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38 The Middle and Inner Ear · The Eustachian Tube

> **EXAM POINT**
>
> **The Tympanum and the Ossicles transmit sound waves to the inner ear.**

The **Tympanum** is the eardrum. The **Ossicles** are the three smallest bones in the body — **Malleus, Incus, Stapes** (Hammer, Anvil, Stirrup). They form a tiny amplifying lever system that converts air pressure waves at the eardrum into mechanical motion at the oval window of the cochlea.

38.1 The Eustachian Tube — pressure equalisation

> **EXAM POINT**
>
> The **Eustachian tube allows pressure in the middle ear to equalize across the ear drum with outside or ambient pressure when climbing or descending**.

> **THE NO-FLY RULE**
>
> **No one should fly if their Eustachian tube is blocked, and they cannot "clear" their ears.**

38.2 Effects of Altitude Change — Clearing the Ears

> **DESCENT IS THE DANGEROUS PHASE**
>
> **It is during DESCENT when difficulty in clearing the ears is most likely to be experienced.**

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WHY DESCENT — THE ONE-WAY-VALVE EFFECT

On **climb**, ambient pressure falls; the higher-pressure middle-ear air *pushes outward* through the Eustachian tube — this happens easily and automatically. On **descent**, ambient pressure rises; the now-relatively-lower-pressure middle-ear air must *draw in* through the Eustachian tube against gravity and tissue swelling. The tube tends to act like a one-way flap valve. If congested (cold, flu, sinusitis), it fails to open and the eardrum is pushed inward by external pressure — **Otis Barotrauma**.

38.3 Otis Barotrauma

DEFINITION

Otis Barotrauma — Stretching of the ear drum caused by the expansion and contraction of gases trapped in the inner ear by a blocked Eustachian tube.

CLEARING TECHNIQUES + "STEP DESCENT"

If you experience problems with pressure equalization during descent:

1. **Swallow deliberately** with the nostrils pinched closed,
2. **Yawn**,
3. **or BLOW DOWN THE NOSE, again with the nostrils pinched closed**. *(This is the Valsalva manoeuvre.)*

If the problem is not resolved, **the rate of descent should be decreased or stopped at intervals to allow more time for pressure to equalize. This is also known as STEP DESCENT.**

Pilot may resort to even CLIMB if pressure persists. Re-climbing reduces external pressure and helps un-block the tube.

Remember, never fly if your Eustachian tube becomes swollen or blocked and you cannot clear your ears.

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39 Audible Range of the Human Ear

SOUND HAS THREE MAIN QUALITIES

- **Pitch** — which depends on the **frequency** of the vibration.
- **Loudness or Intensity** — which depends on the **amplitude** of the vibrations.
- **Quality** — which can be either **harmonious or just plain noisy**.

FREQUENCY RANGE

The range of **pitch or frequency of sounds that a fit young person can hear lies between 16 and 20,000 Hertz**, or cycles per second. **Detectable sound range also depends on loudness.**

16Hz **LOWER LIMIT OF AUDIBLE FREQUENCY**

20,000Hz **UPPER LIMIT (FIT YOUNG PERSON)**

3qualities **PITCH · LOUDNESS · QUALITY**

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40 Noise & Hearing Loss

FOUR CAUSES OF HEARING IMPAIRMENT

Hearing impairment can arise because of:

1. **Exposure to loud noise.**
2. **Physical damage to the hearing mechanism.**
3. **Advanced age.**
4. **A buildup of wax.**

40.1 Conductive Deafness

EXAM POINT

Conductive Deafness — is caused by damage to the ossicles or the ear drum.

40.2 Noise-Induced Hearing Loss — the 80 / 90 dB thresholds

EXACT DB THRESHOLDS

Noise Induced Hearing Loss:

- **> 80 dB — task performance may be impaired**.
- **> 90 dB — measurable impairment of task performance**.

THE COUNTERINTUITIVE "AROUSAL" EFFECT

However, it has been shown that in some situations performance of VIGILANCE tasks can actually be BETTER in high noise levels than in low noise levels. This is because noise increases AROUSAL and can move the individual into the optimum performance area.

Any prolonged exposure to noise in excess of **90 dB** can end up in **noise-induced hearing loss**. This can **damage the very sensitive membrane in the cochlea**.

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40.3 Presbycusis — Hearing loss with age

EXAM POINT

Presbycusis — is the name given to the deterioration of hearing with advanced age. HIGH TONES are cut off FIRST.

Audible range young	16 – 20,000 Hz
Performance impaired above	80 dB
Measurable impairment above	90 dB
Conductive deafness — site	Ossicles / eardrum
Age-related hearing loss term	Presbycusis

Presbycusis — first to go	High tones
Noise damage site	Cochlear membrane
"Optimum" zone of noise-arousal	~ 70 – 80 dB (implied)

41 Protection of Hearing

EXAM POINT

Noise induced hearing loss can be avoided or reduced to a minimum by wearing suitable EAR PROTECTORS.

Always protect your ears if you know you are going to be exposed to excessive noise. In the cockpit, use the best quality HEADSET you can afford in order to reduce background noise.

PRACTICAL COCKPIT APPLICATION

- **Active Noise Reduction (ANR) headsets** — provide both passive (foam seal) and active (anti-phase electronic cancellation) noise attenuation. Reduce cockpit noise by 15–25 dB.
- **Maintenance crews on ramp** — must use over-the-ear muffs near running engines (105–130 dB at engine intakes).
- **Earplugs alone** — useful for passengers; usually inadequate for sustained cockpit duty.
- **Cumulative exposure** — even short bursts of 130+ dB (gunfire, jet blast) can cause permanent damage. Each pilot's hearing is a non-renewable career asset.

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42 The Ear and Balance · Orientation

DEFINITION

"Orientation" refers to a human being's ability to maintain equilibrium and to interpret the body's position in space. The ear also provides us with our sense of balance.

VISION IS PRIMARY

The primary and most reliable sense of spatial orientation is EYESIGHT.

The balance sensors situated in the ear provide us with a **SECONDARY system**.

WHY THIS HIERARCHY MATTERS

This is the single most important fact in spatial-disorientation training. The eyes are primary; the ear's balance organ is secondary. When the two disagree — as happens in IMC (Instrument Meteorological Conditions) where the eyes lose external reference — **the pilot must trust the instruments (which calibrate to true vertical), not the inner ear (which lies)**. This is the foundational principle of IFR (Instrument Flight Rules).

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43 The Balance Mechanism

43.1 Proprioception

DEFINITION

Proprioception: is the **awareness of the body in space**. It is the use of **joint position sense and joint motion sense** to respond to stresses placed upon the body by alteration of posture and movement.

43.2 The Vestibular Apparatus — Otoliths & Semi-Circular Canals

MEMORISE THE COMPONENTS & FUNCTIONS

The **Vestibular apparatus (Otoliths + Semi-circular canals)** helps maintain spatial orientation.

- **The Otoliths detect LINEAR ACCELERATION.**
- **The Semi-Circular Canals detect ANGULAR ACCELERATION.**

OTOLITHS — linear acceleration	The otoliths are tiny **calcium-carbonate crystals embedded in a gelatinous membrane** sitting on top of hair cells in the utricle and saccule. • **Sense linear acceleration**: forward/back (takeoff, braking), up/down (turbulence, climb/dive) • Also sense **gravity at rest** (which way is "down") • Can be tricked by sustained acceleration → "**somatogravic illusion**"
SEMI-CIRCULAR CANALS — angular acceleration	Three fluid-filled loops at right angles to each other (one per axis: roll, pitch, yaw). Each ends in an **ampulla** containing a cupula with hair cells. • **Sense angular acceleration**: turning, pitching, rolling • Only respond to **change** in rotation, not steady rotation • Can be tricked by sustained turns → "**the leans**" / spiral dive illusion

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43.3 The Three Orientation Factors — DGCA's "Big Three"

THE THREE FACTORS

Human beings maintain spatial orientation using a combination of three factors:

1. **The sense of vision.**
2. **The Vestibular Apparatus.**
3. **The Somatosensory system** ("seat of the pants" feeling) / **G-Force**.

RELIABILITY RANKING (MEMORISE VERBATIM)

- **The most reliable sense is the SENSE OF VISION.**
- Our **vestibular apparatus can detect accelerations but cannot determine what position we are in if no acceleration is present**.
- **The somatosensory system is NOT RELIABLE AT ALL.**
- **Neither is the ear's balancing mechanism sufficiently reliable for a pilot to maintain spatial orientation using this sense alone.**

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Source of information	Use and limitation
Vision	Provides orientation from outside references; in instrument conditions, use and cross-check the flight instruments.
Vestibular system	Senses acceleration and can produce compelling false motion sensations.
Somatosensory system	Pressure, muscle and joint cues can be misleading in flight.

Figure 25.22 *Orientation uses several senses. Bodily sensations can conflict with the instruments; use trained instrument cross-check procedures.* *Existing chapter explanation; human-performance sensory-orientation sections.*

43.4 How Angular Acceleration is Sensed

THE FLUID-FLOW MECHANISM

- **Both the Otoliths Organs and the Semi-Circular Canals send signals to our brain by means of impulses arising from the body being subjected to accelerations.**
- **Fluid flow occurs when the body is subject to angular acceleration.**
- The flow takes place in the **OPPOSITE direction to the acceleration**, moving **sensory hairs** which send signals to the brain that the body is in motion.

THE CRUCIAL FAILURE MODE

There is NO FLUID FLOW when the body is at rest or if linear or turning movement is taking place at a STEADY SPEED. In these situations, **the vestibular apparatus alone CANNOT DETECT MOTION**.

Only the eyes and instruments tell a pilot that he is in a steady turn.

WHY THIS IS THE FOUNDATION OF "THE LEANS"

In a steady-rate turn, the canal fluid catches up with the canal walls and stops flowing. The sensory hairs return to centre — the brain thinks **"no turn is taking place"**. If the pilot then rolls out, the fluid *now* flows (in the opposite sense to the roll-out) → the brain interprets this as *a turn in the opposite direction*. The pilot, feeling that "wing-low" sensation, tends to re-roll back into the original turn = **graveyard spiral**. The cure: trust the attitude indicator, not the inner ear. (Illusions are covered later in this chapter.)

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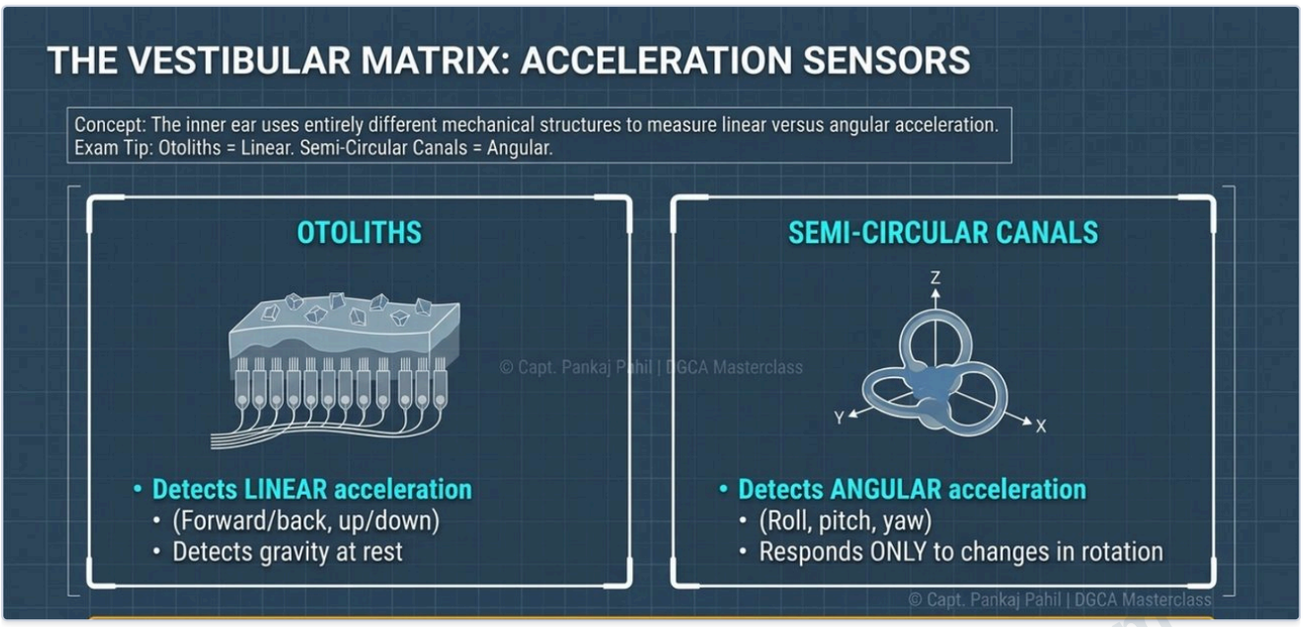

Figure 25.23 *Otoliths sense linear acceleration; canals sense turning*

44 Conflicts Between Ears & Eyes — Illusions & Disorientation / Vertigo

EXAM POINT

Various complex motions and forces and certain visual scenes encountered in flight can create ILLUSIONS of motion and position.

Spatial disorientation from these illusions can be prevented ONLY by visual reference to RELIABLE, FIXED POINTS on the ground or to FLIGHT INSTRUMENTS.

45 Illusions — Introduction

DEFINITION

In aviation any mismatch between what we sense and what we expect is an illusion.

Because of the lack of stable visual references and the erroneous mental models that may be produced, the **pilot is at a disadvantage**.

UNIVERSALITY AND DANGER

- **Illusions may occur during all stages of the flight, and to pilots of every experience and skill level.**
- The pilot, therefore, should be **aware of the possibility of misinterpreting the information received**.
- **Visual illusions are particularly dangerous in aviation, as we normally consider our visual input to be the most reliable of our senses.**

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45.1 Atmospheric Perspective

EXAM POINT

The pilot who has flown mostly in relatively polluted air may use "atmospheric perspective" as a cue to range. If he then flies in a very clear atmosphere he may believe distant objects, because of their clarity, to be much closer than they actually are.

PRACTICAL EXAMPLE — FLYING OVER THE DESERT / OVER THE OCEAN / ABOVE AN INVERSION

A pilot accustomed to hazy/polluted urban airspace develops a mental ruler: "*if the runway looks sharp, it must be close*". On a crystal-clear day above the desert or over an ocean inversion, that same runway looks "close" at 5+ miles — the pilot starts the descent too early, undershoots, or arrives high & fast at the threshold. The illusion runs the other way too: in a polluted layer, the runway looks "farther" → late descent, dive at the runway.

45.2 Laws of Perceptual Organisation — Gestalt Theory

EXAM POINT

The **"Laws of perceptual organization of Gestalt Theory"** deal with factors such as:

- **Proximity**,
- **Continuity**,
- **Similarity**,
- **Symmetry**,
- **Simplicity**, and
- **Closure**.

Gestalt laws formulate basic principles governing **how objects are organised and perceived**.

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WHY GESTALT MATTERS IN COCKPIT DISPLAY DESIGN

The reason a glass cockpit groups *related* data (engine N1/EGT/Fuel-flow on one column; flight-attitude/heading/altitude as a single PFD) is **Gestalt proximity & similarity**. The pilot's brain naturally fuses these into one "engine block" or one "primary flight" gestalt — reducing scan workload. Misdesigned displays violate Gestalt principles and slow the pilot's interpretation under stress.

Master cheat-sheet — numbers & facts

Every numeric / regulatory / definitional fact from this section

Parameter	Exact Value / Fact	Where
Number of DGCA-listed pilot uses of colour	**7**	§34
Cause of colour blindness	**Defect in colour-sensitive cones**	§35
Reaction time depends on	**Closing relative speed**	§36
Two functions of the ear	**Hearing + Balance/accel.**	§37
Three sections of ear	**Outer · Middle · Inner**	§37
Three ossicles (smallest bones)	**Malleus · Incus · Stapes**	§38

Parameter	Exact Value / Fact	Where
Difficulty clearing ears — when	**During DESCENT**	§38.2
Otis Barotrauma — cause	**Blocked Eustachian tube**	§38.3
Three ear-clearing techniques	**Swallow · Yawn · Valsalva**	§38.3
If clearing fails — descent technique	**STEP DESCENT**	§38.3
Three qualities of sound	**Pitch · Loudness · Quality**	§39
Pitch depends on	**Frequency**	§39
Loudness depends on	**Amplitude**	§39
Audible frequency range (young)	**16 – 20,000 Hz**	§39
Number of causes of hearing impairment	**4**	§40
Conductive deafness — site	**Ossicles or eardrum**	§40.1
Task performance impaired above	**80 dB**	§40.2
Measurable impairment above	**90 dB**	§40.2
Noise-induced damage site	**Cochlear membrane**	§40.2
Age-related hearing loss term	**Presbycusis**	§40.3
Presbycusis — first to disappear	**High tones**	§40.3
Primary sense of spatial orientation	**EYESIGHT**	§42
Ear balance system rank	**Secondary**	§42
Proprioception	**Awareness of body in space**	§43.1
Vestibular apparatus components	**Otoliths + Semi-circular canals**	§43.2
Otoliths detect	**LINEAR acceleration**	§43.2
Semi-circular canals detect	**ANGULAR acceleration**	§43.2
Three orientation factors	**Vision · Vestibular · Somatosensory**	§43.3
Most reliable orientation sense	**VISION**	§43.3
Least reliable orientation sense	**Somatosensory ("seat of pants")**	§43.3
Fluid flow direction vs acceleration	**OPPOSITE**	§43.4
Vestibular fails in steady-speed motion	**No fluid flow → no signal**	§43.4
Steady-turn detection requires	**Eyes + instruments only**	§43.4
Definition of an illusion	**Mismatch sense vs expect**	§45
Gestalt principles — count	**6**	§45.2

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46 Illusions When Taxiing — Relative Movement

GENERAL WARNING

We must use extreme caution to ensure that we do not construct our mental model according to our wishes or desires.

TAXI ILLUSION

When taxiing into a headwind the blowing snow will give the illusion that the aircraft is taxiing faster than it actually is.

COCKPIT CROSS-CHECK

The fix is simple — verify against the **ground-speed display on the FMS/GPS**, the **brake-pedal feel**, or by looking at **fixed reference markers** (runway centreline, taxiway edge lights). Never rely on the relative motion of mobile elements (snow, rain droplets, dust) as a speed reference.

47 Illusions on Take-off — Somatogravic Illusion

EXAM POINT

Acceleration gives the pilot an impression of the nose of the aircraft pitching UP.

WHY — THE OTOLITH PHYSICS

The **otoliths** (§43.2) sense linear acceleration. During take-off, strong forward acceleration deflects the otolith hair cells in the same direction *that gravity would deflect them if the head were tilted back*. The brain cannot tell these apart — so it interprets forward acceleration as **nose-up pitch**.

The reflex response of the disoriented pilot is to **push the nose down** — into the ground. This is the classic **Somatogravic Illusion** killer on night/IMC departures, and is the principle behind a string of fatal accidents on dark-night take-offs over water.

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48 Outside References — Six False-Impression Triggers

LEARN THIS LIST

Outside references may give a false impression within the cockpit:

1. **Immediately after take-off.**
2. **Over water.**
3. **In hilly terrain.**
4. **Gently sloping terrain.**
5. **A bank of sloping cloud.**
6. **The ground sloping down on the approach.**

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WHY THESE SIX ARE DANGEROUS — COMMON THREAD

Each of these scenarios **removes or distorts a clear, unambiguous horizon**. The pilot's brain — which is wired to use the horizon as its vertical reference — substitutes **whatever long, straight, dark-light boundary it can find** (a coastline, a hill ridge, a cloud bank top). If that boundary is sloped, the pilot subconsciously aligns the wings with *it* instead of with the true horizon — and ends up in a banked or pitched attitude without realising it.

49 Illusions in the Cruise — Auto-genesis & Vertical Separation

49.1 Auto-genesis (the autokinetic effect)

EXAM POINT

Auto genesis. Staring at an isolated and stationary light when other visual references are inadequate or absent, may cause auto-kinetic movements of the eyes.

In the dark, a static light will appear to move about when stared at for many seconds. The disoriented pilot will lose control of the aircraft in attempting to align it with the light.

PRACTICAL EXAMPLE & DEFENCE

A single distant aircraft light, a star, a ground beacon, or the planet Venus on a clear dark night — fixate on it for 6–12 seconds and it appears to drift, dance, or even rotate. The pilot may attempt to follow this "moving" target with control inputs. The defence: **do not stare**. Scan your visual field continuously, with brief stops on any reference. If a single light source *seems* to move, glance away briefly and verify against the attitude indicator and other lights.

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49.2 Vertical Separation

EXAM POINT

A common problem in flight is the evaluation of the relative altitude of approaching aircraft and the assessment of a potential collision risk.

WHY VISUAL ALTITUDE ASSESSMENT OF ANOTHER AIRCRAFT IS UNRELIABLE

At distance, an aircraft is reduced to a point of light or a tiny silhouette. There is no reliable monocular cue to its altitude relative to your own. A constant-bearing approaching target (no relative motion across your windscreen) is the classic collision-course signature — and the most difficult to detect visually. Modern transport aircraft therefore rely on **TCAS** (Traffic Collision Avoidance System) and ATC altitude verification, not on the pilot's eye, for vertical separation.

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50 Approach and Landing — Three Stages

EXAM POINT

In the final stages of a flight the pilot has to cope with the most critical visual tasks, and these may be divided into 3 stages:

1. **Initial judgment of glide slope**
2. **Maintenance of the glide slope**
3. **Ground proximity judgments.**

50.1 Initial Judgment of Appropriate Glide Slope

THE "VISUAL ANGLE"

Visual Angle — To judge the approach path, the pilot is attempting to establish an angle. **This angle is the "Visual Angle" and is measured at the pilot's eye DOWN FROM THE HORIZON to the visual aiming point on the runway.**

50.2 Width & Slope of Runways

WIDTH OF RUNWAYS ILLUSION

The width of the runway may also cause incorrect height judgments on the final approach.

A pilot used to a **standard width runway** may, when approaching an **unfamiliar NARROW runway**, **judge he is too high and therefore round out too low on approach**.

THE INVERSE — WIDE RUNWAY ILLUSION

By the same physics, an unusually **WIDE** runway looks **closer / lower than it is** → pilot perceives "too low", rounds out too high, lands long and hard.

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50.3 The Black Hole Effect

LEARN THIS LIST

The Black Hole Effect — The absence of visual cues (such as **night-time approaches over desert or unlit water**) leads to an illusion that the **aircraft is TOO HIGH**, as a result the approach path may be flown at **too shallow an angle**, the aircraft may **touch down SHORT of the runway**. This specific illusion is often called the **"black-hole illusion"**, due to the apparent visual "black hole" between the aircraft and the runway.

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50.4 Visual Illusions on Approach

GROUND LIGHTING ILLUSIONS

Lights along a straight path, such as a road, and even **lights on moving trains can be mistaken for runway and approach lights**.

Bright runway and approach lighting systems, especially where few lights illuminate the surrounding terrain, may create the illusion of **less distance to the runway**. The pilot who does not recognize this illusion will **fly a HIGHER approach**.

Conversely, the pilot **overflying terrain which has few lights to provide height cues** may make a **LOWER-than-normal approach**.

(a) Shallow Approaches	Triggers that **create the illusion of being HIGH → resulting in SHALLOWER approach**: • **Up-slope runway or terrain** • **Narrower than usual runway** • **Feature-less terrain** • **Rain on the wind screen** • **Haze** Result: pilot thinks "too high" → reduces descent rate → flies shallow → may land short or hit obstacles.
(b) Steep Approaches	Triggers that **create the illusion of being LOW → resulting in STEEPER (HIGH) approach**: • **Down-sloping runway or terrain** • **Wider than usual runway** • **Bright runway / approach lights** Result: pilot thinks "too low" → increases descent rate → flies steep → may land long or run off the end.

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Runway / approach features → perceived position → pilot's wrong reaction

Visual Feature	Pilot perceives	Reaction	Outcome
Narrower than usual runway	"Too high"	Reduces rate of descent	Shallow approach · land short / low round-out
Wider than usual runway	"Too low"	Increases rate of descent	Steep approach · land long / high round-out
Up-sloping runway/terrain	"Too high"	Shallow approach	Low / land short
Down-sloping runway/terrain	"Too low"	Steep approach	High / land long
Featureless terrain	"Too high"	Shallow	Low / land short (black-hole family)
Rain on windshield / haze	"Too high"	Shallow	Low / land short
Bright runway / approach lights	"Too low" (looks close)	Steep	High / land long

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Visual Feature	Pilot perceives	Reaction	Outcome
Few terrain lights below	"Higher than reality"	Shallow	Lower than normal approach
Black hole (no cues between)	"Too high"	Shallow	Land short

50.5 Maintenance of the Glide Slope

AIMING PILOT & AIRCRAFT ATTITUDE PITCH ANGLE

Once **established on the glide path, it is relatively easy to visually maintain the glide path by keeping the aiming point at a FIXED POSITION on the windscreen**.

INADVERTENT SPEED LOSS TRAP

On the approach, with an **inadvertent speed loss** and a **gradual loss of altitude**, the **runway could remain in the same position on the windscreen**, giving the impression of a safe approach, **until touch down occurs some distance BEFORE the threshold**.

TEXTURE AND TEXTURE FLOW

As long as visual texture flows AWAY FROM the aiming point and the visual angle between this point and the horizon remains constant, the approach will progress normally.

("Texture flow" = the pattern of ground objects (runway markings, taxiway lights, surrounding terrain detail) appearing to flow outward and past you. If the flow point sits ON the aiming spot, you are tracking towards it. If the flow seems to converge ahead of the aiming spot, you are undershooting; behind it, overshooting.)

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50.6 Ground Proximity Judgments

HEIGHT-ASSESSMENT CUES

The pilot will use a number of cues in his height assessment on the final stage of the approach, among which will be:

- **That the apparent speed of objects on the ground will INCREASE as the height reduces.**
- **That the SIZE of objects, such as runway lights etc., will INCREASE with decreasing distance.**
- **That the apparent WIDTH of the runway will INCREASE.**
- **That the TEXTURE of the ground will CHANGE.**

Stage 1 of approach visual task	Initial glide-slope judgment
Stage 2 of approach visual task	Maintenance of glide slope
Stage 3 of approach visual task	Ground-proximity judgments
Visual angle reference	Horizon → aiming point
Aiming-point position rule	Fixed on windscreen

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Number of ground-proximity cues	4 (speed · size · width · texture)

51 Protective Measures against Illusions

LEARN THIS LIST

Organized formal training is the best protective measure against illusions. It is recommended that this should be used to educate pilots to recognize:

1. **The illusions are natural phenomena.**
2. **Know the different types of illusions and their effects.**
3. **That the supplementation of other visual cues with information from other sources is the most effective counter to the effects of illusions.**
4. **The need for comprehensive flight briefing should the occurrence of illusions be known to exist or are anticipated at particular geographic locations.**
5. **Special care must be taken during accelerations and particularly during instrument flying.**
6. **Head movements, fatigue, night and conditions of reduced visibility are all factors that can promote visual illusions.**

52 Disorientation / Vertigo — The Master Rule

THE MOST IMPORTANT SENTENCE IN THIS ENTIRE SECTION

If you suspect disorientation, concentrate on and BELIEVE in aircraft Instruments in IMC. If in VMC, look out at the HORIZON.

WHY THIS IS THE CARDINAL RULE

Every other illusion-management technique in §51 is preparation. **This sentence is the action.** The moment you suspect you are disoriented, you have a binary choice tree:

- **In IMC (cloud, night, no horizon)** → *scan and obey the instruments*. The artificial horizon (attitude indicator), altimeter, VSI, ASI, heading indicator, and turn coordinator are unaffected by illusions and tell you what's true. Override the seat-of-the-pants feeling.
- **In VMC (clear day with horizon)** → *look out at the real horizon* and re-anchor your spatial reference to it. Don't fixate on instruments.

There is no third option. There is no "wait it out". The disorientation will not self-resolve while you ignore both the instruments and the horizon.

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53 Air / Motion Sickness

EXAM POINT

Air/Motion Sickness — this mismatch between vestibular and visual sensory input is the PRIMARY CAUSE of spatial disorientation, and indeed of motion sickness.

Vibrations within the frequency band of 1/10 to 2 Hertz are a factor contributing to airsickness, because they upset the vestibular apparatus.

SYMPTOMS OF MOTION SICKNESS

- **Nausea and fear**
- **Hyperventilation**
- **Vomiting**
- **Pallor**
- **Cold sweating**
- **Headache**
- **Depression**

0.1–2Hz **VIBRATION BAND THAT UPSETS THE VESTIBULAR SYSTEM**

7symptoms **DGCA-LISTED SYMPTOMS OF MOTION SICKNESS**

V vs V **VESTIBULAR VS VISUAL MISMATCH — ROOT CAUSE**

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54 Linear-Acceleration Illusions — Head-Up & Head-Down

(b)	**Head-Up Illusion** **Definition** *This illusion involves a sudden forward LINEAR ACCELERATION during level flight where the pilot perceives that the nose of the aircraft is PITCHING UP.* **Perceived:** nose pitching UP. **Triggers** Night takeoff from a well-lit airport into a dark sky, OR application of full power during a missed instrument approach. **Reflex (wrong) response:** the pilot would be to **push the control forward to pitch the nose of the aircraft DOWN** → flies into the ground.
(c)	**Head-Down Illusion** **Definition** *The head-down illusion involves a sudden linear DECELERATION (e.g., application of air brakes, lowering flaps, decreasing engine power) during level flight.* **Perceived:** nose pitching DOWN. **Reflex (wrong) response:** the pilot's response would be to **RAISE the nose of the aircraft, which may lead to a STALL if executed during a low-speed final approach**.

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UNDERLYING PHYSICS — SAME AS SOMATOGRAVIC ILLUSION (§47)

Both illusions are **somatogravic** — otolith deflection by linear acceleration interpreted as a tilt of the head/body relative to gravity. The general rule:

- **Acceleration FORWARD** → otoliths sense as **HEAD-BACK / NOSE-UP** → reflex push forward (Head-Up illusion / take-off somatogravic).
- **Deceleration (acceleration BACKWARD)** → otoliths sense as **HEAD-FORWARD / NOSE-DOWN** → reflex pull back (Head-Down illusion).

Both kill by reflex-induced loss of aircraft attitude. **Defence in both: scan the attitude indicator.**

55 Elevator Illusion

EXAM POINT

An abrupt UPWARD vertical acceleration, usually by an UPDRAFT, can create the illusion of being in a CLIMB. The disoriented pilot will push the aircraft into a NOSE LOW attitude.

An abrupt DOWNWARD vertical acceleration, usually by a DOWNDRAFT, has the opposite effect, with the disoriented pilot pulling the aircraft into a NOSE UP attitude.

WHEN THIS KILLS

This is the family of illusions that traps pilots in **convective turbulence (CB cells, mountain wave, microbursts)**. The vertical kicks of strong updraft/downdraft trick the vestibular sense into perceiving a sustained climb/dive. The pilot's "correction" — pushing or pulling — superimposes on whatever the vertical motion already is, producing wild pitch excursions and potentially loss of control or wing stall.

Defence: in turbulence, fly the attitude (pitch ~3-5° up, wings level), not the airspeed. Let the altimeter swing. Avoid sudden corrective inputs based on seat-of-pants sensation.

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56 False Horizon

EXAM POINT

Sloping cloud formations, an obscured horizon, a dark scene spread with ground lights and stars, and certain geometric patterns of ground light can create illusions of NOT BEING ALIGNED CORRECTLY with the actual horizon. The disoriented pilot will place the aircraft in a DANGEROUS ATTITUDE.

THE FOUR DGCA-LISTED FALSE HORIZON TRIGGERS

1. **Sloping cloud formations**
2. **An obscured horizon**
3. **A dark scene spread with ground lights and stars**
4. **Certain geometric patterns of ground light**

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THE "GROUND LIGHTS = STARS" TRAP

On a clear dark night over remote terrain, scattered ground lights below merge visually with stars above — the boundary between them (the true horizon) disappears. The pilot's brain then picks the brightest cluster of ground lights and treats it as "ground" or "below" — but if those lights are on a hillside above the aircraft's altitude, the brain has been told the wrong direction is down. This is one of the most lethal night-VFR illusions.

57 Vection Illusions

"Vection" is the sense of **self-motion induced by viewing motion of the surroundings**. Three forms are recognised in the DGCA syllabus:

57.1	**Circular Vection** *Is the sensation of self-rotation induced by viewing a surround rotating about the observer's vertical axis.* *(Example: standing still in a slowly spinning room — you feel as if you are rotating instead.)*
57.2	**Linear Vection** *During linear vection, the observer feels like they have moved forwards or backwards and the stimulus has stayed stationary.* *(Example: seated in a stationary train when the adjacent train moves — you feel your train is moving.)*
57.3	**Roll Vection** *During roll vection, the observer feels like they have rotated around the line of sight and the disk has stayed stationary.* *(Example: the brain interprets a rolling visual surround as the body rolling, even when the body is stationary.)*

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58 Geometric Perspective Illusion

SHORT AND BROAD DEFINITION

Geometric Perspective Illusion — any misinterpretation by the visual system of a figure made of STRAIGHT or CURVED lines.

WHY IT MATTERS IN AVIATION

Runway markings, approach light bars, painted aprons, taxiway centrelines, even cloud-bank edges — all are made of straight or curved lines. The brain extracts depth, distance, and orientation from these lines using assumptions (parallel lines converge, near things look bigger, etc.). Anything that breaks those assumptions (e.g. a runway painted with a converging perspective deception, or an upslope that looks like a level runway from far away) creates a geometric-perspective illusion.

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59 The Stroboscopic Effect (Flicker Vertigo)

DEFINITION

An additional type of vertigo is known as **Flicker vertigo**. Light, flickering at certain frequencies, from **four to twenty times per second (4 – 20 Hz)**, can produce **unpleasant and dangerous reactions** in some people.

REACTIONS

These reactions may include:

- **Nausea**
- **Dizziness**
- **Unconsciousness**
- **Even reactions similar to an EPILEPTIC FIT**

CAUSES IN FLIGHT

- In a **single-engine propeller aeroplane heading into the sun**, the propeller may cut the sun to give this flashing effect, **particularly during landing when the engine is throttled back**.
- These undesirable effects may be avoided by **not staring directly through the prop for more than a moment**, and by making **frequent but small changes in RPM**.
- The **flickering light traversing helicopter blades** has been known to cause this difficulty.
- The **bounce back from rotating beacons on aircraft which have penetrated clouds** — if the beacon is bothersome, **shut it off during these periods**.

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RECOMMENDED PREVENTATIVE ACTIONS

If a member of the crew or a passenger shows symptoms of the Stroboscopic Effect, the recommended preventative actions are:

1. **Turn the aircraft away from the sun.**
2. **Move the person affected to the shade.**
3. **Make the individual close eyes.**

60 Disorientation Summary — Remember

SIX SUMMARY STATEMENTS

1. **Without visual aid, a pilot often interprets CENTRIFUGAL FORCE as a sensation of RISING or FALLING.**
2. **Abrupt head movement during a prolonged constant-rate turn in IMC or simulated instrument conditions can cause pilot disorientation.** *(This is "Coriolis illusion" / somatogyral cross-coupling.)*
3. **A sloping cloud formation, an obscured horizon, and a dark scene spread with ground lights and stars can create an illusion known as FALSE HORIZONS.**
4. **An abrupt change from climb to straight and level flight can create the illusion of TUMBLING BACKWARDS.**
5. **A rapid acceleration during takeoff can create the illusion of being in a NOSE-UP attitude.**

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6. **Symptoms of hypoxia may be difficult to recognize BEFORE the pilot's reactions are affected.**

61 Acceleration and "G" Forces — Introduction

DEFINITION OF 1 G

Flying can expose the human body to conditions for which it is not naturally suited. On the ground, the body is subject to **normal gravitational acceleration: "1G". This is 32 ft/sec² or 9.81 m/sec²**. The reaction of the earth's surface to this acceleration **gives us the sensation we call "weight"**.

A pilot will experience 1 G in straight and level flight.

61.1 Load Factor at Bank Angles

LOAD-FACTOR NUMBERS (MEMORISE)

- At **60° of bank** a pilot is subject to an acceleration of **2 G** acting vertically through his seat. His weight will also **increase by a factor of 2**. This is called the **LOAD FACTOR**.
- In a **70° level turn**, the load factor increases to **3**.

1 G **STRAIGHT & LEVEL FLIGHT (32 FT/S² · 9.81 M/S²)**

2 G **60° LEVEL BANK TURN**

3 G **70° LEVEL BANK TURN**

AEROBATIC CERTIFICATION

A typical **light aircraft cleared for aerobatics** would be **stressed to withstand positive load factors of up to 6**.

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61.2 Counter-measures against adverse G-effects

EXAM POINT

The **adverse effects of increased "G" can be delayed or relieved by:**

- **Tensing the thigh and stomach**, and
- **Easing off the backward pressure on the control column**.

PHYSIOLOGICAL BACKGROUND

Under high positive G, blood pools in the lower body — **cerebral blood pressure drops**. Early symptoms are **greyout** (loss of peripheral vision), then **blackout** (loss of central vision), then **G-LOC** (G-induced Loss of Consciousness). Tensing the lower body's muscles (the "AGSM" — Anti-G Straining Manoeuvre) raises arterial pressure and forces blood back up to the brain. Easing off the control column reduces the G being demanded.

(More detail in continuation of the chapter.)

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Master cheat-sheet — numbers & rules

Every numeric / regulatory / definitional fact from this section

Parameter	Exact Value / Fact	Where
Outside-reference false-impression triggers — count	**6**	§48
Approach-and-landing critical visual stages	**3**	§50
Visual Angle definition	**Horizon → aiming point on runway**	§50.1
Narrow-runway illusion	**"Too high" → round out too low**	§50.2
Black-hole illusion result	**Touch down SHORT**	§50.3
Black-hole illusion typical scenario	**Night over desert / unlit water**	§50.3
Shallow-approach triggers count	**5**	§50.4
Steep-approach triggers count	**3**	§50.4
Ground-proximity height cues	**4 (speed · size · width · texture)**	§50.6
Protective measures against illusions — count	**6**	§51
Master rule — in IMC	**Believe the instruments**	§52
Master rule — in VMC	**Look at the horizon**	§52
Motion-sickness vibration band	**0.1 – 2 Hz**	§53
Motion-sickness symptoms count	**7**	§53
Primary cause of motion sickness	**Vestibular-visual mismatch**	§53
Head-Up illusion = response	**Push forward → dive**	§54 (b)
Head-Down illusion = response	**Pull back → stall**	§54 (c)
Updraft → perceived as	**Climb (pilot pushes nose down)**	§55
Downdraft → perceived as	**Descent (pilot pulls nose up)**	§55
False-horizon triggers count	**4**	§56
Vection types	**3 (Circular · Linear · Roll)**	§57
Flicker-vertigo frequency band	**4 – 20 Hz**	§59
Flicker-vertigo reactions count	**4**	§59
Flicker-vertigo preventative actions	**3 (away from sun · shade · close eyes)**	§59
Disorientation Summary — REMEMBER count	**6**	§60
1 G value	**32 ft/s² · 9.81 m/s²**	§61
60° bank load factor	**2 G**	§61.1
70° bank load factor	**3 G**	§61.1
Aerobatic light aircraft positive G limit	**+ 6 G**	§61.1
G-relief technique	**Tense thigh + stomach · ease stick**	§61.2

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62 Effects of Increased "G" Manoeuvres

THE SIX EFFECTS (MEMORISE THE ORDER)

During increased "G" manoeuvres the pilot will feel the following effects:

1. **Increase in body weight.**
2. **Mobility is impaired.**
3. **Internal organs displaced.**
4. **Onset of tunnel vision.**
5. **Grey out.**
6. **Possible blackout.**

62.1 The Cascade — From Heaviness to Blackout

1 Body weight ↑

2 Mobility impaired

3 Organs displaced

4 Tunnel vision

5 Grey-out

6 Blackout

WHY EACH EFFECT HAPPENS IN THIS ORDER — PHYSIOLOGICAL CONTEXT

- **Effects 1–3 (heaviness, immobility, organ displacement)** are *mechanical*: with apparent weight multiplied by the load factor, limbs become difficult to lift, the head is forced down/sideways, and abdominal organs sag against the diaphragm.
- **Effects 4–6 (tunnel → grey → black)** are *cerebrovascular*: high positive G pulls blood downward, away from the head. Retinal arterial pressure falls first (cones in the fovea drain → tunnel vision = peripheral vision fails first), then central vision goes (grey-out), then cerebral perfusion fails (blackout / G-LOC).
- The ORDER of the six effects is the standard DGCA exam point — they happen in **this sequence as G increases**.

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Effect of sustained positive Gz	Meaning for the pilot
Blood is displaced towards the lower body	Pressure supplying the eyes and brain may decrease.
Visual impairment may develop	Peripheral vision can narrow, followed by grey-out or blackout.
Further loss of cerebral perfusion	G-induced loss of consciousness may occur.

Figure 25.24 *Positive Gz can reduce blood supply to the eyes and brain. Tolerance varies; the sequence is explanatory, not a fixed G threshold or a recovery checklist.* *Human-performance reference, printed pp 89–90.*

62.2 Tilted-Back Seat — the mitigation

EXAM POINT

A tilted back seat can reduce the chance of a black-out during positive G-manoeuvres.

WHY A RECLINED SEAT HELPS

The taller the column of blood from the heart to the brain along the G-vector, the harder it is for the heart to pump against it. By **reclining the seat**, the vertical distance from the heart to the brain (measured along the seat-to-head axis, which is the G axis in a turn) is *reduced*. The blood column is shorter, the heart has less hydrostatic head to overcome, and cerebral perfusion is preserved at higher G loads. This is exactly why **fighter cockpits (F-16, etc.) recline the pilot's seat ≈ 30°** — to push the G-LOC threshold above where the airframe gives up first.

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63 1.41 G at 45° — the DGCA Reference Number

EXACT LOAD-FACTOR REFERENCE

1.41 G is the acceleration experienced in a LEVEL TURN at 45° angle of bank. In normal flying, angles of bank greater than this are not usually necessary.

Any physical disorder or immoderate consumption of alcohol or tobacco will reduce the body's tolerance to accelerations in excess of 1 G.

Load factor (n) vs bank angle in a level turn — n = 1/cos(bank)

Bank Angle	Load Factor (G)	Notes
0° (level flight)	**1.00 G**	Straight & level
15°	**1.04 G**	Shallow turn
30°	**1.15 G**	Standard rate turn (most light a/c)
45°	**1.41 G**	**DGCA reference** · upper limit of "normal flying"
60°	**2.00 G**	Steep turn · weight doubles
70°	**3.00 G**	Aerobatic turn
75°	**3.86 G**	—
80°	**5.76 G**	—
85°	**11.5 G**	Far beyond airframe / human limits

45° MAXIMUM "NORMAL" BANK

1.41 G LOAD FACTOR AT 45° LEVEL TURN

↓ tol. REDUCED BY DISORDER / ALCOHOL / TOBACCO

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64 Factors Adversely Affecting G-Tolerance

THE SIX FACTORS

1. **Alcohol.**
2. **Smoking.**
3. **Fatigue.**
4. **Excessive heat.**
5. **Obesity.**
6. **Sickness.**

Why each factor reduces G-tolerance

Factor	Physiological Effect
Alcohol	Vasodilation + dehydration + histotoxic hypoxia (§14) — blood pressure regulation degraded.

Factor	Physiological Effect
Smoking	5–8 % O_2-capacity loss + vasoconstriction + CO competition (§13) — less reserve when cerebral flow falls.
Fatigue	Cardiovascular reserve drops; muscle tone (for AGSM) is weakened; reaction time slows.
Excessive heat	Vasodilation for cooling + reduced blood volume from sweating → lower BP under G.
Obesity	Reduced cardiovascular fitness; abdominal fat reduces effectiveness of the AGSM straining manoeuvre.
Sickness	Dehydration, low BP, weakness — all the above multiplied.

65 Negative G & Red-out

TOLERANCE ASYMMETRY

Most pilots can learn to tolerate moderate increases in positive "G", but many find even the smallest exposure to negative "G" to be unpleasant.

WHEN NEGATIVE G IS EXPERIENCED IN FLIGHT

During flight negative "G" is experienced if, after **pulling out of a steep dive**, the **control column is instinctively and firmly moved FORWARD** because the pilot might feel that he has his **nose too high in an attitude that may lead to a stall**.

WHAT NEGATIVE G DOES TO THE BODY

Negative "G" manoeuvres INCREASE the flow of blood TO THE HEAD. Blood pressure there increases, the face becomes very FLUSHED, and the EYES BULGE. The combined effect of these symptoms causes what is described as a "RED OUT".

To relieve symptoms select a normal flying attitude.

POSITIVE G → BLACK-OUT	• Blood pulled **away from** head, into legs • Cerebral perfusion falls • Cascade: tunnel → grey → black • Can progress to G-LOC (G-induced LoC) • **Tolerable up to ~6 G** with training (light aerobatic limit)
NEGATIVE G → RED-OUT	• Blood pushed **into** the head • Cerebral & ocular pressure rise • Face flushes red, eyes bulge • "Red-out" — vision tinged red • **Even small exposures unpleasant** • Mitigation: **return to normal flying attitude**

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WHY PILOTS ARE FAR WORSE AT NEGATIVE G THAN POSITIVE

Human cardiovascular anatomy is built for a 1 G "head-up" world. We have valves and reflexes that, under positive G, pump blood upward against gravity (the baroreceptor + sympathetic reflex). Under **negative G the body has NO compensatory mechanism** — blood floods the head, ocular and intracranial pressure rises, and small blood vessels in the eyes and brain can rupture. Even highly trained aerobatic pilots rarely tolerate more than **−3 G** sustained.

66 Physical Characteristics — Individual Differences

EXAM POINT

Differences among individuals are remarkable. Besides obvious physical differences (e.g., height, weight, age, sex, build, sitting height, functional reach, leg length, shoulder width, strength, co-ordination etc.), people also differ with respect to other traits like:

- **Auditory threshold**,
- **Understanding**,
- **Vestibular (ear senses)**,
- **Smell, touch, kin aesthetic (body feelings)**,
- **G-tolerances**, etc.

These traits are relatively stable over time, they differ across individuals, and they influence human behavior.

DGCA-MEDICAL IMPLICATION

This is the underpinning for **cockpit ergonomics** and **pilot-selection standards**: minimum/maximum sitting heights, reach envelopes for switches and rudder pedals, vision and hearing standards, etc. Each pilot is a different statistical sample — and the certification rules are designed so that the *typical* trained pilot can comfortably operate the aircraft, while extreme outliers may be screened out at the medical or flight-school stage.

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67 Exercise and Weight

67.1 Body Mass Index (BMI)

DEFINITION & FORMULA

A person's Body Mass Index, or BMI, is simply a measure of a person's weight in relation to his height.

Body mass Index (BMI) = Weight (kgs) / Height^2 (m).

NORMAL BMI RANGES

Normal BMI for MEN is 20 to 25 and WOMEN 19 to 24.

BMI = kg/m^2 **WEIGHT IN KG ÷ HEIGHT IN M SQUARED**

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20–25 **NORMAL BMI FOR MEN**

19–24 **NORMAL BMI FOR WOMEN**

WORKED EXAMPLE

A pilot weighs 75 kg and is 1.78 m tall. BMI = 75 / $(1.78)^2$ = 75 / 3.168 ≈ **23.7**. Within normal range for men (20–25).

67.2 Exercise Prescription & "No appetite suppressants"

EXERCISE PRESCRIPTION

Regular exercise is beneficial to general health, but the most efficient way to lose weight is by reducing CALORIC CONSUMPTION.

To reduce the risk of coronary artery disease, **exercise should be done to DOUBLE the resting heart rate for at least 20 minutes, three times a week**.

APPETITE SUPPRESSANTS PROHIBITED

Pilots should NOT try to lose weight by taking appetite suppressants.

Best way to lose weight	Reduce calories
Exercise — target HR	2× resting
Exercise — duration	≥ 20 minutes
Exercise — frequency	3 × per week
Appetite suppressants for pilots	Prohibited

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67.3 Effects of Being Badly Overweight — 13 conditions

LEARN THIS LIST

Being **badly overweight INCREASES a pilot's susceptibility** to the following conditions:

1. **Heart attack.**
2. **Hypertension.**
3. **Hypoxia at lower altitudes than normal.**
4. **General circulation problems.**
5. **Gout (swollen joints).**
6. **Osteoarthritis** (a form of arthritis characterized by gradual loss of cartilage of the joints).
7. **Diabetes.**
8. **The inability to tolerate G forces.**
9. **Problems with joints and limbs.**
10. **Decompression sickness.**
11. **Heavy sweating.**

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12. **Chest infections.**
13. **Varicose veins** (a vein that has become swollen and knotted as a result of faulty valves).

CROSS-LINKS — OBESITY IS THE PILOT'S "COMMON ENEMY"

Note how this single list lights up almost every previous part of the chapter:

- Heart attack & hypertension & circulation problems → §9 / §15 / §61
- Hypoxia at lower altitudes → §10 / §13 (smoker comparison)
- Inability to tolerate G forces → §62 / §64
- Decompression sickness → §19 / §20 (DCS predisposing factor)

The DGCA examiner treats obesity as a *multiplier* of every other risk discussed in the chapter — which is why BMI screening is built into every Class-1 / Class-2 medical.

68 Nutrition and Food Hygiene

THE EMPTY-STOMACH RULE

Never fly on an empty stomach. A balanced diet is the foundation of good health.

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68.1 Sources of Carbohydrates

EXAM POINT

Sources of carbohydrates include:

- **Grains**,
- **Vegetables**,
- **Nuts**, and
- **Fruit**.

68.2 Vitamins

EXAM POINT

Vitamins are organic substances required by the body to function properly. They help process other nutrients to form blood cells.

68.3 Minerals — the Three Majors

EXAM POINT

Minerals are essential to many vital body processes — from building strong bones to transmitting nerve impulses.

The three major minerals include:

- **Calcium — for healthy bones and teeth.**
- **Phosphorous — for body's chemical reaction** and

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- **Iron — For Hemoglobin.**

Mineral	Role
Calcium (Ca)	Healthy bones and teeth
Phosphorus (P)	Body’s chemical reactions
Iron (Fe)	Haemoglobin

Figure 25.25 *Examples of mineral functions: calcium, phosphorus and iron.*

69 Hypoglycemia

THE BRAIN NEEDS FUEL

Just as the brain needs oxygen to function, it also needs fuel to combine with the oxygen and produce energy. That fuel is **blood glucose**, which is carried by the bloodstream and easily passes the **blood–brain barrier**.

THE BRAIN CANNOT STORE GLUCOSE

Glucose is a simple sugar and serves as an immediate source of energy for cells.

The brain CANNOT STORE GLUCOSE and requires a CONTINUOUS SUPPLY to function properly.

All foods containing carbohydrates will raise blood glucose levels.

WHAT HYPOGLYCEMIA DOES TO THE PILOT

Failure to eat properly may result in a shortage of glucose (hypoglycemia), which will produce DECREMENTS IN COGNITIVE FUNCTIONING.

Research has proved that hypoglycemia significantly compromises performance, resulting in LONGER RESPONSE TIMES and LOWER SCORES on cognitive tests.

PRACTICAL PILOT RULE

This is the physiological basis of the "**Never fly on an empty stomach**" rule (§68). A pilot who skipped breakfast — or who is on a crash diet — is operating with reduced cognitive performance. This becomes one of the DGCA-listed causes of **subtle incapacitation** (§27.3 — "Temporary hypoglycemia" is listed verbatim).

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70 Gastric Distress and Contaminated Foodstuffs

70.1 Four Major Causes of Food Contamination

LEARN THIS LIST

Major causes of food contamination are:

1. **Unhygienic (i.e. unclean) food preparation.**
2. **Under cooked or stale meats.**
3. **Unwashed salads, fruit or vegetables.**
4. **Seafood and locally made ice-creams and mayonnaise.**

70.2 Gastroenteritis — Pilot Fitness Rule

PILOT UNFIT EVEN WITH MEDICATION

Pilots suffering from Gastroenteritis are NOT FIT TO FLY, even though they may be taking medicine which is relieving the symptoms.

Due to pressure differential, TRAPPED GASES ESCAPE AT HIGH ALTITUDES resulting in extreme discomfort and sickness.

WHY ALTITUDE MAKES GASTROENTERITIS WORSE — BOYLE'S LAW AGAIN

Recall **Boyle's Law** from earlier in this chapter — gas volume increases as pressure decreases. Gut gases that are tolerable at sea level can expand by ~2× by cabin altitude of 18,000 ft (where atmospheric pressure has halved, §19.2). For a gastroenteritis patient already producing excess gas, this expansion → severe abdominal pain, nausea, vomiting, sudden bowel motion — all of which incapacitate the pilot in the cockpit. Anti-diarrhoeal medication may mask the bowel symptoms on the ground but does nothing to prevent the gas-expansion mechanism in flight.

SYMPTOMS OF GASTROENTERITIS

- **Nausea**
- **Vomiting**
- **Abdominal Pain**
- **Diarrhea**, and
- **Loss of appetite**.

Master cheat-sheet — numbers

Every numeric / regulatory / definitional fact from this section

Parameter	Exact Value / Fact	Where
Effects of increased G — number of effects (in order)	**6** (weight ↑ · mobility ↓ · organs displaced · tunnel · grey · black)	§62
Black-out mitigation	**Tilted-back seat**	§62.2
Load factor at 45° level bank	**1.41 G**	§63

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Parameter	Exact Value / Fact	Where
Maximum "normal" bank angle	**45°**	§63
Number of factors that reduce G-tolerance	**6** (alcohol · smoking · fatigue · heat · obesity · sickness)	§64
Negative G — typical trigger	**Instinctive forward push after steep-dive pullout**	§65
Negative-G symptom name	**RED-OUT**	§65
Red-out signs	**Face flushed · eyes bulge · red vision**	§65
Red-out cure	**Return to normal flying attitude**	§65
BMI formula	**kg ÷ m²**	§67.1
Normal BMI — MEN	**20 – 25**	§67.1
Normal BMI — WOMEN	**19 – 24**	§67.1
Best weight-loss method	**Reduce calories**	§67.2
Exercise — heart-rate target	**2 × resting**	§67.2
Exercise — duration / frequency	**≥ 20 min · 3 ×/wk**	§67.2
Appetite suppressants — pilots	**Do not use**	§67.2
Conditions associated with obesity	**13 DGCA-listed**	§67.3
Empty-stomach rule	**Never fly empty**	§68
Carbohydrate sources count	**4** (grains · vegetables · nuts · fruit)	§68.1
Three major minerals	**Ca · P · Fe**	§68.3
Calcium — function	**Bones & teeth**	§68.3
Phosphorous — function	**Chemical reactions**	§68.3
Iron — function	**Hemoglobin**	§68.3
Brain fuel	**Blood glucose**	§69
Can brain store glucose?	**NO — continuous supply needed**	§69
Hypoglycemia effects	**↑ response time · ↓ cognitive scores**	§69
Food-contamination causes count	**4**	§70.1
Pilot with gastroenteritis on meds	**NOT fit to fly**	§70.2
Reason for fly-unfit (gastro)	**Pressure differential — gas expansion**	§70.2
Gastroenteritis symptoms count	**5**	§70.2

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71 Sinus Blockage

EXAM POINT

If the sinuses are congested, then air trapped there can also produce a painful condition. This sinus block occurs most frequently during DESCENT.

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Slowing or stopping the descent until the pressure inside the sinus equalizes with the external pressure is the best course of action once this condition occurs.

Of course, NOT FLYING when the pilot has a congested sinus is even better.

WHY DESCENT — THE SAME ONE-WAY-VALVE PROBLEM AS EARS (§38.2)

The sinuses are air-filled cavities in the skull (frontal, maxillary, ethmoid, sphenoid) that normally drain through narrow ostia into the nasal cavity. On **climb**, sinus air *expands* and pushes outward — easy. On **descent**, ambient pressure rises and sinus air must *contract* by drawing more air in from the nose. If the ostium is blocked by mucus/inflammation (cold, hay fever, sinusitis), incoming air cannot enter — a vacuum forms in the sinus → severe pain over the forehead, cheeks or behind the eyes ("sinus squeeze" / barosinusitis).

Mitigation in flight: stop or slow the descent and allow time for pressure to equalise. **Prevention on the ground:** if you have a head cold, don't fly.

72 Dental Pain — Barodontalgia

DEFINITION

Reduction in atmospheric pressure may also result in BARODONTALGIA — tooth pain.

72.1 Six Dental Conditions That Cause Barodontalgia

LEARN THIS LIST

This may be caused by a number of dental conditions including:

1. **(a) Dental disease**,
2. **(b) Dental caries (a cavity)**,
3. **(c) Defective tooth restoration**,
4. **(d) Pulpitis**,
5. **(e) Cysts**, and
6. **(f) Impacted teeth**.

MECHANISM — GAS TRAPPED IN TOOTH CAVITIES + BOYLE'S LAW

A small gas pocket under a filling, in a decayed pulp, or inside a sealed cyst **expands** as cabin altitude rises (Boyle's Law again — §2.1). The expanding gas pushes outward against the tooth nerve → sharp, throbbing pain that becomes excruciating at typical airline cabin altitude (6,000–8,000 ft). It often resolves on descent — but if the affected tooth is fractured by the expansion, the pain persists on the ground.

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72.2 The 48–72 Hour Post-Dental Rule

THE NO-FLY RULE

This condition may be avoided by NOT FLYING for 48 – 72 hours following a major dental work.

As with the other problems associated with lower atmospheric pressure, **descending to a lower altitude (and correspondingly higher atmospheric pressure) is often the best antidote.**

48–72hr **NO-FLY WINDOW AFTER MAJOR DENTAL WORK**

6 **DGCA-LISTED DENTAL CAUSES OF BARODONTALGIA**

↓ alt **IN-FLIGHT CURE — DESCEND TO HIGHER-PRESSURE ALTITUDE**

73 Dehydration

EXAM POINT

Besides watching what and when they eat, **pilots should also be aware of the effects of dehydration. Low fluid intake and dehydration have ADVERSE EFFECTS ON COGNITIVE FLIGHT PERFORMANCE of pilots.**

CROSS-LINKS — DEHYDRATION IS EVERYWHERE IN THIS CHAPTER

Dehydration has already been flagged repeatedly in earlier parts:

- **§14 (Alcohol)** — alcohol is a diuretic; "masked hangover" includes dehydration.
- **§23 (Humidity)** — cabin RH below 20 % drives over-all dehydration if fluid intake is inadequate; avoid coffee/tea (diuretics).
- **§28.2 (Fainting)** — "standing up quickly after prolonged sitting especially when HOT or DEHYDRATED" is the first DGCA cause of faint.
- **§64 (G-tolerance)** — excessive heat (sweating + lost blood volume) reduces tolerance.

Practical pilot rule: drink ~250 ml of water every hour during a long sector, regardless of thirst.

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74 Personal Hygiene

EXAM POINT

A high standard of personal hygiene must be practiced if the body is to remain healthy and free from infection.

PRACTICAL IMPLICATIONS FOR CREW

- **Frequent hand washing** — particularly before meals and after lavatory use; crew in pressurised, low-humidity cabins are at elevated risk of respiratory infections.
- **Oral hygiene** — directly reduces barodontalgia risk (§72) by reducing the prevalence of caries.
- **Skin care** — low humidity dries skin; lotion/moisturiser reduces cracking that can introduce infection.
- **Foot hygiene & well-fitting shoes** — long sectors with feet on rudder pedals make foot-fungal infections more common in crew.

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75 Common Ailments

LEARN THIS LIST

The in-flight environment can increase the severity of symptoms which may be minor whilst on the ground. If there is any doubt whatsoever in a pilot's mind about his fitness to fly, he should STAY ON THE GROUND.

THE "MINOR ON GROUND, MAJOR IN FLIGHT" PRINCIPLE

A head cold is a nuisance on the ground; in the air it becomes **otic barotrauma + sinus block**. A gastric upset on the ground is a meal-skip; in the air the trapped gases **expand and incapacitate** (§70.2). A mild headache becomes incapacitating under cabin vibration and G-load. The DGCA-codified rule for any doubt is binary: **stay on the ground**. *The cost of a cancelled flight is always less than the cost of a sick pilot in command.*

76 Drugs and Self-Medication

76.1 The Aviation-Medicine-Specialist Clearance Rule

THE ABSOLUTE RULE

It is absolutely essential that pilots DO NOT FLY as part of the operating crew of an aircraft when taking drugs or medication, UNLESS THEY HAVE BEEN CLEARED TO DO SO BY AN AVIATION MEDICINE SPECIALIST.

RISKS OF SELF-PRESCRIBED MEDICATION

A pilot who flies on **self-prescribed medication** runs the risk of:

- Suffering **side effects**, and
- Faces the **hazards associated with the underlying illness in the in-flight environment** which can **make the symptoms of any illness much more debilitating than they might be on the ground**.

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76.2 Four Reasons Medicines Have Flight-Qualification Consequences

FOUR REASONS

The consumption of **medicines or other substances** may have consequences on qualification to fly for the following reasons:

1. **The disease requiring a treatment may be cause for disqualification.**
2. **Flight conditions may modify the reactions of the body to a treatment.**
3. **Drugs may cause adverse side effects impairing flight safety.**
4. **The effects of medicine do not necessarily immediately disappear when the treatment is stopped.**

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77 Caffeine

CAFFEINE AS A DRUG

Caffeine is probably the most widely used drug in the world. It can easily lead to ADDICTION.

Caffeine is present in coffee, tea, cocoa, chocolate, and fizzy drinks such as cola.

RECOMMENDED MAXIMUM DAILY INTAKE

The recommended maximum caffeine intake per day is approximately 250 – 300 mg corresponding to 2 – 3 cups of coffee.

Approximate caffeine content of common beverages

Beverage	Caffeine (mg)	Notes
Filter coffee (250 ml cup)	~ 100 – 120 mg	2–3 cups → DGCA daily max
Espresso (single shot, 30 ml)	~ 60 – 80 mg	—
Black tea (250 ml cup)	~ 40 – 60 mg	Lower than coffee
Cocoa / hot chocolate	~ 5 – 15 mg	Plus theobromine
Cola (330 ml can)	~ 30 – 45 mg	—
Energy drink (250 ml)	~ 80 – 150 mg	Often exceeded
Dark chocolate (50 g)	~ 30 mg	—

250–300mg **DGCA MAX RECOMMENDED DAILY CAFFEINE**

2–3 **CUPS OF COFFEE = APPROXIMATE DGCA LIMIT**

CAFFEINE IS ADDICTIVE — DGCA WORD

WHY THE DGCA BOTHERS TO SET A CAFFEINE LIMIT

Caffeine is a CNS stimulant. In moderate doses it sharpens alertness — useful on long sectors. In excess it:

- Increases heart rate and blood pressure (interacts with §15 hypertension & §43 cardiovascular reserve under G).
- Is a **diuretic** (interacts with §23 humidity / §73 dehydration).
- Disrupts sleep — undermining the very alertness it was meant to provide; classic withdrawal cycle.
- Causes tremor & anxiety at high doses — degrades fine motor control.

The "2-3 cups" guidance is the dose that keeps you alert without these costs.

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78 Anaesthetics & Analgesics

NO-FLY PERIODS

A pilot should NOT fly for at least:

- **12 hours after a LOCAL anaesthetic**, and
- **48 hours following a GENERAL anaesthetic**.

The more potent forms of ANALGESICS (pain killers) may produce a significant deterioration in human performance.

12hr **AFTER LOCAL ANAESTHETIC — NO FLY**

48hr **AFTER GENERAL ANAESTHETIC — NO FLY**

STRONG ANALGESICS — PERFORMANCE DETERIORATION

CROSS-LINK — ADDS TO THE "WAIT-BEFORE-FLYING" MASTER TABLE

You now hold these wait-before-flying numbers in long-term memory:

- **12 hr** — after rapid decompression (§20.4); after local anaesthetic (§78)
- **24 hr** — after alcohol (§14.1); after diving (§21)
- **48 hr** — after blood donation (§17); after general anaesthetic (§78)
- **48–72 hr** — after major dental work (§72.2)

The DGCA examiner pulls these numbers in MCQs — match the event to the number.

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79 Hypothermia

98.6 °F NORMAL

37 °C — baseline core temperature

< 95 °F HYPOTHERMIA

≈ 35 °C — onset of impaired ability

≤ 82 °F SEVERE

≈ 28 °C — life-threatening

WHAT HYPOTHERMIA DOES

Hypothermia AFFECTS PHYSICAL AND MENTAL ABILITIES.

Shivering makes it possible to combat the cold to a certain extent but USES UP A LOT OF ENERGY.

In a PROLONGED exposure, **shivering will tend to CEASE, and be followed by the ONSET OF APATHY**.

Hypothermia is a potentially LIFE-THREATENING condition that needs EMERGENCY MEDICAL ATTENTION.

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OPERATIONAL RELEVANCE — SURVIVAL IN COLD DITCHING / MOUNTAIN CRASH

For commercial pilots flying over cold water (North Atlantic, Bering Sea) or high-terrain routes, hypothermia is the second-most-common post-ditching cause of death (after drowning). Survival suits, life rafts with thermal protection, and immersion-suit drills exist precisely because **cold water kills faster than the absence of food or water**. The "**cessation of shivering + apathy**" is the danger sign that mild hypothermia is becoming severe.

80 Toxic Hazards

OPENING

Even MILD toxic effects can degrade a pilot's performance and lead to an accident.

Prolonged exposure to toxic influences can damage a person's general health.

Anyone who has been exposed to any toxic hazard should seek MEDICAL ASSISTANCE AS SOON AS POSSIBLE.

SIX TOXIC-HAZARD SOURCES

The following materials may produce toxic hazards:

1. **Furnishings and Baggage.**
2. **Acetone and Turpentine.**
3. **Fuels, Lubricants and propellants.**
4. **Anti-icing Fluid.**
5. **Fire Extinguishing Agents.**
6. **Battery Fumes.**

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Toxic-hazard sources in aviation — DGCA list with context

#	Source	Typical Toxin / Mechanism
1	**Furnishings and Baggage**	Off-gassing of plastics, adhesives, treated fabrics — VOCs, formaldehyde. Also packaged chemicals in baggage holds that leak in flight.
2	**Acetone and Turpentine**	Solvents — CNS depressants on inhalation. Respiratory irritation, headache, dizziness.
3	**Fuels, Lubricants and Propellants**	Jet-A vapours, oil mists, hydraulic fluid mists. Skin irritation, respiratory irritation, possible CNS effects.
4	**Anti-icing Fluid**	Glycols (mono-/di-ethylene glycol) — toxic if ingested, irritant if inhaled as mist.
5	**Fire Extinguishing Agents**	Halons / their replacements — high concentrations displace oxygen, can produce hypoxia (§10). Some thermal decomposition products are corrosive.
6	**Battery Fumes**	Sulphuric acid vapours (lead-acid), or KOH (Ni-Cd), or lithium-fire smoke. Strong respiratory irritants, possibly corrosive.

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81 Dangerous Cargo

LEARN THIS LIST

Pilots must be aware that they must NOT carry certain defined items on board their aircraft. Such items are referred to as DANGEROUS CARGO because of the possibility that their discharge, spillage or breakage may endanger the aircraft and/or crew in flight or on the ground.

THE INDIAN DANGEROUS-GOODS FRAMEWORK: DGCA REQUIREMENTS AND ICAO TECHNICAL INSTRUCTIONS

The Indian DGCA enforces dangerous-goods carriage rules in line with **ICAO Technical Instructions for the Safe Transport of Dangerous Goods by Air** and the **IATA Dangerous Goods Regulations (DGR)**. The nine UN dangerous-goods classes are:

1. **Explosives** (ammunition, fireworks)
2. **Gases** (compressed, liquefied, dissolved — including aerosols)
3. **Flammable liquids** (fuel, paint, solvents)
4. **Flammable solids · spontaneously combustible · dangerous when wet**
5. **Oxidizers · organic peroxides**
6. **Toxic & infectious substances**
7. **Radioactive material**
8. **Corrosives** (acids, alkalies)
9. **Miscellaneous** (asbestos, lithium batteries, magnetised material, dry ice)

(For full HPL/exam preparation, learn these nine classes — DGCA RTR & DGR examinations routinely ask which class a given item belongs to and whether it is forbidden, restricted, or permitted in checked/cabin baggage.)

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WHY THIS SECTION ENDS THE CHAPTER

The chapter has built up systematically: **atmosphere → body systems → environmental stressors → orientation & illusions → fitness & diet → drugs & toxins → dangerous cargo**. The last paragraph closes by reminding the pilot that the threat to the aircraft does not come only from the pilot's own body, but also from **what is loaded into the cabin and cargo holds**. The pilot's command authority extends to ensuring forbidden items are not on board — the pre-flight cargo manifest is a flight-safety document, not paperwork.

Master cheat-sheet — numbers

Every numeric / regulatory / definitional fact from this section

Parameter	Exact Value / Fact	Where
Sinus blockage — most common phase	**DESCENT**	§71
Sinus blockage — best in-flight action	**Slow or stop the descent**	§71
Sinus blockage — best prevention	**Don't fly when congested**	§71
Barodontalgia definition	**Tooth pain from pressure reduction**	§72
Dental causes of barodontalgia — count	**6**	§72.1
Post-dental no-fly window	**48 – 72 hours**	§72.2
Dental pain in-flight antidote	**Descend to higher pressure**	§72.2

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Parameter	Exact Value / Fact	Where
Dehydration effect on pilot	**Adverse cognitive flight performance**	§73
Personal hygiene standard	**HIGH — for infection-free body**	§74
If any doubt about fitness to fly	**Stay on the ground**	§75
Self-medication for pilots	**Prohibited without Aviation Med Specialist clearance**	§76.1
Reasons medicines disqualify — count	**4**	§76.2
Most widely used drug worldwide	**Caffeine**	§77
Caffeine sources — DGCA list	**Coffee · Tea · Cocoa · Chocolate · Fizzy/Cola**	§77
Recommended max caffeine/day	**250 – 300 mg**	§77
That equates to	**2 – 3 cups of coffee**	§77
No-fly after LOCAL anaesthetic	**≥ 12 hours**	§78
No-fly after GENERAL anaesthetic	**≥ 48 hours**	§78
Strong analgesics — effect	**Significant performance deterioration**	§78
Normal body temperature	**98.6 °F**	§79
Hypothermia threshold	**< 95 °F**	§79
Severe hypothermia threshold	**82 °F or lower**	§79
Shivering's role	**Combats cold; uses much energy**	§79
Late-stage hypothermia sign	**Shivering CEASES → APATHY**	§79
Toxic-hazard sources — count	**6**	§80
Action after exposure to toxic hazard	**Seek medical assistance ASAP**	§80
UN dangerous-goods classes (regulatory)	**9**	§81

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Closing Master Cross-Reference Summary

Master "Wait-Before-Flying" reference

The full DGCA "Wait-Before-Flying" rules in this chapter

Event	Minimum Wait	Source
After rapid decompression	**12 hr**	§20.4
After local anaesthetic	**12 hr**	§78
After alcohol consumption (bottle-to-throttle)	**24 hr**	§14.1
After diving (compressed air, >30 ft)	**24 hr**	§21
After blood donation	**48 hr**	§17
After general anaesthetic	**48 hr**	§78
After major dental work	**48 – 72 hr**	§72.2

Master "Pressure-Altitude" reference

Key altitude thresholds across the chapter

Threshold Event	Altitude	Source
ISA SL pressure	1013.25 hPa	§3
Maximum altitude without O_2 — no efficiency loss	**8,000 ft**	§19.1
Maximum cabin altitude (commercial pressurised)	**8,000 ft**	§19.4
Typical commercial cabin altitude at FL300	**≈ 6,000 ft**	§19.4
Smoker's effective physiological altitude	**≈ 7,000 ft cabin = 10,000 ft physiology**	§13
Low DCS risk altitude — below	**10,000 ft**	§19.1
Physiologically Deficient Zone begins	12,000 ft	§5
Atmospheric pressure halved at	**18,000 ft**	§19.2
Unpressurised → bends threshold	**> 25,000 ft**	§19.2
Armstrong's line / water boiling at body temp	63,000 ft	§5
Partial Space-Equivalent Zone	50,000 ft → 120 nm	§5
Total Space-Equivalent Zone	> 120 nm	§5
Dark adaptation impaired above cabin altitude	**5,000 ft**	§31.2

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Glossary

Plain meanings of the words used in regulations. Terms are in alphabetical order.

AAI
Airports Authority of India. It runs most Indian airports and provides air traffic services.

AAIB
Aircraft Accident Investigation Bureau. The Indian body that investigates aircraft accidents and serious incidents.

Accede
To join a treaty that other States have already signed.

Accident
An event linked to flying in which a person is killed or seriously injured, the aircraft is seriously damaged, or the aircraft is missing.

ADS-B
Automatic Dependent Surveillance - Broadcast. The aircraft sends its GNSS position and identity, which ground stations can use like radar.

Aerial work
Flying for work such as survey, photography, agriculture or advertising.

Aerodrome
An area of land or water, with its buildings and equipment, used for aircraft arriving, departing and moving on the surface.

Aerodrome traffic
All traffic on the manoeuvring area of an aerodrome and all aircraft flying in the vicinity of it.

AIC
Aeronautical Information Circular. A notice that gives advice or explanations but is not a NOTAM or an AIP change.

AIP
Aeronautical Information Publication. The official book of permanent aviation information for a country. India publishes it online as the eAIP.

AIRAC
Aeronautical Information Regulation and Control. A system in which important changes are published in advance and all take effect on the same fixed dates, every 28 days.

Airspace class
ICAO divides airspace into classes A to G. The class tells you which flights are allowed, which are separated from each other, and what service you get.

Airworthiness
The state of an aircraft being fit to fly safely.

ALERFA
The code word for the alert phase, when there is fear for an aircraft's safety.

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Alerting service
The service that notifies organisations when an aircraft needs search and rescue, and helps them.

Altitude
The vertical distance of a level above mean sea level.

Alveoli
Air sacs where oxygen moves into the blood and carbon dioxide moves out.

Annex
One of the 19 documents attached to the Chicago Convention. Each Annex covers one subject, for example Annex 1 covers licences and Annex 2 covers the Rules of the Air.

Appropriate authority
The government body that is responsible for the matter in question, for example the DGCA for Indian flight operations.

Apron
The area where aircraft stand for loading and unloading, refuelling, parking or maintenance.

ASDA
Accelerate-Stop Distance Available. TORA plus any stopway.

ASHTAM
A special NOTAM series about volcanic activity, an eruption or an ash cloud.

ATC
Air Traffic Control. The service that keeps aircraft safely apart and moving in an orderly way.

ATIS
A continuous recorded broadcast of routine information for arriving and departing aircraft at one aerodrome.

Atrium
A receiving chamber of the heart. The right atrium receives blood returning from the body; the left receives blood from the lungs.

ATS
Air Traffic Services. A general name for air traffic control, flight information service and alerting service.

ATS route
A defined route in airspace designed to channel the flow of traffic.

Barrette
Three or more aeronautical ground lights placed close together in a line so that they look like a short bar of light.

BCAS
Bureau of Civil Aviation Security. The Indian body responsible for aviation security rules.

Bilateral agreement
An agreement between two countries, for example about the air services their airlines may fly between them.

Bronchi
The large airways branching from the trachea into the lungs.

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Bronchioles
Small airways that lead towards the air sacs in the lungs.

Cabotage
The right to carry passengers, mail or cargo for payment between two points in the same country. Most countries keep this right for their own airlines only.

Cancel
To end something officially, for example to cancel a flight plan or a registration.

CAR
Civil Aviation Requirement. A detailed rule issued by the DGCA under the Aircraft Rules. CARs are grouped into sections, for example Section 7 for flight crew.

Certificate of airworthiness
The document that says an aircraft is fit to fly safely. It is issued by the State of Registry.

Certificate of registration
The document that proves an aircraft is entered on a country's register and shows its owner.

Clearance
Permission from air traffic control for an aircraft to proceed under stated conditions.

Clearway
An area beyond the runway, under the airport authority's control, over which an aeroplane may make part of its initial climb.

Compliance
Doing what the rule requires.

Contracting State
A country that has joined the Chicago Convention. India is a Contracting State.

Contravention
Breaking a rule or law. To contravene a rule means to act against it.

Control area
Controlled airspace that begins at a specified height above the earth. Abbreviated CTA.

Control zone
Controlled airspace around an aerodrome, from the surface up to a stated limit. Abbreviated CTR.

Controlled airspace
Airspace in which air traffic control service is provided according to the airspace class.

Convention
A written agreement between States, also called a treaty. The Chicago Convention of 1944 is the most important one in aviation.

Crew member certificate
A card that identifies aircrew at borders. It is not a licence.

Cross-country flight
A flight to a point beyond a stated distance from the aerodrome of departure. The distance depends on the licence rules.

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Dangerous goods
Articles or substances that could endanger health, safety or property when carried by air.

Decompression
A fall in cabin pressure. Slow decompression builds over time. Rapid decompression is sudden.

Derogation
An official permission to depart from a rule.

DETRESFA
The code word for the distress phase, when it is probable that an aircraft is in grave danger.

DGCA
Directorate General of Civil Aviation. India's regulator for civil aviation safety.

Differences
Where a State's own rules do not match an ICAO Standard. Under Article 38 of the Chicago Convention a State must tell ICAO about every difference.

Displaced threshold
A threshold placed further along the runway than its physical beginning. The part before it may be used for take-off and taxiing, but not for landing.

Elevation
The vertical distance of a point on the ground above mean sea level.

Endorsement
An official note written on a licence or certificate that adds a privilege, a rating or a condition, for example a type rating added to a pilot licence.

Essential traffic
Controlled traffic that is not separated from a particular flight by the required minimum, so the pilot must be told about it.

Exemption
Official permission not to follow a rule. An exempt person is excused from it.

Expected approach time
The time at which air traffic control expects an arriving aircraft to leave the holding fix to complete its approach.

Facilitation
Making the passage of aircraft, passengers, crew and cargo across borders smooth and fast.

Flight duty period
The total time that starts when a crew member must report for duty and ends when the aircraft finally stops after the last flight.

Flight information region
A defined block of airspace in which flight information service and alerting service are provided. Abbreviated FIR. India has four: Delhi, Mumbai, Kolkata and Chennai.

Flight level
A level of constant pressure measured with the altimeter set to 1013.2 hPa. FL 100 means 10,000 feet on the standard setting.

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Flight plan
The information about a planned flight that is given to air traffic services.

Flight time
The time from when an aircraft first moves under its own power to take off until it comes to rest at the end of the flight.

FOD
Foreign Object Debris. Loose objects on the runway or apron that could damage an aircraft.

Fuel jettison
Releasing fuel from the aircraft in flight to reduce its weight.

Gazette
The official government publication in which new laws and rules are printed.

General declaration
The standard form that gives border officials the basic details of a flight, such as route, crew and passenger numbers.

Halon
A gas used in aircraft fire extinguishers that stops flames effectively.

Height
The vertical distance of a level above a stated point on the ground.

Holding
Flying a set pattern over a point to wait, for example for clearance to land.

Hydroplaning
When tyres lose contact with the runway because a layer of water lifts them. Braking and steering are then very poor.

Hyperventilation
Breathing more than the body needs removes too much carbon dioxide. Its early symptoms can resemble those of hypoxia.

Hypoxia
A shortage of oxygen reaching body tissues. It can affect judgement and lead to unconsciousness.

ICAO
International Civil Aviation Organization. The United Nations agency, created by the Chicago Convention, that sets worldwide standards for civil aviation.

IFR
Instrument Flight Rules. Rules for flying by reference to instruments, allowed in any weather when the aircraft and pilot are qualified.

IMC
Weather below the minimum for VFR flight.

In lieu of
A formal way of saying "instead of".

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Inadvertent
Not done on purpose. An inadvertent change is a change made by mistake.

Incapacity
Being unable to carry out your duties, for example a pilot who is too ill to fly.

INCERFA
The code word for the uncertainty phase, when there is doubt about an aircraft's safety.

Incident
An event, not an accident, that affects or could affect the safety of flight.

Instrument rating
An entry on a licence that allows the holder to fly under IFR.

Interception
When another aircraft, often military, approaches yours to identify it or to guide it.

Jurisdiction
The legal power of a State or court to make decisions and enforce the law over a person, a place, or an event.

Larynx
The part of the airway containing the vocal cords. It lies before the trachea in the route taken by inhaled air.

Lateral separation
Separation by keeping aircraft on different routes or in different areas.

LDA
Landing Distance Available. The length of runway available for the ground run of a landing aeroplane.

Lessor / lessee
In a lease, the lessor is the party that provides the aircraft and the lessee is the party that uses it.

Liability
Legal responsibility. If someone is liable for a loss, the law can require them to pay for it.

Longitudinal separation
Separation between aircraft one behind the other, measured in time or distance.

Mandatory
Compulsory. You must do it, with no choice.

Manoeuvring area
The part of an aerodrome used for take-off, landing and taxiing. It does not include the apron.

MAYDAY
The radio distress signal, said three times, for an aircraft in grave and imminent danger that needs immediate help.

Microburst
A small, intense downdraught that spreads out near the ground, causing strong windshear.

Minimum equipment list
The operator's list of items that may be unserviceable when the aircraft takes off, and the conditions for doing so. Abbreviated MEL.

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Movement area
The manoeuvring area together with the apron.

Nationality mark
The letters painted on an aircraft that show which State it is registered in. Indian aircraft carry VT.

NOTAM
A notice sent by telecommunication with information about a change in a facility, service, procedure or hazard that pilots must know about in time.

Notwithstanding
A legal word that means "in spite of". "Notwithstanding anything in this rule" means the sentence that follows still applies even if another rule says something different.

Obstacle
Any fixed or moving object that is on, or sticks up above, a surface intended for aircraft, or that could be dangerous to flying.

Offence
An act that the law forbids and punishes.

Operations manual
The book in which an operator writes down how its flights are to be run. The DGCA approves it.

PAN PAN
The radio urgency signal, said three times, for a serious problem that does not need immediate help.

PANS
Procedures for Air Navigation Services. ICAO procedures that are less binding than Annexes, for example Doc 4444 (air traffic management) and Doc 8168 (instrument flight procedures).

PAPI
A row of four lights beside the runway. White and red lights tell the pilot whether the aircraft is above, on or below the correct approach slope.

Penalty
The punishment for breaking a rule, such as a fine or imprisonment.

Pharynx
The passage behind the nose and mouth. Air passes from it through the larynx into the trachea.

Pilot-in-command
The pilot who is responsible for the safe operation of the aircraft during the flight. In everyday speech, the captain.

Prescribed
Laid down by the rules or by the authority. "The prescribed form" means the form that the rules require.

Privileges
The things a licence holder is allowed to do. A private pilot licence, for example, allows flying without payment.

Protocol
A document that changes or adds to a treaty.

Proviso
A short exception or condition added to a rule, usually starting "Provided that". It limits the rule above it.

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Public transport
Carrying passengers or cargo for hire or reward.

Pulmonary
A word meaning connected with the lungs.

Pursuant to
A formal way of saying "according to" or "under".

QFE
The altimeter setting that makes the altimeter read zero on the aerodrome.

QNE
The standard setting of 1013.2 hPa, used for flight levels.

QNH
The altimeter setting that makes the altimeter show height above mean sea level.

Quarantine
Keeping people or goods apart so that a disease cannot spread.

Radar vectoring
Providing aircraft with headings so that they follow a desired track.

Ratify
To approve a treaty formally, so that the State is bound by it. A State may sign a treaty but not ratify it, and then it is not bound.

Recommended practice
A specification that ICAO asks States to follow, but which is not compulsory.

Rectification
Putting right a defect or mistake. When an aircraft part is faulty, the repair is called rectification.

Registration mark
The letters that follow the nationality mark and identify the individual aircraft, for example EPC in VT-EPC.

Regulation
A detailed rule made under the authority of a law.

Rescue coordination centre
A unit that organises search and rescue and coordinates the operation. Abbreviated RCC.

Revoke
To cancel something permanently, for example a licence.

Runway
A rectangular area prepared for aircraft to land and take off.

RVR
Runway Visual Range. How far a pilot on the runway centre line can see the runway markings or lights, measured in metres.

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RVSM
Reduced Vertical Separation Minimum. Between FL 290 and FL 410 aircraft that are approved for RVSM may be separated by 1,000 ft instead of 2,000 ft.

Search and rescue
Looking for aircraft or persons in distress, and helping them.

Separation
The vertical, lateral or longitudinal space or time that air traffic control keeps between aircraft.

Serious incident
An incident in which an accident almost happened.

Shall / should / may
In rules these three words have fixed meanings. "Shall" means it is compulsory. "Should" means it is recommended but not compulsory. "May" means it is allowed but optional.

SNOWTAM
A special NOTAM series giving the state of the runway when there is snow, ice, slush, frost or standing water.

Sovereignty
A State's full and exclusive power over its own land, waters and the air above them.

Special VFR
A VFR flight that air traffic control has cleared to operate within a control zone in weather below the VMC minima.

SSR
Secondary Surveillance Radar. Radar that talks to the aircraft's transponder and displays its code and altitude.

Standard
In ICAO documents, a Standard is a specification that every member State must follow, unless it has told ICAO of a difference.

State of Occurrence
The country in whose territory an accident or incident took place.

State of Registry
The country on whose register the aircraft is entered.

State of the Operator
The country where the operator has its principal place of business.

Statute
A written law made by Parliament.

Stopway
A prepared area at the end of the take-off run where an aircraft can stop if the take-off is abandoned.

Subject to
"Subject to rule 5" means the sentence is limited by rule 5.

Supersede
To replace something older. A new rule supersedes the old one.

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Suspend
To stop something for a period, but not cancel it. A suspended licence cannot be used until it is restored.

Taxiway
A defined path on an aerodrome for aircraft to move from one part to another.

Terminal control area
A control area usually set up where several airways meet near one or more major aerodromes. Abbreviated TMA.

Threshold
The beginning of the part of the runway that can be used for landing.

TODA
Take-off Distance Available. TORA plus any clearway.

TORA
Take-off Run Available. The length of runway available for the ground run of an aeroplane taking off.

Trachea
The airway from the larynx that divides into the main bronchi leading to the lungs.

Transition altitude
The altitude at or below which an aircraft's height is given as an altitude (using QNH). Above it, height is given as a flight level.

Transition layer
The airspace between the transition altitude and the transition level.

Transition level
The lowest flight level available for use above the transition altitude.

Transponder
Equipment in the aircraft that answers an SSR interrogation with a four-digit code and altitude.

Treaty
A formal written agreement between countries.

Type rating
An entry on a licence that allows the holder to fly a particular type of aircraft.

Unlawful interference
Acts or attempts that harm the safety of civil aviation, such as hijacking, sabotage or false bomb threats.

Validity
The period for which a licence, certificate or medical is accepted as good. When the validity ends, the document is no longer valid.

Ventricle
The right ventricle pumps blood to the lungs; the left pumps blood to the body.

Vertical separation
Separation by assigning different levels.

VFR
Visual Flight Rules. Rules for flying with reference to the ground and clear weather.

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VMC
Weather good enough for VFR flight (visibility, distance from cloud and ceiling at or above the stated minimum).

Waiver
An official decision not to apply a requirement in a particular case.

Wake turbulence
The disturbed air trailing a large aircraft, which can upset a smaller aircraft that follows too closely.

Wildlife strike
A collision between an aircraft and a bird or other animal.

Windshear
A sudden change in wind speed or direction over a short distance.

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Syllabus navigation map

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